

Batch 1

Initial Conditions:

Batch: 1.00

X0 (g/L): 1.65

S0 (g/L): 83.34

M0 (g/L): 58.22

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.70

F_vol (L/h): 0.10

S_feed (g/L): 176.62

M_feed (g/L): 123.38

V_final (L): 12.34

Batch 1 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.45
Glucose S (g/L): 83.57
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 57.18
Volume V (L): 4.99

t = 1.00 h

Biomass X (g/L): 0.88
Glucose S (g/L): 84.88
Lactic Acid P (g/L): 2.47
Maltose M (g/L): 59.33
Volume V (L): 5.17

t = 2.00 h

Biomass X (g/L): 9.22
Glucose S (g/L): 87.95
Lactic Acid P (g/L): 3.57
Maltose M (g/L): 57.39
Volume V (L): 5.95

t = 3.00 h

Biomass X (g/L): 18.66
Glucose S (g/L): 84.29
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 58.84
Volume V (L): 4.95

t = 4.00 h

Biomass X (g/L): 2.95
Glucose S (g/L): 90.26
Lactic Acid P (g/L): 19.06
Maltose M (g/L): 58.64
Volume V (L): 5.71

t = 5.00 h

Biomass X (g/L): 4.57
Glucose S (g/L): 92.71

Lactic Acid P (g/L): 8.01
Maltose M (g/L): 57.66
Volume V (L): 5.37

t = 6.00 h

Biomass X (g/L): 24.11
Glucose S (g/L): 96.11
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 61.71
Volume V (L): 5.46

t = 7.00 h

Biomass X (g/L): 18.73
Glucose S (g/L): 90.90
Lactic Acid P (g/L): 12.89
Maltose M (g/L): 59.51
Volume V (L): 6.33

t = 8.00 h

Biomass X (g/L): 9.69
Glucose S (g/L): 89.60
Lactic Acid P (g/L): 0.69
Maltose M (g/L): 55.63
Volume V (L): 6.27

t = 9.00 h

Biomass X (g/L): 22.83
Glucose S (g/L): 89.35
Lactic Acid P (g/L): 18.63
Maltose M (g/L): 58.78
Volume V (L): 6.36

t = 10.00 h

Biomass X (g/L): 16.84
Glucose S (g/L): 91.53
Lactic Acid P (g/L): 24.63
Maltose M (g/L): 57.49
Volume V (L): 6.74

t = 11.00 h

Biomass X (g/L): 20.83

Glucose S (g/L): 88.15
Lactic Acid P (g/L): 10.18
Maltose M (g/L): 59.60
Volume V (L): 6.09

t = 12.00 h

Biomass X (g/L): 31.85
Glucose S (g/L): 83.83
Lactic Acid P (g/L): 28.95
Maltose M (g/L): 57.13
Volume V (L): 6.57

t = 13.00 h

Biomass X (g/L): 15.27
Glucose S (g/L): 84.31
Lactic Acid P (g/L): 27.26
Maltose M (g/L): 58.00
Volume V (L): 6.09

t = 14.00 h

Biomass X (g/L): 21.28
Glucose S (g/L): 80.99
Lactic Acid P (g/L): 33.98
Maltose M (g/L): 58.78
Volume V (L): 6.56

t = 15.00 h

Biomass X (g/L): 36.55
Glucose S (g/L): 81.07
Lactic Acid P (g/L): 46.46
Maltose M (g/L): 59.17
Volume V (L): 6.39

t = 16.00 h

Biomass X (g/L): 36.03
Glucose S (g/L): 74.70
Lactic Acid P (g/L): 30.54
Maltose M (g/L): 56.04
Volume V (L): 6.62

t = 17.00 h

Biomass X (g/L): 52.44
Glucose S (g/L): 73.81
Lactic Acid P (g/L): 28.82
Maltose M (g/L): 57.14
Volume V (L): 7.01

t = 18.00 h

Biomass X (g/L): 43.95
Glucose S (g/L): 71.98
Lactic Acid P (g/L): 30.19
Maltose M (g/L): 56.78
Volume V (L): 6.28

t = 19.00 h

Biomass X (g/L): 42.12
Glucose S (g/L): 67.75
Lactic Acid P (g/L): 33.23
Maltose M (g/L): 56.36
Volume V (L): 8.09

t = 20.00 h

Biomass X (g/L): 72.69
Glucose S (g/L): 71.08
Lactic Acid P (g/L): 41.93
Maltose M (g/L): 59.69
Volume V (L): 6.37

t = 21.00 h

Biomass X (g/L): 58.87
Glucose S (g/L): 69.94
Lactic Acid P (g/L): 47.64
Maltose M (g/L): 57.53
Volume V (L): 6.34

t = 22.00 h

Biomass X (g/L): 63.98
Glucose S (g/L): 68.42
Lactic Acid P (g/L): 48.95
Maltose M (g/L): 54.93
Volume V (L): 7.83

t = 23.00 h

Biomass X (g/L): 51.62
Glucose S (g/L): 67.11
Lactic Acid P (g/L): 55.51
Maltose M (g/L): 57.89
Volume V (L): 7.72

t = 24.00 h

Biomass X (g/L): 61.97
Glucose S (g/L): 63.51
Lactic Acid P (g/L): 49.95
Maltose M (g/L): 59.44
Volume V (L): 7.71

t = 25.00 h

Biomass X (g/L): 69.97
Glucose S (g/L): 65.70
Lactic Acid P (g/L): 64.07
Maltose M (g/L): 57.66
Volume V (L): 7.81

t = 26.00 h

Biomass X (g/L): 59.28
Glucose S (g/L): 65.63
Lactic Acid P (g/L): 50.32
Maltose M (g/L): 53.75
Volume V (L): 7.54

t = 27.00 h

Biomass X (g/L): 75.38
Glucose S (g/L): 64.28
Lactic Acid P (g/L): 77.80
Maltose M (g/L): 55.03
Volume V (L): 7.12

t = 28.00 h

Biomass X (g/L): 67.16
Glucose S (g/L): 65.89
Lactic Acid P (g/L): 60.54
Maltose M (g/L): 57.35
Volume V (L): 7.78

t = 29.00 h

Biomass X (g/L): 71.24
Glucose S (g/L): 67.39
Lactic Acid P (g/L): 48.57
Maltose M (g/L): 54.27
Volume V (L): 7.44

t = 30.00 h

Biomass X (g/L): 69.22
Glucose S (g/L): 71.34
Lactic Acid P (g/L): 47.68
Maltose M (g/L): 55.71
Volume V (L): 8.51

t = 31.00 h

Biomass X (g/L): 93.81
Glucose S (g/L): 67.07
Lactic Acid P (g/L): 62.26
Maltose M (g/L): 55.96
Volume V (L): 7.95

t = 32.00 h

Biomass X (g/L): 76.57
Glucose S (g/L): 67.52
Lactic Acid P (g/L): 56.91
Maltose M (g/L): 53.27
Volume V (L): 7.65

t = 33.00 h

Biomass X (g/L): 67.19
Glucose S (g/L): 66.95
Lactic Acid P (g/L): 65.96
Maltose M (g/L): 54.29
Volume V (L): 8.04

t = 34.00 h

Biomass X (g/L): 85.99
Glucose S (g/L): 62.50
Lactic Acid P (g/L): 64.59
Maltose M (g/L): 49.44

Volume V (L): 8.57

t = 35.00 h

Biomass X (g/L): 66.82

Glucose S (g/L): 67.79

Lactic Acid P (g/L): 60.49

Maltose M (g/L): 52.41

Volume V (L): 8.09

t = 36.00 h

Biomass X (g/L): 81.13

Glucose S (g/L): 68.43

Lactic Acid P (g/L): 54.35

Maltose M (g/L): 53.29

Volume V (L): 8.04

t = 37.00 h

Biomass X (g/L): 60.63

Glucose S (g/L): 75.11

Lactic Acid P (g/L): 49.09

Maltose M (g/L): 51.61

Volume V (L): 8.76

t = 38.00 h

Biomass X (g/L): 67.13

Glucose S (g/L): 68.68

Lactic Acid P (g/L): 59.07

Maltose M (g/L): 55.53

Volume V (L): 8.86

t = 39.00 h

Biomass X (g/L): 82.22

Glucose S (g/L): 70.46

Lactic Acid P (g/L): 71.08

Maltose M (g/L): 50.85

Volume V (L): 8.52

t = 40.00 h

Biomass X (g/L): 87.77

Glucose S (g/L): 70.11

Lactic Acid P (g/L): 65.95

Maltose M (g/L): 52.04

Volume V (L): 8.64

t = 41.00 h

Biomass X (g/L): 82.56

Glucose S (g/L): 67.70

Lactic Acid P (g/L): 53.61

Maltose M (g/L): 52.62

Volume V (L): 9.15

t = 42.00 h

Biomass X (g/L): 80.03

Glucose S (g/L): 74.25

Lactic Acid P (g/L): 66.55

Maltose M (g/L): 54.27

Volume V (L): 8.26

t = 43.00 h

Biomass X (g/L): 78.45

Glucose S (g/L): 73.79

Lactic Acid P (g/L): 68.86

Maltose M (g/L): 49.85

Volume V (L): 8.38

t = 44.00 h

Biomass X (g/L): 67.25

Glucose S (g/L): 74.47

Lactic Acid P (g/L): 58.12

Maltose M (g/L): 53.37

Volume V (L): 8.88

t = 45.00 h

Biomass X (g/L): 74.73

Glucose S (g/L): 70.64

Lactic Acid P (g/L): 67.64

Maltose M (g/L): 51.45

Volume V (L): 9.28

t = 46.00 h

Biomass X (g/L): 77.36

Glucose S (g/L): 77.23

Lactic Acid P (g/L): 67.19

Maltose M (g/L): 49.76

Volume V (L): 9.68

t = 47.00 h

Biomass X (g/L): 92.13

Glucose S (g/L): 70.55

Lactic Acid P (g/L): 56.99

Maltose M (g/L): 51.60

Volume V (L): 10.46

t = 48.00 h

Biomass X (g/L): 85.32

Glucose S (g/L): 76.32

Lactic Acid P (g/L): 70.53

Maltose M (g/L): 50.97

Volume V (L): 10.20

t = 49.00 h

Biomass X (g/L): 65.03

Glucose S (g/L): 81.09

Lactic Acid P (g/L): 72.65

Maltose M (g/L): 49.56

Volume V (L): 9.70

t = 50.00 h

Biomass X (g/L): 85.25

Glucose S (g/L): 73.44

Lactic Acid P (g/L): 77.56

Maltose M (g/L): 50.28

Volume V (L): 9.88

t = 51.00 h

Biomass X (g/L): 78.43

Glucose S (g/L): 75.16

Lactic Acid P (g/L): 77.61

Maltose M (g/L): 49.37

Volume V (L): 9.40

t = 52.00 h

Biomass X (g/L): 75.62

Glucose S (g/L): 77.50
Lactic Acid P (g/L): 56.53
Maltose M (g/L): 49.84
Volume V (L): 10.08

t = 53.00 h

Biomass X (g/L): 88.07
Glucose S (g/L): 76.55
Lactic Acid P (g/L): 60.68
Maltose M (g/L): 50.38
Volume V (L): 10.02

t = 54.00 h

Biomass X (g/L): 92.12
Glucose S (g/L): 74.45
Lactic Acid P (g/L): 73.72
Maltose M (g/L): 51.47
Volume V (L): 10.47

t = 55.00 h

Biomass X (g/L): 91.13
Glucose S (g/L): 79.26
Lactic Acid P (g/L): 73.97
Maltose M (g/L): 47.13
Volume V (L): 9.90

t = 56.00 h

Biomass X (g/L): 73.99
Glucose S (g/L): 76.94
Lactic Acid P (g/L): 74.24
Maltose M (g/L): 51.77
Volume V (L): 10.78

t = 57.00 h

Biomass X (g/L): 79.07
Glucose S (g/L): 81.54
Lactic Acid P (g/L): 103.82
Maltose M (g/L): 45.60
Volume V (L): 11.48

t = 58.00 h

Biomass X (g/L): 85.20
Glucose S (g/L): 78.54
Lactic Acid P (g/L): 75.21
Maltose M (g/L): 47.96
Volume V (L): 10.61

t = 59.00 h

Biomass X (g/L): 91.37
Glucose S (g/L): 85.55
Lactic Acid P (g/L): 80.40
Maltose M (g/L): 48.79
Volume V (L): 11.01

t = 60.00 h

Biomass X (g/L): 77.24
Glucose S (g/L): 80.10
Lactic Acid P (g/L): 79.02
Maltose M (g/L): 48.10
Volume V (L): 11.28

t = 61.00 h

Biomass X (g/L): 80.00
Glucose S (g/L): 81.90
Lactic Acid P (g/L): 76.57
Maltose M (g/L): 46.55
Volume V (L): 10.73

t = 62.00 h

Biomass X (g/L): 71.02
Glucose S (g/L): 85.45
Lactic Acid P (g/L): 68.28
Maltose M (g/L): 46.90
Volume V (L): 11.20

t = 63.00 h

Biomass X (g/L): 70.06
Glucose S (g/L): 80.74
Lactic Acid P (g/L): 77.92
Maltose M (g/L): 46.25
Volume V (L): 11.17

t = 64.00 h

Biomass X (g/L): 89.37
Glucose S (g/L): 85.11
Lactic Acid P (g/L): 64.65
Maltose M (g/L): 45.86
Volume V (L): 11.32

t = 65.00 h

Biomass X (g/L): 94.52
Glucose S (g/L): 88.50
Lactic Acid P (g/L): 69.55
Maltose M (g/L): 47.71
Volume V (L): 10.91

t = 66.00 h

Biomass X (g/L): 80.61
Glucose S (g/L): 81.52
Lactic Acid P (g/L): 67.55
Maltose M (g/L): 46.75
Volume V (L): 11.47

t = 67.00 h

Biomass X (g/L): 90.89
Glucose S (g/L): 86.75
Lactic Acid P (g/L): 72.71
Maltose M (g/L): 44.99
Volume V (L): 11.85

t = 68.00 h

Biomass X (g/L): 84.58
Glucose S (g/L): 87.52
Lactic Acid P (g/L): 92.52
Maltose M (g/L): 47.06
Volume V (L): 12.51

t = 69.00 h

Biomass X (g/L): 74.73
Glucose S (g/L): 89.45
Lactic Acid P (g/L): 55.92
Maltose M (g/L): 45.97
Volume V (L): 12.31

t = 70.00 h

Biomass X (g/L): 84.33
Glucose S (g/L): 84.77
Lactic Acid P (g/L): 78.53
Maltose M (g/L): 46.46
Volume V (L): 13.11

t = 71.00 h

Biomass X (g/L): 95.57
Glucose S (g/L): 85.12
Lactic Acid P (g/L): 58.48
Maltose M (g/L): 45.96
Volume V (L): 11.50

t = 72.00 h

Biomass X (g/L): 80.23
Glucose S (g/L): 90.46
Lactic Acid P (g/L): 68.67
Maltose M (g/L): 43.62
Volume V (L): 12.56

t = 73.00 h

Biomass X (g/L): 95.56
Glucose S (g/L): 90.43
Lactic Acid P (g/L): 82.54
Maltose M (g/L): 43.78
Volume V (L): 12.25

t = 74.00 h

Biomass X (g/L): 54.99
Glucose S (g/L): 90.85
Lactic Acid P (g/L): 73.68
Maltose M (g/L): 45.44
Volume V (L): 13.53

t = 75.00 h

Biomass X (g/L): 88.10
Glucose S (g/L): 91.65
Lactic Acid P (g/L): 63.79
Maltose M (g/L): 45.01

Volume V (L): 11.86

Batch 2

Initial Conditions:

Batch: 2.00

X0 (g/L): 5.85

S0 (g/L): 113.25

M0 (g/L): 37.78

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.33

F_vol (L/h): 0.07

S_feed (g/L): 224.96

M_feed (g/L): 75.04

V_final (L): 10.29

Batch 2 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 112.84
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 35.43
Volume V (L): 4.48

t = 1.00 h

Biomass X (g/L): 2.09
Glucose S (g/L): 109.49
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 39.54
Volume V (L): 5.37

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 110.87
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 38.23
Volume V (L): 5.04

t = 3.00 h

Biomass X (g/L): 8.60
Glucose S (g/L): 107.34
Lactic Acid P (g/L): 6.27
Maltose M (g/L): 37.08
Volume V (L): 5.05

t = 4.00 h

Biomass X (g/L): 10.12
Glucose S (g/L): 117.59
Lactic Acid P (g/L): 0.77
Maltose M (g/L): 37.31
Volume V (L): 5.58

t = 5.00 h

Biomass X (g/L): 23.24
Glucose S (g/L): 107.99

Lactic Acid P (g/L): 8.29
Maltose M (g/L): 35.25
Volume V (L): 5.54

t = 6.00 h

Biomass X (g/L): 29.77
Glucose S (g/L): 102.46
Lactic Acid P (g/L): 24.45
Maltose M (g/L): 39.75
Volume V (L): 5.27

t = 7.00 h

Biomass X (g/L): 49.56
Glucose S (g/L): 102.88
Lactic Acid P (g/L): 13.63
Maltose M (g/L): 36.49
Volume V (L): 5.98

t = 8.00 h

Biomass X (g/L): 45.53
Glucose S (g/L): 98.48
Lactic Acid P (g/L): 30.22
Maltose M (g/L): 36.26
Volume V (L): 5.11

t = 9.00 h

Biomass X (g/L): 35.57
Glucose S (g/L): 94.94
Lactic Acid P (g/L): 15.92
Maltose M (g/L): 36.86
Volume V (L): 5.90

t = 10.00 h

Biomass X (g/L): 37.09
Glucose S (g/L): 95.11
Lactic Acid P (g/L): 34.58
Maltose M (g/L): 36.33
Volume V (L): 5.96

t = 11.00 h

Biomass X (g/L): 54.92

Glucose S (g/L): 92.88
Lactic Acid P (g/L): 46.22
Maltose M (g/L): 35.79
Volume V (L): 5.65

t = 12.00 h

Biomass X (g/L): 41.34
Glucose S (g/L): 85.50
Lactic Acid P (g/L): 13.59
Maltose M (g/L): 34.24
Volume V (L): 5.98

t = 13.00 h

Biomass X (g/L): 75.56
Glucose S (g/L): 83.12
Lactic Acid P (g/L): 31.74
Maltose M (g/L): 35.70
Volume V (L): 5.39

t = 14.00 h

Biomass X (g/L): 72.53
Glucose S (g/L): 82.32
Lactic Acid P (g/L): 46.91
Maltose M (g/L): 37.36
Volume V (L): 6.38

t = 15.00 h

Biomass X (g/L): 59.51
Glucose S (g/L): 72.99
Lactic Acid P (g/L): 42.19
Maltose M (g/L): 36.45
Volume V (L): 5.98

t = 16.00 h

Biomass X (g/L): 50.11
Glucose S (g/L): 74.57
Lactic Acid P (g/L): 49.63
Maltose M (g/L): 36.57
Volume V (L): 5.91

t = 17.00 h

Biomass X (g/L): 82.23
Glucose S (g/L): 84.31
Lactic Acid P (g/L): 42.73
Maltose M (g/L): 33.59
Volume V (L): 6.64

t = 18.00 h

Biomass X (g/L): 70.11
Glucose S (g/L): 84.39
Lactic Acid P (g/L): 50.86
Maltose M (g/L): 32.76
Volume V (L): 5.97

t = 19.00 h

Biomass X (g/L): 85.11
Glucose S (g/L): 76.44
Lactic Acid P (g/L): 50.32
Maltose M (g/L): 33.70
Volume V (L): 5.74

t = 20.00 h

Biomass X (g/L): 59.33
Glucose S (g/L): 78.95
Lactic Acid P (g/L): 63.94
Maltose M (g/L): 33.74
Volume V (L): 5.75

t = 21.00 h

Biomass X (g/L): 70.48
Glucose S (g/L): 77.47
Lactic Acid P (g/L): 57.00
Maltose M (g/L): 34.91
Volume V (L): 6.74

t = 22.00 h

Biomass X (g/L): 77.67
Glucose S (g/L): 87.61
Lactic Acid P (g/L): 59.05
Maltose M (g/L): 31.04
Volume V (L): 6.01

t = 23.00 h

Biomass X (g/L): 79.27
Glucose S (g/L): 79.96
Lactic Acid P (g/L): 53.40
Maltose M (g/L): 35.09
Volume V (L): 7.37

t = 24.00 h

Biomass X (g/L): 75.50
Glucose S (g/L): 75.10
Lactic Acid P (g/L): 53.81
Maltose M (g/L): 32.67
Volume V (L): 5.81

t = 25.00 h

Biomass X (g/L): 86.84
Glucose S (g/L): 71.92
Lactic Acid P (g/L): 55.33
Maltose M (g/L): 32.12
Volume V (L): 7.48

t = 26.00 h

Biomass X (g/L): 71.28
Glucose S (g/L): 69.49
Lactic Acid P (g/L): 63.82
Maltose M (g/L): 31.16
Volume V (L): 6.92

t = 27.00 h

Biomass X (g/L): 80.99
Glucose S (g/L): 76.44
Lactic Acid P (g/L): 57.25
Maltose M (g/L): 30.12
Volume V (L): 6.86

t = 28.00 h

Biomass X (g/L): 84.19
Glucose S (g/L): 72.96
Lactic Acid P (g/L): 64.56
Maltose M (g/L): 33.20
Volume V (L): 6.75

t = 29.00 h

Biomass X (g/L): 88.59
Glucose S (g/L): 70.65
Lactic Acid P (g/L): 81.62
Maltose M (g/L): 32.02
Volume V (L): 7.22

t = 30.00 h

Biomass X (g/L): 91.01
Glucose S (g/L): 73.84
Lactic Acid P (g/L): 71.34
Maltose M (g/L): 30.03
Volume V (L): 7.10

t = 31.00 h

Biomass X (g/L): 73.64
Glucose S (g/L): 72.49
Lactic Acid P (g/L): 61.07
Maltose M (g/L): 29.84
Volume V (L): 7.65

t = 32.00 h

Biomass X (g/L): 70.06
Glucose S (g/L): 83.31
Lactic Acid P (g/L): 75.64
Maltose M (g/L): 30.92
Volume V (L): 7.31

t = 33.00 h

Biomass X (g/L): 97.67
Glucose S (g/L): 80.62
Lactic Acid P (g/L): 61.54
Maltose M (g/L): 33.38
Volume V (L): 7.39

t = 34.00 h

Biomass X (g/L): 88.79
Glucose S (g/L): 77.63
Lactic Acid P (g/L): 47.21
Maltose M (g/L): 30.66

Volume V (L): 7.25

t = 35.00 h

Biomass X (g/L): 78.56

Glucose S (g/L): 83.68

Lactic Acid P (g/L): 57.14

Maltose M (g/L): 28.83

Volume V (L): 7.45

t = 36.00 h

Biomass X (g/L): 101.07

Glucose S (g/L): 78.80

Lactic Acid P (g/L): 49.76

Maltose M (g/L): 30.31

Volume V (L): 7.67

t = 37.00 h

Biomass X (g/L): 87.32

Glucose S (g/L): 75.69

Lactic Acid P (g/L): 64.10

Maltose M (g/L): 30.10

Volume V (L): 6.89

t = 38.00 h

Biomass X (g/L): 97.78

Glucose S (g/L): 85.19

Lactic Acid P (g/L): 67.92

Maltose M (g/L): 26.72

Volume V (L): 7.11

t = 39.00 h

Biomass X (g/L): 87.11

Glucose S (g/L): 81.94

Lactic Acid P (g/L): 83.48

Maltose M (g/L): 30.03

Volume V (L): 8.07

t = 40.00 h

Biomass X (g/L): 106.53

Glucose S (g/L): 76.46

Lactic Acid P (g/L): 71.56

Maltose M (g/L): 29.62

Volume V (L): 7.90

t = 41.00 h

Biomass X (g/L): 103.64

Glucose S (g/L): 80.16

Lactic Acid P (g/L): 66.96

Maltose M (g/L): 30.67

Volume V (L): 7.82

t = 42.00 h

Biomass X (g/L): 84.24

Glucose S (g/L): 78.08

Lactic Acid P (g/L): 76.90

Maltose M (g/L): 32.02

Volume V (L): 7.97

t = 43.00 h

Biomass X (g/L): 96.10

Glucose S (g/L): 76.67

Lactic Acid P (g/L): 49.48

Maltose M (g/L): 30.90

Volume V (L): 8.18

t = 44.00 h

Biomass X (g/L): 92.94

Glucose S (g/L): 85.93

Lactic Acid P (g/L): 72.15

Maltose M (g/L): 28.89

Volume V (L): 7.88

t = 45.00 h

Biomass X (g/L): 99.94

Glucose S (g/L): 90.18

Lactic Acid P (g/L): 77.36

Maltose M (g/L): 27.61

Volume V (L): 7.85

t = 46.00 h

Biomass X (g/L): 77.27

Glucose S (g/L): 78.18

Lactic Acid P (g/L): 57.13

Maltose M (g/L): 32.30

Volume V (L): 8.33

t = 47.00 h

Biomass X (g/L): 93.24

Glucose S (g/L): 86.14

Lactic Acid P (g/L): 81.36

Maltose M (g/L): 28.82

Volume V (L): 7.90

t = 48.00 h

Biomass X (g/L): 96.79

Glucose S (g/L): 82.08

Lactic Acid P (g/L): 74.29

Maltose M (g/L): 28.59

Volume V (L): 8.56

t = 49.00 h

Biomass X (g/L): 69.40

Glucose S (g/L): 83.23

Lactic Acid P (g/L): 67.67

Maltose M (g/L): 28.38

Volume V (L): 7.74

t = 50.00 h

Biomass X (g/L): 74.90

Glucose S (g/L): 83.37

Lactic Acid P (g/L): 77.50

Maltose M (g/L): 28.51

Volume V (L): 8.96

t = 51.00 h

Biomass X (g/L): 66.50

Glucose S (g/L): 82.87

Lactic Acid P (g/L): 92.70

Maltose M (g/L): 27.90

Volume V (L): 8.80

t = 52.00 h

Biomass X (g/L): 83.56

Glucose S (g/L): 86.86
Lactic Acid P (g/L): 73.87
Maltose M (g/L): 27.17
Volume V (L): 8.78

t = 53.00 h

Biomass X (g/L): 93.02
Glucose S (g/L): 84.07
Lactic Acid P (g/L): 74.71
Maltose M (g/L): 27.05
Volume V (L): 9.16

t = 54.00 h

Biomass X (g/L): 100.52
Glucose S (g/L): 88.77
Lactic Acid P (g/L): 68.47
Maltose M (g/L): 27.57
Volume V (L): 9.52

t = 55.00 h

Biomass X (g/L): 86.54
Glucose S (g/L): 88.09
Lactic Acid P (g/L): 65.12
Maltose M (g/L): 26.74
Volume V (L): 9.31

t = 56.00 h

Biomass X (g/L): 101.48
Glucose S (g/L): 87.90
Lactic Acid P (g/L): 80.67
Maltose M (g/L): 26.36
Volume V (L): 8.17

t = 57.00 h

Biomass X (g/L): 72.16
Glucose S (g/L): 85.90
Lactic Acid P (g/L): 65.48
Maltose M (g/L): 27.29
Volume V (L): 8.48

t = 58.00 h

Biomass X (g/L): 68.87
Glucose S (g/L): 87.67
Lactic Acid P (g/L): 74.14
Maltose M (g/L): 26.66
Volume V (L): 8.83

t = 59.00 h

Biomass X (g/L): 84.70
Glucose S (g/L): 93.24
Lactic Acid P (g/L): 69.32
Maltose M (g/L): 28.83
Volume V (L): 9.17

t = 60.00 h

Biomass X (g/L): 88.81
Glucose S (g/L): 92.79
Lactic Acid P (g/L): 78.23
Maltose M (g/L): 23.20
Volume V (L): 9.45

t = 61.00 h

Biomass X (g/L): 84.61
Glucose S (g/L): 87.71
Lactic Acid P (g/L): 77.08
Maltose M (g/L): 27.99
Volume V (L): 9.00

t = 62.00 h

Biomass X (g/L): 64.70
Glucose S (g/L): 92.30
Lactic Acid P (g/L): 83.68
Maltose M (g/L): 28.04
Volume V (L): 9.45

t = 63.00 h

Biomass X (g/L): 83.72
Glucose S (g/L): 95.28
Lactic Acid P (g/L): 69.72
Maltose M (g/L): 23.52
Volume V (L): 9.13

t = 64.00 h

Biomass X (g/L): 71.76
Glucose S (g/L): 86.62
Lactic Acid P (g/L): 72.08
Maltose M (g/L): 25.66
Volume V (L): 9.26

t = 65.00 h

Biomass X (g/L): 90.73
Glucose S (g/L): 95.50
Lactic Acid P (g/L): 65.76
Maltose M (g/L): 25.47
Volume V (L): 8.99

t = 66.00 h

Biomass X (g/L): 87.61
Glucose S (g/L): 91.43
Lactic Acid P (g/L): 70.80
Maltose M (g/L): 23.97
Volume V (L): 9.27

t = 67.00 h

Biomass X (g/L): 74.76
Glucose S (g/L): 96.59
Lactic Acid P (g/L): 78.45
Maltose M (g/L): 24.66
Volume V (L): 9.16

t = 68.00 h

Biomass X (g/L): 78.70
Glucose S (g/L): 92.03
Lactic Acid P (g/L): 82.06
Maltose M (g/L): 24.08
Volume V (L): 9.39

t = 69.00 h

Biomass X (g/L): 73.23
Glucose S (g/L): 88.57
Lactic Acid P (g/L): 66.87
Maltose M (g/L): 27.71
Volume V (L): 10.32

t = 70.00 h

Biomass X (g/L): 82.70
Glucose S (g/L): 89.72
Lactic Acid P (g/L): 83.37
Maltose M (g/L): 26.50
Volume V (L): 9.54

t = 71.00 h

Biomass X (g/L): 92.38
Glucose S (g/L): 96.55
Lactic Acid P (g/L): 87.95
Maltose M (g/L): 26.80
Volume V (L): 11.13

t = 72.00 h

Biomass X (g/L): 73.36
Glucose S (g/L): 97.82
Lactic Acid P (g/L): 79.48
Maltose M (g/L): 25.94
Volume V (L): 10.29

t = 73.00 h

Biomass X (g/L): 87.62
Glucose S (g/L): 93.88
Lactic Acid P (g/L): 92.97
Maltose M (g/L): 23.37
Volume V (L): 10.23

t = 74.00 h

Biomass X (g/L): 77.39
Glucose S (g/L): 100.19
Lactic Acid P (g/L): 68.89
Maltose M (g/L): 24.03
Volume V (L): 9.86

t = 75.00 h

Biomass X (g/L): 74.64
Glucose S (g/L): 91.94
Lactic Acid P (g/L): 64.72
Maltose M (g/L): 25.24

Volume V (L): 10.59

Batch 3

Initial Conditions:

Batch: 3.00

X0 (g/L): 4.14

S0 (g/L): 36.62

M0 (g/L): 94.93

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.59

F_vol (L/h): 0.07

S_feed (g/L): 83.50

M_feed (g/L): 216.50

V_final (L): 10.04

Batch 3 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 41.31

Lactic Acid P (g/L): 2.23

Maltose M (g/L): 94.34

Volume V (L): 4.85

t = 1.00 h

Biomass X (g/L): 6.59

Glucose S (g/L): 34.70

Lactic Acid P (g/L): 15.08

Maltose M (g/L): 94.79

Volume V (L): 4.62

t = 2.00 h

Biomass X (g/L): 31.24

Glucose S (g/L): 41.41

Lactic Acid P (g/L): 1.36

Maltose M (g/L): 95.04

Volume V (L): 4.61

t = 3.00 h

Biomass X (g/L): 8.24

Glucose S (g/L): 38.05

Lactic Acid P (g/L): 6.23

Maltose M (g/L): 88.79

Volume V (L): 5.67

t = 4.00 h

Biomass X (g/L): 22.48

Glucose S (g/L): 38.33

Lactic Acid P (g/L): 11.01

Maltose M (g/L): 93.33

Volume V (L): 5.08

t = 5.00 h

Biomass X (g/L): 7.94

Glucose S (g/L): 39.82

Lactic Acid P (g/L): 9.41
Maltose M (g/L): 91.90
Volume V (L): 5.48

t = 6.00 h

Biomass X (g/L): 17.60
Glucose S (g/L): 33.22
Lactic Acid P (g/L): 6.23
Maltose M (g/L): 90.77
Volume V (L): 5.38

t = 7.00 h

Biomass X (g/L): 38.34
Glucose S (g/L): 36.20
Lactic Acid P (g/L): 14.69
Maltose M (g/L): 92.40
Volume V (L): 5.66

t = 8.00 h

Biomass X (g/L): 19.69
Glucose S (g/L): 33.68
Lactic Acid P (g/L): 25.48
Maltose M (g/L): 97.65
Volume V (L): 5.57

t = 9.00 h

Biomass X (g/L): 46.79
Glucose S (g/L): 32.60
Lactic Acid P (g/L): 9.76
Maltose M (g/L): 94.26
Volume V (L): 5.09

t = 10.00 h

Biomass X (g/L): 40.49
Glucose S (g/L): 29.84
Lactic Acid P (g/L): 23.20
Maltose M (g/L): 90.71
Volume V (L): 6.07

t = 11.00 h

Biomass X (g/L): 32.54

Glucose S (g/L): 34.28
Lactic Acid P (g/L): 18.79
Maltose M (g/L): 93.44
Volume V (L): 5.54

t = 12.00 h

Biomass X (g/L): 47.67
Glucose S (g/L): 22.91
Lactic Acid P (g/L): 38.66
Maltose M (g/L): 95.32
Volume V (L): 5.18

t = 13.00 h

Biomass X (g/L): 54.52
Glucose S (g/L): 21.39
Lactic Acid P (g/L): 17.15
Maltose M (g/L): 93.71
Volume V (L): 5.70

t = 14.00 h

Biomass X (g/L): 54.59
Glucose S (g/L): 19.47
Lactic Acid P (g/L): 28.39
Maltose M (g/L): 91.06
Volume V (L): 6.55

t = 15.00 h

Biomass X (g/L): 37.03
Glucose S (g/L): 18.24
Lactic Acid P (g/L): 39.20
Maltose M (g/L): 88.83
Volume V (L): 6.35

t = 16.00 h

Biomass X (g/L): 47.83
Glucose S (g/L): 16.78
Lactic Acid P (g/L): 52.08
Maltose M (g/L): 92.07
Volume V (L): 5.94

t = 17.00 h

Biomass X (g/L): 54.94
Glucose S (g/L): 21.30
Lactic Acid P (g/L): 39.60
Maltose M (g/L): 90.90
Volume V (L): 6.00

t = 18.00 h

Biomass X (g/L): 58.92
Glucose S (g/L): 21.18
Lactic Acid P (g/L): 37.15
Maltose M (g/L): 91.88
Volume V (L): 6.08

t = 19.00 h

Biomass X (g/L): 67.61
Glucose S (g/L): 14.51
Lactic Acid P (g/L): 60.72
Maltose M (g/L): 90.82
Volume V (L): 7.12

t = 20.00 h

Biomass X (g/L): 65.36
Glucose S (g/L): 13.23
Lactic Acid P (g/L): 32.61
Maltose M (g/L): 91.56
Volume V (L): 6.50

t = 21.00 h

Biomass X (g/L): 52.85
Glucose S (g/L): 16.83
Lactic Acid P (g/L): 27.44
Maltose M (g/L): 86.96
Volume V (L): 6.59

t = 22.00 h

Biomass X (g/L): 70.62
Glucose S (g/L): 13.51
Lactic Acid P (g/L): 52.45
Maltose M (g/L): 91.52
Volume V (L): 6.90

t = 23.00 h

Biomass X (g/L): 74.25
Glucose S (g/L): 16.98
Lactic Acid P (g/L): 45.35
Maltose M (g/L): 88.16
Volume V (L): 6.64

t = 24.00 h

Biomass X (g/L): 75.09
Glucose S (g/L): 15.65
Lactic Acid P (g/L): 41.96
Maltose M (g/L): 92.79
Volume V (L): 6.51

t = 25.00 h

Biomass X (g/L): 81.22
Glucose S (g/L): 10.51
Lactic Acid P (g/L): 58.58
Maltose M (g/L): 85.03
Volume V (L): 6.60

t = 26.00 h

Biomass X (g/L): 80.00
Glucose S (g/L): 19.92
Lactic Acid P (g/L): 55.52
Maltose M (g/L): 91.12
Volume V (L): 6.72

t = 27.00 h

Biomass X (g/L): 77.48
Glucose S (g/L): 20.86
Lactic Acid P (g/L): 49.33
Maltose M (g/L): 86.63
Volume V (L): 6.80

t = 28.00 h

Biomass X (g/L): 73.42
Glucose S (g/L): 13.83
Lactic Acid P (g/L): 43.90
Maltose M (g/L): 83.98
Volume V (L): 7.18

t = 29.00 h

Biomass X (g/L): 59.27
Glucose S (g/L): 14.81
Lactic Acid P (g/L): 64.07
Maltose M (g/L): 84.04
Volume V (L): 6.97

t = 30.00 h

Biomass X (g/L): 79.77
Glucose S (g/L): 17.20
Lactic Acid P (g/L): 62.95
Maltose M (g/L): 84.08
Volume V (L): 7.31

t = 31.00 h

Biomass X (g/L): 78.40
Glucose S (g/L): 17.73
Lactic Acid P (g/L): 57.08
Maltose M (g/L): 85.68
Volume V (L): 7.05

t = 32.00 h

Biomass X (g/L): 79.66
Glucose S (g/L): 19.14
Lactic Acid P (g/L): 75.26
Maltose M (g/L): 85.55
Volume V (L): 7.18

t = 33.00 h

Biomass X (g/L): 65.67
Glucose S (g/L): 23.68
Lactic Acid P (g/L): 49.94
Maltose M (g/L): 82.24
Volume V (L): 6.41

t = 34.00 h

Biomass X (g/L): 68.07
Glucose S (g/L): 17.55
Lactic Acid P (g/L): 46.59
Maltose M (g/L): 83.31

Volume V (L): 7.66

t = 35.00 h

Biomass X (g/L): 88.71

Glucose S (g/L): 16.18

Lactic Acid P (g/L): 54.72

Maltose M (g/L): 88.70

Volume V (L): 7.49

t = 36.00 h

Biomass X (g/L): 78.87

Glucose S (g/L): 14.95

Lactic Acid P (g/L): 61.14

Maltose M (g/L): 87.20

Volume V (L): 7.82

t = 37.00 h

Biomass X (g/L): 86.10

Glucose S (g/L): 17.47

Lactic Acid P (g/L): 64.35

Maltose M (g/L): 83.37

Volume V (L): 6.32

t = 38.00 h

Biomass X (g/L): 80.33

Glucose S (g/L): 20.15

Lactic Acid P (g/L): 60.61

Maltose M (g/L): 81.91

Volume V (L): 8.40

t = 39.00 h

Biomass X (g/L): 80.70

Glucose S (g/L): 26.27

Lactic Acid P (g/L): 66.48

Maltose M (g/L): 81.74

Volume V (L): 7.57

t = 40.00 h

Biomass X (g/L): 89.59

Glucose S (g/L): 19.88

Lactic Acid P (g/L): 52.70

Maltose M (g/L): 82.67

Volume V (L): 8.14

t = 41.00 h

Biomass X (g/L): 76.16

Glucose S (g/L): 22.74

Lactic Acid P (g/L): 77.31

Maltose M (g/L): 77.91

Volume V (L): 7.34

t = 42.00 h

Biomass X (g/L): 82.19

Glucose S (g/L): 22.65

Lactic Acid P (g/L): 64.17

Maltose M (g/L): 79.57

Volume V (L): 8.07

t = 43.00 h

Biomass X (g/L): 77.16

Glucose S (g/L): 26.94

Lactic Acid P (g/L): 75.37

Maltose M (g/L): 75.41

Volume V (L): 7.47

t = 44.00 h

Biomass X (g/L): 77.63

Glucose S (g/L): 31.64

Lactic Acid P (g/L): 68.91

Maltose M (g/L): 80.55

Volume V (L): 7.71

t = 45.00 h

Biomass X (g/L): 79.00

Glucose S (g/L): 23.05

Lactic Acid P (g/L): 70.38

Maltose M (g/L): 80.82

Volume V (L): 8.80

t = 46.00 h

Biomass X (g/L): 84.17

Glucose S (g/L): 23.84

Lactic Acid P (g/L): 71.81
Maltose M (g/L): 77.42
Volume V (L): 8.02

t = 47.00 h

Biomass X (g/L): 77.71
Glucose S (g/L): 29.16
Lactic Acid P (g/L): 71.13
Maltose M (g/L): 82.53
Volume V (L): 8.25

t = 48.00 h

Biomass X (g/L): 94.19
Glucose S (g/L): 28.74
Lactic Acid P (g/L): 73.27
Maltose M (g/L): 74.67
Volume V (L): 8.58

t = 49.00 h

Biomass X (g/L): 74.01
Glucose S (g/L): 32.95
Lactic Acid P (g/L): 79.79
Maltose M (g/L): 73.66
Volume V (L): 8.50

t = 50.00 h

Biomass X (g/L): 80.77
Glucose S (g/L): 29.82
Lactic Acid P (g/L): 70.03
Maltose M (g/L): 77.78
Volume V (L): 8.42

t = 51.00 h

Biomass X (g/L): 78.45
Glucose S (g/L): 27.66
Lactic Acid P (g/L): 74.97
Maltose M (g/L): 79.60
Volume V (L): 8.58

t = 52.00 h

Biomass X (g/L): 96.30

Glucose S (g/L): 31.23
Lactic Acid P (g/L): 68.17
Maltose M (g/L): 80.92
Volume V (L): 9.47

t = 53.00 h

Biomass X (g/L): 84.55
Glucose S (g/L): 33.50
Lactic Acid P (g/L): 82.17
Maltose M (g/L): 74.83
Volume V (L): 8.54

t = 54.00 h

Biomass X (g/L): 92.45
Glucose S (g/L): 33.32
Lactic Acid P (g/L): 63.58
Maltose M (g/L): 78.56
Volume V (L): 8.71

t = 55.00 h

Biomass X (g/L): 68.72
Glucose S (g/L): 33.08
Lactic Acid P (g/L): 86.03
Maltose M (g/L): 75.20
Volume V (L): 9.12

t = 56.00 h

Biomass X (g/L): 85.25
Glucose S (g/L): 35.47
Lactic Acid P (g/L): 69.87
Maltose M (g/L): 74.48
Volume V (L): 9.21

t = 57.00 h

Biomass X (g/L): 91.11
Glucose S (g/L): 36.49
Lactic Acid P (g/L): 57.73
Maltose M (g/L): 76.93
Volume V (L): 9.31

t = 58.00 h

Biomass X (g/L): 83.48
Glucose S (g/L): 37.83
Lactic Acid P (g/L): 71.94
Maltose M (g/L): 75.94
Volume V (L): 9.16

t = 59.00 h

Biomass X (g/L): 92.72
Glucose S (g/L): 36.32
Lactic Acid P (g/L): 64.98
Maltose M (g/L): 69.63
Volume V (L): 8.51

t = 60.00 h

Biomass X (g/L): 77.75
Glucose S (g/L): 34.55
Lactic Acid P (g/L): 78.84
Maltose M (g/L): 77.41
Volume V (L): 9.69

t = 61.00 h

Biomass X (g/L): 95.87
Glucose S (g/L): 36.20
Lactic Acid P (g/L): 65.75
Maltose M (g/L): 76.79
Volume V (L): 8.64

t = 62.00 h

Biomass X (g/L): 104.20
Glucose S (g/L): 40.10
Lactic Acid P (g/L): 67.84
Maltose M (g/L): 71.08
Volume V (L): 9.29

t = 63.00 h

Biomass X (g/L): 79.03
Glucose S (g/L): 34.68
Lactic Acid P (g/L): 55.18
Maltose M (g/L): 73.45
Volume V (L): 8.93

t = 64.00 h

Biomass X (g/L): 78.18
Glucose S (g/L): 38.96
Lactic Acid P (g/L): 68.26
Maltose M (g/L): 73.35
Volume V (L): 9.28

t = 65.00 h

Biomass X (g/L): 96.01
Glucose S (g/L): 37.35
Lactic Acid P (g/L): 50.87
Maltose M (g/L): 72.38
Volume V (L): 9.50

t = 66.00 h

Biomass X (g/L): 97.11
Glucose S (g/L): 39.32
Lactic Acid P (g/L): 58.59
Maltose M (g/L): 69.86
Volume V (L): 9.57

t = 67.00 h

Biomass X (g/L): 77.19
Glucose S (g/L): 43.79
Lactic Acid P (g/L): 79.76
Maltose M (g/L): 69.19
Volume V (L): 9.68

t = 68.00 h

Biomass X (g/L): 78.06
Glucose S (g/L): 41.18
Lactic Acid P (g/L): 80.19
Maltose M (g/L): 70.56
Volume V (L): 10.22

t = 69.00 h

Biomass X (g/L): 79.26
Glucose S (g/L): 41.42
Lactic Acid P (g/L): 77.17
Maltose M (g/L): 73.41
Volume V (L): 9.82

t = 70.00 h

Biomass X (g/L): 69.12
Glucose S (g/L): 37.85
Lactic Acid P (g/L): 64.93
Maltose M (g/L): 71.39
Volume V (L): 9.61

t = 71.00 h

Biomass X (g/L): 73.02
Glucose S (g/L): 44.90
Lactic Acid P (g/L): 73.33
Maltose M (g/L): 68.61
Volume V (L): 10.16

t = 72.00 h

Biomass X (g/L): 72.10
Glucose S (g/L): 45.26
Lactic Acid P (g/L): 73.91
Maltose M (g/L): 71.32
Volume V (L): 10.32

t = 73.00 h

Biomass X (g/L): 74.20
Glucose S (g/L): 50.53
Lactic Acid P (g/L): 63.77
Maltose M (g/L): 61.32
Volume V (L): 9.41

t = 74.00 h

Biomass X (g/L): 80.99
Glucose S (g/L): 43.71
Lactic Acid P (g/L): 87.74
Maltose M (g/L): 71.65
Volume V (L): 10.22

t = 75.00 h

Biomass X (g/L): 83.39
Glucose S (g/L): 45.97
Lactic Acid P (g/L): 82.32
Maltose M (g/L): 63.61

Volume V (L): 10.33

Batch 4

Initial Conditions:

Batch: 4.00

X0 (g/L): 3.19

S0 (g/L): 30.01

M0 (g/L): 166.16

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 5.54

F_vol (L/h): 0.06

S_feed (g/L): 45.90

M_feed (g/L): 254.10

V_final (L): 9.31

Batch 4 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.29

Glucose S (g/L): 27.68

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 162.46

Volume V (L): 5.14

t = 1.00 h

Biomass X (g/L): 17.58

Glucose S (g/L): 33.33

Lactic Acid P (g/L): 8.39

Maltose M (g/L): 156.59

Volume V (L): 4.83

t = 2.00 h

Biomass X (g/L): 3.98

Glucose S (g/L): 31.47

Lactic Acid P (g/L): 2.42

Maltose M (g/L): 160.40

Volume V (L): 4.93

t = 3.00 h

Biomass X (g/L): 8.27

Glucose S (g/L): 43.36

Lactic Acid P (g/L): 1.96

Maltose M (g/L): 174.04

Volume V (L): 5.37

t = 4.00 h

Biomass X (g/L): 7.36

Glucose S (g/L): 32.94

Lactic Acid P (g/L): 12.72

Maltose M (g/L): 146.27

Volume V (L): 5.66

t = 5.00 h

Biomass X (g/L): 11.61

Glucose S (g/L): 38.73

Lactic Acid P (g/L): 3.46
Maltose M (g/L): 142.28
Volume V (L): 5.29

t = 6.00 h

Biomass X (g/L): 3.64
Glucose S (g/L): 39.95
Lactic Acid P (g/L): 10.65
Maltose M (g/L): 166.09
Volume V (L): 5.45

t = 7.00 h

Biomass X (g/L): 3.58
Glucose S (g/L): 38.79
Lactic Acid P (g/L): 13.55
Maltose M (g/L): 150.17
Volume V (L): 5.99

t = 8.00 h

Biomass X (g/L): 36.78
Glucose S (g/L): 35.87
Lactic Acid P (g/L): 7.20
Maltose M (g/L): 144.49
Volume V (L): 5.54

t = 9.00 h

Biomass X (g/L): 16.91
Glucose S (g/L): 34.87
Lactic Acid P (g/L): 25.24
Maltose M (g/L): 156.96
Volume V (L): 6.41

t = 10.00 h

Biomass X (g/L): 19.59
Glucose S (g/L): 34.04
Lactic Acid P (g/L): 17.62
Maltose M (g/L): 141.46
Volume V (L): 5.77

t = 11.00 h

Biomass X (g/L): 12.41

Glucose S (g/L): 31.80
Lactic Acid P (g/L): 15.40
Maltose M (g/L): 160.04
Volume V (L): 5.03

t = 12.00 h

Biomass X (g/L): 31.25
Glucose S (g/L): 34.04
Lactic Acid P (g/L): 30.25
Maltose M (g/L): 148.78
Volume V (L): 5.95

t = 13.00 h

Biomass X (g/L): 28.54
Glucose S (g/L): 36.80
Lactic Acid P (g/L): 20.14
Maltose M (g/L): 137.41
Volume V (L): 5.88

t = 14.00 h

Biomass X (g/L): 61.17
Glucose S (g/L): 25.78
Lactic Acid P (g/L): 31.46
Maltose M (g/L): 157.88
Volume V (L): 6.25

t = 15.00 h

Biomass X (g/L): 58.54
Glucose S (g/L): 21.67
Lactic Acid P (g/L): 33.83
Maltose M (g/L): 133.25
Volume V (L): 6.09

t = 16.00 h

Biomass X (g/L): 45.27
Glucose S (g/L): 25.00
Lactic Acid P (g/L): 46.37
Maltose M (g/L): 142.96
Volume V (L): 5.87

t = 17.00 h

Biomass X (g/L): 76.93
Glucose S (g/L): 33.30
Lactic Acid P (g/L): 43.17
Maltose M (g/L): 137.77
Volume V (L): 5.55

t = 18.00 h

Biomass X (g/L): 67.31
Glucose S (g/L): 20.84
Lactic Acid P (g/L): 40.82
Maltose M (g/L): 144.66
Volume V (L): 5.96

t = 19.00 h

Biomass X (g/L): 57.11
Glucose S (g/L): 28.71
Lactic Acid P (g/L): 50.57
Maltose M (g/L): 145.39
Volume V (L): 5.80

t = 20.00 h

Biomass X (g/L): 38.24
Glucose S (g/L): 28.42
Lactic Acid P (g/L): 57.70
Maltose M (g/L): 131.22
Volume V (L): 5.95

t = 21.00 h

Biomass X (g/L): 87.00
Glucose S (g/L): 20.42
Lactic Acid P (g/L): 51.88
Maltose M (g/L): 123.28
Volume V (L): 6.41

t = 22.00 h

Biomass X (g/L): 53.06
Glucose S (g/L): 24.15
Lactic Acid P (g/L): 48.53
Maltose M (g/L): 132.06
Volume V (L): 6.84

t = 23.00 h

Biomass X (g/L): 52.31
Glucose S (g/L): 27.79
Lactic Acid P (g/L): 37.99
Maltose M (g/L): 121.24
Volume V (L): 6.46

t = 24.00 h

Biomass X (g/L): 79.89
Glucose S (g/L): 20.91
Lactic Acid P (g/L): 41.67
Maltose M (g/L): 142.40
Volume V (L): 5.97

t = 25.00 h

Biomass X (g/L): 95.14
Glucose S (g/L): 21.68
Lactic Acid P (g/L): 49.94
Maltose M (g/L): 125.47
Volume V (L): 6.59

t = 26.00 h

Biomass X (g/L): 86.17
Glucose S (g/L): 20.61
Lactic Acid P (g/L): 48.40
Maltose M (g/L): 120.04
Volume V (L): 6.91

t = 27.00 h

Biomass X (g/L): 79.39
Glucose S (g/L): 19.42
Lactic Acid P (g/L): 53.43
Maltose M (g/L): 123.41
Volume V (L): 6.34

t = 28.00 h

Biomass X (g/L): 80.78
Glucose S (g/L): 21.07
Lactic Acid P (g/L): 56.75
Maltose M (g/L): 109.42
Volume V (L): 6.56

t = 29.00 h

Biomass X (g/L): 84.30
Glucose S (g/L): 19.51
Lactic Acid P (g/L): 62.09
Maltose M (g/L): 111.38
Volume V (L): 6.67

t = 30.00 h

Biomass X (g/L): 84.29
Glucose S (g/L): 24.34
Lactic Acid P (g/L): 57.58
Maltose M (g/L): 128.36
Volume V (L): 6.45

t = 31.00 h

Biomass X (g/L): 80.47
Glucose S (g/L): 22.09
Lactic Acid P (g/L): 63.58
Maltose M (g/L): 111.73
Volume V (L): 7.00

t = 32.00 h

Biomass X (g/L): 79.50
Glucose S (g/L): 26.12
Lactic Acid P (g/L): 60.59
Maltose M (g/L): 107.37
Volume V (L): 6.80

t = 33.00 h

Biomass X (g/L): 78.57
Glucose S (g/L): 20.52
Lactic Acid P (g/L): 42.79
Maltose M (g/L): 119.50
Volume V (L): 7.04

t = 34.00 h

Biomass X (g/L): 72.46
Glucose S (g/L): 27.68
Lactic Acid P (g/L): 53.02
Maltose M (g/L): 123.85

Volume V (L): 6.65

t = 35.00 h

Biomass X (g/L): 77.64

Glucose S (g/L): 30.07

Lactic Acid P (g/L): 61.01

Maltose M (g/L): 118.91

Volume V (L): 6.86

t = 36.00 h

Biomass X (g/L): 64.39

Glucose S (g/L): 20.96

Lactic Acid P (g/L): 51.98

Maltose M (g/L): 120.19

Volume V (L): 7.32

t = 37.00 h

Biomass X (g/L): 80.39

Glucose S (g/L): 29.84

Lactic Acid P (g/L): 69.15

Maltose M (g/L): 104.56

Volume V (L): 7.00

t = 38.00 h

Biomass X (g/L): 72.14

Glucose S (g/L): 33.51

Lactic Acid P (g/L): 78.27

Maltose M (g/L): 98.01

Volume V (L): 7.78

t = 39.00 h

Biomass X (g/L): 71.40

Glucose S (g/L): 25.80

Lactic Acid P (g/L): 76.02

Maltose M (g/L): 121.44

Volume V (L): 7.10

t = 40.00 h

Biomass X (g/L): 84.24

Glucose S (g/L): 36.51

Lactic Acid P (g/L): 63.52

Maltose M (g/L): 105.26
Volume V (L): 7.37

t = 41.00 h

Biomass X (g/L): 96.35
Glucose S (g/L): 32.92
Lactic Acid P (g/L): 79.62
Maltose M (g/L): 104.92
Volume V (L): 7.68

t = 42.00 h

Biomass X (g/L): 92.00
Glucose S (g/L): 29.78
Lactic Acid P (g/L): 67.89
Maltose M (g/L): 117.18
Volume V (L): 6.92

t = 43.00 h

Biomass X (g/L): 68.00
Glucose S (g/L): 32.79
Lactic Acid P (g/L): 63.95
Maltose M (g/L): 109.83
Volume V (L): 6.86

t = 44.00 h

Biomass X (g/L): 73.87
Glucose S (g/L): 35.68
Lactic Acid P (g/L): 61.90
Maltose M (g/L): 108.10
Volume V (L): 7.00

t = 45.00 h

Biomass X (g/L): 92.85
Glucose S (g/L): 34.79
Lactic Acid P (g/L): 65.19
Maltose M (g/L): 121.49
Volume V (L): 8.02

t = 46.00 h

Biomass X (g/L): 74.27
Glucose S (g/L): 37.17

Lactic Acid P (g/L): 64.84
Maltose M (g/L): 97.19
Volume V (L): 7.45

t = 47.00 h

Biomass X (g/L): 81.79
Glucose S (g/L): 32.90
Lactic Acid P (g/L): 70.36
Maltose M (g/L): 109.46
Volume V (L): 8.58

t = 48.00 h

Biomass X (g/L): 89.44
Glucose S (g/L): 37.45
Lactic Acid P (g/L): 70.67
Maltose M (g/L): 101.35
Volume V (L): 7.56

t = 49.00 h

Biomass X (g/L): 74.74
Glucose S (g/L): 32.75
Lactic Acid P (g/L): 69.61
Maltose M (g/L): 102.30
Volume V (L): 7.88

t = 50.00 h

Biomass X (g/L): 85.27
Glucose S (g/L): 38.04
Lactic Acid P (g/L): 73.94
Maltose M (g/L): 112.01
Volume V (L): 8.40

t = 51.00 h

Biomass X (g/L): 71.31
Glucose S (g/L): 36.08
Lactic Acid P (g/L): 81.18
Maltose M (g/L): 102.70
Volume V (L): 8.62

t = 52.00 h

Biomass X (g/L): 78.48

Glucose S (g/L): 37.44
Lactic Acid P (g/L): 79.32
Maltose M (g/L): 102.18
Volume V (L): 8.70

t = 53.00 h

Biomass X (g/L): 87.47
Glucose S (g/L): 45.04
Lactic Acid P (g/L): 57.43
Maltose M (g/L): 105.87
Volume V (L): 8.46

t = 54.00 h

Biomass X (g/L): 68.85
Glucose S (g/L): 45.32
Lactic Acid P (g/L): 47.99
Maltose M (g/L): 93.12
Volume V (L): 8.45

t = 55.00 h

Biomass X (g/L): 88.65
Glucose S (g/L): 44.42
Lactic Acid P (g/L): 80.13
Maltose M (g/L): 88.41
Volume V (L): 8.36

t = 56.00 h

Biomass X (g/L): 84.14
Glucose S (g/L): 39.30
Lactic Acid P (g/L): 59.43
Maltose M (g/L): 85.97
Volume V (L): 8.48

t = 57.00 h

Biomass X (g/L): 89.68
Glucose S (g/L): 42.75
Lactic Acid P (g/L): 70.13
Maltose M (g/L): 77.56
Volume V (L): 8.08

t = 58.00 h

Biomass X (g/L): 86.44
Glucose S (g/L): 48.80
Lactic Acid P (g/L): 61.78
Maltose M (g/L): 110.42
Volume V (L): 8.05

t = 59.00 h

Biomass X (g/L): 97.49
Glucose S (g/L): 45.85
Lactic Acid P (g/L): 56.28
Maltose M (g/L): 83.35
Volume V (L): 8.39

t = 60.00 h

Biomass X (g/L): 85.36
Glucose S (g/L): 54.14
Lactic Acid P (g/L): 77.93
Maltose M (g/L): 90.03
Volume V (L): 8.39

t = 61.00 h

Biomass X (g/L): 79.89
Glucose S (g/L): 48.23
Lactic Acid P (g/L): 80.16
Maltose M (g/L): 88.89
Volume V (L): 8.35

t = 62.00 h

Biomass X (g/L): 85.18
Glucose S (g/L): 48.27
Lactic Acid P (g/L): 68.22
Maltose M (g/L): 93.10
Volume V (L): 8.80

t = 63.00 h

Biomass X (g/L): 89.19
Glucose S (g/L): 47.28
Lactic Acid P (g/L): 50.06
Maltose M (g/L): 96.68
Volume V (L): 8.95

t = 64.00 h

Biomass X (g/L): 84.28
Glucose S (g/L): 49.54
Lactic Acid P (g/L): 68.95
Maltose M (g/L): 96.41
Volume V (L): 8.71

t = 65.00 h

Biomass X (g/L): 84.09
Glucose S (g/L): 49.28
Lactic Acid P (g/L): 77.74
Maltose M (g/L): 83.72
Volume V (L): 9.24

t = 66.00 h

Biomass X (g/L): 76.75
Glucose S (g/L): 52.98
Lactic Acid P (g/L): 60.88
Maltose M (g/L): 86.57
Volume V (L): 8.40

t = 67.00 h

Biomass X (g/L): 77.01
Glucose S (g/L): 54.55
Lactic Acid P (g/L): 76.26
Maltose M (g/L): 89.70
Volume V (L): 8.78

t = 68.00 h

Biomass X (g/L): 69.68
Glucose S (g/L): 49.23
Lactic Acid P (g/L): 74.64
Maltose M (g/L): 84.66
Volume V (L): 8.66

t = 69.00 h

Biomass X (g/L): 100.26
Glucose S (g/L): 48.05
Lactic Acid P (g/L): 80.27
Maltose M (g/L): 99.55
Volume V (L): 8.32

t = 70.00 h

Biomass X (g/L): 70.97
Glucose S (g/L): 47.00
Lactic Acid P (g/L): 76.26
Maltose M (g/L): 89.64
Volume V (L): 8.99

t = 71.00 h

Biomass X (g/L): 76.20
Glucose S (g/L): 55.81
Lactic Acid P (g/L): 66.48
Maltose M (g/L): 91.09
Volume V (L): 9.03

t = 72.00 h

Biomass X (g/L): 75.81
Glucose S (g/L): 50.92
Lactic Acid P (g/L): 86.46
Maltose M (g/L): 89.91
Volume V (L): 9.66

t = 73.00 h

Biomass X (g/L): 91.38
Glucose S (g/L): 47.96
Lactic Acid P (g/L): 74.27
Maltose M (g/L): 93.60
Volume V (L): 9.41

t = 74.00 h

Biomass X (g/L): 79.63
Glucose S (g/L): 57.43
Lactic Acid P (g/L): 75.64
Maltose M (g/L): 84.76
Volume V (L): 8.90

t = 75.00 h

Biomass X (g/L): 94.51
Glucose S (g/L): 65.39
Lactic Acid P (g/L): 67.54
Maltose M (g/L): 71.87

Volume V (L): 9.95

Batch 5

Initial Conditions:

Batch: 5.00

X0 (g/L): 8.42

S0 (g/L): 34.22

M0 (g/L): 121.72

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 3.56

F_vol (L/h): 0.09

S_feed (g/L): 65.83

M_feed (g/L): 234.17

V_final (L): 11.56

Batch 5 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 19.43
Glucose S (g/L): 36.66
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 122.46
Volume V (L): 5.09

t = 1.00 h

Biomass X (g/L): 16.19
Glucose S (g/L): 42.57
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 125.26
Volume V (L): 5.32

t = 2.00 h

Biomass X (g/L): 11.94
Glucose S (g/L): 34.18
Lactic Acid P (g/L): 18.49
Maltose M (g/L): 110.21
Volume V (L): 4.60

t = 3.00 h

Biomass X (g/L): 8.86
Glucose S (g/L): 30.32
Lactic Acid P (g/L): 4.04
Maltose M (g/L): 123.12
Volume V (L): 6.00

t = 4.00 h

Biomass X (g/L): 18.10
Glucose S (g/L): 31.08
Lactic Acid P (g/L): 20.59
Maltose M (g/L): 117.40
Volume V (L): 5.30

t = 5.00 h

Biomass X (g/L): 35.52
Glucose S (g/L): 35.48

Lactic Acid P (g/L): 26.47

Maltose M (g/L): 112.29

Volume V (L): 5.45

t = 6.00 h

Biomass X (g/L): 47.09

Glucose S (g/L): 25.51

Lactic Acid P (g/L): 25.16

Maltose M (g/L): 121.05

Volume V (L): 5.90

t = 7.00 h

Biomass X (g/L): 48.42

Glucose S (g/L): 22.84

Lactic Acid P (g/L): 15.88

Maltose M (g/L): 123.96

Volume V (L): 5.73

t = 8.00 h

Biomass X (g/L): 34.39

Glucose S (g/L): 22.06

Lactic Acid P (g/L): 26.52

Maltose M (g/L): 117.49

Volume V (L): 6.20

t = 9.00 h

Biomass X (g/L): 32.78

Glucose S (g/L): 18.31

Lactic Acid P (g/L): 25.88

Maltose M (g/L): 121.28

Volume V (L): 5.94

t = 10.00 h

Biomass X (g/L): 47.95

Glucose S (g/L): 20.87

Lactic Acid P (g/L): 21.07

Maltose M (g/L): 123.36

Volume V (L): 5.55

t = 11.00 h

Biomass X (g/L): 34.23

Glucose S (g/L): 25.34
Lactic Acid P (g/L): 34.66
Maltose M (g/L): 121.66
Volume V (L): 6.15

t = 12.00 h

Biomass X (g/L): 56.28
Glucose S (g/L): 18.98
Lactic Acid P (g/L): 23.94
Maltose M (g/L): 122.40
Volume V (L): 5.45

t = 13.00 h

Biomass X (g/L): 54.92
Glucose S (g/L): 15.73
Lactic Acid P (g/L): 47.17
Maltose M (g/L): 118.31
Volume V (L): 6.19

t = 14.00 h

Biomass X (g/L): 54.66
Glucose S (g/L): 14.81
Lactic Acid P (g/L): 34.73
Maltose M (g/L): 115.13
Volume V (L): 6.21

t = 15.00 h

Biomass X (g/L): 46.84
Glucose S (g/L): 9.22
Lactic Acid P (g/L): 42.59
Maltose M (g/L): 107.95
Volume V (L): 6.20

t = 16.00 h

Biomass X (g/L): 67.97
Glucose S (g/L): 18.49
Lactic Acid P (g/L): 45.14
Maltose M (g/L): 112.74
Volume V (L): 6.33

t = 17.00 h

Biomass X (g/L): 60.46
Glucose S (g/L): 14.46
Lactic Acid P (g/L): 42.03
Maltose M (g/L): 109.82
Volume V (L): 5.50

t = 18.00 h

Biomass X (g/L): 56.46
Glucose S (g/L): 13.96
Lactic Acid P (g/L): 51.05
Maltose M (g/L): 121.15
Volume V (L): 6.29

t = 19.00 h

Biomass X (g/L): 42.69
Glucose S (g/L): 27.29
Lactic Acid P (g/L): 43.13
Maltose M (g/L): 111.31
Volume V (L): 6.71

t = 20.00 h

Biomass X (g/L): 69.77
Glucose S (g/L): 15.98
Lactic Acid P (g/L): 56.36
Maltose M (g/L): 111.92
Volume V (L): 6.83

t = 21.00 h

Biomass X (g/L): 87.30
Glucose S (g/L): 12.03
Lactic Acid P (g/L): 38.05
Maltose M (g/L): 113.04
Volume V (L): 6.30

t = 22.00 h

Biomass X (g/L): 87.49
Glucose S (g/L): 13.60
Lactic Acid P (g/L): 52.23
Maltose M (g/L): 110.46
Volume V (L): 7.59

t = 23.00 h

Biomass X (g/L): 69.49
Glucose S (g/L): 20.16
Lactic Acid P (g/L): 69.34
Maltose M (g/L): 116.40
Volume V (L): 6.56

t = 24.00 h

Biomass X (g/L): 69.18
Glucose S (g/L): 16.48
Lactic Acid P (g/L): 44.45
Maltose M (g/L): 106.67
Volume V (L): 7.61

t = 25.00 h

Biomass X (g/L): 79.75
Glucose S (g/L): 10.80
Lactic Acid P (g/L): 48.28
Maltose M (g/L): 109.32
Volume V (L): 7.41

t = 26.00 h

Biomass X (g/L): 67.36
Glucose S (g/L): 20.61
Lactic Acid P (g/L): 59.73
Maltose M (g/L): 105.04
Volume V (L): 6.94

t = 27.00 h

Biomass X (g/L): 79.85
Glucose S (g/L): 14.88
Lactic Acid P (g/L): 57.66
Maltose M (g/L): 102.18
Volume V (L): 8.29

t = 28.00 h

Biomass X (g/L): 84.73
Glucose S (g/L): 25.50
Lactic Acid P (g/L): 60.01
Maltose M (g/L): 104.39
Volume V (L): 6.82

t = 29.00 h

Biomass X (g/L): 89.53
Glucose S (g/L): 22.97
Lactic Acid P (g/L): 62.03
Maltose M (g/L): 106.08
Volume V (L): 8.02

t = 30.00 h

Biomass X (g/L): 78.15
Glucose S (g/L): 17.58
Lactic Acid P (g/L): 64.07
Maltose M (g/L): 103.96
Volume V (L): 8.15

t = 31.00 h

Biomass X (g/L): 77.76
Glucose S (g/L): 25.06
Lactic Acid P (g/L): 68.98
Maltose M (g/L): 102.34
Volume V (L): 7.36

t = 32.00 h

Biomass X (g/L): 72.17
Glucose S (g/L): 20.95
Lactic Acid P (g/L): 63.48
Maltose M (g/L): 98.95
Volume V (L): 8.53

t = 33.00 h

Biomass X (g/L): 73.07
Glucose S (g/L): 24.76
Lactic Acid P (g/L): 58.38
Maltose M (g/L): 103.60
Volume V (L): 7.76

t = 34.00 h

Biomass X (g/L): 78.89
Glucose S (g/L): 24.03
Lactic Acid P (g/L): 67.07
Maltose M (g/L): 97.41

Volume V (L): 8.13

t = 35.00 h

Biomass X (g/L): 84.60

Glucose S (g/L): 19.51

Lactic Acid P (g/L): 78.49

Maltose M (g/L): 105.78

Volume V (L): 8.20

t = 36.00 h

Biomass X (g/L): 89.85

Glucose S (g/L): 21.06

Lactic Acid P (g/L): 61.12

Maltose M (g/L): 102.53

Volume V (L): 8.08

t = 37.00 h

Biomass X (g/L): 74.67

Glucose S (g/L): 28.43

Lactic Acid P (g/L): 54.23

Maltose M (g/L): 95.89

Volume V (L): 8.66

t = 38.00 h

Biomass X (g/L): 95.63

Glucose S (g/L): 23.82

Lactic Acid P (g/L): 73.12

Maltose M (g/L): 99.85

Volume V (L): 8.74

t = 39.00 h

Biomass X (g/L): 79.97

Glucose S (g/L): 24.33

Lactic Acid P (g/L): 62.78

Maltose M (g/L): 97.54

Volume V (L): 7.49

t = 40.00 h

Biomass X (g/L): 82.43

Glucose S (g/L): 22.94

Lactic Acid P (g/L): 45.14

Maltose M (g/L): 101.21
Volume V (L): 9.18

t = 41.00 h

Biomass X (g/L): 76.07
Glucose S (g/L): 29.18
Lactic Acid P (g/L): 72.24
Maltose M (g/L): 104.68
Volume V (L): 7.71

t = 42.00 h

Biomass X (g/L): 61.11
Glucose S (g/L): 36.46
Lactic Acid P (g/L): 60.47
Maltose M (g/L): 91.43
Volume V (L): 9.22

t = 43.00 h

Biomass X (g/L): 91.07
Glucose S (g/L): 24.91
Lactic Acid P (g/L): 64.82
Maltose M (g/L): 98.74
Volume V (L): 9.35

t = 44.00 h

Biomass X (g/L): 89.85
Glucose S (g/L): 30.03
Lactic Acid P (g/L): 68.95
Maltose M (g/L): 97.10
Volume V (L): 8.28

t = 45.00 h

Biomass X (g/L): 80.77
Glucose S (g/L): 36.33
Lactic Acid P (g/L): 68.08
Maltose M (g/L): 99.83
Volume V (L): 8.72

t = 46.00 h

Biomass X (g/L): 83.99
Glucose S (g/L): 31.25

Lactic Acid P (g/L): 73.03

Maltose M (g/L): 97.48

Volume V (L): 8.44

t = 47.00 h

Biomass X (g/L): 88.10

Glucose S (g/L): 34.17

Lactic Acid P (g/L): 63.28

Maltose M (g/L): 93.12

Volume V (L): 9.00

t = 48.00 h

Biomass X (g/L): 69.41

Glucose S (g/L): 29.01

Lactic Acid P (g/L): 72.00

Maltose M (g/L): 94.29

Volume V (L): 9.04

t = 49.00 h

Biomass X (g/L): 78.75

Glucose S (g/L): 35.72

Lactic Acid P (g/L): 65.51

Maltose M (g/L): 91.87

Volume V (L): 9.70

t = 50.00 h

Biomass X (g/L): 90.70

Glucose S (g/L): 34.11

Lactic Acid P (g/L): 74.64

Maltose M (g/L): 98.73

Volume V (L): 10.06

t = 51.00 h

Biomass X (g/L): 72.23

Glucose S (g/L): 39.55

Lactic Acid P (g/L): 73.86

Maltose M (g/L): 94.04

Volume V (L): 10.19

t = 52.00 h

Biomass X (g/L): 86.18

Glucose S (g/L): 41.17
Lactic Acid P (g/L): 60.47
Maltose M (g/L): 100.93
Volume V (L): 9.25

t = 53.00 h

Biomass X (g/L): 71.28
Glucose S (g/L): 47.52
Lactic Acid P (g/L): 78.59
Maltose M (g/L): 92.66
Volume V (L): 9.53

t = 54.00 h

Biomass X (g/L): 87.69
Glucose S (g/L): 40.39
Lactic Acid P (g/L): 64.56
Maltose M (g/L): 89.58
Volume V (L): 8.84

t = 55.00 h

Biomass X (g/L): 83.12
Glucose S (g/L): 40.20
Lactic Acid P (g/L): 76.69
Maltose M (g/L): 85.89
Volume V (L): 9.39

t = 56.00 h

Biomass X (g/L): 68.83
Glucose S (g/L): 40.49
Lactic Acid P (g/L): 73.06
Maltose M (g/L): 84.23
Volume V (L): 10.41

t = 57.00 h

Biomass X (g/L): 93.83
Glucose S (g/L): 39.13
Lactic Acid P (g/L): 86.29
Maltose M (g/L): 84.58
Volume V (L): 10.84

t = 58.00 h

Biomass X (g/L): 76.64
Glucose S (g/L): 42.66
Lactic Acid P (g/L): 68.34
Maltose M (g/L): 91.39
Volume V (L): 9.43

t = 59.00 h

Biomass X (g/L): 75.88
Glucose S (g/L): 40.95
Lactic Acid P (g/L): 67.46
Maltose M (g/L): 91.84
Volume V (L): 9.85

t = 60.00 h

Biomass X (g/L): 84.91
Glucose S (g/L): 43.57
Lactic Acid P (g/L): 70.60
Maltose M (g/L): 83.08
Volume V (L): 10.24

t = 61.00 h

Biomass X (g/L): 93.37
Glucose S (g/L): 48.89
Lactic Acid P (g/L): 68.32
Maltose M (g/L): 92.36
Volume V (L): 10.48

t = 62.00 h

Biomass X (g/L): 73.76
Glucose S (g/L): 40.50
Lactic Acid P (g/L): 78.27
Maltose M (g/L): 86.43
Volume V (L): 10.00

t = 63.00 h

Biomass X (g/L): 83.38
Glucose S (g/L): 49.75
Lactic Acid P (g/L): 57.16
Maltose M (g/L): 88.43
Volume V (L): 10.73

t = 64.00 h

Biomass X (g/L): 86.58
Glucose S (g/L): 51.22
Lactic Acid P (g/L): 62.33
Maltose M (g/L): 93.72
Volume V (L): 10.35

t = 65.00 h

Biomass X (g/L): 88.85
Glucose S (g/L): 46.32
Lactic Acid P (g/L): 62.40
Maltose M (g/L): 90.42
Volume V (L): 10.68

t = 66.00 h

Biomass X (g/L): 84.07
Glucose S (g/L): 39.17
Lactic Acid P (g/L): 80.40
Maltose M (g/L): 85.33
Volume V (L): 11.06

t = 67.00 h

Biomass X (g/L): 79.05
Glucose S (g/L): 45.90
Lactic Acid P (g/L): 87.30
Maltose M (g/L): 82.40
Volume V (L): 10.87

t = 68.00 h

Biomass X (g/L): 74.78
Glucose S (g/L): 54.18
Lactic Acid P (g/L): 71.47
Maltose M (g/L): 76.95
Volume V (L): 10.72

t = 69.00 h

Biomass X (g/L): 82.88
Glucose S (g/L): 49.71
Lactic Acid P (g/L): 71.08
Maltose M (g/L): 82.13
Volume V (L): 10.98

t = 70.00 h

Biomass X (g/L): 72.31
Glucose S (g/L): 46.60
Lactic Acid P (g/L): 73.36
Maltose M (g/L): 83.67
Volume V (L): 11.08

t = 71.00 h

Biomass X (g/L): 55.50
Glucose S (g/L): 49.28
Lactic Acid P (g/L): 83.16
Maltose M (g/L): 80.15
Volume V (L): 11.02

t = 72.00 h

Biomass X (g/L): 85.63
Glucose S (g/L): 55.32
Lactic Acid P (g/L): 64.89
Maltose M (g/L): 83.07
Volume V (L): 11.16

t = 73.00 h

Biomass X (g/L): 89.70
Glucose S (g/L): 54.89
Lactic Acid P (g/L): 75.34
Maltose M (g/L): 83.43
Volume V (L): 12.23

t = 74.00 h

Biomass X (g/L): 60.11
Glucose S (g/L): 52.50
Lactic Acid P (g/L): 82.00
Maltose M (g/L): 81.79
Volume V (L): 11.77

t = 75.00 h

Biomass X (g/L): 72.16
Glucose S (g/L): 52.71
Lactic Acid P (g/L): 77.97
Maltose M (g/L): 79.66

Volume V (L): 11.41

Batch 6

Initial Conditions:

Batch: 6.00

X0 (g/L): 9.93

S0 (g/L): 90.34

M0 (g/L): 34.31

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.38

F_vol (L/h): 0.05

S_feed (g/L): 217.42

M_feed (g/L): 82.58

V_final (L): 8.97

Batch 6 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 16.72
Glucose S (g/L): 95.43
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 35.07
Volume V (L): 4.70

t = 1.00 h

Biomass X (g/L): 10.10
Glucose S (g/L): 93.12
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 34.92
Volume V (L): 5.20

t = 2.00 h

Biomass X (g/L): 28.90
Glucose S (g/L): 83.07
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 34.46
Volume V (L): 5.58

t = 3.00 h

Biomass X (g/L): 25.86
Glucose S (g/L): 85.46
Lactic Acid P (g/L): 1.41
Maltose M (g/L): 33.86
Volume V (L): 5.27

t = 4.00 h

Biomass X (g/L): 26.47
Glucose S (g/L): 75.74
Lactic Acid P (g/L): 8.54
Maltose M (g/L): 34.40
Volume V (L): 5.11

t = 5.00 h

Biomass X (g/L): 37.40
Glucose S (g/L): 70.09

Lactic Acid P (g/L): 25.37

Maltose M (g/L): 34.15

Volume V (L): 5.26

t = 6.00 h

Biomass X (g/L): 23.31

Glucose S (g/L): 69.60

Lactic Acid P (g/L): 23.75

Maltose M (g/L): 32.99

Volume V (L): 4.92

t = 7.00 h

Biomass X (g/L): 33.04

Glucose S (g/L): 72.52

Lactic Acid P (g/L): 27.17

Maltose M (g/L): 32.60

Volume V (L): 5.56

t = 8.00 h

Biomass X (g/L): 37.49

Glucose S (g/L): 69.34

Lactic Acid P (g/L): 37.24

Maltose M (g/L): 33.16

Volume V (L): 5.70

t = 9.00 h

Biomass X (g/L): 55.24

Glucose S (g/L): 64.07

Lactic Acid P (g/L): 31.56

Maltose M (g/L): 34.53

Volume V (L): 5.33

t = 10.00 h

Biomass X (g/L): 75.02

Glucose S (g/L): 62.10

Lactic Acid P (g/L): 39.88

Maltose M (g/L): 31.41

Volume V (L): 5.38

t = 11.00 h

Biomass X (g/L): 83.75

Glucose S (g/L): 58.59
Lactic Acid P (g/L): 38.30
Maltose M (g/L): 31.01
Volume V (L): 5.67

t = 12.00 h

Biomass X (g/L): 71.72
Glucose S (g/L): 51.14
Lactic Acid P (g/L): 44.56
Maltose M (g/L): 32.59
Volume V (L): 5.50

t = 13.00 h

Biomass X (g/L): 73.48
Glucose S (g/L): 54.11
Lactic Acid P (g/L): 39.11
Maltose M (g/L): 32.31
Volume V (L): 5.67

t = 14.00 h

Biomass X (g/L): 53.82
Glucose S (g/L): 54.19
Lactic Acid P (g/L): 23.91
Maltose M (g/L): 32.28
Volume V (L): 6.41

t = 15.00 h

Biomass X (g/L): 68.51
Glucose S (g/L): 52.65
Lactic Acid P (g/L): 47.73
Maltose M (g/L): 32.19
Volume V (L): 5.71

t = 16.00 h

Biomass X (g/L): 68.29
Glucose S (g/L): 53.53
Lactic Acid P (g/L): 64.30
Maltose M (g/L): 31.54
Volume V (L): 5.73

t = 17.00 h

Biomass X (g/L): 71.83
Glucose S (g/L): 50.46
Lactic Acid P (g/L): 61.36
Maltose M (g/L): 33.20
Volume V (L): 5.69

t = 18.00 h

Biomass X (g/L): 77.93
Glucose S (g/L): 48.14
Lactic Acid P (g/L): 63.31
Maltose M (g/L): 29.76
Volume V (L): 6.19

t = 19.00 h

Biomass X (g/L): 76.47
Glucose S (g/L): 50.85
Lactic Acid P (g/L): 67.31
Maltose M (g/L): 30.43
Volume V (L): 5.95

t = 20.00 h

Biomass X (g/L): 63.13
Glucose S (g/L): 45.48
Lactic Acid P (g/L): 57.83
Maltose M (g/L): 31.04
Volume V (L): 5.85

t = 21.00 h

Biomass X (g/L): 90.86
Glucose S (g/L): 41.32
Lactic Acid P (g/L): 60.09
Maltose M (g/L): 29.25
Volume V (L): 6.53

t = 22.00 h

Biomass X (g/L): 74.18
Glucose S (g/L): 44.39
Lactic Acid P (g/L): 53.54
Maltose M (g/L): 30.65
Volume V (L): 6.61

t = 23.00 h

Biomass X (g/L): 87.88
Glucose S (g/L): 41.67
Lactic Acid P (g/L): 57.44
Maltose M (g/L): 32.45
Volume V (L): 6.03

t = 24.00 h

Biomass X (g/L): 106.55
Glucose S (g/L): 41.99
Lactic Acid P (g/L): 67.20
Maltose M (g/L): 30.55
Volume V (L): 5.68

t = 25.00 h

Biomass X (g/L): 78.14
Glucose S (g/L): 44.46
Lactic Acid P (g/L): 73.65
Maltose M (g/L): 30.62
Volume V (L): 6.64

t = 26.00 h

Biomass X (g/L): 63.09
Glucose S (g/L): 41.66
Lactic Acid P (g/L): 61.84
Maltose M (g/L): 28.88
Volume V (L): 6.16

t = 27.00 h

Biomass X (g/L): 85.49
Glucose S (g/L): 44.05
Lactic Acid P (g/L): 59.70
Maltose M (g/L): 29.42
Volume V (L): 6.68

t = 28.00 h

Biomass X (g/L): 99.48
Glucose S (g/L): 37.10
Lactic Acid P (g/L): 64.70
Maltose M (g/L): 29.52
Volume V (L): 5.86

t = 29.00 h

Biomass X (g/L): 108.25
Glucose S (g/L): 46.93
Lactic Acid P (g/L): 71.06
Maltose M (g/L): 29.43
Volume V (L): 6.82

t = 30.00 h

Biomass X (g/L): 77.59
Glucose S (g/L): 46.29
Lactic Acid P (g/L): 52.44
Maltose M (g/L): 30.98
Volume V (L): 6.20

t = 31.00 h

Biomass X (g/L): 94.71
Glucose S (g/L): 42.94
Lactic Acid P (g/L): 45.03
Maltose M (g/L): 30.06
Volume V (L): 6.87

t = 32.00 h

Biomass X (g/L): 85.62
Glucose S (g/L): 50.02
Lactic Acid P (g/L): 62.06
Maltose M (g/L): 28.14
Volume V (L): 6.70

t = 33.00 h

Biomass X (g/L): 89.34
Glucose S (g/L): 51.69
Lactic Acid P (g/L): 56.30
Maltose M (g/L): 29.43
Volume V (L): 6.44

t = 34.00 h

Biomass X (g/L): 116.30
Glucose S (g/L): 40.60
Lactic Acid P (g/L): 63.54
Maltose M (g/L): 29.55

Volume V (L): 6.67

t = 35.00 h

Biomass X (g/L): 76.08

Glucose S (g/L): 44.88

Lactic Acid P (g/L): 67.43

Maltose M (g/L): 30.09

Volume V (L): 7.07

t = 36.00 h

Biomass X (g/L): 93.31

Glucose S (g/L): 45.33

Lactic Acid P (g/L): 65.03

Maltose M (g/L): 30.41

Volume V (L): 6.94

t = 37.00 h

Biomass X (g/L): 95.48

Glucose S (g/L): 45.05

Lactic Acid P (g/L): 83.44

Maltose M (g/L): 27.51

Volume V (L): 7.14

t = 38.00 h

Biomass X (g/L): 101.56

Glucose S (g/L): 53.57

Lactic Acid P (g/L): 73.26

Maltose M (g/L): 27.58

Volume V (L): 7.64

t = 39.00 h

Biomass X (g/L): 91.98

Glucose S (g/L): 49.45

Lactic Acid P (g/L): 73.42

Maltose M (g/L): 26.96

Volume V (L): 6.57

t = 40.00 h

Biomass X (g/L): 94.05

Glucose S (g/L): 49.66

Lactic Acid P (g/L): 76.84

Maltose M (g/L): 27.90

Volume V (L): 7.63

t = 41.00 h

Biomass X (g/L): 87.23

Glucose S (g/L): 42.32

Lactic Acid P (g/L): 60.61

Maltose M (g/L): 28.51

Volume V (L): 7.20

t = 42.00 h

Biomass X (g/L): 93.64

Glucose S (g/L): 54.82

Lactic Acid P (g/L): 81.50

Maltose M (g/L): 28.92

Volume V (L): 6.94

t = 43.00 h

Biomass X (g/L): 87.64

Glucose S (g/L): 46.88

Lactic Acid P (g/L): 71.33

Maltose M (g/L): 27.65

Volume V (L): 6.85

t = 44.00 h

Biomass X (g/L): 96.97

Glucose S (g/L): 53.32

Lactic Acid P (g/L): 78.69

Maltose M (g/L): 27.25

Volume V (L): 7.27

t = 45.00 h

Biomass X (g/L): 87.25

Glucose S (g/L): 54.87

Lactic Acid P (g/L): 62.59

Maltose M (g/L): 27.61

Volume V (L): 7.67

t = 46.00 h

Biomass X (g/L): 94.43

Glucose S (g/L): 60.34

Lactic Acid P (g/L): 58.35

Maltose M (g/L): 26.43

Volume V (L): 7.39

t = 47.00 h

Biomass X (g/L): 90.38

Glucose S (g/L): 48.82

Lactic Acid P (g/L): 80.99

Maltose M (g/L): 29.40

Volume V (L): 7.97

t = 48.00 h

Biomass X (g/L): 101.51

Glucose S (g/L): 41.65

Lactic Acid P (g/L): 77.11

Maltose M (g/L): 25.33

Volume V (L): 7.08

t = 49.00 h

Biomass X (g/L): 89.93

Glucose S (g/L): 56.73

Lactic Acid P (g/L): 65.33

Maltose M (g/L): 27.10

Volume V (L): 7.59

t = 50.00 h

Biomass X (g/L): 73.04

Glucose S (g/L): 64.63

Lactic Acid P (g/L): 65.89

Maltose M (g/L): 28.92

Volume V (L): 7.25

t = 51.00 h

Biomass X (g/L): 102.44

Glucose S (g/L): 62.66

Lactic Acid P (g/L): 84.95

Maltose M (g/L): 26.64

Volume V (L): 7.81

t = 52.00 h

Biomass X (g/L): 81.19

Glucose S (g/L): 58.57
Lactic Acid P (g/L): 63.37
Maltose M (g/L): 26.30
Volume V (L): 8.03

t = 53.00 h

Biomass X (g/L): 83.49
Glucose S (g/L): 55.98
Lactic Acid P (g/L): 90.30
Maltose M (g/L): 25.99
Volume V (L): 7.67

t = 54.00 h

Biomass X (g/L): 83.89
Glucose S (g/L): 62.43
Lactic Acid P (g/L): 66.55
Maltose M (g/L): 26.05
Volume V (L): 7.03

t = 55.00 h

Biomass X (g/L): 102.89
Glucose S (g/L): 54.94
Lactic Acid P (g/L): 81.28
Maltose M (g/L): 27.57
Volume V (L): 7.97

t = 56.00 h

Biomass X (g/L): 68.55
Glucose S (g/L): 62.08
Lactic Acid P (g/L): 72.38
Maltose M (g/L): 23.63
Volume V (L): 8.10

t = 57.00 h

Biomass X (g/L): 119.69
Glucose S (g/L): 62.79
Lactic Acid P (g/L): 81.36
Maltose M (g/L): 27.48
Volume V (L): 8.14

t = 58.00 h

Biomass X (g/L): 99.24
Glucose S (g/L): 65.87
Lactic Acid P (g/L): 70.75
Maltose M (g/L): 24.21
Volume V (L): 8.46

t = 59.00 h

Biomass X (g/L): 90.94
Glucose S (g/L): 74.14
Lactic Acid P (g/L): 74.54
Maltose M (g/L): 25.32
Volume V (L): 7.78

t = 60.00 h

Biomass X (g/L): 88.04
Glucose S (g/L): 63.98
Lactic Acid P (g/L): 76.64
Maltose M (g/L): 25.02
Volume V (L): 8.39

t = 61.00 h

Biomass X (g/L): 91.15
Glucose S (g/L): 60.46
Lactic Acid P (g/L): 66.99
Maltose M (g/L): 25.97
Volume V (L): 8.60

t = 62.00 h

Biomass X (g/L): 93.23
Glucose S (g/L): 63.21
Lactic Acid P (g/L): 83.60
Maltose M (g/L): 23.53
Volume V (L): 7.71

t = 63.00 h

Biomass X (g/L): 94.23
Glucose S (g/L): 55.88
Lactic Acid P (g/L): 71.95
Maltose M (g/L): 25.59
Volume V (L): 8.43

t = 64.00 h

Biomass X (g/L): 93.15
Glucose S (g/L): 61.30
Lactic Acid P (g/L): 74.80
Maltose M (g/L): 27.13
Volume V (L): 8.22

t = 65.00 h

Biomass X (g/L): 88.39
Glucose S (g/L): 63.14
Lactic Acid P (g/L): 67.75
Maltose M (g/L): 23.41
Volume V (L): 8.41

t = 66.00 h

Biomass X (g/L): 85.38
Glucose S (g/L): 63.17
Lactic Acid P (g/L): 53.13
Maltose M (g/L): 24.94
Volume V (L): 8.60

t = 67.00 h

Biomass X (g/L): 85.61
Glucose S (g/L): 66.78
Lactic Acid P (g/L): 75.63
Maltose M (g/L): 23.91
Volume V (L): 8.39

t = 68.00 h

Biomass X (g/L): 84.96
Glucose S (g/L): 62.91
Lactic Acid P (g/L): 100.32
Maltose M (g/L): 24.95
Volume V (L): 8.38

t = 69.00 h

Biomass X (g/L): 97.67
Glucose S (g/L): 72.25
Lactic Acid P (g/L): 85.12
Maltose M (g/L): 24.40
Volume V (L): 8.66

t = 70.00 h

Biomass X (g/L): 92.18
Glucose S (g/L): 67.71
Lactic Acid P (g/L): 77.69
Maltose M (g/L): 23.58
Volume V (L): 8.36

t = 71.00 h

Biomass X (g/L): 57.55
Glucose S (g/L): 68.20
Lactic Acid P (g/L): 75.53
Maltose M (g/L): 24.66
Volume V (L): 8.41

t = 72.00 h

Biomass X (g/L): 92.33
Glucose S (g/L): 67.29
Lactic Acid P (g/L): 78.36
Maltose M (g/L): 25.95
Volume V (L): 9.02

t = 73.00 h

Biomass X (g/L): 87.61
Glucose S (g/L): 70.00
Lactic Acid P (g/L): 83.44
Maltose M (g/L): 23.71
Volume V (L): 8.50

t = 74.00 h

Biomass X (g/L): 85.30
Glucose S (g/L): 73.56
Lactic Acid P (g/L): 67.28
Maltose M (g/L): 24.57
Volume V (L): 9.12

t = 75.00 h

Biomass X (g/L): 81.46
Glucose S (g/L): 66.41
Lactic Acid P (g/L): 89.04
Maltose M (g/L): 22.31

Volume V (L): 9.11

Batch 7

Initial Conditions:

Batch: 7.00

X0 (g/L): 7.05

S0 (g/L): 73.77

M0 (g/L): 40.04

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.54

F_vol (L/h): 0.06

S_feed (g/L): 194.46

M_feed (g/L): 105.54

V_final (L): 9.26

Batch 7 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 72.34

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 38.76

Volume V (L): 5.37

t = 1.00 h

Biomass X (g/L): 2.89

Glucose S (g/L): 73.98

Lactic Acid P (g/L): 0.18

Maltose M (g/L): 38.75

Volume V (L): 4.84

t = 2.00 h

Biomass X (g/L): 24.04

Glucose S (g/L): 65.60

Lactic Acid P (g/L): 6.11

Maltose M (g/L): 40.41

Volume V (L): 4.66

t = 3.00 h

Biomass X (g/L): 6.55

Glucose S (g/L): 72.41

Lactic Acid P (g/L): 3.25

Maltose M (g/L): 40.42

Volume V (L): 5.10

t = 4.00 h

Biomass X (g/L): 15.72

Glucose S (g/L): 64.64

Lactic Acid P (g/L): 8.53

Maltose M (g/L): 40.36

Volume V (L): 4.95

t = 5.00 h

Biomass X (g/L): 29.32

Glucose S (g/L): 60.68

Lactic Acid P (g/L): 20.82

Maltose M (g/L): 39.54

Volume V (L): 5.26

t = 6.00 h

Biomass X (g/L): 37.34

Glucose S (g/L): 56.97

Lactic Acid P (g/L): 12.74

Maltose M (g/L): 39.59

Volume V (L): 5.76

t = 7.00 h

Biomass X (g/L): 32.31

Glucose S (g/L): 63.91

Lactic Acid P (g/L): 5.61

Maltose M (g/L): 39.95

Volume V (L): 5.24

t = 8.00 h

Biomass X (g/L): 33.90

Glucose S (g/L): 53.95

Lactic Acid P (g/L): 40.46

Maltose M (g/L): 39.64

Volume V (L): 5.44

t = 9.00 h

Biomass X (g/L): 58.26

Glucose S (g/L): 59.18

Lactic Acid P (g/L): 40.64

Maltose M (g/L): 37.38

Volume V (L): 5.53

t = 10.00 h

Biomass X (g/L): 65.81

Glucose S (g/L): 48.30

Lactic Acid P (g/L): 19.25

Maltose M (g/L): 39.17

Volume V (L): 5.86

t = 11.00 h

Biomass X (g/L): 62.58

Glucose S (g/L): 41.16
Lactic Acid P (g/L): 40.84
Maltose M (g/L): 38.49
Volume V (L): 4.86

t = 12.00 h

Biomass X (g/L): 57.71
Glucose S (g/L): 41.11
Lactic Acid P (g/L): 32.73
Maltose M (g/L): 39.36
Volume V (L): 5.47

t = 13.00 h

Biomass X (g/L): 52.48
Glucose S (g/L): 48.15
Lactic Acid P (g/L): 42.57
Maltose M (g/L): 41.92
Volume V (L): 5.81

t = 14.00 h

Biomass X (g/L): 55.36
Glucose S (g/L): 41.03
Lactic Acid P (g/L): 45.35
Maltose M (g/L): 39.34
Volume V (L): 6.20

t = 15.00 h

Biomass X (g/L): 71.26
Glucose S (g/L): 38.04
Lactic Acid P (g/L): 42.73
Maltose M (g/L): 38.59
Volume V (L): 5.68

t = 16.00 h

Biomass X (g/L): 72.73
Glucose S (g/L): 39.54
Lactic Acid P (g/L): 39.72
Maltose M (g/L): 39.65
Volume V (L): 5.27

t = 17.00 h

Biomass X (g/L): 84.49
Glucose S (g/L): 39.84
Lactic Acid P (g/L): 42.49
Maltose M (g/L): 39.12
Volume V (L): 6.18

t = 18.00 h

Biomass X (g/L): 79.19
Glucose S (g/L): 41.40
Lactic Acid P (g/L): 42.15
Maltose M (g/L): 38.94
Volume V (L): 5.81

t = 19.00 h

Biomass X (g/L): 60.08
Glucose S (g/L): 42.03
Lactic Acid P (g/L): 44.53
Maltose M (g/L): 39.00
Volume V (L): 5.67

t = 20.00 h

Biomass X (g/L): 86.44
Glucose S (g/L): 34.56
Lactic Acid P (g/L): 53.10
Maltose M (g/L): 40.25
Volume V (L): 5.92

t = 21.00 h

Biomass X (g/L): 67.98
Glucose S (g/L): 30.48
Lactic Acid P (g/L): 48.44
Maltose M (g/L): 37.11
Volume V (L): 6.13

t = 22.00 h

Biomass X (g/L): 75.20
Glucose S (g/L): 33.76
Lactic Acid P (g/L): 62.66
Maltose M (g/L): 38.50
Volume V (L): 6.04

t = 23.00 h

Biomass X (g/L): 82.47
Glucose S (g/L): 36.23
Lactic Acid P (g/L): 71.29
Maltose M (g/L): 37.05
Volume V (L): 5.55

t = 24.00 h

Biomass X (g/L): 100.41
Glucose S (g/L): 31.47
Lactic Acid P (g/L): 53.20
Maltose M (g/L): 38.33
Volume V (L): 6.67

t = 25.00 h

Biomass X (g/L): 112.12
Glucose S (g/L): 31.53
Lactic Acid P (g/L): 61.80
Maltose M (g/L): 36.80
Volume V (L): 6.85

t = 26.00 h

Biomass X (g/L): 88.64
Glucose S (g/L): 36.67
Lactic Acid P (g/L): 65.15
Maltose M (g/L): 37.14
Volume V (L): 6.70

t = 27.00 h

Biomass X (g/L): 81.70
Glucose S (g/L): 35.33
Lactic Acid P (g/L): 65.77
Maltose M (g/L): 37.12
Volume V (L): 6.15

t = 28.00 h

Biomass X (g/L): 89.30
Glucose S (g/L): 35.88
Lactic Acid P (g/L): 62.91
Maltose M (g/L): 38.19
Volume V (L): 6.52

t = 29.00 h

Biomass X (g/L): 93.29
Glucose S (g/L): 33.11
Lactic Acid P (g/L): 69.87
Maltose M (g/L): 38.00
Volume V (L): 6.64

t = 30.00 h

Biomass X (g/L): 82.07
Glucose S (g/L): 35.48
Lactic Acid P (g/L): 68.18
Maltose M (g/L): 35.82
Volume V (L): 6.91

t = 31.00 h

Biomass X (g/L): 103.04
Glucose S (g/L): 33.17
Lactic Acid P (g/L): 52.90
Maltose M (g/L): 36.91
Volume V (L): 6.99

t = 32.00 h

Biomass X (g/L): 88.25
Glucose S (g/L): 40.57
Lactic Acid P (g/L): 73.09
Maltose M (g/L): 36.05
Volume V (L): 6.57

t = 33.00 h

Biomass X (g/L): 91.18
Glucose S (g/L): 41.56
Lactic Acid P (g/L): 73.86
Maltose M (g/L): 35.39
Volume V (L): 6.90

t = 34.00 h

Biomass X (g/L): 93.11
Glucose S (g/L): 30.90
Lactic Acid P (g/L): 59.13
Maltose M (g/L): 36.23

Volume V (L): 7.09

t = 35.00 h

Biomass X (g/L): 91.00

Glucose S (g/L): 34.92

Lactic Acid P (g/L): 67.11

Maltose M (g/L): 36.00

Volume V (L): 7.48

t = 36.00 h

Biomass X (g/L): 95.27

Glucose S (g/L): 33.23

Lactic Acid P (g/L): 73.61

Maltose M (g/L): 38.44

Volume V (L): 7.28

t = 37.00 h

Biomass X (g/L): 81.33

Glucose S (g/L): 37.02

Lactic Acid P (g/L): 65.82

Maltose M (g/L): 36.41

Volume V (L): 7.37

t = 38.00 h

Biomass X (g/L): 82.06

Glucose S (g/L): 39.75

Lactic Acid P (g/L): 78.19

Maltose M (g/L): 34.56

Volume V (L): 7.19

t = 39.00 h

Biomass X (g/L): 76.93

Glucose S (g/L): 35.03

Lactic Acid P (g/L): 90.61

Maltose M (g/L): 35.18

Volume V (L): 7.69

t = 40.00 h

Biomass X (g/L): 91.96

Glucose S (g/L): 37.63

Lactic Acid P (g/L): 61.82

Maltose M (g/L): 36.54

Volume V (L): 7.41

t = 41.00 h

Biomass X (g/L): 95.33

Glucose S (g/L): 37.89

Lactic Acid P (g/L): 56.70

Maltose M (g/L): 37.40

Volume V (L): 7.03

t = 42.00 h

Biomass X (g/L): 92.27

Glucose S (g/L): 43.15

Lactic Acid P (g/L): 72.72

Maltose M (g/L): 33.78

Volume V (L): 7.86

t = 43.00 h

Biomass X (g/L): 104.10

Glucose S (g/L): 43.21

Lactic Acid P (g/L): 76.97

Maltose M (g/L): 33.36

Volume V (L): 7.68

t = 44.00 h

Biomass X (g/L): 87.82

Glucose S (g/L): 40.76

Lactic Acid P (g/L): 68.49

Maltose M (g/L): 36.76

Volume V (L): 7.35

t = 45.00 h

Biomass X (g/L): 94.36

Glucose S (g/L): 44.83

Lactic Acid P (g/L): 84.46

Maltose M (g/L): 35.21

Volume V (L): 7.39

t = 46.00 h

Biomass X (g/L): 77.26

Glucose S (g/L): 42.65

Lactic Acid P (g/L): 58.89

Maltose M (g/L): 33.88

Volume V (L): 7.71

t = 47.00 h

Biomass X (g/L): 84.06

Glucose S (g/L): 44.33

Lactic Acid P (g/L): 78.05

Maltose M (g/L): 35.06

Volume V (L): 7.83

t = 48.00 h

Biomass X (g/L): 94.51

Glucose S (g/L): 52.45

Lactic Acid P (g/L): 66.31

Maltose M (g/L): 31.92

Volume V (L): 7.79

t = 49.00 h

Biomass X (g/L): 95.94

Glucose S (g/L): 51.79

Lactic Acid P (g/L): 88.17

Maltose M (g/L): 34.27

Volume V (L): 7.89

t = 50.00 h

Biomass X (g/L): 84.78

Glucose S (g/L): 42.22

Lactic Acid P (g/L): 90.99

Maltose M (g/L): 34.43

Volume V (L): 8.42

t = 51.00 h

Biomass X (g/L): 86.85

Glucose S (g/L): 51.06

Lactic Acid P (g/L): 65.66

Maltose M (g/L): 34.85

Volume V (L): 8.26

t = 52.00 h

Biomass X (g/L): 72.73

Glucose S (g/L): 51.62
Lactic Acid P (g/L): 67.94
Maltose M (g/L): 32.87
Volume V (L): 8.02

t = 53.00 h

Biomass X (g/L): 83.88
Glucose S (g/L): 44.68
Lactic Acid P (g/L): 81.26
Maltose M (g/L): 33.76
Volume V (L): 8.33

t = 54.00 h

Biomass X (g/L): 95.86
Glucose S (g/L): 52.90
Lactic Acid P (g/L): 69.94
Maltose M (g/L): 32.66
Volume V (L): 7.71

t = 55.00 h

Biomass X (g/L): 92.68
Glucose S (g/L): 56.30
Lactic Acid P (g/L): 73.35
Maltose M (g/L): 33.63
Volume V (L): 8.26

t = 56.00 h

Biomass X (g/L): 91.83
Glucose S (g/L): 46.65
Lactic Acid P (g/L): 74.24
Maltose M (g/L): 32.40
Volume V (L): 8.80

t = 57.00 h

Biomass X (g/L): 87.32
Glucose S (g/L): 48.07
Lactic Acid P (g/L): 56.42
Maltose M (g/L): 32.72
Volume V (L): 8.16

t = 58.00 h

Biomass X (g/L): 86.76
Glucose S (g/L): 56.16
Lactic Acid P (g/L): 74.31
Maltose M (g/L): 32.02
Volume V (L): 8.00

t = 59.00 h

Biomass X (g/L): 80.14
Glucose S (g/L): 53.90
Lactic Acid P (g/L): 68.25
Maltose M (g/L): 33.43
Volume V (L): 8.13

t = 60.00 h

Biomass X (g/L): 93.17
Glucose S (g/L): 55.59
Lactic Acid P (g/L): 70.28
Maltose M (g/L): 33.48
Volume V (L): 8.03

t = 61.00 h

Biomass X (g/L): 90.31
Glucose S (g/L): 55.39
Lactic Acid P (g/L): 88.44
Maltose M (g/L): 32.31
Volume V (L): 8.20

t = 62.00 h

Biomass X (g/L): 90.83
Glucose S (g/L): 56.20
Lactic Acid P (g/L): 63.58
Maltose M (g/L): 32.43
Volume V (L): 8.39

t = 63.00 h

Biomass X (g/L): 80.79
Glucose S (g/L): 50.18
Lactic Acid P (g/L): 82.97
Maltose M (g/L): 30.88
Volume V (L): 8.10

t = 64.00 h

Biomass X (g/L): 64.83
Glucose S (g/L): 59.95
Lactic Acid P (g/L): 73.02
Maltose M (g/L): 31.82
Volume V (L): 8.64

t = 65.00 h

Biomass X (g/L): 96.18
Glucose S (g/L): 61.03
Lactic Acid P (g/L): 73.18
Maltose M (g/L): 31.92
Volume V (L): 9.00

t = 66.00 h

Biomass X (g/L): 80.71
Glucose S (g/L): 56.23
Lactic Acid P (g/L): 83.28
Maltose M (g/L): 34.30
Volume V (L): 8.44

t = 67.00 h

Biomass X (g/L): 76.80
Glucose S (g/L): 57.54
Lactic Acid P (g/L): 79.27
Maltose M (g/L): 30.64
Volume V (L): 9.32

t = 68.00 h

Biomass X (g/L): 91.71
Glucose S (g/L): 52.40
Lactic Acid P (g/L): 81.57
Maltose M (g/L): 32.93
Volume V (L): 9.03

t = 69.00 h

Biomass X (g/L): 80.99
Glucose S (g/L): 60.88
Lactic Acid P (g/L): 85.42
Maltose M (g/L): 32.35
Volume V (L): 9.27

t = 70.00 h

Biomass X (g/L): 94.88
Glucose S (g/L): 61.69
Lactic Acid P (g/L): 73.52
Maltose M (g/L): 31.54
Volume V (L): 8.75

t = 71.00 h

Biomass X (g/L): 90.38
Glucose S (g/L): 59.82
Lactic Acid P (g/L): 71.06
Maltose M (g/L): 32.79
Volume V (L): 9.32

t = 72.00 h

Biomass X (g/L): 90.83
Glucose S (g/L): 64.98
Lactic Acid P (g/L): 66.82
Maltose M (g/L): 31.35
Volume V (L): 8.72

t = 73.00 h

Biomass X (g/L): 70.75
Glucose S (g/L): 59.93
Lactic Acid P (g/L): 69.37
Maltose M (g/L): 31.23
Volume V (L): 9.45

t = 74.00 h

Biomass X (g/L): 95.55
Glucose S (g/L): 63.97
Lactic Acid P (g/L): 104.52
Maltose M (g/L): 30.67
Volume V (L): 9.26

t = 75.00 h

Biomass X (g/L): 86.13
Glucose S (g/L): 58.42
Lactic Acid P (g/L): 88.85
Maltose M (g/L): 31.75

Volume V (L): 9.45

Batch 8

Initial Conditions:

Batch: 8.00

X0 (g/L): 4.77

S0 (g/L): 112.83

M0 (g/L): 24.48

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.22

F_vol (L/h): 0.09

S_feed (g/L): 246.51

M_feed (g/L): 53.49

V_final (L): 11.60

Batch 8 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 111.83

Lactic Acid P (g/L): 15.54

Maltose M (g/L): 24.25

Volume V (L): 5.65

t = 1.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 114.91

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 24.75

Volume V (L): 4.95

t = 2.00 h

Biomass X (g/L): 14.27

Glucose S (g/L): 119.04

Lactic Acid P (g/L): 19.22

Maltose M (g/L): 24.62

Volume V (L): 5.36

t = 3.00 h

Biomass X (g/L): 2.12

Glucose S (g/L): 116.86

Lactic Acid P (g/L): 12.09

Maltose M (g/L): 23.25

Volume V (L): 4.96

t = 4.00 h

Biomass X (g/L): 18.83

Glucose S (g/L): 114.82

Lactic Acid P (g/L): 11.49

Maltose M (g/L): 25.01

Volume V (L): 4.39

t = 5.00 h

Biomass X (g/L): 5.03

Glucose S (g/L): 113.64

Lactic Acid P (g/L): 13.22

Maltose M (g/L): 24.67

Volume V (L): 5.58

t = 6.00 h

Biomass X (g/L): 30.42

Glucose S (g/L): 110.53

Lactic Acid P (g/L): 16.53

Maltose M (g/L): 24.84

Volume V (L): 4.52

t = 7.00 h

Biomass X (g/L): 15.00

Glucose S (g/L): 109.80

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 26.89

Volume V (L): 5.58

t = 8.00 h

Biomass X (g/L): 36.77

Glucose S (g/L): 98.04

Lactic Acid P (g/L): 17.34

Maltose M (g/L): 23.22

Volume V (L): 4.98

t = 9.00 h

Biomass X (g/L): 42.18

Glucose S (g/L): 102.06

Lactic Acid P (g/L): 30.90

Maltose M (g/L): 25.50

Volume V (L): 6.84

t = 10.00 h

Biomass X (g/L): 34.12

Glucose S (g/L): 101.04

Lactic Acid P (g/L): 26.37

Maltose M (g/L): 24.25

Volume V (L): 6.36

t = 11.00 h

Biomass X (g/L): 50.76

Glucose S (g/L): 94.38
Lactic Acid P (g/L): 17.58
Maltose M (g/L): 24.55
Volume V (L): 6.02

t = 12.00 h

Biomass X (g/L): 51.80
Glucose S (g/L): 96.66
Lactic Acid P (g/L): 32.82
Maltose M (g/L): 24.30
Volume V (L): 6.72

t = 13.00 h

Biomass X (g/L): 47.94
Glucose S (g/L): 92.82
Lactic Acid P (g/L): 51.46
Maltose M (g/L): 24.02
Volume V (L): 5.70

t = 14.00 h

Biomass X (g/L): 46.92
Glucose S (g/L): 96.85
Lactic Acid P (g/L): 50.09
Maltose M (g/L): 23.88
Volume V (L): 6.52

t = 15.00 h

Biomass X (g/L): 40.93
Glucose S (g/L): 94.73
Lactic Acid P (g/L): 40.90
Maltose M (g/L): 24.37
Volume V (L): 6.43

t = 16.00 h

Biomass X (g/L): 65.90
Glucose S (g/L): 87.31
Lactic Acid P (g/L): 52.25
Maltose M (g/L): 23.18
Volume V (L): 6.55

t = 17.00 h

Biomass X (g/L): 50.12
Glucose S (g/L): 87.15
Lactic Acid P (g/L): 42.57
Maltose M (g/L): 24.49
Volume V (L): 7.17

t = 18.00 h

Biomass X (g/L): 68.30
Glucose S (g/L): 86.47
Lactic Acid P (g/L): 32.54
Maltose M (g/L): 23.71
Volume V (L): 6.97

t = 19.00 h

Biomass X (g/L): 66.90
Glucose S (g/L): 84.63
Lactic Acid P (g/L): 41.26
Maltose M (g/L): 23.21
Volume V (L): 6.82

t = 20.00 h

Biomass X (g/L): 77.75
Glucose S (g/L): 86.01
Lactic Acid P (g/L): 65.10
Maltose M (g/L): 23.55
Volume V (L): 5.89

t = 21.00 h

Biomass X (g/L): 69.98
Glucose S (g/L): 82.77
Lactic Acid P (g/L): 50.15
Maltose M (g/L): 24.50
Volume V (L): 6.34

t = 22.00 h

Biomass X (g/L): 71.16
Glucose S (g/L): 81.38
Lactic Acid P (g/L): 51.77
Maltose M (g/L): 23.34
Volume V (L): 6.87

t = 23.00 h

Biomass X (g/L): 82.62
Glucose S (g/L): 86.46
Lactic Acid P (g/L): 52.51
Maltose M (g/L): 23.80
Volume V (L): 7.07

t = 24.00 h

Biomass X (g/L): 79.97
Glucose S (g/L): 86.42
Lactic Acid P (g/L): 73.35
Maltose M (g/L): 23.05
Volume V (L): 6.52

t = 25.00 h

Biomass X (g/L): 80.81
Glucose S (g/L): 82.17
Lactic Acid P (g/L): 71.92
Maltose M (g/L): 23.87
Volume V (L): 8.47

t = 26.00 h

Biomass X (g/L): 83.19
Glucose S (g/L): 87.35
Lactic Acid P (g/L): 78.20
Maltose M (g/L): 23.74
Volume V (L): 8.09

t = 27.00 h

Biomass X (g/L): 64.80
Glucose S (g/L): 82.96
Lactic Acid P (g/L): 57.85
Maltose M (g/L): 24.35
Volume V (L): 7.69

t = 28.00 h

Biomass X (g/L): 87.47
Glucose S (g/L): 85.83
Lactic Acid P (g/L): 62.76
Maltose M (g/L): 23.03
Volume V (L): 7.50

t = 29.00 h

Biomass X (g/L): 86.73
Glucose S (g/L): 87.81
Lactic Acid P (g/L): 47.35
Maltose M (g/L): 23.45
Volume V (L): 7.44

t = 30.00 h

Biomass X (g/L): 71.69
Glucose S (g/L): 85.58
Lactic Acid P (g/L): 74.83
Maltose M (g/L): 23.01
Volume V (L): 7.14

t = 31.00 h

Biomass X (g/L): 79.59
Glucose S (g/L): 87.41
Lactic Acid P (g/L): 51.88
Maltose M (g/L): 23.65
Volume V (L): 7.77

t = 32.00 h

Biomass X (g/L): 81.99
Glucose S (g/L): 84.90
Lactic Acid P (g/L): 64.90
Maltose M (g/L): 22.33
Volume V (L): 7.95

t = 33.00 h

Biomass X (g/L): 81.46
Glucose S (g/L): 86.29
Lactic Acid P (g/L): 53.76
Maltose M (g/L): 23.12
Volume V (L): 7.25

t = 34.00 h

Biomass X (g/L): 86.28
Glucose S (g/L): 86.71
Lactic Acid P (g/L): 58.87
Maltose M (g/L): 22.45

Volume V (L): 8.17

t = 35.00 h

Biomass X (g/L): 97.08

Glucose S (g/L): 85.57

Lactic Acid P (g/L): 57.18

Maltose M (g/L): 22.43

Volume V (L): 8.00

t = 36.00 h

Biomass X (g/L): 93.98

Glucose S (g/L): 87.76

Lactic Acid P (g/L): 59.73

Maltose M (g/L): 22.43

Volume V (L): 7.16

t = 37.00 h

Biomass X (g/L): 71.27

Glucose S (g/L): 84.17

Lactic Acid P (g/L): 60.82

Maltose M (g/L): 22.43

Volume V (L): 7.80

t = 38.00 h

Biomass X (g/L): 89.97

Glucose S (g/L): 87.84

Lactic Acid P (g/L): 61.19

Maltose M (g/L): 22.10

Volume V (L): 8.13

t = 39.00 h

Biomass X (g/L): 96.89

Glucose S (g/L): 88.99

Lactic Acid P (g/L): 73.97

Maltose M (g/L): 22.20

Volume V (L): 9.43

t = 40.00 h

Biomass X (g/L): 87.99

Glucose S (g/L): 89.62

Lactic Acid P (g/L): 57.05

Maltose M (g/L): 22.93

Volume V (L): 8.81

t = 41.00 h

Biomass X (g/L): 94.83

Glucose S (g/L): 90.21

Lactic Acid P (g/L): 69.28

Maltose M (g/L): 22.47

Volume V (L): 7.90

t = 42.00 h

Biomass X (g/L): 103.73

Glucose S (g/L): 91.29

Lactic Acid P (g/L): 78.56

Maltose M (g/L): 22.30

Volume V (L): 8.95

t = 43.00 h

Biomass X (g/L): 94.31

Glucose S (g/L): 93.47

Lactic Acid P (g/L): 63.44

Maltose M (g/L): 21.39

Volume V (L): 7.97

t = 44.00 h

Biomass X (g/L): 86.93

Glucose S (g/L): 95.21

Lactic Acid P (g/L): 61.94

Maltose M (g/L): 21.02

Volume V (L): 9.44

t = 45.00 h

Biomass X (g/L): 94.57

Glucose S (g/L): 91.22

Lactic Acid P (g/L): 66.62

Maltose M (g/L): 22.04

Volume V (L): 8.71

t = 46.00 h

Biomass X (g/L): 85.94

Glucose S (g/L): 90.13

Lactic Acid P (g/L): 79.53

Maltose M (g/L): 21.89

Volume V (L): 9.00

t = 47.00 h

Biomass X (g/L): 84.77

Glucose S (g/L): 91.18

Lactic Acid P (g/L): 71.01

Maltose M (g/L): 22.15

Volume V (L): 9.84

t = 48.00 h

Biomass X (g/L): 81.14

Glucose S (g/L): 93.40

Lactic Acid P (g/L): 64.81

Maltose M (g/L): 20.79

Volume V (L): 8.55

t = 49.00 h

Biomass X (g/L): 99.44

Glucose S (g/L): 94.37

Lactic Acid P (g/L): 74.15

Maltose M (g/L): 20.91

Volume V (L): 8.58

t = 50.00 h

Biomass X (g/L): 76.57

Glucose S (g/L): 93.68

Lactic Acid P (g/L): 60.24

Maltose M (g/L): 19.96

Volume V (L): 9.09

t = 51.00 h

Biomass X (g/L): 99.02

Glucose S (g/L): 92.67

Lactic Acid P (g/L): 76.43

Maltose M (g/L): 20.61

Volume V (L): 10.04

t = 52.00 h

Biomass X (g/L): 89.05

Glucose S (g/L): 95.74
Lactic Acid P (g/L): 74.23
Maltose M (g/L): 21.35
Volume V (L): 8.77

t = 53.00 h

Biomass X (g/L): 70.78
Glucose S (g/L): 98.20
Lactic Acid P (g/L): 68.06
Maltose M (g/L): 20.36
Volume V (L): 8.17

t = 54.00 h

Biomass X (g/L): 87.61
Glucose S (g/L): 99.99
Lactic Acid P (g/L): 63.54
Maltose M (g/L): 20.93
Volume V (L): 9.52

t = 55.00 h

Biomass X (g/L): 63.88
Glucose S (g/L): 97.10
Lactic Acid P (g/L): 63.32
Maltose M (g/L): 20.32
Volume V (L): 10.13

t = 56.00 h

Biomass X (g/L): 89.87
Glucose S (g/L): 98.07
Lactic Acid P (g/L): 76.24
Maltose M (g/L): 20.54
Volume V (L): 10.56

t = 57.00 h

Biomass X (g/L): 92.24
Glucose S (g/L): 97.08
Lactic Acid P (g/L): 82.64
Maltose M (g/L): 19.54
Volume V (L): 9.77

t = 58.00 h

Biomass X (g/L): 77.97
Glucose S (g/L): 100.29
Lactic Acid P (g/L): 66.16
Maltose M (g/L): 20.19
Volume V (L): 9.89

t = 59.00 h

Biomass X (g/L): 84.70
Glucose S (g/L): 98.39
Lactic Acid P (g/L): 76.14
Maltose M (g/L): 18.22
Volume V (L): 10.80

t = 60.00 h

Biomass X (g/L): 102.10
Glucose S (g/L): 103.16
Lactic Acid P (g/L): 53.58
Maltose M (g/L): 19.82
Volume V (L): 9.29

t = 61.00 h

Biomass X (g/L): 78.41
Glucose S (g/L): 96.64
Lactic Acid P (g/L): 67.18
Maltose M (g/L): 18.85
Volume V (L): 10.16

t = 62.00 h

Biomass X (g/L): 72.97
Glucose S (g/L): 101.42
Lactic Acid P (g/L): 72.31
Maltose M (g/L): 20.47
Volume V (L): 10.56

t = 63.00 h

Biomass X (g/L): 97.15
Glucose S (g/L): 103.91
Lactic Acid P (g/L): 64.62
Maltose M (g/L): 20.11
Volume V (L): 10.78

t = 64.00 h

Biomass X (g/L): 101.60
Glucose S (g/L): 104.07
Lactic Acid P (g/L): 64.55
Maltose M (g/L): 19.82
Volume V (L): 10.36

t = 65.00 h

Biomass X (g/L): 82.70
Glucose S (g/L): 112.07
Lactic Acid P (g/L): 67.11
Maltose M (g/L): 20.36
Volume V (L): 10.74

t = 66.00 h

Biomass X (g/L): 92.19
Glucose S (g/L): 105.58
Lactic Acid P (g/L): 75.84
Maltose M (g/L): 18.77
Volume V (L): 9.50

t = 67.00 h

Biomass X (g/L): 95.19
Glucose S (g/L): 102.03
Lactic Acid P (g/L): 71.71
Maltose M (g/L): 20.30
Volume V (L): 10.55

t = 68.00 h

Biomass X (g/L): 75.08
Glucose S (g/L): 105.23
Lactic Acid P (g/L): 53.53
Maltose M (g/L): 19.71
Volume V (L): 11.23

t = 69.00 h

Biomass X (g/L): 77.37
Glucose S (g/L): 100.70
Lactic Acid P (g/L): 54.72
Maltose M (g/L): 18.91
Volume V (L): 10.55

t = 70.00 h

Biomass X (g/L): 93.64
Glucose S (g/L): 106.25
Lactic Acid P (g/L): 68.39
Maltose M (g/L): 19.64
Volume V (L): 11.19

t = 71.00 h

Biomass X (g/L): 97.82
Glucose S (g/L): 104.47
Lactic Acid P (g/L): 78.04
Maltose M (g/L): 18.19
Volume V (L): 11.49

t = 72.00 h

Biomass X (g/L): 86.91
Glucose S (g/L): 110.17
Lactic Acid P (g/L): 49.53
Maltose M (g/L): 18.26
Volume V (L): 11.16

t = 73.00 h

Biomass X (g/L): 72.92
Glucose S (g/L): 106.26
Lactic Acid P (g/L): 69.75
Maltose M (g/L): 18.30
Volume V (L): 11.52

t = 74.00 h

Biomass X (g/L): 100.20
Glucose S (g/L): 108.01
Lactic Acid P (g/L): 69.65
Maltose M (g/L): 18.25
Volume V (L): 11.01

t = 75.00 h

Biomass X (g/L): 75.92
Glucose S (g/L): 106.92
Lactic Acid P (g/L): 86.88
Maltose M (g/L): 17.84

Volume V (L): 11.49

Batch 9

Initial Conditions:

Batch: 9.00

X0 (g/L): 7.99

S0 (g/L): 36.69

M0 (g/L): 119.94

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 3.27

F_vol (L/h): 0.06

S_feed (g/L): 70.27

M_feed (g/L): 229.73

V_final (L): 9.85

Batch 9 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 13.65
Glucose S (g/L): 35.99
Lactic Acid P (g/L): 20.66
Maltose M (g/L): 121.01
Volume V (L): 4.63

t = 1.00 h

Biomass X (g/L): 11.92
Glucose S (g/L): 34.16
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 122.03
Volume V (L): 5.70

t = 2.00 h

Biomass X (g/L): 8.04
Glucose S (g/L): 40.17
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 117.15
Volume V (L): 5.02

t = 3.00 h

Biomass X (g/L): 19.09
Glucose S (g/L): 41.34
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 113.23
Volume V (L): 5.81

t = 4.00 h

Biomass X (g/L): 27.62
Glucose S (g/L): 32.20
Lactic Acid P (g/L): 16.16
Maltose M (g/L): 120.46
Volume V (L): 5.52

t = 5.00 h

Biomass X (g/L): 6.30
Glucose S (g/L): 29.82

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 118.86
Volume V (L): 5.27

t = 6.00 h

Biomass X (g/L): 40.59
Glucose S (g/L): 31.57
Lactic Acid P (g/L): 12.93
Maltose M (g/L): 110.44
Volume V (L): 5.19

t = 7.00 h

Biomass X (g/L): 31.28
Glucose S (g/L): 26.25
Lactic Acid P (g/L): 23.69
Maltose M (g/L): 120.09
Volume V (L): 5.32

t = 8.00 h

Biomass X (g/L): 42.80
Glucose S (g/L): 24.76
Lactic Acid P (g/L): 24.77
Maltose M (g/L): 118.41
Volume V (L): 5.68

t = 9.00 h

Biomass X (g/L): 54.21
Glucose S (g/L): 25.76
Lactic Acid P (g/L): 30.19
Maltose M (g/L): 112.80
Volume V (L): 5.61

t = 10.00 h

Biomass X (g/L): 34.35
Glucose S (g/L): 14.38
Lactic Acid P (g/L): 31.41
Maltose M (g/L): 111.73
Volume V (L): 5.42

t = 11.00 h

Biomass X (g/L): 44.16

Glucose S (g/L): 24.19
Lactic Acid P (g/L): 43.83
Maltose M (g/L): 113.12
Volume V (L): 5.65

t = 12.00 h

Biomass X (g/L): 51.40
Glucose S (g/L): 18.32
Lactic Acid P (g/L): 26.27
Maltose M (g/L): 105.78
Volume V (L): 5.81

t = 13.00 h

Biomass X (g/L): 49.83
Glucose S (g/L): 15.41
Lactic Acid P (g/L): 31.19
Maltose M (g/L): 109.12
Volume V (L): 5.01

t = 14.00 h

Biomass X (g/L): 75.92
Glucose S (g/L): 15.60
Lactic Acid P (g/L): 32.55
Maltose M (g/L): 98.26
Volume V (L): 6.08

t = 15.00 h

Biomass X (g/L): 54.33
Glucose S (g/L): 15.05
Lactic Acid P (g/L): 49.11
Maltose M (g/L): 111.49
Volume V (L): 5.59

t = 16.00 h

Biomass X (g/L): 59.34
Glucose S (g/L): 8.47
Lactic Acid P (g/L): 51.63
Maltose M (g/L): 99.46
Volume V (L): 5.59

t = 17.00 h

Biomass X (g/L): 68.38
Glucose S (g/L): 19.59
Lactic Acid P (g/L): 49.29
Maltose M (g/L): 112.88
Volume V (L): 5.44

t = 18.00 h

Biomass X (g/L): 57.13
Glucose S (g/L): 14.17
Lactic Acid P (g/L): 57.53
Maltose M (g/L): 103.66
Volume V (L): 5.47

t = 19.00 h

Biomass X (g/L): 66.25
Glucose S (g/L): 15.49
Lactic Acid P (g/L): 37.63
Maltose M (g/L): 103.48
Volume V (L): 5.70

t = 20.00 h

Biomass X (g/L): 74.35
Glucose S (g/L): 17.30
Lactic Acid P (g/L): 64.79
Maltose M (g/L): 100.83
Volume V (L): 6.02

t = 21.00 h

Biomass X (g/L): 69.66
Glucose S (g/L): 10.36
Lactic Acid P (g/L): 64.78
Maltose M (g/L): 101.68
Volume V (L): 7.12

t = 22.00 h

Biomass X (g/L): 73.11
Glucose S (g/L): 15.86
Lactic Acid P (g/L): 58.86
Maltose M (g/L): 108.86
Volume V (L): 6.22

t = 23.00 h

Biomass X (g/L): 70.19
Glucose S (g/L): 13.46
Lactic Acid P (g/L): 69.47
Maltose M (g/L): 90.11
Volume V (L): 6.56

t = 24.00 h

Biomass X (g/L): 62.08
Glucose S (g/L): 17.73
Lactic Acid P (g/L): 39.41
Maltose M (g/L): 106.81
Volume V (L): 6.71

t = 25.00 h

Biomass X (g/L): 80.06
Glucose S (g/L): 7.50
Lactic Acid P (g/L): 50.68
Maltose M (g/L): 101.79
Volume V (L): 6.67

t = 26.00 h

Biomass X (g/L): 94.21
Glucose S (g/L): 15.13
Lactic Acid P (g/L): 49.73
Maltose M (g/L): 93.97
Volume V (L): 6.65

t = 27.00 h

Biomass X (g/L): 78.89
Glucose S (g/L): 13.78
Lactic Acid P (g/L): 63.83
Maltose M (g/L): 93.91
Volume V (L): 6.28

t = 28.00 h

Biomass X (g/L): 81.54
Glucose S (g/L): 17.95
Lactic Acid P (g/L): 48.65
Maltose M (g/L): 105.96
Volume V (L): 7.37

t = 29.00 h

Biomass X (g/L): 86.53
Glucose S (g/L): 18.60
Lactic Acid P (g/L): 65.84
Maltose M (g/L): 93.19
Volume V (L): 7.58

t = 30.00 h

Biomass X (g/L): 68.94
Glucose S (g/L): 14.96
Lactic Acid P (g/L): 72.17
Maltose M (g/L): 91.58
Volume V (L): 6.29

t = 31.00 h

Biomass X (g/L): 97.06
Glucose S (g/L): 15.51
Lactic Acid P (g/L): 66.79
Maltose M (g/L): 101.52
Volume V (L): 6.61

t = 32.00 h

Biomass X (g/L): 83.64
Glucose S (g/L): 18.25
Lactic Acid P (g/L): 63.77
Maltose M (g/L): 97.20
Volume V (L): 6.96

t = 33.00 h

Biomass X (g/L): 97.36
Glucose S (g/L): 16.07
Lactic Acid P (g/L): 52.93
Maltose M (g/L): 90.67
Volume V (L): 6.76

t = 34.00 h

Biomass X (g/L): 75.84
Glucose S (g/L): 22.91
Lactic Acid P (g/L): 54.85
Maltose M (g/L): 95.38

Volume V (L): 7.39

t = 35.00 h

Biomass X (g/L): 99.88

Glucose S (g/L): 19.25

Lactic Acid P (g/L): 74.45

Maltose M (g/L): 89.03

Volume V (L): 6.98

t = 36.00 h

Biomass X (g/L): 90.54

Glucose S (g/L): 10.88

Lactic Acid P (g/L): 70.78

Maltose M (g/L): 98.25

Volume V (L): 8.17

t = 37.00 h

Biomass X (g/L): 88.78

Glucose S (g/L): 20.58

Lactic Acid P (g/L): 63.93

Maltose M (g/L): 97.46

Volume V (L): 7.85

t = 38.00 h

Biomass X (g/L): 86.10

Glucose S (g/L): 18.91

Lactic Acid P (g/L): 61.03

Maltose M (g/L): 94.42

Volume V (L): 7.54

t = 39.00 h

Biomass X (g/L): 81.86

Glucose S (g/L): 17.51

Lactic Acid P (g/L): 46.99

Maltose M (g/L): 93.04

Volume V (L): 7.92

t = 40.00 h

Biomass X (g/L): 95.52

Glucose S (g/L): 19.63

Lactic Acid P (g/L): 56.13

Maltose M (g/L): 98.93

Volume V (L): 8.01

t = 41.00 h

Biomass X (g/L): 85.87

Glucose S (g/L): 25.15

Lactic Acid P (g/L): 83.61

Maltose M (g/L): 90.21

Volume V (L): 7.43

t = 42.00 h

Biomass X (g/L): 66.48

Glucose S (g/L): 30.22

Lactic Acid P (g/L): 55.16

Maltose M (g/L): 84.18

Volume V (L): 7.40

t = 43.00 h

Biomass X (g/L): 74.50

Glucose S (g/L): 27.34

Lactic Acid P (g/L): 68.90

Maltose M (g/L): 91.16

Volume V (L): 7.81

t = 44.00 h

Biomass X (g/L): 83.26

Glucose S (g/L): 24.81

Lactic Acid P (g/L): 76.73

Maltose M (g/L): 98.16

Volume V (L): 8.18

t = 45.00 h

Biomass X (g/L): 87.95

Glucose S (g/L): 29.34

Lactic Acid P (g/L): 56.64

Maltose M (g/L): 83.28

Volume V (L): 7.96

t = 46.00 h

Biomass X (g/L): 84.63

Glucose S (g/L): 26.83

Lactic Acid P (g/L): 70.07

Maltose M (g/L): 85.75

Volume V (L): 7.36

t = 47.00 h

Biomass X (g/L): 96.52

Glucose S (g/L): 26.74

Lactic Acid P (g/L): 77.26

Maltose M (g/L): 88.80

Volume V (L): 7.73

t = 48.00 h

Biomass X (g/L): 67.16

Glucose S (g/L): 25.86

Lactic Acid P (g/L): 69.24

Maltose M (g/L): 86.09

Volume V (L): 8.07

t = 49.00 h

Biomass X (g/L): 86.62

Glucose S (g/L): 24.48

Lactic Acid P (g/L): 95.45

Maltose M (g/L): 81.27

Volume V (L): 8.14

t = 50.00 h

Biomass X (g/L): 68.43

Glucose S (g/L): 30.09

Lactic Acid P (g/L): 54.70

Maltose M (g/L): 90.47

Volume V (L): 9.00

t = 51.00 h

Biomass X (g/L): 92.34

Glucose S (g/L): 32.80

Lactic Acid P (g/L): 79.72

Maltose M (g/L): 95.49

Volume V (L): 7.76

t = 52.00 h

Biomass X (g/L): 84.40

Glucose S (g/L): 32.64
Lactic Acid P (g/L): 68.87
Maltose M (g/L): 85.16
Volume V (L): 8.56

t = 53.00 h

Biomass X (g/L): 89.14
Glucose S (g/L): 32.26
Lactic Acid P (g/L): 77.73
Maltose M (g/L): 79.56
Volume V (L): 9.00

t = 54.00 h

Biomass X (g/L): 77.20
Glucose S (g/L): 31.12
Lactic Acid P (g/L): 75.03
Maltose M (g/L): 76.08
Volume V (L): 9.37

t = 55.00 h

Biomass X (g/L): 97.03
Glucose S (g/L): 37.41
Lactic Acid P (g/L): 64.67
Maltose M (g/L): 83.93
Volume V (L): 8.40

t = 56.00 h

Biomass X (g/L): 94.76
Glucose S (g/L): 31.30
Lactic Acid P (g/L): 73.65
Maltose M (g/L): 87.35
Volume V (L): 8.81

t = 57.00 h

Biomass X (g/L): 90.44
Glucose S (g/L): 39.50
Lactic Acid P (g/L): 55.02
Maltose M (g/L): 83.95
Volume V (L): 8.91

t = 58.00 h

Biomass X (g/L): 83.75
Glucose S (g/L): 33.91
Lactic Acid P (g/L): 77.56
Maltose M (g/L): 81.26
Volume V (L): 8.83

t = 59.00 h

Biomass X (g/L): 90.25
Glucose S (g/L): 35.54
Lactic Acid P (g/L): 75.04
Maltose M (g/L): 78.07
Volume V (L): 8.78

t = 60.00 h

Biomass X (g/L): 78.22
Glucose S (g/L): 33.74
Lactic Acid P (g/L): 78.37
Maltose M (g/L): 75.27
Volume V (L): 9.05

t = 61.00 h

Biomass X (g/L): 92.61
Glucose S (g/L): 37.91
Lactic Acid P (g/L): 82.40
Maltose M (g/L): 83.70
Volume V (L): 7.96

t = 62.00 h

Biomass X (g/L): 99.86
Glucose S (g/L): 43.90
Lactic Acid P (g/L): 76.37
Maltose M (g/L): 83.60
Volume V (L): 9.27

t = 63.00 h

Biomass X (g/L): 94.38
Glucose S (g/L): 42.99
Lactic Acid P (g/L): 74.60
Maltose M (g/L): 71.78
Volume V (L): 9.12

t = 64.00 h

Biomass X (g/L): 77.56
Glucose S (g/L): 45.36
Lactic Acid P (g/L): 65.49
Maltose M (g/L): 85.81
Volume V (L): 9.05

t = 65.00 h

Biomass X (g/L): 87.55
Glucose S (g/L): 44.46
Lactic Acid P (g/L): 67.23
Maltose M (g/L): 80.66
Volume V (L): 10.00

t = 66.00 h

Biomass X (g/L): 72.07
Glucose S (g/L): 39.38
Lactic Acid P (g/L): 84.62
Maltose M (g/L): 84.03
Volume V (L): 9.00

t = 67.00 h

Biomass X (g/L): 89.24
Glucose S (g/L): 42.84
Lactic Acid P (g/L): 87.08
Maltose M (g/L): 70.92
Volume V (L): 9.18

t = 68.00 h

Biomass X (g/L): 87.97
Glucose S (g/L): 41.95
Lactic Acid P (g/L): 72.11
Maltose M (g/L): 70.37
Volume V (L): 10.17

t = 69.00 h

Biomass X (g/L): 66.30
Glucose S (g/L): 50.17
Lactic Acid P (g/L): 74.86
Maltose M (g/L): 75.56
Volume V (L): 9.14

t = 70.00 h

Biomass X (g/L): 72.72
Glucose S (g/L): 41.86
Lactic Acid P (g/L): 66.43
Maltose M (g/L): 78.14
Volume V (L): 9.31

t = 71.00 h

Biomass X (g/L): 80.71
Glucose S (g/L): 50.00
Lactic Acid P (g/L): 85.42
Maltose M (g/L): 83.69
Volume V (L): 9.53

t = 72.00 h

Biomass X (g/L): 80.39
Glucose S (g/L): 50.69
Lactic Acid P (g/L): 81.00
Maltose M (g/L): 78.23
Volume V (L): 9.98

t = 73.00 h

Biomass X (g/L): 71.80
Glucose S (g/L): 54.71
Lactic Acid P (g/L): 92.72
Maltose M (g/L): 73.43
Volume V (L): 9.80

t = 74.00 h

Biomass X (g/L): 86.47
Glucose S (g/L): 47.41
Lactic Acid P (g/L): 67.35
Maltose M (g/L): 69.39
Volume V (L): 9.28

t = 75.00 h

Biomass X (g/L): 85.02
Glucose S (g/L): 52.08
Lactic Acid P (g/L): 72.04
Maltose M (g/L): 79.41

Volume V (L): 9.64

Batch 10

Initial Conditions:

Batch: 10.00

X0 (g/L): 9.63

S0 (g/L): 102.47

M0 (g/L): 63.06

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.62

F_vol (L/h): 0.07

S_feed (g/L): 185.71

M_feed (g/L): 114.29

V_final (L): 9.94

Batch 10 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.80

Glucose S (g/L): 105.77

Lactic Acid P (g/L): 2.96

Maltose M (g/L): 65.08

Volume V (L): 4.75

t = 1.00 h

Biomass X (g/L): 12.20

Glucose S (g/L): 101.15

Lactic Acid P (g/L): 4.27

Maltose M (g/L): 61.40

Volume V (L): 5.30

t = 2.00 h

Biomass X (g/L): 21.04

Glucose S (g/L): 98.73

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 63.03

Volume V (L): 5.33

t = 3.00 h

Biomass X (g/L): 25.63

Glucose S (g/L): 95.44

Lactic Acid P (g/L): 11.58

Maltose M (g/L): 62.87

Volume V (L): 5.62

t = 4.00 h

Biomass X (g/L): 9.74

Glucose S (g/L): 98.54

Lactic Acid P (g/L): 5.07

Maltose M (g/L): 62.07

Volume V (L): 4.79

t = 5.00 h

Biomass X (g/L): 42.41

Glucose S (g/L): 87.73

Lactic Acid P (g/L): 10.91
Maltose M (g/L): 58.29
Volume V (L): 4.41

t = 6.00 h

Biomass X (g/L): 40.12
Glucose S (g/L): 86.84
Lactic Acid P (g/L): 13.38
Maltose M (g/L): 60.30
Volume V (L): 5.73

t = 7.00 h

Biomass X (g/L): 56.33
Glucose S (g/L): 83.30
Lactic Acid P (g/L): 15.10
Maltose M (g/L): 60.35
Volume V (L): 5.13

t = 8.00 h

Biomass X (g/L): 52.40
Glucose S (g/L): 77.69
Lactic Acid P (g/L): 48.44
Maltose M (g/L): 58.94
Volume V (L): 4.73

t = 9.00 h

Biomass X (g/L): 63.41
Glucose S (g/L): 71.09
Lactic Acid P (g/L): 44.73
Maltose M (g/L): 53.07
Volume V (L): 5.65

t = 10.00 h

Biomass X (g/L): 61.62
Glucose S (g/L): 75.23
Lactic Acid P (g/L): 53.52
Maltose M (g/L): 58.38
Volume V (L): 5.97

t = 11.00 h

Biomass X (g/L): 73.94

Glucose S (g/L): 71.28
Lactic Acid P (g/L): 27.55
Maltose M (g/L): 63.20
Volume V (L): 5.68

t = 12.00 h

Biomass X (g/L): 69.51
Glucose S (g/L): 71.51
Lactic Acid P (g/L): 42.29
Maltose M (g/L): 56.11
Volume V (L): 5.61

t = 13.00 h

Biomass X (g/L): 66.65
Glucose S (g/L): 65.59
Lactic Acid P (g/L): 32.22
Maltose M (g/L): 54.71
Volume V (L): 5.79

t = 14.00 h

Biomass X (g/L): 92.13
Glucose S (g/L): 71.26
Lactic Acid P (g/L): 40.04
Maltose M (g/L): 57.10
Volume V (L): 5.78

t = 15.00 h

Biomass X (g/L): 81.29
Glucose S (g/L): 61.58
Lactic Acid P (g/L): 40.07
Maltose M (g/L): 56.51
Volume V (L): 5.46

t = 16.00 h

Biomass X (g/L): 80.42
Glucose S (g/L): 71.12
Lactic Acid P (g/L): 61.89
Maltose M (g/L): 58.34
Volume V (L): 5.77

t = 17.00 h

Biomass X (g/L): 64.58
Glucose S (g/L): 56.35
Lactic Acid P (g/L): 48.51
Maltose M (g/L): 51.60
Volume V (L): 6.33

t = 18.00 h

Biomass X (g/L): 66.15
Glucose S (g/L): 64.62
Lactic Acid P (g/L): 63.61
Maltose M (g/L): 56.30
Volume V (L): 6.31

t = 19.00 h

Biomass X (g/L): 82.80
Glucose S (g/L): 64.37
Lactic Acid P (g/L): 60.75
Maltose M (g/L): 55.52
Volume V (L): 6.32

t = 20.00 h

Biomass X (g/L): 82.70
Glucose S (g/L): 58.94
Lactic Acid P (g/L): 77.77
Maltose M (g/L): 53.22
Volume V (L): 5.92

t = 21.00 h

Biomass X (g/L): 76.57
Glucose S (g/L): 63.72
Lactic Acid P (g/L): 44.04
Maltose M (g/L): 54.05
Volume V (L): 6.31

t = 22.00 h

Biomass X (g/L): 84.84
Glucose S (g/L): 67.12
Lactic Acid P (g/L): 70.44
Maltose M (g/L): 55.43
Volume V (L): 5.59

t = 23.00 h

Biomass X (g/L): 99.63
Glucose S (g/L): 52.98
Lactic Acid P (g/L): 70.43
Maltose M (g/L): 54.64
Volume V (L): 6.59

t = 24.00 h

Biomass X (g/L): 91.85
Glucose S (g/L): 64.12
Lactic Acid P (g/L): 51.81
Maltose M (g/L): 49.05
Volume V (L): 7.15

t = 25.00 h

Biomass X (g/L): 74.94
Glucose S (g/L): 52.15
Lactic Acid P (g/L): 70.98
Maltose M (g/L): 53.73
Volume V (L): 6.06

t = 26.00 h

Biomass X (g/L): 85.42
Glucose S (g/L): 58.58
Lactic Acid P (g/L): 65.61
Maltose M (g/L): 51.07
Volume V (L): 6.48

t = 27.00 h

Biomass X (g/L): 79.31
Glucose S (g/L): 69.39
Lactic Acid P (g/L): 61.17
Maltose M (g/L): 51.05
Volume V (L): 6.91

t = 28.00 h

Biomass X (g/L): 83.78
Glucose S (g/L): 60.76
Lactic Acid P (g/L): 56.38
Maltose M (g/L): 50.68
Volume V (L): 7.34

t = 29.00 h

Biomass X (g/L): 83.93
Glucose S (g/L): 58.74
Lactic Acid P (g/L): 58.02
Maltose M (g/L): 49.74
Volume V (L): 6.73

t = 30.00 h

Biomass X (g/L): 83.66
Glucose S (g/L): 63.10
Lactic Acid P (g/L): 84.22
Maltose M (g/L): 52.84
Volume V (L): 6.70

t = 31.00 h

Biomass X (g/L): 76.50
Glucose S (g/L): 72.73
Lactic Acid P (g/L): 65.89
Maltose M (g/L): 49.20
Volume V (L): 6.26

t = 32.00 h

Biomass X (g/L): 87.56
Glucose S (g/L): 57.39
Lactic Acid P (g/L): 73.67
Maltose M (g/L): 46.18
Volume V (L): 7.15

t = 33.00 h

Biomass X (g/L): 83.62
Glucose S (g/L): 61.04
Lactic Acid P (g/L): 59.67
Maltose M (g/L): 47.70
Volume V (L): 7.98

t = 34.00 h

Biomass X (g/L): 61.07
Glucose S (g/L): 71.88
Lactic Acid P (g/L): 60.27
Maltose M (g/L): 48.83

Volume V (L): 7.35

t = 35.00 h

Biomass X (g/L): 108.99

Glucose S (g/L): 73.87

Lactic Acid P (g/L): 77.93

Maltose M (g/L): 51.20

Volume V (L): 7.30

t = 36.00 h

Biomass X (g/L): 99.85

Glucose S (g/L): 66.46

Lactic Acid P (g/L): 55.92

Maltose M (g/L): 49.25

Volume V (L): 7.50

t = 37.00 h

Biomass X (g/L): 85.87

Glucose S (g/L): 62.41

Lactic Acid P (g/L): 64.98

Maltose M (g/L): 45.18

Volume V (L): 7.43

t = 38.00 h

Biomass X (g/L): 89.71

Glucose S (g/L): 76.49

Lactic Acid P (g/L): 69.53

Maltose M (g/L): 53.45

Volume V (L): 7.18

t = 39.00 h

Biomass X (g/L): 78.32

Glucose S (g/L): 75.03

Lactic Acid P (g/L): 70.62

Maltose M (g/L): 47.41

Volume V (L): 7.89

t = 40.00 h

Biomass X (g/L): 105.82

Glucose S (g/L): 71.90

Lactic Acid P (g/L): 68.31

Maltose M (g/L): 49.69

Volume V (L): 7.72

t = 41.00 h

Biomass X (g/L): 107.74

Glucose S (g/L): 66.45

Lactic Acid P (g/L): 73.97

Maltose M (g/L): 45.70

Volume V (L): 7.71

t = 42.00 h

Biomass X (g/L): 91.01

Glucose S (g/L): 70.13

Lactic Acid P (g/L): 77.04

Maltose M (g/L): 47.16

Volume V (L): 7.38

t = 43.00 h

Biomass X (g/L): 99.27

Glucose S (g/L): 70.28

Lactic Acid P (g/L): 77.75

Maltose M (g/L): 47.64

Volume V (L): 8.35

t = 44.00 h

Biomass X (g/L): 91.07

Glucose S (g/L): 72.87

Lactic Acid P (g/L): 62.16

Maltose M (g/L): 48.31

Volume V (L): 8.16

t = 45.00 h

Biomass X (g/L): 76.49

Glucose S (g/L): 78.01

Lactic Acid P (g/L): 68.93

Maltose M (g/L): 47.60

Volume V (L): 7.75

t = 46.00 h

Biomass X (g/L): 85.26

Glucose S (g/L): 76.29

Lactic Acid P (g/L): 82.87
Maltose M (g/L): 46.82
Volume V (L): 8.43

t = 47.00 h

Biomass X (g/L): 90.37
Glucose S (g/L): 68.96
Lactic Acid P (g/L): 80.37
Maltose M (g/L): 43.47
Volume V (L): 7.90

t = 48.00 h

Biomass X (g/L): 84.63
Glucose S (g/L): 79.51
Lactic Acid P (g/L): 59.69
Maltose M (g/L): 42.52
Volume V (L): 7.85

t = 49.00 h

Biomass X (g/L): 86.77
Glucose S (g/L): 75.80
Lactic Acid P (g/L): 75.86
Maltose M (g/L): 44.57
Volume V (L): 7.93

t = 50.00 h

Biomass X (g/L): 87.85
Glucose S (g/L): 77.92
Lactic Acid P (g/L): 54.59
Maltose M (g/L): 45.83
Volume V (L): 7.59

t = 51.00 h

Biomass X (g/L): 87.81
Glucose S (g/L): 81.68
Lactic Acid P (g/L): 73.52
Maltose M (g/L): 42.57
Volume V (L): 8.55

t = 52.00 h

Biomass X (g/L): 88.93

Glucose S (g/L): 71.83
Lactic Acid P (g/L): 90.84
Maltose M (g/L): 45.62
Volume V (L): 7.70

t = 53.00 h

Biomass X (g/L): 78.10
Glucose S (g/L): 76.78
Lactic Acid P (g/L): 64.77
Maltose M (g/L): 46.34
Volume V (L): 8.01

t = 54.00 h

Biomass X (g/L): 81.23
Glucose S (g/L): 75.37
Lactic Acid P (g/L): 66.03
Maltose M (g/L): 39.97
Volume V (L): 7.74

t = 55.00 h

Biomass X (g/L): 75.62
Glucose S (g/L): 71.75
Lactic Acid P (g/L): 80.99
Maltose M (g/L): 39.90
Volume V (L): 8.59

t = 56.00 h

Biomass X (g/L): 97.14
Glucose S (g/L): 80.10
Lactic Acid P (g/L): 59.89
Maltose M (g/L): 41.44
Volume V (L): 8.63

t = 57.00 h

Biomass X (g/L): 67.38
Glucose S (g/L): 89.46
Lactic Acid P (g/L): 68.48
Maltose M (g/L): 42.33
Volume V (L): 8.63

t = 58.00 h

Biomass X (g/L): 98.46
Glucose S (g/L): 72.40
Lactic Acid P (g/L): 80.01
Maltose M (g/L): 39.93
Volume V (L): 8.41

t = 59.00 h

Biomass X (g/L): 100.50
Glucose S (g/L): 80.30
Lactic Acid P (g/L): 68.83
Maltose M (g/L): 44.51
Volume V (L): 8.42

t = 60.00 h

Biomass X (g/L): 84.11
Glucose S (g/L): 83.60
Lactic Acid P (g/L): 72.68
Maltose M (g/L): 41.86
Volume V (L): 9.14

t = 61.00 h

Biomass X (g/L): 61.84
Glucose S (g/L): 82.07
Lactic Acid P (g/L): 81.90
Maltose M (g/L): 40.62
Volume V (L): 8.80

t = 62.00 h

Biomass X (g/L): 83.14
Glucose S (g/L): 81.68
Lactic Acid P (g/L): 79.08
Maltose M (g/L): 36.57
Volume V (L): 9.21

t = 63.00 h

Biomass X (g/L): 85.78
Glucose S (g/L): 77.43
Lactic Acid P (g/L): 77.81
Maltose M (g/L): 36.64
Volume V (L): 8.80

t = 64.00 h

Biomass X (g/L): 76.50
Glucose S (g/L): 84.19
Lactic Acid P (g/L): 72.37
Maltose M (g/L): 40.51
Volume V (L): 9.29

t = 65.00 h

Biomass X (g/L): 81.04
Glucose S (g/L): 80.73
Lactic Acid P (g/L): 79.07
Maltose M (g/L): 34.05
Volume V (L): 9.80

t = 66.00 h

Biomass X (g/L): 100.45
Glucose S (g/L): 75.47
Lactic Acid P (g/L): 74.65
Maltose M (g/L): 41.92
Volume V (L): 9.58

t = 67.00 h

Biomass X (g/L): 91.93
Glucose S (g/L): 89.73
Lactic Acid P (g/L): 80.16
Maltose M (g/L): 38.67
Volume V (L): 9.25

t = 68.00 h

Biomass X (g/L): 76.47
Glucose S (g/L): 89.31
Lactic Acid P (g/L): 71.49
Maltose M (g/L): 47.20
Volume V (L): 9.58

t = 69.00 h

Biomass X (g/L): 69.80
Glucose S (g/L): 86.17
Lactic Acid P (g/L): 74.52
Maltose M (g/L): 44.56
Volume V (L): 9.45

t = 70.00 h

Biomass X (g/L): 81.77
Glucose S (g/L): 87.41
Lactic Acid P (g/L): 67.20
Maltose M (g/L): 37.92
Volume V (L): 9.61

t = 71.00 h

Biomass X (g/L): 82.64
Glucose S (g/L): 91.36
Lactic Acid P (g/L): 76.15
Maltose M (g/L): 37.89
Volume V (L): 9.55

t = 72.00 h

Biomass X (g/L): 89.22
Glucose S (g/L): 88.43
Lactic Acid P (g/L): 69.11
Maltose M (g/L): 40.51
Volume V (L): 9.68

t = 73.00 h

Biomass X (g/L): 79.11
Glucose S (g/L): 91.70
Lactic Acid P (g/L): 66.43
Maltose M (g/L): 42.11
Volume V (L): 9.89

t = 74.00 h

Biomass X (g/L): 88.92
Glucose S (g/L): 87.34
Lactic Acid P (g/L): 79.23
Maltose M (g/L): 33.28
Volume V (L): 9.48

t = 75.00 h

Biomass X (g/L): 102.36
Glucose S (g/L): 94.37
Lactic Acid P (g/L): 78.92
Maltose M (g/L): 32.69

Volume V (L): 9.54

Batch 11

Initial Conditions:

Batch: 11.00

X0 (g/L): 8.07

S0 (g/L): 33.22

M0 (g/L): 92.95

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.80

F_vol (L/h): 0.09

S_feed (g/L): 78.99

M_feed (g/L): 221.01

V_final (L): 11.40

Batch 11 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 36.68
Lactic Acid P (g/L): 19.21
Maltose M (g/L): 96.12
Volume V (L): 4.97

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 34.13
Lactic Acid P (g/L): 6.86
Maltose M (g/L): 94.89
Volume V (L): 5.21

t = 2.00 h

Biomass X (g/L): 2.48
Glucose S (g/L): 27.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 95.32
Volume V (L): 5.02

t = 3.00 h

Biomass X (g/L): 18.21
Glucose S (g/L): 30.97
Lactic Acid P (g/L): 1.21
Maltose M (g/L): 91.06
Volume V (L): 5.95

t = 4.00 h

Biomass X (g/L): 12.58
Glucose S (g/L): 31.27
Lactic Acid P (g/L): 8.63
Maltose M (g/L): 91.81
Volume V (L): 5.24

t = 5.00 h

Biomass X (g/L): 29.30
Glucose S (g/L): 28.94

Lactic Acid P (g/L): 10.42

Maltose M (g/L): 88.64

Volume V (L): 5.55

t = 6.00 h

Biomass X (g/L): 29.28

Glucose S (g/L): 19.50

Lactic Acid P (g/L): 25.57

Maltose M (g/L): 92.81

Volume V (L): 6.10

t = 7.00 h

Biomass X (g/L): 15.62

Glucose S (g/L): 20.78

Lactic Acid P (g/L): 20.64

Maltose M (g/L): 96.42

Volume V (L): 5.20

t = 8.00 h

Biomass X (g/L): 43.35

Glucose S (g/L): 21.64

Lactic Acid P (g/L): 15.76

Maltose M (g/L): 95.63

Volume V (L): 6.48

t = 9.00 h

Biomass X (g/L): 48.74

Glucose S (g/L): 21.82

Lactic Acid P (g/L): 41.89

Maltose M (g/L): 94.21

Volume V (L): 4.98

t = 10.00 h

Biomass X (g/L): 36.29

Glucose S (g/L): 10.07

Lactic Acid P (g/L): 28.59

Maltose M (g/L): 94.62

Volume V (L): 6.69

t = 11.00 h

Biomass X (g/L): 52.40

Glucose S (g/L): 15.91
Lactic Acid P (g/L): 37.61
Maltose M (g/L): 93.29
Volume V (L): 5.75

t = 12.00 h

Biomass X (g/L): 44.08
Glucose S (g/L): 15.70
Lactic Acid P (g/L): 27.99
Maltose M (g/L): 95.03
Volume V (L): 5.90

t = 13.00 h

Biomass X (g/L): 55.46
Glucose S (g/L): 17.19
Lactic Acid P (g/L): 38.82
Maltose M (g/L): 96.08
Volume V (L): 5.14

t = 14.00 h

Biomass X (g/L): 45.79
Glucose S (g/L): 15.78
Lactic Acid P (g/L): 40.46
Maltose M (g/L): 91.27
Volume V (L): 5.96

t = 15.00 h

Biomass X (g/L): 43.34
Glucose S (g/L): 14.73
Lactic Acid P (g/L): 23.03
Maltose M (g/L): 94.97
Volume V (L): 5.75

t = 16.00 h

Biomass X (g/L): 58.06
Glucose S (g/L): 9.24
Lactic Acid P (g/L): 59.14
Maltose M (g/L): 96.10
Volume V (L): 6.05

t = 17.00 h

Biomass X (g/L): 62.25
Glucose S (g/L): 4.90
Lactic Acid P (g/L): 54.41
Maltose M (g/L): 93.29
Volume V (L): 6.38

t = 18.00 h

Biomass X (g/L): 61.49
Glucose S (g/L): 10.24
Lactic Acid P (g/L): 41.08
Maltose M (g/L): 95.35
Volume V (L): 7.03

t = 19.00 h

Biomass X (g/L): 55.99
Glucose S (g/L): 15.25
Lactic Acid P (g/L): 36.29
Maltose M (g/L): 99.37
Volume V (L): 6.73

t = 20.00 h

Biomass X (g/L): 64.43
Glucose S (g/L): 16.72
Lactic Acid P (g/L): 50.37
Maltose M (g/L): 93.38
Volume V (L): 6.86

t = 21.00 h

Biomass X (g/L): 71.72
Glucose S (g/L): 14.36
Lactic Acid P (g/L): 49.31
Maltose M (g/L): 94.17
Volume V (L): 6.23

t = 22.00 h

Biomass X (g/L): 70.10
Glucose S (g/L): 13.53
Lactic Acid P (g/L): 52.85
Maltose M (g/L): 91.71
Volume V (L): 6.43

t = 23.00 h

Biomass X (g/L): 73.23
Glucose S (g/L): 13.60
Lactic Acid P (g/L): 52.51
Maltose M (g/L): 93.30
Volume V (L): 7.88

t = 24.00 h

Biomass X (g/L): 66.59
Glucose S (g/L): 10.51
Lactic Acid P (g/L): 56.32
Maltose M (g/L): 93.61
Volume V (L): 6.67

t = 25.00 h

Biomass X (g/L): 84.05
Glucose S (g/L): 12.43
Lactic Acid P (g/L): 65.59
Maltose M (g/L): 92.83
Volume V (L): 7.98

t = 26.00 h

Biomass X (g/L): 63.85
Glucose S (g/L): 19.38
Lactic Acid P (g/L): 58.80
Maltose M (g/L): 90.16
Volume V (L): 7.09

t = 27.00 h

Biomass X (g/L): 79.57
Glucose S (g/L): 10.32
Lactic Acid P (g/L): 45.47
Maltose M (g/L): 92.32
Volume V (L): 7.31

t = 28.00 h

Biomass X (g/L): 86.54
Glucose S (g/L): 20.10
Lactic Acid P (g/L): 67.54
Maltose M (g/L): 94.53
Volume V (L): 7.94

t = 29.00 h

Biomass X (g/L): 80.93
Glucose S (g/L): 13.57
Lactic Acid P (g/L): 70.80
Maltose M (g/L): 92.23
Volume V (L): 8.13

t = 30.00 h

Biomass X (g/L): 75.44
Glucose S (g/L): 20.10
Lactic Acid P (g/L): 84.87
Maltose M (g/L): 92.95
Volume V (L): 7.50

t = 31.00 h

Biomass X (g/L): 73.10
Glucose S (g/L): 10.72
Lactic Acid P (g/L): 48.01
Maltose M (g/L): 90.27
Volume V (L): 7.45

t = 32.00 h

Biomass X (g/L): 87.66
Glucose S (g/L): 18.56
Lactic Acid P (g/L): 60.74
Maltose M (g/L): 91.91
Volume V (L): 7.70

t = 33.00 h

Biomass X (g/L): 66.11
Glucose S (g/L): 14.36
Lactic Acid P (g/L): 40.97
Maltose M (g/L): 90.66
Volume V (L): 7.99

t = 34.00 h

Biomass X (g/L): 70.62
Glucose S (g/L): 12.52
Lactic Acid P (g/L): 64.80
Maltose M (g/L): 87.90

Volume V (L): 8.36

t = 35.00 h

Biomass X (g/L): 76.17

Glucose S (g/L): 19.11

Lactic Acid P (g/L): 55.73

Maltose M (g/L): 90.09

Volume V (L): 7.99

t = 36.00 h

Biomass X (g/L): 91.22

Glucose S (g/L): 22.65

Lactic Acid P (g/L): 78.07

Maltose M (g/L): 87.15

Volume V (L): 8.39

t = 37.00 h

Biomass X (g/L): 83.88

Glucose S (g/L): 16.58

Lactic Acid P (g/L): 54.91

Maltose M (g/L): 87.52

Volume V (L): 9.00

t = 38.00 h

Biomass X (g/L): 72.11

Glucose S (g/L): 14.89

Lactic Acid P (g/L): 47.42

Maltose M (g/L): 88.81

Volume V (L): 8.44

t = 39.00 h

Biomass X (g/L): 78.29

Glucose S (g/L): 20.71

Lactic Acid P (g/L): 58.66

Maltose M (g/L): 88.68

Volume V (L): 8.49

t = 40.00 h

Biomass X (g/L): 75.83

Glucose S (g/L): 21.44

Lactic Acid P (g/L): 45.93

Maltose M (g/L): 85.38

Volume V (L): 7.64

t = 41.00 h

Biomass X (g/L): 75.20

Glucose S (g/L): 24.47

Lactic Acid P (g/L): 70.40

Maltose M (g/L): 89.48

Volume V (L): 7.15

t = 42.00 h

Biomass X (g/L): 90.56

Glucose S (g/L): 26.66

Lactic Acid P (g/L): 46.10

Maltose M (g/L): 88.16

Volume V (L): 8.78

t = 43.00 h

Biomass X (g/L): 69.66

Glucose S (g/L): 23.73

Lactic Acid P (g/L): 64.96

Maltose M (g/L): 85.27

Volume V (L): 8.62

t = 44.00 h

Biomass X (g/L): 96.56

Glucose S (g/L): 31.44

Lactic Acid P (g/L): 46.86

Maltose M (g/L): 83.25

Volume V (L): 8.49

t = 45.00 h

Biomass X (g/L): 89.70

Glucose S (g/L): 26.26

Lactic Acid P (g/L): 62.74

Maltose M (g/L): 85.65

Volume V (L): 8.97

t = 46.00 h

Biomass X (g/L): 92.18

Glucose S (g/L): 24.94

Lactic Acid P (g/L): 80.41
Maltose M (g/L): 84.69
Volume V (L): 8.90

t = 47.00 h

Biomass X (g/L): 78.35
Glucose S (g/L): 21.10
Lactic Acid P (g/L): 69.51
Maltose M (g/L): 85.43
Volume V (L): 9.59

t = 48.00 h

Biomass X (g/L): 81.24
Glucose S (g/L): 26.77
Lactic Acid P (g/L): 59.11
Maltose M (g/L): 84.11
Volume V (L): 9.14

t = 49.00 h

Biomass X (g/L): 69.81
Glucose S (g/L): 27.27
Lactic Acid P (g/L): 66.79
Maltose M (g/L): 80.48
Volume V (L): 8.80

t = 50.00 h

Biomass X (g/L): 78.01
Glucose S (g/L): 32.75
Lactic Acid P (g/L): 67.57
Maltose M (g/L): 83.32
Volume V (L): 9.77

t = 51.00 h

Biomass X (g/L): 87.19
Glucose S (g/L): 31.71
Lactic Acid P (g/L): 71.90
Maltose M (g/L): 83.11
Volume V (L): 9.24

t = 52.00 h

Biomass X (g/L): 94.41

Glucose S (g/L): 26.51
Lactic Acid P (g/L): 56.05
Maltose M (g/L): 84.60
Volume V (L): 9.71

t = 53.00 h

Biomass X (g/L): 72.87
Glucose S (g/L): 32.20
Lactic Acid P (g/L): 73.59
Maltose M (g/L): 81.58
Volume V (L): 9.28

t = 54.00 h

Biomass X (g/L): 79.53
Glucose S (g/L): 34.79
Lactic Acid P (g/L): 44.89
Maltose M (g/L): 85.93
Volume V (L): 9.28

t = 55.00 h

Biomass X (g/L): 81.44
Glucose S (g/L): 38.67
Lactic Acid P (g/L): 63.43
Maltose M (g/L): 78.43
Volume V (L): 9.74

t = 56.00 h

Biomass X (g/L): 70.26
Glucose S (g/L): 37.11
Lactic Acid P (g/L): 64.96
Maltose M (g/L): 84.45
Volume V (L): 9.79

t = 57.00 h

Biomass X (g/L): 88.06
Glucose S (g/L): 37.42
Lactic Acid P (g/L): 75.10
Maltose M (g/L): 81.02
Volume V (L): 10.11

t = 58.00 h

Biomass X (g/L): 76.22
Glucose S (g/L): 35.64
Lactic Acid P (g/L): 69.99
Maltose M (g/L): 82.80
Volume V (L): 9.81

t = 59.00 h

Biomass X (g/L): 80.92
Glucose S (g/L): 37.60
Lactic Acid P (g/L): 57.68
Maltose M (g/L): 80.88
Volume V (L): 9.10

t = 60.00 h

Biomass X (g/L): 78.56
Glucose S (g/L): 41.49
Lactic Acid P (g/L): 61.63
Maltose M (g/L): 82.61
Volume V (L): 9.90

t = 61.00 h

Biomass X (g/L): 75.47
Glucose S (g/L): 32.86
Lactic Acid P (g/L): 72.28
Maltose M (g/L): 79.20
Volume V (L): 10.61

t = 62.00 h

Biomass X (g/L): 82.42
Glucose S (g/L): 37.20
Lactic Acid P (g/L): 70.07
Maltose M (g/L): 77.42
Volume V (L): 10.14

t = 63.00 h

Biomass X (g/L): 80.47
Glucose S (g/L): 36.41
Lactic Acid P (g/L): 65.90
Maltose M (g/L): 82.04
Volume V (L): 10.25

t = 64.00 h

Biomass X (g/L): 78.99
Glucose S (g/L): 41.35
Lactic Acid P (g/L): 66.99
Maltose M (g/L): 78.62
Volume V (L): 10.97

t = 65.00 h

Biomass X (g/L): 75.70
Glucose S (g/L): 45.20
Lactic Acid P (g/L): 77.55
Maltose M (g/L): 78.92
Volume V (L): 10.18

t = 66.00 h

Biomass X (g/L): 85.00
Glucose S (g/L): 39.30
Lactic Acid P (g/L): 67.09
Maltose M (g/L): 80.56
Volume V (L): 9.84

t = 67.00 h

Biomass X (g/L): 96.87
Glucose S (g/L): 33.51
Lactic Acid P (g/L): 71.07
Maltose M (g/L): 75.69
Volume V (L): 10.64

t = 68.00 h

Biomass X (g/L): 83.18
Glucose S (g/L): 41.45
Lactic Acid P (g/L): 91.13
Maltose M (g/L): 78.55
Volume V (L): 10.56

t = 69.00 h

Biomass X (g/L): 90.75
Glucose S (g/L): 47.00
Lactic Acid P (g/L): 59.71
Maltose M (g/L): 75.83
Volume V (L): 10.39

t = 70.00 h

Biomass X (g/L): 69.23
Glucose S (g/L): 46.36
Lactic Acid P (g/L): 56.13
Maltose M (g/L): 76.37
Volume V (L): 10.88

t = 71.00 h

Biomass X (g/L): 83.00
Glucose S (g/L): 46.69
Lactic Acid P (g/L): 79.50
Maltose M (g/L): 76.41
Volume V (L): 11.31

t = 72.00 h

Biomass X (g/L): 71.62
Glucose S (g/L): 44.83
Lactic Acid P (g/L): 66.81
Maltose M (g/L): 72.93
Volume V (L): 11.14

t = 73.00 h

Biomass X (g/L): 68.09
Glucose S (g/L): 41.91
Lactic Acid P (g/L): 70.16
Maltose M (g/L): 74.00
Volume V (L): 11.39

t = 74.00 h

Biomass X (g/L): 74.57
Glucose S (g/L): 49.15
Lactic Acid P (g/L): 66.26
Maltose M (g/L): 77.86
Volume V (L): 12.24

t = 75.00 h

Biomass X (g/L): 91.76
Glucose S (g/L): 44.95
Lactic Acid P (g/L): 71.01
Maltose M (g/L): 76.58

Volume V (L): 11.23

Batch 12

Initial Conditions:

Batch: 12.00

X0 (g/L): 2.55

S0 (g/L): 21.71

M0 (g/L): 93.48

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 4.31

F_vol (L/h): 0.10

S_feed (g/L): 56.55

M_feed (g/L): 243.45

V_final (L): 12.30

Batch 12 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 5.11
Glucose S (g/L): 19.19
Lactic Acid P (g/L): 6.46
Maltose M (g/L): 94.41
Volume V (L): 4.98

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 27.74
Lactic Acid P (g/L): 2.25
Maltose M (g/L): 96.05
Volume V (L): 4.17

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 24.65
Lactic Acid P (g/L): 5.28
Maltose M (g/L): 94.35
Volume V (L): 5.44

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 23.47
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 96.27
Volume V (L): 4.95

t = 4.00 h

Biomass X (g/L): 2.28
Glucose S (g/L): 30.07
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 95.35
Volume V (L): 5.90

t = 5.00 h

Biomass X (g/L): 11.73
Glucose S (g/L): 26.76

Lactic Acid P (g/L): 8.72
Maltose M (g/L): 93.58
Volume V (L): 5.49

t = 6.00 h

Biomass X (g/L): 32.14
Glucose S (g/L): 30.11
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 98.88
Volume V (L): 5.82

t = 7.00 h

Biomass X (g/L): 6.28
Glucose S (g/L): 26.74
Lactic Acid P (g/L): 5.00
Maltose M (g/L): 100.16
Volume V (L): 5.12

t = 8.00 h

Biomass X (g/L): 11.80
Glucose S (g/L): 26.86
Lactic Acid P (g/L): 17.81
Maltose M (g/L): 100.08
Volume V (L): 5.43

t = 9.00 h

Biomass X (g/L): 18.12
Glucose S (g/L): 27.61
Lactic Acid P (g/L): 13.65
Maltose M (g/L): 96.73
Volume V (L): 5.61

t = 10.00 h

Biomass X (g/L): 25.64
Glucose S (g/L): 25.80
Lactic Acid P (g/L): 31.12
Maltose M (g/L): 99.89
Volume V (L): 6.47

t = 11.00 h

Biomass X (g/L): 27.88

Glucose S (g/L): 17.49
Lactic Acid P (g/L): 25.79
Maltose M (g/L): 97.02
Volume V (L): 7.05

t = 12.00 h

Biomass X (g/L): 40.81
Glucose S (g/L): 18.61
Lactic Acid P (g/L): 14.04
Maltose M (g/L): 103.51
Volume V (L): 5.70

t = 13.00 h

Biomass X (g/L): 10.09
Glucose S (g/L): 23.25
Lactic Acid P (g/L): 22.50
Maltose M (g/L): 99.54
Volume V (L): 6.89

t = 14.00 h

Biomass X (g/L): 13.88
Glucose S (g/L): 18.53
Lactic Acid P (g/L): 24.44
Maltose M (g/L): 103.79
Volume V (L): 5.10

t = 15.00 h

Biomass X (g/L): 34.59
Glucose S (g/L): 15.93
Lactic Acid P (g/L): 13.80
Maltose M (g/L): 103.43
Volume V (L): 5.76

t = 16.00 h

Biomass X (g/L): 52.37
Glucose S (g/L): 12.13
Lactic Acid P (g/L): 21.66
Maltose M (g/L): 100.39
Volume V (L): 6.00

t = 17.00 h

Biomass X (g/L): 30.69
Glucose S (g/L): 11.85
Lactic Acid P (g/L): 27.15
Maltose M (g/L): 100.09
Volume V (L): 5.87

t = 18.00 h

Biomass X (g/L): 20.05
Glucose S (g/L): 8.05
Lactic Acid P (g/L): 16.61
Maltose M (g/L): 102.72
Volume V (L): 5.83

t = 19.00 h

Biomass X (g/L): 55.51
Glucose S (g/L): 13.06
Lactic Acid P (g/L): 36.40
Maltose M (g/L): 100.70
Volume V (L): 7.09

t = 20.00 h

Biomass X (g/L): 54.74
Glucose S (g/L): 10.93
Lactic Acid P (g/L): 35.69
Maltose M (g/L): 102.34
Volume V (L): 6.82

t = 21.00 h

Biomass X (g/L): 44.51
Glucose S (g/L): 9.34
Lactic Acid P (g/L): 25.19
Maltose M (g/L): 101.31
Volume V (L): 7.27

t = 22.00 h

Biomass X (g/L): 47.37
Glucose S (g/L): 11.71
Lactic Acid P (g/L): 44.56
Maltose M (g/L): 99.03
Volume V (L): 7.55

t = 23.00 h

Biomass X (g/L): 49.26
Glucose S (g/L): 12.78
Lactic Acid P (g/L): 48.70
Maltose M (g/L): 98.19
Volume V (L): 7.39

t = 24.00 h

Biomass X (g/L): 60.83
Glucose S (g/L): 17.87
Lactic Acid P (g/L): 45.91
Maltose M (g/L): 102.33
Volume V (L): 7.72

t = 25.00 h

Biomass X (g/L): 55.51
Glucose S (g/L): 6.96
Lactic Acid P (g/L): 42.49
Maltose M (g/L): 100.86
Volume V (L): 7.39

t = 26.00 h

Biomass X (g/L): 67.22
Glucose S (g/L): 15.06
Lactic Acid P (g/L): 51.56
Maltose M (g/L): 100.74
Volume V (L): 6.84

t = 27.00 h

Biomass X (g/L): 37.60
Glucose S (g/L): 14.62
Lactic Acid P (g/L): 53.49
Maltose M (g/L): 102.74
Volume V (L): 8.25

t = 28.00 h

Biomass X (g/L): 57.21
Glucose S (g/L): 9.50
Lactic Acid P (g/L): 51.40
Maltose M (g/L): 103.40
Volume V (L): 7.38

t = 29.00 h

Biomass X (g/L): 55.83
Glucose S (g/L): 12.70
Lactic Acid P (g/L): 42.76
Maltose M (g/L): 100.72
Volume V (L): 7.28

t = 30.00 h

Biomass X (g/L): 64.88
Glucose S (g/L): 10.98
Lactic Acid P (g/L): 53.18
Maltose M (g/L): 100.40
Volume V (L): 7.82

t = 31.00 h

Biomass X (g/L): 63.24
Glucose S (g/L): 13.25
Lactic Acid P (g/L): 62.00
Maltose M (g/L): 101.63
Volume V (L): 7.49

t = 32.00 h

Biomass X (g/L): 76.46
Glucose S (g/L): 17.61
Lactic Acid P (g/L): 47.57
Maltose M (g/L): 98.19
Volume V (L): 7.87

t = 33.00 h

Biomass X (g/L): 52.79
Glucose S (g/L): 15.28
Lactic Acid P (g/L): 71.50
Maltose M (g/L): 100.67
Volume V (L): 8.41

t = 34.00 h

Biomass X (g/L): 71.15
Glucose S (g/L): 15.41
Lactic Acid P (g/L): 49.88
Maltose M (g/L): 100.98

Volume V (L): 8.51

t = 35.00 h

Biomass X (g/L): 76.61

Glucose S (g/L): 13.04

Lactic Acid P (g/L): 55.45

Maltose M (g/L): 100.94

Volume V (L): 7.88

t = 36.00 h

Biomass X (g/L): 89.27

Glucose S (g/L): 10.34

Lactic Acid P (g/L): 50.18

Maltose M (g/L): 96.15

Volume V (L): 8.22

t = 37.00 h

Biomass X (g/L): 68.34

Glucose S (g/L): 14.70

Lactic Acid P (g/L): 58.58

Maltose M (g/L): 97.88

Volume V (L): 9.23

t = 38.00 h

Biomass X (g/L): 76.37

Glucose S (g/L): 19.51

Lactic Acid P (g/L): 42.46

Maltose M (g/L): 97.38

Volume V (L): 8.99

t = 39.00 h

Biomass X (g/L): 57.30

Glucose S (g/L): 11.81

Lactic Acid P (g/L): 53.37

Maltose M (g/L): 98.76

Volume V (L): 10.01

t = 40.00 h

Biomass X (g/L): 52.39

Glucose S (g/L): 13.81

Lactic Acid P (g/L): 49.35

Maltose M (g/L): 99.10

Volume V (L): 9.58

t = 41.00 h

Biomass X (g/L): 82.73

Glucose S (g/L): 17.97

Lactic Acid P (g/L): 60.47

Maltose M (g/L): 100.53

Volume V (L): 9.66

t = 42.00 h

Biomass X (g/L): 75.34

Glucose S (g/L): 15.22

Lactic Acid P (g/L): 68.43

Maltose M (g/L): 96.76

Volume V (L): 8.28

t = 43.00 h

Biomass X (g/L): 86.67

Glucose S (g/L): 16.42

Lactic Acid P (g/L): 66.02

Maltose M (g/L): 96.42

Volume V (L): 9.63

t = 44.00 h

Biomass X (g/L): 63.52

Glucose S (g/L): 18.17

Lactic Acid P (g/L): 40.74

Maltose M (g/L): 98.92

Volume V (L): 9.25

t = 45.00 h

Biomass X (g/L): 81.34

Glucose S (g/L): 15.98

Lactic Acid P (g/L): 79.05

Maltose M (g/L): 97.10

Volume V (L): 9.06

t = 46.00 h

Biomass X (g/L): 94.11

Glucose S (g/L): 17.82

Lactic Acid P (g/L): 64.61
Maltose M (g/L): 98.42
Volume V (L): 9.58

t = 47.00 h

Biomass X (g/L): 81.05
Glucose S (g/L): 23.56
Lactic Acid P (g/L): 55.93
Maltose M (g/L): 96.37
Volume V (L): 10.13

t = 48.00 h

Biomass X (g/L): 80.01
Glucose S (g/L): 18.17
Lactic Acid P (g/L): 64.40
Maltose M (g/L): 93.51
Volume V (L): 9.97

t = 49.00 h

Biomass X (g/L): 77.79
Glucose S (g/L): 17.98
Lactic Acid P (g/L): 65.02
Maltose M (g/L): 92.84
Volume V (L): 9.26

t = 50.00 h

Biomass X (g/L): 63.45
Glucose S (g/L): 21.60
Lactic Acid P (g/L): 58.24
Maltose M (g/L): 92.09
Volume V (L): 10.30

t = 51.00 h

Biomass X (g/L): 90.66
Glucose S (g/L): 26.42
Lactic Acid P (g/L): 66.07
Maltose M (g/L): 95.53
Volume V (L): 9.59

t = 52.00 h

Biomass X (g/L): 72.71

Glucose S (g/L): 24.32
Lactic Acid P (g/L): 59.59
Maltose M (g/L): 96.06
Volume V (L): 10.19

t = 53.00 h

Biomass X (g/L): 83.17
Glucose S (g/L): 24.23
Lactic Acid P (g/L): 61.72
Maltose M (g/L): 96.82
Volume V (L): 9.98

t = 54.00 h

Biomass X (g/L): 64.25
Glucose S (g/L): 26.49
Lactic Acid P (g/L): 59.69
Maltose M (g/L): 94.70
Volume V (L): 10.13

t = 55.00 h

Biomass X (g/L): 67.93
Glucose S (g/L): 21.81
Lactic Acid P (g/L): 53.67
Maltose M (g/L): 94.86
Volume V (L): 10.80

t = 56.00 h

Biomass X (g/L): 92.16
Glucose S (g/L): 29.30
Lactic Acid P (g/L): 67.54
Maltose M (g/L): 91.47
Volume V (L): 9.84

t = 57.00 h

Biomass X (g/L): 75.83
Glucose S (g/L): 29.70
Lactic Acid P (g/L): 71.65
Maltose M (g/L): 90.09
Volume V (L): 11.69

t = 58.00 h

Biomass X (g/L): 92.53
Glucose S (g/L): 28.04
Lactic Acid P (g/L): 71.38
Maltose M (g/L): 93.15
Volume V (L): 9.58

t = 59.00 h

Biomass X (g/L): 78.48
Glucose S (g/L): 33.05
Lactic Acid P (g/L): 74.31
Maltose M (g/L): 92.01
Volume V (L): 10.40

t = 60.00 h

Biomass X (g/L): 62.27
Glucose S (g/L): 27.34
Lactic Acid P (g/L): 70.38
Maltose M (g/L): 91.08
Volume V (L): 11.06

t = 61.00 h

Biomass X (g/L): 87.54
Glucose S (g/L): 34.89
Lactic Acid P (g/L): 63.35
Maltose M (g/L): 91.17
Volume V (L): 11.28

t = 62.00 h

Biomass X (g/L): 73.91
Glucose S (g/L): 32.69
Lactic Acid P (g/L): 75.39
Maltose M (g/L): 90.66
Volume V (L): 10.98

t = 63.00 h

Biomass X (g/L): 85.83
Glucose S (g/L): 31.94
Lactic Acid P (g/L): 56.87
Maltose M (g/L): 91.92
Volume V (L): 11.10

t = 64.00 h

Biomass X (g/L): 79.64
Glucose S (g/L): 33.64
Lactic Acid P (g/L): 61.06
Maltose M (g/L): 91.54
Volume V (L): 11.47

t = 65.00 h

Biomass X (g/L): 65.18
Glucose S (g/L): 34.11
Lactic Acid P (g/L): 60.78
Maltose M (g/L): 89.85
Volume V (L): 11.19

t = 66.00 h

Biomass X (g/L): 80.93
Glucose S (g/L): 38.86
Lactic Acid P (g/L): 78.54
Maltose M (g/L): 89.97
Volume V (L): 12.24

t = 67.00 h

Biomass X (g/L): 99.66
Glucose S (g/L): 32.66
Lactic Acid P (g/L): 77.61
Maltose M (g/L): 89.00
Volume V (L): 12.01

t = 68.00 h

Biomass X (g/L): 83.93
Glucose S (g/L): 34.34
Lactic Acid P (g/L): 73.96
Maltose M (g/L): 86.92
Volume V (L): 12.13

t = 69.00 h

Biomass X (g/L): 78.83
Glucose S (g/L): 40.50
Lactic Acid P (g/L): 79.20
Maltose M (g/L): 88.62
Volume V (L): 11.15

t = 70.00 h

Biomass X (g/L): 89.25
Glucose S (g/L): 36.30
Lactic Acid P (g/L): 62.73
Maltose M (g/L): 84.51
Volume V (L): 11.77

t = 71.00 h

Biomass X (g/L): 81.23
Glucose S (g/L): 41.55
Lactic Acid P (g/L): 73.47
Maltose M (g/L): 88.33
Volume V (L): 11.74

t = 72.00 h

Biomass X (g/L): 81.17
Glucose S (g/L): 44.68
Lactic Acid P (g/L): 69.63
Maltose M (g/L): 84.06
Volume V (L): 11.42

t = 73.00 h

Biomass X (g/L): 69.12
Glucose S (g/L): 39.06
Lactic Acid P (g/L): 78.17
Maltose M (g/L): 86.21
Volume V (L): 12.32

t = 74.00 h

Biomass X (g/L): 81.29
Glucose S (g/L): 43.05
Lactic Acid P (g/L): 70.14
Maltose M (g/L): 83.87
Volume V (L): 11.89

t = 75.00 h

Biomass X (g/L): 88.62
Glucose S (g/L): 38.23
Lactic Acid P (g/L): 68.03
Maltose M (g/L): 85.88

Volume V (L): 12.69

Batch 13

Initial Conditions:

Batch: 13.00

X0 (g/L): 9.49

S0 (g/L): 98.68

M0 (g/L): 29.18

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.30

F_vol (L/h): 0.03

S_feed (g/L): 231.54

M_feed (g/L): 68.46

V_final (L): 7.52

Batch 13 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 101.70

Lactic Acid P (g/L): 4.04

Maltose M (g/L): 26.55

Volume V (L): 4.98

t = 1.00 h

Biomass X (g/L): 30.23

Glucose S (g/L): 89.72

Lactic Acid P (g/L): 5.97

Maltose M (g/L): 28.66

Volume V (L): 4.69

t = 2.00 h

Biomass X (g/L): 7.83

Glucose S (g/L): 82.43

Lactic Acid P (g/L): 6.25

Maltose M (g/L): 28.25

Volume V (L): 5.11

t = 3.00 h

Biomass X (g/L): 23.89

Glucose S (g/L): 74.33

Lactic Acid P (g/L): 8.56

Maltose M (g/L): 27.79

Volume V (L): 5.09

t = 4.00 h

Biomass X (g/L): 34.98

Glucose S (g/L): 91.38

Lactic Acid P (g/L): 36.40

Maltose M (g/L): 27.87

Volume V (L): 5.16

t = 5.00 h

Biomass X (g/L): 26.78

Glucose S (g/L): 83.10

Lactic Acid P (g/L): 28.64
Maltose M (g/L): 28.07
Volume V (L): 5.06

t = 6.00 h

Biomass X (g/L): 46.87
Glucose S (g/L): 76.60
Lactic Acid P (g/L): 28.30
Maltose M (g/L): 27.28
Volume V (L): 5.82

t = 7.00 h

Biomass X (g/L): 31.35
Glucose S (g/L): 81.52
Lactic Acid P (g/L): 26.42
Maltose M (g/L): 25.61
Volume V (L): 5.51

t = 8.00 h

Biomass X (g/L): 63.78
Glucose S (g/L): 73.02
Lactic Acid P (g/L): 39.24
Maltose M (g/L): 27.55
Volume V (L): 5.02

t = 9.00 h

Biomass X (g/L): 54.12
Glucose S (g/L): 74.81
Lactic Acid P (g/L): 38.15
Maltose M (g/L): 24.23
Volume V (L): 5.35

t = 10.00 h

Biomass X (g/L): 52.42
Glucose S (g/L): 53.97
Lactic Acid P (g/L): 36.75
Maltose M (g/L): 23.85
Volume V (L): 5.21

t = 11.00 h

Biomass X (g/L): 59.49

Glucose S (g/L): 58.69
Lactic Acid P (g/L): 36.67
Maltose M (g/L): 27.11
Volume V (L): 5.17

t = 12.00 h

Biomass X (g/L): 74.35
Glucose S (g/L): 46.82
Lactic Acid P (g/L): 41.76
Maltose M (g/L): 26.58
Volume V (L): 5.13

t = 13.00 h

Biomass X (g/L): 63.48
Glucose S (g/L): 52.41
Lactic Acid P (g/L): 13.90
Maltose M (g/L): 25.63
Volume V (L): 5.52

t = 14.00 h

Biomass X (g/L): 64.70
Glucose S (g/L): 50.20
Lactic Acid P (g/L): 49.67
Maltose M (g/L): 23.72
Volume V (L): 5.33

t = 15.00 h

Biomass X (g/L): 72.62
Glucose S (g/L): 53.44
Lactic Acid P (g/L): 44.94
Maltose M (g/L): 24.35
Volume V (L): 5.70

t = 16.00 h

Biomass X (g/L): 65.23
Glucose S (g/L): 48.52
Lactic Acid P (g/L): 67.47
Maltose M (g/L): 24.69
Volume V (L): 5.74

t = 17.00 h

Biomass X (g/L): 97.62
Glucose S (g/L): 50.51
Lactic Acid P (g/L): 53.76
Maltose M (g/L): 22.79
Volume V (L): 5.85

t = 18.00 h

Biomass X (g/L): 75.25
Glucose S (g/L): 49.83
Lactic Acid P (g/L): 43.49
Maltose M (g/L): 23.65
Volume V (L): 5.43

t = 19.00 h

Biomass X (g/L): 91.01
Glucose S (g/L): 57.22
Lactic Acid P (g/L): 56.26
Maltose M (g/L): 25.19
Volume V (L): 5.73

t = 20.00 h

Biomass X (g/L): 86.14
Glucose S (g/L): 42.15
Lactic Acid P (g/L): 47.84
Maltose M (g/L): 25.96
Volume V (L): 5.80

t = 21.00 h

Biomass X (g/L): 75.32
Glucose S (g/L): 37.85
Lactic Acid P (g/L): 64.14
Maltose M (g/L): 23.43
Volume V (L): 5.61

t = 22.00 h

Biomass X (g/L): 86.36
Glucose S (g/L): 44.12
Lactic Acid P (g/L): 63.54
Maltose M (g/L): 24.71
Volume V (L): 5.58

t = 23.00 h

Biomass X (g/L): 95.37
Glucose S (g/L): 37.61
Lactic Acid P (g/L): 71.24
Maltose M (g/L): 21.15
Volume V (L): 6.10

t = 24.00 h

Biomass X (g/L): 86.42
Glucose S (g/L): 50.40
Lactic Acid P (g/L): 55.02
Maltose M (g/L): 24.19
Volume V (L): 5.85

t = 25.00 h

Biomass X (g/L): 101.20
Glucose S (g/L): 39.93
Lactic Acid P (g/L): 64.13
Maltose M (g/L): 22.16
Volume V (L): 5.97

t = 26.00 h

Biomass X (g/L): 100.17
Glucose S (g/L): 38.08
Lactic Acid P (g/L): 51.24
Maltose M (g/L): 23.37
Volume V (L): 5.71

t = 27.00 h

Biomass X (g/L): 99.75
Glucose S (g/L): 38.46
Lactic Acid P (g/L): 72.49
Maltose M (g/L): 22.49
Volume V (L): 5.66

t = 28.00 h

Biomass X (g/L): 86.34
Glucose S (g/L): 45.22
Lactic Acid P (g/L): 71.49
Maltose M (g/L): 24.28
Volume V (L): 5.83

t = 29.00 h

Biomass X (g/L): 80.55
Glucose S (g/L): 40.12
Lactic Acid P (g/L): 72.02
Maltose M (g/L): 21.34
Volume V (L): 5.95

t = 30.00 h

Biomass X (g/L): 85.70
Glucose S (g/L): 37.33
Lactic Acid P (g/L): 67.86
Maltose M (g/L): 23.05
Volume V (L): 5.79

t = 31.00 h

Biomass X (g/L): 94.27
Glucose S (g/L): 50.66
Lactic Acid P (g/L): 72.61
Maltose M (g/L): 21.74
Volume V (L): 6.04

t = 32.00 h

Biomass X (g/L): 98.13
Glucose S (g/L): 36.20
Lactic Acid P (g/L): 84.99
Maltose M (g/L): 21.72
Volume V (L): 5.73

t = 33.00 h

Biomass X (g/L): 70.89
Glucose S (g/L): 41.45
Lactic Acid P (g/L): 66.95
Maltose M (g/L): 21.52
Volume V (L): 5.88

t = 34.00 h

Biomass X (g/L): 93.04
Glucose S (g/L): 45.14
Lactic Acid P (g/L): 71.71
Maltose M (g/L): 20.52

Volume V (L): 6.28

t = 35.00 h

Biomass X (g/L): 94.92

Glucose S (g/L): 32.14

Lactic Acid P (g/L): 60.17

Maltose M (g/L): 21.41

Volume V (L): 6.30

t = 36.00 h

Biomass X (g/L): 90.67

Glucose S (g/L): 44.51

Lactic Acid P (g/L): 62.85

Maltose M (g/L): 21.19

Volume V (L): 6.13

t = 37.00 h

Biomass X (g/L): 106.69

Glucose S (g/L): 35.87

Lactic Acid P (g/L): 72.89

Maltose M (g/L): 21.41

Volume V (L): 6.04

t = 38.00 h

Biomass X (g/L): 97.99

Glucose S (g/L): 46.92

Lactic Acid P (g/L): 83.30

Maltose M (g/L): 21.06

Volume V (L): 6.07

t = 39.00 h

Biomass X (g/L): 107.06

Glucose S (g/L): 36.88

Lactic Acid P (g/L): 70.33

Maltose M (g/L): 21.81

Volume V (L): 6.23

t = 40.00 h

Biomass X (g/L): 91.66

Glucose S (g/L): 46.69

Lactic Acid P (g/L): 70.97

Maltose M (g/L): 21.82

Volume V (L): 6.47

t = 41.00 h

Biomass X (g/L): 103.37

Glucose S (g/L): 49.35

Lactic Acid P (g/L): 69.71

Maltose M (g/L): 19.19

Volume V (L): 6.56

t = 42.00 h

Biomass X (g/L): 110.44

Glucose S (g/L): 48.23

Lactic Acid P (g/L): 68.85

Maltose M (g/L): 21.46

Volume V (L): 6.31

t = 43.00 h

Biomass X (g/L): 107.16

Glucose S (g/L): 45.02

Lactic Acid P (g/L): 77.43

Maltose M (g/L): 18.73

Volume V (L): 6.21

t = 44.00 h

Biomass X (g/L): 82.78

Glucose S (g/L): 42.55

Lactic Acid P (g/L): 60.12

Maltose M (g/L): 20.82

Volume V (L): 6.53

t = 45.00 h

Biomass X (g/L): 91.23

Glucose S (g/L): 40.89

Lactic Acid P (g/L): 87.90

Maltose M (g/L): 20.72

Volume V (L): 6.35

t = 46.00 h

Biomass X (g/L): 90.24

Glucose S (g/L): 53.03

Lactic Acid P (g/L): 69.04

Maltose M (g/L): 17.40

Volume V (L): 6.44

t = 47.00 h

Biomass X (g/L): 98.62

Glucose S (g/L): 40.86

Lactic Acid P (g/L): 83.07

Maltose M (g/L): 19.79

Volume V (L): 6.85

t = 48.00 h

Biomass X (g/L): 100.52

Glucose S (g/L): 39.26

Lactic Acid P (g/L): 63.73

Maltose M (g/L): 18.95

Volume V (L): 6.78

t = 49.00 h

Biomass X (g/L): 84.98

Glucose S (g/L): 49.43

Lactic Acid P (g/L): 83.18

Maltose M (g/L): 18.46

Volume V (L): 6.68

t = 50.00 h

Biomass X (g/L): 94.05

Glucose S (g/L): 41.36

Lactic Acid P (g/L): 83.52

Maltose M (g/L): 20.98

Volume V (L): 6.42

t = 51.00 h

Biomass X (g/L): 72.78

Glucose S (g/L): 40.84

Lactic Acid P (g/L): 80.97

Maltose M (g/L): 18.33

Volume V (L): 6.76

t = 52.00 h

Biomass X (g/L): 107.71

Glucose S (g/L): 38.37
Lactic Acid P (g/L): 67.65
Maltose M (g/L): 18.02
Volume V (L): 6.96

t = 53.00 h

Biomass X (g/L): 87.77
Glucose S (g/L): 44.28
Lactic Acid P (g/L): 80.20
Maltose M (g/L): 19.00
Volume V (L): 6.70

t = 54.00 h

Biomass X (g/L): 93.37
Glucose S (g/L): 46.97
Lactic Acid P (g/L): 66.33
Maltose M (g/L): 19.65
Volume V (L): 7.09

t = 55.00 h

Biomass X (g/L): 89.51
Glucose S (g/L): 49.23
Lactic Acid P (g/L): 75.77
Maltose M (g/L): 18.49
Volume V (L): 7.11

t = 56.00 h

Biomass X (g/L): 104.09
Glucose S (g/L): 53.69
Lactic Acid P (g/L): 88.68
Maltose M (g/L): 20.02
Volume V (L): 7.22

t = 57.00 h

Biomass X (g/L): 63.57
Glucose S (g/L): 51.63
Lactic Acid P (g/L): 86.63
Maltose M (g/L): 19.10
Volume V (L): 6.98

t = 58.00 h

Biomass X (g/L): 91.96
Glucose S (g/L): 47.88
Lactic Acid P (g/L): 79.15
Maltose M (g/L): 15.58
Volume V (L): 7.07

t = 59.00 h

Biomass X (g/L): 90.87
Glucose S (g/L): 48.23
Lactic Acid P (g/L): 69.61
Maltose M (g/L): 17.54
Volume V (L): 7.27

t = 60.00 h

Biomass X (g/L): 87.16
Glucose S (g/L): 52.70
Lactic Acid P (g/L): 85.83
Maltose M (g/L): 17.39
Volume V (L): 7.11

t = 61.00 h

Biomass X (g/L): 109.04
Glucose S (g/L): 57.49
Lactic Acid P (g/L): 79.87
Maltose M (g/L): 18.59
Volume V (L): 6.95

t = 62.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 51.27
Lactic Acid P (g/L): 56.99
Maltose M (g/L): 14.58
Volume V (L): 6.99

t = 63.00 h

Biomass X (g/L): 98.96
Glucose S (g/L): 48.23
Lactic Acid P (g/L): 87.83
Maltose M (g/L): 20.98
Volume V (L): 7.16

t = 64.00 h

Biomass X (g/L): 94.55
Glucose S (g/L): 45.61
Lactic Acid P (g/L): 80.07
Maltose M (g/L): 18.33
Volume V (L): 6.89

t = 65.00 h

Biomass X (g/L): 103.50
Glucose S (g/L): 43.39
Lactic Acid P (g/L): 92.37
Maltose M (g/L): 17.75
Volume V (L): 7.15

t = 66.00 h

Biomass X (g/L): 83.75
Glucose S (g/L): 55.96
Lactic Acid P (g/L): 69.89
Maltose M (g/L): 18.95
Volume V (L): 7.44

t = 67.00 h

Biomass X (g/L): 82.86
Glucose S (g/L): 55.85
Lactic Acid P (g/L): 49.53
Maltose M (g/L): 20.31
Volume V (L): 6.81

t = 68.00 h

Biomass X (g/L): 101.16
Glucose S (g/L): 56.23
Lactic Acid P (g/L): 83.83
Maltose M (g/L): 18.35
Volume V (L): 7.59

t = 69.00 h

Biomass X (g/L): 94.14
Glucose S (g/L): 52.56
Lactic Acid P (g/L): 84.13
Maltose M (g/L): 18.35
Volume V (L): 7.54

t = 70.00 h

Biomass X (g/L): 73.96
Glucose S (g/L): 61.80
Lactic Acid P (g/L): 64.74
Maltose M (g/L): 17.23
Volume V (L): 7.26

t = 71.00 h

Biomass X (g/L): 83.91
Glucose S (g/L): 51.29
Lactic Acid P (g/L): 71.94
Maltose M (g/L): 16.79
Volume V (L): 7.49

t = 72.00 h

Biomass X (g/L): 93.05
Glucose S (g/L): 59.98
Lactic Acid P (g/L): 108.96
Maltose M (g/L): 16.27
Volume V (L): 7.42

t = 73.00 h

Biomass X (g/L): 97.39
Glucose S (g/L): 54.64
Lactic Acid P (g/L): 78.92
Maltose M (g/L): 15.74
Volume V (L): 7.70

t = 74.00 h

Biomass X (g/L): 81.79
Glucose S (g/L): 51.74
Lactic Acid P (g/L): 89.31
Maltose M (g/L): 15.82
Volume V (L): 7.68

t = 75.00 h

Biomass X (g/L): 58.91
Glucose S (g/L): 59.05
Lactic Acid P (g/L): 75.83
Maltose M (g/L): 14.90

Volume V (L): 7.24

Batch 14

Initial Conditions:

Batch: 14.00

X0 (g/L): 6.15

S0 (g/L): 20.83

M0 (g/L): 121.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 5.83

F_vol (L/h): 0.06

S_feed (g/L): 43.90

M_feed (g/L): 256.10

V_final (L): 9.63

Batch 14 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 11.53
Glucose S (g/L): 14.62
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 118.29
Volume V (L): 5.36

t = 1.00 h

Biomass X (g/L): 5.98
Glucose S (g/L): 27.69
Lactic Acid P (g/L): 16.43
Maltose M (g/L): 116.07
Volume V (L): 5.06

t = 2.00 h

Biomass X (g/L): 10.72
Glucose S (g/L): 22.53
Lactic Acid P (g/L): 1.84
Maltose M (g/L): 121.41
Volume V (L): 4.90

t = 3.00 h

Biomass X (g/L): 5.80
Glucose S (g/L): 23.20
Lactic Acid P (g/L): 1.66
Maltose M (g/L): 114.08
Volume V (L): 4.61

t = 4.00 h

Biomass X (g/L): 31.37
Glucose S (g/L): 21.36
Lactic Acid P (g/L): 14.79
Maltose M (g/L): 116.99
Volume V (L): 5.55

t = 5.00 h

Biomass X (g/L): 28.14
Glucose S (g/L): 18.47

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 113.86
Volume V (L): 5.47

t = 6.00 h

Biomass X (g/L): 15.38
Glucose S (g/L): 18.85
Lactic Acid P (g/L): 5.13
Maltose M (g/L): 119.66
Volume V (L): 5.01

t = 7.00 h

Biomass X (g/L): 13.64
Glucose S (g/L): 13.22
Lactic Acid P (g/L): 15.32
Maltose M (g/L): 124.23
Volume V (L): 5.23

t = 8.00 h

Biomass X (g/L): 26.09
Glucose S (g/L): 23.07
Lactic Acid P (g/L): 26.68
Maltose M (g/L): 110.53
Volume V (L): 5.32

t = 9.00 h

Biomass X (g/L): 37.32
Glucose S (g/L): 11.12
Lactic Acid P (g/L): 24.81
Maltose M (g/L): 107.54
Volume V (L): 5.01

t = 10.00 h

Biomass X (g/L): 39.12
Glucose S (g/L): 10.20
Lactic Acid P (g/L): 25.28
Maltose M (g/L): 114.12
Volume V (L): 5.31

t = 11.00 h

Biomass X (g/L): 51.77

Glucose S (g/L): 10.57
Lactic Acid P (g/L): 26.36
Maltose M (g/L): 114.04
Volume V (L): 5.26

t = 12.00 h

Biomass X (g/L): 50.66
Glucose S (g/L): 14.47
Lactic Acid P (g/L): 31.66
Maltose M (g/L): 114.23
Volume V (L): 5.59

t = 13.00 h

Biomass X (g/L): 55.18
Glucose S (g/L): 9.71
Lactic Acid P (g/L): 32.16
Maltose M (g/L): 110.77
Volume V (L): 5.53

t = 14.00 h

Biomass X (g/L): 51.92
Glucose S (g/L): 5.23
Lactic Acid P (g/L): 23.32
Maltose M (g/L): 111.14
Volume V (L): 5.26

t = 15.00 h

Biomass X (g/L): 38.60
Glucose S (g/L): 11.95
Lactic Acid P (g/L): 34.82
Maltose M (g/L): 112.87
Volume V (L): 6.68

t = 16.00 h

Biomass X (g/L): 51.71
Glucose S (g/L): 9.95
Lactic Acid P (g/L): 26.67
Maltose M (g/L): 115.04
Volume V (L): 6.45

t = 17.00 h

Biomass X (g/L): 65.20
Glucose S (g/L): 9.07
Lactic Acid P (g/L): 38.61
Maltose M (g/L): 101.82
Volume V (L): 5.76

t = 18.00 h

Biomass X (g/L): 56.86
Glucose S (g/L): 9.39
Lactic Acid P (g/L): 35.19
Maltose M (g/L): 107.84
Volume V (L): 6.21

t = 19.00 h

Biomass X (g/L): 63.63
Glucose S (g/L): 14.24
Lactic Acid P (g/L): 63.50
Maltose M (g/L): 109.64
Volume V (L): 6.33

t = 20.00 h

Biomass X (g/L): 57.68
Glucose S (g/L): 13.59
Lactic Acid P (g/L): 41.98
Maltose M (g/L): 104.26
Volume V (L): 6.68

t = 21.00 h

Biomass X (g/L): 52.64
Glucose S (g/L): 5.17
Lactic Acid P (g/L): 31.19
Maltose M (g/L): 114.74
Volume V (L): 5.99

t = 22.00 h

Biomass X (g/L): 62.32
Glucose S (g/L): 12.08
Lactic Acid P (g/L): 43.74
Maltose M (g/L): 104.18
Volume V (L): 6.76

t = 23.00 h

Biomass X (g/L): 67.57
Glucose S (g/L): 8.57
Lactic Acid P (g/L): 50.76
Maltose M (g/L): 106.05
Volume V (L): 6.22

t = 24.00 h

Biomass X (g/L): 59.96
Glucose S (g/L): 9.10
Lactic Acid P (g/L): 62.17
Maltose M (g/L): 107.94
Volume V (L): 5.99

t = 25.00 h

Biomass X (g/L): 64.96
Glucose S (g/L): 2.85
Lactic Acid P (g/L): 43.19
Maltose M (g/L): 102.51
Volume V (L): 6.81

t = 26.00 h

Biomass X (g/L): 59.61
Glucose S (g/L): 7.60
Lactic Acid P (g/L): 35.44
Maltose M (g/L): 96.63
Volume V (L): 7.08

t = 27.00 h

Biomass X (g/L): 79.49
Glucose S (g/L): 5.99
Lactic Acid P (g/L): 57.79
Maltose M (g/L): 101.97
Volume V (L): 7.63

t = 28.00 h

Biomass X (g/L): 67.53
Glucose S (g/L): 11.14
Lactic Acid P (g/L): 39.27
Maltose M (g/L): 107.49
Volume V (L): 6.29

t = 29.00 h

Biomass X (g/L): 84.44
Glucose S (g/L): 12.20
Lactic Acid P (g/L): 48.62
Maltose M (g/L): 101.16
Volume V (L): 7.05

t = 30.00 h

Biomass X (g/L): 86.01
Glucose S (g/L): 3.28
Lactic Acid P (g/L): 51.14
Maltose M (g/L): 98.42
Volume V (L): 6.70

t = 31.00 h

Biomass X (g/L): 72.10
Glucose S (g/L): 9.71
Lactic Acid P (g/L): 53.71
Maltose M (g/L): 105.27
Volume V (L): 7.21

t = 32.00 h

Biomass X (g/L): 64.39
Glucose S (g/L): 13.49
Lactic Acid P (g/L): 61.31
Maltose M (g/L): 99.41
Volume V (L): 6.43

t = 33.00 h

Biomass X (g/L): 55.26
Glucose S (g/L): 10.05
Lactic Acid P (g/L): 49.95
Maltose M (g/L): 98.08
Volume V (L): 6.57

t = 34.00 h

Biomass X (g/L): 61.57
Glucose S (g/L): 13.20
Lactic Acid P (g/L): 54.40
Maltose M (g/L): 100.27

Volume V (L): 6.81

t = 35.00 h

Biomass X (g/L): 63.11

Glucose S (g/L): 12.54

Lactic Acid P (g/L): 57.48

Maltose M (g/L): 102.31

Volume V (L): 7.19

t = 36.00 h

Biomass X (g/L): 88.92

Glucose S (g/L): 12.99

Lactic Acid P (g/L): 59.68

Maltose M (g/L): 108.50

Volume V (L): 6.99

t = 37.00 h

Biomass X (g/L): 77.93

Glucose S (g/L): 11.68

Lactic Acid P (g/L): 59.95

Maltose M (g/L): 101.41

Volume V (L): 7.53

t = 38.00 h

Biomass X (g/L): 80.98

Glucose S (g/L): 9.40

Lactic Acid P (g/L): 63.65

Maltose M (g/L): 102.25

Volume V (L): 7.73

t = 39.00 h

Biomass X (g/L): 57.58

Glucose S (g/L): 10.21

Lactic Acid P (g/L): 67.94

Maltose M (g/L): 94.70

Volume V (L): 7.63

t = 40.00 h

Biomass X (g/L): 67.40

Glucose S (g/L): 16.14

Lactic Acid P (g/L): 64.58

Maltose M (g/L): 91.97

Volume V (L): 8.07

t = 41.00 h

Biomass X (g/L): 86.93

Glucose S (g/L): 14.66

Lactic Acid P (g/L): 65.75

Maltose M (g/L): 95.99

Volume V (L): 7.69

t = 42.00 h

Biomass X (g/L): 81.83

Glucose S (g/L): 14.76

Lactic Acid P (g/L): 48.26

Maltose M (g/L): 95.63

Volume V (L): 7.14

t = 43.00 h

Biomass X (g/L): 78.23

Glucose S (g/L): 14.49

Lactic Acid P (g/L): 63.35

Maltose M (g/L): 90.12

Volume V (L): 8.16

t = 44.00 h

Biomass X (g/L): 85.09

Glucose S (g/L): 15.43

Lactic Acid P (g/L): 67.72

Maltose M (g/L): 93.38

Volume V (L): 7.49

t = 45.00 h

Biomass X (g/L): 95.21

Glucose S (g/L): 20.01

Lactic Acid P (g/L): 70.48

Maltose M (g/L): 95.26

Volume V (L): 8.12

t = 46.00 h

Biomass X (g/L): 78.50

Glucose S (g/L): 16.54

Lactic Acid P (g/L): 58.37

Maltose M (g/L): 94.57

Volume V (L): 8.42

t = 47.00 h

Biomass X (g/L): 81.42

Glucose S (g/L): 14.19

Lactic Acid P (g/L): 59.17

Maltose M (g/L): 94.65

Volume V (L): 8.07

t = 48.00 h

Biomass X (g/L): 82.79

Glucose S (g/L): 20.96

Lactic Acid P (g/L): 60.41

Maltose M (g/L): 93.24

Volume V (L): 7.41

t = 49.00 h

Biomass X (g/L): 103.15

Glucose S (g/L): 18.79

Lactic Acid P (g/L): 70.75

Maltose M (g/L): 89.38

Volume V (L): 7.69

t = 50.00 h

Biomass X (g/L): 62.22

Glucose S (g/L): 20.13

Lactic Acid P (g/L): 75.62

Maltose M (g/L): 88.84

Volume V (L): 8.58

t = 51.00 h

Biomass X (g/L): 87.87

Glucose S (g/L): 25.42

Lactic Acid P (g/L): 63.25

Maltose M (g/L): 85.46

Volume V (L): 8.09

t = 52.00 h

Biomass X (g/L): 80.72

Glucose S (g/L): 21.26
Lactic Acid P (g/L): 75.36
Maltose M (g/L): 97.48
Volume V (L): 8.24

t = 53.00 h

Biomass X (g/L): 87.10
Glucose S (g/L): 22.83
Lactic Acid P (g/L): 68.79
Maltose M (g/L): 89.55
Volume V (L): 8.56

t = 54.00 h

Biomass X (g/L): 76.01
Glucose S (g/L): 19.16
Lactic Acid P (g/L): 66.99
Maltose M (g/L): 88.06
Volume V (L): 8.59

t = 55.00 h

Biomass X (g/L): 81.68
Glucose S (g/L): 22.64
Lactic Acid P (g/L): 66.59
Maltose M (g/L): 84.61
Volume V (L): 8.50

t = 56.00 h

Biomass X (g/L): 88.73
Glucose S (g/L): 25.75
Lactic Acid P (g/L): 75.47
Maltose M (g/L): 100.35
Volume V (L): 8.38

t = 57.00 h

Biomass X (g/L): 110.79
Glucose S (g/L): 20.03
Lactic Acid P (g/L): 72.09
Maltose M (g/L): 83.04
Volume V (L): 8.41

t = 58.00 h

Biomass X (g/L): 89.74
Glucose S (g/L): 25.88
Lactic Acid P (g/L): 89.76
Maltose M (g/L): 85.27
Volume V (L): 8.90

t = 59.00 h

Biomass X (g/L): 74.63
Glucose S (g/L): 26.26
Lactic Acid P (g/L): 62.30
Maltose M (g/L): 88.34
Volume V (L): 8.85

t = 60.00 h

Biomass X (g/L): 78.54
Glucose S (g/L): 26.75
Lactic Acid P (g/L): 74.80
Maltose M (g/L): 88.65
Volume V (L): 8.61

t = 61.00 h

Biomass X (g/L): 78.29
Glucose S (g/L): 30.36
Lactic Acid P (g/L): 65.81
Maltose M (g/L): 89.68
Volume V (L): 9.17

t = 62.00 h

Biomass X (g/L): 66.00
Glucose S (g/L): 28.96
Lactic Acid P (g/L): 63.09
Maltose M (g/L): 92.79
Volume V (L): 8.78

t = 63.00 h

Biomass X (g/L): 79.42
Glucose S (g/L): 33.01
Lactic Acid P (g/L): 71.01
Maltose M (g/L): 86.87
Volume V (L): 8.58

t = 64.00 h

Biomass X (g/L): 89.13
Glucose S (g/L): 26.39
Lactic Acid P (g/L): 72.61
Maltose M (g/L): 79.86
Volume V (L): 8.67

t = 65.00 h

Biomass X (g/L): 70.73
Glucose S (g/L): 30.40
Lactic Acid P (g/L): 97.59
Maltose M (g/L): 82.76
Volume V (L): 8.65

t = 66.00 h

Biomass X (g/L): 91.96
Glucose S (g/L): 30.92
Lactic Acid P (g/L): 70.30
Maltose M (g/L): 84.68
Volume V (L): 8.98

t = 67.00 h

Biomass X (g/L): 86.79
Glucose S (g/L): 28.18
Lactic Acid P (g/L): 62.16
Maltose M (g/L): 91.63
Volume V (L): 9.60

t = 68.00 h

Biomass X (g/L): 75.55
Glucose S (g/L): 34.83
Lactic Acid P (g/L): 74.88
Maltose M (g/L): 76.64
Volume V (L): 8.90

t = 69.00 h

Biomass X (g/L): 80.91
Glucose S (g/L): 39.53
Lactic Acid P (g/L): 54.26
Maltose M (g/L): 87.12
Volume V (L): 9.33

t = 70.00 h

Biomass X (g/L): 96.77
Glucose S (g/L): 33.29
Lactic Acid P (g/L): 72.12
Maltose M (g/L): 82.40
Volume V (L): 9.08

t = 71.00 h

Biomass X (g/L): 88.78
Glucose S (g/L): 35.50
Lactic Acid P (g/L): 73.31
Maltose M (g/L): 84.73
Volume V (L): 9.29

t = 72.00 h

Biomass X (g/L): 82.07
Glucose S (g/L): 34.92
Lactic Acid P (g/L): 80.45
Maltose M (g/L): 80.36
Volume V (L): 8.89

t = 73.00 h

Biomass X (g/L): 79.72
Glucose S (g/L): 33.23
Lactic Acid P (g/L): 82.26
Maltose M (g/L): 76.83
Volume V (L): 9.43

t = 74.00 h

Biomass X (g/L): 82.50
Glucose S (g/L): 30.62
Lactic Acid P (g/L): 69.35
Maltose M (g/L): 83.10
Volume V (L): 9.50

t = 75.00 h

Biomass X (g/L): 77.02
Glucose S (g/L): 34.56
Lactic Acid P (g/L): 72.70
Maltose M (g/L): 82.28

Volume V (L): 9.83

Batch 15

Initial Conditions:

Batch: 15.00

X0 (g/L): 8.61

S0 (g/L): 83.39

M0 (g/L): 80.04

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.96

F_vol (L/h): 0.10

S_feed (g/L): 153.08

M_feed (g/L): 146.92

V_final (L): 12.19

Batch 15 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 10.67
Glucose S (g/L): 84.21
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 78.72
Volume V (L): 5.99

t = 1.00 h

Biomass X (g/L): 0.40
Glucose S (g/L): 85.75
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 83.31
Volume V (L): 5.04

t = 2.00 h

Biomass X (g/L): 24.97
Glucose S (g/L): 75.88
Lactic Acid P (g/L): 3.30
Maltose M (g/L): 76.87
Volume V (L): 5.38

t = 3.00 h

Biomass X (g/L): 20.98
Glucose S (g/L): 81.92
Lactic Acid P (g/L): 22.42
Maltose M (g/L): 80.25
Volume V (L): 5.84

t = 4.00 h

Biomass X (g/L): 25.45
Glucose S (g/L): 74.67
Lactic Acid P (g/L): 9.67
Maltose M (g/L): 78.39
Volume V (L): 4.92

t = 5.00 h

Biomass X (g/L): 28.43
Glucose S (g/L): 74.87

Lactic Acid P (g/L): 14.80

Maltose M (g/L): 80.24

Volume V (L): 5.69

t = 6.00 h

Biomass X (g/L): 28.78

Glucose S (g/L): 75.31

Lactic Acid P (g/L): 14.48

Maltose M (g/L): 78.02

Volume V (L): 5.15

t = 7.00 h

Biomass X (g/L): 40.59

Glucose S (g/L): 70.98

Lactic Acid P (g/L): 32.32

Maltose M (g/L): 73.97

Volume V (L): 6.17

t = 8.00 h

Biomass X (g/L): 47.81

Glucose S (g/L): 65.03

Lactic Acid P (g/L): 43.65

Maltose M (g/L): 77.71

Volume V (L): 5.85

t = 9.00 h

Biomass X (g/L): 31.83

Glucose S (g/L): 65.45

Lactic Acid P (g/L): 33.74

Maltose M (g/L): 79.96

Volume V (L): 6.03

t = 10.00 h

Biomass X (g/L): 47.20

Glucose S (g/L): 57.90

Lactic Acid P (g/L): 33.74

Maltose M (g/L): 76.71

Volume V (L): 5.51

t = 11.00 h

Biomass X (g/L): 53.65

Glucose S (g/L): 65.13
Lactic Acid P (g/L): 36.35
Maltose M (g/L): 78.50
Volume V (L): 6.46

t = 12.00 h

Biomass X (g/L): 61.84
Glucose S (g/L): 58.80
Lactic Acid P (g/L): 17.60
Maltose M (g/L): 73.26
Volume V (L): 7.35

t = 13.00 h

Biomass X (g/L): 73.60
Glucose S (g/L): 61.94
Lactic Acid P (g/L): 47.93
Maltose M (g/L): 75.61
Volume V (L): 5.67

t = 14.00 h

Biomass X (g/L): 58.22
Glucose S (g/L): 55.16
Lactic Acid P (g/L): 44.80
Maltose M (g/L): 73.63
Volume V (L): 6.15

t = 15.00 h

Biomass X (g/L): 63.21
Glucose S (g/L): 54.28
Lactic Acid P (g/L): 49.50
Maltose M (g/L): 76.92
Volume V (L): 6.57

t = 16.00 h

Biomass X (g/L): 65.27
Glucose S (g/L): 52.78
Lactic Acid P (g/L): 41.27
Maltose M (g/L): 74.62
Volume V (L): 6.99

t = 17.00 h

Biomass X (g/L): 64.54
Glucose S (g/L): 57.93
Lactic Acid P (g/L): 45.91
Maltose M (g/L): 76.00
Volume V (L): 6.31

t = 18.00 h

Biomass X (g/L): 70.47
Glucose S (g/L): 52.17
Lactic Acid P (g/L): 47.25
Maltose M (g/L): 76.67
Volume V (L): 5.10

t = 19.00 h

Biomass X (g/L): 82.32
Glucose S (g/L): 56.65
Lactic Acid P (g/L): 61.42
Maltose M (g/L): 75.22
Volume V (L): 7.60

t = 20.00 h

Biomass X (g/L): 66.82
Glucose S (g/L): 53.09
Lactic Acid P (g/L): 50.29
Maltose M (g/L): 73.71
Volume V (L): 7.36

t = 21.00 h

Biomass X (g/L): 81.71
Glucose S (g/L): 47.70
Lactic Acid P (g/L): 73.38
Maltose M (g/L): 69.91
Volume V (L): 6.01

t = 22.00 h

Biomass X (g/L): 63.10
Glucose S (g/L): 52.69
Lactic Acid P (g/L): 59.81
Maltose M (g/L): 72.18
Volume V (L): 6.76

t = 23.00 h

Biomass X (g/L): 74.58
Glucose S (g/L): 50.86
Lactic Acid P (g/L): 45.68
Maltose M (g/L): 70.37
Volume V (L): 8.11

t = 24.00 h

Biomass X (g/L): 77.55
Glucose S (g/L): 55.35
Lactic Acid P (g/L): 56.82
Maltose M (g/L): 70.48
Volume V (L): 8.13

t = 25.00 h

Biomass X (g/L): 96.80
Glucose S (g/L): 51.54
Lactic Acid P (g/L): 53.06
Maltose M (g/L): 70.00
Volume V (L): 7.94

t = 26.00 h

Biomass X (g/L): 79.44
Glucose S (g/L): 59.25
Lactic Acid P (g/L): 40.17
Maltose M (g/L): 67.99
Volume V (L): 7.39

t = 27.00 h

Biomass X (g/L): 62.22
Glucose S (g/L): 56.26
Lactic Acid P (g/L): 74.64
Maltose M (g/L): 64.98
Volume V (L): 7.64

t = 28.00 h

Biomass X (g/L): 64.59
Glucose S (g/L): 56.50
Lactic Acid P (g/L): 67.39
Maltose M (g/L): 68.03
Volume V (L): 7.61

t = 29.00 h

Biomass X (g/L): 83.80
Glucose S (g/L): 59.93
Lactic Acid P (g/L): 66.55
Maltose M (g/L): 69.94
Volume V (L): 8.43

t = 30.00 h

Biomass X (g/L): 89.54
Glucose S (g/L): 61.59
Lactic Acid P (g/L): 63.90
Maltose M (g/L): 65.43
Volume V (L): 7.37

t = 31.00 h

Biomass X (g/L): 87.67
Glucose S (g/L): 58.84
Lactic Acid P (g/L): 50.74
Maltose M (g/L): 68.68
Volume V (L): 7.51

t = 32.00 h

Biomass X (g/L): 86.03
Glucose S (g/L): 56.91
Lactic Acid P (g/L): 59.27
Maltose M (g/L): 73.34
Volume V (L): 8.65

t = 33.00 h

Biomass X (g/L): 95.46
Glucose S (g/L): 54.14
Lactic Acid P (g/L): 71.57
Maltose M (g/L): 70.60
Volume V (L): 8.14

t = 34.00 h

Biomass X (g/L): 79.95
Glucose S (g/L): 58.55
Lactic Acid P (g/L): 55.24
Maltose M (g/L): 71.87

Volume V (L): 9.05

t = 35.00 h

Biomass X (g/L): 93.37

Glucose S (g/L): 59.49

Lactic Acid P (g/L): 42.55

Maltose M (g/L): 68.13

Volume V (L): 7.97

t = 36.00 h

Biomass X (g/L): 70.17

Glucose S (g/L): 62.68

Lactic Acid P (g/L): 63.47

Maltose M (g/L): 69.82

Volume V (L): 8.42

t = 37.00 h

Biomass X (g/L): 77.00

Glucose S (g/L): 61.84

Lactic Acid P (g/L): 64.64

Maltose M (g/L): 64.37

Volume V (L): 8.02

t = 38.00 h

Biomass X (g/L): 68.76

Glucose S (g/L): 66.11

Lactic Acid P (g/L): 67.50

Maltose M (g/L): 64.20

Volume V (L): 8.65

t = 39.00 h

Biomass X (g/L): 82.97

Glucose S (g/L): 62.38

Lactic Acid P (g/L): 73.15

Maltose M (g/L): 58.22

Volume V (L): 7.89

t = 40.00 h

Biomass X (g/L): 85.13

Glucose S (g/L): 65.02

Lactic Acid P (g/L): 72.70

Maltose M (g/L): 68.34

Volume V (L): 8.56

t = 41.00 h

Biomass X (g/L): 86.63

Glucose S (g/L): 70.01

Lactic Acid P (g/L): 69.68

Maltose M (g/L): 68.91

Volume V (L): 8.11

t = 42.00 h

Biomass X (g/L): 86.15

Glucose S (g/L): 67.86

Lactic Acid P (g/L): 78.33

Maltose M (g/L): 62.93

Volume V (L): 9.43

t = 43.00 h

Biomass X (g/L): 77.00

Glucose S (g/L): 67.21

Lactic Acid P (g/L): 54.52

Maltose M (g/L): 63.36

Volume V (L): 9.54

t = 44.00 h

Biomass X (g/L): 83.78

Glucose S (g/L): 66.09

Lactic Acid P (g/L): 72.80

Maltose M (g/L): 61.65

Volume V (L): 8.22

t = 45.00 h

Biomass X (g/L): 71.67

Glucose S (g/L): 66.72

Lactic Acid P (g/L): 74.18

Maltose M (g/L): 64.32

Volume V (L): 10.08

t = 46.00 h

Biomass X (g/L): 98.04

Glucose S (g/L): 66.70

Lactic Acid P (g/L): 57.87

Maltose M (g/L): 65.94

Volume V (L): 10.85

t = 47.00 h

Biomass X (g/L): 82.58

Glucose S (g/L): 68.50

Lactic Acid P (g/L): 55.80

Maltose M (g/L): 57.90

Volume V (L): 9.53

t = 48.00 h

Biomass X (g/L): 80.79

Glucose S (g/L): 73.68

Lactic Acid P (g/L): 62.78

Maltose M (g/L): 63.30

Volume V (L): 9.23

t = 49.00 h

Biomass X (g/L): 99.90

Glucose S (g/L): 71.04

Lactic Acid P (g/L): 63.78

Maltose M (g/L): 61.10

Volume V (L): 11.32

t = 50.00 h

Biomass X (g/L): 73.66

Glucose S (g/L): 67.89

Lactic Acid P (g/L): 76.59

Maltose M (g/L): 61.79

Volume V (L): 9.04

t = 51.00 h

Biomass X (g/L): 74.63

Glucose S (g/L): 79.31

Lactic Acid P (g/L): 64.66

Maltose M (g/L): 62.35

Volume V (L): 9.90

t = 52.00 h

Biomass X (g/L): 74.79

Glucose S (g/L): 75.05
Lactic Acid P (g/L): 62.74
Maltose M (g/L): 60.27
Volume V (L): 10.09

t = 53.00 h

Biomass X (g/L): 84.18
Glucose S (g/L): 69.83
Lactic Acid P (g/L): 65.13
Maltose M (g/L): 60.14
Volume V (L): 10.86

t = 54.00 h

Biomass X (g/L): 88.67
Glucose S (g/L): 73.85
Lactic Acid P (g/L): 80.71
Maltose M (g/L): 59.50
Volume V (L): 10.39

t = 55.00 h

Biomass X (g/L): 91.38
Glucose S (g/L): 76.44
Lactic Acid P (g/L): 71.09
Maltose M (g/L): 62.32
Volume V (L): 9.99

t = 56.00 h

Biomass X (g/L): 108.18
Glucose S (g/L): 82.60
Lactic Acid P (g/L): 69.57
Maltose M (g/L): 60.50
Volume V (L): 9.82

t = 57.00 h

Biomass X (g/L): 63.50
Glucose S (g/L): 73.63
Lactic Acid P (g/L): 70.38
Maltose M (g/L): 59.62
Volume V (L): 10.27

t = 58.00 h

Biomass X (g/L): 89.76
Glucose S (g/L): 81.46
Lactic Acid P (g/L): 83.02
Maltose M (g/L): 55.29
Volume V (L): 9.98

t = 59.00 h

Biomass X (g/L): 78.84
Glucose S (g/L): 83.59
Lactic Acid P (g/L): 65.06
Maltose M (g/L): 57.34
Volume V (L): 10.69

t = 60.00 h

Biomass X (g/L): 78.42
Glucose S (g/L): 77.30
Lactic Acid P (g/L): 83.58
Maltose M (g/L): 60.43
Volume V (L): 10.43

t = 61.00 h

Biomass X (g/L): 77.96
Glucose S (g/L): 79.16
Lactic Acid P (g/L): 73.93
Maltose M (g/L): 57.79
Volume V (L): 9.99

t = 62.00 h

Biomass X (g/L): 75.13
Glucose S (g/L): 77.21
Lactic Acid P (g/L): 70.80
Maltose M (g/L): 59.12
Volume V (L): 11.67

t = 63.00 h

Biomass X (g/L): 98.92
Glucose S (g/L): 80.72
Lactic Acid P (g/L): 94.61
Maltose M (g/L): 61.76
Volume V (L): 10.76

t = 64.00 h

Biomass X (g/L): 90.98
Glucose S (g/L): 81.94
Lactic Acid P (g/L): 63.16
Maltose M (g/L): 59.48
Volume V (L): 10.41

t = 65.00 h

Biomass X (g/L): 83.25
Glucose S (g/L): 84.96
Lactic Acid P (g/L): 69.55
Maltose M (g/L): 56.63
Volume V (L): 11.14

t = 66.00 h

Biomass X (g/L): 95.34
Glucose S (g/L): 87.07
Lactic Acid P (g/L): 61.89
Maltose M (g/L): 54.69
Volume V (L): 11.37

t = 67.00 h

Biomass X (g/L): 68.72
Glucose S (g/L): 81.39
Lactic Acid P (g/L): 81.45
Maltose M (g/L): 48.90
Volume V (L): 10.42

t = 68.00 h

Biomass X (g/L): 70.23
Glucose S (g/L): 77.82
Lactic Acid P (g/L): 88.14
Maltose M (g/L): 54.97
Volume V (L): 11.71

t = 69.00 h

Biomass X (g/L): 62.23
Glucose S (g/L): 85.11
Lactic Acid P (g/L): 88.11
Maltose M (g/L): 54.24
Volume V (L): 11.02

t = 70.00 h

Biomass X (g/L): 75.71
Glucose S (g/L): 86.80
Lactic Acid P (g/L): 84.28
Maltose M (g/L): 55.84
Volume V (L): 12.70

t = 71.00 h

Biomass X (g/L): 83.49
Glucose S (g/L): 87.24
Lactic Acid P (g/L): 71.78
Maltose M (g/L): 54.08
Volume V (L): 12.11

t = 72.00 h

Biomass X (g/L): 82.77
Glucose S (g/L): 85.86
Lactic Acid P (g/L): 70.43
Maltose M (g/L): 53.32
Volume V (L): 11.68

t = 73.00 h

Biomass X (g/L): 83.02
Glucose S (g/L): 87.99
Lactic Acid P (g/L): 58.95
Maltose M (g/L): 49.11
Volume V (L): 11.57

t = 74.00 h

Biomass X (g/L): 68.43
Glucose S (g/L): 93.35
Lactic Acid P (g/L): 92.01
Maltose M (g/L): 51.06
Volume V (L): 12.04

t = 75.00 h

Biomass X (g/L): 88.70
Glucose S (g/L): 90.03
Lactic Acid P (g/L): 78.21
Maltose M (g/L): 54.55

Volume V (L): 12.55

Batch 16

Initial Conditions:

Batch: 16.00

X0 (g/L): 7.15

S0 (g/L): 40.69

M0 (g/L): 154.84

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 3.81

F_vol (L/h): 0.04

S_feed (g/L): 62.43

M_feed (g/L): 237.57

V_final (L): 7.71

Batch 16 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.90
Glucose S (g/L): 42.24
Lactic Acid P (g/L): 5.90
Maltose M (g/L): 165.17
Volume V (L): 4.95

t = 1.00 h

Biomass X (g/L): 7.31
Glucose S (g/L): 40.24
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 155.99
Volume V (L): 5.22

t = 2.00 h

Biomass X (g/L): 12.91
Glucose S (g/L): 41.72
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 141.69
Volume V (L): 5.00

t = 3.00 h

Biomass X (g/L): 9.78
Glucose S (g/L): 43.54
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 140.91
Volume V (L): 4.76

t = 4.00 h

Biomass X (g/L): 16.32
Glucose S (g/L): 35.81
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 146.57
Volume V (L): 5.39

t = 5.00 h

Biomass X (g/L): 3.42
Glucose S (g/L): 36.11

Lactic Acid P (g/L): 7.49
Maltose M (g/L): 123.27
Volume V (L): 5.00

t = 6.00 h

Biomass X (g/L): 38.47
Glucose S (g/L): 38.08
Lactic Acid P (g/L): 13.04
Maltose M (g/L): 156.04
Volume V (L): 5.20

t = 7.00 h

Biomass X (g/L): 36.57
Glucose S (g/L): 41.57
Lactic Acid P (g/L): 13.42
Maltose M (g/L): 143.21
Volume V (L): 5.35

t = 8.00 h

Biomass X (g/L): 30.70
Glucose S (g/L): 31.64
Lactic Acid P (g/L): 35.06
Maltose M (g/L): 148.07
Volume V (L): 5.33

t = 9.00 h

Biomass X (g/L): 61.85
Glucose S (g/L): 31.99
Lactic Acid P (g/L): 38.39
Maltose M (g/L): 121.56
Volume V (L): 5.56

t = 10.00 h

Biomass X (g/L): 48.23
Glucose S (g/L): 27.59
Lactic Acid P (g/L): 26.65
Maltose M (g/L): 144.35
Volume V (L): 5.47

t = 11.00 h

Biomass X (g/L): 41.16

Glucose S (g/L): 32.85
Lactic Acid P (g/L): 15.71
Maltose M (g/L): 151.12
Volume V (L): 5.21

t = 12.00 h

Biomass X (g/L): 63.69
Glucose S (g/L): 18.61
Lactic Acid P (g/L): 30.46
Maltose M (g/L): 118.22
Volume V (L): 5.91

t = 13.00 h

Biomass X (g/L): 62.56
Glucose S (g/L): 23.65
Lactic Acid P (g/L): 32.84
Maltose M (g/L): 134.74
Volume V (L): 5.51

t = 14.00 h

Biomass X (g/L): 65.63
Glucose S (g/L): 16.67
Lactic Acid P (g/L): 34.77
Maltose M (g/L): 126.26
Volume V (L): 5.47

t = 15.00 h

Biomass X (g/L): 60.30
Glucose S (g/L): 12.79
Lactic Acid P (g/L): 36.68
Maltose M (g/L): 111.04
Volume V (L): 5.75

t = 16.00 h

Biomass X (g/L): 71.54
Glucose S (g/L): 21.81
Lactic Acid P (g/L): 39.22
Maltose M (g/L): 127.20
Volume V (L): 5.63

t = 17.00 h

Biomass X (g/L): 66.76
Glucose S (g/L): 20.80
Lactic Acid P (g/L): 43.74
Maltose M (g/L): 128.55
Volume V (L): 5.70

t = 18.00 h

Biomass X (g/L): 76.95
Glucose S (g/L): 19.04
Lactic Acid P (g/L): 43.14
Maltose M (g/L): 114.42
Volume V (L): 6.19

t = 19.00 h

Biomass X (g/L): 54.40
Glucose S (g/L): 19.50
Lactic Acid P (g/L): 52.95
Maltose M (g/L): 128.32
Volume V (L): 5.63

t = 20.00 h

Biomass X (g/L): 70.71
Glucose S (g/L): 30.49
Lactic Acid P (g/L): 55.97
Maltose M (g/L): 102.03
Volume V (L): 5.65

t = 21.00 h

Biomass X (g/L): 93.65
Glucose S (g/L): 19.00
Lactic Acid P (g/L): 45.16
Maltose M (g/L): 102.05
Volume V (L): 5.79

t = 22.00 h

Biomass X (g/L): 70.34
Glucose S (g/L): 21.81
Lactic Acid P (g/L): 67.54
Maltose M (g/L): 114.87
Volume V (L): 6.03

t = 23.00 h

Biomass X (g/L): 73.23
Glucose S (g/L): 18.05
Lactic Acid P (g/L): 47.10
Maltose M (g/L): 112.47
Volume V (L): 5.51

t = 24.00 h

Biomass X (g/L): 75.93
Glucose S (g/L): 21.62
Lactic Acid P (g/L): 42.14
Maltose M (g/L): 104.60
Volume V (L): 5.44

t = 25.00 h

Biomass X (g/L): 69.37
Glucose S (g/L): 20.15
Lactic Acid P (g/L): 62.51
Maltose M (g/L): 116.59
Volume V (L): 5.78

t = 26.00 h

Biomass X (g/L): 60.46
Glucose S (g/L): 19.98
Lactic Acid P (g/L): 52.01
Maltose M (g/L): 112.82
Volume V (L): 5.59

t = 27.00 h

Biomass X (g/L): 80.96
Glucose S (g/L): 28.12
Lactic Acid P (g/L): 63.98
Maltose M (g/L): 114.39
Volume V (L): 5.88

t = 28.00 h

Biomass X (g/L): 85.38
Glucose S (g/L): 26.43
Lactic Acid P (g/L): 38.64
Maltose M (g/L): 104.53
Volume V (L): 6.12

t = 29.00 h

Biomass X (g/L): 84.23
Glucose S (g/L): 23.39
Lactic Acid P (g/L): 76.12
Maltose M (g/L): 109.64
Volume V (L): 5.86

t = 30.00 h

Biomass X (g/L): 83.96
Glucose S (g/L): 15.57
Lactic Acid P (g/L): 70.01
Maltose M (g/L): 99.25
Volume V (L): 5.78

t = 31.00 h

Biomass X (g/L): 86.41
Glucose S (g/L): 22.86
Lactic Acid P (g/L): 49.86
Maltose M (g/L): 119.92
Volume V (L): 6.14

t = 32.00 h

Biomass X (g/L): 96.66
Glucose S (g/L): 34.04
Lactic Acid P (g/L): 50.83
Maltose M (g/L): 93.68
Volume V (L): 6.35

t = 33.00 h

Biomass X (g/L): 85.10
Glucose S (g/L): 24.59
Lactic Acid P (g/L): 81.92
Maltose M (g/L): 118.06
Volume V (L): 5.83

t = 34.00 h

Biomass X (g/L): 81.72
Glucose S (g/L): 29.85
Lactic Acid P (g/L): 59.37
Maltose M (g/L): 98.83

Volume V (L): 6.41

t = 35.00 h

Biomass X (g/L): 85.01

Glucose S (g/L): 31.69

Lactic Acid P (g/L): 59.09

Maltose M (g/L): 85.40

Volume V (L): 6.44

t = 36.00 h

Biomass X (g/L): 93.89

Glucose S (g/L): 28.18

Lactic Acid P (g/L): 45.91

Maltose M (g/L): 108.33

Volume V (L): 6.51

t = 37.00 h

Biomass X (g/L): 102.46

Glucose S (g/L): 25.23

Lactic Acid P (g/L): 64.97

Maltose M (g/L): 96.41

Volume V (L): 6.60

t = 38.00 h

Biomass X (g/L): 82.47

Glucose S (g/L): 20.46

Lactic Acid P (g/L): 66.73

Maltose M (g/L): 101.73

Volume V (L): 6.85

t = 39.00 h

Biomass X (g/L): 89.92

Glucose S (g/L): 31.92

Lactic Acid P (g/L): 76.36

Maltose M (g/L): 103.98

Volume V (L): 6.33

t = 40.00 h

Biomass X (g/L): 92.50

Glucose S (g/L): 27.48

Lactic Acid P (g/L): 56.50

Maltose M (g/L): 95.24

Volume V (L): 6.47

t = 41.00 h

Biomass X (g/L): 79.46

Glucose S (g/L): 31.63

Lactic Acid P (g/L): 55.83

Maltose M (g/L): 86.14

Volume V (L): 6.39

t = 42.00 h

Biomass X (g/L): 89.67

Glucose S (g/L): 27.45

Lactic Acid P (g/L): 75.31

Maltose M (g/L): 98.90

Volume V (L): 6.52

t = 43.00 h

Biomass X (g/L): 75.04

Glucose S (g/L): 31.58

Lactic Acid P (g/L): 81.13

Maltose M (g/L): 89.45

Volume V (L): 6.89

t = 44.00 h

Biomass X (g/L): 72.12

Glucose S (g/L): 33.34

Lactic Acid P (g/L): 56.72

Maltose M (g/L): 94.53

Volume V (L): 6.30

t = 45.00 h

Biomass X (g/L): 93.40

Glucose S (g/L): 34.89

Lactic Acid P (g/L): 82.72

Maltose M (g/L): 68.80

Volume V (L): 6.44

t = 46.00 h

Biomass X (g/L): 86.00

Glucose S (g/L): 30.87

Lactic Acid P (g/L): 73.07

Maltose M (g/L): 90.67

Volume V (L): 6.96

t = 47.00 h

Biomass X (g/L): 91.86

Glucose S (g/L): 39.56

Lactic Acid P (g/L): 83.63

Maltose M (g/L): 80.32

Volume V (L): 6.92

t = 48.00 h

Biomass X (g/L): 89.69

Glucose S (g/L): 34.12

Lactic Acid P (g/L): 72.81

Maltose M (g/L): 99.52

Volume V (L): 6.66

t = 49.00 h

Biomass X (g/L): 80.18

Glucose S (g/L): 31.15

Lactic Acid P (g/L): 49.54

Maltose M (g/L): 89.69

Volume V (L): 6.96

t = 50.00 h

Biomass X (g/L): 89.41

Glucose S (g/L): 37.26

Lactic Acid P (g/L): 72.74

Maltose M (g/L): 85.53

Volume V (L): 6.88

t = 51.00 h

Biomass X (g/L): 77.59

Glucose S (g/L): 38.48

Lactic Acid P (g/L): 84.77

Maltose M (g/L): 86.89

Volume V (L): 7.17

t = 52.00 h

Biomass X (g/L): 103.60

Glucose S (g/L): 39.64
Lactic Acid P (g/L): 66.27
Maltose M (g/L): 73.12
Volume V (L): 6.90

t = 53.00 h

Biomass X (g/L): 76.27
Glucose S (g/L): 44.50
Lactic Acid P (g/L): 71.46
Maltose M (g/L): 95.29
Volume V (L): 7.14

t = 54.00 h

Biomass X (g/L): 87.72
Glucose S (g/L): 39.71
Lactic Acid P (g/L): 73.82
Maltose M (g/L): 97.76
Volume V (L): 7.53

t = 55.00 h

Biomass X (g/L): 79.24
Glucose S (g/L): 40.82
Lactic Acid P (g/L): 47.86
Maltose M (g/L): 83.04
Volume V (L): 7.12

t = 56.00 h

Biomass X (g/L): 95.16
Glucose S (g/L): 38.14
Lactic Acid P (g/L): 92.22
Maltose M (g/L): 92.79
Volume V (L): 7.04

t = 57.00 h

Biomass X (g/L): 86.11
Glucose S (g/L): 45.46
Lactic Acid P (g/L): 67.70
Maltose M (g/L): 73.20
Volume V (L): 7.22

t = 58.00 h

Biomass X (g/L): 84.15
Glucose S (g/L): 43.97
Lactic Acid P (g/L): 81.54
Maltose M (g/L): 70.51
Volume V (L): 7.02

t = 59.00 h

Biomass X (g/L): 97.65
Glucose S (g/L): 42.58
Lactic Acid P (g/L): 76.09
Maltose M (g/L): 77.16
Volume V (L): 7.65

t = 60.00 h

Biomass X (g/L): 89.35
Glucose S (g/L): 38.15
Lactic Acid P (g/L): 89.31
Maltose M (g/L): 79.93
Volume V (L): 7.21

t = 61.00 h

Biomass X (g/L): 94.05
Glucose S (g/L): 50.36
Lactic Acid P (g/L): 81.35
Maltose M (g/L): 72.02
Volume V (L): 7.15

t = 62.00 h

Biomass X (g/L): 79.04
Glucose S (g/L): 41.83
Lactic Acid P (g/L): 75.42
Maltose M (g/L): 79.53
Volume V (L): 7.06

t = 63.00 h

Biomass X (g/L): 74.69
Glucose S (g/L): 44.69
Lactic Acid P (g/L): 69.31
Maltose M (g/L): 76.87
Volume V (L): 7.52

t = 64.00 h

Biomass X (g/L): 89.55
Glucose S (g/L): 48.62
Lactic Acid P (g/L): 85.84
Maltose M (g/L): 64.83
Volume V (L): 7.50

t = 65.00 h

Biomass X (g/L): 85.16
Glucose S (g/L): 43.92
Lactic Acid P (g/L): 60.31
Maltose M (g/L): 77.49
Volume V (L): 7.24

t = 66.00 h

Biomass X (g/L): 94.38
Glucose S (g/L): 43.55
Lactic Acid P (g/L): 84.82
Maltose M (g/L): 79.90
Volume V (L): 7.51

t = 67.00 h

Biomass X (g/L): 87.20
Glucose S (g/L): 42.02
Lactic Acid P (g/L): 68.21
Maltose M (g/L): 73.26
Volume V (L): 7.10

t = 68.00 h

Biomass X (g/L): 89.86
Glucose S (g/L): 52.45
Lactic Acid P (g/L): 91.09
Maltose M (g/L): 81.26
Volume V (L): 7.50

t = 69.00 h

Biomass X (g/L): 71.53
Glucose S (g/L): 48.72
Lactic Acid P (g/L): 80.77
Maltose M (g/L): 59.96
Volume V (L): 7.60

t = 70.00 h

Biomass X (g/L): 106.21
Glucose S (g/L): 61.29
Lactic Acid P (g/L): 85.62
Maltose M (g/L): 74.14
Volume V (L): 7.90

t = 71.00 h

Biomass X (g/L): 85.65
Glucose S (g/L): 48.41
Lactic Acid P (g/L): 80.40
Maltose M (g/L): 59.26
Volume V (L): 7.40

t = 72.00 h

Biomass X (g/L): 94.01
Glucose S (g/L): 49.27
Lactic Acid P (g/L): 62.90
Maltose M (g/L): 59.21
Volume V (L): 7.33

t = 73.00 h

Biomass X (g/L): 78.59
Glucose S (g/L): 51.10
Lactic Acid P (g/L): 71.68
Maltose M (g/L): 62.90
Volume V (L): 7.45

t = 74.00 h

Biomass X (g/L): 89.75
Glucose S (g/L): 54.49
Lactic Acid P (g/L): 86.64
Maltose M (g/L): 59.66
Volume V (L): 7.39

t = 75.00 h

Biomass X (g/L): 73.56
Glucose S (g/L): 52.62
Lactic Acid P (g/L): 81.94
Maltose M (g/L): 62.59

Volume V (L): 8.00

Batch 17

Initial Conditions:

Batch: 17.00

X0 (g/L): 9.04

S0 (g/L): 143.12

M0 (g/L): 35.33

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.25

F_vol (L/h): 0.07

S_feed (g/L): 240.61

M_feed (g/L): 59.39

V_final (L): 9.90

Batch 17 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 24.03
Glucose S (g/L): 142.36
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 35.22
Volume V (L): 5.17

t = 1.00 h

Biomass X (g/L): 15.32
Glucose S (g/L): 137.36
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 35.00
Volume V (L): 5.56

t = 2.00 h

Biomass X (g/L): 27.94
Glucose S (g/L): 140.13
Lactic Acid P (g/L): 3.22
Maltose M (g/L): 34.31
Volume V (L): 5.12

t = 3.00 h

Biomass X (g/L): 21.98
Glucose S (g/L): 129.32
Lactic Acid P (g/L): 17.63
Maltose M (g/L): 36.15
Volume V (L): 4.64

t = 4.00 h

Biomass X (g/L): 24.86
Glucose S (g/L): 134.15
Lactic Acid P (g/L): 6.59
Maltose M (g/L): 31.96
Volume V (L): 5.59

t = 5.00 h

Biomass X (g/L): 27.99
Glucose S (g/L): 133.86

Lactic Acid P (g/L): 16.95

Maltose M (g/L): 34.52

Volume V (L): 5.62

t = 6.00 h

Biomass X (g/L): 33.74

Glucose S (g/L): 119.66

Lactic Acid P (g/L): 13.59

Maltose M (g/L): 34.96

Volume V (L): 4.63

t = 7.00 h

Biomass X (g/L): 27.19

Glucose S (g/L): 114.35

Lactic Acid P (g/L): 28.01

Maltose M (g/L): 34.03

Volume V (L): 5.46

t = 8.00 h

Biomass X (g/L): 49.91

Glucose S (g/L): 125.59

Lactic Acid P (g/L): 26.59

Maltose M (g/L): 36.23

Volume V (L): 4.90

t = 9.00 h

Biomass X (g/L): 51.78

Glucose S (g/L): 111.55

Lactic Acid P (g/L): 23.59

Maltose M (g/L): 31.44

Volume V (L): 5.33

t = 10.00 h

Biomass X (g/L): 63.79

Glucose S (g/L): 116.83

Lactic Acid P (g/L): 48.04

Maltose M (g/L): 33.47

Volume V (L): 5.30

t = 11.00 h

Biomass X (g/L): 61.46

Glucose S (g/L): 107.83
Lactic Acid P (g/L): 47.44
Maltose M (g/L): 31.50
Volume V (L): 6.04

t = 12.00 h

Biomass X (g/L): 62.55
Glucose S (g/L): 105.35
Lactic Acid P (g/L): 36.08
Maltose M (g/L): 33.31
Volume V (L): 5.83

t = 13.00 h

Biomass X (g/L): 58.87
Glucose S (g/L): 106.15
Lactic Acid P (g/L): 48.89
Maltose M (g/L): 31.07
Volume V (L): 5.35

t = 14.00 h

Biomass X (g/L): 70.82
Glucose S (g/L): 97.97
Lactic Acid P (g/L): 51.60
Maltose M (g/L): 30.19
Volume V (L): 6.26

t = 15.00 h

Biomass X (g/L): 79.77
Glucose S (g/L): 101.34
Lactic Acid P (g/L): 54.70
Maltose M (g/L): 32.56
Volume V (L): 5.42

t = 16.00 h

Biomass X (g/L): 63.77
Glucose S (g/L): 97.58
Lactic Acid P (g/L): 34.99
Maltose M (g/L): 29.47
Volume V (L): 6.18

t = 17.00 h

Biomass X (g/L): 77.63
Glucose S (g/L): 91.98
Lactic Acid P (g/L): 61.76
Maltose M (g/L): 27.75
Volume V (L): 5.79

t = 18.00 h

Biomass X (g/L): 80.76
Glucose S (g/L): 99.09
Lactic Acid P (g/L): 48.75
Maltose M (g/L): 28.89
Volume V (L): 6.03

t = 19.00 h

Biomass X (g/L): 73.08
Glucose S (g/L): 95.48
Lactic Acid P (g/L): 52.17
Maltose M (g/L): 30.42
Volume V (L): 6.44

t = 20.00 h

Biomass X (g/L): 75.68
Glucose S (g/L): 95.42
Lactic Acid P (g/L): 81.25
Maltose M (g/L): 31.37
Volume V (L): 6.51

t = 21.00 h

Biomass X (g/L): 84.79
Glucose S (g/L): 89.90
Lactic Acid P (g/L): 60.84
Maltose M (g/L): 29.11
Volume V (L): 6.90

t = 22.00 h

Biomass X (g/L): 72.63
Glucose S (g/L): 103.07
Lactic Acid P (g/L): 55.59
Maltose M (g/L): 30.76
Volume V (L): 6.41

t = 23.00 h

Biomass X (g/L): 87.91
Glucose S (g/L): 99.38
Lactic Acid P (g/L): 39.79
Maltose M (g/L): 26.31
Volume V (L): 6.62

t = 24.00 h

Biomass X (g/L): 76.66
Glucose S (g/L): 99.65
Lactic Acid P (g/L): 74.04
Maltose M (g/L): 27.51
Volume V (L): 6.40

t = 25.00 h

Biomass X (g/L): 96.33
Glucose S (g/L): 93.45
Lactic Acid P (g/L): 49.60
Maltose M (g/L): 27.52
Volume V (L): 6.71

t = 26.00 h

Biomass X (g/L): 109.29
Glucose S (g/L): 86.57
Lactic Acid P (g/L): 53.60
Maltose M (g/L): 27.56
Volume V (L): 6.87

t = 27.00 h

Biomass X (g/L): 78.87
Glucose S (g/L): 104.37
Lactic Acid P (g/L): 60.59
Maltose M (g/L): 27.64
Volume V (L): 6.73

t = 28.00 h

Biomass X (g/L): 92.84
Glucose S (g/L): 94.75
Lactic Acid P (g/L): 77.50
Maltose M (g/L): 28.37
Volume V (L): 6.12

t = 29.00 h

Biomass X (g/L): 81.04
Glucose S (g/L): 91.58
Lactic Acid P (g/L): 72.21
Maltose M (g/L): 26.37
Volume V (L): 7.03

t = 30.00 h

Biomass X (g/L): 98.50
Glucose S (g/L): 97.37
Lactic Acid P (g/L): 64.22
Maltose M (g/L): 27.06
Volume V (L): 7.10

t = 31.00 h

Biomass X (g/L): 76.47
Glucose S (g/L): 101.20
Lactic Acid P (g/L): 69.13
Maltose M (g/L): 26.56
Volume V (L): 6.76

t = 32.00 h

Biomass X (g/L): 93.62
Glucose S (g/L): 95.71
Lactic Acid P (g/L): 61.27
Maltose M (g/L): 24.97
Volume V (L): 6.59

t = 33.00 h

Biomass X (g/L): 80.72
Glucose S (g/L): 102.25
Lactic Acid P (g/L): 68.65
Maltose M (g/L): 26.03
Volume V (L): 7.74

t = 34.00 h

Biomass X (g/L): 87.34
Glucose S (g/L): 101.60
Lactic Acid P (g/L): 68.53
Maltose M (g/L): 25.78

Volume V (L): 7.41

t = 35.00 h

Biomass X (g/L): 79.49

Glucose S (g/L): 91.50

Lactic Acid P (g/L): 77.29

Maltose M (g/L): 25.55

Volume V (L): 7.20

t = 36.00 h

Biomass X (g/L): 97.53

Glucose S (g/L): 93.77

Lactic Acid P (g/L): 77.72

Maltose M (g/L): 25.66

Volume V (L): 6.91

t = 37.00 h

Biomass X (g/L): 81.54

Glucose S (g/L): 101.98

Lactic Acid P (g/L): 73.75

Maltose M (g/L): 25.53

Volume V (L): 7.10

t = 38.00 h

Biomass X (g/L): 95.03

Glucose S (g/L): 84.65

Lactic Acid P (g/L): 66.22

Maltose M (g/L): 24.54

Volume V (L): 7.99

t = 39.00 h

Biomass X (g/L): 86.38

Glucose S (g/L): 97.37

Lactic Acid P (g/L): 76.83

Maltose M (g/L): 27.03

Volume V (L): 7.60

t = 40.00 h

Biomass X (g/L): 88.60

Glucose S (g/L): 91.58

Lactic Acid P (g/L): 69.47

Maltose M (g/L): 23.01

Volume V (L): 7.40

t = 41.00 h

Biomass X (g/L): 91.12

Glucose S (g/L): 96.57

Lactic Acid P (g/L): 60.64

Maltose M (g/L): 23.53

Volume V (L): 7.88

t = 42.00 h

Biomass X (g/L): 82.28

Glucose S (g/L): 97.82

Lactic Acid P (g/L): 67.26

Maltose M (g/L): 28.84

Volume V (L): 7.43

t = 43.00 h

Biomass X (g/L): 91.63

Glucose S (g/L): 97.94

Lactic Acid P (g/L): 72.09

Maltose M (g/L): 22.67

Volume V (L): 8.40

t = 44.00 h

Biomass X (g/L): 104.34

Glucose S (g/L): 97.41

Lactic Acid P (g/L): 80.04

Maltose M (g/L): 21.99

Volume V (L): 8.41

t = 45.00 h

Biomass X (g/L): 81.91

Glucose S (g/L): 95.20

Lactic Acid P (g/L): 70.19

Maltose M (g/L): 25.10

Volume V (L): 7.76

t = 46.00 h

Biomass X (g/L): 100.68

Glucose S (g/L): 111.95

Lactic Acid P (g/L): 60.97

Maltose M (g/L): 20.51

Volume V (L): 7.76

t = 47.00 h

Biomass X (g/L): 103.73

Glucose S (g/L): 110.33

Lactic Acid P (g/L): 72.96

Maltose M (g/L): 23.23

Volume V (L): 7.05

t = 48.00 h

Biomass X (g/L): 81.21

Glucose S (g/L): 96.14

Lactic Acid P (g/L): 73.74

Maltose M (g/L): 26.80

Volume V (L): 7.56

t = 49.00 h

Biomass X (g/L): 74.89

Glucose S (g/L): 109.80

Lactic Acid P (g/L): 74.66

Maltose M (g/L): 20.84

Volume V (L): 7.68

t = 50.00 h

Biomass X (g/L): 91.76

Glucose S (g/L): 100.67

Lactic Acid P (g/L): 81.69

Maltose M (g/L): 19.99

Volume V (L): 8.32

t = 51.00 h

Biomass X (g/L): 83.39

Glucose S (g/L): 103.64

Lactic Acid P (g/L): 82.83

Maltose M (g/L): 20.59

Volume V (L): 8.21

t = 52.00 h

Biomass X (g/L): 106.62

Glucose S (g/L): 103.14
Lactic Acid P (g/L): 68.11
Maltose M (g/L): 24.20
Volume V (L): 7.87

t = 53.00 h

Biomass X (g/L): 88.70
Glucose S (g/L): 100.68
Lactic Acid P (g/L): 68.12
Maltose M (g/L): 24.39
Volume V (L): 8.35

t = 54.00 h

Biomass X (g/L): 80.43
Glucose S (g/L): 109.03
Lactic Acid P (g/L): 64.40
Maltose M (g/L): 21.67
Volume V (L): 8.73

t = 55.00 h

Biomass X (g/L): 76.92
Glucose S (g/L): 105.32
Lactic Acid P (g/L): 80.70
Maltose M (g/L): 23.39
Volume V (L): 8.80

t = 56.00 h

Biomass X (g/L): 98.39
Glucose S (g/L): 102.83
Lactic Acid P (g/L): 87.84
Maltose M (g/L): 24.35
Volume V (L): 7.92

t = 57.00 h

Biomass X (g/L): 82.12
Glucose S (g/L): 99.26
Lactic Acid P (g/L): 71.69
Maltose M (g/L): 24.83
Volume V (L): 8.68

t = 58.00 h

Biomass X (g/L): 87.89
Glucose S (g/L): 101.66
Lactic Acid P (g/L): 72.55
Maltose M (g/L): 21.49
Volume V (L): 8.25

t = 59.00 h

Biomass X (g/L): 96.52
Glucose S (g/L): 98.47
Lactic Acid P (g/L): 88.54
Maltose M (g/L): 18.12
Volume V (L): 9.21

t = 60.00 h

Biomass X (g/L): 82.38
Glucose S (g/L): 106.86
Lactic Acid P (g/L): 79.81
Maltose M (g/L): 20.30
Volume V (L): 9.23

t = 61.00 h

Biomass X (g/L): 86.42
Glucose S (g/L): 101.44
Lactic Acid P (g/L): 79.33
Maltose M (g/L): 20.77
Volume V (L): 8.63

t = 62.00 h

Biomass X (g/L): 86.50
Glucose S (g/L): 106.34
Lactic Acid P (g/L): 64.42
Maltose M (g/L): 20.90
Volume V (L): 8.75

t = 63.00 h

Biomass X (g/L): 93.06
Glucose S (g/L): 114.63
Lactic Acid P (g/L): 81.89
Maltose M (g/L): 20.87
Volume V (L): 8.94

t = 64.00 h

Biomass X (g/L): 91.96
Glucose S (g/L): 114.21
Lactic Acid P (g/L): 67.15
Maltose M (g/L): 22.79
Volume V (L): 8.30

t = 65.00 h

Biomass X (g/L): 83.89
Glucose S (g/L): 108.67
Lactic Acid P (g/L): 65.40
Maltose M (g/L): 22.90
Volume V (L): 9.38

t = 66.00 h

Biomass X (g/L): 86.43
Glucose S (g/L): 108.81
Lactic Acid P (g/L): 72.83
Maltose M (g/L): 20.49
Volume V (L): 9.21

t = 67.00 h

Biomass X (g/L): 81.99
Glucose S (g/L): 118.69
Lactic Acid P (g/L): 70.76
Maltose M (g/L): 22.35
Volume V (L): 9.76

t = 68.00 h

Biomass X (g/L): 71.07
Glucose S (g/L): 112.53
Lactic Acid P (g/L): 74.27
Maltose M (g/L): 21.06
Volume V (L): 9.01

t = 69.00 h

Biomass X (g/L): 87.65
Glucose S (g/L): 108.82
Lactic Acid P (g/L): 63.11
Maltose M (g/L): 20.47
Volume V (L): 9.27

t = 70.00 h

Biomass X (g/L): 103.25
Glucose S (g/L): 103.91
Lactic Acid P (g/L): 85.46
Maltose M (g/L): 21.67
Volume V (L): 8.98

t = 71.00 h

Biomass X (g/L): 76.15
Glucose S (g/L): 113.07
Lactic Acid P (g/L): 87.15
Maltose M (g/L): 20.23
Volume V (L): 9.25

t = 72.00 h

Biomass X (g/L): 77.00
Glucose S (g/L): 115.55
Lactic Acid P (g/L): 58.79
Maltose M (g/L): 23.23
Volume V (L): 9.24

t = 73.00 h

Biomass X (g/L): 75.83
Glucose S (g/L): 108.88
Lactic Acid P (g/L): 86.01
Maltose M (g/L): 17.62
Volume V (L): 10.16

t = 74.00 h

Biomass X (g/L): 91.05
Glucose S (g/L): 113.70
Lactic Acid P (g/L): 69.56
Maltose M (g/L): 18.94
Volume V (L): 9.56

t = 75.00 h

Biomass X (g/L): 88.15
Glucose S (g/L): 110.91
Lactic Acid P (g/L): 85.12
Maltose M (g/L): 19.43

Volume V (L): 9.95

Batch 18

Initial Conditions:

Batch: 18.00

X0 (g/L): 6.76

S0 (g/L): 64.31

M0 (g/L): 79.26

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.23

F_vol (L/h): 0.08

S_feed (g/L): 134.38

M_feed (g/L): 165.62

V_final (L): 11.22

Batch 18 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.18
Glucose S (g/L): 61.04
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 75.27
Volume V (L): 3.94

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 67.04
Lactic Acid P (g/L): 21.34
Maltose M (g/L): 77.56
Volume V (L): 5.29

t = 2.00 h

Biomass X (g/L): 10.19
Glucose S (g/L): 62.61
Lactic Acid P (g/L): 10.64
Maltose M (g/L): 80.08
Volume V (L): 6.23

t = 3.00 h

Biomass X (g/L): 20.73
Glucose S (g/L): 59.98
Lactic Acid P (g/L): 6.25
Maltose M (g/L): 77.25
Volume V (L): 5.39

t = 4.00 h

Biomass X (g/L): 7.63
Glucose S (g/L): 62.99
Lactic Acid P (g/L): 18.88
Maltose M (g/L): 78.14
Volume V (L): 6.15

t = 5.00 h

Biomass X (g/L): 30.86
Glucose S (g/L): 50.87

Lactic Acid P (g/L): 33.49

Maltose M (g/L): 79.55

Volume V (L): 5.10

t = 6.00 h

Biomass X (g/L): 19.88

Glucose S (g/L): 51.70

Lactic Acid P (g/L): 2.88

Maltose M (g/L): 79.48

Volume V (L): 5.48

t = 7.00 h

Biomass X (g/L): 28.06

Glucose S (g/L): 54.56

Lactic Acid P (g/L): 17.27

Maltose M (g/L): 75.63

Volume V (L): 5.60

t = 8.00 h

Biomass X (g/L): 42.29

Glucose S (g/L): 52.84

Lactic Acid P (g/L): 1.56

Maltose M (g/L): 80.13

Volume V (L): 4.56

t = 9.00 h

Biomass X (g/L): 35.66

Glucose S (g/L): 48.74

Lactic Acid P (g/L): 28.02

Maltose M (g/L): 80.19

Volume V (L): 5.16

t = 10.00 h

Biomass X (g/L): 46.57

Glucose S (g/L): 42.20

Lactic Acid P (g/L): 32.88

Maltose M (g/L): 74.82

Volume V (L): 5.28

t = 11.00 h

Biomass X (g/L): 47.31

Glucose S (g/L): 42.66
Lactic Acid P (g/L): 32.38
Maltose M (g/L): 75.26
Volume V (L): 6.32

t = 12.00 h

Biomass X (g/L): 55.18
Glucose S (g/L): 44.91
Lactic Acid P (g/L): 29.77
Maltose M (g/L): 79.57
Volume V (L): 6.43

t = 13.00 h

Biomass X (g/L): 35.01
Glucose S (g/L): 43.93
Lactic Acid P (g/L): 19.97
Maltose M (g/L): 76.27
Volume V (L): 6.24

t = 14.00 h

Biomass X (g/L): 43.23
Glucose S (g/L): 40.50
Lactic Acid P (g/L): 44.10
Maltose M (g/L): 71.28
Volume V (L): 6.57

t = 15.00 h

Biomass X (g/L): 68.93
Glucose S (g/L): 43.40
Lactic Acid P (g/L): 56.03
Maltose M (g/L): 72.86
Volume V (L): 6.16

t = 16.00 h

Biomass X (g/L): 75.35
Glucose S (g/L): 39.88
Lactic Acid P (g/L): 39.79
Maltose M (g/L): 75.18
Volume V (L): 6.92

t = 17.00 h

Biomass X (g/L): 59.28
Glucose S (g/L): 35.25
Lactic Acid P (g/L): 50.90
Maltose M (g/L): 72.81
Volume V (L): 6.41

t = 18.00 h

Biomass X (g/L): 67.51
Glucose S (g/L): 36.33
Lactic Acid P (g/L): 34.66
Maltose M (g/L): 74.21
Volume V (L): 6.52

t = 19.00 h

Biomass X (g/L): 74.34
Glucose S (g/L): 32.12
Lactic Acid P (g/L): 54.36
Maltose M (g/L): 74.91
Volume V (L): 6.44

t = 20.00 h

Biomass X (g/L): 79.66
Glucose S (g/L): 36.36
Lactic Acid P (g/L): 48.21
Maltose M (g/L): 77.85
Volume V (L): 7.00

t = 21.00 h

Biomass X (g/L): 55.54
Glucose S (g/L): 40.47
Lactic Acid P (g/L): 46.24
Maltose M (g/L): 73.55
Volume V (L): 6.53

t = 22.00 h

Biomass X (g/L): 67.51
Glucose S (g/L): 39.03
Lactic Acid P (g/L): 47.50
Maltose M (g/L): 76.33
Volume V (L): 7.31

t = 23.00 h

Biomass X (g/L): 73.84
Glucose S (g/L): 41.61
Lactic Acid P (g/L): 51.92
Maltose M (g/L): 76.81
Volume V (L): 6.34

t = 24.00 h

Biomass X (g/L): 84.56
Glucose S (g/L): 35.96
Lactic Acid P (g/L): 59.93
Maltose M (g/L): 76.50
Volume V (L): 6.44

t = 25.00 h

Biomass X (g/L): 102.31
Glucose S (g/L): 41.91
Lactic Acid P (g/L): 45.75
Maltose M (g/L): 73.56
Volume V (L): 7.41

t = 26.00 h

Biomass X (g/L): 77.53
Glucose S (g/L): 36.97
Lactic Acid P (g/L): 54.85
Maltose M (g/L): 68.12
Volume V (L): 7.11

t = 27.00 h

Biomass X (g/L): 67.19
Glucose S (g/L): 43.10
Lactic Acid P (g/L): 58.89
Maltose M (g/L): 73.30
Volume V (L): 7.01

t = 28.00 h

Biomass X (g/L): 84.81
Glucose S (g/L): 38.43
Lactic Acid P (g/L): 50.23
Maltose M (g/L): 74.29
Volume V (L): 7.76

t = 29.00 h

Biomass X (g/L): 94.00
Glucose S (g/L): 41.43
Lactic Acid P (g/L): 76.88
Maltose M (g/L): 71.39
Volume V (L): 7.10

t = 30.00 h

Biomass X (g/L): 83.12
Glucose S (g/L): 44.61
Lactic Acid P (g/L): 53.88
Maltose M (g/L): 75.20
Volume V (L): 7.39

t = 31.00 h

Biomass X (g/L): 74.05
Glucose S (g/L): 39.45
Lactic Acid P (g/L): 56.19
Maltose M (g/L): 72.85
Volume V (L): 8.26

t = 32.00 h

Biomass X (g/L): 91.94
Glucose S (g/L): 41.47
Lactic Acid P (g/L): 54.05
Maltose M (g/L): 71.85
Volume V (L): 8.42

t = 33.00 h

Biomass X (g/L): 89.53
Glucose S (g/L): 45.77
Lactic Acid P (g/L): 79.48
Maltose M (g/L): 72.68
Volume V (L): 8.52

t = 34.00 h

Biomass X (g/L): 73.98
Glucose S (g/L): 41.54
Lactic Acid P (g/L): 68.89
Maltose M (g/L): 70.63

Volume V (L): 7.68

t = 35.00 h

Biomass X (g/L): 87.45

Glucose S (g/L): 38.12

Lactic Acid P (g/L): 56.09

Maltose M (g/L): 63.87

Volume V (L): 8.51

t = 36.00 h

Biomass X (g/L): 93.46

Glucose S (g/L): 50.71

Lactic Acid P (g/L): 74.86

Maltose M (g/L): 68.05

Volume V (L): 7.51

t = 37.00 h

Biomass X (g/L): 95.39

Glucose S (g/L): 42.43

Lactic Acid P (g/L): 67.23

Maltose M (g/L): 71.56

Volume V (L): 7.62

t = 38.00 h

Biomass X (g/L): 83.84

Glucose S (g/L): 45.30

Lactic Acid P (g/L): 68.19

Maltose M (g/L): 66.67

Volume V (L): 6.98

t = 39.00 h

Biomass X (g/L): 81.24

Glucose S (g/L): 49.44

Lactic Acid P (g/L): 63.24

Maltose M (g/L): 67.65

Volume V (L): 9.06

t = 40.00 h

Biomass X (g/L): 96.23

Glucose S (g/L): 44.11

Lactic Acid P (g/L): 60.02

Maltose M (g/L): 65.44

Volume V (L): 7.89

t = 41.00 h

Biomass X (g/L): 73.45

Glucose S (g/L): 43.47

Lactic Acid P (g/L): 70.14

Maltose M (g/L): 71.52

Volume V (L): 8.33

t = 42.00 h

Biomass X (g/L): 88.98

Glucose S (g/L): 48.55

Lactic Acid P (g/L): 62.61

Maltose M (g/L): 70.95

Volume V (L): 8.03

t = 43.00 h

Biomass X (g/L): 72.06

Glucose S (g/L): 42.26

Lactic Acid P (g/L): 84.10

Maltose M (g/L): 64.78

Volume V (L): 8.91

t = 44.00 h

Biomass X (g/L): 94.94

Glucose S (g/L): 48.17

Lactic Acid P (g/L): 56.83

Maltose M (g/L): 66.78

Volume V (L): 8.46

t = 45.00 h

Biomass X (g/L): 82.54

Glucose S (g/L): 50.75

Lactic Acid P (g/L): 82.25

Maltose M (g/L): 64.56

Volume V (L): 8.08

t = 46.00 h

Biomass X (g/L): 96.86

Glucose S (g/L): 49.73

Lactic Acid P (g/L): 67.83

Maltose M (g/L): 67.17

Volume V (L): 9.03

t = 47.00 h

Biomass X (g/L): 89.55

Glucose S (g/L): 46.11

Lactic Acid P (g/L): 71.80

Maltose M (g/L): 66.59

Volume V (L): 8.18

t = 48.00 h

Biomass X (g/L): 73.43

Glucose S (g/L): 57.64

Lactic Acid P (g/L): 76.87

Maltose M (g/L): 66.94

Volume V (L): 8.56

t = 49.00 h

Biomass X (g/L): 98.23

Glucose S (g/L): 53.27

Lactic Acid P (g/L): 64.05

Maltose M (g/L): 65.51

Volume V (L): 8.01

t = 50.00 h

Biomass X (g/L): 99.36

Glucose S (g/L): 58.58

Lactic Acid P (g/L): 77.69

Maltose M (g/L): 71.09

Volume V (L): 8.44

t = 51.00 h

Biomass X (g/L): 93.02

Glucose S (g/L): 53.51

Lactic Acid P (g/L): 46.00

Maltose M (g/L): 61.76

Volume V (L): 9.32

t = 52.00 h

Biomass X (g/L): 77.52

Glucose S (g/L): 52.13
Lactic Acid P (g/L): 64.03
Maltose M (g/L): 59.87
Volume V (L): 8.33

t = 53.00 h

Biomass X (g/L): 81.71
Glucose S (g/L): 56.81
Lactic Acid P (g/L): 62.44
Maltose M (g/L): 62.53
Volume V (L): 8.66

t = 54.00 h

Biomass X (g/L): 85.85
Glucose S (g/L): 55.89
Lactic Acid P (g/L): 59.73
Maltose M (g/L): 59.54
Volume V (L): 9.47

t = 55.00 h

Biomass X (g/L): 77.79
Glucose S (g/L): 53.31
Lactic Acid P (g/L): 69.25
Maltose M (g/L): 59.38
Volume V (L): 9.29

t = 56.00 h

Biomass X (g/L): 100.16
Glucose S (g/L): 63.14
Lactic Acid P (g/L): 74.84
Maltose M (g/L): 60.17
Volume V (L): 10.63

t = 57.00 h

Biomass X (g/L): 85.52
Glucose S (g/L): 56.68
Lactic Acid P (g/L): 77.88
Maltose M (g/L): 60.02
Volume V (L): 10.64

t = 58.00 h

Biomass X (g/L): 58.68
Glucose S (g/L): 53.21
Lactic Acid P (g/L): 73.96
Maltose M (g/L): 59.62
Volume V (L): 9.38

t = 59.00 h

Biomass X (g/L): 87.11
Glucose S (g/L): 64.82
Lactic Acid P (g/L): 66.05
Maltose M (g/L): 62.65
Volume V (L): 10.17

t = 60.00 h

Biomass X (g/L): 80.53
Glucose S (g/L): 59.23
Lactic Acid P (g/L): 75.50
Maltose M (g/L): 60.00
Volume V (L): 10.34

t = 61.00 h

Biomass X (g/L): 82.95
Glucose S (g/L): 59.58
Lactic Acid P (g/L): 73.49
Maltose M (g/L): 60.42
Volume V (L): 9.76

t = 62.00 h

Biomass X (g/L): 80.06
Glucose S (g/L): 56.31
Lactic Acid P (g/L): 80.94
Maltose M (g/L): 62.54
Volume V (L): 9.39

t = 63.00 h

Biomass X (g/L): 90.95
Glucose S (g/L): 69.06
Lactic Acid P (g/L): 79.69
Maltose M (g/L): 58.46
Volume V (L): 10.21

t = 64.00 h

Biomass X (g/L): 93.73
Glucose S (g/L): 59.33
Lactic Acid P (g/L): 73.19
Maltose M (g/L): 59.28
Volume V (L): 10.22

t = 65.00 h

Biomass X (g/L): 85.73
Glucose S (g/L): 64.65
Lactic Acid P (g/L): 65.05
Maltose M (g/L): 60.40
Volume V (L): 10.17

t = 66.00 h

Biomass X (g/L): 79.73
Glucose S (g/L): 68.27
Lactic Acid P (g/L): 67.40
Maltose M (g/L): 54.85
Volume V (L): 10.69

t = 67.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 67.26
Lactic Acid P (g/L): 68.49
Maltose M (g/L): 53.54
Volume V (L): 10.18

t = 68.00 h

Biomass X (g/L): 82.78
Glucose S (g/L): 66.27
Lactic Acid P (g/L): 70.74
Maltose M (g/L): 59.43
Volume V (L): 10.06

t = 69.00 h

Biomass X (g/L): 74.90
Glucose S (g/L): 61.29
Lactic Acid P (g/L): 76.11
Maltose M (g/L): 59.46
Volume V (L): 10.77

t = 70.00 h

Biomass X (g/L): 87.70
Glucose S (g/L): 65.02
Lactic Acid P (g/L): 80.26
Maltose M (g/L): 62.73
Volume V (L): 10.99

t = 71.00 h

Biomass X (g/L): 86.42
Glucose S (g/L): 69.10
Lactic Acid P (g/L): 73.55
Maltose M (g/L): 58.10
Volume V (L): 10.30

t = 72.00 h

Biomass X (g/L): 96.39
Glucose S (g/L): 66.40
Lactic Acid P (g/L): 74.29
Maltose M (g/L): 59.30
Volume V (L): 11.43

t = 73.00 h

Biomass X (g/L): 93.54
Glucose S (g/L): 70.44
Lactic Acid P (g/L): 75.57
Maltose M (g/L): 56.70
Volume V (L): 10.17

t = 74.00 h

Biomass X (g/L): 75.15
Glucose S (g/L): 72.72
Lactic Acid P (g/L): 69.81
Maltose M (g/L): 57.52
Volume V (L): 10.85

t = 75.00 h

Biomass X (g/L): 89.11
Glucose S (g/L): 74.32
Lactic Acid P (g/L): 66.27
Maltose M (g/L): 60.34

Volume V (L): 11.09

Batch 19

Initial Conditions:

Batch: 19.00

X0 (g/L): 7.22

S0 (g/L): 77.51

M0 (g/L): 107.25

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.38

F_vol (L/h): 0.09

S_feed (g/L): 125.86

M_feed (g/L): 174.14

V_final (L): 12.12

Batch 19 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 1.30
Glucose S (g/L): 73.75
Lactic Acid P (g/L): 8.39
Maltose M (g/L): 114.17
Volume V (L): 5.18

t = 1.00 h

Biomass X (g/L): 4.37
Glucose S (g/L): 86.81
Lactic Acid P (g/L): 11.36
Maltose M (g/L): 97.48
Volume V (L): 5.37

t = 2.00 h

Biomass X (g/L): 30.02
Glucose S (g/L): 83.79
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 110.84
Volume V (L): 5.53

t = 3.00 h

Biomass X (g/L): 8.23
Glucose S (g/L): 76.75
Lactic Acid P (g/L): 4.09
Maltose M (g/L): 100.81
Volume V (L): 3.77

t = 4.00 h

Biomass X (g/L): 42.95
Glucose S (g/L): 76.27
Lactic Acid P (g/L): 16.60
Maltose M (g/L): 98.15
Volume V (L): 6.16

t = 5.00 h

Biomass X (g/L): 28.79
Glucose S (g/L): 71.23

Lactic Acid P (g/L): 8.44
Maltose M (g/L): 108.36
Volume V (L): 5.05

t = 6.00 h

Biomass X (g/L): 30.32
Glucose S (g/L): 72.58
Lactic Acid P (g/L): 16.63
Maltose M (g/L): 105.66
Volume V (L): 5.96

t = 7.00 h

Biomass X (g/L): 38.74
Glucose S (g/L): 66.10
Lactic Acid P (g/L): 23.85
Maltose M (g/L): 107.14
Volume V (L): 6.09

t = 8.00 h

Biomass X (g/L): 62.64
Glucose S (g/L): 59.01
Lactic Acid P (g/L): 30.89
Maltose M (g/L): 104.43
Volume V (L): 5.26

t = 9.00 h

Biomass X (g/L): 34.44
Glucose S (g/L): 59.03
Lactic Acid P (g/L): 11.64
Maltose M (g/L): 109.10
Volume V (L): 6.23

t = 10.00 h

Biomass X (g/L): 37.58
Glucose S (g/L): 61.42
Lactic Acid P (g/L): 37.07
Maltose M (g/L): 101.89
Volume V (L): 5.73

t = 11.00 h

Biomass X (g/L): 44.98

Glucose S (g/L): 55.42
Lactic Acid P (g/L): 31.76
Maltose M (g/L): 98.07
Volume V (L): 4.86

t = 12.00 h

Biomass X (g/L): 55.95
Glucose S (g/L): 57.35
Lactic Acid P (g/L): 42.98
Maltose M (g/L): 99.49
Volume V (L): 6.81

t = 13.00 h

Biomass X (g/L): 42.90
Glucose S (g/L): 52.38
Lactic Acid P (g/L): 46.24
Maltose M (g/L): 102.65
Volume V (L): 5.79

t = 14.00 h

Biomass X (g/L): 62.15
Glucose S (g/L): 52.17
Lactic Acid P (g/L): 56.03
Maltose M (g/L): 93.22
Volume V (L): 5.84

t = 15.00 h

Biomass X (g/L): 80.37
Glucose S (g/L): 56.84
Lactic Acid P (g/L): 55.97
Maltose M (g/L): 93.06
Volume V (L): 6.19

t = 16.00 h

Biomass X (g/L): 68.40
Glucose S (g/L): 53.99
Lactic Acid P (g/L): 57.92
Maltose M (g/L): 93.83
Volume V (L): 6.67

t = 17.00 h

Biomass X (g/L): 66.39
Glucose S (g/L): 52.53
Lactic Acid P (g/L): 60.62
Maltose M (g/L): 85.40
Volume V (L): 6.58

t = 18.00 h

Biomass X (g/L): 65.25
Glucose S (g/L): 55.22
Lactic Acid P (g/L): 52.07
Maltose M (g/L): 94.38
Volume V (L): 7.02

t = 19.00 h

Biomass X (g/L): 81.06
Glucose S (g/L): 53.61
Lactic Acid P (g/L): 51.30
Maltose M (g/L): 84.98
Volume V (L): 6.04

t = 20.00 h

Biomass X (g/L): 66.51
Glucose S (g/L): 56.85
Lactic Acid P (g/L): 60.17
Maltose M (g/L): 92.66
Volume V (L): 6.48

t = 21.00 h

Biomass X (g/L): 92.72
Glucose S (g/L): 50.28
Lactic Acid P (g/L): 60.21
Maltose M (g/L): 95.49
Volume V (L): 6.93

t = 22.00 h

Biomass X (g/L): 94.71
Glucose S (g/L): 50.52
Lactic Acid P (g/L): 52.16
Maltose M (g/L): 89.04
Volume V (L): 8.17

t = 23.00 h

Biomass X (g/L): 80.17
Glucose S (g/L): 47.72
Lactic Acid P (g/L): 59.59
Maltose M (g/L): 97.67
Volume V (L): 7.12

t = 24.00 h

Biomass X (g/L): 78.61
Glucose S (g/L): 51.03
Lactic Acid P (g/L): 56.70
Maltose M (g/L): 83.51
Volume V (L): 7.90

t = 25.00 h

Biomass X (g/L): 89.14
Glucose S (g/L): 56.22
Lactic Acid P (g/L): 50.37
Maltose M (g/L): 94.29
Volume V (L): 7.40

t = 26.00 h

Biomass X (g/L): 90.48
Glucose S (g/L): 57.81
Lactic Acid P (g/L): 70.27
Maltose M (g/L): 91.43
Volume V (L): 7.19

t = 27.00 h

Biomass X (g/L): 73.45
Glucose S (g/L): 57.11
Lactic Acid P (g/L): 60.12
Maltose M (g/L): 80.50
Volume V (L): 7.42

t = 28.00 h

Biomass X (g/L): 76.63
Glucose S (g/L): 48.11
Lactic Acid P (g/L): 78.57
Maltose M (g/L): 98.95
Volume V (L): 7.56

t = 29.00 h

Biomass X (g/L): 69.73
Glucose S (g/L): 58.62
Lactic Acid P (g/L): 46.67
Maltose M (g/L): 86.38
Volume V (L): 8.10

t = 30.00 h

Biomass X (g/L): 86.30
Glucose S (g/L): 49.11
Lactic Acid P (g/L): 56.82
Maltose M (g/L): 82.27
Volume V (L): 8.63

t = 31.00 h

Biomass X (g/L): 89.09
Glucose S (g/L): 59.20
Lactic Acid P (g/L): 62.67
Maltose M (g/L): 79.38
Volume V (L): 8.15

t = 32.00 h

Biomass X (g/L): 70.67
Glucose S (g/L): 55.47
Lactic Acid P (g/L): 73.06
Maltose M (g/L): 83.90
Volume V (L): 8.72

t = 33.00 h

Biomass X (g/L): 79.30
Glucose S (g/L): 59.84
Lactic Acid P (g/L): 67.06
Maltose M (g/L): 83.71
Volume V (L): 7.58

t = 34.00 h

Biomass X (g/L): 71.25
Glucose S (g/L): 53.49
Lactic Acid P (g/L): 66.79
Maltose M (g/L): 84.12

Volume V (L): 7.99

t = 35.00 h

Biomass X (g/L): 75.65

Glucose S (g/L): 62.84

Lactic Acid P (g/L): 54.53

Maltose M (g/L): 81.81

Volume V (L): 7.91

t = 36.00 h

Biomass X (g/L): 89.33

Glucose S (g/L): 54.44

Lactic Acid P (g/L): 54.73

Maltose M (g/L): 85.25

Volume V (L): 8.60

t = 37.00 h

Biomass X (g/L): 77.53

Glucose S (g/L): 57.92

Lactic Acid P (g/L): 68.30

Maltose M (g/L): 80.29

Volume V (L): 8.31

t = 38.00 h

Biomass X (g/L): 81.69

Glucose S (g/L): 57.51

Lactic Acid P (g/L): 53.75

Maltose M (g/L): 73.06

Volume V (L): 9.02

t = 39.00 h

Biomass X (g/L): 69.44

Glucose S (g/L): 55.63

Lactic Acid P (g/L): 61.53

Maltose M (g/L): 80.20

Volume V (L): 8.17

t = 40.00 h

Biomass X (g/L): 86.41

Glucose S (g/L): 62.76

Lactic Acid P (g/L): 71.08

Maltose M (g/L): 70.09

Volume V (L): 8.82

t = 41.00 h

Biomass X (g/L): 84.31

Glucose S (g/L): 69.43

Lactic Acid P (g/L): 58.28

Maltose M (g/L): 81.27

Volume V (L): 9.29

t = 42.00 h

Biomass X (g/L): 77.98

Glucose S (g/L): 73.15

Lactic Acid P (g/L): 81.59

Maltose M (g/L): 74.99

Volume V (L): 7.81

t = 43.00 h

Biomass X (g/L): 83.02

Glucose S (g/L): 65.01

Lactic Acid P (g/L): 78.53

Maltose M (g/L): 79.38

Volume V (L): 10.29

t = 44.00 h

Biomass X (g/L): 86.47

Glucose S (g/L): 69.29

Lactic Acid P (g/L): 54.04

Maltose M (g/L): 83.81

Volume V (L): 9.02

t = 45.00 h

Biomass X (g/L): 77.19

Glucose S (g/L): 66.97

Lactic Acid P (g/L): 60.02

Maltose M (g/L): 73.54

Volume V (L): 8.12

t = 46.00 h

Biomass X (g/L): 71.01

Glucose S (g/L): 68.64

Lactic Acid P (g/L): 48.30
Maltose M (g/L): 81.68
Volume V (L): 9.46

t = 47.00 h

Biomass X (g/L): 101.51
Glucose S (g/L): 66.48
Lactic Acid P (g/L): 61.92
Maltose M (g/L): 74.73
Volume V (L): 9.48

t = 48.00 h

Biomass X (g/L): 78.76
Glucose S (g/L): 69.08
Lactic Acid P (g/L): 75.93
Maltose M (g/L): 83.64
Volume V (L): 9.51

t = 49.00 h

Biomass X (g/L): 83.70
Glucose S (g/L): 71.00
Lactic Acid P (g/L): 65.72
Maltose M (g/L): 74.02
Volume V (L): 10.22

t = 50.00 h

Biomass X (g/L): 93.93
Glucose S (g/L): 67.88
Lactic Acid P (g/L): 73.73
Maltose M (g/L): 71.47
Volume V (L): 10.17

t = 51.00 h

Biomass X (g/L): 117.29
Glucose S (g/L): 70.34
Lactic Acid P (g/L): 56.37
Maltose M (g/L): 77.56
Volume V (L): 10.01

t = 52.00 h

Biomass X (g/L): 72.77

Glucose S (g/L): 71.13
Lactic Acid P (g/L): 77.19
Maltose M (g/L): 72.24
Volume V (L): 10.20

t = 53.00 h

Biomass X (g/L): 88.44
Glucose S (g/L): 74.91
Lactic Acid P (g/L): 65.41
Maltose M (g/L): 70.01
Volume V (L): 9.97

t = 54.00 h

Biomass X (g/L): 77.57
Glucose S (g/L): 69.08
Lactic Acid P (g/L): 70.66
Maltose M (g/L): 71.10
Volume V (L): 11.14

t = 55.00 h

Biomass X (g/L): 79.03
Glucose S (g/L): 73.30
Lactic Acid P (g/L): 69.94
Maltose M (g/L): 72.49
Volume V (L): 9.38

t = 56.00 h

Biomass X (g/L): 77.22
Glucose S (g/L): 80.80
Lactic Acid P (g/L): 59.68
Maltose M (g/L): 78.00
Volume V (L): 10.40

t = 57.00 h

Biomass X (g/L): 73.32
Glucose S (g/L): 78.91
Lactic Acid P (g/L): 60.93
Maltose M (g/L): 63.42
Volume V (L): 10.45

t = 58.00 h

Biomass X (g/L): 83.29
Glucose S (g/L): 80.25
Lactic Acid P (g/L): 79.94
Maltose M (g/L): 71.40
Volume V (L): 10.97

t = 59.00 h

Biomass X (g/L): 69.81
Glucose S (g/L): 76.35
Lactic Acid P (g/L): 75.08
Maltose M (g/L): 73.54
Volume V (L): 10.12

t = 60.00 h

Biomass X (g/L): 92.21
Glucose S (g/L): 78.06
Lactic Acid P (g/L): 81.04
Maltose M (g/L): 71.23
Volume V (L): 9.70

t = 61.00 h

Biomass X (g/L): 83.77
Glucose S (g/L): 81.09
Lactic Acid P (g/L): 74.78
Maltose M (g/L): 75.59
Volume V (L): 10.88

t = 62.00 h

Biomass X (g/L): 87.05
Glucose S (g/L): 83.69
Lactic Acid P (g/L): 73.34
Maltose M (g/L): 69.19
Volume V (L): 11.05

t = 63.00 h

Biomass X (g/L): 78.63
Glucose S (g/L): 72.21
Lactic Acid P (g/L): 56.92
Maltose M (g/L): 68.28
Volume V (L): 11.83

t = 64.00 h

Biomass X (g/L): 79.39
Glucose S (g/L): 75.07
Lactic Acid P (g/L): 70.00
Maltose M (g/L): 71.20
Volume V (L): 10.35

t = 65.00 h

Biomass X (g/L): 85.33
Glucose S (g/L): 84.80
Lactic Acid P (g/L): 72.31
Maltose M (g/L): 68.64
Volume V (L): 11.83

t = 66.00 h

Biomass X (g/L): 71.67
Glucose S (g/L): 79.96
Lactic Acid P (g/L): 90.04
Maltose M (g/L): 66.11
Volume V (L): 12.62

t = 67.00 h

Biomass X (g/L): 79.01
Glucose S (g/L): 79.23
Lactic Acid P (g/L): 78.96
Maltose M (g/L): 69.60
Volume V (L): 11.15

t = 68.00 h

Biomass X (g/L): 74.18
Glucose S (g/L): 77.85
Lactic Acid P (g/L): 86.64
Maltose M (g/L): 64.15
Volume V (L): 10.72

t = 69.00 h

Biomass X (g/L): 73.72
Glucose S (g/L): 80.89
Lactic Acid P (g/L): 78.45
Maltose M (g/L): 63.92
Volume V (L): 11.32

t = 70.00 h

Biomass X (g/L): 62.21
Glucose S (g/L): 78.58
Lactic Acid P (g/L): 66.98
Maltose M (g/L): 63.87
Volume V (L): 11.66

t = 71.00 h

Biomass X (g/L): 94.14
Glucose S (g/L): 91.27
Lactic Acid P (g/L): 74.47
Maltose M (g/L): 71.57
Volume V (L): 12.06

t = 72.00 h

Biomass X (g/L): 75.12
Glucose S (g/L): 84.57
Lactic Acid P (g/L): 75.66
Maltose M (g/L): 51.47
Volume V (L): 11.50

t = 73.00 h

Biomass X (g/L): 92.17
Glucose S (g/L): 85.05
Lactic Acid P (g/L): 65.36
Maltose M (g/L): 66.65
Volume V (L): 12.82

t = 74.00 h

Biomass X (g/L): 91.11
Glucose S (g/L): 87.88
Lactic Acid P (g/L): 70.41
Maltose M (g/L): 64.71
Volume V (L): 11.39

t = 75.00 h

Biomass X (g/L): 69.02
Glucose S (g/L): 89.76
Lactic Acid P (g/L): 72.13
Maltose M (g/L): 56.93

Volume V (L): 11.13

Batch 20

Initial Conditions:

Batch: 20.00

X0 (g/L): 9.79

S0 (g/L): 70.43

M0 (g/L): 119.91

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.70

F_vol (L/h): 0.08

S_feed (g/L): 111.00

M_feed (g/L): 189.00

V_final (L): 10.71

Batch 20 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 20.16
Glucose S (g/L): 67.36
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 126.57
Volume V (L): 4.97

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 66.03
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 117.28
Volume V (L): 4.71

t = 2.00 h

Biomass X (g/L): 15.03
Glucose S (g/L): 66.11
Lactic Acid P (g/L): 9.64
Maltose M (g/L): 114.95
Volume V (L): 5.62

t = 3.00 h

Biomass X (g/L): 50.63
Glucose S (g/L): 71.08
Lactic Acid P (g/L): 26.63
Maltose M (g/L): 118.76
Volume V (L): 5.04

t = 4.00 h

Biomass X (g/L): 26.27
Glucose S (g/L): 62.05
Lactic Acid P (g/L): 16.64
Maltose M (g/L): 109.29
Volume V (L): 4.63

t = 5.00 h

Biomass X (g/L): 21.10
Glucose S (g/L): 59.85

Lactic Acid P (g/L): 21.85
Maltose M (g/L): 119.35
Volume V (L): 4.44

t = 6.00 h

Biomass X (g/L): 36.32
Glucose S (g/L): 57.17
Lactic Acid P (g/L): 25.47
Maltose M (g/L): 107.04
Volume V (L): 6.03

t = 7.00 h

Biomass X (g/L): 41.52
Glucose S (g/L): 57.27
Lactic Acid P (g/L): 27.77
Maltose M (g/L): 105.96
Volume V (L): 5.82

t = 8.00 h

Biomass X (g/L): 48.38
Glucose S (g/L): 55.01
Lactic Acid P (g/L): 36.35
Maltose M (g/L): 105.62
Volume V (L): 5.65

t = 9.00 h

Biomass X (g/L): 57.61
Glucose S (g/L): 39.55
Lactic Acid P (g/L): 34.98
Maltose M (g/L): 113.53
Volume V (L): 5.48

t = 10.00 h

Biomass X (g/L): 45.83
Glucose S (g/L): 52.54
Lactic Acid P (g/L): 31.79
Maltose M (g/L): 105.07
Volume V (L): 5.52

t = 11.00 h

Biomass X (g/L): 75.12

Glucose S (g/L): 52.25
Lactic Acid P (g/L): 30.38
Maltose M (g/L): 100.11
Volume V (L): 5.35

t = 12.00 h

Biomass X (g/L): 64.98
Glucose S (g/L): 41.48
Lactic Acid P (g/L): 38.07
Maltose M (g/L): 103.09
Volume V (L): 5.72

t = 13.00 h

Biomass X (g/L): 72.50
Glucose S (g/L): 40.53
Lactic Acid P (g/L): 45.65
Maltose M (g/L): 105.19
Volume V (L): 5.02

t = 14.00 h

Biomass X (g/L): 62.49
Glucose S (g/L): 41.50
Lactic Acid P (g/L): 42.18
Maltose M (g/L): 98.03
Volume V (L): 5.73

t = 15.00 h

Biomass X (g/L): 82.80
Glucose S (g/L): 39.63
Lactic Acid P (g/L): 49.94
Maltose M (g/L): 97.23
Volume V (L): 6.00

t = 16.00 h

Biomass X (g/L): 70.46
Glucose S (g/L): 42.74
Lactic Acid P (g/L): 52.94
Maltose M (g/L): 113.85
Volume V (L): 6.50

t = 17.00 h

Biomass X (g/L): 77.34
Glucose S (g/L): 34.32
Lactic Acid P (g/L): 58.21
Maltose M (g/L): 96.93
Volume V (L): 6.28

t = 18.00 h

Biomass X (g/L): 67.67
Glucose S (g/L): 37.22
Lactic Acid P (g/L): 63.84
Maltose M (g/L): 98.92
Volume V (L): 6.73

t = 19.00 h

Biomass X (g/L): 68.55
Glucose S (g/L): 47.06
Lactic Acid P (g/L): 54.33
Maltose M (g/L): 97.79
Volume V (L): 7.06

t = 20.00 h

Biomass X (g/L): 85.63
Glucose S (g/L): 37.95
Lactic Acid P (g/L): 43.05
Maltose M (g/L): 96.62
Volume V (L): 7.01

t = 21.00 h

Biomass X (g/L): 76.20
Glucose S (g/L): 46.21
Lactic Acid P (g/L): 52.99
Maltose M (g/L): 101.37
Volume V (L): 6.81

t = 22.00 h

Biomass X (g/L): 73.38
Glucose S (g/L): 43.17
Lactic Acid P (g/L): 57.63
Maltose M (g/L): 102.28
Volume V (L): 6.68

t = 23.00 h

Biomass X (g/L): 76.27
Glucose S (g/L): 35.46
Lactic Acid P (g/L): 41.60
Maltose M (g/L): 94.70
Volume V (L): 6.30

t = 24.00 h

Biomass X (g/L): 89.64
Glucose S (g/L): 44.97
Lactic Acid P (g/L): 71.81
Maltose M (g/L): 88.87
Volume V (L): 7.17

t = 25.00 h

Biomass X (g/L): 77.92
Glucose S (g/L): 42.64
Lactic Acid P (g/L): 61.39
Maltose M (g/L): 101.58
Volume V (L): 7.67

t = 26.00 h

Biomass X (g/L): 83.33
Glucose S (g/L): 41.91
Lactic Acid P (g/L): 64.38
Maltose M (g/L): 93.41
Volume V (L): 7.14

t = 27.00 h

Biomass X (g/L): 85.13
Glucose S (g/L): 45.08
Lactic Acid P (g/L): 74.08
Maltose M (g/L): 102.46
Volume V (L): 7.93

t = 28.00 h

Biomass X (g/L): 67.52
Glucose S (g/L): 46.54
Lactic Acid P (g/L): 56.69
Maltose M (g/L): 80.29
Volume V (L): 6.44

t = 29.00 h

Biomass X (g/L): 72.12
Glucose S (g/L): 48.05
Lactic Acid P (g/L): 68.30
Maltose M (g/L): 92.67
Volume V (L): 6.68

t = 30.00 h

Biomass X (g/L): 89.41
Glucose S (g/L): 50.46
Lactic Acid P (g/L): 62.73
Maltose M (g/L): 101.20
Volume V (L): 7.68

t = 31.00 h

Biomass X (g/L): 79.68
Glucose S (g/L): 48.29
Lactic Acid P (g/L): 62.65
Maltose M (g/L): 88.24
Volume V (L): 7.48

t = 32.00 h

Biomass X (g/L): 96.75
Glucose S (g/L): 42.36
Lactic Acid P (g/L): 57.88
Maltose M (g/L): 88.21
Volume V (L): 8.04

t = 33.00 h

Biomass X (g/L): 96.14
Glucose S (g/L): 46.44
Lactic Acid P (g/L): 80.53
Maltose M (g/L): 90.30
Volume V (L): 7.51

t = 34.00 h

Biomass X (g/L): 94.01
Glucose S (g/L): 59.90
Lactic Acid P (g/L): 62.51
Maltose M (g/L): 85.00

Volume V (L): 7.19

t = 35.00 h

Biomass X (g/L): 104.48

Glucose S (g/L): 52.39

Lactic Acid P (g/L): 72.75

Maltose M (g/L): 85.64

Volume V (L): 7.57

t = 36.00 h

Biomass X (g/L): 86.79

Glucose S (g/L): 59.27

Lactic Acid P (g/L): 49.83

Maltose M (g/L): 84.82

Volume V (L): 7.04

t = 37.00 h

Biomass X (g/L): 93.74

Glucose S (g/L): 56.42

Lactic Acid P (g/L): 77.99

Maltose M (g/L): 82.59

Volume V (L): 7.30

t = 38.00 h

Biomass X (g/L): 82.27

Glucose S (g/L): 52.90

Lactic Acid P (g/L): 77.24

Maltose M (g/L): 91.91

Volume V (L): 8.34

t = 39.00 h

Biomass X (g/L): 110.75

Glucose S (g/L): 53.90

Lactic Acid P (g/L): 68.96

Maltose M (g/L): 87.13

Volume V (L): 8.37

t = 40.00 h

Biomass X (g/L): 106.39

Glucose S (g/L): 53.43

Lactic Acid P (g/L): 66.39

Maltose M (g/L): 85.36

Volume V (L): 8.21

t = 41.00 h

Biomass X (g/L): 87.93

Glucose S (g/L): 47.88

Lactic Acid P (g/L): 72.43

Maltose M (g/L): 87.72

Volume V (L): 7.34

t = 42.00 h

Biomass X (g/L): 91.43

Glucose S (g/L): 55.20

Lactic Acid P (g/L): 89.92

Maltose M (g/L): 81.33

Volume V (L): 8.04

t = 43.00 h

Biomass X (g/L): 100.51

Glucose S (g/L): 60.42

Lactic Acid P (g/L): 53.47

Maltose M (g/L): 77.97

Volume V (L): 7.72

t = 44.00 h

Biomass X (g/L): 86.37

Glucose S (g/L): 54.47

Lactic Acid P (g/L): 76.88

Maltose M (g/L): 78.28

Volume V (L): 8.89

t = 45.00 h

Biomass X (g/L): 81.54

Glucose S (g/L): 59.02

Lactic Acid P (g/L): 61.80

Maltose M (g/L): 80.93

Volume V (L): 8.16

t = 46.00 h

Biomass X (g/L): 95.80

Glucose S (g/L): 61.43

Lactic Acid P (g/L): 70.18

Maltose M (g/L): 74.63

Volume V (L): 8.01

t = 47.00 h

Biomass X (g/L): 88.27

Glucose S (g/L): 55.65

Lactic Acid P (g/L): 68.39

Maltose M (g/L): 77.06

Volume V (L): 8.59

t = 48.00 h

Biomass X (g/L): 89.16

Glucose S (g/L): 58.13

Lactic Acid P (g/L): 67.02

Maltose M (g/L): 76.40

Volume V (L): 8.36

t = 49.00 h

Biomass X (g/L): 79.53

Glucose S (g/L): 56.79

Lactic Acid P (g/L): 71.06

Maltose M (g/L): 90.04

Volume V (L): 9.41

t = 50.00 h

Biomass X (g/L): 71.14

Glucose S (g/L): 58.29

Lactic Acid P (g/L): 61.11

Maltose M (g/L): 63.12

Volume V (L): 8.47

t = 51.00 h

Biomass X (g/L): 89.32

Glucose S (g/L): 57.32

Lactic Acid P (g/L): 80.98

Maltose M (g/L): 76.53

Volume V (L): 8.25

t = 52.00 h

Biomass X (g/L): 95.45

Glucose S (g/L): 59.51
Lactic Acid P (g/L): 55.59
Maltose M (g/L): 86.23
Volume V (L): 7.98

t = 53.00 h

Biomass X (g/L): 84.73
Glucose S (g/L): 68.89
Lactic Acid P (g/L): 76.99
Maltose M (g/L): 77.06
Volume V (L): 9.02

t = 54.00 h

Biomass X (g/L): 100.15
Glucose S (g/L): 60.95
Lactic Acid P (g/L): 69.34
Maltose M (g/L): 79.93
Volume V (L): 8.86

t = 55.00 h

Biomass X (g/L): 79.96
Glucose S (g/L): 65.57
Lactic Acid P (g/L): 61.89
Maltose M (g/L): 75.89
Volume V (L): 8.54

t = 56.00 h

Biomass X (g/L): 76.52
Glucose S (g/L): 63.96
Lactic Acid P (g/L): 74.55
Maltose M (g/L): 71.24
Volume V (L): 9.81

t = 57.00 h

Biomass X (g/L): 85.60
Glucose S (g/L): 70.94
Lactic Acid P (g/L): 58.78
Maltose M (g/L): 62.26
Volume V (L): 9.65

t = 58.00 h

Biomass X (g/L): 85.30
Glucose S (g/L): 72.16
Lactic Acid P (g/L): 73.46
Maltose M (g/L): 74.92
Volume V (L): 9.41

t = 59.00 h

Biomass X (g/L): 93.11
Glucose S (g/L): 71.21
Lactic Acid P (g/L): 60.36
Maltose M (g/L): 69.14
Volume V (L): 10.12

t = 60.00 h

Biomass X (g/L): 78.47
Glucose S (g/L): 69.84
Lactic Acid P (g/L): 77.64
Maltose M (g/L): 68.52
Volume V (L): 9.87

t = 61.00 h

Biomass X (g/L): 98.36
Glucose S (g/L): 68.54
Lactic Acid P (g/L): 65.24
Maltose M (g/L): 69.65
Volume V (L): 9.27

t = 62.00 h

Biomass X (g/L): 103.79
Glucose S (g/L): 65.66
Lactic Acid P (g/L): 69.70
Maltose M (g/L): 81.24
Volume V (L): 9.25

t = 63.00 h

Biomass X (g/L): 84.82
Glucose S (g/L): 78.02
Lactic Acid P (g/L): 76.63
Maltose M (g/L): 84.32
Volume V (L): 10.17

t = 64.00 h

Biomass X (g/L): 96.70
Glucose S (g/L): 68.64
Lactic Acid P (g/L): 67.11
Maltose M (g/L): 72.08
Volume V (L): 10.06

t = 65.00 h

Biomass X (g/L): 74.92
Glucose S (g/L): 74.61
Lactic Acid P (g/L): 73.75
Maltose M (g/L): 67.94
Volume V (L): 10.03

t = 66.00 h

Biomass X (g/L): 94.68
Glucose S (g/L): 76.76
Lactic Acid P (g/L): 60.68
Maltose M (g/L): 63.65
Volume V (L): 9.64

t = 67.00 h

Biomass X (g/L): 94.23
Glucose S (g/L): 71.41
Lactic Acid P (g/L): 80.30
Maltose M (g/L): 72.06
Volume V (L): 10.38

t = 68.00 h

Biomass X (g/L): 92.31
Glucose S (g/L): 77.99
Lactic Acid P (g/L): 90.73
Maltose M (g/L): 78.68
Volume V (L): 9.83

t = 69.00 h

Biomass X (g/L): 83.88
Glucose S (g/L): 77.62
Lactic Acid P (g/L): 100.19
Maltose M (g/L): 67.31
Volume V (L): 10.21

t = 70.00 h

Biomass X (g/L): 56.17
Glucose S (g/L): 81.85
Lactic Acid P (g/L): 75.82
Maltose M (g/L): 55.40
Volume V (L): 10.99

t = 71.00 h

Biomass X (g/L): 77.62
Glucose S (g/L): 74.31
Lactic Acid P (g/L): 82.28
Maltose M (g/L): 71.38
Volume V (L): 10.11

t = 72.00 h

Biomass X (g/L): 77.57
Glucose S (g/L): 71.90
Lactic Acid P (g/L): 59.09
Maltose M (g/L): 72.65
Volume V (L): 9.77

t = 73.00 h

Biomass X (g/L): 74.06
Glucose S (g/L): 75.43
Lactic Acid P (g/L): 89.02
Maltose M (g/L): 58.39
Volume V (L): 10.08

t = 74.00 h

Biomass X (g/L): 84.17
Glucose S (g/L): 78.71
Lactic Acid P (g/L): 82.55
Maltose M (g/L): 67.46
Volume V (L): 11.09

t = 75.00 h

Biomass X (g/L): 78.89
Glucose S (g/L): 78.76
Lactic Acid P (g/L): 71.24
Maltose M (g/L): 68.52

Volume V (L): 10.83

Batch 21

Initial Conditions:

Batch: 21.00

X0 (g/L): 7.51

S0 (g/L): 62.33

M0 (g/L): 45.26

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.73

F_vol (L/h): 0.03

S_feed (g/L): 173.80

M_feed (g/L): 126.20

V_final (L): 6.97

Batch 21 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.24

Glucose S (g/L): 61.23

Lactic Acid P (g/L): 7.90

Maltose M (g/L): 45.62

Volume V (L): 5.18

t = 1.00 h

Biomass X (g/L): 5.06

Glucose S (g/L): 42.70

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 42.64

Volume V (L): 5.02

t = 2.00 h

Biomass X (g/L): 18.08

Glucose S (g/L): 55.80

Lactic Acid P (g/L): 4.75

Maltose M (g/L): 44.00

Volume V (L): 5.07

t = 3.00 h

Biomass X (g/L): 8.73

Glucose S (g/L): 52.47

Lactic Acid P (g/L): 8.25

Maltose M (g/L): 45.35

Volume V (L): 5.03

t = 4.00 h

Biomass X (g/L): 24.52

Glucose S (g/L): 56.68

Lactic Acid P (g/L): 0.91

Maltose M (g/L): 44.73

Volume V (L): 5.23

t = 5.00 h

Biomass X (g/L): 19.09

Glucose S (g/L): 46.32

Lactic Acid P (g/L): 36.38

Maltose M (g/L): 42.59

Volume V (L): 5.25

t = 6.00 h

Biomass X (g/L): 22.52

Glucose S (g/L): 45.57

Lactic Acid P (g/L): 3.77

Maltose M (g/L): 43.10

Volume V (L): 5.21

t = 7.00 h

Biomass X (g/L): 30.21

Glucose S (g/L): 50.71

Lactic Acid P (g/L): 8.63

Maltose M (g/L): 40.73

Volume V (L): 5.27

t = 8.00 h

Biomass X (g/L): 37.81

Glucose S (g/L): 44.06

Lactic Acid P (g/L): 37.87

Maltose M (g/L): 43.23

Volume V (L): 5.18

t = 9.00 h

Biomass X (g/L): 45.82

Glucose S (g/L): 39.94

Lactic Acid P (g/L): 35.61

Maltose M (g/L): 41.37

Volume V (L): 5.32

t = 10.00 h

Biomass X (g/L): 53.09

Glucose S (g/L): 29.35

Lactic Acid P (g/L): 22.03

Maltose M (g/L): 40.61

Volume V (L): 5.21

t = 11.00 h

Biomass X (g/L): 18.47

Glucose S (g/L): 31.64
Lactic Acid P (g/L): 41.13
Maltose M (g/L): 40.08
Volume V (L): 5.29

t = 12.00 h

Biomass X (g/L): 50.62
Glucose S (g/L): 24.90
Lactic Acid P (g/L): 52.43
Maltose M (g/L): 40.28
Volume V (L): 5.27

t = 13.00 h

Biomass X (g/L): 58.07
Glucose S (g/L): 22.68
Lactic Acid P (g/L): 40.48
Maltose M (g/L): 39.19
Volume V (L): 5.27

t = 14.00 h

Biomass X (g/L): 79.79
Glucose S (g/L): 23.68
Lactic Acid P (g/L): 68.56
Maltose M (g/L): 40.94
Volume V (L): 5.26

t = 15.00 h

Biomass X (g/L): 56.95
Glucose S (g/L): 16.68
Lactic Acid P (g/L): 28.59
Maltose M (g/L): 38.60
Volume V (L): 5.47

t = 16.00 h

Biomass X (g/L): 71.60
Glucose S (g/L): 25.84
Lactic Acid P (g/L): 54.20
Maltose M (g/L): 42.02
Volume V (L): 5.63

t = 17.00 h

Biomass X (g/L): 86.87
Glucose S (g/L): 16.01
Lactic Acid P (g/L): 60.65
Maltose M (g/L): 41.45
Volume V (L): 5.63

t = 18.00 h

Biomass X (g/L): 61.05
Glucose S (g/L): 18.86
Lactic Acid P (g/L): 59.64
Maltose M (g/L): 36.66
Volume V (L): 5.27

t = 19.00 h

Biomass X (g/L): 79.58
Glucose S (g/L): 22.19
Lactic Acid P (g/L): 50.50
Maltose M (g/L): 38.16
Volume V (L): 5.70

t = 20.00 h

Biomass X (g/L): 69.27
Glucose S (g/L): 19.20
Lactic Acid P (g/L): 51.08
Maltose M (g/L): 36.48
Volume V (L): 5.48

t = 21.00 h

Biomass X (g/L): 73.55
Glucose S (g/L): 5.99
Lactic Acid P (g/L): 59.56
Maltose M (g/L): 36.79
Volume V (L): 5.65

t = 22.00 h

Biomass X (g/L): 71.81
Glucose S (g/L): 14.95
Lactic Acid P (g/L): 64.41
Maltose M (g/L): 40.89
Volume V (L): 5.44

t = 23.00 h

Biomass X (g/L): 73.59
Glucose S (g/L): 16.58
Lactic Acid P (g/L): 63.46
Maltose M (g/L): 36.22
Volume V (L): 5.63

t = 24.00 h

Biomass X (g/L): 108.64
Glucose S (g/L): 16.43
Lactic Acid P (g/L): 68.79
Maltose M (g/L): 37.19
Volume V (L): 5.68

t = 25.00 h

Biomass X (g/L): 90.99
Glucose S (g/L): 5.84
Lactic Acid P (g/L): 49.70
Maltose M (g/L): 37.24
Volume V (L): 5.42

t = 26.00 h

Biomass X (g/L): 94.31
Glucose S (g/L): 7.96
Lactic Acid P (g/L): 58.35
Maltose M (g/L): 34.04
Volume V (L): 5.78

t = 27.00 h

Biomass X (g/L): 89.00
Glucose S (g/L): 16.28
Lactic Acid P (g/L): 52.29
Maltose M (g/L): 34.04
Volume V (L): 5.92

t = 28.00 h

Biomass X (g/L): 68.94
Glucose S (g/L): 7.15
Lactic Acid P (g/L): 58.53
Maltose M (g/L): 33.10
Volume V (L): 5.71

t = 29.00 h

Biomass X (g/L): 89.27
Glucose S (g/L): 9.46
Lactic Acid P (g/L): 73.00
Maltose M (g/L): 33.44
Volume V (L): 5.98

t = 30.00 h

Biomass X (g/L): 96.48
Glucose S (g/L): 19.07
Lactic Acid P (g/L): 64.90
Maltose M (g/L): 29.46
Volume V (L): 5.76

t = 31.00 h

Biomass X (g/L): 77.95
Glucose S (g/L): 10.75
Lactic Acid P (g/L): 90.01
Maltose M (g/L): 35.62
Volume V (L): 5.95

t = 32.00 h

Biomass X (g/L): 96.32
Glucose S (g/L): 7.31
Lactic Acid P (g/L): 67.15
Maltose M (g/L): 38.22
Volume V (L): 5.49

t = 33.00 h

Biomass X (g/L): 94.14
Glucose S (g/L): 6.61
Lactic Acid P (g/L): 66.51
Maltose M (g/L): 34.36
Volume V (L): 6.25

t = 34.00 h

Biomass X (g/L): 105.81
Glucose S (g/L): 12.93
Lactic Acid P (g/L): 67.60
Maltose M (g/L): 31.16

Volume V (L): 6.13

t = 35.00 h

Biomass X (g/L): 83.73

Glucose S (g/L): 4.50

Lactic Acid P (g/L): 60.96

Maltose M (g/L): 32.38

Volume V (L): 5.87

t = 36.00 h

Biomass X (g/L): 77.31

Glucose S (g/L): 10.22

Lactic Acid P (g/L): 67.20

Maltose M (g/L): 32.48

Volume V (L): 5.63

t = 37.00 h

Biomass X (g/L): 84.82

Glucose S (g/L): 13.37

Lactic Acid P (g/L): 79.34

Maltose M (g/L): 31.19

Volume V (L): 6.02

t = 38.00 h

Biomass X (g/L): 102.79

Glucose S (g/L): 18.51

Lactic Acid P (g/L): 70.00

Maltose M (g/L): 34.51

Volume V (L): 5.77

t = 39.00 h

Biomass X (g/L): 70.78

Glucose S (g/L): 15.46

Lactic Acid P (g/L): 61.67

Maltose M (g/L): 34.19

Volume V (L): 6.09

t = 40.00 h

Biomass X (g/L): 68.59

Glucose S (g/L): 8.58

Lactic Acid P (g/L): 60.90

Maltose M (g/L): 30.24

Volume V (L): 6.12

t = 41.00 h

Biomass X (g/L): 90.04

Glucose S (g/L): 18.53

Lactic Acid P (g/L): 66.03

Maltose M (g/L): 33.58

Volume V (L): 5.93

t = 42.00 h

Biomass X (g/L): 84.92

Glucose S (g/L): 11.17

Lactic Acid P (g/L): 71.62

Maltose M (g/L): 34.88

Volume V (L): 6.28

t = 43.00 h

Biomass X (g/L): 89.48

Glucose S (g/L): 6.62

Lactic Acid P (g/L): 83.01

Maltose M (g/L): 35.38

Volume V (L): 6.26

t = 44.00 h

Biomass X (g/L): 98.83

Glucose S (g/L): 12.34

Lactic Acid P (g/L): 58.74

Maltose M (g/L): 36.99

Volume V (L): 6.31

t = 45.00 h

Biomass X (g/L): 84.13

Glucose S (g/L): 20.87

Lactic Acid P (g/L): 69.14

Maltose M (g/L): 29.65

Volume V (L): 6.18

t = 46.00 h

Biomass X (g/L): 71.16

Glucose S (g/L): 14.61

Lactic Acid P (g/L): 74.09

Maltose M (g/L): 36.50

Volume V (L): 5.99

t = 47.00 h

Biomass X (g/L): 109.87

Glucose S (g/L): 14.12

Lactic Acid P (g/L): 55.20

Maltose M (g/L): 30.28

Volume V (L): 6.39

t = 48.00 h

Biomass X (g/L): 79.88

Glucose S (g/L): 22.75

Lactic Acid P (g/L): 65.36

Maltose M (g/L): 29.63

Volume V (L): 6.22

t = 49.00 h

Biomass X (g/L): 79.29

Glucose S (g/L): 7.29

Lactic Acid P (g/L): 67.25

Maltose M (g/L): 31.18

Volume V (L): 6.07

t = 50.00 h

Biomass X (g/L): 100.64

Glucose S (g/L): 8.35

Lactic Acid P (g/L): 66.50

Maltose M (g/L): 29.26

Volume V (L): 6.53

t = 51.00 h

Biomass X (g/L): 85.95

Glucose S (g/L): 6.96

Lactic Acid P (g/L): 76.51

Maltose M (g/L): 30.93

Volume V (L): 6.67

t = 52.00 h

Biomass X (g/L): 89.10

Glucose S (g/L): 17.49
Lactic Acid P (g/L): 73.43
Maltose M (g/L): 27.49
Volume V (L): 6.36

t = 53.00 h

Biomass X (g/L): 94.46
Glucose S (g/L): 16.75
Lactic Acid P (g/L): 69.93
Maltose M (g/L): 29.35
Volume V (L): 6.73

t = 54.00 h

Biomass X (g/L): 100.35
Glucose S (g/L): 24.15
Lactic Acid P (g/L): 79.87
Maltose M (g/L): 32.06
Volume V (L): 6.43

t = 55.00 h

Biomass X (g/L): 86.31
Glucose S (g/L): 17.81
Lactic Acid P (g/L): 92.84
Maltose M (g/L): 29.34
Volume V (L): 6.38

t = 56.00 h

Biomass X (g/L): 104.54
Glucose S (g/L): 14.81
Lactic Acid P (g/L): 92.34
Maltose M (g/L): 29.54
Volume V (L): 6.37

t = 57.00 h

Biomass X (g/L): 68.40
Glucose S (g/L): 17.24
Lactic Acid P (g/L): 57.91
Maltose M (g/L): 29.99
Volume V (L): 6.76

t = 58.00 h

Biomass X (g/L): 96.90
Glucose S (g/L): 18.17
Lactic Acid P (g/L): 82.61
Maltose M (g/L): 31.14
Volume V (L): 6.40

t = 59.00 h

Biomass X (g/L): 81.03
Glucose S (g/L): 9.33
Lactic Acid P (g/L): 75.54
Maltose M (g/L): 25.62
Volume V (L): 6.67

t = 60.00 h

Biomass X (g/L): 102.76
Glucose S (g/L): 32.18
Lactic Acid P (g/L): 79.09
Maltose M (g/L): 29.69
Volume V (L): 6.49

t = 61.00 h

Biomass X (g/L): 90.52
Glucose S (g/L): 21.28
Lactic Acid P (g/L): 73.98
Maltose M (g/L): 27.92
Volume V (L): 6.53

t = 62.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 20.04
Lactic Acid P (g/L): 79.29
Maltose M (g/L): 28.58
Volume V (L): 6.79

t = 63.00 h

Biomass X (g/L): 95.44
Glucose S (g/L): 22.47
Lactic Acid P (g/L): 79.90
Maltose M (g/L): 30.11
Volume V (L): 6.59

t = 64.00 h

Biomass X (g/L): 97.60
Glucose S (g/L): 23.31
Lactic Acid P (g/L): 73.04
Maltose M (g/L): 25.68
Volume V (L): 6.62

t = 65.00 h

Biomass X (g/L): 94.99
Glucose S (g/L): 19.30
Lactic Acid P (g/L): 70.21
Maltose M (g/L): 25.87
Volume V (L): 6.89

t = 66.00 h

Biomass X (g/L): 78.36
Glucose S (g/L): 17.14
Lactic Acid P (g/L): 74.83
Maltose M (g/L): 23.97
Volume V (L): 6.99

t = 67.00 h

Biomass X (g/L): 83.61
Glucose S (g/L): 23.82
Lactic Acid P (g/L): 81.75
Maltose M (g/L): 26.44
Volume V (L): 6.86

t = 68.00 h

Biomass X (g/L): 88.21
Glucose S (g/L): 23.08
Lactic Acid P (g/L): 85.55
Maltose M (g/L): 29.85
Volume V (L): 6.68

t = 69.00 h

Biomass X (g/L): 87.69
Glucose S (g/L): 17.30
Lactic Acid P (g/L): 88.28
Maltose M (g/L): 26.96
Volume V (L): 6.73

t = 70.00 h

Biomass X (g/L): 95.35
Glucose S (g/L): 29.11
Lactic Acid P (g/L): 75.37
Maltose M (g/L): 24.51
Volume V (L): 6.85

t = 71.00 h

Biomass X (g/L): 67.91
Glucose S (g/L): 23.69
Lactic Acid P (g/L): 83.19
Maltose M (g/L): 25.98
Volume V (L): 6.92

t = 72.00 h

Biomass X (g/L): 90.93
Glucose S (g/L): 26.29
Lactic Acid P (g/L): 90.04
Maltose M (g/L): 23.94
Volume V (L): 6.74

t = 73.00 h

Biomass X (g/L): 84.61
Glucose S (g/L): 30.05
Lactic Acid P (g/L): 92.25
Maltose M (g/L): 24.49
Volume V (L): 7.03

t = 74.00 h

Biomass X (g/L): 79.03
Glucose S (g/L): 23.38
Lactic Acid P (g/L): 80.52
Maltose M (g/L): 27.21
Volume V (L): 6.96

t = 75.00 h

Biomass X (g/L): 79.81
Glucose S (g/L): 25.26
Lactic Acid P (g/L): 79.15
Maltose M (g/L): 24.12

Volume V (L): 6.70

Batch 22

Initial Conditions:

Batch: 22.00

X0 (g/L): 1.45

S0 (g/L): 21.37

M0 (g/L): 108.54

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 5.08

F_vol (L/h): 0.02

S_feed (g/L): 49.35

M_feed (g/L): 250.65

V_final (L): 6.83

Batch 22 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 21.56
Lactic Acid P (g/L): 2.57
Maltose M (g/L): 102.65
Volume V (L): 4.85

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 22.49
Lactic Acid P (g/L): 17.04
Maltose M (g/L): 109.02
Volume V (L): 5.16

t = 2.00 h

Biomass X (g/L): 1.73
Glucose S (g/L): 25.22
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 109.18
Volume V (L): 5.07

t = 3.00 h

Biomass X (g/L): 21.45
Glucose S (g/L): 23.88
Lactic Acid P (g/L): 0.14
Maltose M (g/L): 107.94
Volume V (L): 4.92

t = 4.00 h

Biomass X (g/L): 6.40
Glucose S (g/L): 27.63
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 98.82
Volume V (L): 5.00

t = 5.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 28.63

Lactic Acid P (g/L): 8.31
Maltose M (g/L): 96.49
Volume V (L): 4.84

t = 6.00 h

Biomass X (g/L): 8.89
Glucose S (g/L): 25.31
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 97.82
Volume V (L): 5.13

t = 7.00 h

Biomass X (g/L): 5.16
Glucose S (g/L): 31.36
Lactic Acid P (g/L): 8.35
Maltose M (g/L): 100.67
Volume V (L): 5.50

t = 8.00 h

Biomass X (g/L): 13.91
Glucose S (g/L): 26.73
Lactic Acid P (g/L): 1.04
Maltose M (g/L): 106.27
Volume V (L): 5.31

t = 9.00 h

Biomass X (g/L): 18.96
Glucose S (g/L): 28.37
Lactic Acid P (g/L): 1.97
Maltose M (g/L): 85.79
Volume V (L): 5.54

t = 10.00 h

Biomass X (g/L): 28.00
Glucose S (g/L): 29.84
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 94.62
Volume V (L): 5.37

t = 11.00 h

Biomass X (g/L): 19.32

Glucose S (g/L): 27.86
Lactic Acid P (g/L): 8.97
Maltose M (g/L): 87.41
Volume V (L): 5.42

t = 12.00 h

Biomass X (g/L): 38.94
Glucose S (g/L): 24.06
Lactic Acid P (g/L): 25.70
Maltose M (g/L): 84.56
Volume V (L): 5.58

t = 13.00 h

Biomass X (g/L): 35.54
Glucose S (g/L): 21.21
Lactic Acid P (g/L): 18.28
Maltose M (g/L): 97.79
Volume V (L): 5.37

t = 14.00 h

Biomass X (g/L): 23.63
Glucose S (g/L): 22.27
Lactic Acid P (g/L): 27.00
Maltose M (g/L): 94.43
Volume V (L): 5.31

t = 15.00 h

Biomass X (g/L): 55.56
Glucose S (g/L): 19.60
Lactic Acid P (g/L): 39.22
Maltose M (g/L): 92.87
Volume V (L): 5.52

t = 16.00 h

Biomass X (g/L): 40.89
Glucose S (g/L): 14.61
Lactic Acid P (g/L): 29.11
Maltose M (g/L): 94.43
Volume V (L): 5.60

t = 17.00 h

Biomass X (g/L): 33.01
Glucose S (g/L): 12.37
Lactic Acid P (g/L): 24.17
Maltose M (g/L): 85.78
Volume V (L): 5.34

t = 18.00 h

Biomass X (g/L): 60.23
Glucose S (g/L): 7.72
Lactic Acid P (g/L): 35.95
Maltose M (g/L): 85.06
Volume V (L): 5.33

t = 19.00 h

Biomass X (g/L): 67.24
Glucose S (g/L): 11.51
Lactic Acid P (g/L): 28.97
Maltose M (g/L): 98.41
Volume V (L): 5.48

t = 20.00 h

Biomass X (g/L): 49.44
Glucose S (g/L): 9.71
Lactic Acid P (g/L): 31.41
Maltose M (g/L): 84.18
Volume V (L): 5.61

t = 21.00 h

Biomass X (g/L): 47.27
Glucose S (g/L): 11.54
Lactic Acid P (g/L): 24.06
Maltose M (g/L): 88.63
Volume V (L): 5.63

t = 22.00 h

Biomass X (g/L): 52.63
Glucose S (g/L): 8.86
Lactic Acid P (g/L): 48.15
Maltose M (g/L): 83.75
Volume V (L): 5.33

t = 23.00 h

Biomass X (g/L): 63.17
Glucose S (g/L): 10.46
Lactic Acid P (g/L): 37.80
Maltose M (g/L): 72.73
Volume V (L): 5.84

t = 24.00 h

Biomass X (g/L): 70.38
Glucose S (g/L): 5.15
Lactic Acid P (g/L): 44.49
Maltose M (g/L): 70.80
Volume V (L): 5.49

t = 25.00 h

Biomass X (g/L): 65.13
Glucose S (g/L): 9.05
Lactic Acid P (g/L): 34.77
Maltose M (g/L): 77.96
Volume V (L): 5.57

t = 26.00 h

Biomass X (g/L): 62.06
Glucose S (g/L): 4.82
Lactic Acid P (g/L): 49.51
Maltose M (g/L): 86.07
Volume V (L): 5.50

t = 27.00 h

Biomass X (g/L): 76.92
Glucose S (g/L): 7.67
Lactic Acid P (g/L): 39.80
Maltose M (g/L): 83.79
Volume V (L): 6.06

t = 28.00 h

Biomass X (g/L): 60.63
Glucose S (g/L): 2.26
Lactic Acid P (g/L): 46.62
Maltose M (g/L): 83.68
Volume V (L): 5.72

t = 29.00 h

Biomass X (g/L): 74.79
Glucose S (g/L): 6.14
Lactic Acid P (g/L): 57.03
Maltose M (g/L): 72.51
Volume V (L): 5.58

t = 30.00 h

Biomass X (g/L): 63.10
Glucose S (g/L): 0.88
Lactic Acid P (g/L): 57.29
Maltose M (g/L): 85.97
Volume V (L): 5.86

t = 31.00 h

Biomass X (g/L): 71.07
Glucose S (g/L): 4.91
Lactic Acid P (g/L): 50.88
Maltose M (g/L): 71.11
Volume V (L): 5.87

t = 32.00 h

Biomass X (g/L): 66.74
Glucose S (g/L): 6.60
Lactic Acid P (g/L): 35.91
Maltose M (g/L): 76.81
Volume V (L): 5.75

t = 33.00 h

Biomass X (g/L): 68.20
Glucose S (g/L): 4.40
Lactic Acid P (g/L): 80.30
Maltose M (g/L): 74.87
Volume V (L): 5.93

t = 34.00 h

Biomass X (g/L): 65.93
Glucose S (g/L): 3.23
Lactic Acid P (g/L): 69.82
Maltose M (g/L): 73.26

Volume V (L): 5.71

t = 35.00 h

Biomass X (g/L): 83.41

Glucose S (g/L): 7.79

Lactic Acid P (g/L): 61.35

Maltose M (g/L): 80.83

Volume V (L): 5.78

t = 36.00 h

Biomass X (g/L): 64.06

Glucose S (g/L): 5.67

Lactic Acid P (g/L): 48.73

Maltose M (g/L): 68.98

Volume V (L): 5.58

t = 37.00 h

Biomass X (g/L): 64.71

Glucose S (g/L): 4.90

Lactic Acid P (g/L): 55.39

Maltose M (g/L): 64.79

Volume V (L): 5.71

t = 38.00 h

Biomass X (g/L): 93.10

Glucose S (g/L): 6.30

Lactic Acid P (g/L): 52.27

Maltose M (g/L): 71.62

Volume V (L): 6.19

t = 39.00 h

Biomass X (g/L): 82.69

Glucose S (g/L): 12.55

Lactic Acid P (g/L): 69.55

Maltose M (g/L): 68.68

Volume V (L): 6.19

t = 40.00 h

Biomass X (g/L): 50.75

Glucose S (g/L): 9.24

Lactic Acid P (g/L): 61.16

Maltose M (g/L): 88.69

Volume V (L): 5.99

t = 41.00 h

Biomass X (g/L): 68.97

Glucose S (g/L): 9.45

Lactic Acid P (g/L): 52.35

Maltose M (g/L): 73.31

Volume V (L): 6.25

t = 42.00 h

Biomass X (g/L): 57.46

Glucose S (g/L): 7.93

Lactic Acid P (g/L): 55.93

Maltose M (g/L): 66.76

Volume V (L): 6.00

t = 43.00 h

Biomass X (g/L): 66.33

Glucose S (g/L): 7.51

Lactic Acid P (g/L): 71.16

Maltose M (g/L): 62.63

Volume V (L): 6.01

t = 44.00 h

Biomass X (g/L): 45.65

Glucose S (g/L): 1.70

Lactic Acid P (g/L): 76.21

Maltose M (g/L): 57.54

Volume V (L): 6.08

t = 45.00 h

Biomass X (g/L): 75.19

Glucose S (g/L): 5.96

Lactic Acid P (g/L): 76.22

Maltose M (g/L): 61.64

Volume V (L): 6.08

t = 46.00 h

Biomass X (g/L): 65.92

Glucose S (g/L): 6.66

Lactic Acid P (g/L): 69.82

Maltose M (g/L): 67.30

Volume V (L): 6.09

t = 47.00 h

Biomass X (g/L): 77.99

Glucose S (g/L): 3.62

Lactic Acid P (g/L): 61.24

Maltose M (g/L): 73.92

Volume V (L): 6.32

t = 48.00 h

Biomass X (g/L): 62.08

Glucose S (g/L): 5.55

Lactic Acid P (g/L): 34.57

Maltose M (g/L): 71.70

Volume V (L): 6.19

t = 49.00 h

Biomass X (g/L): 79.00

Glucose S (g/L): 3.88

Lactic Acid P (g/L): 67.47

Maltose M (g/L): 66.28

Volume V (L): 6.30

t = 50.00 h

Biomass X (g/L): 74.54

Glucose S (g/L): 8.70

Lactic Acid P (g/L): 65.32

Maltose M (g/L): 62.74

Volume V (L): 6.37

t = 51.00 h

Biomass X (g/L): 93.43

Glucose S (g/L): 10.98

Lactic Acid P (g/L): 67.19

Maltose M (g/L): 65.46

Volume V (L): 6.30

t = 52.00 h

Biomass X (g/L): 92.88

Glucose S (g/L): 11.93
Lactic Acid P (g/L): 66.96
Maltose M (g/L): 60.65
Volume V (L): 6.13

t = 53.00 h

Biomass X (g/L): 87.11
Glucose S (g/L): 10.59
Lactic Acid P (g/L): 54.90
Maltose M (g/L): 72.37
Volume V (L): 6.10

t = 54.00 h

Biomass X (g/L): 76.62
Glucose S (g/L): 7.64
Lactic Acid P (g/L): 72.62
Maltose M (g/L): 61.46
Volume V (L): 6.34

t = 55.00 h

Biomass X (g/L): 99.27
Glucose S (g/L): 5.66
Lactic Acid P (g/L): 73.81
Maltose M (g/L): 61.54
Volume V (L): 6.32

t = 56.00 h

Biomass X (g/L): 65.25
Glucose S (g/L): 6.37
Lactic Acid P (g/L): 74.77
Maltose M (g/L): 49.42
Volume V (L): 6.38

t = 57.00 h

Biomass X (g/L): 66.92
Glucose S (g/L): 7.82
Lactic Acid P (g/L): 77.67
Maltose M (g/L): 60.00
Volume V (L): 6.39

t = 58.00 h

Biomass X (g/L): 79.90
Glucose S (g/L): 9.02
Lactic Acid P (g/L): 75.02
Maltose M (g/L): 65.18
Volume V (L): 6.80

t = 59.00 h

Biomass X (g/L): 102.73
Glucose S (g/L): 11.05
Lactic Acid P (g/L): 66.23
Maltose M (g/L): 54.32
Volume V (L): 6.25

t = 60.00 h

Biomass X (g/L): 79.12
Glucose S (g/L): 9.36
Lactic Acid P (g/L): 69.78
Maltose M (g/L): 55.61
Volume V (L): 6.17

t = 61.00 h

Biomass X (g/L): 79.46
Glucose S (g/L): 12.86
Lactic Acid P (g/L): 71.70
Maltose M (g/L): 63.61
Volume V (L): 6.54

t = 62.00 h

Biomass X (g/L): 67.42
Glucose S (g/L): 11.84
Lactic Acid P (g/L): 72.28
Maltose M (g/L): 50.78
Volume V (L): 6.46

t = 63.00 h

Biomass X (g/L): 90.67
Glucose S (g/L): 12.74
Lactic Acid P (g/L): 80.83
Maltose M (g/L): 64.21
Volume V (L): 6.59

t = 64.00 h

Biomass X (g/L): 90.85
Glucose S (g/L): 15.73
Lactic Acid P (g/L): 78.00
Maltose M (g/L): 55.11
Volume V (L): 6.65

t = 65.00 h

Biomass X (g/L): 89.37
Glucose S (g/L): 10.61
Lactic Acid P (g/L): 91.49
Maltose M (g/L): 55.49
Volume V (L): 6.78

t = 66.00 h

Biomass X (g/L): 75.72
Glucose S (g/L): 14.71
Lactic Acid P (g/L): 93.06
Maltose M (g/L): 51.07
Volume V (L): 6.39

t = 67.00 h

Biomass X (g/L): 75.05
Glucose S (g/L): 14.03
Lactic Acid P (g/L): 74.03
Maltose M (g/L): 58.89
Volume V (L): 6.56

t = 68.00 h

Biomass X (g/L): 69.44
Glucose S (g/L): 13.62
Lactic Acid P (g/L): 38.66
Maltose M (g/L): 69.88
Volume V (L): 6.72

t = 69.00 h

Biomass X (g/L): 82.83
Glucose S (g/L): 16.83
Lactic Acid P (g/L): 85.33
Maltose M (g/L): 60.32
Volume V (L): 6.80

t = 70.00 h

Biomass X (g/L): 71.60
Glucose S (g/L): 17.17
Lactic Acid P (g/L): 69.46
Maltose M (g/L): 50.95
Volume V (L): 7.03

t = 71.00 h

Biomass X (g/L): 70.69
Glucose S (g/L): 13.23
Lactic Acid P (g/L): 67.43
Maltose M (g/L): 61.51
Volume V (L): 6.87

t = 72.00 h

Biomass X (g/L): 78.01
Glucose S (g/L): 15.38
Lactic Acid P (g/L): 71.45
Maltose M (g/L): 53.21
Volume V (L): 6.75

t = 73.00 h

Biomass X (g/L): 77.39
Glucose S (g/L): 16.82
Lactic Acid P (g/L): 71.34
Maltose M (g/L): 62.10
Volume V (L): 6.65

t = 74.00 h

Biomass X (g/L): 76.92
Glucose S (g/L): 15.87
Lactic Acid P (g/L): 81.33
Maltose M (g/L): 58.64
Volume V (L): 6.73

t = 75.00 h

Biomass X (g/L): 86.64
Glucose S (g/L): 18.24
Lactic Acid P (g/L): 61.82
Maltose M (g/L): 37.81

Volume V (L): 6.75

Batch 23

Initial Conditions:

Batch: 23.00

X0 (g/L): 8.73

S0 (g/L): 136.85

M0 (g/L): 15.65

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.11

F_vol (L/h): 0.06

S_feed (g/L): 269.21

M_feed (g/L): 30.79

V_final (L): 9.58

Batch 23 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 15.11
Glucose S (g/L): 137.61
Lactic Acid P (g/L): 2.22
Maltose M (g/L): 15.23
Volume V (L): 4.88

t = 1.00 h

Biomass X (g/L): 14.31
Glucose S (g/L): 136.54
Lactic Acid P (g/L): 6.76
Maltose M (g/L): 15.41
Volume V (L): 5.38

t = 2.00 h

Biomass X (g/L): 2.68
Glucose S (g/L): 133.46
Lactic Acid P (g/L): 1.07
Maltose M (g/L): 16.29
Volume V (L): 4.76

t = 3.00 h

Biomass X (g/L): 15.80
Glucose S (g/L): 133.32
Lactic Acid P (g/L): 10.56
Maltose M (g/L): 14.32
Volume V (L): 4.93

t = 4.00 h

Biomass X (g/L): 33.60
Glucose S (g/L): 123.65
Lactic Acid P (g/L): 3.46
Maltose M (g/L): 16.35
Volume V (L): 5.39

t = 5.00 h

Biomass X (g/L): 33.84
Glucose S (g/L): 130.56

Lactic Acid P (g/L): 28.08

Maltose M (g/L): 15.15

Volume V (L): 5.60

t = 6.00 h

Biomass X (g/L): 41.45

Glucose S (g/L): 118.67

Lactic Acid P (g/L): 14.26

Maltose M (g/L): 14.90

Volume V (L): 5.78

t = 7.00 h

Biomass X (g/L): 45.07

Glucose S (g/L): 109.50

Lactic Acid P (g/L): 34.76

Maltose M (g/L): 14.73

Volume V (L): 6.00

t = 8.00 h

Biomass X (g/L): 62.38

Glucose S (g/L): 113.00

Lactic Acid P (g/L): 25.99

Maltose M (g/L): 14.48

Volume V (L): 5.46

t = 9.00 h

Biomass X (g/L): 53.69

Glucose S (g/L): 104.43

Lactic Acid P (g/L): 33.75

Maltose M (g/L): 15.26

Volume V (L): 5.99

t = 10.00 h

Biomass X (g/L): 52.08

Glucose S (g/L): 108.60

Lactic Acid P (g/L): 30.71

Maltose M (g/L): 13.89

Volume V (L): 5.49

t = 11.00 h

Biomass X (g/L): 50.96

Glucose S (g/L): 113.25
Lactic Acid P (g/L): 29.02
Maltose M (g/L): 14.63
Volume V (L): 6.00

t = 12.00 h

Biomass X (g/L): 65.95
Glucose S (g/L): 96.22
Lactic Acid P (g/L): 25.11
Maltose M (g/L): 14.09
Volume V (L): 5.81

t = 13.00 h

Biomass X (g/L): 42.97
Glucose S (g/L): 99.49
Lactic Acid P (g/L): 34.16
Maltose M (g/L): 13.76
Volume V (L): 5.96

t = 14.00 h

Biomass X (g/L): 67.84
Glucose S (g/L): 87.50
Lactic Acid P (g/L): 43.36
Maltose M (g/L): 15.67
Volume V (L): 6.01

t = 15.00 h

Biomass X (g/L): 70.19
Glucose S (g/L): 96.64
Lactic Acid P (g/L): 50.52
Maltose M (g/L): 14.27
Volume V (L): 5.33

t = 16.00 h

Biomass X (g/L): 61.16
Glucose S (g/L): 88.81
Lactic Acid P (g/L): 59.92
Maltose M (g/L): 14.29
Volume V (L): 5.56

t = 17.00 h

Biomass X (g/L): 67.41
Glucose S (g/L): 88.96
Lactic Acid P (g/L): 48.14
Maltose M (g/L): 14.50
Volume V (L): 6.08

t = 18.00 h

Biomass X (g/L): 86.12
Glucose S (g/L): 85.72
Lactic Acid P (g/L): 45.70
Maltose M (g/L): 12.87
Volume V (L): 5.48

t = 19.00 h

Biomass X (g/L): 82.05
Glucose S (g/L): 93.87
Lactic Acid P (g/L): 41.96
Maltose M (g/L): 14.68
Volume V (L): 6.46

t = 20.00 h

Biomass X (g/L): 82.17
Glucose S (g/L): 94.60
Lactic Acid P (g/L): 61.24
Maltose M (g/L): 12.03
Volume V (L): 6.54

t = 21.00 h

Biomass X (g/L): 85.39
Glucose S (g/L): 89.35
Lactic Acid P (g/L): 70.91
Maltose M (g/L): 12.98
Volume V (L): 6.40

t = 22.00 h

Biomass X (g/L): 84.39
Glucose S (g/L): 84.67
Lactic Acid P (g/L): 70.97
Maltose M (g/L): 13.08
Volume V (L): 5.93

t = 23.00 h

Biomass X (g/L): 96.55
Glucose S (g/L): 96.32
Lactic Acid P (g/L): 66.85
Maltose M (g/L): 13.61
Volume V (L): 6.91

t = 24.00 h

Biomass X (g/L): 86.48
Glucose S (g/L): 84.79
Lactic Acid P (g/L): 67.16
Maltose M (g/L): 13.30
Volume V (L): 7.21

t = 25.00 h

Biomass X (g/L): 71.80
Glucose S (g/L): 77.84
Lactic Acid P (g/L): 77.94
Maltose M (g/L): 12.98
Volume V (L): 6.58

t = 26.00 h

Biomass X (g/L): 102.55
Glucose S (g/L): 90.53
Lactic Acid P (g/L): 66.06
Maltose M (g/L): 12.99
Volume V (L): 6.77

t = 27.00 h

Biomass X (g/L): 90.84
Glucose S (g/L): 82.97
Lactic Acid P (g/L): 69.78
Maltose M (g/L): 12.32
Volume V (L): 6.72

t = 28.00 h

Biomass X (g/L): 81.98
Glucose S (g/L): 93.21
Lactic Acid P (g/L): 61.51
Maltose M (g/L): 11.92
Volume V (L): 6.90

t = 29.00 h

Biomass X (g/L): 75.93
Glucose S (g/L): 86.69
Lactic Acid P (g/L): 52.49
Maltose M (g/L): 13.17
Volume V (L): 6.73

t = 30.00 h

Biomass X (g/L): 92.38
Glucose S (g/L): 91.64
Lactic Acid P (g/L): 59.46
Maltose M (g/L): 13.07
Volume V (L): 6.82

t = 31.00 h

Biomass X (g/L): 87.69
Glucose S (g/L): 95.71
Lactic Acid P (g/L): 66.35
Maltose M (g/L): 13.52
Volume V (L): 7.22

t = 32.00 h

Biomass X (g/L): 104.45
Glucose S (g/L): 83.28
Lactic Acid P (g/L): 67.57
Maltose M (g/L): 12.01
Volume V (L): 7.81

t = 33.00 h

Biomass X (g/L): 85.95
Glucose S (g/L): 86.66
Lactic Acid P (g/L): 85.50
Maltose M (g/L): 12.82
Volume V (L): 6.50

t = 34.00 h

Biomass X (g/L): 81.35
Glucose S (g/L): 89.33
Lactic Acid P (g/L): 68.81
Maltose M (g/L): 12.15

Volume V (L): 7.04

t = 35.00 h

Biomass X (g/L): 98.39

Glucose S (g/L): 97.47

Lactic Acid P (g/L): 79.52

Maltose M (g/L): 12.96

Volume V (L): 6.70

t = 36.00 h

Biomass X (g/L): 98.79

Glucose S (g/L): 87.14

Lactic Acid P (g/L): 64.94

Maltose M (g/L): 12.78

Volume V (L): 7.78

t = 37.00 h

Biomass X (g/L): 91.50

Glucose S (g/L): 87.79

Lactic Acid P (g/L): 66.52

Maltose M (g/L): 12.21

Volume V (L): 7.04

t = 38.00 h

Biomass X (g/L): 96.51

Glucose S (g/L): 93.36

Lactic Acid P (g/L): 57.45

Maltose M (g/L): 11.81

Volume V (L): 6.79

t = 39.00 h

Biomass X (g/L): 89.58

Glucose S (g/L): 88.61

Lactic Acid P (g/L): 63.31

Maltose M (g/L): 12.51

Volume V (L): 7.61

t = 40.00 h

Biomass X (g/L): 92.24

Glucose S (g/L): 92.53

Lactic Acid P (g/L): 75.50

Maltose M (g/L): 12.25

Volume V (L): 8.08

t = 41.00 h

Biomass X (g/L): 101.30

Glucose S (g/L): 84.65

Lactic Acid P (g/L): 71.02

Maltose M (g/L): 11.37

Volume V (L): 7.75

t = 42.00 h

Biomass X (g/L): 96.01

Glucose S (g/L): 103.34

Lactic Acid P (g/L): 80.07

Maltose M (g/L): 10.91

Volume V (L): 7.64

t = 43.00 h

Biomass X (g/L): 88.22

Glucose S (g/L): 86.45

Lactic Acid P (g/L): 73.30

Maltose M (g/L): 11.46

Volume V (L): 7.77

t = 44.00 h

Biomass X (g/L): 102.99

Glucose S (g/L): 91.85

Lactic Acid P (g/L): 68.63

Maltose M (g/L): 12.47

Volume V (L): 8.20

t = 45.00 h

Biomass X (g/L): 66.67

Glucose S (g/L): 93.67

Lactic Acid P (g/L): 92.23

Maltose M (g/L): 12.18

Volume V (L): 7.43

t = 46.00 h

Biomass X (g/L): 109.64

Glucose S (g/L): 103.54

Lactic Acid P (g/L): 71.62

Maltose M (g/L): 10.75

Volume V (L): 7.77

t = 47.00 h

Biomass X (g/L): 72.90

Glucose S (g/L): 88.25

Lactic Acid P (g/L): 83.61

Maltose M (g/L): 10.65

Volume V (L): 7.70

t = 48.00 h

Biomass X (g/L): 83.30

Glucose S (g/L): 104.74

Lactic Acid P (g/L): 69.30

Maltose M (g/L): 12.05

Volume V (L): 7.50

t = 49.00 h

Biomass X (g/L): 88.89

Glucose S (g/L): 99.22

Lactic Acid P (g/L): 79.90

Maltose M (g/L): 10.53

Volume V (L): 7.87

t = 50.00 h

Biomass X (g/L): 87.72

Glucose S (g/L): 98.00

Lactic Acid P (g/L): 82.18

Maltose M (g/L): 11.45

Volume V (L): 8.46

t = 51.00 h

Biomass X (g/L): 98.82

Glucose S (g/L): 91.63

Lactic Acid P (g/L): 75.71

Maltose M (g/L): 10.71

Volume V (L): 8.32

t = 52.00 h

Biomass X (g/L): 78.54

Glucose S (g/L): 99.62
Lactic Acid P (g/L): 52.05
Maltose M (g/L): 11.98
Volume V (L): 8.41

t = 53.00 h

Biomass X (g/L): 91.23
Glucose S (g/L): 93.77
Lactic Acid P (g/L): 74.38
Maltose M (g/L): 10.48
Volume V (L): 8.29

t = 54.00 h

Biomass X (g/L): 82.89
Glucose S (g/L): 103.86
Lactic Acid P (g/L): 85.88
Maltose M (g/L): 11.57
Volume V (L): 8.49

t = 55.00 h

Biomass X (g/L): 91.23
Glucose S (g/L): 93.43
Lactic Acid P (g/L): 79.54
Maltose M (g/L): 10.93
Volume V (L): 8.06

t = 56.00 h

Biomass X (g/L): 87.54
Glucose S (g/L): 98.03
Lactic Acid P (g/L): 78.97
Maltose M (g/L): 10.90
Volume V (L): 8.31

t = 57.00 h

Biomass X (g/L): 88.89
Glucose S (g/L): 96.32
Lactic Acid P (g/L): 58.96
Maltose M (g/L): 9.51
Volume V (L): 8.55

t = 58.00 h

Biomass X (g/L): 82.44
Glucose S (g/L): 94.65
Lactic Acid P (g/L): 68.48
Maltose M (g/L): 9.30
Volume V (L): 8.64

t = 59.00 h

Biomass X (g/L): 82.63
Glucose S (g/L): 103.47
Lactic Acid P (g/L): 79.20
Maltose M (g/L): 10.43
Volume V (L): 8.72

t = 60.00 h

Biomass X (g/L): 90.21
Glucose S (g/L): 102.90
Lactic Acid P (g/L): 79.37
Maltose M (g/L): 10.65
Volume V (L): 8.61

t = 61.00 h

Biomass X (g/L): 96.40
Glucose S (g/L): 106.59
Lactic Acid P (g/L): 87.03
Maltose M (g/L): 9.77
Volume V (L): 8.59

t = 62.00 h

Biomass X (g/L): 81.17
Glucose S (g/L): 92.50
Lactic Acid P (g/L): 74.53
Maltose M (g/L): 9.73
Volume V (L): 9.94

t = 63.00 h

Biomass X (g/L): 78.32
Glucose S (g/L): 106.73
Lactic Acid P (g/L): 79.82
Maltose M (g/L): 10.66
Volume V (L): 9.18

t = 64.00 h

Biomass X (g/L): 92.56
Glucose S (g/L): 114.15
Lactic Acid P (g/L): 76.34
Maltose M (g/L): 10.60
Volume V (L): 9.06

t = 65.00 h

Biomass X (g/L): 86.05
Glucose S (g/L): 100.20
Lactic Acid P (g/L): 67.14
Maltose M (g/L): 10.12
Volume V (L): 8.88

t = 66.00 h

Biomass X (g/L): 94.24
Glucose S (g/L): 112.01
Lactic Acid P (g/L): 55.47
Maltose M (g/L): 10.61
Volume V (L): 8.86

t = 67.00 h

Biomass X (g/L): 77.10
Glucose S (g/L): 101.94
Lactic Acid P (g/L): 80.64
Maltose M (g/L): 9.38
Volume V (L): 9.04

t = 68.00 h

Biomass X (g/L): 68.36
Glucose S (g/L): 96.96
Lactic Acid P (g/L): 78.39
Maltose M (g/L): 9.82
Volume V (L): 9.26

t = 69.00 h

Biomass X (g/L): 80.29
Glucose S (g/L): 102.55
Lactic Acid P (g/L): 71.14
Maltose M (g/L): 10.97
Volume V (L): 8.52

t = 70.00 h

Biomass X (g/L): 77.79
Glucose S (g/L): 109.52
Lactic Acid P (g/L): 61.95
Maltose M (g/L): 9.96
Volume V (L): 8.70

t = 71.00 h

Biomass X (g/L): 84.27
Glucose S (g/L): 100.76
Lactic Acid P (g/L): 83.03
Maltose M (g/L): 9.94
Volume V (L): 9.10

t = 72.00 h

Biomass X (g/L): 78.60
Glucose S (g/L): 102.31
Lactic Acid P (g/L): 74.81
Maltose M (g/L): 11.19
Volume V (L): 9.47

t = 73.00 h

Biomass X (g/L): 91.81
Glucose S (g/L): 108.50
Lactic Acid P (g/L): 82.60
Maltose M (g/L): 8.92
Volume V (L): 9.37

t = 74.00 h

Biomass X (g/L): 81.55
Glucose S (g/L): 106.02
Lactic Acid P (g/L): 94.90
Maltose M (g/L): 10.18
Volume V (L): 10.06

t = 75.00 h

Biomass X (g/L): 85.60
Glucose S (g/L): 109.23
Lactic Acid P (g/L): 57.70
Maltose M (g/L): 10.65

Volume V (L): 9.04

Batch 24

Initial Conditions:

Batch: 24.00

X0 (g/L): 4.38

S0 (g/L): 126.94

M0 (g/L): 18.68

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.15

F_vol (L/h): 0.04

S_feed (g/L): 261.51

M_feed (g/L): 38.49

V_final (L): 7.84

Batch 24 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 125.41
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 19.14
Volume V (L): 4.93

t = 1.00 h

Biomass X (g/L): 21.55
Glucose S (g/L): 127.16
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 19.75
Volume V (L): 4.70

t = 2.00 h

Biomass X (g/L): 9.39
Glucose S (g/L): 128.71
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 17.93
Volume V (L): 5.37

t = 3.00 h

Biomass X (g/L): 19.74
Glucose S (g/L): 126.16
Lactic Acid P (g/L): 24.40
Maltose M (g/L): 19.33
Volume V (L): 5.10

t = 4.00 h

Biomass X (g/L): 34.90
Glucose S (g/L): 124.88
Lactic Acid P (g/L): 12.09
Maltose M (g/L): 17.30
Volume V (L): 5.11

t = 5.00 h

Biomass X (g/L): 10.58
Glucose S (g/L): 127.81

Lactic Acid P (g/L): 12.68

Maltose M (g/L): 18.81

Volume V (L): 5.39

t = 6.00 h

Biomass X (g/L): 20.98

Glucose S (g/L): 118.61

Lactic Acid P (g/L): 2.65

Maltose M (g/L): 18.18

Volume V (L): 5.32

t = 7.00 h

Biomass X (g/L): 23.91

Glucose S (g/L): 109.91

Lactic Acid P (g/L): 22.37

Maltose M (g/L): 16.51

Volume V (L): 5.01

t = 8.00 h

Biomass X (g/L): 11.61

Glucose S (g/L): 114.84

Lactic Acid P (g/L): 18.84

Maltose M (g/L): 17.38

Volume V (L): 5.29

t = 9.00 h

Biomass X (g/L): 33.44

Glucose S (g/L): 98.17

Lactic Acid P (g/L): 25.74

Maltose M (g/L): 16.89

Volume V (L): 5.26

t = 10.00 h

Biomass X (g/L): 50.55

Glucose S (g/L): 95.94

Lactic Acid P (g/L): 17.88

Maltose M (g/L): 17.30

Volume V (L): 4.98

t = 11.00 h

Biomass X (g/L): 51.76

Glucose S (g/L): 100.30
Lactic Acid P (g/L): 32.41
Maltose M (g/L): 16.40
Volume V (L): 5.26

t = 12.00 h

Biomass X (g/L): 56.98
Glucose S (g/L): 97.87
Lactic Acid P (g/L): 30.11
Maltose M (g/L): 16.16
Volume V (L): 5.41

t = 13.00 h

Biomass X (g/L): 50.88
Glucose S (g/L): 84.68
Lactic Acid P (g/L): 29.95
Maltose M (g/L): 17.43
Volume V (L): 5.47

t = 14.00 h

Biomass X (g/L): 51.74
Glucose S (g/L): 86.69
Lactic Acid P (g/L): 47.37
Maltose M (g/L): 16.02
Volume V (L): 5.19

t = 15.00 h

Biomass X (g/L): 74.87
Glucose S (g/L): 82.83
Lactic Acid P (g/L): 28.07
Maltose M (g/L): 16.65
Volume V (L): 5.83

t = 16.00 h

Biomass X (g/L): 59.84
Glucose S (g/L): 81.20
Lactic Acid P (g/L): 50.88
Maltose M (g/L): 15.59
Volume V (L): 5.58

t = 17.00 h

Biomass X (g/L): 70.71
Glucose S (g/L): 87.28
Lactic Acid P (g/L): 54.31
Maltose M (g/L): 17.06
Volume V (L): 5.65

t = 18.00 h

Biomass X (g/L): 71.85
Glucose S (g/L): 84.14
Lactic Acid P (g/L): 41.25
Maltose M (g/L): 14.91
Volume V (L): 5.77

t = 19.00 h

Biomass X (g/L): 87.38
Glucose S (g/L): 78.44
Lactic Acid P (g/L): 43.72
Maltose M (g/L): 14.81
Volume V (L): 5.54

t = 20.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 74.07
Lactic Acid P (g/L): 51.25
Maltose M (g/L): 16.50
Volume V (L): 5.86

t = 21.00 h

Biomass X (g/L): 62.92
Glucose S (g/L): 68.30
Lactic Acid P (g/L): 57.75
Maltose M (g/L): 14.68
Volume V (L): 5.71

t = 22.00 h

Biomass X (g/L): 65.63
Glucose S (g/L): 83.00
Lactic Acid P (g/L): 55.02
Maltose M (g/L): 16.29
Volume V (L): 5.63

t = 23.00 h

Biomass X (g/L): 85.16
Glucose S (g/L): 77.40
Lactic Acid P (g/L): 62.07
Maltose M (g/L): 15.55
Volume V (L): 6.00

t = 24.00 h

Biomass X (g/L): 79.86
Glucose S (g/L): 69.25
Lactic Acid P (g/L): 76.20
Maltose M (g/L): 14.52
Volume V (L): 6.25

t = 25.00 h

Biomass X (g/L): 77.01
Glucose S (g/L): 76.70
Lactic Acid P (g/L): 55.40
Maltose M (g/L): 12.97
Volume V (L): 6.11

t = 26.00 h

Biomass X (g/L): 93.11
Glucose S (g/L): 71.65
Lactic Acid P (g/L): 44.81
Maltose M (g/L): 14.37
Volume V (L): 6.20

t = 27.00 h

Biomass X (g/L): 85.54
Glucose S (g/L): 69.95
Lactic Acid P (g/L): 70.42
Maltose M (g/L): 14.35
Volume V (L): 5.90

t = 28.00 h

Biomass X (g/L): 84.99
Glucose S (g/L): 73.67
Lactic Acid P (g/L): 79.67
Maltose M (g/L): 15.14
Volume V (L): 5.70

t = 29.00 h

Biomass X (g/L): 93.17
Glucose S (g/L): 66.66
Lactic Acid P (g/L): 68.08
Maltose M (g/L): 15.38
Volume V (L): 6.44

t = 30.00 h

Biomass X (g/L): 82.00
Glucose S (g/L): 68.67
Lactic Acid P (g/L): 54.59
Maltose M (g/L): 13.31
Volume V (L): 6.60

t = 31.00 h

Biomass X (g/L): 102.49
Glucose S (g/L): 71.12
Lactic Acid P (g/L): 64.70
Maltose M (g/L): 14.47
Volume V (L): 6.53

t = 32.00 h

Biomass X (g/L): 78.87
Glucose S (g/L): 67.45
Lactic Acid P (g/L): 55.82
Maltose M (g/L): 14.32
Volume V (L): 6.14

t = 33.00 h

Biomass X (g/L): 94.13
Glucose S (g/L): 73.99
Lactic Acid P (g/L): 73.56
Maltose M (g/L): 14.47
Volume V (L): 5.99

t = 34.00 h

Biomass X (g/L): 102.49
Glucose S (g/L): 80.75
Lactic Acid P (g/L): 54.92
Maltose M (g/L): 13.87

Volume V (L): 6.05

t = 35.00 h

Biomass X (g/L): 74.05

Glucose S (g/L): 65.83

Lactic Acid P (g/L): 65.72

Maltose M (g/L): 11.67

Volume V (L): 5.94

t = 36.00 h

Biomass X (g/L): 84.95

Glucose S (g/L): 60.67

Lactic Acid P (g/L): 59.50

Maltose M (g/L): 13.09

Volume V (L): 6.34

t = 37.00 h

Biomass X (g/L): 100.44

Glucose S (g/L): 58.81

Lactic Acid P (g/L): 71.01

Maltose M (g/L): 12.13

Volume V (L): 6.77

t = 38.00 h

Biomass X (g/L): 100.82

Glucose S (g/L): 69.40

Lactic Acid P (g/L): 66.77

Maltose M (g/L): 12.84

Volume V (L): 6.41

t = 39.00 h

Biomass X (g/L): 87.40

Glucose S (g/L): 66.59

Lactic Acid P (g/L): 72.54

Maltose M (g/L): 13.75

Volume V (L): 6.55

t = 40.00 h

Biomass X (g/L): 81.80

Glucose S (g/L): 71.67

Lactic Acid P (g/L): 71.53

Maltose M (g/L): 13.08

Volume V (L): 6.83

t = 41.00 h

Biomass X (g/L): 99.25

Glucose S (g/L): 71.38

Lactic Acid P (g/L): 62.23

Maltose M (g/L): 13.70

Volume V (L): 6.46

t = 42.00 h

Biomass X (g/L): 101.69

Glucose S (g/L): 61.68

Lactic Acid P (g/L): 78.29

Maltose M (g/L): 12.91

Volume V (L): 7.05

t = 43.00 h

Biomass X (g/L): 77.71

Glucose S (g/L): 60.66

Lactic Acid P (g/L): 72.17

Maltose M (g/L): 11.41

Volume V (L): 7.08

t = 44.00 h

Biomass X (g/L): 77.37

Glucose S (g/L): 72.85

Lactic Acid P (g/L): 63.48

Maltose M (g/L): 10.27

Volume V (L): 6.64

t = 45.00 h

Biomass X (g/L): 92.05

Glucose S (g/L): 72.15

Lactic Acid P (g/L): 95.12

Maltose M (g/L): 12.46

Volume V (L): 6.57

t = 46.00 h

Biomass X (g/L): 96.20

Glucose S (g/L): 69.18

Lactic Acid P (g/L): 78.42

Maltose M (g/L): 12.06

Volume V (L): 6.90

t = 47.00 h

Biomass X (g/L): 86.69

Glucose S (g/L): 62.42

Lactic Acid P (g/L): 72.69

Maltose M (g/L): 11.32

Volume V (L): 6.70

t = 48.00 h

Biomass X (g/L): 96.69

Glucose S (g/L): 74.76

Lactic Acid P (g/L): 88.21

Maltose M (g/L): 10.61

Volume V (L): 6.56

t = 49.00 h

Biomass X (g/L): 82.42

Glucose S (g/L): 79.56

Lactic Acid P (g/L): 92.68

Maltose M (g/L): 11.35

Volume V (L): 7.15

t = 50.00 h

Biomass X (g/L): 67.98

Glucose S (g/L): 74.18

Lactic Acid P (g/L): 73.77

Maltose M (g/L): 11.73

Volume V (L): 7.17

t = 51.00 h

Biomass X (g/L): 85.24

Glucose S (g/L): 68.05

Lactic Acid P (g/L): 78.84

Maltose M (g/L): 12.38

Volume V (L): 6.97

t = 52.00 h

Biomass X (g/L): 90.16

Glucose S (g/L): 73.40
Lactic Acid P (g/L): 66.22
Maltose M (g/L): 11.89
Volume V (L): 6.68

t = 53.00 h

Biomass X (g/L): 86.60
Glucose S (g/L): 65.67
Lactic Acid P (g/L): 74.65
Maltose M (g/L): 13.02
Volume V (L): 6.73

t = 54.00 h

Biomass X (g/L): 77.45
Glucose S (g/L): 79.73
Lactic Acid P (g/L): 74.42
Maltose M (g/L): 10.68
Volume V (L): 7.07

t = 55.00 h

Biomass X (g/L): 70.61
Glucose S (g/L): 74.48
Lactic Acid P (g/L): 78.41
Maltose M (g/L): 12.24
Volume V (L): 6.97

t = 56.00 h

Biomass X (g/L): 83.83
Glucose S (g/L): 64.26
Lactic Acid P (g/L): 80.10
Maltose M (g/L): 11.45
Volume V (L): 7.04

t = 57.00 h

Biomass X (g/L): 98.17
Glucose S (g/L): 72.90
Lactic Acid P (g/L): 82.55
Maltose M (g/L): 11.93
Volume V (L): 7.39

t = 58.00 h

Biomass X (g/L): 101.70
Glucose S (g/L): 77.55
Lactic Acid P (g/L): 79.68
Maltose M (g/L): 10.88
Volume V (L): 7.36

t = 59.00 h

Biomass X (g/L): 100.86
Glucose S (g/L): 73.34
Lactic Acid P (g/L): 65.90
Maltose M (g/L): 12.10
Volume V (L): 7.55

t = 60.00 h

Biomass X (g/L): 80.18
Glucose S (g/L): 73.97
Lactic Acid P (g/L): 68.76
Maltose M (g/L): 11.36
Volume V (L): 7.05

t = 61.00 h

Biomass X (g/L): 76.57
Glucose S (g/L): 68.90
Lactic Acid P (g/L): 72.78
Maltose M (g/L): 10.45
Volume V (L): 7.23

t = 62.00 h

Biomass X (g/L): 93.31
Glucose S (g/L): 79.39
Lactic Acid P (g/L): 75.55
Maltose M (g/L): 10.95
Volume V (L): 7.52

t = 63.00 h

Biomass X (g/L): 94.56
Glucose S (g/L): 72.81
Lactic Acid P (g/L): 88.66
Maltose M (g/L): 10.69
Volume V (L): 7.49

t = 64.00 h

Biomass X (g/L): 91.98
Glucose S (g/L): 64.56
Lactic Acid P (g/L): 67.45
Maltose M (g/L): 11.24
Volume V (L): 7.77

t = 65.00 h

Biomass X (g/L): 67.07
Glucose S (g/L): 75.02
Lactic Acid P (g/L): 75.44
Maltose M (g/L): 9.74
Volume V (L): 7.56

t = 66.00 h

Biomass X (g/L): 78.64
Glucose S (g/L): 83.03
Lactic Acid P (g/L): 88.32
Maltose M (g/L): 10.66
Volume V (L): 7.92

t = 67.00 h

Biomass X (g/L): 87.03
Glucose S (g/L): 75.58
Lactic Acid P (g/L): 76.38
Maltose M (g/L): 10.96
Volume V (L): 7.66

t = 68.00 h

Biomass X (g/L): 92.13
Glucose S (g/L): 87.98
Lactic Acid P (g/L): 83.92
Maltose M (g/L): 10.58
Volume V (L): 7.82

t = 69.00 h

Biomass X (g/L): 94.80
Glucose S (g/L): 80.86
Lactic Acid P (g/L): 78.30
Maltose M (g/L): 11.64
Volume V (L): 7.54

t = 70.00 h

Biomass X (g/L): 88.22
Glucose S (g/L): 80.67
Lactic Acid P (g/L): 82.31
Maltose M (g/L): 11.09
Volume V (L): 7.66

t = 71.00 h

Biomass X (g/L): 89.93
Glucose S (g/L): 78.34
Lactic Acid P (g/L): 77.78
Maltose M (g/L): 10.36
Volume V (L): 7.90

t = 72.00 h

Biomass X (g/L): 87.21
Glucose S (g/L): 79.01
Lactic Acid P (g/L): 86.62
Maltose M (g/L): 10.48
Volume V (L): 8.10

t = 73.00 h

Biomass X (g/L): 93.15
Glucose S (g/L): 78.23
Lactic Acid P (g/L): 65.37
Maltose M (g/L): 10.49
Volume V (L): 7.93

t = 74.00 h

Biomass X (g/L): 89.13
Glucose S (g/L): 78.33
Lactic Acid P (g/L): 76.97
Maltose M (g/L): 10.52
Volume V (L): 7.79

t = 75.00 h

Biomass X (g/L): 86.38
Glucose S (g/L): 83.39
Lactic Acid P (g/L): 83.51
Maltose M (g/L): 8.55

Volume V (L): 8.16

Batch 25

Initial Conditions:

Batch: 25.00

X0 (g/L): 5.92

S0 (g/L): 46.00

M0 (g/L): 65.91

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.43

F_vol (L/h): 0.02

S_feed (g/L): 123.32

M_feed (g/L): 176.68

V_final (L): 6.62

Batch 25 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 52.31
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 67.83
Volume V (L): 5.11

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 51.42
Lactic Acid P (g/L): 11.24
Maltose M (g/L): 64.07
Volume V (L): 5.02

t = 2.00 h

Biomass X (g/L): 16.88
Glucose S (g/L): 34.48
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 65.36
Volume V (L): 4.92

t = 3.00 h

Biomass X (g/L): 13.55
Glucose S (g/L): 41.32
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 65.09
Volume V (L): 4.99

t = 4.00 h

Biomass X (g/L): 22.76
Glucose S (g/L): 47.97
Lactic Acid P (g/L): 4.06
Maltose M (g/L): 59.66
Volume V (L): 4.83

t = 5.00 h

Biomass X (g/L): 21.64
Glucose S (g/L): 36.63

Lactic Acid P (g/L): 14.81
Maltose M (g/L): 62.64
Volume V (L): 5.37

t = 6.00 h

Biomass X (g/L): 30.87
Glucose S (g/L): 33.61
Lactic Acid P (g/L): 9.88
Maltose M (g/L): 54.58
Volume V (L): 5.24

t = 7.00 h

Biomass X (g/L): 22.27
Glucose S (g/L): 31.26
Lactic Acid P (g/L): 14.75
Maltose M (g/L): 62.22
Volume V (L): 5.04

t = 8.00 h

Biomass X (g/L): 40.12
Glucose S (g/L): 24.51
Lactic Acid P (g/L): 31.95
Maltose M (g/L): 62.09
Volume V (L): 5.14

t = 9.00 h

Biomass X (g/L): 48.72
Glucose S (g/L): 18.86
Lactic Acid P (g/L): 25.52
Maltose M (g/L): 61.69
Volume V (L): 5.11

t = 10.00 h

Biomass X (g/L): 55.21
Glucose S (g/L): 27.08
Lactic Acid P (g/L): 19.79
Maltose M (g/L): 57.43
Volume V (L): 5.42

t = 11.00 h

Biomass X (g/L): 66.82

Glucose S (g/L): 12.33
Lactic Acid P (g/L): 37.52
Maltose M (g/L): 59.01
Volume V (L): 5.05

t = 12.00 h

Biomass X (g/L): 53.67
Glucose S (g/L): 18.34
Lactic Acid P (g/L): 35.60
Maltose M (g/L): 61.90
Volume V (L): 5.27

t = 13.00 h

Biomass X (g/L): 57.31
Glucose S (g/L): 18.70
Lactic Acid P (g/L): 12.45
Maltose M (g/L): 61.24
Volume V (L): 5.27

t = 14.00 h

Biomass X (g/L): 43.19
Glucose S (g/L): 7.14
Lactic Acid P (g/L): 42.81
Maltose M (g/L): 56.09
Volume V (L): 5.41

t = 15.00 h

Biomass X (g/L): 55.31
Glucose S (g/L): 7.63
Lactic Acid P (g/L): 44.63
Maltose M (g/L): 56.03
Volume V (L): 5.31

t = 16.00 h

Biomass X (g/L): 60.89
Glucose S (g/L): 15.01
Lactic Acid P (g/L): 43.06
Maltose M (g/L): 58.51
Volume V (L): 5.12

t = 17.00 h

Biomass X (g/L): 74.81
Glucose S (g/L): 11.06
Lactic Acid P (g/L): 53.09
Maltose M (g/L): 49.70
Volume V (L): 5.34

t = 18.00 h

Biomass X (g/L): 60.81
Glucose S (g/L): 5.40
Lactic Acid P (g/L): 39.87
Maltose M (g/L): 52.86
Volume V (L): 5.26

t = 19.00 h

Biomass X (g/L): 75.83
Glucose S (g/L): 12.10
Lactic Acid P (g/L): 50.89
Maltose M (g/L): 57.54
Volume V (L): 5.42

t = 20.00 h

Biomass X (g/L): 69.65
Glucose S (g/L): 11.52
Lactic Acid P (g/L): 34.17
Maltose M (g/L): 55.86
Volume V (L): 5.62

t = 21.00 h

Biomass X (g/L): 69.24
Glucose S (g/L): 7.20
Lactic Acid P (g/L): 54.11
Maltose M (g/L): 53.43
Volume V (L): 5.23

t = 22.00 h

Biomass X (g/L): 74.44
Glucose S (g/L): 2.03
Lactic Acid P (g/L): 69.90
Maltose M (g/L): 47.00
Volume V (L): 5.37

t = 23.00 h

Biomass X (g/L): 69.36
Glucose S (g/L): 3.76
Lactic Acid P (g/L): 43.01
Maltose M (g/L): 51.08
Volume V (L): 5.43

t = 24.00 h

Biomass X (g/L): 69.92
Glucose S (g/L): 6.43
Lactic Acid P (g/L): 46.23
Maltose M (g/L): 48.88
Volume V (L): 5.47

t = 25.00 h

Biomass X (g/L): 58.61
Glucose S (g/L): 4.34
Lactic Acid P (g/L): 47.64
Maltose M (g/L): 51.30
Volume V (L): 5.64

t = 26.00 h

Biomass X (g/L): 73.26
Glucose S (g/L): 8.88
Lactic Acid P (g/L): 56.50
Maltose M (g/L): 47.22
Volume V (L): 5.54

t = 27.00 h

Biomass X (g/L): 71.77
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 46.04
Maltose M (g/L): 46.07
Volume V (L): 5.64

t = 28.00 h

Biomass X (g/L): 72.09
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 60.93
Maltose M (g/L): 42.34
Volume V (L): 5.52

t = 29.00 h

Biomass X (g/L): 81.59
Glucose S (g/L): 5.71
Lactic Acid P (g/L): 65.75
Maltose M (g/L): 52.25
Volume V (L): 5.66

t = 30.00 h

Biomass X (g/L): 84.47
Glucose S (g/L): 2.99
Lactic Acid P (g/L): 63.55
Maltose M (g/L): 46.69
Volume V (L): 5.75

t = 31.00 h

Biomass X (g/L): 80.95
Glucose S (g/L): 2.49
Lactic Acid P (g/L): 42.45
Maltose M (g/L): 48.01
Volume V (L): 5.58

t = 32.00 h

Biomass X (g/L): 88.72
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 69.36
Maltose M (g/L): 47.08
Volume V (L): 5.79

t = 33.00 h

Biomass X (g/L): 85.20
Glucose S (g/L): 5.31
Lactic Acid P (g/L): 69.51
Maltose M (g/L): 43.31
Volume V (L): 5.72

t = 34.00 h

Biomass X (g/L): 77.35
Glucose S (g/L): 5.12
Lactic Acid P (g/L): 69.16
Maltose M (g/L): 42.29

Volume V (L): 5.87

t = 35.00 h

Biomass X (g/L): 76.17

Glucose S (g/L): 8.13

Lactic Acid P (g/L): 65.48

Maltose M (g/L): 42.69

Volume V (L): 5.73

t = 36.00 h

Biomass X (g/L): 83.96

Glucose S (g/L): 5.86

Lactic Acid P (g/L): 64.68

Maltose M (g/L): 43.18

Volume V (L): 5.83

t = 37.00 h

Biomass X (g/L): 86.82

Glucose S (g/L): 5.17

Lactic Acid P (g/L): 73.00

Maltose M (g/L): 46.23

Volume V (L): 5.80

t = 38.00 h

Biomass X (g/L): 95.03

Glucose S (g/L): 10.69

Lactic Acid P (g/L): 82.06

Maltose M (g/L): 42.18

Volume V (L): 5.81

t = 39.00 h

Biomass X (g/L): 93.39

Glucose S (g/L): 0.63

Lactic Acid P (g/L): 55.38

Maltose M (g/L): 49.83

Volume V (L): 5.88

t = 40.00 h

Biomass X (g/L): 77.19

Glucose S (g/L): 8.38

Lactic Acid P (g/L): 75.21

Maltose M (g/L): 47.19

Volume V (L): 5.93

t = 41.00 h

Biomass X (g/L): 81.67

Glucose S (g/L): 0.00

Lactic Acid P (g/L): 64.74

Maltose M (g/L): 46.14

Volume V (L): 5.72

t = 42.00 h

Biomass X (g/L): 89.31

Glucose S (g/L): 3.08

Lactic Acid P (g/L): 64.88

Maltose M (g/L): 42.86

Volume V (L): 6.09

t = 43.00 h

Biomass X (g/L): 93.35

Glucose S (g/L): 1.73

Lactic Acid P (g/L): 52.09

Maltose M (g/L): 47.66

Volume V (L): 5.69

t = 44.00 h

Biomass X (g/L): 92.35

Glucose S (g/L): 10.11

Lactic Acid P (g/L): 64.48

Maltose M (g/L): 41.86

Volume V (L): 5.86

t = 45.00 h

Biomass X (g/L): 83.02

Glucose S (g/L): 6.82

Lactic Acid P (g/L): 58.25

Maltose M (g/L): 42.11

Volume V (L): 5.96

t = 46.00 h

Biomass X (g/L): 71.88

Glucose S (g/L): 7.24

Lactic Acid P (g/L): 63.81

Maltose M (g/L): 42.42

Volume V (L): 6.05

t = 47.00 h

Biomass X (g/L): 74.51

Glucose S (g/L): 2.96

Lactic Acid P (g/L): 58.29

Maltose M (g/L): 43.77

Volume V (L): 6.22

t = 48.00 h

Biomass X (g/L): 86.80

Glucose S (g/L): 6.48

Lactic Acid P (g/L): 64.93

Maltose M (g/L): 43.24

Volume V (L): 5.80

t = 49.00 h

Biomass X (g/L): 87.62

Glucose S (g/L): 4.80

Lactic Acid P (g/L): 65.81

Maltose M (g/L): 39.42

Volume V (L): 5.99

t = 50.00 h

Biomass X (g/L): 74.04

Glucose S (g/L): 9.73

Lactic Acid P (g/L): 71.04

Maltose M (g/L): 38.93

Volume V (L): 5.81

t = 51.00 h

Biomass X (g/L): 85.07

Glucose S (g/L): 4.31

Lactic Acid P (g/L): 55.26

Maltose M (g/L): 42.33

Volume V (L): 6.11

t = 52.00 h

Biomass X (g/L): 90.78

Glucose S (g/L): 14.07
Lactic Acid P (g/L): 57.31
Maltose M (g/L): 42.27
Volume V (L): 6.16

t = 53.00 h

Biomass X (g/L): 73.90
Glucose S (g/L): 1.97
Lactic Acid P (g/L): 74.88
Maltose M (g/L): 38.48
Volume V (L): 6.35

t = 54.00 h

Biomass X (g/L): 95.49
Glucose S (g/L): 5.78
Lactic Acid P (g/L): 64.31
Maltose M (g/L): 34.43
Volume V (L): 6.18

t = 55.00 h

Biomass X (g/L): 81.45
Glucose S (g/L): 16.22
Lactic Acid P (g/L): 82.48
Maltose M (g/L): 38.50
Volume V (L): 6.33

t = 56.00 h

Biomass X (g/L): 97.10
Glucose S (g/L): 7.79
Lactic Acid P (g/L): 74.83
Maltose M (g/L): 40.84
Volume V (L): 6.29

t = 57.00 h

Biomass X (g/L): 80.10
Glucose S (g/L): 4.43
Lactic Acid P (g/L): 78.84
Maltose M (g/L): 44.30
Volume V (L): 6.18

t = 58.00 h

Biomass X (g/L): 75.45
Glucose S (g/L): 10.98
Lactic Acid P (g/L): 86.86
Maltose M (g/L): 34.56
Volume V (L): 6.07

t = 59.00 h

Biomass X (g/L): 80.90
Glucose S (g/L): 8.39
Lactic Acid P (g/L): 70.92
Maltose M (g/L): 36.73
Volume V (L): 6.31

t = 60.00 h

Biomass X (g/L): 84.39
Glucose S (g/L): 7.17
Lactic Acid P (g/L): 68.79
Maltose M (g/L): 36.58
Volume V (L): 6.10

t = 61.00 h

Biomass X (g/L): 81.62
Glucose S (g/L): 6.37
Lactic Acid P (g/L): 82.86
Maltose M (g/L): 41.83
Volume V (L): 6.49

t = 62.00 h

Biomass X (g/L): 83.27
Glucose S (g/L): 22.49
Lactic Acid P (g/L): 79.79
Maltose M (g/L): 37.19
Volume V (L): 6.32

t = 63.00 h

Biomass X (g/L): 88.51
Glucose S (g/L): 10.37
Lactic Acid P (g/L): 73.53
Maltose M (g/L): 34.20
Volume V (L): 6.52

t = 64.00 h

Biomass X (g/L): 83.69
Glucose S (g/L): 3.77
Lactic Acid P (g/L): 88.84
Maltose M (g/L): 38.57
Volume V (L): 6.32

t = 65.00 h

Biomass X (g/L): 84.68
Glucose S (g/L): 18.08
Lactic Acid P (g/L): 73.97
Maltose M (g/L): 35.30
Volume V (L): 6.44

t = 66.00 h

Biomass X (g/L): 69.00
Glucose S (g/L): 14.73
Lactic Acid P (g/L): 76.63
Maltose M (g/L): 32.48
Volume V (L): 6.45

t = 67.00 h

Biomass X (g/L): 88.99
Glucose S (g/L): 9.35
Lactic Acid P (g/L): 76.84
Maltose M (g/L): 33.29
Volume V (L): 6.44

t = 68.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 8.95
Lactic Acid P (g/L): 68.10
Maltose M (g/L): 41.35
Volume V (L): 6.41

t = 69.00 h

Biomass X (g/L): 100.51
Glucose S (g/L): 15.82
Lactic Acid P (g/L): 78.00
Maltose M (g/L): 36.87
Volume V (L): 6.54

t = 70.00 h

Biomass X (g/L): 76.16
Glucose S (g/L): 12.33
Lactic Acid P (g/L): 90.81
Maltose M (g/L): 35.77
Volume V (L): 6.39

t = 71.00 h

Biomass X (g/L): 67.83
Glucose S (g/L): 5.72
Lactic Acid P (g/L): 82.09
Maltose M (g/L): 34.43
Volume V (L): 6.38

t = 72.00 h

Biomass X (g/L): 90.88
Glucose S (g/L): 21.25
Lactic Acid P (g/L): 78.46
Maltose M (g/L): 35.92
Volume V (L): 6.72

t = 73.00 h

Biomass X (g/L): 88.22
Glucose S (g/L): 12.42
Lactic Acid P (g/L): 76.92
Maltose M (g/L): 35.67
Volume V (L): 6.67

t = 74.00 h

Biomass X (g/L): 75.23
Glucose S (g/L): 20.62
Lactic Acid P (g/L): 81.87
Maltose M (g/L): 35.50
Volume V (L): 6.61

t = 75.00 h

Biomass X (g/L): 70.18
Glucose S (g/L): 20.02
Lactic Acid P (g/L): 86.12
Maltose M (g/L): 34.74

Volume V (L): 6.84

Batch 26

Initial Conditions:

Batch: 26.00

X0 (g/L): 5.42

S0 (g/L): 117.99

M0 (g/L): 75.31

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.64

F_vol (L/h): 0.08

S_feed (g/L): 183.12

M_feed (g/L): 116.88

V_final (L): 11.12

Batch 26 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 2.85
Glucose S (g/L): 112.44
Lactic Acid P (g/L): 1.74
Maltose M (g/L): 73.76
Volume V (L): 5.40

t = 1.00 h

Biomass X (g/L): 20.02
Glucose S (g/L): 121.11
Lactic Acid P (g/L): 15.10
Maltose M (g/L): 75.84
Volume V (L): 4.44

t = 2.00 h

Biomass X (g/L): 11.41
Glucose S (g/L): 114.73
Lactic Acid P (g/L): 7.01
Maltose M (g/L): 69.49
Volume V (L): 5.24

t = 3.00 h

Biomass X (g/L): 2.85
Glucose S (g/L): 116.60
Lactic Acid P (g/L): 17.19
Maltose M (g/L): 76.91
Volume V (L): 5.72

t = 4.00 h

Biomass X (g/L): 34.00
Glucose S (g/L): 121.49
Lactic Acid P (g/L): 13.53
Maltose M (g/L): 76.56
Volume V (L): 4.31

t = 5.00 h

Biomass X (g/L): 29.83
Glucose S (g/L): 109.67

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 67.41
Volume V (L): 5.31

t = 6.00 h

Biomass X (g/L): 23.63
Glucose S (g/L): 106.84
Lactic Acid P (g/L): 26.13
Maltose M (g/L): 75.44
Volume V (L): 5.21

t = 7.00 h

Biomass X (g/L): 36.52
Glucose S (g/L): 110.31
Lactic Acid P (g/L): 19.12
Maltose M (g/L): 69.72
Volume V (L): 5.23

t = 8.00 h

Biomass X (g/L): 30.51
Glucose S (g/L): 110.42
Lactic Acid P (g/L): 21.04
Maltose M (g/L): 70.93
Volume V (L): 5.84

t = 9.00 h

Biomass X (g/L): 47.14
Glucose S (g/L): 98.10
Lactic Acid P (g/L): 27.64
Maltose M (g/L): 67.20
Volume V (L): 6.28

t = 10.00 h

Biomass X (g/L): 20.83
Glucose S (g/L): 102.07
Lactic Acid P (g/L): 25.48
Maltose M (g/L): 71.32
Volume V (L): 5.75

t = 11.00 h

Biomass X (g/L): 38.35

Glucose S (g/L): 100.01
Lactic Acid P (g/L): 33.95
Maltose M (g/L): 68.95
Volume V (L): 6.00

t = 12.00 h

Biomass X (g/L): 61.58
Glucose S (g/L): 88.50
Lactic Acid P (g/L): 29.68
Maltose M (g/L): 64.23
Volume V (L): 6.04

t = 13.00 h

Biomass X (g/L): 53.47
Glucose S (g/L): 96.18
Lactic Acid P (g/L): 49.37
Maltose M (g/L): 62.43
Volume V (L): 6.06

t = 14.00 h

Biomass X (g/L): 61.22
Glucose S (g/L): 94.57
Lactic Acid P (g/L): 39.53
Maltose M (g/L): 60.84
Volume V (L): 5.61

t = 15.00 h

Biomass X (g/L): 49.31
Glucose S (g/L): 88.86
Lactic Acid P (g/L): 39.53
Maltose M (g/L): 65.78
Volume V (L): 6.77

t = 16.00 h

Biomass X (g/L): 82.70
Glucose S (g/L): 92.88
Lactic Acid P (g/L): 36.87
Maltose M (g/L): 66.49
Volume V (L): 6.45

t = 17.00 h

Biomass X (g/L): 78.82
Glucose S (g/L): 91.34
Lactic Acid P (g/L): 49.65
Maltose M (g/L): 66.18
Volume V (L): 6.58

t = 18.00 h

Biomass X (g/L): 67.62
Glucose S (g/L): 86.84
Lactic Acid P (g/L): 50.77
Maltose M (g/L): 63.04
Volume V (L): 6.07

t = 19.00 h

Biomass X (g/L): 67.59
Glucose S (g/L): 84.18
Lactic Acid P (g/L): 54.27
Maltose M (g/L): 59.00
Volume V (L): 6.33

t = 20.00 h

Biomass X (g/L): 83.09
Glucose S (g/L): 81.80
Lactic Acid P (g/L): 43.07
Maltose M (g/L): 60.05
Volume V (L): 6.55

t = 21.00 h

Biomass X (g/L): 93.78
Glucose S (g/L): 84.58
Lactic Acid P (g/L): 39.32
Maltose M (g/L): 65.67
Volume V (L): 5.96

t = 22.00 h

Biomass X (g/L): 85.26
Glucose S (g/L): 86.36
Lactic Acid P (g/L): 60.68
Maltose M (g/L): 59.65
Volume V (L): 5.97

t = 23.00 h

Biomass X (g/L): 98.18
Glucose S (g/L): 83.08
Lactic Acid P (g/L): 69.61
Maltose M (g/L): 61.24
Volume V (L): 6.58

t = 24.00 h

Biomass X (g/L): 69.53
Glucose S (g/L): 76.53
Lactic Acid P (g/L): 48.10
Maltose M (g/L): 56.63
Volume V (L): 7.21

t = 25.00 h

Biomass X (g/L): 71.48
Glucose S (g/L): 88.45
Lactic Acid P (g/L): 72.68
Maltose M (g/L): 60.87
Volume V (L): 6.72

t = 26.00 h

Biomass X (g/L): 68.79
Glucose S (g/L): 84.79
Lactic Acid P (g/L): 61.87
Maltose M (g/L): 57.71
Volume V (L): 6.62

t = 27.00 h

Biomass X (g/L): 85.20
Glucose S (g/L): 82.96
Lactic Acid P (g/L): 60.13
Maltose M (g/L): 62.27
Volume V (L): 6.80

t = 28.00 h

Biomass X (g/L): 83.32
Glucose S (g/L): 87.40
Lactic Acid P (g/L): 57.43
Maltose M (g/L): 60.88
Volume V (L): 7.06

t = 29.00 h

Biomass X (g/L): 80.76
Glucose S (g/L): 89.93
Lactic Acid P (g/L): 49.30
Maltose M (g/L): 56.15
Volume V (L): 7.19

t = 30.00 h

Biomass X (g/L): 90.52
Glucose S (g/L): 75.32
Lactic Acid P (g/L): 71.38
Maltose M (g/L): 56.92
Volume V (L): 8.51

t = 31.00 h

Biomass X (g/L): 95.04
Glucose S (g/L): 86.78
Lactic Acid P (g/L): 56.16
Maltose M (g/L): 55.04
Volume V (L): 8.05

t = 32.00 h

Biomass X (g/L): 68.95
Glucose S (g/L): 80.72
Lactic Acid P (g/L): 56.75
Maltose M (g/L): 51.67
Volume V (L): 8.11

t = 33.00 h

Biomass X (g/L): 78.22
Glucose S (g/L): 87.73
Lactic Acid P (g/L): 73.29
Maltose M (g/L): 53.11
Volume V (L): 8.27

t = 34.00 h

Biomass X (g/L): 92.13
Glucose S (g/L): 86.86
Lactic Acid P (g/L): 47.95
Maltose M (g/L): 56.22

Volume V (L): 7.29

t = 35.00 h

Biomass X (g/L): 80.02

Glucose S (g/L): 89.04

Lactic Acid P (g/L): 51.71

Maltose M (g/L): 56.21

Volume V (L): 7.26

t = 36.00 h

Biomass X (g/L): 77.06

Glucose S (g/L): 85.58

Lactic Acid P (g/L): 65.76

Maltose M (g/L): 55.19

Volume V (L): 7.40

t = 37.00 h

Biomass X (g/L): 94.70

Glucose S (g/L): 81.31

Lactic Acid P (g/L): 69.03

Maltose M (g/L): 53.26

Volume V (L): 8.67

t = 38.00 h

Biomass X (g/L): 72.69

Glucose S (g/L): 86.69

Lactic Acid P (g/L): 71.19

Maltose M (g/L): 56.99

Volume V (L): 7.83

t = 39.00 h

Biomass X (g/L): 82.30

Glucose S (g/L): 90.13

Lactic Acid P (g/L): 64.52

Maltose M (g/L): 53.66

Volume V (L): 7.57

t = 40.00 h

Biomass X (g/L): 88.45

Glucose S (g/L): 92.73

Lactic Acid P (g/L): 90.43

Maltose M (g/L): 55.41

Volume V (L): 8.05

t = 41.00 h

Biomass X (g/L): 79.86

Glucose S (g/L): 85.70

Lactic Acid P (g/L): 65.96

Maltose M (g/L): 46.70

Volume V (L): 8.23

t = 42.00 h

Biomass X (g/L): 92.94

Glucose S (g/L): 93.79

Lactic Acid P (g/L): 66.30

Maltose M (g/L): 39.10

Volume V (L): 8.56

t = 43.00 h

Biomass X (g/L): 79.99

Glucose S (g/L): 88.55

Lactic Acid P (g/L): 61.54

Maltose M (g/L): 52.82

Volume V (L): 8.63

t = 44.00 h

Biomass X (g/L): 98.99

Glucose S (g/L): 92.55

Lactic Acid P (g/L): 70.16

Maltose M (g/L): 46.71

Volume V (L): 8.19

t = 45.00 h

Biomass X (g/L): 84.25

Glucose S (g/L): 90.08

Lactic Acid P (g/L): 74.90

Maltose M (g/L): 50.40

Volume V (L): 9.01

t = 46.00 h

Biomass X (g/L): 91.89

Glucose S (g/L): 92.99

Lactic Acid P (g/L): 80.50
Maltose M (g/L): 49.73
Volume V (L): 8.44

t = 47.00 h

Biomass X (g/L): 92.57
Glucose S (g/L): 95.97
Lactic Acid P (g/L): 56.53
Maltose M (g/L): 55.70
Volume V (L): 8.76

t = 48.00 h

Biomass X (g/L): 79.33
Glucose S (g/L): 89.21
Lactic Acid P (g/L): 68.01
Maltose M (g/L): 51.35
Volume V (L): 9.25

t = 49.00 h

Biomass X (g/L): 65.57
Glucose S (g/L): 91.63
Lactic Acid P (g/L): 74.51
Maltose M (g/L): 50.21
Volume V (L): 8.69

t = 50.00 h

Biomass X (g/L): 88.69
Glucose S (g/L): 95.14
Lactic Acid P (g/L): 76.03
Maltose M (g/L): 52.11
Volume V (L): 8.80

t = 51.00 h

Biomass X (g/L): 85.84
Glucose S (g/L): 103.30
Lactic Acid P (g/L): 71.05
Maltose M (g/L): 50.19
Volume V (L): 9.45

t = 52.00 h

Biomass X (g/L): 90.39

Glucose S (g/L): 92.60
Lactic Acid P (g/L): 76.11
Maltose M (g/L): 47.08
Volume V (L): 8.82

t = 53.00 h

Biomass X (g/L): 82.77
Glucose S (g/L): 103.12
Lactic Acid P (g/L): 64.17
Maltose M (g/L): 51.87
Volume V (L): 9.14

t = 54.00 h

Biomass X (g/L): 80.71
Glucose S (g/L): 103.53
Lactic Acid P (g/L): 85.70
Maltose M (g/L): 44.21
Volume V (L): 9.68

t = 55.00 h

Biomass X (g/L): 104.02
Glucose S (g/L): 96.82
Lactic Acid P (g/L): 80.82
Maltose M (g/L): 48.69
Volume V (L): 10.50

t = 56.00 h

Biomass X (g/L): 70.46
Glucose S (g/L): 96.65
Lactic Acid P (g/L): 59.98
Maltose M (g/L): 42.09
Volume V (L): 9.87

t = 57.00 h

Biomass X (g/L): 91.52
Glucose S (g/L): 95.81
Lactic Acid P (g/L): 62.61
Maltose M (g/L): 47.64
Volume V (L): 9.88

t = 58.00 h

Biomass X (g/L): 78.82
Glucose S (g/L): 98.48
Lactic Acid P (g/L): 63.05
Maltose M (g/L): 47.28
Volume V (L): 9.99

t = 59.00 h

Biomass X (g/L): 80.97
Glucose S (g/L): 94.38
Lactic Acid P (g/L): 78.77
Maltose M (g/L): 51.95
Volume V (L): 9.29

t = 60.00 h

Biomass X (g/L): 81.01
Glucose S (g/L): 99.48
Lactic Acid P (g/L): 61.28
Maltose M (g/L): 40.76
Volume V (L): 9.95

t = 61.00 h

Biomass X (g/L): 76.19
Glucose S (g/L): 105.40
Lactic Acid P (g/L): 70.04
Maltose M (g/L): 43.69
Volume V (L): 9.96

t = 62.00 h

Biomass X (g/L): 89.24
Glucose S (g/L): 98.64
Lactic Acid P (g/L): 77.29
Maltose M (g/L): 48.16
Volume V (L): 10.22

t = 63.00 h

Biomass X (g/L): 96.95
Glucose S (g/L): 100.42
Lactic Acid P (g/L): 72.76
Maltose M (g/L): 46.22
Volume V (L): 10.20

t = 64.00 h

Biomass X (g/L): 87.94
Glucose S (g/L): 95.88
Lactic Acid P (g/L): 75.85
Maltose M (g/L): 45.41
Volume V (L): 10.41

t = 65.00 h

Biomass X (g/L): 84.26
Glucose S (g/L): 103.52
Lactic Acid P (g/L): 64.08
Maltose M (g/L): 41.30
Volume V (L): 10.15

t = 66.00 h

Biomass X (g/L): 73.62
Glucose S (g/L): 103.63
Lactic Acid P (g/L): 79.19
Maltose M (g/L): 42.24
Volume V (L): 10.42

t = 67.00 h

Biomass X (g/L): 87.89
Glucose S (g/L): 103.42
Lactic Acid P (g/L): 68.46
Maltose M (g/L): 48.11
Volume V (L): 9.87

t = 68.00 h

Biomass X (g/L): 103.47
Glucose S (g/L): 102.93
Lactic Acid P (g/L): 71.20
Maltose M (g/L): 43.48
Volume V (L): 10.39

t = 69.00 h

Biomass X (g/L): 78.92
Glucose S (g/L): 103.14
Lactic Acid P (g/L): 76.41
Maltose M (g/L): 44.63
Volume V (L): 11.31

t = 70.00 h

Biomass X (g/L): 78.79
Glucose S (g/L): 106.34
Lactic Acid P (g/L): 75.33
Maltose M (g/L): 42.71
Volume V (L): 10.87

t = 71.00 h

Biomass X (g/L): 76.83
Glucose S (g/L): 104.98
Lactic Acid P (g/L): 74.76
Maltose M (g/L): 38.85
Volume V (L): 10.79

t = 72.00 h

Biomass X (g/L): 77.34
Glucose S (g/L): 106.62
Lactic Acid P (g/L): 67.91
Maltose M (g/L): 45.68
Volume V (L): 11.81

t = 73.00 h

Biomass X (g/L): 55.49
Glucose S (g/L): 106.73
Lactic Acid P (g/L): 72.11
Maltose M (g/L): 43.97
Volume V (L): 10.29

t = 74.00 h

Biomass X (g/L): 85.57
Glucose S (g/L): 110.31
Lactic Acid P (g/L): 74.95
Maltose M (g/L): 49.75
Volume V (L): 10.50

t = 75.00 h

Biomass X (g/L): 73.37
Glucose S (g/L): 109.22
Lactic Acid P (g/L): 62.67
Maltose M (g/L): 38.35

Volume V (L): 11.96

Batch 27

Initial Conditions:

Batch: 27.00

X0 (g/L): 6.96

S0 (g/L): 168.26

M0 (g/L): 15.02

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.09

F_vol (L/h): 0.04

S_feed (g/L): 275.41

M_feed (g/L): 24.59

V_final (L): 7.98

Batch 27 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 7.43
Glucose S (g/L): 170.86
Lactic Acid P (g/L): 4.90
Maltose M (g/L): 14.96
Volume V (L): 4.85

t = 1.00 h

Biomass X (g/L): 7.02
Glucose S (g/L): 159.97
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 15.86
Volume V (L): 5.74

t = 2.00 h

Biomass X (g/L): 0.48
Glucose S (g/L): 169.02
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 14.04
Volume V (L): 4.77

t = 3.00 h

Biomass X (g/L): 21.16
Glucose S (g/L): 164.69
Lactic Acid P (g/L): 2.25
Maltose M (g/L): 15.89
Volume V (L): 4.86

t = 4.00 h

Biomass X (g/L): 26.67
Glucose S (g/L): 151.28
Lactic Acid P (g/L): 7.43
Maltose M (g/L): 13.66
Volume V (L): 5.21

t = 5.00 h

Biomass X (g/L): 14.99
Glucose S (g/L): 149.15

Lactic Acid P (g/L): 11.03

Maltose M (g/L): 14.08

Volume V (L): 5.70

t = 6.00 h

Biomass X (g/L): 30.05

Glucose S (g/L): 154.54

Lactic Acid P (g/L): 24.02

Maltose M (g/L): 13.59

Volume V (L): 5.04

t = 7.00 h

Biomass X (g/L): 33.50

Glucose S (g/L): 143.89

Lactic Acid P (g/L): 41.35

Maltose M (g/L): 13.73

Volume V (L): 5.22

t = 8.00 h

Biomass X (g/L): 44.88

Glucose S (g/L): 151.49

Lactic Acid P (g/L): 27.23

Maltose M (g/L): 13.81

Volume V (L): 5.58

t = 9.00 h

Biomass X (g/L): 35.98

Glucose S (g/L): 139.97

Lactic Acid P (g/L): 27.77

Maltose M (g/L): 13.62

Volume V (L): 5.35

t = 10.00 h

Biomass X (g/L): 64.94

Glucose S (g/L): 129.38

Lactic Acid P (g/L): 31.00

Maltose M (g/L): 12.50

Volume V (L): 5.32

t = 11.00 h

Biomass X (g/L): 65.91

Glucose S (g/L): 128.84
Lactic Acid P (g/L): 38.11
Maltose M (g/L): 12.04
Volume V (L): 5.57

t = 12.00 h

Biomass X (g/L): 56.88
Glucose S (g/L): 125.97
Lactic Acid P (g/L): 31.97
Maltose M (g/L): 14.77
Volume V (L): 5.54

t = 13.00 h

Biomass X (g/L): 61.98
Glucose S (g/L): 117.72
Lactic Acid P (g/L): 44.05
Maltose M (g/L): 12.22
Volume V (L): 5.76

t = 14.00 h

Biomass X (g/L): 66.04
Glucose S (g/L): 113.82
Lactic Acid P (g/L): 49.87
Maltose M (g/L): 13.01
Volume V (L): 5.87

t = 15.00 h

Biomass X (g/L): 62.38
Glucose S (g/L): 122.14
Lactic Acid P (g/L): 49.90
Maltose M (g/L): 11.02
Volume V (L): 5.67

t = 16.00 h

Biomass X (g/L): 68.37
Glucose S (g/L): 121.00
Lactic Acid P (g/L): 54.48
Maltose M (g/L): 11.93
Volume V (L): 5.71

t = 17.00 h

Biomass X (g/L): 72.43
Glucose S (g/L): 117.55
Lactic Acid P (g/L): 56.22
Maltose M (g/L): 11.90
Volume V (L): 5.73

t = 18.00 h

Biomass X (g/L): 93.67
Glucose S (g/L): 121.97
Lactic Acid P (g/L): 36.58
Maltose M (g/L): 12.24
Volume V (L): 5.83

t = 19.00 h

Biomass X (g/L): 88.70
Glucose S (g/L): 106.90
Lactic Acid P (g/L): 61.65
Maltose M (g/L): 9.54
Volume V (L): 5.77

t = 20.00 h

Biomass X (g/L): 90.19
Glucose S (g/L): 120.33
Lactic Acid P (g/L): 50.45
Maltose M (g/L): 11.72
Volume V (L): 5.80

t = 21.00 h

Biomass X (g/L): 85.87
Glucose S (g/L): 102.16
Lactic Acid P (g/L): 47.72
Maltose M (g/L): 10.53
Volume V (L): 6.12

t = 22.00 h

Biomass X (g/L): 72.00
Glucose S (g/L): 100.46
Lactic Acid P (g/L): 48.92
Maltose M (g/L): 11.58
Volume V (L): 6.24

t = 23.00 h

Biomass X (g/L): 84.10
Glucose S (g/L): 94.66
Lactic Acid P (g/L): 65.06
Maltose M (g/L): 10.36
Volume V (L): 5.56

t = 24.00 h

Biomass X (g/L): 95.42
Glucose S (g/L): 102.64
Lactic Acid P (g/L): 59.79
Maltose M (g/L): 10.38
Volume V (L): 5.96

t = 25.00 h

Biomass X (g/L): 73.61
Glucose S (g/L): 101.62
Lactic Acid P (g/L): 46.62
Maltose M (g/L): 10.72
Volume V (L): 5.83

t = 26.00 h

Biomass X (g/L): 93.71
Glucose S (g/L): 107.89
Lactic Acid P (g/L): 68.58
Maltose M (g/L): 9.98
Volume V (L): 5.96

t = 27.00 h

Biomass X (g/L): 111.60
Glucose S (g/L): 105.54
Lactic Acid P (g/L): 50.83
Maltose M (g/L): 10.32
Volume V (L): 6.17

t = 28.00 h

Biomass X (g/L): 99.38
Glucose S (g/L): 95.61
Lactic Acid P (g/L): 67.02
Maltose M (g/L): 9.63
Volume V (L): 6.27

t = 29.00 h

Biomass X (g/L): 103.51
Glucose S (g/L): 100.57
Lactic Acid P (g/L): 68.76
Maltose M (g/L): 9.01
Volume V (L): 6.13

t = 30.00 h

Biomass X (g/L): 91.81
Glucose S (g/L): 100.42
Lactic Acid P (g/L): 75.78
Maltose M (g/L): 11.17
Volume V (L): 5.74

t = 31.00 h

Biomass X (g/L): 99.87
Glucose S (g/L): 90.86
Lactic Acid P (g/L): 85.19
Maltose M (g/L): 9.22
Volume V (L): 6.00

t = 32.00 h

Biomass X (g/L): 83.06
Glucose S (g/L): 95.94
Lactic Acid P (g/L): 55.25
Maltose M (g/L): 10.00
Volume V (L): 6.49

t = 33.00 h

Biomass X (g/L): 98.80
Glucose S (g/L): 97.17
Lactic Acid P (g/L): 68.62
Maltose M (g/L): 10.64
Volume V (L): 5.66

t = 34.00 h

Biomass X (g/L): 97.63
Glucose S (g/L): 111.66
Lactic Acid P (g/L): 67.03
Maltose M (g/L): 9.45

Volume V (L): 6.64

t = 35.00 h

Biomass X (g/L): 94.14

Glucose S (g/L): 103.34

Lactic Acid P (g/L): 74.49

Maltose M (g/L): 9.18

Volume V (L): 6.52

t = 36.00 h

Biomass X (g/L): 104.79

Glucose S (g/L): 106.40

Lactic Acid P (g/L): 72.89

Maltose M (g/L): 9.92

Volume V (L): 6.75

t = 37.00 h

Biomass X (g/L): 86.04

Glucose S (g/L): 92.04

Lactic Acid P (g/L): 67.86

Maltose M (g/L): 10.54

Volume V (L): 6.42

t = 38.00 h

Biomass X (g/L): 89.75

Glucose S (g/L): 109.89

Lactic Acid P (g/L): 67.48

Maltose M (g/L): 8.61

Volume V (L): 6.02

t = 39.00 h

Biomass X (g/L): 92.73

Glucose S (g/L): 110.61

Lactic Acid P (g/L): 61.36

Maltose M (g/L): 9.34

Volume V (L): 6.65

t = 40.00 h

Biomass X (g/L): 81.33

Glucose S (g/L): 88.77

Lactic Acid P (g/L): 82.26

Maltose M (g/L): 7.98

Volume V (L): 6.90

t = 41.00 h

Biomass X (g/L): 94.18

Glucose S (g/L): 97.39

Lactic Acid P (g/L): 77.53

Maltose M (g/L): 8.96

Volume V (L): 6.85

t = 42.00 h

Biomass X (g/L): 74.37

Glucose S (g/L): 105.42

Lactic Acid P (g/L): 82.38

Maltose M (g/L): 9.22

Volume V (L): 6.51

t = 43.00 h

Biomass X (g/L): 102.56

Glucose S (g/L): 106.49

Lactic Acid P (g/L): 70.82

Maltose M (g/L): 8.42

Volume V (L): 6.73

t = 44.00 h

Biomass X (g/L): 92.34

Glucose S (g/L): 97.82

Lactic Acid P (g/L): 67.82

Maltose M (g/L): 8.65

Volume V (L): 6.62

t = 45.00 h

Biomass X (g/L): 86.52

Glucose S (g/L): 100.40

Lactic Acid P (g/L): 64.07

Maltose M (g/L): 7.81

Volume V (L): 6.59

t = 46.00 h

Biomass X (g/L): 83.27

Glucose S (g/L): 98.59

Lactic Acid P (g/L): 78.37

Maltose M (g/L): 8.92

Volume V (L): 7.06

t = 47.00 h

Biomass X (g/L): 89.32

Glucose S (g/L): 103.41

Lactic Acid P (g/L): 65.10

Maltose M (g/L): 9.62

Volume V (L): 6.58

t = 48.00 h

Biomass X (g/L): 98.64

Glucose S (g/L): 110.75

Lactic Acid P (g/L): 93.49

Maltose M (g/L): 7.82

Volume V (L): 6.70

t = 49.00 h

Biomass X (g/L): 97.11

Glucose S (g/L): 97.86

Lactic Acid P (g/L): 63.10

Maltose M (g/L): 8.12

Volume V (L): 6.69

t = 50.00 h

Biomass X (g/L): 84.23

Glucose S (g/L): 109.76

Lactic Acid P (g/L): 78.76

Maltose M (g/L): 8.95

Volume V (L): 6.63

t = 51.00 h

Biomass X (g/L): 82.68

Glucose S (g/L): 98.11

Lactic Acid P (g/L): 91.61

Maltose M (g/L): 9.44

Volume V (L): 7.06

t = 52.00 h

Biomass X (g/L): 97.87

Glucose S (g/L): 97.84
Lactic Acid P (g/L): 82.47
Maltose M (g/L): 7.20
Volume V (L): 7.35

t = 53.00 h

Biomass X (g/L): 83.67
Glucose S (g/L): 109.06
Lactic Acid P (g/L): 78.61
Maltose M (g/L): 8.19
Volume V (L): 7.06

t = 54.00 h

Biomass X (g/L): 91.50
Glucose S (g/L): 109.93
Lactic Acid P (g/L): 67.74
Maltose M (g/L): 8.48
Volume V (L): 6.79

t = 55.00 h

Biomass X (g/L): 91.99
Glucose S (g/L): 109.65
Lactic Acid P (g/L): 63.96
Maltose M (g/L): 8.11
Volume V (L): 7.39

t = 56.00 h

Biomass X (g/L): 93.75
Glucose S (g/L): 95.48
Lactic Acid P (g/L): 85.12
Maltose M (g/L): 6.14
Volume V (L): 7.63

t = 57.00 h

Biomass X (g/L): 99.14
Glucose S (g/L): 101.30
Lactic Acid P (g/L): 92.50
Maltose M (g/L): 8.45
Volume V (L): 7.11

t = 58.00 h

Biomass X (g/L): 77.35
Glucose S (g/L): 110.05
Lactic Acid P (g/L): 73.81
Maltose M (g/L): 9.20
Volume V (L): 7.19

t = 59.00 h

Biomass X (g/L): 70.56
Glucose S (g/L): 105.07
Lactic Acid P (g/L): 67.22
Maltose M (g/L): 8.02
Volume V (L): 7.47

t = 60.00 h

Biomass X (g/L): 84.06
Glucose S (g/L): 97.88
Lactic Acid P (g/L): 54.60
Maltose M (g/L): 7.04
Volume V (L): 7.77

t = 61.00 h

Biomass X (g/L): 108.01
Glucose S (g/L): 108.25
Lactic Acid P (g/L): 69.17
Maltose M (g/L): 7.39
Volume V (L): 7.30

t = 62.00 h

Biomass X (g/L): 110.06
Glucose S (g/L): 100.68
Lactic Acid P (g/L): 77.79
Maltose M (g/L): 9.15
Volume V (L): 7.15

t = 63.00 h

Biomass X (g/L): 107.91
Glucose S (g/L): 117.53
Lactic Acid P (g/L): 94.64
Maltose M (g/L): 8.40
Volume V (L): 7.66

t = 64.00 h

Biomass X (g/L): 87.07
Glucose S (g/L): 116.08
Lactic Acid P (g/L): 85.82
Maltose M (g/L): 8.13
Volume V (L): 7.74

t = 65.00 h

Biomass X (g/L): 70.66
Glucose S (g/L): 110.06
Lactic Acid P (g/L): 82.42
Maltose M (g/L): 7.85
Volume V (L): 7.59

t = 66.00 h

Biomass X (g/L): 64.73
Glucose S (g/L): 115.30
Lactic Acid P (g/L): 75.74
Maltose M (g/L): 6.97
Volume V (L): 7.49

t = 67.00 h

Biomass X (g/L): 82.14
Glucose S (g/L): 106.92
Lactic Acid P (g/L): 74.13
Maltose M (g/L): 7.29
Volume V (L): 7.90

t = 68.00 h

Biomass X (g/L): 86.15
Glucose S (g/L): 111.79
Lactic Acid P (g/L): 93.63
Maltose M (g/L): 7.33
Volume V (L): 7.83

t = 69.00 h

Biomass X (g/L): 101.20
Glucose S (g/L): 116.73
Lactic Acid P (g/L): 82.08
Maltose M (g/L): 7.78
Volume V (L): 7.56

t = 70.00 h

Biomass X (g/L): 96.57
Glucose S (g/L): 108.78
Lactic Acid P (g/L): 59.84
Maltose M (g/L): 8.49
Volume V (L): 7.80

t = 71.00 h

Biomass X (g/L): 89.81
Glucose S (g/L): 105.68
Lactic Acid P (g/L): 73.89
Maltose M (g/L): 8.12
Volume V (L): 7.50

t = 72.00 h

Biomass X (g/L): 78.19
Glucose S (g/L): 96.29
Lactic Acid P (g/L): 72.25
Maltose M (g/L): 6.04
Volume V (L): 8.03

t = 73.00 h

Biomass X (g/L): 77.24
Glucose S (g/L): 107.26
Lactic Acid P (g/L): 75.43
Maltose M (g/L): 7.31
Volume V (L): 7.84

t = 74.00 h

Biomass X (g/L): 89.38
Glucose S (g/L): 113.50
Lactic Acid P (g/L): 86.36
Maltose M (g/L): 7.67
Volume V (L): 7.84

t = 75.00 h

Biomass X (g/L): 93.66
Glucose S (g/L): 114.53
Lactic Acid P (g/L): 79.98
Maltose M (g/L): 7.52

Volume V (L): 7.98

Batch 28

Initial Conditions:

Batch: 28.00

X0 (g/L): 2.07

S0 (g/L): 42.52

M0 (g/L): 78.58

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.85

F_vol (L/h): 0.09

S_feed (g/L): 105.33

M_feed (g/L): 194.67

V_final (L): 11.47

Batch 28 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 2.46
Glucose S (g/L): 40.09
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 77.04
Volume V (L): 6.19

t = 1.00 h

Biomass X (g/L): 11.27
Glucose S (g/L): 49.21
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 77.82
Volume V (L): 4.34

t = 2.00 h

Biomass X (g/L): 9.23
Glucose S (g/L): 44.48
Lactic Acid P (g/L): 1.18
Maltose M (g/L): 77.86
Volume V (L): 5.84

t = 3.00 h

Biomass X (g/L): 3.22
Glucose S (g/L): 44.05
Lactic Acid P (g/L): 0.27
Maltose M (g/L): 79.92
Volume V (L): 5.60

t = 4.00 h

Biomass X (g/L): 13.41
Glucose S (g/L): 47.05
Lactic Acid P (g/L): 14.20
Maltose M (g/L): 78.40
Volume V (L): 5.10

t = 5.00 h

Biomass X (g/L): 11.71
Glucose S (g/L): 43.05

Lactic Acid P (g/L): 6.23
Maltose M (g/L): 79.31
Volume V (L): 6.27

t = 6.00 h

Biomass X (g/L): 5.98
Glucose S (g/L): 47.20
Lactic Acid P (g/L): 13.42
Maltose M (g/L): 79.51
Volume V (L): 4.56

t = 7.00 h

Biomass X (g/L): 11.44
Glucose S (g/L): 45.96
Lactic Acid P (g/L): 7.88
Maltose M (g/L): 80.40
Volume V (L): 5.33

t = 8.00 h

Biomass X (g/L): 14.21
Glucose S (g/L): 49.88
Lactic Acid P (g/L): 19.27
Maltose M (g/L): 78.65
Volume V (L): 5.06

t = 9.00 h

Biomass X (g/L): 23.80
Glucose S (g/L): 44.87
Lactic Acid P (g/L): 15.95
Maltose M (g/L): 81.24
Volume V (L): 5.99

t = 10.00 h

Biomass X (g/L): 19.17
Glucose S (g/L): 47.93
Lactic Acid P (g/L): 15.29
Maltose M (g/L): 83.16
Volume V (L): 5.29

t = 11.00 h

Biomass X (g/L): 22.83

Glucose S (g/L): 47.62
Lactic Acid P (g/L): 17.92
Maltose M (g/L): 81.50
Volume V (L): 6.17

t = 12.00 h

Biomass X (g/L): 49.32
Glucose S (g/L): 43.81
Lactic Acid P (g/L): 12.62
Maltose M (g/L): 83.75
Volume V (L): 5.94

t = 13.00 h

Biomass X (g/L): 43.52
Glucose S (g/L): 38.81
Lactic Acid P (g/L): 28.44
Maltose M (g/L): 83.45
Volume V (L): 5.87

t = 14.00 h

Biomass X (g/L): 31.22
Glucose S (g/L): 37.85
Lactic Acid P (g/L): 26.57
Maltose M (g/L): 82.59
Volume V (L): 6.41

t = 15.00 h

Biomass X (g/L): 47.67
Glucose S (g/L): 31.00
Lactic Acid P (g/L): 25.36
Maltose M (g/L): 81.47
Volume V (L): 6.15

t = 16.00 h

Biomass X (g/L): 46.90
Glucose S (g/L): 33.34
Lactic Acid P (g/L): 42.19
Maltose M (g/L): 81.35
Volume V (L): 5.82

t = 17.00 h

Biomass X (g/L): 52.82
Glucose S (g/L): 29.60
Lactic Acid P (g/L): 24.05
Maltose M (g/L): 82.45
Volume V (L): 7.92

t = 18.00 h

Biomass X (g/L): 53.46
Glucose S (g/L): 32.80
Lactic Acid P (g/L): 33.71
Maltose M (g/L): 80.58
Volume V (L): 6.30

t = 19.00 h

Biomass X (g/L): 66.30
Glucose S (g/L): 37.16
Lactic Acid P (g/L): 48.67
Maltose M (g/L): 78.16
Volume V (L): 5.49

t = 20.00 h

Biomass X (g/L): 60.82
Glucose S (g/L): 30.02
Lactic Acid P (g/L): 44.48
Maltose M (g/L): 81.75
Volume V (L): 6.29

t = 21.00 h

Biomass X (g/L): 59.27
Glucose S (g/L): 21.82
Lactic Acid P (g/L): 43.20
Maltose M (g/L): 79.10
Volume V (L): 6.16

t = 22.00 h

Biomass X (g/L): 71.93
Glucose S (g/L): 25.99
Lactic Acid P (g/L): 47.59
Maltose M (g/L): 79.23
Volume V (L): 7.32

t = 23.00 h

Biomass X (g/L): 66.15
Glucose S (g/L): 26.92
Lactic Acid P (g/L): 48.60
Maltose M (g/L): 80.58
Volume V (L): 7.15

t = 24.00 h

Biomass X (g/L): 66.83
Glucose S (g/L): 26.45
Lactic Acid P (g/L): 35.22
Maltose M (g/L): 80.27
Volume V (L): 7.28

t = 25.00 h

Biomass X (g/L): 70.26
Glucose S (g/L): 23.80
Lactic Acid P (g/L): 46.88
Maltose M (g/L): 78.04
Volume V (L): 6.96

t = 26.00 h

Biomass X (g/L): 67.82
Glucose S (g/L): 26.56
Lactic Acid P (g/L): 65.45
Maltose M (g/L): 80.31
Volume V (L): 7.10

t = 27.00 h

Biomass X (g/L): 70.10
Glucose S (g/L): 22.13
Lactic Acid P (g/L): 46.67
Maltose M (g/L): 81.37
Volume V (L): 7.14

t = 28.00 h

Biomass X (g/L): 65.61
Glucose S (g/L): 23.08
Lactic Acid P (g/L): 50.80
Maltose M (g/L): 77.90
Volume V (L): 7.37

t = 29.00 h

Biomass X (g/L): 65.12
Glucose S (g/L): 27.93
Lactic Acid P (g/L): 49.90
Maltose M (g/L): 81.42
Volume V (L): 7.93

t = 30.00 h

Biomass X (g/L): 63.75
Glucose S (g/L): 23.23
Lactic Acid P (g/L): 54.88
Maltose M (g/L): 76.79
Volume V (L): 7.76

t = 31.00 h

Biomass X (g/L): 60.89
Glucose S (g/L): 28.10
Lactic Acid P (g/L): 51.45
Maltose M (g/L): 82.29
Volume V (L): 7.93

t = 32.00 h

Biomass X (g/L): 66.58
Glucose S (g/L): 22.79
Lactic Acid P (g/L): 61.72
Maltose M (g/L): 78.11
Volume V (L): 8.02

t = 33.00 h

Biomass X (g/L): 74.04
Glucose S (g/L): 32.44
Lactic Acid P (g/L): 60.00
Maltose M (g/L): 78.73
Volume V (L): 8.30

t = 34.00 h

Biomass X (g/L): 77.36
Glucose S (g/L): 25.36
Lactic Acid P (g/L): 65.83
Maltose M (g/L): 75.30

Volume V (L): 8.08

t = 35.00 h

Biomass X (g/L): 74.39

Glucose S (g/L): 27.45

Lactic Acid P (g/L): 50.54

Maltose M (g/L): 77.90

Volume V (L): 8.20

t = 36.00 h

Biomass X (g/L): 55.07

Glucose S (g/L): 29.59

Lactic Acid P (g/L): 57.47

Maltose M (g/L): 78.09

Volume V (L): 8.72

t = 37.00 h

Biomass X (g/L): 93.04

Glucose S (g/L): 27.47

Lactic Acid P (g/L): 59.13

Maltose M (g/L): 77.33

Volume V (L): 7.85

t = 38.00 h

Biomass X (g/L): 88.84

Glucose S (g/L): 30.12

Lactic Acid P (g/L): 65.74

Maltose M (g/L): 76.31

Volume V (L): 7.83

t = 39.00 h

Biomass X (g/L): 83.14

Glucose S (g/L): 31.98

Lactic Acid P (g/L): 55.35

Maltose M (g/L): 75.44

Volume V (L): 7.75

t = 40.00 h

Biomass X (g/L): 97.44

Glucose S (g/L): 33.72

Lactic Acid P (g/L): 70.92

Maltose M (g/L): 78.39

Volume V (L): 8.54

t = 41.00 h

Biomass X (g/L): 70.26

Glucose S (g/L): 33.85

Lactic Acid P (g/L): 73.83

Maltose M (g/L): 76.39

Volume V (L): 7.58

t = 42.00 h

Biomass X (g/L): 97.87

Glucose S (g/L): 31.64

Lactic Acid P (g/L): 55.24

Maltose M (g/L): 78.46

Volume V (L): 8.86

t = 43.00 h

Biomass X (g/L): 70.94

Glucose S (g/L): 32.36

Lactic Acid P (g/L): 60.09

Maltose M (g/L): 74.94

Volume V (L): 8.65

t = 44.00 h

Biomass X (g/L): 82.65

Glucose S (g/L): 40.41

Lactic Acid P (g/L): 60.20

Maltose M (g/L): 74.79

Volume V (L): 9.38

t = 45.00 h

Biomass X (g/L): 68.44

Glucose S (g/L): 36.95

Lactic Acid P (g/L): 88.43

Maltose M (g/L): 76.44

Volume V (L): 8.53

t = 46.00 h

Biomass X (g/L): 80.90

Glucose S (g/L): 32.10

Lactic Acid P (g/L): 61.96

Maltose M (g/L): 72.34

Volume V (L): 8.86

t = 47.00 h

Biomass X (g/L): 84.60

Glucose S (g/L): 35.82

Lactic Acid P (g/L): 80.87

Maltose M (g/L): 72.67

Volume V (L): 9.63

t = 48.00 h

Biomass X (g/L): 79.58

Glucose S (g/L): 35.23

Lactic Acid P (g/L): 55.40

Maltose M (g/L): 75.97

Volume V (L): 9.86

t = 49.00 h

Biomass X (g/L): 87.98

Glucose S (g/L): 36.53

Lactic Acid P (g/L): 58.21

Maltose M (g/L): 72.12

Volume V (L): 8.87

t = 50.00 h

Biomass X (g/L): 82.21

Glucose S (g/L): 34.01

Lactic Acid P (g/L): 61.23

Maltose M (g/L): 70.81

Volume V (L): 9.41

t = 51.00 h

Biomass X (g/L): 76.37

Glucose S (g/L): 37.14

Lactic Acid P (g/L): 60.18

Maltose M (g/L): 76.24

Volume V (L): 9.55

t = 52.00 h

Biomass X (g/L): 76.01

Glucose S (g/L): 44.44
Lactic Acid P (g/L): 65.58
Maltose M (g/L): 70.85
Volume V (L): 9.68

t = 53.00 h

Biomass X (g/L): 89.19
Glucose S (g/L): 41.52
Lactic Acid P (g/L): 62.03
Maltose M (g/L): 72.99
Volume V (L): 9.43

t = 54.00 h

Biomass X (g/L): 75.26
Glucose S (g/L): 33.16
Lactic Acid P (g/L): 61.37
Maltose M (g/L): 72.50
Volume V (L): 10.32

t = 55.00 h

Biomass X (g/L): 90.24
Glucose S (g/L): 44.86
Lactic Acid P (g/L): 79.03
Maltose M (g/L): 67.70
Volume V (L): 8.77

t = 56.00 h

Biomass X (g/L): 93.18
Glucose S (g/L): 45.21
Lactic Acid P (g/L): 79.89
Maltose M (g/L): 71.35
Volume V (L): 9.61

t = 57.00 h

Biomass X (g/L): 65.61
Glucose S (g/L): 41.65
Lactic Acid P (g/L): 77.76
Maltose M (g/L): 72.15
Volume V (L): 10.15

t = 58.00 h

Biomass X (g/L): 92.30
Glucose S (g/L): 45.56
Lactic Acid P (g/L): 74.29
Maltose M (g/L): 71.68
Volume V (L): 10.24

t = 59.00 h

Biomass X (g/L): 101.12
Glucose S (g/L): 41.62
Lactic Acid P (g/L): 70.76
Maltose M (g/L): 72.22
Volume V (L): 10.79

t = 60.00 h

Biomass X (g/L): 83.85
Glucose S (g/L): 43.49
Lactic Acid P (g/L): 65.83
Maltose M (g/L): 69.89
Volume V (L): 10.79

t = 61.00 h

Biomass X (g/L): 87.11
Glucose S (g/L): 50.04
Lactic Acid P (g/L): 62.07
Maltose M (g/L): 71.29
Volume V (L): 10.11

t = 62.00 h

Biomass X (g/L): 91.23
Glucose S (g/L): 45.52
Lactic Acid P (g/L): 73.86
Maltose M (g/L): 68.79
Volume V (L): 11.02

t = 63.00 h

Biomass X (g/L): 78.66
Glucose S (g/L): 41.51
Lactic Acid P (g/L): 60.60
Maltose M (g/L): 67.35
Volume V (L): 10.06

t = 64.00 h

Biomass X (g/L): 82.96
Glucose S (g/L): 47.57
Lactic Acid P (g/L): 62.40
Maltose M (g/L): 67.20
Volume V (L): 11.00

t = 65.00 h

Biomass X (g/L): 68.62
Glucose S (g/L): 48.24
Lactic Acid P (g/L): 81.47
Maltose M (g/L): 66.63
Volume V (L): 11.54

t = 66.00 h

Biomass X (g/L): 82.39
Glucose S (g/L): 52.72
Lactic Acid P (g/L): 88.24
Maltose M (g/L): 70.17
Volume V (L): 11.32

t = 67.00 h

Biomass X (g/L): 85.73
Glucose S (g/L): 49.93
Lactic Acid P (g/L): 75.21
Maltose M (g/L): 65.92
Volume V (L): 10.83

t = 68.00 h

Biomass X (g/L): 85.81
Glucose S (g/L): 49.34
Lactic Acid P (g/L): 81.67
Maltose M (g/L): 68.39
Volume V (L): 10.69

t = 69.00 h

Biomass X (g/L): 90.95
Glucose S (g/L): 54.38
Lactic Acid P (g/L): 69.98
Maltose M (g/L): 67.24
Volume V (L): 10.16

t = 70.00 h

Biomass X (g/L): 73.30
Glucose S (g/L): 52.34
Lactic Acid P (g/L): 60.58
Maltose M (g/L): 67.18
Volume V (L): 11.20

t = 71.00 h

Biomass X (g/L): 75.51
Glucose S (g/L): 50.45
Lactic Acid P (g/L): 77.89
Maltose M (g/L): 69.15
Volume V (L): 10.55

t = 72.00 h

Biomass X (g/L): 98.54
Glucose S (g/L): 56.77
Lactic Acid P (g/L): 77.69
Maltose M (g/L): 66.03
Volume V (L): 10.80

t = 73.00 h

Biomass X (g/L): 78.92
Glucose S (g/L): 54.61
Lactic Acid P (g/L): 73.49
Maltose M (g/L): 67.36
Volume V (L): 10.68

t = 74.00 h

Biomass X (g/L): 82.15
Glucose S (g/L): 54.73
Lactic Acid P (g/L): 70.90
Maltose M (g/L): 66.15
Volume V (L): 11.14

t = 75.00 h

Biomass X (g/L): 79.48
Glucose S (g/L): 55.92
Lactic Acid P (g/L): 81.64
Maltose M (g/L): 65.55

Volume V (L): 11.45

Batch 29

Initial Conditions:

Batch: 29.00

X0 (g/L): 3.76

S0 (g/L): 11.92

M0 (g/L): 111.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 9.35

F_vol (L/h): 0.07

S_feed (g/L): 28.98

M_feed (g/L): 271.02

V_final (L): 10.41

Batch 29 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.25

Glucose S (g/L): 12.03

Lactic Acid P (g/L): 0.47

Maltose M (g/L): 109.71

Volume V (L): 5.40

t = 1.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 15.20

Lactic Acid P (g/L): 4.47

Maltose M (g/L): 113.11

Volume V (L): 5.60

t = 2.00 h

Biomass X (g/L): 15.49

Glucose S (g/L): 18.21

Lactic Acid P (g/L): 8.84

Maltose M (g/L): 114.96

Volume V (L): 4.87

t = 3.00 h

Biomass X (g/L): 5.22

Glucose S (g/L): 15.98

Lactic Acid P (g/L): 0.16

Maltose M (g/L): 112.45

Volume V (L): 5.56

t = 4.00 h

Biomass X (g/L): 1.58

Glucose S (g/L): 15.62

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 113.49

Volume V (L): 4.96

t = 5.00 h

Biomass X (g/L): 12.76

Glucose S (g/L): 11.54

Lactic Acid P (g/L): 1.09
Maltose M (g/L): 113.28
Volume V (L): 6.33

t = 6.00 h

Biomass X (g/L): 30.84
Glucose S (g/L): 16.34
Lactic Acid P (g/L): 7.28
Maltose M (g/L): 115.27
Volume V (L): 5.13

t = 7.00 h

Biomass X (g/L): 18.54
Glucose S (g/L): 16.05
Lactic Acid P (g/L): 19.27
Maltose M (g/L): 106.73
Volume V (L): 5.95

t = 8.00 h

Biomass X (g/L): 14.97
Glucose S (g/L): 16.74
Lactic Acid P (g/L): 5.99
Maltose M (g/L): 113.22
Volume V (L): 5.94

t = 9.00 h

Biomass X (g/L): 7.33
Glucose S (g/L): 12.56
Lactic Acid P (g/L): 17.46
Maltose M (g/L): 113.90
Volume V (L): 5.79

t = 10.00 h

Biomass X (g/L): 22.32
Glucose S (g/L): 12.16
Lactic Acid P (g/L): 17.86
Maltose M (g/L): 107.92
Volume V (L): 5.43

t = 11.00 h

Biomass X (g/L): 29.56

Glucose S (g/L): 8.49
Lactic Acid P (g/L): 18.08
Maltose M (g/L): 108.90
Volume V (L): 5.73

t = 12.00 h

Biomass X (g/L): 26.94
Glucose S (g/L): 5.38
Lactic Acid P (g/L): 27.99
Maltose M (g/L): 108.81
Volume V (L): 4.93

t = 13.00 h

Biomass X (g/L): 18.25
Glucose S (g/L): 7.19
Lactic Acid P (g/L): 16.59
Maltose M (g/L): 108.92
Volume V (L): 6.03

t = 14.00 h

Biomass X (g/L): 28.47
Glucose S (g/L): 5.10
Lactic Acid P (g/L): 19.48
Maltose M (g/L): 109.33
Volume V (L): 6.16

t = 15.00 h

Biomass X (g/L): 52.07
Glucose S (g/L): 8.92
Lactic Acid P (g/L): 26.34
Maltose M (g/L): 110.92
Volume V (L): 6.14

t = 16.00 h

Biomass X (g/L): 54.31
Glucose S (g/L): 6.15
Lactic Acid P (g/L): 22.86
Maltose M (g/L): 112.24
Volume V (L): 4.87

t = 17.00 h

Biomass X (g/L): 35.41
Glucose S (g/L): 5.57
Lactic Acid P (g/L): 34.49
Maltose M (g/L): 112.53
Volume V (L): 5.68

t = 18.00 h

Biomass X (g/L): 51.68
Glucose S (g/L): 4.07
Lactic Acid P (g/L): 25.69
Maltose M (g/L): 107.43
Volume V (L): 5.97

t = 19.00 h

Biomass X (g/L): 36.74
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 35.32
Maltose M (g/L): 107.13
Volume V (L): 6.50

t = 20.00 h

Biomass X (g/L): 62.04
Glucose S (g/L): 9.64
Lactic Acid P (g/L): 34.00
Maltose M (g/L): 112.26
Volume V (L): 6.57

t = 21.00 h

Biomass X (g/L): 53.15
Glucose S (g/L): 3.42
Lactic Acid P (g/L): 38.12
Maltose M (g/L): 110.27
Volume V (L): 6.44

t = 22.00 h

Biomass X (g/L): 61.14
Glucose S (g/L): 4.80
Lactic Acid P (g/L): 33.68
Maltose M (g/L): 108.54
Volume V (L): 6.86

t = 23.00 h

Biomass X (g/L): 38.36
Glucose S (g/L): 2.27
Lactic Acid P (g/L): 36.00
Maltose M (g/L): 109.19
Volume V (L): 6.21

t = 24.00 h

Biomass X (g/L): 61.55
Glucose S (g/L): 8.23
Lactic Acid P (g/L): 40.97
Maltose M (g/L): 107.59
Volume V (L): 6.99

t = 25.00 h

Biomass X (g/L): 56.65
Glucose S (g/L): 6.30
Lactic Acid P (g/L): 43.08
Maltose M (g/L): 108.64
Volume V (L): 7.01

t = 26.00 h

Biomass X (g/L): 65.76
Glucose S (g/L): 4.06
Lactic Acid P (g/L): 42.46
Maltose M (g/L): 106.12
Volume V (L): 7.04

t = 27.00 h

Biomass X (g/L): 67.56
Glucose S (g/L): 5.27
Lactic Acid P (g/L): 58.24
Maltose M (g/L): 107.28
Volume V (L): 6.87

t = 28.00 h

Biomass X (g/L): 61.00
Glucose S (g/L): 10.48
Lactic Acid P (g/L): 36.21
Maltose M (g/L): 102.44
Volume V (L): 7.81

t = 29.00 h

Biomass X (g/L): 56.05
Glucose S (g/L): 6.42
Lactic Acid P (g/L): 45.21
Maltose M (g/L): 104.13
Volume V (L): 7.30

t = 30.00 h

Biomass X (g/L): 69.70
Glucose S (g/L): 6.37
Lactic Acid P (g/L): 33.30
Maltose M (g/L): 103.78
Volume V (L): 6.74

t = 31.00 h

Biomass X (g/L): 63.09
Glucose S (g/L): 4.76
Lactic Acid P (g/L): 50.98
Maltose M (g/L): 104.15
Volume V (L): 7.34

t = 32.00 h

Biomass X (g/L): 56.67
Glucose S (g/L): 2.70
Lactic Acid P (g/L): 39.19
Maltose M (g/L): 102.38
Volume V (L): 7.63

t = 33.00 h

Biomass X (g/L): 57.19
Glucose S (g/L): 7.20
Lactic Acid P (g/L): 47.31
Maltose M (g/L): 102.97
Volume V (L): 7.13

t = 34.00 h

Biomass X (g/L): 53.03
Glucose S (g/L): 8.68
Lactic Acid P (g/L): 47.42
Maltose M (g/L): 100.10

Volume V (L): 7.92

t = 35.00 h

Biomass X (g/L): 62.70

Glucose S (g/L): 10.88

Lactic Acid P (g/L): 53.91

Maltose M (g/L): 104.89

Volume V (L): 7.69

t = 36.00 h

Biomass X (g/L): 78.22

Glucose S (g/L): 3.06

Lactic Acid P (g/L): 47.63

Maltose M (g/L): 102.89

Volume V (L): 6.58

t = 37.00 h

Biomass X (g/L): 70.59

Glucose S (g/L): 6.14

Lactic Acid P (g/L): 48.10

Maltose M (g/L): 98.40

Volume V (L): 7.80

t = 38.00 h

Biomass X (g/L): 64.90

Glucose S (g/L): 7.83

Lactic Acid P (g/L): 66.33

Maltose M (g/L): 100.18

Volume V (L): 7.56

t = 39.00 h

Biomass X (g/L): 77.77

Glucose S (g/L): 9.62

Lactic Acid P (g/L): 57.25

Maltose M (g/L): 102.71

Volume V (L): 8.04

t = 40.00 h

Biomass X (g/L): 63.54

Glucose S (g/L): 9.67

Lactic Acid P (g/L): 50.77

Maltose M (g/L): 100.08
Volume V (L): 8.29

t = 41.00 h

Biomass X (g/L): 72.64
Glucose S (g/L): 10.14
Lactic Acid P (g/L): 54.17
Maltose M (g/L): 99.27
Volume V (L): 8.04

t = 42.00 h

Biomass X (g/L): 70.76
Glucose S (g/L): 11.86
Lactic Acid P (g/L): 71.15
Maltose M (g/L): 100.41
Volume V (L): 8.75

t = 43.00 h

Biomass X (g/L): 80.07
Glucose S (g/L): 9.11
Lactic Acid P (g/L): 55.72
Maltose M (g/L): 101.04
Volume V (L): 8.02

t = 44.00 h

Biomass X (g/L): 59.16
Glucose S (g/L): 14.03
Lactic Acid P (g/L): 52.50
Maltose M (g/L): 96.14
Volume V (L): 8.44

t = 45.00 h

Biomass X (g/L): 66.30
Glucose S (g/L): 13.29
Lactic Acid P (g/L): 53.19
Maltose M (g/L): 100.70
Volume V (L): 8.59

t = 46.00 h

Biomass X (g/L): 59.62
Glucose S (g/L): 11.94

Lactic Acid P (g/L): 54.99

Maltose M (g/L): 96.88

Volume V (L): 8.28

t = 47.00 h

Biomass X (g/L): 72.64

Glucose S (g/L): 17.11

Lactic Acid P (g/L): 55.22

Maltose M (g/L): 98.42

Volume V (L): 8.73

t = 48.00 h

Biomass X (g/L): 86.47

Glucose S (g/L): 17.51

Lactic Acid P (g/L): 49.94

Maltose M (g/L): 97.94

Volume V (L): 8.20

t = 49.00 h

Biomass X (g/L): 71.24

Glucose S (g/L): 16.51

Lactic Acid P (g/L): 51.85

Maltose M (g/L): 96.28

Volume V (L): 8.35

t = 50.00 h

Biomass X (g/L): 71.57

Glucose S (g/L): 7.43

Lactic Acid P (g/L): 54.93

Maltose M (g/L): 94.60

Volume V (L): 8.04

t = 51.00 h

Biomass X (g/L): 84.51

Glucose S (g/L): 19.16

Lactic Acid P (g/L): 67.65

Maltose M (g/L): 96.02

Volume V (L): 8.78

t = 52.00 h

Biomass X (g/L): 86.64

Glucose S (g/L): 17.80
Lactic Acid P (g/L): 59.83
Maltose M (g/L): 100.40
Volume V (L): 8.89

t = 53.00 h

Biomass X (g/L): 76.40
Glucose S (g/L): 21.31
Lactic Acid P (g/L): 61.83
Maltose M (g/L): 92.46
Volume V (L): 9.33

t = 54.00 h

Biomass X (g/L): 83.73
Glucose S (g/L): 20.01
Lactic Acid P (g/L): 74.98
Maltose M (g/L): 99.70
Volume V (L): 9.24

t = 55.00 h

Biomass X (g/L): 67.95
Glucose S (g/L): 21.42
Lactic Acid P (g/L): 75.67
Maltose M (g/L): 94.36
Volume V (L): 8.84

t = 56.00 h

Biomass X (g/L): 90.55
Glucose S (g/L): 20.30
Lactic Acid P (g/L): 62.60
Maltose M (g/L): 94.68
Volume V (L): 8.53

t = 57.00 h

Biomass X (g/L): 97.79
Glucose S (g/L): 22.04
Lactic Acid P (g/L): 60.05
Maltose M (g/L): 91.43
Volume V (L): 9.30

t = 58.00 h

Biomass X (g/L): 83.17
Glucose S (g/L): 24.21
Lactic Acid P (g/L): 58.48
Maltose M (g/L): 91.79
Volume V (L): 9.66

t = 59.00 h

Biomass X (g/L): 59.62
Glucose S (g/L): 20.49
Lactic Acid P (g/L): 72.24
Maltose M (g/L): 92.04
Volume V (L): 9.58

t = 60.00 h

Biomass X (g/L): 73.69
Glucose S (g/L): 16.75
Lactic Acid P (g/L): 64.95
Maltose M (g/L): 90.24
Volume V (L): 9.80

t = 61.00 h

Biomass X (g/L): 84.50
Glucose S (g/L): 20.45
Lactic Acid P (g/L): 63.78
Maltose M (g/L): 91.92
Volume V (L): 9.31

t = 62.00 h

Biomass X (g/L): 75.22
Glucose S (g/L): 22.77
Lactic Acid P (g/L): 60.49
Maltose M (g/L): 91.00
Volume V (L): 9.27

t = 63.00 h

Biomass X (g/L): 90.76
Glucose S (g/L): 22.63
Lactic Acid P (g/L): 61.24
Maltose M (g/L): 96.25
Volume V (L): 9.22

t = 64.00 h

Biomass X (g/L): 84.63
Glucose S (g/L): 20.04
Lactic Acid P (g/L): 70.93
Maltose M (g/L): 88.24
Volume V (L): 9.59

t = 65.00 h

Biomass X (g/L): 72.93
Glucose S (g/L): 19.74
Lactic Acid P (g/L): 57.25
Maltose M (g/L): 91.01
Volume V (L): 9.65

t = 66.00 h

Biomass X (g/L): 79.74
Glucose S (g/L): 22.54
Lactic Acid P (g/L): 77.39
Maltose M (g/L): 90.96
Volume V (L): 9.85

t = 67.00 h

Biomass X (g/L): 71.37
Glucose S (g/L): 26.95
Lactic Acid P (g/L): 64.61
Maltose M (g/L): 87.22
Volume V (L): 9.78

t = 68.00 h

Biomass X (g/L): 77.90
Glucose S (g/L): 29.47
Lactic Acid P (g/L): 68.21
Maltose M (g/L): 91.22
Volume V (L): 9.85

t = 69.00 h

Biomass X (g/L): 74.55
Glucose S (g/L): 28.50
Lactic Acid P (g/L): 58.34
Maltose M (g/L): 88.08
Volume V (L): 9.64

t = 70.00 h

Biomass X (g/L): 90.33
Glucose S (g/L): 29.48
Lactic Acid P (g/L): 69.99
Maltose M (g/L): 82.40
Volume V (L): 10.46

t = 71.00 h

Biomass X (g/L): 98.92
Glucose S (g/L): 27.36
Lactic Acid P (g/L): 67.57
Maltose M (g/L): 89.91
Volume V (L): 10.15

t = 72.00 h

Biomass X (g/L): 66.62
Glucose S (g/L): 23.96
Lactic Acid P (g/L): 70.23
Maltose M (g/L): 85.03
Volume V (L): 10.12

t = 73.00 h

Biomass X (g/L): 62.14
Glucose S (g/L): 27.74
Lactic Acid P (g/L): 83.55
Maltose M (g/L): 84.80
Volume V (L): 9.94

t = 74.00 h

Biomass X (g/L): 77.41
Glucose S (g/L): 28.86
Lactic Acid P (g/L): 72.71
Maltose M (g/L): 88.75
Volume V (L): 9.76

t = 75.00 h

Biomass X (g/L): 65.80
Glucose S (g/L): 28.93
Lactic Acid P (g/L): 64.63
Maltose M (g/L): 85.74

Volume V (L): 10.97

Batch 30

Initial Conditions:

Batch: 30.00

X0 (g/L): 5.61

S0 (g/L): 93.42

M0 (g/L): 15.86

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.17

F_vol (L/h): 0.09

S_feed (g/L): 256.47

M_feed (g/L): 43.53

V_final (L): 11.92

Batch 30 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 10.86
Glucose S (g/L): 89.37
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 16.20
Volume V (L): 5.25

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 95.26
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 15.98
Volume V (L): 5.60

t = 2.00 h

Biomass X (g/L): 17.17
Glucose S (g/L): 97.08
Lactic Acid P (g/L): 9.16
Maltose M (g/L): 16.58
Volume V (L): 5.26

t = 3.00 h

Biomass X (g/L): 21.65
Glucose S (g/L): 94.72
Lactic Acid P (g/L): 10.03
Maltose M (g/L): 16.37
Volume V (L): 5.65

t = 4.00 h

Biomass X (g/L): 17.62
Glucose S (g/L): 95.84
Lactic Acid P (g/L): 10.43
Maltose M (g/L): 16.31
Volume V (L): 4.93

t = 5.00 h

Biomass X (g/L): 35.59
Glucose S (g/L): 90.38

Lactic Acid P (g/L): 19.39

Maltose M (g/L): 16.87

Volume V (L): 5.21

t = 6.00 h

Biomass X (g/L): 26.92

Glucose S (g/L): 88.73

Lactic Acid P (g/L): 11.13

Maltose M (g/L): 16.72

Volume V (L): 5.75

t = 7.00 h

Biomass X (g/L): 16.10

Glucose S (g/L): 90.38

Lactic Acid P (g/L): 19.41

Maltose M (g/L): 17.08

Volume V (L): 5.84

t = 8.00 h

Biomass X (g/L): 37.72

Glucose S (g/L): 89.49

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 16.27

Volume V (L): 6.02

t = 9.00 h

Biomass X (g/L): 41.27

Glucose S (g/L): 79.26

Lactic Acid P (g/L): 20.88

Maltose M (g/L): 17.11

Volume V (L): 6.44

t = 10.00 h

Biomass X (g/L): 37.75

Glucose S (g/L): 84.45

Lactic Acid P (g/L): 35.10

Maltose M (g/L): 17.06

Volume V (L): 5.13

t = 11.00 h

Biomass X (g/L): 53.14

Glucose S (g/L): 79.07
Lactic Acid P (g/L): 28.07
Maltose M (g/L): 16.97
Volume V (L): 5.48

t = 12.00 h

Biomass X (g/L): 35.86
Glucose S (g/L): 78.29
Lactic Acid P (g/L): 30.44
Maltose M (g/L): 17.17
Volume V (L): 6.23

t = 13.00 h

Biomass X (g/L): 64.65
Glucose S (g/L): 73.11
Lactic Acid P (g/L): 28.15
Maltose M (g/L): 17.22
Volume V (L): 5.80

t = 14.00 h

Biomass X (g/L): 50.24
Glucose S (g/L): 73.25
Lactic Acid P (g/L): 48.95
Maltose M (g/L): 17.41
Volume V (L): 8.16

t = 15.00 h

Biomass X (g/L): 52.19
Glucose S (g/L): 77.74
Lactic Acid P (g/L): 47.29
Maltose M (g/L): 17.10
Volume V (L): 6.67

t = 16.00 h

Biomass X (g/L): 63.10
Glucose S (g/L): 73.93
Lactic Acid P (g/L): 51.94
Maltose M (g/L): 17.59
Volume V (L): 6.43

t = 17.00 h

Biomass X (g/L): 70.66
Glucose S (g/L): 71.94
Lactic Acid P (g/L): 31.22
Maltose M (g/L): 17.90
Volume V (L): 6.28

t = 18.00 h

Biomass X (g/L): 70.24
Glucose S (g/L): 72.24
Lactic Acid P (g/L): 46.99
Maltose M (g/L): 17.63
Volume V (L): 6.40

t = 19.00 h

Biomass X (g/L): 63.56
Glucose S (g/L): 73.76
Lactic Acid P (g/L): 47.59
Maltose M (g/L): 17.79
Volume V (L): 6.77

t = 20.00 h

Biomass X (g/L): 67.49
Glucose S (g/L): 73.40
Lactic Acid P (g/L): 49.11
Maltose M (g/L): 17.88
Volume V (L): 6.91

t = 21.00 h

Biomass X (g/L): 63.07
Glucose S (g/L): 71.72
Lactic Acid P (g/L): 47.52
Maltose M (g/L): 17.92
Volume V (L): 6.69

t = 22.00 h

Biomass X (g/L): 73.02
Glucose S (g/L): 75.38
Lactic Acid P (g/L): 51.21
Maltose M (g/L): 18.10
Volume V (L): 6.88

t = 23.00 h

Biomass X (g/L): 86.53
Glucose S (g/L): 75.13
Lactic Acid P (g/L): 63.04
Maltose M (g/L): 17.45
Volume V (L): 7.87

t = 24.00 h

Biomass X (g/L): 77.37
Glucose S (g/L): 73.62
Lactic Acid P (g/L): 39.54
Maltose M (g/L): 17.38
Volume V (L): 7.98

t = 25.00 h

Biomass X (g/L): 86.50
Glucose S (g/L): 72.80
Lactic Acid P (g/L): 73.40
Maltose M (g/L): 17.78
Volume V (L): 7.45

t = 26.00 h

Biomass X (g/L): 92.99
Glucose S (g/L): 75.02
Lactic Acid P (g/L): 62.82
Maltose M (g/L): 17.74
Volume V (L): 6.66

t = 27.00 h

Biomass X (g/L): 74.39
Glucose S (g/L): 74.37
Lactic Acid P (g/L): 59.00
Maltose M (g/L): 17.74
Volume V (L): 7.79

t = 28.00 h

Biomass X (g/L): 78.86
Glucose S (g/L): 77.14
Lactic Acid P (g/L): 68.26
Maltose M (g/L): 18.03
Volume V (L): 8.45

t = 29.00 h

Biomass X (g/L): 101.77
Glucose S (g/L): 77.14
Lactic Acid P (g/L): 48.01
Maltose M (g/L): 17.49
Volume V (L): 8.44

t = 30.00 h

Biomass X (g/L): 95.41
Glucose S (g/L): 71.73
Lactic Acid P (g/L): 55.72
Maltose M (g/L): 17.57
Volume V (L): 8.18

t = 31.00 h

Biomass X (g/L): 66.35
Glucose S (g/L): 72.08
Lactic Acid P (g/L): 61.22
Maltose M (g/L): 17.16
Volume V (L): 8.65

t = 32.00 h

Biomass X (g/L): 86.26
Glucose S (g/L): 73.87
Lactic Acid P (g/L): 75.07
Maltose M (g/L): 17.98
Volume V (L): 7.98

t = 33.00 h

Biomass X (g/L): 82.95
Glucose S (g/L): 79.68
Lactic Acid P (g/L): 63.08
Maltose M (g/L): 17.26
Volume V (L): 8.39

t = 34.00 h

Biomass X (g/L): 91.07
Glucose S (g/L): 72.39
Lactic Acid P (g/L): 68.05
Maltose M (g/L): 17.23

Volume V (L): 7.76

t = 35.00 h

Biomass X (g/L): 76.46

Glucose S (g/L): 79.15

Lactic Acid P (g/L): 59.31

Maltose M (g/L): 17.18

Volume V (L): 8.39

t = 36.00 h

Biomass X (g/L): 75.79

Glucose S (g/L): 79.71

Lactic Acid P (g/L): 69.38

Maltose M (g/L): 17.06

Volume V (L): 9.06

t = 37.00 h

Biomass X (g/L): 77.53

Glucose S (g/L): 82.86

Lactic Acid P (g/L): 80.44

Maltose M (g/L): 16.99

Volume V (L): 8.09

t = 38.00 h

Biomass X (g/L): 78.67

Glucose S (g/L): 80.48

Lactic Acid P (g/L): 74.80

Maltose M (g/L): 17.00

Volume V (L): 8.03

t = 39.00 h

Biomass X (g/L): 91.94

Glucose S (g/L): 76.44

Lactic Acid P (g/L): 65.63

Maltose M (g/L): 16.73

Volume V (L): 8.65

t = 40.00 h

Biomass X (g/L): 73.14

Glucose S (g/L): 78.31

Lactic Acid P (g/L): 72.76

Maltose M (g/L): 16.83

Volume V (L): 8.57

t = 41.00 h

Biomass X (g/L): 78.51

Glucose S (g/L): 78.31

Lactic Acid P (g/L): 72.27

Maltose M (g/L): 17.53

Volume V (L): 9.14

t = 42.00 h

Biomass X (g/L): 67.52

Glucose S (g/L): 81.00

Lactic Acid P (g/L): 71.99

Maltose M (g/L): 17.15

Volume V (L): 8.63

t = 43.00 h

Biomass X (g/L): 94.04

Glucose S (g/L): 82.79

Lactic Acid P (g/L): 77.72

Maltose M (g/L): 17.15

Volume V (L): 9.92

t = 44.00 h

Biomass X (g/L): 96.56

Glucose S (g/L): 80.43

Lactic Acid P (g/L): 77.98

Maltose M (g/L): 17.16

Volume V (L): 8.35

t = 45.00 h

Biomass X (g/L): 81.43

Glucose S (g/L): 89.29

Lactic Acid P (g/L): 67.33

Maltose M (g/L): 17.28

Volume V (L): 9.85

t = 46.00 h

Biomass X (g/L): 81.70

Glucose S (g/L): 81.31

Lactic Acid P (g/L): 74.47

Maltose M (g/L): 16.24

Volume V (L): 8.82

t = 47.00 h

Biomass X (g/L): 88.79

Glucose S (g/L): 90.28

Lactic Acid P (g/L): 65.64

Maltose M (g/L): 16.33

Volume V (L): 8.34

t = 48.00 h

Biomass X (g/L): 98.42

Glucose S (g/L): 90.46

Lactic Acid P (g/L): 61.10

Maltose M (g/L): 16.39

Volume V (L): 9.56

t = 49.00 h

Biomass X (g/L): 88.28

Glucose S (g/L): 90.24

Lactic Acid P (g/L): 67.19

Maltose M (g/L): 16.42

Volume V (L): 9.38

t = 50.00 h

Biomass X (g/L): 77.74

Glucose S (g/L): 85.97

Lactic Acid P (g/L): 69.16

Maltose M (g/L): 16.41

Volume V (L): 9.00

t = 51.00 h

Biomass X (g/L): 68.94

Glucose S (g/L): 89.86

Lactic Acid P (g/L): 55.29

Maltose M (g/L): 16.31

Volume V (L): 8.78

t = 52.00 h

Biomass X (g/L): 73.63

Glucose S (g/L): 85.49
Lactic Acid P (g/L): 77.98
Maltose M (g/L): 16.43
Volume V (L): 9.79

t = 53.00 h

Biomass X (g/L): 84.80
Glucose S (g/L): 92.94
Lactic Acid P (g/L): 45.91
Maltose M (g/L): 16.31
Volume V (L): 10.43

t = 54.00 h

Biomass X (g/L): 77.80
Glucose S (g/L): 90.53
Lactic Acid P (g/L): 94.93
Maltose M (g/L): 16.04
Volume V (L): 10.31

t = 55.00 h

Biomass X (g/L): 85.75
Glucose S (g/L): 90.14
Lactic Acid P (g/L): 84.35
Maltose M (g/L): 15.58
Volume V (L): 10.11

t = 56.00 h

Biomass X (g/L): 78.55
Glucose S (g/L): 88.25
Lactic Acid P (g/L): 68.86
Maltose M (g/L): 15.84
Volume V (L): 10.43

t = 57.00 h

Biomass X (g/L): 73.65
Glucose S (g/L): 85.00
Lactic Acid P (g/L): 71.47
Maltose M (g/L): 15.90
Volume V (L): 10.25

t = 58.00 h

Biomass X (g/L): 72.40
Glucose S (g/L): 90.09
Lactic Acid P (g/L): 74.28
Maltose M (g/L): 15.92
Volume V (L): 9.56

t = 59.00 h

Biomass X (g/L): 98.22
Glucose S (g/L): 97.38
Lactic Acid P (g/L): 66.52
Maltose M (g/L): 15.61
Volume V (L): 9.66

t = 60.00 h

Biomass X (g/L): 91.71
Glucose S (g/L): 93.29
Lactic Acid P (g/L): 73.66
Maltose M (g/L): 15.72
Volume V (L): 10.43

t = 61.00 h

Biomass X (g/L): 99.54
Glucose S (g/L): 94.63
Lactic Acid P (g/L): 70.84
Maltose M (g/L): 15.48
Volume V (L): 10.54

t = 62.00 h

Biomass X (g/L): 88.12
Glucose S (g/L): 97.78
Lactic Acid P (g/L): 71.46
Maltose M (g/L): 15.69
Volume V (L): 11.19

t = 63.00 h

Biomass X (g/L): 78.79
Glucose S (g/L): 101.95
Lactic Acid P (g/L): 74.08
Maltose M (g/L): 15.36
Volume V (L): 11.23

t = 64.00 h

Biomass X (g/L): 95.25
Glucose S (g/L): 101.04
Lactic Acid P (g/L): 60.94
Maltose M (g/L): 15.36
Volume V (L): 11.24

t = 65.00 h

Biomass X (g/L): 82.25
Glucose S (g/L): 94.50
Lactic Acid P (g/L): 73.14
Maltose M (g/L): 15.17
Volume V (L): 10.84

t = 66.00 h

Biomass X (g/L): 84.50
Glucose S (g/L): 92.07
Lactic Acid P (g/L): 74.40
Maltose M (g/L): 15.42
Volume V (L): 10.33

t = 67.00 h

Biomass X (g/L): 81.84
Glucose S (g/L): 105.98
Lactic Acid P (g/L): 74.24
Maltose M (g/L): 15.15
Volume V (L): 11.41

t = 68.00 h

Biomass X (g/L): 54.18
Glucose S (g/L): 106.26
Lactic Acid P (g/L): 71.32
Maltose M (g/L): 15.57
Volume V (L): 11.57

t = 69.00 h

Biomass X (g/L): 97.12
Glucose S (g/L): 100.36
Lactic Acid P (g/L): 83.51
Maltose M (g/L): 15.37
Volume V (L): 10.05

t = 70.00 h

Biomass X (g/L): 86.41
Glucose S (g/L): 96.39
Lactic Acid P (g/L): 70.37
Maltose M (g/L): 15.17
Volume V (L): 11.00

t = 71.00 h

Biomass X (g/L): 84.72
Glucose S (g/L): 103.86
Lactic Acid P (g/L): 75.68
Maltose M (g/L): 14.91
Volume V (L): 11.17

t = 72.00 h

Biomass X (g/L): 80.84
Glucose S (g/L): 101.07
Lactic Acid P (g/L): 85.34
Maltose M (g/L): 15.16
Volume V (L): 11.71

t = 73.00 h

Biomass X (g/L): 82.80
Glucose S (g/L): 101.58
Lactic Acid P (g/L): 73.22
Maltose M (g/L): 15.18
Volume V (L): 11.29

t = 74.00 h

Biomass X (g/L): 75.43
Glucose S (g/L): 105.23
Lactic Acid P (g/L): 75.96
Maltose M (g/L): 14.96
Volume V (L): 11.81

t = 75.00 h

Biomass X (g/L): 86.75
Glucose S (g/L): 104.01
Lactic Acid P (g/L): 73.32
Maltose M (g/L): 15.29

Volume V (L): 11.81

Batch 31

Initial Conditions:

Batch: 31.00

X0 (g/L): 1.22

S0 (g/L): 72.46

M0 (g/L): 74.89

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.03

F_vol (L/h): 0.05

S_feed (g/L): 147.53

M_feed (g/L): 152.47

V_final (L): 8.64

Batch 31 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 76.69

Lactic Acid P (g/L): 10.92

Maltose M (g/L): 82.66

Volume V (L): 5.15

t = 1.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 74.57

Lactic Acid P (g/L): 10.85

Maltose M (g/L): 79.79

Volume V (L): 5.27

t = 2.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 71.11

Lactic Acid P (g/L): 4.73

Maltose M (g/L): 78.14

Volume V (L): 4.88

t = 3.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 72.61

Lactic Acid P (g/L): 6.02

Maltose M (g/L): 71.81

Volume V (L): 5.33

t = 4.00 h

Biomass X (g/L): 7.82

Glucose S (g/L): 78.13

Lactic Acid P (g/L): 5.95

Maltose M (g/L): 79.70

Volume V (L): 5.52

t = 5.00 h

Biomass X (g/L): 11.05

Glucose S (g/L): 79.99

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 72.67
Volume V (L): 5.21

t = 6.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 77.73
Lactic Acid P (g/L): 5.28
Maltose M (g/L): 73.93
Volume V (L): 5.87

t = 7.00 h

Biomass X (g/L): 24.37
Glucose S (g/L): 77.21
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 70.83
Volume V (L): 5.35

t = 8.00 h

Biomass X (g/L): 8.38
Glucose S (g/L): 76.16
Lactic Acid P (g/L): 9.12
Maltose M (g/L): 69.51
Volume V (L): 5.72

t = 9.00 h

Biomass X (g/L): 30.00
Glucose S (g/L): 77.07
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 68.27
Volume V (L): 5.52

t = 10.00 h

Biomass X (g/L): 11.53
Glucose S (g/L): 66.92
Lactic Acid P (g/L): 1.23
Maltose M (g/L): 67.81
Volume V (L): 5.65

t = 11.00 h

Biomass X (g/L): 1.40

Glucose S (g/L): 69.48
Lactic Acid P (g/L): 29.95
Maltose M (g/L): 69.77
Volume V (L): 5.05

t = 12.00 h

Biomass X (g/L): 30.05
Glucose S (g/L): 68.78
Lactic Acid P (g/L): 28.01
Maltose M (g/L): 68.45
Volume V (L): 5.69

t = 13.00 h

Biomass X (g/L): 32.81
Glucose S (g/L): 72.78
Lactic Acid P (g/L): 21.21
Maltose M (g/L): 66.65
Volume V (L): 5.42

t = 14.00 h

Biomass X (g/L): 35.00
Glucose S (g/L): 62.75
Lactic Acid P (g/L): 26.19
Maltose M (g/L): 70.87
Volume V (L): 5.43

t = 15.00 h

Biomass X (g/L): 32.46
Glucose S (g/L): 63.61
Lactic Acid P (g/L): 25.37
Maltose M (g/L): 56.45
Volume V (L): 6.23

t = 16.00 h

Biomass X (g/L): 42.74
Glucose S (g/L): 61.14
Lactic Acid P (g/L): 36.08
Maltose M (g/L): 66.69
Volume V (L): 5.71

t = 17.00 h

Biomass X (g/L): 41.86
Glucose S (g/L): 62.53
Lactic Acid P (g/L): 27.49
Maltose M (g/L): 66.81
Volume V (L): 5.53

t = 18.00 h

Biomass X (g/L): 47.64
Glucose S (g/L): 60.12
Lactic Acid P (g/L): 45.81
Maltose M (g/L): 67.62
Volume V (L): 5.77

t = 19.00 h

Biomass X (g/L): 62.59
Glucose S (g/L): 53.30
Lactic Acid P (g/L): 36.88
Maltose M (g/L): 63.35
Volume V (L): 5.54

t = 20.00 h

Biomass X (g/L): 59.08
Glucose S (g/L): 49.95
Lactic Acid P (g/L): 37.79
Maltose M (g/L): 63.69
Volume V (L): 6.39

t = 21.00 h

Biomass X (g/L): 66.23
Glucose S (g/L): 47.66
Lactic Acid P (g/L): 44.16
Maltose M (g/L): 65.61
Volume V (L): 5.82

t = 22.00 h

Biomass X (g/L): 60.33
Glucose S (g/L): 47.79
Lactic Acid P (g/L): 40.58
Maltose M (g/L): 66.69
Volume V (L): 6.22

t = 23.00 h

Biomass X (g/L): 61.58
Glucose S (g/L): 45.89
Lactic Acid P (g/L): 51.06
Maltose M (g/L): 60.66
Volume V (L): 6.51

t = 24.00 h

Biomass X (g/L): 55.52
Glucose S (g/L): 48.43
Lactic Acid P (g/L): 44.82
Maltose M (g/L): 61.94
Volume V (L): 6.02

t = 25.00 h

Biomass X (g/L): 65.36
Glucose S (g/L): 47.09
Lactic Acid P (g/L): 55.82
Maltose M (g/L): 65.85
Volume V (L): 6.37

t = 26.00 h

Biomass X (g/L): 37.31
Glucose S (g/L): 47.00
Lactic Acid P (g/L): 64.30
Maltose M (g/L): 53.61
Volume V (L): 5.96

t = 27.00 h

Biomass X (g/L): 66.45
Glucose S (g/L): 48.84
Lactic Acid P (g/L): 47.75
Maltose M (g/L): 57.60
Volume V (L): 7.02

t = 28.00 h

Biomass X (g/L): 73.01
Glucose S (g/L): 50.57
Lactic Acid P (g/L): 54.54
Maltose M (g/L): 63.84
Volume V (L): 6.63

t = 29.00 h

Biomass X (g/L): 80.81
Glucose S (g/L): 43.95
Lactic Acid P (g/L): 61.26
Maltose M (g/L): 62.24
Volume V (L): 6.35

t = 30.00 h

Biomass X (g/L): 73.57
Glucose S (g/L): 47.22
Lactic Acid P (g/L): 54.08
Maltose M (g/L): 60.04
Volume V (L): 6.23

t = 31.00 h

Biomass X (g/L): 80.84
Glucose S (g/L): 41.82
Lactic Acid P (g/L): 63.50
Maltose M (g/L): 56.61
Volume V (L): 6.47

t = 32.00 h

Biomass X (g/L): 56.28
Glucose S (g/L): 47.28
Lactic Acid P (g/L): 65.34
Maltose M (g/L): 54.85
Volume V (L): 6.45

t = 33.00 h

Biomass X (g/L): 78.56
Glucose S (g/L): 46.82
Lactic Acid P (g/L): 71.18
Maltose M (g/L): 55.33
Volume V (L): 7.17

t = 34.00 h

Biomass X (g/L): 62.95
Glucose S (g/L): 46.67
Lactic Acid P (g/L): 56.37
Maltose M (g/L): 61.56

Volume V (L): 6.29

t = 35.00 h

Biomass X (g/L): 68.40

Glucose S (g/L): 43.47

Lactic Acid P (g/L): 54.07

Maltose M (g/L): 54.41

Volume V (L): 6.74

t = 36.00 h

Biomass X (g/L): 79.33

Glucose S (g/L): 40.86

Lactic Acid P (g/L): 60.74

Maltose M (g/L): 56.27

Volume V (L): 6.74

t = 37.00 h

Biomass X (g/L): 94.66

Glucose S (g/L): 38.43

Lactic Acid P (g/L): 55.34

Maltose M (g/L): 53.23

Volume V (L): 6.76

t = 38.00 h

Biomass X (g/L): 86.03

Glucose S (g/L): 43.55

Lactic Acid P (g/L): 70.26

Maltose M (g/L): 57.80

Volume V (L): 6.81

t = 39.00 h

Biomass X (g/L): 82.91

Glucose S (g/L): 49.50

Lactic Acid P (g/L): 54.88

Maltose M (g/L): 53.13

Volume V (L): 6.86

t = 40.00 h

Biomass X (g/L): 82.93

Glucose S (g/L): 43.14

Lactic Acid P (g/L): 87.95

Maltose M (g/L): 54.13

Volume V (L): 7.19

t = 41.00 h

Biomass X (g/L): 76.28

Glucose S (g/L): 47.42

Lactic Acid P (g/L): 78.91

Maltose M (g/L): 48.84

Volume V (L): 7.16

t = 42.00 h

Biomass X (g/L): 78.31

Glucose S (g/L): 47.98

Lactic Acid P (g/L): 67.55

Maltose M (g/L): 48.69

Volume V (L): 7.16

t = 43.00 h

Biomass X (g/L): 88.39

Glucose S (g/L): 44.28

Lactic Acid P (g/L): 76.02

Maltose M (g/L): 53.36

Volume V (L): 6.20

t = 44.00 h

Biomass X (g/L): 70.34

Glucose S (g/L): 50.44

Lactic Acid P (g/L): 53.28

Maltose M (g/L): 48.87

Volume V (L): 7.22

t = 45.00 h

Biomass X (g/L): 90.52

Glucose S (g/L): 35.60

Lactic Acid P (g/L): 92.38

Maltose M (g/L): 59.15

Volume V (L): 6.70

t = 46.00 h

Biomass X (g/L): 80.79

Glucose S (g/L): 42.11

Lactic Acid P (g/L): 63.94

Maltose M (g/L): 45.89

Volume V (L): 7.42

t = 47.00 h

Biomass X (g/L): 93.20

Glucose S (g/L): 43.66

Lactic Acid P (g/L): 68.74

Maltose M (g/L): 49.46

Volume V (L): 7.71

t = 48.00 h

Biomass X (g/L): 94.70

Glucose S (g/L): 44.45

Lactic Acid P (g/L): 53.05

Maltose M (g/L): 51.65

Volume V (L): 7.06

t = 49.00 h

Biomass X (g/L): 68.21

Glucose S (g/L): 49.09

Lactic Acid P (g/L): 75.20

Maltose M (g/L): 50.14

Volume V (L): 7.30

t = 50.00 h

Biomass X (g/L): 79.31

Glucose S (g/L): 49.39

Lactic Acid P (g/L): 60.80

Maltose M (g/L): 47.52

Volume V (L): 7.25

t = 51.00 h

Biomass X (g/L): 76.61

Glucose S (g/L): 50.25

Lactic Acid P (g/L): 72.38

Maltose M (g/L): 47.65

Volume V (L): 7.83

t = 52.00 h

Biomass X (g/L): 66.99

Glucose S (g/L): 51.46
Lactic Acid P (g/L): 66.74
Maltose M (g/L): 47.22
Volume V (L): 7.55

t = 53.00 h

Biomass X (g/L): 74.90
Glucose S (g/L): 46.38
Lactic Acid P (g/L): 62.13
Maltose M (g/L): 49.09
Volume V (L): 8.07

t = 54.00 h

Biomass X (g/L): 109.97
Glucose S (g/L): 45.91
Lactic Acid P (g/L): 65.00
Maltose M (g/L): 55.72
Volume V (L): 7.74

t = 55.00 h

Biomass X (g/L): 91.48
Glucose S (g/L): 52.24
Lactic Acid P (g/L): 74.89
Maltose M (g/L): 49.60
Volume V (L): 7.74

t = 56.00 h

Biomass X (g/L): 104.66
Glucose S (g/L): 49.04
Lactic Acid P (g/L): 61.20
Maltose M (g/L): 52.72
Volume V (L): 7.73

t = 57.00 h

Biomass X (g/L): 93.64
Glucose S (g/L): 46.16
Lactic Acid P (g/L): 66.44
Maltose M (g/L): 50.32
Volume V (L): 8.00

t = 58.00 h

Biomass X (g/L): 77.74
Glucose S (g/L): 55.85
Lactic Acid P (g/L): 79.84
Maltose M (g/L): 51.90
Volume V (L): 7.31

t = 59.00 h

Biomass X (g/L): 81.32
Glucose S (g/L): 57.90
Lactic Acid P (g/L): 98.12
Maltose M (g/L): 39.48
Volume V (L): 7.88

t = 60.00 h

Biomass X (g/L): 91.05
Glucose S (g/L): 59.00
Lactic Acid P (g/L): 78.65
Maltose M (g/L): 46.92
Volume V (L): 7.89

t = 61.00 h

Biomass X (g/L): 89.45
Glucose S (g/L): 54.57
Lactic Acid P (g/L): 65.96
Maltose M (g/L): 51.63
Volume V (L): 8.17

t = 62.00 h

Biomass X (g/L): 79.12
Glucose S (g/L): 56.68
Lactic Acid P (g/L): 71.32
Maltose M (g/L): 45.52
Volume V (L): 7.95

t = 63.00 h

Biomass X (g/L): 69.61
Glucose S (g/L): 59.89
Lactic Acid P (g/L): 75.26
Maltose M (g/L): 52.48
Volume V (L): 7.75

t = 64.00 h

Biomass X (g/L): 117.94
Glucose S (g/L): 53.15
Lactic Acid P (g/L): 71.96
Maltose M (g/L): 51.34
Volume V (L): 8.10

t = 65.00 h

Biomass X (g/L): 83.22
Glucose S (g/L): 59.99
Lactic Acid P (g/L): 83.81
Maltose M (g/L): 45.01
Volume V (L): 7.84

t = 66.00 h

Biomass X (g/L): 81.24
Glucose S (g/L): 57.50
Lactic Acid P (g/L): 77.67
Maltose M (g/L): 44.63
Volume V (L): 7.64

t = 67.00 h

Biomass X (g/L): 85.55
Glucose S (g/L): 60.32
Lactic Acid P (g/L): 84.29
Maltose M (g/L): 42.78
Volume V (L): 8.46

t = 68.00 h

Biomass X (g/L): 81.82
Glucose S (g/L): 58.22
Lactic Acid P (g/L): 73.15
Maltose M (g/L): 43.71
Volume V (L): 8.46

t = 69.00 h

Biomass X (g/L): 85.38
Glucose S (g/L): 62.74
Lactic Acid P (g/L): 88.58
Maltose M (g/L): 41.58
Volume V (L): 8.48

t = 70.00 h

Biomass X (g/L): 96.99
Glucose S (g/L): 59.38
Lactic Acid P (g/L): 74.92
Maltose M (g/L): 46.47
Volume V (L): 8.77

t = 71.00 h

Biomass X (g/L): 82.60
Glucose S (g/L): 53.74
Lactic Acid P (g/L): 67.64
Maltose M (g/L): 40.37
Volume V (L): 8.25

t = 72.00 h

Biomass X (g/L): 71.97
Glucose S (g/L): 64.85
Lactic Acid P (g/L): 97.06
Maltose M (g/L): 46.50
Volume V (L): 8.07

t = 73.00 h

Biomass X (g/L): 81.92
Glucose S (g/L): 56.70
Lactic Acid P (g/L): 60.08
Maltose M (g/L): 46.33
Volume V (L): 8.52

t = 74.00 h

Biomass X (g/L): 78.70
Glucose S (g/L): 65.15
Lactic Acid P (g/L): 70.34
Maltose M (g/L): 38.99
Volume V (L): 8.49

t = 75.00 h

Biomass X (g/L): 97.60
Glucose S (g/L): 60.81
Lactic Acid P (g/L): 71.55
Maltose M (g/L): 48.70

Volume V (L): 8.39

Batch 32

Initial Conditions:

Batch: 32.00

X0 (g/L): 8.92

S0 (g/L): 104.99

M0 (g/L): 80.52

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.77

F_vol (L/h): 0.07

S_feed (g/L): 169.79

M_feed (g/L): 130.21

V_final (L): 10.21

Batch 32 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 8.15
Glucose S (g/L): 104.09
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 79.18
Volume V (L): 4.66

t = 1.00 h

Biomass X (g/L): 6.13
Glucose S (g/L): 108.15
Lactic Acid P (g/L): 10.63
Maltose M (g/L): 78.17
Volume V (L): 5.72

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 101.28
Lactic Acid P (g/L): 1.65
Maltose M (g/L): 76.58
Volume V (L): 5.47

t = 3.00 h

Biomass X (g/L): 15.93
Glucose S (g/L): 100.68
Lactic Acid P (g/L): 10.77
Maltose M (g/L): 80.84
Volume V (L): 5.43

t = 4.00 h

Biomass X (g/L): 14.19
Glucose S (g/L): 99.40
Lactic Acid P (g/L): 19.30
Maltose M (g/L): 74.82
Volume V (L): 5.79

t = 5.00 h

Biomass X (g/L): 32.54
Glucose S (g/L): 90.76

Lactic Acid P (g/L): 10.69

Maltose M (g/L): 79.46

Volume V (L): 4.51

t = 6.00 h

Biomass X (g/L): 29.42

Glucose S (g/L): 93.59

Lactic Acid P (g/L): 14.07

Maltose M (g/L): 75.67

Volume V (L): 5.66

t = 7.00 h

Biomass X (g/L): 51.65

Glucose S (g/L): 94.05

Lactic Acid P (g/L): 39.17

Maltose M (g/L): 72.95

Volume V (L): 4.94

t = 8.00 h

Biomass X (g/L): 61.44

Glucose S (g/L): 83.93

Lactic Acid P (g/L): 36.83

Maltose M (g/L): 76.34

Volume V (L): 5.55

t = 9.00 h

Biomass X (g/L): 57.33

Glucose S (g/L): 77.31

Lactic Acid P (g/L): 38.68

Maltose M (g/L): 70.72

Volume V (L): 5.50

t = 10.00 h

Biomass X (g/L): 40.32

Glucose S (g/L): 78.65

Lactic Acid P (g/L): 42.02

Maltose M (g/L): 74.97

Volume V (L): 5.97

t = 11.00 h

Biomass X (g/L): 73.54

Glucose S (g/L): 68.10
Lactic Acid P (g/L): 42.12
Maltose M (g/L): 72.72
Volume V (L): 5.86

t = 12.00 h

Biomass X (g/L): 77.77
Glucose S (g/L): 75.78
Lactic Acid P (g/L): 59.26
Maltose M (g/L): 69.09
Volume V (L): 6.39

t = 13.00 h

Biomass X (g/L): 66.03
Glucose S (g/L): 74.50
Lactic Acid P (g/L): 43.71
Maltose M (g/L): 72.50
Volume V (L): 5.65

t = 14.00 h

Biomass X (g/L): 81.34
Glucose S (g/L): 77.70
Lactic Acid P (g/L): 50.95
Maltose M (g/L): 71.91
Volume V (L): 6.27

t = 15.00 h

Biomass X (g/L): 65.90
Glucose S (g/L): 69.43
Lactic Acid P (g/L): 34.46
Maltose M (g/L): 73.48
Volume V (L): 6.11

t = 16.00 h

Biomass X (g/L): 55.53
Glucose S (g/L): 70.77
Lactic Acid P (g/L): 42.14
Maltose M (g/L): 72.14
Volume V (L): 5.43

t = 17.00 h

Biomass X (g/L): 68.09
Glucose S (g/L): 65.21
Lactic Acid P (g/L): 40.45
Maltose M (g/L): 69.79
Volume V (L): 5.80

t = 18.00 h

Biomass X (g/L): 93.67
Glucose S (g/L): 66.57
Lactic Acid P (g/L): 64.77
Maltose M (g/L): 63.29
Volume V (L): 6.19

t = 19.00 h

Biomass X (g/L): 64.37
Glucose S (g/L): 61.73
Lactic Acid P (g/L): 31.10
Maltose M (g/L): 75.68
Volume V (L): 6.78

t = 20.00 h

Biomass X (g/L): 80.01
Glucose S (g/L): 66.51
Lactic Acid P (g/L): 52.19
Maltose M (g/L): 61.08
Volume V (L): 6.58

t = 21.00 h

Biomass X (g/L): 81.54
Glucose S (g/L): 65.94
Lactic Acid P (g/L): 29.63
Maltose M (g/L): 69.05
Volume V (L): 5.97

t = 22.00 h

Biomass X (g/L): 77.37
Glucose S (g/L): 68.19
Lactic Acid P (g/L): 50.61
Maltose M (g/L): 66.59
Volume V (L): 6.47

t = 23.00 h

Biomass X (g/L): 78.97
Glucose S (g/L): 64.73
Lactic Acid P (g/L): 54.79
Maltose M (g/L): 60.19
Volume V (L): 7.44

t = 24.00 h

Biomass X (g/L): 84.74
Glucose S (g/L): 67.54
Lactic Acid P (g/L): 72.98
Maltose M (g/L): 63.45
Volume V (L): 6.71

t = 25.00 h

Biomass X (g/L): 79.39
Glucose S (g/L): 66.15
Lactic Acid P (g/L): 64.80
Maltose M (g/L): 63.07
Volume V (L): 6.60

t = 26.00 h

Biomass X (g/L): 71.80
Glucose S (g/L): 66.41
Lactic Acid P (g/L): 54.35
Maltose M (g/L): 67.57
Volume V (L): 7.01

t = 27.00 h

Biomass X (g/L): 88.04
Glucose S (g/L): 68.28
Lactic Acid P (g/L): 72.56
Maltose M (g/L): 59.70
Volume V (L): 7.19

t = 28.00 h

Biomass X (g/L): 96.74
Glucose S (g/L): 68.48
Lactic Acid P (g/L): 63.66
Maltose M (g/L): 60.82
Volume V (L): 6.76

t = 29.00 h

Biomass X (g/L): 95.22
Glucose S (g/L): 71.61
Lactic Acid P (g/L): 64.52
Maltose M (g/L): 52.77
Volume V (L): 7.22

t = 30.00 h

Biomass X (g/L): 79.50
Glucose S (g/L): 68.69
Lactic Acid P (g/L): 53.77
Maltose M (g/L): 67.07
Volume V (L): 6.95

t = 31.00 h

Biomass X (g/L): 75.59
Glucose S (g/L): 68.28
Lactic Acid P (g/L): 56.32
Maltose M (g/L): 56.37
Volume V (L): 7.45

t = 32.00 h

Biomass X (g/L): 85.12
Glucose S (g/L): 69.40
Lactic Acid P (g/L): 58.40
Maltose M (g/L): 60.97
Volume V (L): 7.29

t = 33.00 h

Biomass X (g/L): 103.08
Glucose S (g/L): 67.88
Lactic Acid P (g/L): 66.94
Maltose M (g/L): 58.93
Volume V (L): 7.49

t = 34.00 h

Biomass X (g/L): 94.43
Glucose S (g/L): 69.75
Lactic Acid P (g/L): 67.72
Maltose M (g/L): 58.47

Volume V (L): 7.05

t = 35.00 h

Biomass X (g/L): 95.14

Glucose S (g/L): 79.36

Lactic Acid P (g/L): 81.66

Maltose M (g/L): 61.30

Volume V (L): 7.09

t = 36.00 h

Biomass X (g/L): 97.04

Glucose S (g/L): 68.74

Lactic Acid P (g/L): 71.47

Maltose M (g/L): 63.98

Volume V (L): 7.90

t = 37.00 h

Biomass X (g/L): 92.64

Glucose S (g/L): 71.71

Lactic Acid P (g/L): 43.50

Maltose M (g/L): 58.36

Volume V (L): 7.54

t = 38.00 h

Biomass X (g/L): 89.82

Glucose S (g/L): 71.88

Lactic Acid P (g/L): 71.17

Maltose M (g/L): 59.39

Volume V (L): 7.76

t = 39.00 h

Biomass X (g/L): 95.05

Glucose S (g/L): 72.99

Lactic Acid P (g/L): 55.83

Maltose M (g/L): 57.76

Volume V (L): 7.58

t = 40.00 h

Biomass X (g/L): 83.49

Glucose S (g/L): 73.26

Lactic Acid P (g/L): 60.65

Maltose M (g/L): 59.83

Volume V (L): 8.15

t = 41.00 h

Biomass X (g/L): 75.17

Glucose S (g/L): 76.18

Lactic Acid P (g/L): 62.41

Maltose M (g/L): 54.03

Volume V (L): 7.60

t = 42.00 h

Biomass X (g/L): 95.62

Glucose S (g/L): 69.23

Lactic Acid P (g/L): 64.11

Maltose M (g/L): 59.20

Volume V (L): 7.86

t = 43.00 h

Biomass X (g/L): 100.51

Glucose S (g/L): 80.25

Lactic Acid P (g/L): 57.89

Maltose M (g/L): 58.91

Volume V (L): 8.32

t = 44.00 h

Biomass X (g/L): 77.89

Glucose S (g/L): 79.32

Lactic Acid P (g/L): 72.69

Maltose M (g/L): 58.40

Volume V (L): 8.55

t = 45.00 h

Biomass X (g/L): 96.40

Glucose S (g/L): 72.81

Lactic Acid P (g/L): 58.16

Maltose M (g/L): 51.67

Volume V (L): 7.91

t = 46.00 h

Biomass X (g/L): 104.54

Glucose S (g/L): 69.53

Lactic Acid P (g/L): 74.67

Maltose M (g/L): 53.06

Volume V (L): 9.01

t = 47.00 h

Biomass X (g/L): 88.33

Glucose S (g/L): 75.04

Lactic Acid P (g/L): 92.20

Maltose M (g/L): 54.15

Volume V (L): 8.40

t = 48.00 h

Biomass X (g/L): 80.41

Glucose S (g/L): 78.43

Lactic Acid P (g/L): 59.72

Maltose M (g/L): 47.91

Volume V (L): 7.90

t = 49.00 h

Biomass X (g/L): 89.30

Glucose S (g/L): 89.74

Lactic Acid P (g/L): 69.56

Maltose M (g/L): 58.31

Volume V (L): 8.56

t = 50.00 h

Biomass X (g/L): 82.73

Glucose S (g/L): 82.62

Lactic Acid P (g/L): 49.25

Maltose M (g/L): 52.09

Volume V (L): 8.31

t = 51.00 h

Biomass X (g/L): 81.96

Glucose S (g/L): 74.30

Lactic Acid P (g/L): 69.88

Maltose M (g/L): 52.80

Volume V (L): 8.72

t = 52.00 h

Biomass X (g/L): 91.87

Glucose S (g/L): 79.14
Lactic Acid P (g/L): 78.77
Maltose M (g/L): 49.30
Volume V (L): 8.60

t = 53.00 h

Biomass X (g/L): 72.21
Glucose S (g/L): 82.15
Lactic Acid P (g/L): 79.70
Maltose M (g/L): 46.22
Volume V (L): 8.71

t = 54.00 h

Biomass X (g/L): 73.59
Glucose S (g/L): 83.91
Lactic Acid P (g/L): 74.00
Maltose M (g/L): 48.49
Volume V (L): 8.37

t = 55.00 h

Biomass X (g/L): 87.44
Glucose S (g/L): 78.55
Lactic Acid P (g/L): 81.66
Maltose M (g/L): 51.13
Volume V (L): 9.23

t = 56.00 h

Biomass X (g/L): 86.87
Glucose S (g/L): 83.61
Lactic Acid P (g/L): 97.06
Maltose M (g/L): 52.43
Volume V (L): 8.41

t = 57.00 h

Biomass X (g/L): 90.20
Glucose S (g/L): 81.98
Lactic Acid P (g/L): 89.60
Maltose M (g/L): 50.23
Volume V (L): 8.50

t = 58.00 h

Biomass X (g/L): 104.69
Glucose S (g/L): 81.09
Lactic Acid P (g/L): 69.26
Maltose M (g/L): 50.57
Volume V (L): 9.18

t = 59.00 h

Biomass X (g/L): 87.55
Glucose S (g/L): 86.84
Lactic Acid P (g/L): 60.91
Maltose M (g/L): 47.75
Volume V (L): 9.77

t = 60.00 h

Biomass X (g/L): 92.09
Glucose S (g/L): 89.35
Lactic Acid P (g/L): 70.68
Maltose M (g/L): 50.45
Volume V (L): 9.02

t = 61.00 h

Biomass X (g/L): 76.83
Glucose S (g/L): 90.17
Lactic Acid P (g/L): 65.88
Maltose M (g/L): 53.71
Volume V (L): 9.53

t = 62.00 h

Biomass X (g/L): 93.33
Glucose S (g/L): 89.72
Lactic Acid P (g/L): 78.88
Maltose M (g/L): 50.56
Volume V (L): 9.76

t = 63.00 h

Biomass X (g/L): 80.74
Glucose S (g/L): 86.21
Lactic Acid P (g/L): 64.05
Maltose M (g/L): 50.63
Volume V (L): 9.72

t = 64.00 h

Biomass X (g/L): 86.26
Glucose S (g/L): 91.36
Lactic Acid P (g/L): 74.38
Maltose M (g/L): 48.44
Volume V (L): 9.32

t = 65.00 h

Biomass X (g/L): 84.33
Glucose S (g/L): 90.37
Lactic Acid P (g/L): 82.89
Maltose M (g/L): 41.60
Volume V (L): 9.06

t = 66.00 h

Biomass X (g/L): 95.07
Glucose S (g/L): 90.24
Lactic Acid P (g/L): 74.42
Maltose M (g/L): 42.96
Volume V (L): 8.67

t = 67.00 h

Biomass X (g/L): 72.42
Glucose S (g/L): 87.95
Lactic Acid P (g/L): 66.80
Maltose M (g/L): 48.11
Volume V (L): 9.55

t = 68.00 h

Biomass X (g/L): 62.89
Glucose S (g/L): 88.91
Lactic Acid P (g/L): 86.41
Maltose M (g/L): 46.12
Volume V (L): 10.22

t = 69.00 h

Biomass X (g/L): 82.50
Glucose S (g/L): 91.61
Lactic Acid P (g/L): 58.51
Maltose M (g/L): 47.46
Volume V (L): 9.33

t = 70.00 h

Biomass X (g/L): 88.82
Glucose S (g/L): 101.10
Lactic Acid P (g/L): 72.88
Maltose M (g/L): 51.66
Volume V (L): 10.16

t = 71.00 h

Biomass X (g/L): 85.58
Glucose S (g/L): 93.50
Lactic Acid P (g/L): 75.64
Maltose M (g/L): 47.92
Volume V (L): 9.54

t = 72.00 h

Biomass X (g/L): 95.71
Glucose S (g/L): 90.55
Lactic Acid P (g/L): 76.30
Maltose M (g/L): 49.58
Volume V (L): 10.05

t = 73.00 h

Biomass X (g/L): 87.89
Glucose S (g/L): 90.22
Lactic Acid P (g/L): 77.93
Maltose M (g/L): 45.06
Volume V (L): 10.37

t = 74.00 h

Biomass X (g/L): 78.08
Glucose S (g/L): 92.01
Lactic Acid P (g/L): 82.00
Maltose M (g/L): 41.51
Volume V (L): 9.87

t = 75.00 h

Biomass X (g/L): 60.99
Glucose S (g/L): 102.64
Lactic Acid P (g/L): 76.19
Maltose M (g/L): 43.96

Volume V (L): 10.28

Batch 33

Initial Conditions:

Batch: 33.00

X0 (g/L): 9.45

S0 (g/L): 47.95

M0 (g/L): 105.47

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.20

F_vol (L/h): 0.02

S_feed (g/L): 93.76

M_feed (g/L): 206.24

V_final (L): 6.56

Batch 33 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 5.31
Glucose S (g/L): 56.59
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 102.14
Volume V (L): 5.03

t = 1.00 h

Biomass X (g/L): 19.62
Glucose S (g/L): 44.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 103.26
Volume V (L): 4.98

t = 2.00 h

Biomass X (g/L): 18.73
Glucose S (g/L): 49.14
Lactic Acid P (g/L): 1.18
Maltose M (g/L): 109.76
Volume V (L): 5.01

t = 3.00 h

Biomass X (g/L): 21.35
Glucose S (g/L): 41.59
Lactic Acid P (g/L): 15.84
Maltose M (g/L): 114.15
Volume V (L): 5.12

t = 4.00 h

Biomass X (g/L): 41.23
Glucose S (g/L): 36.15
Lactic Acid P (g/L): 30.19
Maltose M (g/L): 107.37
Volume V (L): 5.27

t = 5.00 h

Biomass X (g/L): 37.46
Glucose S (g/L): 37.43

Lactic Acid P (g/L): 19.87

Maltose M (g/L): 93.04

Volume V (L): 5.20

t = 6.00 h

Biomass X (g/L): 29.23

Glucose S (g/L): 36.13

Lactic Acid P (g/L): 31.30

Maltose M (g/L): 92.09

Volume V (L): 5.31

t = 7.00 h

Biomass X (g/L): 40.43

Glucose S (g/L): 33.84

Lactic Acid P (g/L): 20.97

Maltose M (g/L): 93.95

Volume V (L): 5.02

t = 8.00 h

Biomass X (g/L): 45.45

Glucose S (g/L): 33.10

Lactic Acid P (g/L): 33.38

Maltose M (g/L): 104.40

Volume V (L): 5.21

t = 9.00 h

Biomass X (g/L): 51.60

Glucose S (g/L): 35.41

Lactic Acid P (g/L): 38.88

Maltose M (g/L): 91.18

Volume V (L): 5.21

t = 10.00 h

Biomass X (g/L): 80.19

Glucose S (g/L): 22.72

Lactic Acid P (g/L): 14.00

Maltose M (g/L): 82.79

Volume V (L): 5.25

t = 11.00 h

Biomass X (g/L): 62.49

Glucose S (g/L): 17.06
Lactic Acid P (g/L): 32.67
Maltose M (g/L): 86.53
Volume V (L): 5.40

t = 12.00 h

Biomass X (g/L): 41.88
Glucose S (g/L): 19.34
Lactic Acid P (g/L): 49.05
Maltose M (g/L): 92.93
Volume V (L): 5.22

t = 13.00 h

Biomass X (g/L): 59.83
Glucose S (g/L): 15.03
Lactic Acid P (g/L): 44.74
Maltose M (g/L): 91.29
Volume V (L): 5.22

t = 14.00 h

Biomass X (g/L): 48.67
Glucose S (g/L): 18.17
Lactic Acid P (g/L): 32.23
Maltose M (g/L): 84.66
Volume V (L): 5.36

t = 15.00 h

Biomass X (g/L): 67.08
Glucose S (g/L): 8.73
Lactic Acid P (g/L): 21.90
Maltose M (g/L): 88.86
Volume V (L): 5.09

t = 16.00 h

Biomass X (g/L): 52.70
Glucose S (g/L): 21.62
Lactic Acid P (g/L): 44.47
Maltose M (g/L): 69.71
Volume V (L): 5.30

t = 17.00 h

Biomass X (g/L): 65.76
Glucose S (g/L): 15.24
Lactic Acid P (g/L): 49.14
Maltose M (g/L): 83.05
Volume V (L): 5.34

t = 18.00 h

Biomass X (g/L): 61.01
Glucose S (g/L): 5.16
Lactic Acid P (g/L): 42.58
Maltose M (g/L): 86.55
Volume V (L): 5.26

t = 19.00 h

Biomass X (g/L): 81.58
Glucose S (g/L): 8.41
Lactic Acid P (g/L): 54.34
Maltose M (g/L): 75.15
Volume V (L): 5.53

t = 20.00 h

Biomass X (g/L): 79.16
Glucose S (g/L): 13.84
Lactic Acid P (g/L): 59.35
Maltose M (g/L): 74.07
Volume V (L): 5.28

t = 21.00 h

Biomass X (g/L): 83.45
Glucose S (g/L): 14.99
Lactic Acid P (g/L): 62.28
Maltose M (g/L): 66.76
Volume V (L): 5.42

t = 22.00 h

Biomass X (g/L): 80.54
Glucose S (g/L): 5.67
Lactic Acid P (g/L): 33.13
Maltose M (g/L): 86.79
Volume V (L): 5.50

t = 23.00 h

Biomass X (g/L): 68.72
Glucose S (g/L): 13.07
Lactic Acid P (g/L): 44.70
Maltose M (g/L): 71.73
Volume V (L): 5.65

t = 24.00 h

Biomass X (g/L): 88.73
Glucose S (g/L): 3.08
Lactic Acid P (g/L): 55.90
Maltose M (g/L): 76.61
Volume V (L): 5.50

t = 25.00 h

Biomass X (g/L): 103.67
Glucose S (g/L): 10.12
Lactic Acid P (g/L): 65.71
Maltose M (g/L): 77.87
Volume V (L): 5.64

t = 26.00 h

Biomass X (g/L): 89.38
Glucose S (g/L): 14.73
Lactic Acid P (g/L): 65.85
Maltose M (g/L): 82.46
Volume V (L): 5.63

t = 27.00 h

Biomass X (g/L): 81.79
Glucose S (g/L): 8.62
Lactic Acid P (g/L): 66.59
Maltose M (g/L): 71.41
Volume V (L): 5.58

t = 28.00 h

Biomass X (g/L): 72.47
Glucose S (g/L): 5.20
Lactic Acid P (g/L): 58.90
Maltose M (g/L): 68.17
Volume V (L): 5.45

t = 29.00 h

Biomass X (g/L): 78.28
Glucose S (g/L): 11.61
Lactic Acid P (g/L): 60.30
Maltose M (g/L): 71.63
Volume V (L): 5.43

t = 30.00 h

Biomass X (g/L): 92.45
Glucose S (g/L): 9.02
Lactic Acid P (g/L): 57.09
Maltose M (g/L): 77.54
Volume V (L): 5.64

t = 31.00 h

Biomass X (g/L): 99.16
Glucose S (g/L): 13.90
Lactic Acid P (g/L): 72.88
Maltose M (g/L): 67.84
Volume V (L): 5.58

t = 32.00 h

Biomass X (g/L): 91.54
Glucose S (g/L): 8.48
Lactic Acid P (g/L): 56.69
Maltose M (g/L): 62.24
Volume V (L): 5.78

t = 33.00 h

Biomass X (g/L): 97.37
Glucose S (g/L): 9.20
Lactic Acid P (g/L): 75.17
Maltose M (g/L): 57.87
Volume V (L): 5.78

t = 34.00 h

Biomass X (g/L): 96.85
Glucose S (g/L): 4.81
Lactic Acid P (g/L): 69.42
Maltose M (g/L): 59.03

Volume V (L): 5.69

t = 35.00 h

Biomass X (g/L): 95.22

Glucose S (g/L): 13.36

Lactic Acid P (g/L): 67.55

Maltose M (g/L): 70.95

Volume V (L): 5.59

t = 36.00 h

Biomass X (g/L): 83.93

Glucose S (g/L): 12.45

Lactic Acid P (g/L): 62.06

Maltose M (g/L): 62.98

Volume V (L): 5.81

t = 37.00 h

Biomass X (g/L): 92.54

Glucose S (g/L): 8.93

Lactic Acid P (g/L): 55.68

Maltose M (g/L): 60.50

Volume V (L): 5.64

t = 38.00 h

Biomass X (g/L): 82.61

Glucose S (g/L): 14.48

Lactic Acid P (g/L): 63.38

Maltose M (g/L): 59.11

Volume V (L): 5.70

t = 39.00 h

Biomass X (g/L): 91.50

Glucose S (g/L): 17.97

Lactic Acid P (g/L): 67.15

Maltose M (g/L): 63.27

Volume V (L): 5.69

t = 40.00 h

Biomass X (g/L): 94.85

Glucose S (g/L): 13.68

Lactic Acid P (g/L): 47.26

Maltose M (g/L): 60.60

Volume V (L): 5.82

t = 41.00 h

Biomass X (g/L): 108.51

Glucose S (g/L): 15.87

Lactic Acid P (g/L): 69.59

Maltose M (g/L): 59.71

Volume V (L): 6.00

t = 42.00 h

Biomass X (g/L): 78.85

Glucose S (g/L): 16.28

Lactic Acid P (g/L): 81.36

Maltose M (g/L): 68.92

Volume V (L): 5.88

t = 43.00 h

Biomass X (g/L): 96.88

Glucose S (g/L): 11.12

Lactic Acid P (g/L): 78.29

Maltose M (g/L): 69.41

Volume V (L): 5.84

t = 44.00 h

Biomass X (g/L): 106.99

Glucose S (g/L): 23.76

Lactic Acid P (g/L): 68.13

Maltose M (g/L): 49.61

Volume V (L): 5.73

t = 45.00 h

Biomass X (g/L): 99.89

Glucose S (g/L): 14.67

Lactic Acid P (g/L): 74.05

Maltose M (g/L): 60.77

Volume V (L): 6.01

t = 46.00 h

Biomass X (g/L): 100.72

Glucose S (g/L): 19.31

Lactic Acid P (g/L): 68.47
Maltose M (g/L): 53.43
Volume V (L): 5.86

t = 47.00 h

Biomass X (g/L): 98.35
Glucose S (g/L): 20.14
Lactic Acid P (g/L): 76.00
Maltose M (g/L): 56.04
Volume V (L): 6.00

t = 48.00 h

Biomass X (g/L): 105.50
Glucose S (g/L): 17.14
Lactic Acid P (g/L): 90.39
Maltose M (g/L): 58.84
Volume V (L): 5.96

t = 49.00 h

Biomass X (g/L): 81.73
Glucose S (g/L): 20.36
Lactic Acid P (g/L): 85.09
Maltose M (g/L): 55.08
Volume V (L): 6.13

t = 50.00 h

Biomass X (g/L): 78.78
Glucose S (g/L): 19.42
Lactic Acid P (g/L): 81.02
Maltose M (g/L): 57.21
Volume V (L): 6.02

t = 51.00 h

Biomass X (g/L): 92.56
Glucose S (g/L): 13.59
Lactic Acid P (g/L): 71.26
Maltose M (g/L): 59.34
Volume V (L): 6.06

t = 52.00 h

Biomass X (g/L): 71.05

Glucose S (g/L): 18.08
Lactic Acid P (g/L): 84.12
Maltose M (g/L): 56.49
Volume V (L): 6.15

t = 53.00 h

Biomass X (g/L): 104.90
Glucose S (g/L): 17.45
Lactic Acid P (g/L): 60.07
Maltose M (g/L): 50.22
Volume V (L): 6.37

t = 54.00 h

Biomass X (g/L): 104.77
Glucose S (g/L): 18.43
Lactic Acid P (g/L): 74.83
Maltose M (g/L): 49.80
Volume V (L): 6.24

t = 55.00 h

Biomass X (g/L): 88.23
Glucose S (g/L): 15.69
Lactic Acid P (g/L): 83.58
Maltose M (g/L): 44.35
Volume V (L): 6.14

t = 56.00 h

Biomass X (g/L): 88.80
Glucose S (g/L): 25.23
Lactic Acid P (g/L): 85.51
Maltose M (g/L): 47.41
Volume V (L): 6.17

t = 57.00 h

Biomass X (g/L): 91.72
Glucose S (g/L): 27.43
Lactic Acid P (g/L): 85.15
Maltose M (g/L): 64.06
Volume V (L): 6.12

t = 58.00 h

Biomass X (g/L): 92.90
Glucose S (g/L): 25.51
Lactic Acid P (g/L): 81.73
Maltose M (g/L): 46.80
Volume V (L): 6.09

t = 59.00 h

Biomass X (g/L): 93.57
Glucose S (g/L): 27.02
Lactic Acid P (g/L): 77.14
Maltose M (g/L): 43.88
Volume V (L): 6.16

t = 60.00 h

Biomass X (g/L): 97.79
Glucose S (g/L): 21.37
Lactic Acid P (g/L): 84.83
Maltose M (g/L): 52.62
Volume V (L): 6.15

t = 61.00 h

Biomass X (g/L): 90.40
Glucose S (g/L): 20.69
Lactic Acid P (g/L): 70.19
Maltose M (g/L): 49.54
Volume V (L): 6.35

t = 62.00 h

Biomass X (g/L): 101.44
Glucose S (g/L): 20.57
Lactic Acid P (g/L): 71.59
Maltose M (g/L): 48.57
Volume V (L): 6.16

t = 63.00 h

Biomass X (g/L): 96.37
Glucose S (g/L): 26.32
Lactic Acid P (g/L): 80.98
Maltose M (g/L): 57.90
Volume V (L): 6.48

t = 64.00 h

Biomass X (g/L): 85.74
Glucose S (g/L): 25.16
Lactic Acid P (g/L): 82.40
Maltose M (g/L): 52.23
Volume V (L): 6.43

t = 65.00 h

Biomass X (g/L): 105.00
Glucose S (g/L): 24.42
Lactic Acid P (g/L): 62.65
Maltose M (g/L): 45.23
Volume V (L): 6.36

t = 66.00 h

Biomass X (g/L): 77.99
Glucose S (g/L): 29.98
Lactic Acid P (g/L): 85.80
Maltose M (g/L): 45.45
Volume V (L): 6.38

t = 67.00 h

Biomass X (g/L): 90.50
Glucose S (g/L): 27.04
Lactic Acid P (g/L): 89.27
Maltose M (g/L): 44.56
Volume V (L): 6.33

t = 68.00 h

Biomass X (g/L): 100.93
Glucose S (g/L): 22.77
Lactic Acid P (g/L): 81.02
Maltose M (g/L): 52.54
Volume V (L): 6.45

t = 69.00 h

Biomass X (g/L): 83.91
Glucose S (g/L): 23.76
Lactic Acid P (g/L): 105.87
Maltose M (g/L): 44.95
Volume V (L): 6.58

t = 70.00 h

Biomass X (g/L): 94.07
Glucose S (g/L): 29.27
Lactic Acid P (g/L): 75.48
Maltose M (g/L): 35.52
Volume V (L): 6.37

t = 71.00 h

Biomass X (g/L): 96.25
Glucose S (g/L): 31.01
Lactic Acid P (g/L): 78.96
Maltose M (g/L): 45.76
Volume V (L): 6.29

t = 72.00 h

Biomass X (g/L): 82.08
Glucose S (g/L): 29.56
Lactic Acid P (g/L): 78.88
Maltose M (g/L): 36.32
Volume V (L): 6.59

t = 73.00 h

Biomass X (g/L): 89.78
Glucose S (g/L): 35.00
Lactic Acid P (g/L): 82.32
Maltose M (g/L): 58.60
Volume V (L): 6.56

t = 74.00 h

Biomass X (g/L): 88.78
Glucose S (g/L): 34.40
Lactic Acid P (g/L): 80.03
Maltose M (g/L): 39.06
Volume V (L): 6.40

t = 75.00 h

Biomass X (g/L): 84.66
Glucose S (g/L): 30.30
Lactic Acid P (g/L): 69.46
Maltose M (g/L): 43.60

Volume V (L): 6.47

Batch 34

Initial Conditions:

Batch: 34.00

X0 (g/L): 8.15

S0 (g/L): 46.46

M0 (g/L): 73.19

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.58

F_vol (L/h): 0.08

S_feed (g/L): 116.50

M_feed (g/L): 183.50

V_final (L): 10.64

Batch 34 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.24
Glucose S (g/L): 52.35
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 72.83
Volume V (L): 5.97

t = 1.00 h

Biomass X (g/L): 4.07
Glucose S (g/L): 48.19
Lactic Acid P (g/L): 2.23
Maltose M (g/L): 73.58
Volume V (L): 5.09

t = 2.00 h

Biomass X (g/L): 10.91
Glucose S (g/L): 46.77
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 75.54
Volume V (L): 5.97

t = 3.00 h

Biomass X (g/L): 15.13
Glucose S (g/L): 44.78
Lactic Acid P (g/L): 17.02
Maltose M (g/L): 74.73
Volume V (L): 4.70

t = 4.00 h

Biomass X (g/L): 15.75
Glucose S (g/L): 40.27
Lactic Acid P (g/L): 22.94
Maltose M (g/L): 74.94
Volume V (L): 4.59

t = 5.00 h

Biomass X (g/L): 19.26
Glucose S (g/L): 39.52

Lactic Acid P (g/L): 12.34

Maltose M (g/L): 75.08

Volume V (L): 4.87

t = 6.00 h

Biomass X (g/L): 28.81

Glucose S (g/L): 34.51

Lactic Acid P (g/L): 4.61

Maltose M (g/L): 73.85

Volume V (L): 5.99

t = 7.00 h

Biomass X (g/L): 37.16

Glucose S (g/L): 34.19

Lactic Acid P (g/L): 15.88

Maltose M (g/L): 74.62

Volume V (L): 6.32

t = 8.00 h

Biomass X (g/L): 37.27

Glucose S (g/L): 23.53

Lactic Acid P (g/L): 22.39

Maltose M (g/L): 72.55

Volume V (L): 5.25

t = 9.00 h

Biomass X (g/L): 55.19

Glucose S (g/L): 30.78

Lactic Acid P (g/L): 31.88

Maltose M (g/L): 76.09

Volume V (L): 5.47

t = 10.00 h

Biomass X (g/L): 53.25

Glucose S (g/L): 26.63

Lactic Acid P (g/L): 21.44

Maltose M (g/L): 70.02

Volume V (L): 5.22

t = 11.00 h

Biomass X (g/L): 37.63

Glucose S (g/L): 23.92
Lactic Acid P (g/L): 26.49
Maltose M (g/L): 74.04
Volume V (L): 5.01

t = 12.00 h

Biomass X (g/L): 60.48
Glucose S (g/L): 24.19
Lactic Acid P (g/L): 40.64
Maltose M (g/L): 75.84
Volume V (L): 6.23

t = 13.00 h

Biomass X (g/L): 64.28
Glucose S (g/L): 16.98
Lactic Acid P (g/L): 46.93
Maltose M (g/L): 72.03
Volume V (L): 6.08

t = 14.00 h

Biomass X (g/L): 57.44
Glucose S (g/L): 27.64
Lactic Acid P (g/L): 35.19
Maltose M (g/L): 74.60
Volume V (L): 6.06

t = 15.00 h

Biomass X (g/L): 50.13
Glucose S (g/L): 19.58
Lactic Acid P (g/L): 30.87
Maltose M (g/L): 72.90
Volume V (L): 5.50

t = 16.00 h

Biomass X (g/L): 66.17
Glucose S (g/L): 18.72
Lactic Acid P (g/L): 74.14
Maltose M (g/L): 73.27
Volume V (L): 6.94

t = 17.00 h

Biomass X (g/L): 67.99
Glucose S (g/L): 15.31
Lactic Acid P (g/L): 53.45
Maltose M (g/L): 75.13
Volume V (L): 6.05

t = 18.00 h

Biomass X (g/L): 59.51
Glucose S (g/L): 18.42
Lactic Acid P (g/L): 45.62
Maltose M (g/L): 73.88
Volume V (L): 6.59

t = 19.00 h

Biomass X (g/L): 58.06
Glucose S (g/L): 22.98
Lactic Acid P (g/L): 57.27
Maltose M (g/L): 70.41
Volume V (L): 5.77

t = 20.00 h

Biomass X (g/L): 72.05
Glucose S (g/L): 21.65
Lactic Acid P (g/L): 49.72
Maltose M (g/L): 71.85
Volume V (L): 6.42

t = 21.00 h

Biomass X (g/L): 71.99
Glucose S (g/L): 21.02
Lactic Acid P (g/L): 59.87
Maltose M (g/L): 74.61
Volume V (L): 6.71

t = 22.00 h

Biomass X (g/L): 71.46
Glucose S (g/L): 20.04
Lactic Acid P (g/L): 54.70
Maltose M (g/L): 73.66
Volume V (L): 6.80

t = 23.00 h

Biomass X (g/L): 75.59
Glucose S (g/L): 17.04
Lactic Acid P (g/L): 55.18
Maltose M (g/L): 75.43
Volume V (L): 6.80

t = 24.00 h

Biomass X (g/L): 67.15
Glucose S (g/L): 13.46
Lactic Acid P (g/L): 54.37
Maltose M (g/L): 75.15
Volume V (L): 7.03

t = 25.00 h

Biomass X (g/L): 74.16
Glucose S (g/L): 18.16
Lactic Acid P (g/L): 43.37
Maltose M (g/L): 71.93
Volume V (L): 6.37

t = 26.00 h

Biomass X (g/L): 83.86
Glucose S (g/L): 18.41
Lactic Acid P (g/L): 74.62
Maltose M (g/L): 75.99
Volume V (L): 7.04

t = 27.00 h

Biomass X (g/L): 76.43
Glucose S (g/L): 23.34
Lactic Acid P (g/L): 49.66
Maltose M (g/L): 72.08
Volume V (L): 7.08

t = 28.00 h

Biomass X (g/L): 64.90
Glucose S (g/L): 21.20
Lactic Acid P (g/L): 46.72
Maltose M (g/L): 70.39
Volume V (L): 7.21

t = 29.00 h

Biomass X (g/L): 76.52
Glucose S (g/L): 26.27
Lactic Acid P (g/L): 57.76
Maltose M (g/L): 70.66
Volume V (L): 7.21

t = 30.00 h

Biomass X (g/L): 71.54
Glucose S (g/L): 25.35
Lactic Acid P (g/L): 51.77
Maltose M (g/L): 68.01
Volume V (L): 6.68

t = 31.00 h

Biomass X (g/L): 84.32
Glucose S (g/L): 20.10
Lactic Acid P (g/L): 60.60
Maltose M (g/L): 73.57
Volume V (L): 6.74

t = 32.00 h

Biomass X (g/L): 85.22
Glucose S (g/L): 15.99
Lactic Acid P (g/L): 67.88
Maltose M (g/L): 73.32
Volume V (L): 6.41

t = 33.00 h

Biomass X (g/L): 67.39
Glucose S (g/L): 27.77
Lactic Acid P (g/L): 71.25
Maltose M (g/L): 69.17
Volume V (L): 7.85

t = 34.00 h

Biomass X (g/L): 90.00
Glucose S (g/L): 28.31
Lactic Acid P (g/L): 70.22
Maltose M (g/L): 70.11

Volume V (L): 7.93

t = 35.00 h

Biomass X (g/L): 73.07

Glucose S (g/L): 19.39

Lactic Acid P (g/L): 60.74

Maltose M (g/L): 70.35

Volume V (L): 8.01

t = 36.00 h

Biomass X (g/L): 97.94

Glucose S (g/L): 30.16

Lactic Acid P (g/L): 64.46

Maltose M (g/L): 68.52

Volume V (L): 7.84

t = 37.00 h

Biomass X (g/L): 83.37

Glucose S (g/L): 26.88

Lactic Acid P (g/L): 68.18

Maltose M (g/L): 67.07

Volume V (L): 6.82

t = 38.00 h

Biomass X (g/L): 72.55

Glucose S (g/L): 26.79

Lactic Acid P (g/L): 49.26

Maltose M (g/L): 69.13

Volume V (L): 7.39

t = 39.00 h

Biomass X (g/L): 73.69

Glucose S (g/L): 32.61

Lactic Acid P (g/L): 70.59

Maltose M (g/L): 68.71

Volume V (L): 8.01

t = 40.00 h

Biomass X (g/L): 87.57

Glucose S (g/L): 32.37

Lactic Acid P (g/L): 52.86

Maltose M (g/L): 68.34

Volume V (L): 8.07

t = 41.00 h

Biomass X (g/L): 80.89

Glucose S (g/L): 30.66

Lactic Acid P (g/L): 84.64

Maltose M (g/L): 68.99

Volume V (L): 7.98

t = 42.00 h

Biomass X (g/L): 83.72

Glucose S (g/L): 30.76

Lactic Acid P (g/L): 63.14

Maltose M (g/L): 70.54

Volume V (L): 8.32

t = 43.00 h

Biomass X (g/L): 99.55

Glucose S (g/L): 32.00

Lactic Acid P (g/L): 65.50

Maltose M (g/L): 67.67

Volume V (L): 8.24

t = 44.00 h

Biomass X (g/L): 78.33

Glucose S (g/L): 29.17

Lactic Acid P (g/L): 60.63

Maltose M (g/L): 68.11

Volume V (L): 8.36

t = 45.00 h

Biomass X (g/L): 85.37

Glucose S (g/L): 31.13

Lactic Acid P (g/L): 62.25

Maltose M (g/L): 68.67

Volume V (L): 8.09

t = 46.00 h

Biomass X (g/L): 97.93

Glucose S (g/L): 28.90

Lactic Acid P (g/L): 88.63

Maltose M (g/L): 68.14

Volume V (L): 8.90

t = 47.00 h

Biomass X (g/L): 87.37

Glucose S (g/L): 31.65

Lactic Acid P (g/L): 77.01

Maltose M (g/L): 66.11

Volume V (L): 8.34

t = 48.00 h

Biomass X (g/L): 70.09

Glucose S (g/L): 32.29

Lactic Acid P (g/L): 55.02

Maltose M (g/L): 66.90

Volume V (L): 8.01

t = 49.00 h

Biomass X (g/L): 89.53

Glucose S (g/L): 32.10

Lactic Acid P (g/L): 82.70

Maltose M (g/L): 67.30

Volume V (L): 8.34

t = 50.00 h

Biomass X (g/L): 74.14

Glucose S (g/L): 35.57

Lactic Acid P (g/L): 80.22

Maltose M (g/L): 64.51

Volume V (L): 8.60

t = 51.00 h

Biomass X (g/L): 97.77

Glucose S (g/L): 34.70

Lactic Acid P (g/L): 67.29

Maltose M (g/L): 67.81

Volume V (L): 9.28

t = 52.00 h

Biomass X (g/L): 77.20

Glucose S (g/L): 38.36
Lactic Acid P (g/L): 72.84
Maltose M (g/L): 67.78
Volume V (L): 8.89

t = 53.00 h

Biomass X (g/L): 112.07
Glucose S (g/L): 31.65
Lactic Acid P (g/L): 70.23
Maltose M (g/L): 64.28
Volume V (L): 8.90

t = 54.00 h

Biomass X (g/L): 98.17
Glucose S (g/L): 37.70
Lactic Acid P (g/L): 88.31
Maltose M (g/L): 65.37
Volume V (L): 8.50

t = 55.00 h

Biomass X (g/L): 77.25
Glucose S (g/L): 39.31
Lactic Acid P (g/L): 63.75
Maltose M (g/L): 65.76
Volume V (L): 9.49

t = 56.00 h

Biomass X (g/L): 84.69
Glucose S (g/L): 38.51
Lactic Acid P (g/L): 72.44
Maltose M (g/L): 63.70
Volume V (L): 8.92

t = 57.00 h

Biomass X (g/L): 91.36
Glucose S (g/L): 39.74
Lactic Acid P (g/L): 67.53
Maltose M (g/L): 64.95
Volume V (L): 9.24

t = 58.00 h

Biomass X (g/L): 91.54
Glucose S (g/L): 35.52
Lactic Acid P (g/L): 58.76
Maltose M (g/L): 64.19
Volume V (L): 10.03

t = 59.00 h

Biomass X (g/L): 88.29
Glucose S (g/L): 43.08
Lactic Acid P (g/L): 91.47
Maltose M (g/L): 61.47
Volume V (L): 9.67

t = 60.00 h

Biomass X (g/L): 95.64
Glucose S (g/L): 44.84
Lactic Acid P (g/L): 61.45
Maltose M (g/L): 64.07
Volume V (L): 8.78

t = 61.00 h

Biomass X (g/L): 82.86
Glucose S (g/L): 45.03
Lactic Acid P (g/L): 74.23
Maltose M (g/L): 61.46
Volume V (L): 8.79

t = 62.00 h

Biomass X (g/L): 80.50
Glucose S (g/L): 47.68
Lactic Acid P (g/L): 56.86
Maltose M (g/L): 64.23
Volume V (L): 9.63

t = 63.00 h

Biomass X (g/L): 80.55
Glucose S (g/L): 45.46
Lactic Acid P (g/L): 69.08
Maltose M (g/L): 61.54
Volume V (L): 9.19

t = 64.00 h

Biomass X (g/L): 91.52
Glucose S (g/L): 47.16
Lactic Acid P (g/L): 76.63
Maltose M (g/L): 62.02
Volume V (L): 10.10

t = 65.00 h

Biomass X (g/L): 86.84
Glucose S (g/L): 46.75
Lactic Acid P (g/L): 65.65
Maltose M (g/L): 64.04
Volume V (L): 9.91

t = 66.00 h

Biomass X (g/L): 86.02
Glucose S (g/L): 49.33
Lactic Acid P (g/L): 78.49
Maltose M (g/L): 62.44
Volume V (L): 10.04

t = 67.00 h

Biomass X (g/L): 90.52
Glucose S (g/L): 42.08
Lactic Acid P (g/L): 70.99
Maltose M (g/L): 59.80
Volume V (L): 9.72

t = 68.00 h

Biomass X (g/L): 64.16
Glucose S (g/L): 45.19
Lactic Acid P (g/L): 83.98
Maltose M (g/L): 59.62
Volume V (L): 10.90

t = 69.00 h

Biomass X (g/L): 85.75
Glucose S (g/L): 54.60
Lactic Acid P (g/L): 85.96
Maltose M (g/L): 59.97
Volume V (L): 11.47

t = 70.00 h

Biomass X (g/L): 84.75
Glucose S (g/L): 41.08
Lactic Acid P (g/L): 68.66
Maltose M (g/L): 61.18
Volume V (L): 10.14

t = 71.00 h

Biomass X (g/L): 97.95
Glucose S (g/L): 49.93
Lactic Acid P (g/L): 74.77
Maltose M (g/L): 60.85
Volume V (L): 10.50

t = 72.00 h

Biomass X (g/L): 68.35
Glucose S (g/L): 55.22
Lactic Acid P (g/L): 75.88
Maltose M (g/L): 58.77
Volume V (L): 10.46

t = 73.00 h

Biomass X (g/L): 96.60
Glucose S (g/L): 51.46
Lactic Acid P (g/L): 79.94
Maltose M (g/L): 58.01
Volume V (L): 10.37

t = 74.00 h

Biomass X (g/L): 82.79
Glucose S (g/L): 51.04
Lactic Acid P (g/L): 74.67
Maltose M (g/L): 59.91
Volume V (L): 10.19

t = 75.00 h

Biomass X (g/L): 86.43
Glucose S (g/L): 53.59
Lactic Acid P (g/L): 76.39
Maltose M (g/L): 58.94

Volume V (L): 10.15

Batch 35

Initial Conditions:

Batch: 35.00

X0 (g/L): 4.96

S0 (g/L): 10.40

M0 (g/L): 128.12

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 12.33

F_vol (L/h): 0.04

S_feed (g/L): 22.51

M_feed (g/L): 277.49

V_final (L): 8.36

Batch 35 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 10.24
Lactic Acid P (g/L): 6.19
Maltose M (g/L): 125.13
Volume V (L): 5.28

t = 1.00 h

Biomass X (g/L): 9.84
Glucose S (g/L): 10.71
Lactic Acid P (g/L): 4.93
Maltose M (g/L): 124.62
Volume V (L): 5.97

t = 2.00 h

Biomass X (g/L): 2.72
Glucose S (g/L): 16.01
Lactic Acid P (g/L): 1.98
Maltose M (g/L): 129.95
Volume V (L): 5.12

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 11.07
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 126.06
Volume V (L): 4.78

t = 4.00 h

Biomass X (g/L): 15.36
Glucose S (g/L): 13.97
Lactic Acid P (g/L): 8.26
Maltose M (g/L): 119.00
Volume V (L): 5.02

t = 5.00 h

Biomass X (g/L): 13.25
Glucose S (g/L): 15.02

Lactic Acid P (g/L): 22.87
Maltose M (g/L): 129.35
Volume V (L): 5.39

t = 6.00 h

Biomass X (g/L): 28.67
Glucose S (g/L): 10.32
Lactic Acid P (g/L): 17.44
Maltose M (g/L): 112.80
Volume V (L): 4.84

t = 7.00 h

Biomass X (g/L): 22.57
Glucose S (g/L): 11.09
Lactic Acid P (g/L): 15.29
Maltose M (g/L): 124.58
Volume V (L): 5.36

t = 8.00 h

Biomass X (g/L): 8.55
Glucose S (g/L): 11.81
Lactic Acid P (g/L): 21.95
Maltose M (g/L): 112.99
Volume V (L): 5.46

t = 9.00 h

Biomass X (g/L): 4.77
Glucose S (g/L): 9.30
Lactic Acid P (g/L): 11.90
Maltose M (g/L): 128.26
Volume V (L): 5.81

t = 10.00 h

Biomass X (g/L): 21.85
Glucose S (g/L): 8.39
Lactic Acid P (g/L): 13.05
Maltose M (g/L): 109.58
Volume V (L): 5.80

t = 11.00 h

Biomass X (g/L): 25.79

Glucose S (g/L): 8.30
Lactic Acid P (g/L): 13.26
Maltose M (g/L): 122.53
Volume V (L): 5.72

t = 12.00 h

Biomass X (g/L): 43.19
Glucose S (g/L): 9.38
Lactic Acid P (g/L): 21.02
Maltose M (g/L): 115.22
Volume V (L): 4.96

t = 13.00 h

Biomass X (g/L): 37.17
Glucose S (g/L): 6.52
Lactic Acid P (g/L): 20.52
Maltose M (g/L): 112.82
Volume V (L): 5.83

t = 14.00 h

Biomass X (g/L): 50.27
Glucose S (g/L): 8.72
Lactic Acid P (g/L): 36.97
Maltose M (g/L): 114.75
Volume V (L): 5.82

t = 15.00 h

Biomass X (g/L): 29.96
Glucose S (g/L): 3.93
Lactic Acid P (g/L): 23.93
Maltose M (g/L): 116.15
Volume V (L): 5.74

t = 16.00 h

Biomass X (g/L): 53.28
Glucose S (g/L): 2.99
Lactic Acid P (g/L): 22.80
Maltose M (g/L): 107.33
Volume V (L): 5.25

t = 17.00 h

Biomass X (g/L): 53.06
Glucose S (g/L): 9.16
Lactic Acid P (g/L): 48.77
Maltose M (g/L): 106.92
Volume V (L): 5.55

t = 18.00 h

Biomass X (g/L): 29.68
Glucose S (g/L): 3.68
Lactic Acid P (g/L): 29.13
Maltose M (g/L): 113.62
Volume V (L): 5.49

t = 19.00 h

Biomass X (g/L): 62.27
Glucose S (g/L): 2.21
Lactic Acid P (g/L): 35.97
Maltose M (g/L): 100.61
Volume V (L): 5.31

t = 20.00 h

Biomass X (g/L): 52.60
Glucose S (g/L): 1.25
Lactic Acid P (g/L): 40.03
Maltose M (g/L): 105.57
Volume V (L): 5.89

t = 21.00 h

Biomass X (g/L): 56.93
Glucose S (g/L): 6.97
Lactic Acid P (g/L): 27.33
Maltose M (g/L): 123.45
Volume V (L): 5.73

t = 22.00 h

Biomass X (g/L): 46.84
Glucose S (g/L): 4.10
Lactic Acid P (g/L): 49.65
Maltose M (g/L): 101.96
Volume V (L): 6.69

t = 23.00 h

Biomass X (g/L): 68.02
Glucose S (g/L): 4.33
Lactic Acid P (g/L): 45.62
Maltose M (g/L): 112.08
Volume V (L): 5.98

t = 24.00 h

Biomass X (g/L): 46.68
Glucose S (g/L): 4.60
Lactic Acid P (g/L): 55.01
Maltose M (g/L): 102.35
Volume V (L): 6.42

t = 25.00 h

Biomass X (g/L): 66.61
Glucose S (g/L): 7.09
Lactic Acid P (g/L): 36.39
Maltose M (g/L): 107.23
Volume V (L): 5.70

t = 26.00 h

Biomass X (g/L): 67.77
Glucose S (g/L): 4.77
Lactic Acid P (g/L): 44.45
Maltose M (g/L): 107.21
Volume V (L): 6.02

t = 27.00 h

Biomass X (g/L): 52.42
Glucose S (g/L): 10.06
Lactic Acid P (g/L): 45.05
Maltose M (g/L): 108.91
Volume V (L): 6.23

t = 28.00 h

Biomass X (g/L): 80.87
Glucose S (g/L): 5.71
Lactic Acid P (g/L): 48.14
Maltose M (g/L): 101.31
Volume V (L): 6.01

t = 29.00 h

Biomass X (g/L): 65.82
Glucose S (g/L): 7.77
Lactic Acid P (g/L): 33.99
Maltose M (g/L): 111.88
Volume V (L): 6.18

t = 30.00 h

Biomass X (g/L): 70.79
Glucose S (g/L): 7.43
Lactic Acid P (g/L): 48.90
Maltose M (g/L): 97.85
Volume V (L): 6.28

t = 31.00 h

Biomass X (g/L): 51.70
Glucose S (g/L): 5.92
Lactic Acid P (g/L): 36.30
Maltose M (g/L): 101.39
Volume V (L): 6.32

t = 32.00 h

Biomass X (g/L): 86.31
Glucose S (g/L): 5.01
Lactic Acid P (g/L): 53.09
Maltose M (g/L): 100.75
Volume V (L): 6.93

t = 33.00 h

Biomass X (g/L): 56.43
Glucose S (g/L): 7.11
Lactic Acid P (g/L): 32.03
Maltose M (g/L): 93.01
Volume V (L): 6.58

t = 34.00 h

Biomass X (g/L): 78.09
Glucose S (g/L): 7.72
Lactic Acid P (g/L): 35.42
Maltose M (g/L): 96.64

Volume V (L): 6.65

t = 35.00 h

Biomass X (g/L): 60.33

Glucose S (g/L): 5.77

Lactic Acid P (g/L): 59.98

Maltose M (g/L): 98.28

Volume V (L): 6.71

t = 36.00 h

Biomass X (g/L): 64.63

Glucose S (g/L): 7.55

Lactic Acid P (g/L): 48.78

Maltose M (g/L): 94.55

Volume V (L): 6.77

t = 37.00 h

Biomass X (g/L): 78.41

Glucose S (g/L): 10.01

Lactic Acid P (g/L): 45.74

Maltose M (g/L): 108.73

Volume V (L): 6.63

t = 38.00 h

Biomass X (g/L): 81.37

Glucose S (g/L): 12.01

Lactic Acid P (g/L): 66.58

Maltose M (g/L): 91.16

Volume V (L): 6.61

t = 39.00 h

Biomass X (g/L): 60.88

Glucose S (g/L): 9.84

Lactic Acid P (g/L): 59.05

Maltose M (g/L): 100.94

Volume V (L): 6.79

t = 40.00 h

Biomass X (g/L): 83.36

Glucose S (g/L): 12.22

Lactic Acid P (g/L): 64.66

Maltose M (g/L): 98.64

Volume V (L): 7.10

t = 41.00 h

Biomass X (g/L): 83.59

Glucose S (g/L): 7.47

Lactic Acid P (g/L): 56.24

Maltose M (g/L): 93.53

Volume V (L): 7.14

t = 42.00 h

Biomass X (g/L): 65.92

Glucose S (g/L): 10.54

Lactic Acid P (g/L): 82.19

Maltose M (g/L): 88.53

Volume V (L): 6.81

t = 43.00 h

Biomass X (g/L): 89.90

Glucose S (g/L): 12.20

Lactic Acid P (g/L): 53.83

Maltose M (g/L): 88.43

Volume V (L): 7.13

t = 44.00 h

Biomass X (g/L): 62.27

Glucose S (g/L): 8.01

Lactic Acid P (g/L): 75.34

Maltose M (g/L): 89.12

Volume V (L): 6.92

t = 45.00 h

Biomass X (g/L): 70.47

Glucose S (g/L): 9.23

Lactic Acid P (g/L): 60.03

Maltose M (g/L): 81.99

Volume V (L): 7.02

t = 46.00 h

Biomass X (g/L): 81.15

Glucose S (g/L): 14.16

Lactic Acid P (g/L): 46.58

Maltose M (g/L): 85.45

Volume V (L): 6.93

t = 47.00 h

Biomass X (g/L): 69.32

Glucose S (g/L): 9.03

Lactic Acid P (g/L): 66.16

Maltose M (g/L): 82.41

Volume V (L): 7.05

t = 48.00 h

Biomass X (g/L): 80.07

Glucose S (g/L): 8.71

Lactic Acid P (g/L): 65.29

Maltose M (g/L): 96.43

Volume V (L): 7.51

t = 49.00 h

Biomass X (g/L): 80.27

Glucose S (g/L): 14.99

Lactic Acid P (g/L): 53.80

Maltose M (g/L): 92.12

Volume V (L): 7.22

t = 50.00 h

Biomass X (g/L): 76.70

Glucose S (g/L): 14.75

Lactic Acid P (g/L): 55.53

Maltose M (g/L): 92.44

Volume V (L): 7.27

t = 51.00 h

Biomass X (g/L): 86.46

Glucose S (g/L): 13.63

Lactic Acid P (g/L): 64.05

Maltose M (g/L): 90.87

Volume V (L): 6.89

t = 52.00 h

Biomass X (g/L): 77.01

Glucose S (g/L): 13.12
Lactic Acid P (g/L): 67.91
Maltose M (g/L): 88.16
Volume V (L): 7.24

t = 53.00 h

Biomass X (g/L): 81.56
Glucose S (g/L): 12.78
Lactic Acid P (g/L): 43.47
Maltose M (g/L): 94.30
Volume V (L): 7.31

t = 54.00 h

Biomass X (g/L): 90.29
Glucose S (g/L): 14.72
Lactic Acid P (g/L): 75.19
Maltose M (g/L): 83.09
Volume V (L): 7.19

t = 55.00 h

Biomass X (g/L): 87.27
Glucose S (g/L): 17.85
Lactic Acid P (g/L): 61.99
Maltose M (g/L): 86.27
Volume V (L): 7.47

t = 56.00 h

Biomass X (g/L): 87.47
Glucose S (g/L): 15.60
Lactic Acid P (g/L): 55.06
Maltose M (g/L): 89.07
Volume V (L): 7.39

t = 57.00 h

Biomass X (g/L): 69.84
Glucose S (g/L): 20.03
Lactic Acid P (g/L): 75.66
Maltose M (g/L): 77.89
Volume V (L): 7.57

t = 58.00 h

Biomass X (g/L): 85.49
Glucose S (g/L): 14.27
Lactic Acid P (g/L): 67.58
Maltose M (g/L): 100.18
Volume V (L): 7.63

t = 59.00 h

Biomass X (g/L): 72.73
Glucose S (g/L): 16.57
Lactic Acid P (g/L): 79.00
Maltose M (g/L): 83.86
Volume V (L): 7.38

t = 60.00 h

Biomass X (g/L): 69.18
Glucose S (g/L): 16.36
Lactic Acid P (g/L): 72.54
Maltose M (g/L): 83.95
Volume V (L): 7.58

t = 61.00 h

Biomass X (g/L): 80.49
Glucose S (g/L): 15.42
Lactic Acid P (g/L): 69.06
Maltose M (g/L): 89.22
Volume V (L): 7.75

t = 62.00 h

Biomass X (g/L): 72.94
Glucose S (g/L): 17.85
Lactic Acid P (g/L): 68.13
Maltose M (g/L): 86.98
Volume V (L): 8.09

t = 63.00 h

Biomass X (g/L): 88.69
Glucose S (g/L): 19.53
Lactic Acid P (g/L): 74.15
Maltose M (g/L): 81.98
Volume V (L): 7.89

t = 64.00 h

Biomass X (g/L): 89.01
Glucose S (g/L): 20.28
Lactic Acid P (g/L): 66.60
Maltose M (g/L): 81.99
Volume V (L): 7.69

t = 65.00 h

Biomass X (g/L): 64.03
Glucose S (g/L): 24.17
Lactic Acid P (g/L): 63.08
Maltose M (g/L): 73.10
Volume V (L): 8.01

t = 66.00 h

Biomass X (g/L): 74.02
Glucose S (g/L): 21.23
Lactic Acid P (g/L): 76.39
Maltose M (g/L): 77.53
Volume V (L): 8.41

t = 67.00 h

Biomass X (g/L): 68.25
Glucose S (g/L): 23.16
Lactic Acid P (g/L): 70.06
Maltose M (g/L): 82.62
Volume V (L): 8.24

t = 68.00 h

Biomass X (g/L): 77.50
Glucose S (g/L): 21.49
Lactic Acid P (g/L): 71.97
Maltose M (g/L): 81.80
Volume V (L): 8.19

t = 69.00 h

Biomass X (g/L): 86.00
Glucose S (g/L): 25.32
Lactic Acid P (g/L): 81.65
Maltose M (g/L): 85.37
Volume V (L): 8.28

t = 70.00 h

Biomass X (g/L): 79.59
Glucose S (g/L): 27.34
Lactic Acid P (g/L): 71.32
Maltose M (g/L): 78.01
Volume V (L): 8.52

t = 71.00 h

Biomass X (g/L): 76.39
Glucose S (g/L): 24.07
Lactic Acid P (g/L): 72.77
Maltose M (g/L): 71.04
Volume V (L): 8.46

t = 72.00 h

Biomass X (g/L): 93.83
Glucose S (g/L): 21.51
Lactic Acid P (g/L): 67.63
Maltose M (g/L): 83.23
Volume V (L): 8.37

t = 73.00 h

Biomass X (g/L): 60.57
Glucose S (g/L): 22.96
Lactic Acid P (g/L): 77.05
Maltose M (g/L): 84.76
Volume V (L): 8.55

t = 74.00 h

Biomass X (g/L): 93.22
Glucose S (g/L): 25.21
Lactic Acid P (g/L): 76.62
Maltose M (g/L): 81.57
Volume V (L): 8.02

t = 75.00 h

Biomass X (g/L): 76.76
Glucose S (g/L): 26.11
Lactic Acid P (g/L): 81.78
Maltose M (g/L): 75.68

Volume V (L): 7.95

Batch 36

Initial Conditions:

Batch: 36.00

X0 (g/L): 1.95

S0 (g/L): 76.24

M0 (g/L): 88.27

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.16

F_vol (L/h): 0.07

S_feed (g/L): 139.03

M_feed (g/L): 160.97

V_final (L): 10.13

Batch 36 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.49
Glucose S (g/L): 74.96
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 94.54
Volume V (L): 5.37

t = 1.00 h

Biomass X (g/L): 3.30
Glucose S (g/L): 76.63
Lactic Acid P (g/L): 1.24
Maltose M (g/L): 89.55
Volume V (L): 5.39

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 83.78
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 89.09
Volume V (L): 5.42

t = 3.00 h

Biomass X (g/L): 12.83
Glucose S (g/L): 82.57
Lactic Acid P (g/L): 3.40
Maltose M (g/L): 89.22
Volume V (L): 5.68

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 84.22
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 91.49
Volume V (L): 5.09

t = 5.00 h

Biomass X (g/L): 2.83
Glucose S (g/L): 85.82

Lactic Acid P (g/L): 4.78
Maltose M (g/L): 89.05
Volume V (L): 4.79

t = 6.00 h

Biomass X (g/L): 5.67
Glucose S (g/L): 83.63
Lactic Acid P (g/L): 13.04
Maltose M (g/L): 85.67
Volume V (L): 5.44

t = 7.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 84.21
Lactic Acid P (g/L): 7.83
Maltose M (g/L): 86.64
Volume V (L): 5.14

t = 8.00 h

Biomass X (g/L): 31.29
Glucose S (g/L): 75.58
Lactic Acid P (g/L): 7.18
Maltose M (g/L): 79.69
Volume V (L): 5.11

t = 9.00 h

Biomass X (g/L): 6.88
Glucose S (g/L): 79.29
Lactic Acid P (g/L): 15.27
Maltose M (g/L): 84.90
Volume V (L): 5.32

t = 10.00 h

Biomass X (g/L): 18.52
Glucose S (g/L): 76.13
Lactic Acid P (g/L): 22.71
Maltose M (g/L): 77.99
Volume V (L): 6.70

t = 11.00 h

Biomass X (g/L): 37.53

Glucose S (g/L): 79.41
Lactic Acid P (g/L): 9.52
Maltose M (g/L): 79.62
Volume V (L): 5.81

t = 12.00 h

Biomass X (g/L): 39.31
Glucose S (g/L): 74.72
Lactic Acid P (g/L): 25.12
Maltose M (g/L): 69.44
Volume V (L): 5.97

t = 13.00 h

Biomass X (g/L): 26.35
Glucose S (g/L): 67.58
Lactic Acid P (g/L): 37.91
Maltose M (g/L): 81.10
Volume V (L): 5.39

t = 14.00 h

Biomass X (g/L): 48.21
Glucose S (g/L): 68.15
Lactic Acid P (g/L): 34.34
Maltose M (g/L): 76.82
Volume V (L): 5.20

t = 15.00 h

Biomass X (g/L): 48.34
Glucose S (g/L): 65.25
Lactic Acid P (g/L): 32.70
Maltose M (g/L): 82.41
Volume V (L): 5.69

t = 16.00 h

Biomass X (g/L): 72.24
Glucose S (g/L): 63.99
Lactic Acid P (g/L): 40.84
Maltose M (g/L): 79.28
Volume V (L): 6.68

t = 17.00 h

Biomass X (g/L): 56.72
Glucose S (g/L): 57.69
Lactic Acid P (g/L): 41.84
Maltose M (g/L): 81.29
Volume V (L): 6.30

t = 18.00 h

Biomass X (g/L): 54.55
Glucose S (g/L): 60.13
Lactic Acid P (g/L): 47.08
Maltose M (g/L): 72.94
Volume V (L): 6.48

t = 19.00 h

Biomass X (g/L): 50.43
Glucose S (g/L): 56.91
Lactic Acid P (g/L): 24.26
Maltose M (g/L): 79.85
Volume V (L): 6.60

t = 20.00 h

Biomass X (g/L): 54.19
Glucose S (g/L): 57.19
Lactic Acid P (g/L): 30.67
Maltose M (g/L): 83.05
Volume V (L): 6.12

t = 21.00 h

Biomass X (g/L): 55.80
Glucose S (g/L): 53.14
Lactic Acid P (g/L): 55.80
Maltose M (g/L): 77.78
Volume V (L): 7.00

t = 22.00 h

Biomass X (g/L): 79.62
Glucose S (g/L): 52.70
Lactic Acid P (g/L): 34.90
Maltose M (g/L): 77.33
Volume V (L): 6.24

t = 23.00 h

Biomass X (g/L): 68.25
Glucose S (g/L): 57.26
Lactic Acid P (g/L): 53.84
Maltose M (g/L): 74.60
Volume V (L): 7.15

t = 24.00 h

Biomass X (g/L): 75.41
Glucose S (g/L): 50.07
Lactic Acid P (g/L): 48.57
Maltose M (g/L): 71.63
Volume V (L): 6.82

t = 25.00 h

Biomass X (g/L): 86.36
Glucose S (g/L): 49.88
Lactic Acid P (g/L): 44.17
Maltose M (g/L): 75.63
Volume V (L): 5.97

t = 26.00 h

Biomass X (g/L): 49.63
Glucose S (g/L): 52.36
Lactic Acid P (g/L): 57.67
Maltose M (g/L): 67.13
Volume V (L): 6.47

t = 27.00 h

Biomass X (g/L): 86.77
Glucose S (g/L): 50.03
Lactic Acid P (g/L): 57.55
Maltose M (g/L): 70.45
Volume V (L): 6.11

t = 28.00 h

Biomass X (g/L): 75.88
Glucose S (g/L): 49.61
Lactic Acid P (g/L): 75.97
Maltose M (g/L): 65.18
Volume V (L): 7.39

t = 29.00 h

Biomass X (g/L): 80.66
Glucose S (g/L): 52.58
Lactic Acid P (g/L): 52.27
Maltose M (g/L): 68.44
Volume V (L): 6.97

t = 30.00 h

Biomass X (g/L): 69.16
Glucose S (g/L): 47.84
Lactic Acid P (g/L): 71.94
Maltose M (g/L): 72.69
Volume V (L): 6.14

t = 31.00 h

Biomass X (g/L): 70.72
Glucose S (g/L): 49.47
Lactic Acid P (g/L): 63.01
Maltose M (g/L): 71.72
Volume V (L): 7.27

t = 32.00 h

Biomass X (g/L): 83.74
Glucose S (g/L): 58.83
Lactic Acid P (g/L): 54.70
Maltose M (g/L): 69.64
Volume V (L): 6.97

t = 33.00 h

Biomass X (g/L): 103.10
Glucose S (g/L): 54.32
Lactic Acid P (g/L): 68.72
Maltose M (g/L): 69.58
Volume V (L): 6.65

t = 34.00 h

Biomass X (g/L): 90.10
Glucose S (g/L): 51.02
Lactic Acid P (g/L): 67.36
Maltose M (g/L): 61.87

Volume V (L): 8.01

t = 35.00 h

Biomass X (g/L): 93.23

Glucose S (g/L): 54.85

Lactic Acid P (g/L): 63.27

Maltose M (g/L): 66.73

Volume V (L): 7.44

t = 36.00 h

Biomass X (g/L): 79.35

Glucose S (g/L): 55.68

Lactic Acid P (g/L): 66.02

Maltose M (g/L): 65.18

Volume V (L): 7.34

t = 37.00 h

Biomass X (g/L): 75.76

Glucose S (g/L): 57.37

Lactic Acid P (g/L): 58.84

Maltose M (g/L): 64.81

Volume V (L): 7.36

t = 38.00 h

Biomass X (g/L): 74.37

Glucose S (g/L): 54.46

Lactic Acid P (g/L): 57.36

Maltose M (g/L): 68.79

Volume V (L): 6.98

t = 39.00 h

Biomass X (g/L): 77.59

Glucose S (g/L): 58.22

Lactic Acid P (g/L): 72.27

Maltose M (g/L): 63.20

Volume V (L): 7.13

t = 40.00 h

Biomass X (g/L): 62.47

Glucose S (g/L): 49.70

Lactic Acid P (g/L): 63.90

Maltose M (g/L): 63.66

Volume V (L): 8.26

t = 41.00 h

Biomass X (g/L): 75.82

Glucose S (g/L): 53.14

Lactic Acid P (g/L): 65.71

Maltose M (g/L): 58.01

Volume V (L): 7.61

t = 42.00 h

Biomass X (g/L): 77.67

Glucose S (g/L): 57.31

Lactic Acid P (g/L): 69.44

Maltose M (g/L): 63.89

Volume V (L): 7.88

t = 43.00 h

Biomass X (g/L): 81.65

Glucose S (g/L): 58.34

Lactic Acid P (g/L): 69.77

Maltose M (g/L): 60.67

Volume V (L): 7.10

t = 44.00 h

Biomass X (g/L): 84.85

Glucose S (g/L): 55.19

Lactic Acid P (g/L): 54.86

Maltose M (g/L): 64.36

Volume V (L): 7.47

t = 45.00 h

Biomass X (g/L): 87.26

Glucose S (g/L): 59.25

Lactic Acid P (g/L): 79.86

Maltose M (g/L): 60.32

Volume V (L): 8.33

t = 46.00 h

Biomass X (g/L): 83.65

Glucose S (g/L): 57.18

Lactic Acid P (g/L): 69.71
Maltose M (g/L): 63.63
Volume V (L): 8.30

t = 47.00 h

Biomass X (g/L): 78.16
Glucose S (g/L): 58.80
Lactic Acid P (g/L): 67.15
Maltose M (g/L): 62.34
Volume V (L): 8.93

t = 48.00 h

Biomass X (g/L): 64.62
Glucose S (g/L): 63.27
Lactic Acid P (g/L): 73.25
Maltose M (g/L): 68.34
Volume V (L): 8.83

t = 49.00 h

Biomass X (g/L): 78.35
Glucose S (g/L): 63.52
Lactic Acid P (g/L): 71.83
Maltose M (g/L): 63.80
Volume V (L): 8.25

t = 50.00 h

Biomass X (g/L): 74.46
Glucose S (g/L): 58.42
Lactic Acid P (g/L): 76.97
Maltose M (g/L): 55.18
Volume V (L): 8.61

t = 51.00 h

Biomass X (g/L): 88.66
Glucose S (g/L): 58.84
Lactic Acid P (g/L): 75.21
Maltose M (g/L): 56.81
Volume V (L): 8.36

t = 52.00 h

Biomass X (g/L): 93.03

Glucose S (g/L): 61.37
Lactic Acid P (g/L): 77.14
Maltose M (g/L): 61.08
Volume V (L): 9.40

t = 53.00 h

Biomass X (g/L): 93.30
Glucose S (g/L): 63.13
Lactic Acid P (g/L): 83.17
Maltose M (g/L): 57.23
Volume V (L): 9.09

t = 54.00 h

Biomass X (g/L): 91.48
Glucose S (g/L): 61.84
Lactic Acid P (g/L): 79.22
Maltose M (g/L): 49.67
Volume V (L): 8.76

t = 55.00 h

Biomass X (g/L): 82.67
Glucose S (g/L): 67.53
Lactic Acid P (g/L): 70.04
Maltose M (g/L): 64.50
Volume V (L): 8.84

t = 56.00 h

Biomass X (g/L): 78.79
Glucose S (g/L): 66.06
Lactic Acid P (g/L): 61.59
Maltose M (g/L): 61.92
Volume V (L): 9.42

t = 57.00 h

Biomass X (g/L): 76.51
Glucose S (g/L): 66.85
Lactic Acid P (g/L): 87.10
Maltose M (g/L): 60.85
Volume V (L): 9.13

t = 58.00 h

Biomass X (g/L): 68.95
Glucose S (g/L): 62.45
Lactic Acid P (g/L): 70.10
Maltose M (g/L): 56.52
Volume V (L): 9.60

t = 59.00 h

Biomass X (g/L): 93.50
Glucose S (g/L): 70.33
Lactic Acid P (g/L): 68.21
Maltose M (g/L): 59.80
Volume V (L): 8.13

t = 60.00 h

Biomass X (g/L): 81.92
Glucose S (g/L): 59.02
Lactic Acid P (g/L): 57.00
Maltose M (g/L): 55.46
Volume V (L): 9.29

t = 61.00 h

Biomass X (g/L): 71.26
Glucose S (g/L): 70.54
Lactic Acid P (g/L): 76.00
Maltose M (g/L): 58.32
Volume V (L): 8.96

t = 62.00 h

Biomass X (g/L): 75.29
Glucose S (g/L): 71.44
Lactic Acid P (g/L): 74.50
Maltose M (g/L): 54.01
Volume V (L): 9.96

t = 63.00 h

Biomass X (g/L): 74.78
Glucose S (g/L): 67.95
Lactic Acid P (g/L): 59.64
Maltose M (g/L): 62.24
Volume V (L): 9.94

t = 64.00 h

Biomass X (g/L): 83.63
Glucose S (g/L): 73.64
Lactic Acid P (g/L): 77.66
Maltose M (g/L): 51.03
Volume V (L): 9.39

t = 65.00 h

Biomass X (g/L): 93.61
Glucose S (g/L): 70.24
Lactic Acid P (g/L): 69.24
Maltose M (g/L): 60.57
Volume V (L): 10.07

t = 66.00 h

Biomass X (g/L): 79.53
Glucose S (g/L): 71.20
Lactic Acid P (g/L): 98.29
Maltose M (g/L): 56.03
Volume V (L): 10.24

t = 67.00 h

Biomass X (g/L): 60.63
Glucose S (g/L): 72.62
Lactic Acid P (g/L): 68.97
Maltose M (g/L): 53.84
Volume V (L): 9.61

t = 68.00 h

Biomass X (g/L): 80.82
Glucose S (g/L): 81.71
Lactic Acid P (g/L): 70.36
Maltose M (g/L): 55.83
Volume V (L): 10.66

t = 69.00 h

Biomass X (g/L): 81.01
Glucose S (g/L): 76.78
Lactic Acid P (g/L): 66.56
Maltose M (g/L): 57.50
Volume V (L): 9.70

t = 70.00 h

Biomass X (g/L): 80.18
Glucose S (g/L): 75.43
Lactic Acid P (g/L): 86.21
Maltose M (g/L): 59.41
Volume V (L): 9.70

t = 71.00 h

Biomass X (g/L): 76.07
Glucose S (g/L): 71.18
Lactic Acid P (g/L): 71.44
Maltose M (g/L): 55.39
Volume V (L): 10.21

t = 72.00 h

Biomass X (g/L): 84.24
Glucose S (g/L): 74.90
Lactic Acid P (g/L): 68.41
Maltose M (g/L): 50.83
Volume V (L): 9.78

t = 73.00 h

Biomass X (g/L): 82.59
Glucose S (g/L): 72.60
Lactic Acid P (g/L): 81.52
Maltose M (g/L): 53.86
Volume V (L): 10.02

t = 74.00 h

Biomass X (g/L): 81.53
Glucose S (g/L): 78.05
Lactic Acid P (g/L): 62.78
Maltose M (g/L): 59.75
Volume V (L): 10.64

t = 75.00 h

Biomass X (g/L): 85.68
Glucose S (g/L): 76.57
Lactic Acid P (g/L): 81.08
Maltose M (g/L): 57.80

Volume V (L): 10.13

Batch 37

Initial Conditions:

Batch: 37.00

X0 (g/L): 4.56

S0 (g/L): 160.47

M0 (g/L): 16.26

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.10

F_vol (L/h): 0.08

S_feed (g/L): 272.40

M_feed (g/L): 27.60

V_final (L): 11.03

Batch 37 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 167.70

Lactic Acid P (g/L): 0.21

Maltose M (g/L): 15.82

Volume V (L): 5.21

t = 1.00 h

Biomass X (g/L): 9.75

Glucose S (g/L): 154.48

Lactic Acid P (g/L): 10.50

Maltose M (g/L): 17.00

Volume V (L): 4.76

t = 2.00 h

Biomass X (g/L): 1.53

Glucose S (g/L): 163.82

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 15.43

Volume V (L): 5.21

t = 3.00 h

Biomass X (g/L): 19.31

Glucose S (g/L): 159.58

Lactic Acid P (g/L): 19.49

Maltose M (g/L): 16.80

Volume V (L): 5.78

t = 4.00 h

Biomass X (g/L): 20.87

Glucose S (g/L): 166.83

Lactic Acid P (g/L): 15.24

Maltose M (g/L): 16.81

Volume V (L): 4.87

t = 5.00 h

Biomass X (g/L): 32.14

Glucose S (g/L): 158.17

Lactic Acid P (g/L): 9.23
Maltose M (g/L): 14.33
Volume V (L): 4.76

t = 6.00 h

Biomass X (g/L): 30.90
Glucose S (g/L): 152.66
Lactic Acid P (g/L): 11.64
Maltose M (g/L): 15.55
Volume V (L): 4.86

t = 7.00 h

Biomass X (g/L): 37.26
Glucose S (g/L): 150.22
Lactic Acid P (g/L): 20.70
Maltose M (g/L): 15.11
Volume V (L): 5.00

t = 8.00 h

Biomass X (g/L): 24.97
Glucose S (g/L): 148.28
Lactic Acid P (g/L): 33.64
Maltose M (g/L): 13.33
Volume V (L): 5.93

t = 9.00 h

Biomass X (g/L): 44.08
Glucose S (g/L): 139.91
Lactic Acid P (g/L): 24.41
Maltose M (g/L): 14.18
Volume V (L): 5.05

t = 10.00 h

Biomass X (g/L): 60.08
Glucose S (g/L): 138.26
Lactic Acid P (g/L): 28.33
Maltose M (g/L): 15.52
Volume V (L): 5.66

t = 11.00 h

Biomass X (g/L): 38.81

Glucose S (g/L): 130.77
Lactic Acid P (g/L): 30.60
Maltose M (g/L): 16.23
Volume V (L): 6.18

t = 12.00 h

Biomass X (g/L): 67.16
Glucose S (g/L): 131.57
Lactic Acid P (g/L): 49.74
Maltose M (g/L): 15.38
Volume V (L): 6.17

t = 13.00 h

Biomass X (g/L): 48.26
Glucose S (g/L): 133.29
Lactic Acid P (g/L): 33.32
Maltose M (g/L): 14.88
Volume V (L): 5.86

t = 14.00 h

Biomass X (g/L): 48.88
Glucose S (g/L): 131.80
Lactic Acid P (g/L): 39.16
Maltose M (g/L): 14.54
Volume V (L): 6.22

t = 15.00 h

Biomass X (g/L): 63.22
Glucose S (g/L): 119.91
Lactic Acid P (g/L): 55.19
Maltose M (g/L): 15.55
Volume V (L): 6.26

t = 16.00 h

Biomass X (g/L): 70.19
Glucose S (g/L): 125.98
Lactic Acid P (g/L): 45.65
Maltose M (g/L): 15.75
Volume V (L): 6.18

t = 17.00 h

Biomass X (g/L): 74.68
Glucose S (g/L): 127.39
Lactic Acid P (g/L): 56.72
Maltose M (g/L): 13.84
Volume V (L): 6.26

t = 18.00 h

Biomass X (g/L): 58.83
Glucose S (g/L): 122.90
Lactic Acid P (g/L): 40.18
Maltose M (g/L): 12.80
Volume V (L): 7.53

t = 19.00 h

Biomass X (g/L): 74.44
Glucose S (g/L): 119.71
Lactic Acid P (g/L): 48.33
Maltose M (g/L): 12.63
Volume V (L): 6.92

t = 20.00 h

Biomass X (g/L): 65.43
Glucose S (g/L): 119.67
Lactic Acid P (g/L): 65.21
Maltose M (g/L): 14.78
Volume V (L): 7.80

t = 21.00 h

Biomass X (g/L): 59.28
Glucose S (g/L): 118.29
Lactic Acid P (g/L): 47.20
Maltose M (g/L): 13.77
Volume V (L): 6.45

t = 22.00 h

Biomass X (g/L): 66.18
Glucose S (g/L): 117.20
Lactic Acid P (g/L): 53.30
Maltose M (g/L): 13.83
Volume V (L): 6.73

t = 23.00 h

Biomass X (g/L): 69.21
Glucose S (g/L): 120.12
Lactic Acid P (g/L): 58.24
Maltose M (g/L): 12.42
Volume V (L): 6.21

t = 24.00 h

Biomass X (g/L): 82.71
Glucose S (g/L): 128.83
Lactic Acid P (g/L): 48.15
Maltose M (g/L): 13.01
Volume V (L): 6.54

t = 25.00 h

Biomass X (g/L): 89.86
Glucose S (g/L): 115.38
Lactic Acid P (g/L): 54.41
Maltose M (g/L): 13.21
Volume V (L): 7.93

t = 26.00 h

Biomass X (g/L): 65.32
Glucose S (g/L): 116.94
Lactic Acid P (g/L): 60.15
Maltose M (g/L): 13.45
Volume V (L): 6.84

t = 27.00 h

Biomass X (g/L): 83.83
Glucose S (g/L): 117.89
Lactic Acid P (g/L): 60.47
Maltose M (g/L): 12.95
Volume V (L): 8.10

t = 28.00 h

Biomass X (g/L): 69.73
Glucose S (g/L): 115.04
Lactic Acid P (g/L): 72.45
Maltose M (g/L): 12.43
Volume V (L): 6.65

t = 29.00 h

Biomass X (g/L): 87.88
Glucose S (g/L): 122.65
Lactic Acid P (g/L): 75.60
Maltose M (g/L): 12.51
Volume V (L): 6.82

t = 30.00 h

Biomass X (g/L): 76.30
Glucose S (g/L): 107.74
Lactic Acid P (g/L): 54.70
Maltose M (g/L): 13.73
Volume V (L): 7.25

t = 31.00 h

Biomass X (g/L): 72.64
Glucose S (g/L): 115.67
Lactic Acid P (g/L): 64.12
Maltose M (g/L): 13.00
Volume V (L): 7.46

t = 32.00 h

Biomass X (g/L): 91.03
Glucose S (g/L): 121.54
Lactic Acid P (g/L): 69.84
Maltose M (g/L): 13.15
Volume V (L): 8.02

t = 33.00 h

Biomass X (g/L): 77.96
Glucose S (g/L): 116.88
Lactic Acid P (g/L): 78.05
Maltose M (g/L): 12.07
Volume V (L): 7.87

t = 34.00 h

Biomass X (g/L): 80.92
Glucose S (g/L): 114.76
Lactic Acid P (g/L): 71.42
Maltose M (g/L): 10.62

Volume V (L): 7.53

t = 35.00 h

Biomass X (g/L): 62.74

Glucose S (g/L): 120.35

Lactic Acid P (g/L): 68.09

Maltose M (g/L): 12.14

Volume V (L): 7.75

t = 36.00 h

Biomass X (g/L): 93.69

Glucose S (g/L): 122.54

Lactic Acid P (g/L): 71.99

Maltose M (g/L): 11.83

Volume V (L): 7.74

t = 37.00 h

Biomass X (g/L): 92.68

Glucose S (g/L): 119.77

Lactic Acid P (g/L): 72.53

Maltose M (g/L): 12.54

Volume V (L): 7.53

t = 38.00 h

Biomass X (g/L): 84.95

Glucose S (g/L): 115.66

Lactic Acid P (g/L): 58.52

Maltose M (g/L): 14.40

Volume V (L): 7.63

t = 39.00 h

Biomass X (g/L): 70.22

Glucose S (g/L): 121.31

Lactic Acid P (g/L): 69.07

Maltose M (g/L): 12.60

Volume V (L): 8.63

t = 40.00 h

Biomass X (g/L): 96.88

Glucose S (g/L): 117.89

Lactic Acid P (g/L): 80.90

Maltose M (g/L): 13.87

Volume V (L): 7.33

t = 41.00 h

Biomass X (g/L): 78.78

Glucose S (g/L): 123.09

Lactic Acid P (g/L): 66.25

Maltose M (g/L): 12.36

Volume V (L): 9.08

t = 42.00 h

Biomass X (g/L): 80.91

Glucose S (g/L): 126.55

Lactic Acid P (g/L): 69.65

Maltose M (g/L): 11.48

Volume V (L): 8.21

t = 43.00 h

Biomass X (g/L): 107.77

Glucose S (g/L): 109.29

Lactic Acid P (g/L): 66.65

Maltose M (g/L): 12.05

Volume V (L): 8.08

t = 44.00 h

Biomass X (g/L): 78.54

Glucose S (g/L): 121.26

Lactic Acid P (g/L): 48.28

Maltose M (g/L): 12.92

Volume V (L): 8.89

t = 45.00 h

Biomass X (g/L): 92.57

Glucose S (g/L): 113.64

Lactic Acid P (g/L): 69.67

Maltose M (g/L): 12.87

Volume V (L): 9.33

t = 46.00 h

Biomass X (g/L): 90.81

Glucose S (g/L): 131.90

Lactic Acid P (g/L): 72.52

Maltose M (g/L): 11.20

Volume V (L): 9.24

t = 47.00 h

Biomass X (g/L): 89.12

Glucose S (g/L): 119.73

Lactic Acid P (g/L): 65.46

Maltose M (g/L): 11.67

Volume V (L): 9.02

t = 48.00 h

Biomass X (g/L): 88.03

Glucose S (g/L): 125.27

Lactic Acid P (g/L): 59.94

Maltose M (g/L): 10.49

Volume V (L): 9.38

t = 49.00 h

Biomass X (g/L): 88.89

Glucose S (g/L): 122.24

Lactic Acid P (g/L): 62.68

Maltose M (g/L): 11.90

Volume V (L): 8.11

t = 50.00 h

Biomass X (g/L): 87.81

Glucose S (g/L): 119.28

Lactic Acid P (g/L): 77.79

Maltose M (g/L): 12.31

Volume V (L): 8.77

t = 51.00 h

Biomass X (g/L): 84.60

Glucose S (g/L): 119.10

Lactic Acid P (g/L): 88.94

Maltose M (g/L): 11.24

Volume V (L): 8.66

t = 52.00 h

Biomass X (g/L): 99.67

Glucose S (g/L): 130.21
Lactic Acid P (g/L): 83.86
Maltose M (g/L): 10.70
Volume V (L): 9.77

t = 53.00 h

Biomass X (g/L): 72.19
Glucose S (g/L): 127.84
Lactic Acid P (g/L): 59.47
Maltose M (g/L): 11.30
Volume V (L): 9.08

t = 54.00 h

Biomass X (g/L): 94.72
Glucose S (g/L): 132.42
Lactic Acid P (g/L): 79.54
Maltose M (g/L): 11.27
Volume V (L): 9.75

t = 55.00 h

Biomass X (g/L): 92.67
Glucose S (g/L): 124.65
Lactic Acid P (g/L): 66.06
Maltose M (g/L): 10.98
Volume V (L): 9.68

t = 56.00 h

Biomass X (g/L): 91.35
Glucose S (g/L): 125.99
Lactic Acid P (g/L): 66.99
Maltose M (g/L): 10.97
Volume V (L): 9.09

t = 57.00 h

Biomass X (g/L): 95.29
Glucose S (g/L): 123.94
Lactic Acid P (g/L): 68.28
Maltose M (g/L): 10.67
Volume V (L): 9.30

t = 58.00 h

Biomass X (g/L): 83.42
Glucose S (g/L): 124.19
Lactic Acid P (g/L): 79.02
Maltose M (g/L): 11.37
Volume V (L): 10.29

t = 59.00 h

Biomass X (g/L): 66.06
Glucose S (g/L): 121.29
Lactic Acid P (g/L): 81.49
Maltose M (g/L): 11.14
Volume V (L): 9.37

t = 60.00 h

Biomass X (g/L): 83.34
Glucose S (g/L): 126.45
Lactic Acid P (g/L): 72.13
Maltose M (g/L): 9.93
Volume V (L): 9.88

t = 61.00 h

Biomass X (g/L): 74.28
Glucose S (g/L): 123.04
Lactic Acid P (g/L): 72.46
Maltose M (g/L): 9.85
Volume V (L): 9.53

t = 62.00 h

Biomass X (g/L): 76.64
Glucose S (g/L): 128.65
Lactic Acid P (g/L): 77.55
Maltose M (g/L): 10.28
Volume V (L): 9.83

t = 63.00 h

Biomass X (g/L): 71.54
Glucose S (g/L): 127.81
Lactic Acid P (g/L): 78.66
Maltose M (g/L): 10.87
Volume V (L): 10.04

t = 64.00 h

Biomass X (g/L): 78.80
Glucose S (g/L): 128.65
Lactic Acid P (g/L): 84.31
Maltose M (g/L): 10.42
Volume V (L): 9.77

t = 65.00 h

Biomass X (g/L): 77.48
Glucose S (g/L): 134.93
Lactic Acid P (g/L): 62.25
Maltose M (g/L): 9.84
Volume V (L): 9.76

t = 66.00 h

Biomass X (g/L): 60.04
Glucose S (g/L): 127.89
Lactic Acid P (g/L): 74.18
Maltose M (g/L): 10.37
Volume V (L): 10.44

t = 67.00 h

Biomass X (g/L): 89.72
Glucose S (g/L): 131.63
Lactic Acid P (g/L): 64.51
Maltose M (g/L): 9.92
Volume V (L): 9.98

t = 68.00 h

Biomass X (g/L): 104.29
Glucose S (g/L): 137.50
Lactic Acid P (g/L): 73.84
Maltose M (g/L): 10.66
Volume V (L): 10.08

t = 69.00 h

Biomass X (g/L): 88.46
Glucose S (g/L): 134.62
Lactic Acid P (g/L): 72.40
Maltose M (g/L): 10.23
Volume V (L): 10.09

t = 70.00 h

Biomass X (g/L): 80.44
Glucose S (g/L): 134.27
Lactic Acid P (g/L): 74.15
Maltose M (g/L): 9.75
Volume V (L): 10.59

t = 71.00 h

Biomass X (g/L): 81.69
Glucose S (g/L): 128.31
Lactic Acid P (g/L): 67.49
Maltose M (g/L): 10.05
Volume V (L): 11.16

t = 72.00 h

Biomass X (g/L): 77.96
Glucose S (g/L): 130.92
Lactic Acid P (g/L): 61.42
Maltose M (g/L): 10.70
Volume V (L): 11.51

t = 73.00 h

Biomass X (g/L): 82.74
Glucose S (g/L): 124.57
Lactic Acid P (g/L): 56.07
Maltose M (g/L): 9.40
Volume V (L): 10.52

t = 74.00 h

Biomass X (g/L): 81.32
Glucose S (g/L): 132.68
Lactic Acid P (g/L): 73.91
Maltose M (g/L): 9.98
Volume V (L): 11.99

t = 75.00 h

Biomass X (g/L): 73.34
Glucose S (g/L): 132.01
Lactic Acid P (g/L): 71.33
Maltose M (g/L): 9.61

Volume V (L): 10.54

Batch 38

Initial Conditions:

Batch: 38.00

X0 (g/L): 5.19

S0 (g/L): 11.82

M0 (g/L): 166.15

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 14.05

F_vol (L/h): 0.09

S_feed (g/L): 19.93

M_feed (g/L): 280.07

V_final (L): 11.79

Batch 38 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 15.06
Glucose S (g/L): 13.45
Lactic Acid P (g/L): 10.10
Maltose M (g/L): 164.90
Volume V (L): 4.31

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 10.68
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 173.28
Volume V (L): 5.74

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 6.72
Lactic Acid P (g/L): 8.30
Maltose M (g/L): 154.86
Volume V (L): 4.85

t = 3.00 h

Biomass X (g/L): 2.93
Glucose S (g/L): 18.06
Lactic Acid P (g/L): 0.07
Maltose M (g/L): 158.82
Volume V (L): 5.09

t = 4.00 h

Biomass X (g/L): 11.41
Glucose S (g/L): 17.05
Lactic Acid P (g/L): 5.42
Maltose M (g/L): 156.03
Volume V (L): 6.15

t = 5.00 h

Biomass X (g/L): 19.69
Glucose S (g/L): 15.94

Lactic Acid P (g/L): 3.88
Maltose M (g/L): 158.60
Volume V (L): 4.84

t = 6.00 h

Biomass X (g/L): 23.36
Glucose S (g/L): 20.86
Lactic Acid P (g/L): 15.80
Maltose M (g/L): 159.72
Volume V (L): 5.15

t = 7.00 h

Biomass X (g/L): 11.77
Glucose S (g/L): 17.55
Lactic Acid P (g/L): 23.92
Maltose M (g/L): 160.95
Volume V (L): 5.22

t = 8.00 h

Biomass X (g/L): 14.67
Glucose S (g/L): 12.26
Lactic Acid P (g/L): 20.90
Maltose M (g/L): 154.80
Volume V (L): 6.60

t = 9.00 h

Biomass X (g/L): 18.76
Glucose S (g/L): 9.21
Lactic Acid P (g/L): 24.55
Maltose M (g/L): 150.26
Volume V (L): 6.90

t = 10.00 h

Biomass X (g/L): 29.56
Glucose S (g/L): 11.26
Lactic Acid P (g/L): 27.18
Maltose M (g/L): 165.79
Volume V (L): 6.31

t = 11.00 h

Biomass X (g/L): 32.07

Glucose S (g/L): 23.31
Lactic Acid P (g/L): 26.54
Maltose M (g/L): 169.41
Volume V (L): 6.42

t = 12.00 h

Biomass X (g/L): 28.62
Glucose S (g/L): 16.27
Lactic Acid P (g/L): 14.39
Maltose M (g/L): 153.62
Volume V (L): 5.77

t = 13.00 h

Biomass X (g/L): 42.88
Glucose S (g/L): 13.49
Lactic Acid P (g/L): 41.48
Maltose M (g/L): 146.72
Volume V (L): 7.23

t = 14.00 h

Biomass X (g/L): 63.99
Glucose S (g/L): 11.43
Lactic Acid P (g/L): 44.68
Maltose M (g/L): 160.13
Volume V (L): 6.41

t = 15.00 h

Biomass X (g/L): 41.55
Glucose S (g/L): 10.62
Lactic Acid P (g/L): 29.17
Maltose M (g/L): 140.13
Volume V (L): 6.82

t = 16.00 h

Biomass X (g/L): 44.77
Glucose S (g/L): 15.39
Lactic Acid P (g/L): 30.04
Maltose M (g/L): 146.30
Volume V (L): 6.39

t = 17.00 h

Biomass X (g/L): 56.21
Glucose S (g/L): 8.96
Lactic Acid P (g/L): 40.51
Maltose M (g/L): 142.93
Volume V (L): 6.39

t = 18.00 h

Biomass X (g/L): 48.23
Glucose S (g/L): 12.69
Lactic Acid P (g/L): 41.79
Maltose M (g/L): 138.90
Volume V (L): 7.53

t = 19.00 h

Biomass X (g/L): 61.47
Glucose S (g/L): 9.18
Lactic Acid P (g/L): 36.97
Maltose M (g/L): 143.58
Volume V (L): 7.01

t = 20.00 h

Biomass X (g/L): 50.74
Glucose S (g/L): 17.59
Lactic Acid P (g/L): 29.48
Maltose M (g/L): 138.79
Volume V (L): 7.60

t = 21.00 h

Biomass X (g/L): 75.75
Glucose S (g/L): 13.25
Lactic Acid P (g/L): 44.51
Maltose M (g/L): 143.48
Volume V (L): 7.04

t = 22.00 h

Biomass X (g/L): 63.86
Glucose S (g/L): 15.25
Lactic Acid P (g/L): 48.09
Maltose M (g/L): 142.58
Volume V (L): 6.53

t = 23.00 h

Biomass X (g/L): 56.62
Glucose S (g/L): 8.72
Lactic Acid P (g/L): 50.98
Maltose M (g/L): 137.40
Volume V (L): 6.76

t = 24.00 h

Biomass X (g/L): 58.35
Glucose S (g/L): 15.03
Lactic Acid P (g/L): 49.01
Maltose M (g/L): 138.05
Volume V (L): 7.23

t = 25.00 h

Biomass X (g/L): 60.38
Glucose S (g/L): 10.30
Lactic Acid P (g/L): 44.87
Maltose M (g/L): 137.99
Volume V (L): 6.92

t = 26.00 h

Biomass X (g/L): 78.83
Glucose S (g/L): 12.21
Lactic Acid P (g/L): 57.03
Maltose M (g/L): 143.79
Volume V (L): 6.75

t = 27.00 h

Biomass X (g/L): 55.95
Glucose S (g/L): 18.23
Lactic Acid P (g/L): 50.10
Maltose M (g/L): 136.63
Volume V (L): 8.10

t = 28.00 h

Biomass X (g/L): 62.53
Glucose S (g/L): 8.81
Lactic Acid P (g/L): 46.57
Maltose M (g/L): 142.00
Volume V (L): 7.24

t = 29.00 h

Biomass X (g/L): 64.53
Glucose S (g/L): 13.87
Lactic Acid P (g/L): 50.06
Maltose M (g/L): 139.53
Volume V (L): 7.70

t = 30.00 h

Biomass X (g/L): 70.47
Glucose S (g/L): 15.24
Lactic Acid P (g/L): 37.46
Maltose M (g/L): 138.51
Volume V (L): 7.88

t = 31.00 h

Biomass X (g/L): 69.05
Glucose S (g/L): 16.08
Lactic Acid P (g/L): 51.97
Maltose M (g/L): 142.90
Volume V (L): 7.49

t = 32.00 h

Biomass X (g/L): 58.78
Glucose S (g/L): 13.07
Lactic Acid P (g/L): 60.42
Maltose M (g/L): 125.74
Volume V (L): 8.68

t = 33.00 h

Biomass X (g/L): 69.51
Glucose S (g/L): 23.00
Lactic Acid P (g/L): 58.25
Maltose M (g/L): 129.03
Volume V (L): 7.38

t = 34.00 h

Biomass X (g/L): 72.98
Glucose S (g/L): 11.16
Lactic Acid P (g/L): 69.32
Maltose M (g/L): 142.65

Volume V (L): 8.22

t = 35.00 h

Biomass X (g/L): 68.85

Glucose S (g/L): 21.94

Lactic Acid P (g/L): 54.62

Maltose M (g/L): 128.67

Volume V (L): 8.30

t = 36.00 h

Biomass X (g/L): 81.46

Glucose S (g/L): 21.70

Lactic Acid P (g/L): 61.89

Maltose M (g/L): 111.22

Volume V (L): 8.23

t = 37.00 h

Biomass X (g/L): 84.17

Glucose S (g/L): 24.89

Lactic Acid P (g/L): 65.26

Maltose M (g/L): 127.29

Volume V (L): 7.64

t = 38.00 h

Biomass X (g/L): 75.63

Glucose S (g/L): 12.73

Lactic Acid P (g/L): 79.40

Maltose M (g/L): 128.63

Volume V (L): 8.91

t = 39.00 h

Biomass X (g/L): 78.95

Glucose S (g/L): 16.09

Lactic Acid P (g/L): 60.98

Maltose M (g/L): 109.94

Volume V (L): 8.93

t = 40.00 h

Biomass X (g/L): 57.63

Glucose S (g/L): 25.50

Lactic Acid P (g/L): 51.29

Maltose M (g/L): 134.72
Volume V (L): 8.25

t = 41.00 h

Biomass X (g/L): 73.91
Glucose S (g/L): 13.82
Lactic Acid P (g/L): 68.32
Maltose M (g/L): 120.62
Volume V (L): 9.09

t = 42.00 h

Biomass X (g/L): 87.29
Glucose S (g/L): 15.56
Lactic Acid P (g/L): 77.73
Maltose M (g/L): 126.55
Volume V (L): 9.00

t = 43.00 h

Biomass X (g/L): 74.64
Glucose S (g/L): 20.43
Lactic Acid P (g/L): 59.29
Maltose M (g/L): 122.15
Volume V (L): 8.20

t = 44.00 h

Biomass X (g/L): 72.12
Glucose S (g/L): 23.70
Lactic Acid P (g/L): 69.93
Maltose M (g/L): 114.45
Volume V (L): 8.52

t = 45.00 h

Biomass X (g/L): 99.88
Glucose S (g/L): 29.63
Lactic Acid P (g/L): 73.80
Maltose M (g/L): 115.07
Volume V (L): 9.69

t = 46.00 h

Biomass X (g/L): 95.98
Glucose S (g/L): 23.86

Lactic Acid P (g/L): 72.46
Maltose M (g/L): 115.16
Volume V (L): 9.33

t = 47.00 h

Biomass X (g/L): 83.92
Glucose S (g/L): 23.09
Lactic Acid P (g/L): 65.19
Maltose M (g/L): 127.48
Volume V (L): 8.95

t = 48.00 h

Biomass X (g/L): 90.26
Glucose S (g/L): 27.26
Lactic Acid P (g/L): 58.49
Maltose M (g/L): 118.70
Volume V (L): 8.88

t = 49.00 h

Biomass X (g/L): 59.80
Glucose S (g/L): 28.64
Lactic Acid P (g/L): 64.29
Maltose M (g/L): 115.51
Volume V (L): 9.15

t = 50.00 h

Biomass X (g/L): 73.09
Glucose S (g/L): 29.33
Lactic Acid P (g/L): 69.39
Maltose M (g/L): 112.07
Volume V (L): 9.28

t = 51.00 h

Biomass X (g/L): 76.14
Glucose S (g/L): 27.85
Lactic Acid P (g/L): 73.70
Maltose M (g/L): 125.14
Volume V (L): 9.16

t = 52.00 h

Biomass X (g/L): 78.41

Glucose S (g/L): 22.10
Lactic Acid P (g/L): 69.10
Maltose M (g/L): 126.13
Volume V (L): 8.76

t = 53.00 h

Biomass X (g/L): 76.22
Glucose S (g/L): 26.73
Lactic Acid P (g/L): 63.28
Maltose M (g/L): 123.58
Volume V (L): 10.41

t = 54.00 h

Biomass X (g/L): 57.55
Glucose S (g/L): 36.14
Lactic Acid P (g/L): 71.76
Maltose M (g/L): 118.36
Volume V (L): 10.41

t = 55.00 h

Biomass X (g/L): 84.39
Glucose S (g/L): 32.40
Lactic Acid P (g/L): 74.75
Maltose M (g/L): 116.32
Volume V (L): 9.29

t = 56.00 h

Biomass X (g/L): 85.89
Glucose S (g/L): 31.33
Lactic Acid P (g/L): 70.92
Maltose M (g/L): 105.84
Volume V (L): 10.14

t = 57.00 h

Biomass X (g/L): 95.59
Glucose S (g/L): 37.44
Lactic Acid P (g/L): 60.69
Maltose M (g/L): 111.42
Volume V (L): 9.67

t = 58.00 h

Biomass X (g/L): 73.86
Glucose S (g/L): 29.96
Lactic Acid P (g/L): 66.69
Maltose M (g/L): 107.87
Volume V (L): 10.11

t = 59.00 h

Biomass X (g/L): 73.15
Glucose S (g/L): 34.09
Lactic Acid P (g/L): 76.79
Maltose M (g/L): 99.77
Volume V (L): 10.15

t = 60.00 h

Biomass X (g/L): 56.68
Glucose S (g/L): 36.23
Lactic Acid P (g/L): 56.86
Maltose M (g/L): 117.77
Volume V (L): 10.20

t = 61.00 h

Biomass X (g/L): 78.43
Glucose S (g/L): 34.89
Lactic Acid P (g/L): 50.97
Maltose M (g/L): 105.72
Volume V (L): 11.61

t = 62.00 h

Biomass X (g/L): 89.31
Glucose S (g/L): 37.68
Lactic Acid P (g/L): 82.51
Maltose M (g/L): 117.31
Volume V (L): 10.37

t = 63.00 h

Biomass X (g/L): 75.51
Glucose S (g/L): 37.41
Lactic Acid P (g/L): 74.13
Maltose M (g/L): 114.60
Volume V (L): 10.26

t = 64.00 h

Biomass X (g/L): 100.59
Glucose S (g/L): 33.74
Lactic Acid P (g/L): 75.21
Maltose M (g/L): 111.83
Volume V (L): 10.38

t = 65.00 h

Biomass X (g/L): 78.69
Glucose S (g/L): 40.34
Lactic Acid P (g/L): 60.85
Maltose M (g/L): 111.54
Volume V (L): 11.45

t = 66.00 h

Biomass X (g/L): 82.33
Glucose S (g/L): 35.56
Lactic Acid P (g/L): 76.84
Maltose M (g/L): 102.92
Volume V (L): 10.07

t = 67.00 h

Biomass X (g/L): 68.53
Glucose S (g/L): 41.84
Lactic Acid P (g/L): 73.06
Maltose M (g/L): 118.31
Volume V (L): 10.65

t = 68.00 h

Biomass X (g/L): 83.50
Glucose S (g/L): 47.06
Lactic Acid P (g/L): 57.15
Maltose M (g/L): 105.91
Volume V (L): 11.94

t = 69.00 h

Biomass X (g/L): 62.80
Glucose S (g/L): 41.73
Lactic Acid P (g/L): 83.49
Maltose M (g/L): 106.95
Volume V (L): 11.38

t = 70.00 h

Biomass X (g/L): 87.08
Glucose S (g/L): 39.22
Lactic Acid P (g/L): 72.72
Maltose M (g/L): 100.16
Volume V (L): 11.44

t = 71.00 h

Biomass X (g/L): 87.28
Glucose S (g/L): 43.82
Lactic Acid P (g/L): 68.11
Maltose M (g/L): 97.71
Volume V (L): 12.58

t = 72.00 h

Biomass X (g/L): 92.98
Glucose S (g/L): 41.98
Lactic Acid P (g/L): 68.47
Maltose M (g/L): 106.22
Volume V (L): 11.65

t = 73.00 h

Biomass X (g/L): 71.22
Glucose S (g/L): 47.98
Lactic Acid P (g/L): 54.46
Maltose M (g/L): 103.93
Volume V (L): 11.22

t = 74.00 h

Biomass X (g/L): 81.88
Glucose S (g/L): 45.99
Lactic Acid P (g/L): 77.30
Maltose M (g/L): 96.79
Volume V (L): 11.54

t = 75.00 h

Biomass X (g/L): 85.46
Glucose S (g/L): 53.57
Lactic Acid P (g/L): 74.61
Maltose M (g/L): 91.99

Volume V (L): 11.17

Batch 39

Initial Conditions:

Batch: 39.00

X0 (g/L): 4.32

S0 (g/L): 89.32

M0 (g/L): 99.13

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.11

F_vol (L/h): 0.04

S_feed (g/L): 142.19

M_feed (g/L): 157.81

V_final (L): 8.20

Batch 39 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 17.02
Glucose S (g/L): 92.75
Lactic Acid P (g/L): 11.66
Maltose M (g/L): 96.82
Volume V (L): 5.12

t = 1.00 h

Biomass X (g/L): 11.66
Glucose S (g/L): 88.43
Lactic Acid P (g/L): 1.72
Maltose M (g/L): 92.98
Volume V (L): 5.07

t = 2.00 h

Biomass X (g/L): 19.21
Glucose S (g/L): 95.24
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 98.55
Volume V (L): 5.26

t = 3.00 h

Biomass X (g/L): 22.61
Glucose S (g/L): 93.53
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 91.45
Volume V (L): 4.51

t = 4.00 h

Biomass X (g/L): 9.21
Glucose S (g/L): 91.07
Lactic Acid P (g/L): 8.21
Maltose M (g/L): 96.31
Volume V (L): 5.20

t = 5.00 h

Biomass X (g/L): 7.47
Glucose S (g/L): 88.31

Lactic Acid P (g/L): 17.54

Maltose M (g/L): 88.04

Volume V (L): 4.95

t = 6.00 h

Biomass X (g/L): 20.63

Glucose S (g/L): 86.86

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 89.07

Volume V (L): 5.55

t = 7.00 h

Biomass X (g/L): 16.84

Glucose S (g/L): 87.51

Lactic Acid P (g/L): 19.94

Maltose M (g/L): 99.91

Volume V (L): 5.26

t = 8.00 h

Biomass X (g/L): 32.09

Glucose S (g/L): 84.84

Lactic Acid P (g/L): 31.85

Maltose M (g/L): 86.64

Volume V (L): 5.35

t = 9.00 h

Biomass X (g/L): 46.72

Glucose S (g/L): 82.28

Lactic Acid P (g/L): 40.75

Maltose M (g/L): 83.41

Volume V (L): 5.00

t = 10.00 h

Biomass X (g/L): 38.35

Glucose S (g/L): 74.63

Lactic Acid P (g/L): 29.48

Maltose M (g/L): 92.64

Volume V (L): 5.69

t = 11.00 h

Biomass X (g/L): 46.18

Glucose S (g/L): 73.62
Lactic Acid P (g/L): 34.62
Maltose M (g/L): 79.84
Volume V (L): 5.63

t = 12.00 h

Biomass X (g/L): 48.07
Glucose S (g/L): 73.79
Lactic Acid P (g/L): 12.74
Maltose M (g/L): 80.65
Volume V (L): 5.65

t = 13.00 h

Biomass X (g/L): 61.18
Glucose S (g/L): 64.23
Lactic Acid P (g/L): 56.22
Maltose M (g/L): 86.27
Volume V (L): 5.35

t = 14.00 h

Biomass X (g/L): 50.03
Glucose S (g/L): 63.15
Lactic Acid P (g/L): 33.60
Maltose M (g/L): 87.15
Volume V (L): 5.40

t = 15.00 h

Biomass X (g/L): 37.96
Glucose S (g/L): 62.44
Lactic Acid P (g/L): 32.86
Maltose M (g/L): 87.62
Volume V (L): 5.70

t = 16.00 h

Biomass X (g/L): 68.01
Glucose S (g/L): 63.98
Lactic Acid P (g/L): 41.79
Maltose M (g/L): 76.28
Volume V (L): 5.50

t = 17.00 h

Biomass X (g/L): 76.01
Glucose S (g/L): 62.61
Lactic Acid P (g/L): 60.90
Maltose M (g/L): 82.38
Volume V (L): 5.84

t = 18.00 h

Biomass X (g/L): 73.04
Glucose S (g/L): 57.27
Lactic Acid P (g/L): 69.62
Maltose M (g/L): 75.85
Volume V (L): 5.21

t = 19.00 h

Biomass X (g/L): 64.29
Glucose S (g/L): 51.65
Lactic Acid P (g/L): 51.67
Maltose M (g/L): 74.19
Volume V (L): 5.69

t = 20.00 h

Biomass X (g/L): 61.30
Glucose S (g/L): 57.18
Lactic Acid P (g/L): 61.74
Maltose M (g/L): 82.81
Volume V (L): 6.31

t = 21.00 h

Biomass X (g/L): 79.60
Glucose S (g/L): 52.31
Lactic Acid P (g/L): 54.25
Maltose M (g/L): 72.70
Volume V (L): 6.01

t = 22.00 h

Biomass X (g/L): 77.65
Glucose S (g/L): 49.74
Lactic Acid P (g/L): 45.43
Maltose M (g/L): 76.19
Volume V (L): 5.98

t = 23.00 h

Biomass X (g/L): 76.41
Glucose S (g/L): 50.17
Lactic Acid P (g/L): 70.79
Maltose M (g/L): 74.70
Volume V (L): 6.49

t = 24.00 h

Biomass X (g/L): 97.42
Glucose S (g/L): 55.66
Lactic Acid P (g/L): 52.05
Maltose M (g/L): 67.42
Volume V (L): 5.43

t = 25.00 h

Biomass X (g/L): 70.59
Glucose S (g/L): 59.23
Lactic Acid P (g/L): 73.82
Maltose M (g/L): 77.33
Volume V (L): 5.86

t = 26.00 h

Biomass X (g/L): 99.17
Glucose S (g/L): 54.36
Lactic Acid P (g/L): 72.66
Maltose M (g/L): 74.40
Volume V (L): 6.01

t = 27.00 h

Biomass X (g/L): 82.48
Glucose S (g/L): 54.98
Lactic Acid P (g/L): 67.16
Maltose M (g/L): 60.22
Volume V (L): 6.37

t = 28.00 h

Biomass X (g/L): 64.44
Glucose S (g/L): 58.56
Lactic Acid P (g/L): 79.70
Maltose M (g/L): 71.37
Volume V (L): 6.42

t = 29.00 h

Biomass X (g/L): 81.95
Glucose S (g/L): 48.81
Lactic Acid P (g/L): 84.45
Maltose M (g/L): 65.10
Volume V (L): 6.02

t = 30.00 h

Biomass X (g/L): 94.14
Glucose S (g/L): 56.46
Lactic Acid P (g/L): 62.33
Maltose M (g/L): 74.21
Volume V (L): 6.54

t = 31.00 h

Biomass X (g/L): 81.35
Glucose S (g/L): 54.76
Lactic Acid P (g/L): 57.08
Maltose M (g/L): 60.12
Volume V (L): 6.22

t = 32.00 h

Biomass X (g/L): 70.66
Glucose S (g/L): 57.12
Lactic Acid P (g/L): 47.76
Maltose M (g/L): 65.15
Volume V (L): 6.68

t = 33.00 h

Biomass X (g/L): 84.39
Glucose S (g/L): 53.62
Lactic Acid P (g/L): 87.65
Maltose M (g/L): 68.24
Volume V (L): 6.14

t = 34.00 h

Biomass X (g/L): 77.81
Glucose S (g/L): 52.74
Lactic Acid P (g/L): 62.71
Maltose M (g/L): 65.95

Volume V (L): 6.42

t = 35.00 h

Biomass X (g/L): 91.83

Glucose S (g/L): 56.57

Lactic Acid P (g/L): 78.36

Maltose M (g/L): 69.02

Volume V (L): 6.37

t = 36.00 h

Biomass X (g/L): 95.00

Glucose S (g/L): 62.42

Lactic Acid P (g/L): 61.06

Maltose M (g/L): 56.37

Volume V (L): 6.81

t = 37.00 h

Biomass X (g/L): 76.78

Glucose S (g/L): 59.98

Lactic Acid P (g/L): 83.37

Maltose M (g/L): 72.42

Volume V (L): 6.44

t = 38.00 h

Biomass X (g/L): 80.57

Glucose S (g/L): 55.45

Lactic Acid P (g/L): 62.56

Maltose M (g/L): 65.86

Volume V (L): 6.61

t = 39.00 h

Biomass X (g/L): 95.57

Glucose S (g/L): 52.88

Lactic Acid P (g/L): 58.38

Maltose M (g/L): 69.65

Volume V (L): 6.79

t = 40.00 h

Biomass X (g/L): 79.40

Glucose S (g/L): 52.10

Lactic Acid P (g/L): 68.24

Maltose M (g/L): 66.49

Volume V (L): 6.85

t = 41.00 h

Biomass X (g/L): 77.29

Glucose S (g/L): 63.43

Lactic Acid P (g/L): 65.63

Maltose M (g/L): 61.25

Volume V (L): 6.98

t = 42.00 h

Biomass X (g/L): 76.44

Glucose S (g/L): 55.69

Lactic Acid P (g/L): 76.83

Maltose M (g/L): 53.97

Volume V (L): 6.85

t = 43.00 h

Biomass X (g/L): 92.01

Glucose S (g/L): 58.46

Lactic Acid P (g/L): 71.08

Maltose M (g/L): 57.20

Volume V (L): 6.73

t = 44.00 h

Biomass X (g/L): 91.65

Glucose S (g/L): 64.92

Lactic Acid P (g/L): 77.96

Maltose M (g/L): 67.98

Volume V (L): 6.79

t = 45.00 h

Biomass X (g/L): 81.65

Glucose S (g/L): 54.81

Lactic Acid P (g/L): 69.59

Maltose M (g/L): 66.36

Volume V (L): 6.93

t = 46.00 h

Biomass X (g/L): 88.65

Glucose S (g/L): 66.56

Lactic Acid P (g/L): 66.39

Maltose M (g/L): 55.27

Volume V (L): 7.03

t = 47.00 h

Biomass X (g/L): 104.06

Glucose S (g/L): 60.76

Lactic Acid P (g/L): 67.53

Maltose M (g/L): 43.18

Volume V (L): 6.97

t = 48.00 h

Biomass X (g/L): 78.75

Glucose S (g/L): 58.04

Lactic Acid P (g/L): 92.94

Maltose M (g/L): 56.38

Volume V (L): 7.05

t = 49.00 h

Biomass X (g/L): 68.84

Glucose S (g/L): 61.83

Lactic Acid P (g/L): 75.92

Maltose M (g/L): 55.48

Volume V (L): 6.70

t = 50.00 h

Biomass X (g/L): 93.14

Glucose S (g/L): 60.64

Lactic Acid P (g/L): 55.17

Maltose M (g/L): 55.18

Volume V (L): 7.30

t = 51.00 h

Biomass X (g/L): 87.37

Glucose S (g/L): 65.37

Lactic Acid P (g/L): 69.44

Maltose M (g/L): 58.79

Volume V (L): 7.22

t = 52.00 h

Biomass X (g/L): 81.24

Glucose S (g/L): 61.96
Lactic Acid P (g/L): 75.62
Maltose M (g/L): 60.14
Volume V (L): 7.47

t = 53.00 h

Biomass X (g/L): 94.94
Glucose S (g/L): 57.92
Lactic Acid P (g/L): 92.58
Maltose M (g/L): 60.27
Volume V (L): 7.26

t = 54.00 h

Biomass X (g/L): 96.74
Glucose S (g/L): 64.57
Lactic Acid P (g/L): 85.07
Maltose M (g/L): 56.94
Volume V (L): 6.86

t = 55.00 h

Biomass X (g/L): 65.82
Glucose S (g/L): 60.99
Lactic Acid P (g/L): 66.73
Maltose M (g/L): 44.67
Volume V (L): 7.25

t = 56.00 h

Biomass X (g/L): 81.45
Glucose S (g/L): 67.17
Lactic Acid P (g/L): 98.85
Maltose M (g/L): 51.90
Volume V (L): 7.34

t = 57.00 h

Biomass X (g/L): 111.37
Glucose S (g/L): 62.66
Lactic Acid P (g/L): 75.73
Maltose M (g/L): 65.39
Volume V (L): 7.62

t = 58.00 h

Biomass X (g/L): 93.70
Glucose S (g/L): 65.71
Lactic Acid P (g/L): 88.16
Maltose M (g/L): 54.80
Volume V (L): 6.94

t = 59.00 h

Biomass X (g/L): 94.51
Glucose S (g/L): 68.66
Lactic Acid P (g/L): 65.49
Maltose M (g/L): 54.34
Volume V (L): 7.53

t = 60.00 h

Biomass X (g/L): 99.39
Glucose S (g/L): 67.76
Lactic Acid P (g/L): 53.86
Maltose M (g/L): 53.88
Volume V (L): 7.69

t = 61.00 h

Biomass X (g/L): 87.55
Glucose S (g/L): 69.95
Lactic Acid P (g/L): 63.52
Maltose M (g/L): 53.77
Volume V (L): 7.43

t = 62.00 h

Biomass X (g/L): 95.28
Glucose S (g/L): 72.02
Lactic Acid P (g/L): 46.17
Maltose M (g/L): 38.63
Volume V (L): 7.66

t = 63.00 h

Biomass X (g/L): 94.33
Glucose S (g/L): 65.89
Lactic Acid P (g/L): 97.52
Maltose M (g/L): 49.52
Volume V (L): 7.53

t = 64.00 h

Biomass X (g/L): 92.35
Glucose S (g/L): 70.36
Lactic Acid P (g/L): 79.63
Maltose M (g/L): 50.19
Volume V (L): 7.54

t = 65.00 h

Biomass X (g/L): 83.47
Glucose S (g/L): 59.05
Lactic Acid P (g/L): 68.96
Maltose M (g/L): 56.75
Volume V (L): 7.67

t = 66.00 h

Biomass X (g/L): 88.30
Glucose S (g/L): 72.50
Lactic Acid P (g/L): 82.28
Maltose M (g/L): 49.88
Volume V (L): 7.63

t = 67.00 h

Biomass X (g/L): 80.42
Glucose S (g/L): 71.02
Lactic Acid P (g/L): 79.96
Maltose M (g/L): 40.60
Volume V (L): 8.03

t = 68.00 h

Biomass X (g/L): 86.32
Glucose S (g/L): 69.04
Lactic Acid P (g/L): 84.26
Maltose M (g/L): 51.16
Volume V (L): 7.71

t = 69.00 h

Biomass X (g/L): 74.18
Glucose S (g/L): 74.24
Lactic Acid P (g/L): 90.04
Maltose M (g/L): 44.31
Volume V (L): 7.71

t = 70.00 h

Biomass X (g/L): 77.82
Glucose S (g/L): 77.84
Lactic Acid P (g/L): 75.92
Maltose M (g/L): 41.78
Volume V (L): 8.44

t = 71.00 h

Biomass X (g/L): 82.76
Glucose S (g/L): 72.36
Lactic Acid P (g/L): 70.46
Maltose M (g/L): 43.45
Volume V (L): 8.27

t = 72.00 h

Biomass X (g/L): 75.88
Glucose S (g/L): 71.14
Lactic Acid P (g/L): 86.91
Maltose M (g/L): 50.35
Volume V (L): 7.60

t = 73.00 h

Biomass X (g/L): 80.11
Glucose S (g/L): 75.64
Lactic Acid P (g/L): 86.82
Maltose M (g/L): 53.78
Volume V (L): 7.92

t = 74.00 h

Biomass X (g/L): 83.36
Glucose S (g/L): 70.69
Lactic Acid P (g/L): 80.29
Maltose M (g/L): 49.37
Volume V (L): 8.22

t = 75.00 h

Biomass X (g/L): 88.88
Glucose S (g/L): 73.61
Lactic Acid P (g/L): 100.25
Maltose M (g/L): 49.82

Volume V (L): 8.28

Batch 40

Initial Conditions:

Batch: 40.00

X0 (g/L): 5.74

S0 (g/L): 87.78

M0 (g/L): 79.56

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.91

F_vol (L/h): 0.10

S_feed (g/L): 157.36

M_feed (g/L): 142.64

V_final (L): 12.45

Batch 40 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 90.00
Lactic Acid P (g/L): 11.43
Maltose M (g/L): 82.39
Volume V (L): 4.97

t = 1.00 h

Biomass X (g/L): 18.55
Glucose S (g/L): 86.62
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 80.34
Volume V (L): 4.96

t = 2.00 h

Biomass X (g/L): 17.86
Glucose S (g/L): 88.48
Lactic Acid P (g/L): 12.64
Maltose M (g/L): 78.62
Volume V (L): 5.47

t = 3.00 h

Biomass X (g/L): 17.63
Glucose S (g/L): 85.85
Lactic Acid P (g/L): 2.60
Maltose M (g/L): 81.15
Volume V (L): 5.46

t = 4.00 h

Biomass X (g/L): 29.35
Glucose S (g/L): 85.87
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 75.69
Volume V (L): 5.24

t = 5.00 h

Biomass X (g/L): 24.88
Glucose S (g/L): 86.84

Lactic Acid P (g/L): 14.38

Maltose M (g/L): 77.35

Volume V (L): 4.57

t = 6.00 h

Biomass X (g/L): 29.22

Glucose S (g/L): 82.48

Lactic Acid P (g/L): 4.64

Maltose M (g/L): 77.06

Volume V (L): 5.32

t = 7.00 h

Biomass X (g/L): 44.22

Glucose S (g/L): 80.97

Lactic Acid P (g/L): 22.57

Maltose M (g/L): 73.89

Volume V (L): 5.01

t = 8.00 h

Biomass X (g/L): 44.63

Glucose S (g/L): 74.74

Lactic Acid P (g/L): 10.80

Maltose M (g/L): 75.75

Volume V (L): 5.40

t = 9.00 h

Biomass X (g/L): 31.03

Glucose S (g/L): 80.24

Lactic Acid P (g/L): 22.00

Maltose M (g/L): 70.88

Volume V (L): 5.91

t = 10.00 h

Biomass X (g/L): 49.30

Glucose S (g/L): 75.93

Lactic Acid P (g/L): 15.54

Maltose M (g/L): 81.34

Volume V (L): 6.21

t = 11.00 h

Biomass X (g/L): 41.32

Glucose S (g/L): 74.57
Lactic Acid P (g/L): 56.34
Maltose M (g/L): 73.22
Volume V (L): 7.15

t = 12.00 h

Biomass X (g/L): 56.15
Glucose S (g/L): 74.10
Lactic Acid P (g/L): 19.78
Maltose M (g/L): 79.70
Volume V (L): 5.87

t = 13.00 h

Biomass X (g/L): 59.44
Glucose S (g/L): 65.83
Lactic Acid P (g/L): 37.26
Maltose M (g/L): 75.10
Volume V (L): 6.75

t = 14.00 h

Biomass X (g/L): 45.23
Glucose S (g/L): 65.77
Lactic Acid P (g/L): 49.84
Maltose M (g/L): 76.14
Volume V (L): 6.57

t = 15.00 h

Biomass X (g/L): 59.97
Glucose S (g/L): 59.78
Lactic Acid P (g/L): 51.25
Maltose M (g/L): 72.89
Volume V (L): 7.26

t = 16.00 h

Biomass X (g/L): 66.21
Glucose S (g/L): 74.25
Lactic Acid P (g/L): 51.76
Maltose M (g/L): 74.38
Volume V (L): 6.93

t = 17.00 h

Biomass X (g/L): 61.98
Glucose S (g/L): 63.31
Lactic Acid P (g/L): 43.41
Maltose M (g/L): 75.19
Volume V (L): 6.09

t = 18.00 h

Biomass X (g/L): 71.11
Glucose S (g/L): 66.34
Lactic Acid P (g/L): 59.94
Maltose M (g/L): 74.83
Volume V (L): 5.76

t = 19.00 h

Biomass X (g/L): 77.59
Glucose S (g/L): 58.88
Lactic Acid P (g/L): 57.72
Maltose M (g/L): 74.73
Volume V (L): 7.07

t = 20.00 h

Biomass X (g/L): 71.41
Glucose S (g/L): 65.46
Lactic Acid P (g/L): 53.03
Maltose M (g/L): 71.22
Volume V (L): 6.67

t = 21.00 h

Biomass X (g/L): 75.29
Glucose S (g/L): 67.94
Lactic Acid P (g/L): 54.66
Maltose M (g/L): 70.51
Volume V (L): 6.50

t = 22.00 h

Biomass X (g/L): 87.49
Glucose S (g/L): 61.99
Lactic Acid P (g/L): 51.11
Maltose M (g/L): 67.09
Volume V (L): 6.43

t = 23.00 h

Biomass X (g/L): 80.13
Glucose S (g/L): 59.12
Lactic Acid P (g/L): 46.93
Maltose M (g/L): 72.44
Volume V (L): 7.38

t = 24.00 h

Biomass X (g/L): 78.63
Glucose S (g/L): 61.94
Lactic Acid P (g/L): 66.97
Maltose M (g/L): 73.55
Volume V (L): 8.28

t = 25.00 h

Biomass X (g/L): 68.71
Glucose S (g/L): 55.74
Lactic Acid P (g/L): 68.63
Maltose M (g/L): 74.60
Volume V (L): 7.24

t = 26.00 h

Biomass X (g/L): 84.81
Glucose S (g/L): 62.77
Lactic Acid P (g/L): 50.15
Maltose M (g/L): 70.64
Volume V (L): 7.94

t = 27.00 h

Biomass X (g/L): 82.36
Glucose S (g/L): 65.76
Lactic Acid P (g/L): 57.46
Maltose M (g/L): 67.32
Volume V (L): 7.62

t = 28.00 h

Biomass X (g/L): 91.96
Glucose S (g/L): 63.53
Lactic Acid P (g/L): 58.35
Maltose M (g/L): 64.39
Volume V (L): 7.01

t = 29.00 h

Biomass X (g/L): 80.95
Glucose S (g/L): 63.21
Lactic Acid P (g/L): 60.08
Maltose M (g/L): 71.44
Volume V (L): 8.32

t = 30.00 h

Biomass X (g/L): 64.00
Glucose S (g/L): 66.20
Lactic Acid P (g/L): 61.63
Maltose M (g/L): 63.33
Volume V (L): 7.33

t = 31.00 h

Biomass X (g/L): 65.54
Glucose S (g/L): 66.95
Lactic Acid P (g/L): 70.08
Maltose M (g/L): 66.88
Volume V (L): 8.09

t = 32.00 h

Biomass X (g/L): 88.78
Glucose S (g/L): 60.54
Lactic Acid P (g/L): 55.73
Maltose M (g/L): 66.80
Volume V (L): 7.98

t = 33.00 h

Biomass X (g/L): 76.89
Glucose S (g/L): 66.49
Lactic Acid P (g/L): 55.43
Maltose M (g/L): 66.83
Volume V (L): 8.81

t = 34.00 h

Biomass X (g/L): 86.63
Glucose S (g/L): 74.31
Lactic Acid P (g/L): 74.37
Maltose M (g/L): 66.72

Volume V (L): 7.75

t = 35.00 h

Biomass X (g/L): 83.87

Glucose S (g/L): 63.32

Lactic Acid P (g/L): 59.87

Maltose M (g/L): 66.61

Volume V (L): 8.71

t = 36.00 h

Biomass X (g/L): 81.47

Glucose S (g/L): 67.80

Lactic Acid P (g/L): 53.28

Maltose M (g/L): 59.28

Volume V (L): 8.08

t = 37.00 h

Biomass X (g/L): 82.32

Glucose S (g/L): 74.68

Lactic Acid P (g/L): 60.61

Maltose M (g/L): 65.49

Volume V (L): 8.37

t = 38.00 h

Biomass X (g/L): 75.28

Glucose S (g/L): 72.76

Lactic Acid P (g/L): 62.42

Maltose M (g/L): 59.01

Volume V (L): 6.61

t = 39.00 h

Biomass X (g/L): 83.38

Glucose S (g/L): 68.44

Lactic Acid P (g/L): 79.48

Maltose M (g/L): 66.67

Volume V (L): 10.12

t = 40.00 h

Biomass X (g/L): 78.16

Glucose S (g/L): 71.33

Lactic Acid P (g/L): 74.31

Maltose M (g/L): 62.79

Volume V (L): 9.30

t = 41.00 h

Biomass X (g/L): 80.23

Glucose S (g/L): 71.36

Lactic Acid P (g/L): 76.65

Maltose M (g/L): 58.06

Volume V (L): 8.62

t = 42.00 h

Biomass X (g/L): 80.76

Glucose S (g/L): 69.75

Lactic Acid P (g/L): 74.11

Maltose M (g/L): 61.33

Volume V (L): 8.50

t = 43.00 h

Biomass X (g/L): 81.71

Glucose S (g/L): 72.50

Lactic Acid P (g/L): 75.57

Maltose M (g/L): 61.61

Volume V (L): 8.98

t = 44.00 h

Biomass X (g/L): 92.51

Glucose S (g/L): 72.55

Lactic Acid P (g/L): 61.45

Maltose M (g/L): 66.03

Volume V (L): 9.80

t = 45.00 h

Biomass X (g/L): 82.66

Glucose S (g/L): 74.45

Lactic Acid P (g/L): 66.09

Maltose M (g/L): 62.95

Volume V (L): 8.34

t = 46.00 h

Biomass X (g/L): 93.47

Glucose S (g/L): 72.48

Lactic Acid P (g/L): 59.40

Maltose M (g/L): 62.98

Volume V (L): 9.49

t = 47.00 h

Biomass X (g/L): 98.27

Glucose S (g/L): 71.54

Lactic Acid P (g/L): 66.57

Maltose M (g/L): 64.89

Volume V (L): 10.73

t = 48.00 h

Biomass X (g/L): 91.84

Glucose S (g/L): 76.03

Lactic Acid P (g/L): 74.75

Maltose M (g/L): 61.05

Volume V (L): 9.09

t = 49.00 h

Biomass X (g/L): 88.79

Glucose S (g/L): 82.11

Lactic Acid P (g/L): 70.16

Maltose M (g/L): 63.00

Volume V (L): 9.92

t = 50.00 h

Biomass X (g/L): 86.79

Glucose S (g/L): 77.77

Lactic Acid P (g/L): 80.81

Maltose M (g/L): 62.45

Volume V (L): 11.28

t = 51.00 h

Biomass X (g/L): 91.78

Glucose S (g/L): 77.17

Lactic Acid P (g/L): 59.39

Maltose M (g/L): 56.74

Volume V (L): 10.45

t = 52.00 h

Biomass X (g/L): 84.43

Glucose S (g/L): 81.18
Lactic Acid P (g/L): 74.13
Maltose M (g/L): 63.88
Volume V (L): 10.24

t = 53.00 h

Biomass X (g/L): 92.74
Glucose S (g/L): 79.05
Lactic Acid P (g/L): 58.91
Maltose M (g/L): 55.67
Volume V (L): 10.57

t = 54.00 h

Biomass X (g/L): 89.18
Glucose S (g/L): 85.95
Lactic Acid P (g/L): 70.99
Maltose M (g/L): 61.67
Volume V (L): 10.13

t = 55.00 h

Biomass X (g/L): 79.95
Glucose S (g/L): 80.85
Lactic Acid P (g/L): 76.91
Maltose M (g/L): 54.70
Volume V (L): 11.24

t = 56.00 h

Biomass X (g/L): 91.62
Glucose S (g/L): 83.67
Lactic Acid P (g/L): 68.69
Maltose M (g/L): 60.11
Volume V (L): 10.33

t = 57.00 h

Biomass X (g/L): 83.97
Glucose S (g/L): 79.84
Lactic Acid P (g/L): 75.65
Maltose M (g/L): 63.20
Volume V (L): 9.50

t = 58.00 h

Biomass X (g/L): 82.01
Glucose S (g/L): 84.36
Lactic Acid P (g/L): 62.50
Maltose M (g/L): 57.22
Volume V (L): 10.53

t = 59.00 h

Biomass X (g/L): 78.42
Glucose S (g/L): 84.30
Lactic Acid P (g/L): 56.47
Maltose M (g/L): 56.76
Volume V (L): 10.29

t = 60.00 h

Biomass X (g/L): 84.32
Glucose S (g/L): 83.92
Lactic Acid P (g/L): 76.27
Maltose M (g/L): 58.78
Volume V (L): 10.80

t = 61.00 h

Biomass X (g/L): 91.97
Glucose S (g/L): 83.18
Lactic Acid P (g/L): 86.93
Maltose M (g/L): 55.89
Volume V (L): 10.91

t = 62.00 h

Biomass X (g/L): 90.83
Glucose S (g/L): 83.38
Lactic Acid P (g/L): 62.90
Maltose M (g/L): 61.40
Volume V (L): 11.88

t = 63.00 h

Biomass X (g/L): 85.67
Glucose S (g/L): 86.83
Lactic Acid P (g/L): 82.67
Maltose M (g/L): 56.65
Volume V (L): 11.68

t = 64.00 h

Biomass X (g/L): 87.84
Glucose S (g/L): 88.51
Lactic Acid P (g/L): 72.42
Maltose M (g/L): 54.93
Volume V (L): 11.52

t = 65.00 h

Biomass X (g/L): 94.00
Glucose S (g/L): 86.79
Lactic Acid P (g/L): 60.40
Maltose M (g/L): 54.01
Volume V (L): 11.46

t = 66.00 h

Biomass X (g/L): 80.53
Glucose S (g/L): 88.67
Lactic Acid P (g/L): 62.36
Maltose M (g/L): 52.62
Volume V (L): 10.93

t = 67.00 h

Biomass X (g/L): 71.32
Glucose S (g/L): 88.27
Lactic Acid P (g/L): 72.65
Maltose M (g/L): 53.14
Volume V (L): 10.36

t = 68.00 h

Biomass X (g/L): 93.83
Glucose S (g/L): 89.28
Lactic Acid P (g/L): 69.08
Maltose M (g/L): 53.57
Volume V (L): 11.83

t = 69.00 h

Biomass X (g/L): 76.51
Glucose S (g/L): 88.94
Lactic Acid P (g/L): 61.20
Maltose M (g/L): 52.89
Volume V (L): 12.33

t = 70.00 h

Biomass X (g/L): 74.85
Glucose S (g/L): 90.19
Lactic Acid P (g/L): 57.98
Maltose M (g/L): 52.11
Volume V (L): 11.89

t = 71.00 h

Biomass X (g/L): 64.32
Glucose S (g/L): 99.02
Lactic Acid P (g/L): 65.61
Maltose M (g/L): 59.30
Volume V (L): 11.74

t = 72.00 h

Biomass X (g/L): 86.36
Glucose S (g/L): 92.24
Lactic Acid P (g/L): 74.93
Maltose M (g/L): 50.49
Volume V (L): 12.43

t = 73.00 h

Biomass X (g/L): 65.64
Glucose S (g/L): 93.83
Lactic Acid P (g/L): 73.30
Maltose M (g/L): 53.38
Volume V (L): 12.21

t = 74.00 h

Biomass X (g/L): 73.65
Glucose S (g/L): 93.75
Lactic Acid P (g/L): 69.06
Maltose M (g/L): 54.29
Volume V (L): 13.36

t = 75.00 h

Biomass X (g/L): 86.03
Glucose S (g/L): 92.14
Lactic Acid P (g/L): 75.40
Maltose M (g/L): 54.10

Volume V (L): 13.19

Batch 41

Initial Conditions:

Batch: 41.00

X0 (g/L): 9.27

S0 (g/L): 106.05

M0 (g/L): 91.04

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.86

F_vol (L/h): 0.10

S_feed (g/L): 161.42

M_feed (g/L): 138.58

V_final (L): 12.26

Batch 41 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 18.14
Glucose S (g/L): 101.82
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 90.43
Volume V (L): 4.38

t = 1.00 h

Biomass X (g/L): 19.40
Glucose S (g/L): 98.53
Lactic Acid P (g/L): 0.42
Maltose M (g/L): 94.86
Volume V (L): 5.06

t = 2.00 h

Biomass X (g/L): 4.26
Glucose S (g/L): 104.89
Lactic Acid P (g/L): 11.65
Maltose M (g/L): 86.81
Volume V (L): 5.35

t = 3.00 h

Biomass X (g/L): 26.72
Glucose S (g/L): 103.18
Lactic Acid P (g/L): 4.91
Maltose M (g/L): 90.86
Volume V (L): 5.14

t = 4.00 h

Biomass X (g/L): 28.94
Glucose S (g/L): 100.83
Lactic Acid P (g/L): 4.86
Maltose M (g/L): 84.75
Volume V (L): 5.61

t = 5.00 h

Biomass X (g/L): 36.23
Glucose S (g/L): 93.96

Lactic Acid P (g/L): 12.94

Maltose M (g/L): 87.42

Volume V (L): 5.96

t = 6.00 h

Biomass X (g/L): 40.43

Glucose S (g/L): 96.17

Lactic Acid P (g/L): 13.18

Maltose M (g/L): 81.52

Volume V (L): 6.16

t = 7.00 h

Biomass X (g/L): 45.21

Glucose S (g/L): 93.26

Lactic Acid P (g/L): 15.61

Maltose M (g/L): 82.84

Volume V (L): 6.66

t = 8.00 h

Biomass X (g/L): 70.90

Glucose S (g/L): 84.02

Lactic Acid P (g/L): 13.32

Maltose M (g/L): 84.37

Volume V (L): 5.58

t = 9.00 h

Biomass X (g/L): 49.80

Glucose S (g/L): 80.69

Lactic Acid P (g/L): 31.83

Maltose M (g/L): 91.07

Volume V (L): 5.08

t = 10.00 h

Biomass X (g/L): 63.34

Glucose S (g/L): 83.05

Lactic Acid P (g/L): 29.41

Maltose M (g/L): 83.63

Volume V (L): 6.18

t = 11.00 h

Biomass X (g/L): 54.12

Glucose S (g/L): 86.77
Lactic Acid P (g/L): 46.08
Maltose M (g/L): 89.00
Volume V (L): 7.16

t = 12.00 h

Biomass X (g/L): 69.18
Glucose S (g/L): 84.74
Lactic Acid P (g/L): 27.32
Maltose M (g/L): 82.68
Volume V (L): 5.88

t = 13.00 h

Biomass X (g/L): 73.58
Glucose S (g/L): 79.65
Lactic Acid P (g/L): 37.87
Maltose M (g/L): 79.60
Volume V (L): 6.66

t = 14.00 h

Biomass X (g/L): 70.95
Glucose S (g/L): 74.32
Lactic Acid P (g/L): 61.09
Maltose M (g/L): 86.47
Volume V (L): 6.76

t = 15.00 h

Biomass X (g/L): 85.39
Glucose S (g/L): 78.60
Lactic Acid P (g/L): 38.62
Maltose M (g/L): 81.48
Volume V (L): 6.56

t = 16.00 h

Biomass X (g/L): 63.89
Glucose S (g/L): 73.58
Lactic Acid P (g/L): 46.73
Maltose M (g/L): 76.59
Volume V (L): 6.55

t = 17.00 h

Biomass X (g/L): 69.42
Glucose S (g/L): 72.05
Lactic Acid P (g/L): 39.69
Maltose M (g/L): 80.30
Volume V (L): 6.11

t = 18.00 h

Biomass X (g/L): 77.53
Glucose S (g/L): 67.27
Lactic Acid P (g/L): 40.04
Maltose M (g/L): 76.66
Volume V (L): 5.42

t = 19.00 h

Biomass X (g/L): 85.15
Glucose S (g/L): 74.29
Lactic Acid P (g/L): 67.22
Maltose M (g/L): 69.38
Volume V (L): 6.19

t = 20.00 h

Biomass X (g/L): 79.95
Glucose S (g/L): 71.86
Lactic Acid P (g/L): 56.24
Maltose M (g/L): 80.21
Volume V (L): 6.71

t = 21.00 h

Biomass X (g/L): 91.30
Glucose S (g/L): 74.22
Lactic Acid P (g/L): 57.08
Maltose M (g/L): 80.22
Volume V (L): 7.67

t = 22.00 h

Biomass X (g/L): 74.29
Glucose S (g/L): 66.64
Lactic Acid P (g/L): 41.85
Maltose M (g/L): 82.52
Volume V (L): 7.07

t = 23.00 h

Biomass X (g/L): 70.74
Glucose S (g/L): 74.78
Lactic Acid P (g/L): 61.15
Maltose M (g/L): 76.37
Volume V (L): 6.92

t = 24.00 h

Biomass X (g/L): 84.90
Glucose S (g/L): 66.69
Lactic Acid P (g/L): 69.17
Maltose M (g/L): 73.15
Volume V (L): 6.81

t = 25.00 h

Biomass X (g/L): 82.01
Glucose S (g/L): 78.34
Lactic Acid P (g/L): 69.95
Maltose M (g/L): 76.82
Volume V (L): 7.80

t = 26.00 h

Biomass X (g/L): 88.33
Glucose S (g/L): 76.91
Lactic Acid P (g/L): 66.45
Maltose M (g/L): 76.30
Volume V (L): 8.30

t = 27.00 h

Biomass X (g/L): 83.58
Glucose S (g/L): 74.77
Lactic Acid P (g/L): 51.17
Maltose M (g/L): 78.82
Volume V (L): 8.23

t = 28.00 h

Biomass X (g/L): 81.13
Glucose S (g/L): 77.94
Lactic Acid P (g/L): 64.70
Maltose M (g/L): 71.66
Volume V (L): 8.06

t = 29.00 h

Biomass X (g/L): 83.97
Glucose S (g/L): 73.49
Lactic Acid P (g/L): 68.49
Maltose M (g/L): 70.63
Volume V (L): 6.95

t = 30.00 h

Biomass X (g/L): 85.26
Glucose S (g/L): 77.40
Lactic Acid P (g/L): 47.16
Maltose M (g/L): 67.93
Volume V (L): 7.70

t = 31.00 h

Biomass X (g/L): 82.46
Glucose S (g/L): 80.28
Lactic Acid P (g/L): 53.72
Maltose M (g/L): 72.30
Volume V (L): 9.13

t = 32.00 h

Biomass X (g/L): 83.65
Glucose S (g/L): 68.35
Lactic Acid P (g/L): 72.14
Maltose M (g/L): 63.59
Volume V (L): 8.75

t = 33.00 h

Biomass X (g/L): 82.00
Glucose S (g/L): 77.46
Lactic Acid P (g/L): 72.53
Maltose M (g/L): 72.18
Volume V (L): 7.30

t = 34.00 h

Biomass X (g/L): 93.81
Glucose S (g/L): 79.43
Lactic Acid P (g/L): 51.40
Maltose M (g/L): 70.92

Volume V (L): 8.26

t = 35.00 h

Biomass X (g/L): 87.82

Glucose S (g/L): 79.50

Lactic Acid P (g/L): 72.72

Maltose M (g/L): 70.40

Volume V (L): 8.60

t = 36.00 h

Biomass X (g/L): 106.99

Glucose S (g/L): 85.44

Lactic Acid P (g/L): 66.90

Maltose M (g/L): 68.14

Volume V (L): 8.04

t = 37.00 h

Biomass X (g/L): 87.70

Glucose S (g/L): 77.78

Lactic Acid P (g/L): 73.41

Maltose M (g/L): 73.10

Volume V (L): 7.24

t = 38.00 h

Biomass X (g/L): 81.17

Glucose S (g/L): 80.04

Lactic Acid P (g/L): 61.18

Maltose M (g/L): 70.02

Volume V (L): 8.73

t = 39.00 h

Biomass X (g/L): 110.98

Glucose S (g/L): 76.52

Lactic Acid P (g/L): 72.93

Maltose M (g/L): 63.73

Volume V (L): 8.32

t = 40.00 h

Biomass X (g/L): 90.13

Glucose S (g/L): 88.19

Lactic Acid P (g/L): 75.05

Maltose M (g/L): 63.32

Volume V (L): 8.42

t = 41.00 h

Biomass X (g/L): 95.45

Glucose S (g/L): 83.20

Lactic Acid P (g/L): 64.73

Maltose M (g/L): 61.71

Volume V (L): 8.76

t = 42.00 h

Biomass X (g/L): 91.75

Glucose S (g/L): 83.23

Lactic Acid P (g/L): 68.33

Maltose M (g/L): 66.84

Volume V (L): 8.99

t = 43.00 h

Biomass X (g/L): 92.00

Glucose S (g/L): 85.13

Lactic Acid P (g/L): 62.69

Maltose M (g/L): 60.30

Volume V (L): 9.16

t = 44.00 h

Biomass X (g/L): 91.14

Glucose S (g/L): 87.99

Lactic Acid P (g/L): 83.46

Maltose M (g/L): 68.01

Volume V (L): 8.74

t = 45.00 h

Biomass X (g/L): 75.94

Glucose S (g/L): 87.72

Lactic Acid P (g/L): 67.96

Maltose M (g/L): 68.49

Volume V (L): 9.23

t = 46.00 h

Biomass X (g/L): 89.45

Glucose S (g/L): 95.76

Lactic Acid P (g/L): 61.08

Maltose M (g/L): 59.41

Volume V (L): 8.43

t = 47.00 h

Biomass X (g/L): 73.71

Glucose S (g/L): 89.35

Lactic Acid P (g/L): 63.17

Maltose M (g/L): 61.69

Volume V (L): 9.73

t = 48.00 h

Biomass X (g/L): 84.25

Glucose S (g/L): 87.04

Lactic Acid P (g/L): 70.92

Maltose M (g/L): 62.04

Volume V (L): 7.99

t = 49.00 h

Biomass X (g/L): 79.05

Glucose S (g/L): 87.37

Lactic Acid P (g/L): 76.73

Maltose M (g/L): 65.01

Volume V (L): 9.57

t = 50.00 h

Biomass X (g/L): 88.95

Glucose S (g/L): 88.55

Lactic Acid P (g/L): 58.35

Maltose M (g/L): 67.42

Volume V (L): 10.37

t = 51.00 h

Biomass X (g/L): 88.61

Glucose S (g/L): 85.61

Lactic Acid P (g/L): 86.92

Maltose M (g/L): 54.09

Volume V (L): 9.32

t = 52.00 h

Biomass X (g/L): 89.85

Glucose S (g/L): 91.20
Lactic Acid P (g/L): 73.43
Maltose M (g/L): 59.03
Volume V (L): 10.45

t = 53.00 h

Biomass X (g/L): 87.92
Glucose S (g/L): 94.21
Lactic Acid P (g/L): 75.04
Maltose M (g/L): 58.82
Volume V (L): 10.80

t = 54.00 h

Biomass X (g/L): 79.52
Glucose S (g/L): 90.34
Lactic Acid P (g/L): 70.22
Maltose M (g/L): 53.96
Volume V (L): 8.30

t = 55.00 h

Biomass X (g/L): 82.57
Glucose S (g/L): 90.32
Lactic Acid P (g/L): 71.44
Maltose M (g/L): 57.15
Volume V (L): 10.23

t = 56.00 h

Biomass X (g/L): 93.34
Glucose S (g/L): 98.40
Lactic Acid P (g/L): 70.14
Maltose M (g/L): 60.33
Volume V (L): 8.75

t = 57.00 h

Biomass X (g/L): 94.55
Glucose S (g/L): 89.54
Lactic Acid P (g/L): 80.70
Maltose M (g/L): 57.16
Volume V (L): 10.45

t = 58.00 h

Biomass X (g/L): 73.92
Glucose S (g/L): 88.07
Lactic Acid P (g/L): 71.74
Maltose M (g/L): 51.81
Volume V (L): 10.61

t = 59.00 h

Biomass X (g/L): 70.21
Glucose S (g/L): 91.60
Lactic Acid P (g/L): 78.82
Maltose M (g/L): 68.74
Volume V (L): 9.27

t = 60.00 h

Biomass X (g/L): 77.48
Glucose S (g/L): 90.94
Lactic Acid P (g/L): 68.29
Maltose M (g/L): 58.14
Volume V (L): 11.92

t = 61.00 h

Biomass X (g/L): 93.69
Glucose S (g/L): 97.71
Lactic Acid P (g/L): 74.05
Maltose M (g/L): 57.16
Volume V (L): 11.07

t = 62.00 h

Biomass X (g/L): 95.09
Glucose S (g/L): 95.88
Lactic Acid P (g/L): 65.31
Maltose M (g/L): 52.04
Volume V (L): 10.51

t = 63.00 h

Biomass X (g/L): 75.14
Glucose S (g/L): 96.90
Lactic Acid P (g/L): 83.46
Maltose M (g/L): 57.00
Volume V (L): 11.95

t = 64.00 h

Biomass X (g/L): 86.35
Glucose S (g/L): 101.88
Lactic Acid P (g/L): 65.35
Maltose M (g/L): 54.97
Volume V (L): 10.15

t = 65.00 h

Biomass X (g/L): 100.29
Glucose S (g/L): 96.50
Lactic Acid P (g/L): 70.02
Maltose M (g/L): 47.85
Volume V (L): 10.58

t = 66.00 h

Biomass X (g/L): 86.99
Glucose S (g/L): 101.77
Lactic Acid P (g/L): 70.26
Maltose M (g/L): 50.70
Volume V (L): 10.05

t = 67.00 h

Biomass X (g/L): 96.12
Glucose S (g/L): 104.68
Lactic Acid P (g/L): 70.05
Maltose M (g/L): 57.66
Volume V (L): 12.53

t = 68.00 h

Biomass X (g/L): 92.03
Glucose S (g/L): 98.89
Lactic Acid P (g/L): 79.77
Maltose M (g/L): 61.81
Volume V (L): 11.77

t = 69.00 h

Biomass X (g/L): 74.20
Glucose S (g/L): 103.25
Lactic Acid P (g/L): 70.65
Maltose M (g/L): 55.28
Volume V (L): 12.06

t = 70.00 h

Biomass X (g/L): 80.67
Glucose S (g/L): 100.63
Lactic Acid P (g/L): 86.01
Maltose M (g/L): 57.83
Volume V (L): 11.95

t = 71.00 h

Biomass X (g/L): 82.56
Glucose S (g/L): 104.90
Lactic Acid P (g/L): 66.44
Maltose M (g/L): 52.55
Volume V (L): 12.20

t = 72.00 h

Biomass X (g/L): 83.02
Glucose S (g/L): 97.38
Lactic Acid P (g/L): 75.36
Maltose M (g/L): 52.49
Volume V (L): 12.01

t = 73.00 h

Biomass X (g/L): 90.34
Glucose S (g/L): 97.65
Lactic Acid P (g/L): 67.07
Maltose M (g/L): 56.77
Volume V (L): 12.22

t = 74.00 h

Biomass X (g/L): 85.46
Glucose S (g/L): 102.19
Lactic Acid P (g/L): 77.85
Maltose M (g/L): 58.69
Volume V (L): 12.22

t = 75.00 h

Biomass X (g/L): 54.35
Glucose S (g/L): 101.08
Lactic Acid P (g/L): 74.07
Maltose M (g/L): 56.95

Volume V (L): 11.90

Batch 42

Initial Conditions:

Batch: 42.00

X0 (g/L): 3.29

S0 (g/L): 47.91

M0 (g/L): 53.71

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.12

F_vol (L/h): 0.05

S_feed (g/L): 141.44

M_feed (g/L): 158.56

V_final (L): 8.91

Batch 42 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 2.84

Glucose S (g/L): 43.32

Lactic Acid P (g/L): 5.71

Maltose M (g/L): 54.62

Volume V (L): 4.85

t = 1.00 h

Biomass X (g/L): 7.03

Glucose S (g/L): 46.89

Lactic Acid P (g/L): 1.09

Maltose M (g/L): 53.79

Volume V (L): 5.20

t = 2.00 h

Biomass X (g/L): 6.87

Glucose S (g/L): 49.50

Lactic Acid P (g/L): 16.22

Maltose M (g/L): 51.96

Volume V (L): 5.72

t = 3.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 47.05

Lactic Acid P (g/L): 17.29

Maltose M (g/L): 55.58

Volume V (L): 5.13

t = 4.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 51.95

Lactic Acid P (g/L): 8.76

Maltose M (g/L): 53.28

Volume V (L): 5.56

t = 5.00 h

Biomass X (g/L): 22.98

Glucose S (g/L): 46.15

Lactic Acid P (g/L): 9.64
Maltose M (g/L): 53.41
Volume V (L): 5.78

t = 6.00 h

Biomass X (g/L): 12.68
Glucose S (g/L): 47.70
Lactic Acid P (g/L): 9.69
Maltose M (g/L): 54.06
Volume V (L): 5.63

t = 7.00 h

Biomass X (g/L): 24.89
Glucose S (g/L): 45.59
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 54.42
Volume V (L): 5.36

t = 8.00 h

Biomass X (g/L): 12.59
Glucose S (g/L): 45.49
Lactic Acid P (g/L): 1.86
Maltose M (g/L): 55.74
Volume V (L): 5.62

t = 9.00 h

Biomass X (g/L): 13.06
Glucose S (g/L): 43.56
Lactic Acid P (g/L): 18.58
Maltose M (g/L): 52.03
Volume V (L): 5.07

t = 10.00 h

Biomass X (g/L): 23.42
Glucose S (g/L): 40.28
Lactic Acid P (g/L): 26.45
Maltose M (g/L): 54.50
Volume V (L): 5.28

t = 11.00 h

Biomass X (g/L): 33.68

Glucose S (g/L): 28.48
Lactic Acid P (g/L): 26.65
Maltose M (g/L): 54.36
Volume V (L): 5.38

t = 12.00 h

Biomass X (g/L): 33.54
Glucose S (g/L): 36.71
Lactic Acid P (g/L): 18.96
Maltose M (g/L): 53.41
Volume V (L): 6.33

t = 13.00 h

Biomass X (g/L): 56.44
Glucose S (g/L): 35.37
Lactic Acid P (g/L): 38.67
Maltose M (g/L): 54.27
Volume V (L): 5.42

t = 14.00 h

Biomass X (g/L): 49.37
Glucose S (g/L): 26.55
Lactic Acid P (g/L): 46.48
Maltose M (g/L): 54.35
Volume V (L): 5.47

t = 15.00 h

Biomass X (g/L): 68.95
Glucose S (g/L): 26.89
Lactic Acid P (g/L): 44.17
Maltose M (g/L): 51.91
Volume V (L): 5.46

t = 16.00 h

Biomass X (g/L): 58.13
Glucose S (g/L): 21.41
Lactic Acid P (g/L): 47.11
Maltose M (g/L): 55.00
Volume V (L): 6.01

t = 17.00 h

Biomass X (g/L): 33.75
Glucose S (g/L): 16.40
Lactic Acid P (g/L): 35.25
Maltose M (g/L): 51.55
Volume V (L): 5.93

t = 18.00 h

Biomass X (g/L): 69.91
Glucose S (g/L): 19.03
Lactic Acid P (g/L): 52.30
Maltose M (g/L): 51.88
Volume V (L): 6.28

t = 19.00 h

Biomass X (g/L): 47.45
Glucose S (g/L): 18.56
Lactic Acid P (g/L): 45.75
Maltose M (g/L): 52.66
Volume V (L): 6.20

t = 20.00 h

Biomass X (g/L): 66.42
Glucose S (g/L): 18.23
Lactic Acid P (g/L): 50.89
Maltose M (g/L): 54.77
Volume V (L): 5.61

t = 21.00 h

Biomass X (g/L): 58.62
Glucose S (g/L): 15.49
Lactic Acid P (g/L): 51.21
Maltose M (g/L): 52.38
Volume V (L): 5.84

t = 22.00 h

Biomass X (g/L): 88.58
Glucose S (g/L): 15.62
Lactic Acid P (g/L): 48.29
Maltose M (g/L): 52.54
Volume V (L): 6.57

t = 23.00 h

Biomass X (g/L): 83.36
Glucose S (g/L): 17.51
Lactic Acid P (g/L): 42.97
Maltose M (g/L): 53.91
Volume V (L): 6.14

t = 24.00 h

Biomass X (g/L): 63.68
Glucose S (g/L): 14.67
Lactic Acid P (g/L): 51.39
Maltose M (g/L): 52.02
Volume V (L): 6.62

t = 25.00 h

Biomass X (g/L): 70.92
Glucose S (g/L): 18.73
Lactic Acid P (g/L): 62.38
Maltose M (g/L): 50.99
Volume V (L): 6.26

t = 26.00 h

Biomass X (g/L): 75.70
Glucose S (g/L): 19.02
Lactic Acid P (g/L): 50.49
Maltose M (g/L): 51.63
Volume V (L): 6.05

t = 27.00 h

Biomass X (g/L): 82.75
Glucose S (g/L): 24.51
Lactic Acid P (g/L): 63.76
Maltose M (g/L): 53.31
Volume V (L): 6.31

t = 28.00 h

Biomass X (g/L): 75.65
Glucose S (g/L): 20.40
Lactic Acid P (g/L): 63.34
Maltose M (g/L): 51.85
Volume V (L): 6.16

t = 29.00 h

Biomass X (g/L): 66.34
Glucose S (g/L): 17.58
Lactic Acid P (g/L): 54.83
Maltose M (g/L): 53.59
Volume V (L): 6.34

t = 30.00 h

Biomass X (g/L): 70.12
Glucose S (g/L): 14.87
Lactic Acid P (g/L): 71.14
Maltose M (g/L): 52.23
Volume V (L): 6.63

t = 31.00 h

Biomass X (g/L): 77.76
Glucose S (g/L): 18.04
Lactic Acid P (g/L): 56.06
Maltose M (g/L): 50.68
Volume V (L): 6.78

t = 32.00 h

Biomass X (g/L): 90.41
Glucose S (g/L): 18.36
Lactic Acid P (g/L): 73.90
Maltose M (g/L): 51.23
Volume V (L): 6.17

t = 33.00 h

Biomass X (g/L): 83.22
Glucose S (g/L): 15.87
Lactic Acid P (g/L): 64.84
Maltose M (g/L): 50.15
Volume V (L): 6.72

t = 34.00 h

Biomass X (g/L): 72.16
Glucose S (g/L): 20.41
Lactic Acid P (g/L): 61.45
Maltose M (g/L): 50.18

Volume V (L): 6.59

t = 35.00 h

Biomass X (g/L): 86.42

Glucose S (g/L): 13.59

Lactic Acid P (g/L): 54.18

Maltose M (g/L): 55.25

Volume V (L): 6.62

t = 36.00 h

Biomass X (g/L): 76.87

Glucose S (g/L): 17.73

Lactic Acid P (g/L): 62.14

Maltose M (g/L): 51.58

Volume V (L): 6.92

t = 37.00 h

Biomass X (g/L): 89.65

Glucose S (g/L): 20.73

Lactic Acid P (g/L): 61.72

Maltose M (g/L): 51.25

Volume V (L): 6.76

t = 38.00 h

Biomass X (g/L): 105.86

Glucose S (g/L): 18.38

Lactic Acid P (g/L): 52.69

Maltose M (g/L): 49.58

Volume V (L): 7.41

t = 39.00 h

Biomass X (g/L): 78.29

Glucose S (g/L): 19.07

Lactic Acid P (g/L): 62.77

Maltose M (g/L): 51.64

Volume V (L): 7.41

t = 40.00 h

Biomass X (g/L): 89.83

Glucose S (g/L): 19.74

Lactic Acid P (g/L): 57.96

Maltose M (g/L): 50.99

Volume V (L): 7.46

t = 41.00 h

Biomass X (g/L): 75.14

Glucose S (g/L): 17.92

Lactic Acid P (g/L): 71.18

Maltose M (g/L): 51.62

Volume V (L): 6.98

t = 42.00 h

Biomass X (g/L): 81.52

Glucose S (g/L): 24.98

Lactic Acid P (g/L): 78.27

Maltose M (g/L): 51.40

Volume V (L): 6.89

t = 43.00 h

Biomass X (g/L): 83.11

Glucose S (g/L): 18.30

Lactic Acid P (g/L): 55.62

Maltose M (g/L): 47.86

Volume V (L): 7.27

t = 44.00 h

Biomass X (g/L): 81.15

Glucose S (g/L): 23.19

Lactic Acid P (g/L): 82.93

Maltose M (g/L): 48.06

Volume V (L): 7.25

t = 45.00 h

Biomass X (g/L): 107.03

Glucose S (g/L): 24.02

Lactic Acid P (g/L): 66.57

Maltose M (g/L): 48.00

Volume V (L): 7.16

t = 46.00 h

Biomass X (g/L): 80.39

Glucose S (g/L): 20.09

Lactic Acid P (g/L): 59.27

Maltose M (g/L): 47.88

Volume V (L): 7.03

t = 47.00 h

Biomass X (g/L): 89.16

Glucose S (g/L): 23.98

Lactic Acid P (g/L): 75.51

Maltose M (g/L): 48.99

Volume V (L): 7.44

t = 48.00 h

Biomass X (g/L): 64.92

Glucose S (g/L): 21.31

Lactic Acid P (g/L): 64.91

Maltose M (g/L): 47.91

Volume V (L): 7.38

t = 49.00 h

Biomass X (g/L): 87.86

Glucose S (g/L): 29.40

Lactic Acid P (g/L): 68.63

Maltose M (g/L): 50.42

Volume V (L): 7.77

t = 50.00 h

Biomass X (g/L): 94.81

Glucose S (g/L): 21.68

Lactic Acid P (g/L): 66.06

Maltose M (g/L): 50.07

Volume V (L): 7.82

t = 51.00 h

Biomass X (g/L): 78.71

Glucose S (g/L): 23.84

Lactic Acid P (g/L): 62.65

Maltose M (g/L): 47.95

Volume V (L): 8.00

t = 52.00 h

Biomass X (g/L): 59.58

Glucose S (g/L): 22.54
Lactic Acid P (g/L): 77.07
Maltose M (g/L): 49.76
Volume V (L): 7.32

t = 53.00 h

Biomass X (g/L): 83.52
Glucose S (g/L): 22.69
Lactic Acid P (g/L): 68.18
Maltose M (g/L): 49.11
Volume V (L): 7.89

t = 54.00 h

Biomass X (g/L): 84.07
Glucose S (g/L): 32.93
Lactic Acid P (g/L): 58.00
Maltose M (g/L): 46.96
Volume V (L): 7.59

t = 55.00 h

Biomass X (g/L): 90.91
Glucose S (g/L): 30.08
Lactic Acid P (g/L): 76.47
Maltose M (g/L): 45.20
Volume V (L): 7.49

t = 56.00 h

Biomass X (g/L): 103.35
Glucose S (g/L): 29.35
Lactic Acid P (g/L): 60.73
Maltose M (g/L): 46.84
Volume V (L): 7.79

t = 57.00 h

Biomass X (g/L): 80.20
Glucose S (g/L): 27.34
Lactic Acid P (g/L): 61.50
Maltose M (g/L): 46.22
Volume V (L): 8.31

t = 58.00 h

Biomass X (g/L): 75.56
Glucose S (g/L): 33.05
Lactic Acid P (g/L): 83.21
Maltose M (g/L): 46.91
Volume V (L): 7.91

t = 59.00 h

Biomass X (g/L): 72.51
Glucose S (g/L): 27.98
Lactic Acid P (g/L): 61.67
Maltose M (g/L): 46.61
Volume V (L): 8.57

t = 60.00 h

Biomass X (g/L): 85.67
Glucose S (g/L): 33.88
Lactic Acid P (g/L): 51.71
Maltose M (g/L): 46.40
Volume V (L): 8.30

t = 61.00 h

Biomass X (g/L): 88.33
Glucose S (g/L): 29.02
Lactic Acid P (g/L): 78.82
Maltose M (g/L): 46.69
Volume V (L): 8.06

t = 62.00 h

Biomass X (g/L): 78.34
Glucose S (g/L): 30.36
Lactic Acid P (g/L): 70.60
Maltose M (g/L): 44.75
Volume V (L): 8.60

t = 63.00 h

Biomass X (g/L): 91.08
Glucose S (g/L): 35.56
Lactic Acid P (g/L): 71.64
Maltose M (g/L): 46.22
Volume V (L): 8.43

t = 64.00 h

Biomass X (g/L): 87.96
Glucose S (g/L): 36.50
Lactic Acid P (g/L): 70.99
Maltose M (g/L): 45.17
Volume V (L): 8.05

t = 65.00 h

Biomass X (g/L): 83.78
Glucose S (g/L): 36.23
Lactic Acid P (g/L): 76.21
Maltose M (g/L): 45.64
Volume V (L): 8.97

t = 66.00 h

Biomass X (g/L): 74.54
Glucose S (g/L): 31.43
Lactic Acid P (g/L): 67.45
Maltose M (g/L): 46.23
Volume V (L): 8.52

t = 67.00 h

Biomass X (g/L): 78.50
Glucose S (g/L): 34.01
Lactic Acid P (g/L): 88.02
Maltose M (g/L): 43.84
Volume V (L): 8.50

t = 68.00 h

Biomass X (g/L): 59.55
Glucose S (g/L): 36.58
Lactic Acid P (g/L): 61.27
Maltose M (g/L): 44.61
Volume V (L): 8.28

t = 69.00 h

Biomass X (g/L): 85.22
Glucose S (g/L): 33.88
Lactic Acid P (g/L): 79.03
Maltose M (g/L): 44.60
Volume V (L): 8.97

t = 70.00 h

Biomass X (g/L): 83.07
Glucose S (g/L): 41.75
Lactic Acid P (g/L): 81.03
Maltose M (g/L): 44.02
Volume V (L): 8.81

t = 71.00 h

Biomass X (g/L): 84.51
Glucose S (g/L): 29.99
Lactic Acid P (g/L): 68.43
Maltose M (g/L): 45.28
Volume V (L): 8.90

t = 72.00 h

Biomass X (g/L): 87.44
Glucose S (g/L): 35.89
Lactic Acid P (g/L): 66.21
Maltose M (g/L): 43.46
Volume V (L): 8.60

t = 73.00 h

Biomass X (g/L): 86.16
Glucose S (g/L): 37.36
Lactic Acid P (g/L): 85.03
Maltose M (g/L): 43.45
Volume V (L): 9.07

t = 74.00 h

Biomass X (g/L): 81.22
Glucose S (g/L): 30.31
Lactic Acid P (g/L): 66.88
Maltose M (g/L): 43.11
Volume V (L): 9.14

t = 75.00 h

Biomass X (g/L): 82.61
Glucose S (g/L): 35.63
Lactic Acid P (g/L): 70.56
Maltose M (g/L): 44.77

Volume V (L): 8.70

Batch 43

Initial Conditions:

Batch: 43.00

X0 (g/L): 4.48

S0 (g/L): 131.01

M0 (g/L): 49.04

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.37

F_vol (L/h): 0.05

S_feed (g/L): 218.29

M_feed (g/L): 81.71

V_final (L): 8.57

Batch 43 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 133.30
Lactic Acid P (g/L): 11.41
Maltose M (g/L): 49.03
Volume V (L): 5.16

t = 1.00 h

Biomass X (g/L): 11.97
Glucose S (g/L): 131.37
Lactic Acid P (g/L): 0.16
Maltose M (g/L): 50.21
Volume V (L): 5.32

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 127.32
Lactic Acid P (g/L): 17.28
Maltose M (g/L): 48.98
Volume V (L): 5.19

t = 3.00 h

Biomass X (g/L): 12.40
Glucose S (g/L): 129.27
Lactic Acid P (g/L): 19.19
Maltose M (g/L): 43.50
Volume V (L): 5.46

t = 4.00 h

Biomass X (g/L): 9.59
Glucose S (g/L): 129.04
Lactic Acid P (g/L): 15.09
Maltose M (g/L): 43.27
Volume V (L): 5.11

t = 5.00 h

Biomass X (g/L): 30.22
Glucose S (g/L): 130.88

Lactic Acid P (g/L): 4.97
Maltose M (g/L): 45.11
Volume V (L): 4.95

t = 6.00 h

Biomass X (g/L): 18.28
Glucose S (g/L): 118.85
Lactic Acid P (g/L): 7.23
Maltose M (g/L): 49.93
Volume V (L): 5.27

t = 7.00 h

Biomass X (g/L): 38.11
Glucose S (g/L): 118.67
Lactic Acid P (g/L): 8.78
Maltose M (g/L): 42.82
Volume V (L): 5.39

t = 8.00 h

Biomass X (g/L): 26.66
Glucose S (g/L): 116.70
Lactic Acid P (g/L): 29.09
Maltose M (g/L): 43.86
Volume V (L): 5.04

t = 9.00 h

Biomass X (g/L): 46.89
Glucose S (g/L): 116.12
Lactic Acid P (g/L): 12.30
Maltose M (g/L): 42.53
Volume V (L): 5.05

t = 10.00 h

Biomass X (g/L): 62.45
Glucose S (g/L): 105.00
Lactic Acid P (g/L): 33.82
Maltose M (g/L): 41.97
Volume V (L): 5.60

t = 11.00 h

Biomass X (g/L): 35.52

Glucose S (g/L): 99.65
Lactic Acid P (g/L): 20.66
Maltose M (g/L): 43.26
Volume V (L): 5.41

t = 12.00 h

Biomass X (g/L): 53.60
Glucose S (g/L): 106.12
Lactic Acid P (g/L): 42.03
Maltose M (g/L): 45.76
Volume V (L): 5.52

t = 13.00 h

Biomass X (g/L): 82.42
Glucose S (g/L): 92.66
Lactic Acid P (g/L): 37.49
Maltose M (g/L): 42.97
Volume V (L): 5.52

t = 14.00 h

Biomass X (g/L): 59.04
Glucose S (g/L): 103.61
Lactic Acid P (g/L): 52.83
Maltose M (g/L): 39.42
Volume V (L): 6.10

t = 15.00 h

Biomass X (g/L): 62.33
Glucose S (g/L): 94.35
Lactic Acid P (g/L): 53.65
Maltose M (g/L): 47.66
Volume V (L): 6.07

t = 16.00 h

Biomass X (g/L): 72.67
Glucose S (g/L): 85.60
Lactic Acid P (g/L): 58.86
Maltose M (g/L): 40.66
Volume V (L): 5.41

t = 17.00 h

Biomass X (g/L): 67.84
Glucose S (g/L): 88.98
Lactic Acid P (g/L): 61.33
Maltose M (g/L): 38.62
Volume V (L): 5.66

t = 18.00 h

Biomass X (g/L): 52.63
Glucose S (g/L): 91.60
Lactic Acid P (g/L): 50.89
Maltose M (g/L): 43.05
Volume V (L): 6.10

t = 19.00 h

Biomass X (g/L): 83.46
Glucose S (g/L): 85.87
Lactic Acid P (g/L): 40.41
Maltose M (g/L): 38.03
Volume V (L): 6.08

t = 20.00 h

Biomass X (g/L): 53.77
Glucose S (g/L): 87.41
Lactic Acid P (g/L): 43.13
Maltose M (g/L): 42.83
Volume V (L): 5.94

t = 21.00 h

Biomass X (g/L): 71.52
Glucose S (g/L): 89.93
Lactic Acid P (g/L): 34.90
Maltose M (g/L): 36.23
Volume V (L): 5.67

t = 22.00 h

Biomass X (g/L): 89.47
Glucose S (g/L): 84.57
Lactic Acid P (g/L): 45.85
Maltose M (g/L): 41.46
Volume V (L): 5.85

t = 23.00 h

Biomass X (g/L): 75.08
Glucose S (g/L): 80.34
Lactic Acid P (g/L): 58.11
Maltose M (g/L): 38.06
Volume V (L): 5.95

t = 24.00 h

Biomass X (g/L): 72.92
Glucose S (g/L): 82.87
Lactic Acid P (g/L): 53.39
Maltose M (g/L): 40.35
Volume V (L): 6.22

t = 25.00 h

Biomass X (g/L): 89.90
Glucose S (g/L): 90.70
Lactic Acid P (g/L): 58.13
Maltose M (g/L): 41.47
Volume V (L): 6.68

t = 26.00 h

Biomass X (g/L): 73.44
Glucose S (g/L): 91.65
Lactic Acid P (g/L): 65.56
Maltose M (g/L): 33.68
Volume V (L): 6.04

t = 27.00 h

Biomass X (g/L): 70.58
Glucose S (g/L): 73.48
Lactic Acid P (g/L): 59.00
Maltose M (g/L): 39.21
Volume V (L): 5.99

t = 28.00 h

Biomass X (g/L): 86.47
Glucose S (g/L): 88.31
Lactic Acid P (g/L): 52.17
Maltose M (g/L): 32.29
Volume V (L): 6.30

t = 29.00 h

Biomass X (g/L): 82.56
Glucose S (g/L): 84.11
Lactic Acid P (g/L): 63.16
Maltose M (g/L): 32.83
Volume V (L): 6.67

t = 30.00 h

Biomass X (g/L): 72.48
Glucose S (g/L): 87.29
Lactic Acid P (g/L): 55.96
Maltose M (g/L): 33.09
Volume V (L): 6.01

t = 31.00 h

Biomass X (g/L): 92.00
Glucose S (g/L): 80.15
Lactic Acid P (g/L): 62.66
Maltose M (g/L): 35.95
Volume V (L): 6.89

t = 32.00 h

Biomass X (g/L): 90.75
Glucose S (g/L): 74.29
Lactic Acid P (g/L): 78.66
Maltose M (g/L): 34.48
Volume V (L): 6.47

t = 33.00 h

Biomass X (g/L): 82.87
Glucose S (g/L): 85.72
Lactic Acid P (g/L): 64.04
Maltose M (g/L): 33.67
Volume V (L): 6.36

t = 34.00 h

Biomass X (g/L): 83.93
Glucose S (g/L): 78.45
Lactic Acid P (g/L): 84.01
Maltose M (g/L): 34.12

Volume V (L): 6.16

t = 35.00 h

Biomass X (g/L): 91.95

Glucose S (g/L): 76.43

Lactic Acid P (g/L): 81.39

Maltose M (g/L): 34.90

Volume V (L): 6.87

t = 36.00 h

Biomass X (g/L): 99.50

Glucose S (g/L): 76.93

Lactic Acid P (g/L): 62.90

Maltose M (g/L): 34.79

Volume V (L): 6.66

t = 37.00 h

Biomass X (g/L): 84.87

Glucose S (g/L): 86.39

Lactic Acid P (g/L): 58.40

Maltose M (g/L): 35.53

Volume V (L): 6.81

t = 38.00 h

Biomass X (g/L): 94.54

Glucose S (g/L): 84.37

Lactic Acid P (g/L): 68.94

Maltose M (g/L): 37.01

Volume V (L): 6.55

t = 39.00 h

Biomass X (g/L): 93.81

Glucose S (g/L): 87.71

Lactic Acid P (g/L): 80.84

Maltose M (g/L): 32.22

Volume V (L): 6.76

t = 40.00 h

Biomass X (g/L): 97.12

Glucose S (g/L): 87.42

Lactic Acid P (g/L): 65.73

Maltose M (g/L): 30.72

Volume V (L): 6.76

t = 41.00 h

Biomass X (g/L): 88.76

Glucose S (g/L): 82.42

Lactic Acid P (g/L): 79.41

Maltose M (g/L): 33.22

Volume V (L): 6.92

t = 42.00 h

Biomass X (g/L): 79.24

Glucose S (g/L): 84.40

Lactic Acid P (g/L): 82.68

Maltose M (g/L): 28.65

Volume V (L): 6.81

t = 43.00 h

Biomass X (g/L): 83.84

Glucose S (g/L): 82.57

Lactic Acid P (g/L): 64.29

Maltose M (g/L): 29.52

Volume V (L): 6.61

t = 44.00 h

Biomass X (g/L): 92.01

Glucose S (g/L): 84.33

Lactic Acid P (g/L): 65.60

Maltose M (g/L): 33.36

Volume V (L): 7.87

t = 45.00 h

Biomass X (g/L): 106.92

Glucose S (g/L): 80.99

Lactic Acid P (g/L): 79.12

Maltose M (g/L): 30.36

Volume V (L): 6.75

t = 46.00 h

Biomass X (g/L): 83.69

Glucose S (g/L): 87.88

Lactic Acid P (g/L): 74.16

Maltose M (g/L): 27.45

Volume V (L): 7.24

t = 47.00 h

Biomass X (g/L): 96.67

Glucose S (g/L): 91.53

Lactic Acid P (g/L): 66.55

Maltose M (g/L): 26.51

Volume V (L): 6.73

t = 48.00 h

Biomass X (g/L): 85.21

Glucose S (g/L): 82.73

Lactic Acid P (g/L): 70.04

Maltose M (g/L): 30.03

Volume V (L): 6.71

t = 49.00 h

Biomass X (g/L): 89.59

Glucose S (g/L): 93.22

Lactic Acid P (g/L): 73.01

Maltose M (g/L): 32.47

Volume V (L): 7.36

t = 50.00 h

Biomass X (g/L): 103.70

Glucose S (g/L): 87.93

Lactic Acid P (g/L): 59.15

Maltose M (g/L): 32.20

Volume V (L): 7.43

t = 51.00 h

Biomass X (g/L): 96.05

Glucose S (g/L): 88.91

Lactic Acid P (g/L): 81.52

Maltose M (g/L): 29.47

Volume V (L): 7.79

t = 52.00 h

Biomass X (g/L): 75.72

Glucose S (g/L): 79.89
Lactic Acid P (g/L): 70.01
Maltose M (g/L): 33.89
Volume V (L): 7.59

t = 53.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 79.74
Lactic Acid P (g/L): 48.43
Maltose M (g/L): 28.49
Volume V (L): 7.48

t = 54.00 h

Biomass X (g/L): 100.72
Glucose S (g/L): 89.54
Lactic Acid P (g/L): 74.84
Maltose M (g/L): 28.19
Volume V (L): 7.30

t = 55.00 h

Biomass X (g/L): 89.36
Glucose S (g/L): 93.03
Lactic Acid P (g/L): 68.28
Maltose M (g/L): 22.03
Volume V (L): 8.05

t = 56.00 h

Biomass X (g/L): 76.20
Glucose S (g/L): 86.34
Lactic Acid P (g/L): 91.55
Maltose M (g/L): 27.00
Volume V (L): 7.93

t = 57.00 h

Biomass X (g/L): 79.40
Glucose S (g/L): 98.03
Lactic Acid P (g/L): 70.23
Maltose M (g/L): 29.97
Volume V (L): 7.36

t = 58.00 h

Biomass X (g/L): 94.62
Glucose S (g/L): 93.89
Lactic Acid P (g/L): 65.60
Maltose M (g/L): 31.50
Volume V (L): 7.77

t = 59.00 h

Biomass X (g/L): 97.41
Glucose S (g/L): 86.33
Lactic Acid P (g/L): 81.03
Maltose M (g/L): 25.44
Volume V (L): 7.85

t = 60.00 h

Biomass X (g/L): 87.40
Glucose S (g/L): 91.24
Lactic Acid P (g/L): 70.17
Maltose M (g/L): 28.40
Volume V (L): 7.76

t = 61.00 h

Biomass X (g/L): 112.03
Glucose S (g/L): 103.44
Lactic Acid P (g/L): 69.25
Maltose M (g/L): 29.63
Volume V (L): 8.12

t = 62.00 h

Biomass X (g/L): 95.20
Glucose S (g/L): 85.85
Lactic Acid P (g/L): 81.66
Maltose M (g/L): 26.90
Volume V (L): 8.19

t = 63.00 h

Biomass X (g/L): 84.26
Glucose S (g/L): 94.78
Lactic Acid P (g/L): 67.16
Maltose M (g/L): 21.10
Volume V (L): 7.94

t = 64.00 h

Biomass X (g/L): 78.30
Glucose S (g/L): 91.05
Lactic Acid P (g/L): 63.40
Maltose M (g/L): 27.10
Volume V (L): 7.97

t = 65.00 h

Biomass X (g/L): 76.52
Glucose S (g/L): 96.40
Lactic Acid P (g/L): 93.74
Maltose M (g/L): 28.17
Volume V (L): 8.52

t = 66.00 h

Biomass X (g/L): 89.55
Glucose S (g/L): 99.48
Lactic Acid P (g/L): 83.06
Maltose M (g/L): 23.81
Volume V (L): 8.41

t = 67.00 h

Biomass X (g/L): 81.05
Glucose S (g/L): 91.86
Lactic Acid P (g/L): 75.23
Maltose M (g/L): 25.32
Volume V (L): 8.21

t = 68.00 h

Biomass X (g/L): 102.67
Glucose S (g/L): 99.51
Lactic Acid P (g/L): 77.87
Maltose M (g/L): 22.94
Volume V (L): 7.77

t = 69.00 h

Biomass X (g/L): 100.38
Glucose S (g/L): 105.20
Lactic Acid P (g/L): 82.86
Maltose M (g/L): 19.30
Volume V (L): 8.38

t = 70.00 h

Biomass X (g/L): 91.78
Glucose S (g/L): 95.16
Lactic Acid P (g/L): 87.44
Maltose M (g/L): 27.93
Volume V (L): 8.53

t = 71.00 h

Biomass X (g/L): 73.29
Glucose S (g/L): 99.00
Lactic Acid P (g/L): 71.21
Maltose M (g/L): 25.23
Volume V (L): 7.59

t = 72.00 h

Biomass X (g/L): 78.20
Glucose S (g/L): 92.73
Lactic Acid P (g/L): 81.99
Maltose M (g/L): 28.98
Volume V (L): 8.50

t = 73.00 h

Biomass X (g/L): 97.44
Glucose S (g/L): 96.03
Lactic Acid P (g/L): 73.87
Maltose M (g/L): 27.01
Volume V (L): 8.35

t = 74.00 h

Biomass X (g/L): 78.88
Glucose S (g/L): 95.45
Lactic Acid P (g/L): 77.02
Maltose M (g/L): 23.88
Volume V (L): 8.46

t = 75.00 h

Biomass X (g/L): 95.68
Glucose S (g/L): 100.51
Lactic Acid P (g/L): 80.48
Maltose M (g/L): 24.38

Volume V (L): 8.31

Batch 44

Initial Conditions:

Batch: 44.00

X0 (g/L): 2.24

S0 (g/L): 82.94

M0 (g/L): 34.13

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.41

F_vol (L/h): 0.05

S_feed (g/L): 212.54

M_feed (g/L): 87.46

V_final (L): 8.54

Batch 44 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.74
Glucose S (g/L): 75.11
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 34.79
Volume V (L): 5.46

t = 1.00 h

Biomass X (g/L): 17.05
Glucose S (g/L): 83.17
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 35.72
Volume V (L): 5.30

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 78.79
Lactic Acid P (g/L): 3.32
Maltose M (g/L): 34.19
Volume V (L): 5.48

t = 3.00 h

Biomass X (g/L): 15.18
Glucose S (g/L): 84.53
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 34.19
Volume V (L): 4.99

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 85.34
Lactic Acid P (g/L): 9.05
Maltose M (g/L): 34.94
Volume V (L): 5.31

t = 5.00 h

Biomass X (g/L): 17.07
Glucose S (g/L): 89.32

Lactic Acid P (g/L): 12.00
Maltose M (g/L): 32.67
Volume V (L): 5.29

t = 6.00 h

Biomass X (g/L): 12.15
Glucose S (g/L): 77.55
Lactic Acid P (g/L): 12.25
Maltose M (g/L): 31.62
Volume V (L): 5.60

t = 7.00 h

Biomass X (g/L): 18.88
Glucose S (g/L): 85.94
Lactic Acid P (g/L): 19.16
Maltose M (g/L): 33.21
Volume V (L): 5.66

t = 8.00 h

Biomass X (g/L): 19.07
Glucose S (g/L): 80.59
Lactic Acid P (g/L): 3.78
Maltose M (g/L): 31.37
Volume V (L): 5.43

t = 9.00 h

Biomass X (g/L): 29.52
Glucose S (g/L): 81.95
Lactic Acid P (g/L): 23.95
Maltose M (g/L): 35.24
Volume V (L): 4.99

t = 10.00 h

Biomass X (g/L): 1.54
Glucose S (g/L): 73.17
Lactic Acid P (g/L): 11.90
Maltose M (g/L): 34.04
Volume V (L): 5.98

t = 11.00 h

Biomass X (g/L): 17.58

Glucose S (g/L): 66.50
Lactic Acid P (g/L): 12.92
Maltose M (g/L): 34.62
Volume V (L): 5.68

t = 12.00 h

Biomass X (g/L): 40.58
Glucose S (g/L): 63.93
Lactic Acid P (g/L): 30.04
Maltose M (g/L): 32.28
Volume V (L): 5.39

t = 13.00 h

Biomass X (g/L): 20.20
Glucose S (g/L): 60.63
Lactic Acid P (g/L): 39.73
Maltose M (g/L): 32.63
Volume V (L): 5.73

t = 14.00 h

Biomass X (g/L): 39.66
Glucose S (g/L): 65.03
Lactic Acid P (g/L): 36.10
Maltose M (g/L): 31.42
Volume V (L): 5.23

t = 15.00 h

Biomass X (g/L): 48.42
Glucose S (g/L): 66.46
Lactic Acid P (g/L): 34.75
Maltose M (g/L): 32.66
Volume V (L): 6.01

t = 16.00 h

Biomass X (g/L): 44.32
Glucose S (g/L): 55.64
Lactic Acid P (g/L): 40.29
Maltose M (g/L): 30.32
Volume V (L): 5.34

t = 17.00 h

Biomass X (g/L): 49.09
Glucose S (g/L): 58.34
Lactic Acid P (g/L): 41.32
Maltose M (g/L): 28.79
Volume V (L): 5.74

t = 18.00 h

Biomass X (g/L): 61.89
Glucose S (g/L): 47.55
Lactic Acid P (g/L): 31.96
Maltose M (g/L): 30.45
Volume V (L): 5.92

t = 19.00 h

Biomass X (g/L): 72.15
Glucose S (g/L): 52.82
Lactic Acid P (g/L): 33.14
Maltose M (g/L): 29.61
Volume V (L): 6.19

t = 20.00 h

Biomass X (g/L): 54.88
Glucose S (g/L): 53.33
Lactic Acid P (g/L): 46.12
Maltose M (g/L): 32.50
Volume V (L): 5.97

t = 21.00 h

Biomass X (g/L): 61.06
Glucose S (g/L): 43.35
Lactic Acid P (g/L): 49.53
Maltose M (g/L): 32.70
Volume V (L): 6.17

t = 22.00 h

Biomass X (g/L): 61.57
Glucose S (g/L): 47.36
Lactic Acid P (g/L): 50.67
Maltose M (g/L): 29.20
Volume V (L): 6.03

t = 23.00 h

Biomass X (g/L): 46.43
Glucose S (g/L): 50.20
Lactic Acid P (g/L): 58.52
Maltose M (g/L): 31.85
Volume V (L): 6.08

t = 24.00 h

Biomass X (g/L): 73.94
Glucose S (g/L): 34.44
Lactic Acid P (g/L): 51.70
Maltose M (g/L): 31.35
Volume V (L): 6.16

t = 25.00 h

Biomass X (g/L): 83.46
Glucose S (g/L): 51.32
Lactic Acid P (g/L): 60.38
Maltose M (g/L): 28.58
Volume V (L): 6.13

t = 26.00 h

Biomass X (g/L): 85.66
Glucose S (g/L): 46.77
Lactic Acid P (g/L): 55.06
Maltose M (g/L): 29.65
Volume V (L): 5.75

t = 27.00 h

Biomass X (g/L): 67.89
Glucose S (g/L): 47.85
Lactic Acid P (g/L): 74.68
Maltose M (g/L): 29.31
Volume V (L): 6.06

t = 28.00 h

Biomass X (g/L): 65.17
Glucose S (g/L): 42.91
Lactic Acid P (g/L): 59.50
Maltose M (g/L): 30.54
Volume V (L): 6.39

t = 29.00 h

Biomass X (g/L): 79.00
Glucose S (g/L): 37.31
Lactic Acid P (g/L): 81.69
Maltose M (g/L): 28.34
Volume V (L): 6.30

t = 30.00 h

Biomass X (g/L): 88.04
Glucose S (g/L): 46.43
Lactic Acid P (g/L): 64.06
Maltose M (g/L): 29.37
Volume V (L): 6.37

t = 31.00 h

Biomass X (g/L): 82.06
Glucose S (g/L): 43.82
Lactic Acid P (g/L): 59.69
Maltose M (g/L): 28.61
Volume V (L): 6.37

t = 32.00 h

Biomass X (g/L): 84.15
Glucose S (g/L): 44.28
Lactic Acid P (g/L): 54.13
Maltose M (g/L): 29.67
Volume V (L): 6.26

t = 33.00 h

Biomass X (g/L): 93.39
Glucose S (g/L): 48.65
Lactic Acid P (g/L): 49.85
Maltose M (g/L): 28.81
Volume V (L): 6.42

t = 34.00 h

Biomass X (g/L): 75.04
Glucose S (g/L): 47.41
Lactic Acid P (g/L): 54.84
Maltose M (g/L): 28.33

Volume V (L): 6.52

t = 35.00 h

Biomass X (g/L): 80.79

Glucose S (g/L): 52.31

Lactic Acid P (g/L): 56.60

Maltose M (g/L): 28.53

Volume V (L): 6.65

t = 36.00 h

Biomass X (g/L): 99.94

Glucose S (g/L): 45.23

Lactic Acid P (g/L): 69.20

Maltose M (g/L): 30.72

Volume V (L): 6.81

t = 37.00 h

Biomass X (g/L): 81.92

Glucose S (g/L): 41.97

Lactic Acid P (g/L): 61.22

Maltose M (g/L): 28.03

Volume V (L): 6.69

t = 38.00 h

Biomass X (g/L): 71.51

Glucose S (g/L): 44.28

Lactic Acid P (g/L): 68.50

Maltose M (g/L): 27.24

Volume V (L): 6.50

t = 39.00 h

Biomass X (g/L): 77.86

Glucose S (g/L): 47.04

Lactic Acid P (g/L): 79.32

Maltose M (g/L): 28.79

Volume V (L): 6.44

t = 40.00 h

Biomass X (g/L): 75.44

Glucose S (g/L): 49.80

Lactic Acid P (g/L): 67.05

Maltose M (g/L): 27.90

Volume V (L): 7.62

t = 41.00 h

Biomass X (g/L): 88.74

Glucose S (g/L): 39.93

Lactic Acid P (g/L): 87.85

Maltose M (g/L): 26.44

Volume V (L): 6.61

t = 42.00 h

Biomass X (g/L): 76.33

Glucose S (g/L): 46.34

Lactic Acid P (g/L): 57.89

Maltose M (g/L): 30.09

Volume V (L): 6.83

t = 43.00 h

Biomass X (g/L): 88.27

Glucose S (g/L): 53.27

Lactic Acid P (g/L): 67.96

Maltose M (g/L): 27.27

Volume V (L): 7.17

t = 44.00 h

Biomass X (g/L): 78.29

Glucose S (g/L): 46.06

Lactic Acid P (g/L): 64.77

Maltose M (g/L): 27.11

Volume V (L): 7.18

t = 45.00 h

Biomass X (g/L): 64.45

Glucose S (g/L): 51.26

Lactic Acid P (g/L): 79.01

Maltose M (g/L): 27.53

Volume V (L): 7.27

t = 46.00 h

Biomass X (g/L): 89.03

Glucose S (g/L): 49.87

Lactic Acid P (g/L): 49.71

Maltose M (g/L): 29.03

Volume V (L): 7.04

t = 47.00 h

Biomass X (g/L): 87.52

Glucose S (g/L): 52.19

Lactic Acid P (g/L): 57.56

Maltose M (g/L): 25.36

Volume V (L): 7.28

t = 48.00 h

Biomass X (g/L): 91.30

Glucose S (g/L): 45.38

Lactic Acid P (g/L): 70.43

Maltose M (g/L): 27.26

Volume V (L): 7.28

t = 49.00 h

Biomass X (g/L): 79.35

Glucose S (g/L): 45.84

Lactic Acid P (g/L): 86.31

Maltose M (g/L): 29.01

Volume V (L): 7.22

t = 50.00 h

Biomass X (g/L): 78.31

Glucose S (g/L): 49.68

Lactic Acid P (g/L): 74.26

Maltose M (g/L): 26.84

Volume V (L): 7.43

t = 51.00 h

Biomass X (g/L): 90.21

Glucose S (g/L): 43.76

Lactic Acid P (g/L): 59.84

Maltose M (g/L): 25.30

Volume V (L): 7.70

t = 52.00 h

Biomass X (g/L): 101.46

Glucose S (g/L): 49.41
Lactic Acid P (g/L): 61.61
Maltose M (g/L): 25.81
Volume V (L): 7.45

t = 53.00 h

Biomass X (g/L): 86.25
Glucose S (g/L): 52.02
Lactic Acid P (g/L): 88.78
Maltose M (g/L): 26.54
Volume V (L): 7.50

t = 54.00 h

Biomass X (g/L): 90.38
Glucose S (g/L): 58.78
Lactic Acid P (g/L): 67.61
Maltose M (g/L): 27.58
Volume V (L): 7.85

t = 55.00 h

Biomass X (g/L): 82.82
Glucose S (g/L): 53.59
Lactic Acid P (g/L): 64.89
Maltose M (g/L): 25.72
Volume V (L): 7.87

t = 56.00 h

Biomass X (g/L): 70.45
Glucose S (g/L): 54.98
Lactic Acid P (g/L): 87.82
Maltose M (g/L): 26.81
Volume V (L): 8.09

t = 57.00 h

Biomass X (g/L): 95.67
Glucose S (g/L): 57.68
Lactic Acid P (g/L): 83.73
Maltose M (g/L): 24.33
Volume V (L): 7.19

t = 58.00 h

Biomass X (g/L): 95.00
Glucose S (g/L): 49.43
Lactic Acid P (g/L): 86.45
Maltose M (g/L): 24.20
Volume V (L): 7.99

t = 59.00 h

Biomass X (g/L): 92.92
Glucose S (g/L): 53.20
Lactic Acid P (g/L): 65.89
Maltose M (g/L): 25.08
Volume V (L): 7.95

t = 60.00 h

Biomass X (g/L): 83.83
Glucose S (g/L): 64.33
Lactic Acid P (g/L): 60.76
Maltose M (g/L): 26.95
Volume V (L): 7.58

t = 61.00 h

Biomass X (g/L): 91.57
Glucose S (g/L): 54.54
Lactic Acid P (g/L): 76.51
Maltose M (g/L): 25.41
Volume V (L): 7.95

t = 62.00 h

Biomass X (g/L): 94.44
Glucose S (g/L): 59.94
Lactic Acid P (g/L): 78.63
Maltose M (g/L): 25.52
Volume V (L): 8.22

t = 63.00 h

Biomass X (g/L): 101.63
Glucose S (g/L): 59.81
Lactic Acid P (g/L): 79.13
Maltose M (g/L): 25.39
Volume V (L): 7.74

t = 64.00 h

Biomass X (g/L): 95.14
Glucose S (g/L): 50.28
Lactic Acid P (g/L): 89.76
Maltose M (g/L): 26.13
Volume V (L): 8.14

t = 65.00 h

Biomass X (g/L): 85.70
Glucose S (g/L): 58.12
Lactic Acid P (g/L): 59.93
Maltose M (g/L): 24.23
Volume V (L): 8.31

t = 66.00 h

Biomass X (g/L): 71.74
Glucose S (g/L): 57.94
Lactic Acid P (g/L): 84.39
Maltose M (g/L): 25.46
Volume V (L): 7.81

t = 67.00 h

Biomass X (g/L): 86.34
Glucose S (g/L): 59.19
Lactic Acid P (g/L): 66.29
Maltose M (g/L): 23.77
Volume V (L): 8.12

t = 68.00 h

Biomass X (g/L): 87.20
Glucose S (g/L): 55.23
Lactic Acid P (g/L): 84.75
Maltose M (g/L): 26.58
Volume V (L): 8.29

t = 69.00 h

Biomass X (g/L): 91.60
Glucose S (g/L): 54.23
Lactic Acid P (g/L): 72.49
Maltose M (g/L): 23.57
Volume V (L): 8.03

t = 70.00 h

Biomass X (g/L): 97.94
Glucose S (g/L): 57.35
Lactic Acid P (g/L): 69.50
Maltose M (g/L): 23.37
Volume V (L): 8.05

t = 71.00 h

Biomass X (g/L): 83.18
Glucose S (g/L): 59.40
Lactic Acid P (g/L): 71.58
Maltose M (g/L): 24.75
Volume V (L): 8.31

t = 72.00 h

Biomass X (g/L): 105.13
Glucose S (g/L): 65.15
Lactic Acid P (g/L): 63.94
Maltose M (g/L): 24.15
Volume V (L): 8.62

t = 73.00 h

Biomass X (g/L): 66.08
Glucose S (g/L): 68.23
Lactic Acid P (g/L): 78.01
Maltose M (g/L): 24.29
Volume V (L): 8.63

t = 74.00 h

Biomass X (g/L): 87.84
Glucose S (g/L): 66.99
Lactic Acid P (g/L): 70.57
Maltose M (g/L): 24.08
Volume V (L): 8.25

t = 75.00 h

Biomass X (g/L): 75.40
Glucose S (g/L): 65.27
Lactic Acid P (g/L): 85.31
Maltose M (g/L): 22.60

Volume V (L): 8.72

Batch 45

Initial Conditions:

Batch: 45.00

X0 (g/L): 4.92

S0 (g/L): 41.24

M0 (g/L): 91.77

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.23

F_vol (L/h): 0.04

S_feed (g/L): 93.01

M_feed (g/L): 206.99

V_final (L): 8.02

Batch 45 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 36.96
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 89.71
Volume V (L): 4.88

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 40.26
Lactic Acid P (g/L): 13.80
Maltose M (g/L): 91.12
Volume V (L): 5.11

t = 2.00 h

Biomass X (g/L): 5.01
Glucose S (g/L): 44.02
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 95.07
Volume V (L): 5.02

t = 3.00 h

Biomass X (g/L): 21.92
Glucose S (g/L): 37.88
Lactic Acid P (g/L): 2.07
Maltose M (g/L): 88.72
Volume V (L): 5.09

t = 4.00 h

Biomass X (g/L): 20.41
Glucose S (g/L): 46.97
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 88.14
Volume V (L): 5.08

t = 5.00 h

Biomass X (g/L): 17.54
Glucose S (g/L): 39.03

Lactic Acid P (g/L): 1.04
Maltose M (g/L): 94.34
Volume V (L): 5.31

t = 6.00 h

Biomass X (g/L): 17.43
Glucose S (g/L): 37.20
Lactic Acid P (g/L): 24.98
Maltose M (g/L): 90.19
Volume V (L): 5.26

t = 7.00 h

Biomass X (g/L): 21.88
Glucose S (g/L): 32.73
Lactic Acid P (g/L): 17.12
Maltose M (g/L): 88.30
Volume V (L): 5.38

t = 8.00 h

Biomass X (g/L): 31.07
Glucose S (g/L): 33.75
Lactic Acid P (g/L): 9.48
Maltose M (g/L): 92.73
Volume V (L): 5.32

t = 9.00 h

Biomass X (g/L): 24.10
Glucose S (g/L): 26.12
Lactic Acid P (g/L): 16.26
Maltose M (g/L): 86.50
Volume V (L): 4.97

t = 10.00 h

Biomass X (g/L): 35.37
Glucose S (g/L): 33.64
Lactic Acid P (g/L): 17.86
Maltose M (g/L): 76.51
Volume V (L): 5.41

t = 11.00 h

Biomass X (g/L): 40.78

Glucose S (g/L): 28.62
Lactic Acid P (g/L): 27.71
Maltose M (g/L): 86.35
Volume V (L): 5.46

t = 12.00 h

Biomass X (g/L): 33.69
Glucose S (g/L): 25.35
Lactic Acid P (g/L): 32.68
Maltose M (g/L): 76.58
Volume V (L): 5.54

t = 13.00 h

Biomass X (g/L): 42.23
Glucose S (g/L): 16.11
Lactic Acid P (g/L): 51.51
Maltose M (g/L): 83.55
Volume V (L): 5.38

t = 14.00 h

Biomass X (g/L): 57.57
Glucose S (g/L): 19.19
Lactic Acid P (g/L): 18.55
Maltose M (g/L): 74.76
Volume V (L): 5.27

t = 15.00 h

Biomass X (g/L): 56.68
Glucose S (g/L): 20.71
Lactic Acid P (g/L): 38.46
Maltose M (g/L): 78.77
Volume V (L): 5.75

t = 16.00 h

Biomass X (g/L): 56.30
Glucose S (g/L): 17.54
Lactic Acid P (g/L): 38.03
Maltose M (g/L): 78.50
Volume V (L): 5.58

t = 17.00 h

Biomass X (g/L): 52.91
Glucose S (g/L): 11.62
Lactic Acid P (g/L): 51.79
Maltose M (g/L): 76.34
Volume V (L): 5.72

t = 18.00 h

Biomass X (g/L): 79.14
Glucose S (g/L): 11.82
Lactic Acid P (g/L): 48.12
Maltose M (g/L): 77.29
Volume V (L): 5.66

t = 19.00 h

Biomass X (g/L): 57.78
Glucose S (g/L): 11.72
Lactic Acid P (g/L): 52.01
Maltose M (g/L): 81.84
Volume V (L): 5.76

t = 20.00 h

Biomass X (g/L): 70.09
Glucose S (g/L): 10.71
Lactic Acid P (g/L): 51.30
Maltose M (g/L): 74.69
Volume V (L): 5.70

t = 21.00 h

Biomass X (g/L): 64.56
Glucose S (g/L): 16.77
Lactic Acid P (g/L): 40.85
Maltose M (g/L): 78.81
Volume V (L): 5.61

t = 22.00 h

Biomass X (g/L): 71.09
Glucose S (g/L): 5.99
Lactic Acid P (g/L): 62.18
Maltose M (g/L): 72.56
Volume V (L): 6.00

t = 23.00 h

Biomass X (g/L): 76.49
Glucose S (g/L): 13.62
Lactic Acid P (g/L): 59.60
Maltose M (g/L): 74.88
Volume V (L): 5.78

t = 24.00 h

Biomass X (g/L): 60.04
Glucose S (g/L): 7.59
Lactic Acid P (g/L): 61.88
Maltose M (g/L): 76.23
Volume V (L): 6.00

t = 25.00 h

Biomass X (g/L): 74.32
Glucose S (g/L): 8.56
Lactic Acid P (g/L): 66.24
Maltose M (g/L): 70.73
Volume V (L): 6.10

t = 26.00 h

Biomass X (g/L): 86.91
Glucose S (g/L): 5.40
Lactic Acid P (g/L): 59.41
Maltose M (g/L): 68.99
Volume V (L): 5.90

t = 27.00 h

Biomass X (g/L): 93.35
Glucose S (g/L): 9.92
Lactic Acid P (g/L): 70.32
Maltose M (g/L): 75.20
Volume V (L): 5.80

t = 28.00 h

Biomass X (g/L): 75.94
Glucose S (g/L): 16.32
Lactic Acid P (g/L): 60.98
Maltose M (g/L): 77.02
Volume V (L): 6.71

t = 29.00 h

Biomass X (g/L): 78.39
Glucose S (g/L): 12.71
Lactic Acid P (g/L): 56.32
Maltose M (g/L): 71.99
Volume V (L): 6.04

t = 30.00 h

Biomass X (g/L): 100.08
Glucose S (g/L): 8.48
Lactic Acid P (g/L): 59.16
Maltose M (g/L): 70.05
Volume V (L): 6.36

t = 31.00 h

Biomass X (g/L): 90.35
Glucose S (g/L): 15.46
Lactic Acid P (g/L): 59.82
Maltose M (g/L): 71.83
Volume V (L): 6.71

t = 32.00 h

Biomass X (g/L): 74.04
Glucose S (g/L): 16.40
Lactic Acid P (g/L): 55.98
Maltose M (g/L): 70.95
Volume V (L): 6.48

t = 33.00 h

Biomass X (g/L): 74.60
Glucose S (g/L): 13.28
Lactic Acid P (g/L): 76.78
Maltose M (g/L): 71.15
Volume V (L): 6.25

t = 34.00 h

Biomass X (g/L): 90.34
Glucose S (g/L): 19.48
Lactic Acid P (g/L): 65.05
Maltose M (g/L): 60.29

Volume V (L): 6.35

t = 35.00 h

Biomass X (g/L): 71.82

Glucose S (g/L): 16.15

Lactic Acid P (g/L): 63.86

Maltose M (g/L): 65.26

Volume V (L): 6.21

t = 36.00 h

Biomass X (g/L): 92.54

Glucose S (g/L): 16.11

Lactic Acid P (g/L): 67.26

Maltose M (g/L): 65.62

Volume V (L): 6.66

t = 37.00 h

Biomass X (g/L): 85.67

Glucose S (g/L): 11.99

Lactic Acid P (g/L): 92.05

Maltose M (g/L): 68.88

Volume V (L): 6.62

t = 38.00 h

Biomass X (g/L): 87.81

Glucose S (g/L): 16.75

Lactic Acid P (g/L): 54.34

Maltose M (g/L): 68.55

Volume V (L): 6.40

t = 39.00 h

Biomass X (g/L): 80.67

Glucose S (g/L): 15.93

Lactic Acid P (g/L): 57.60

Maltose M (g/L): 69.78

Volume V (L): 6.32

t = 40.00 h

Biomass X (g/L): 77.38

Glucose S (g/L): 17.30

Lactic Acid P (g/L): 53.18

Maltose M (g/L): 67.91

Volume V (L): 6.29

t = 41.00 h

Biomass X (g/L): 97.87

Glucose S (g/L): 14.10

Lactic Acid P (g/L): 64.43

Maltose M (g/L): 61.63

Volume V (L): 6.89

t = 42.00 h

Biomass X (g/L): 81.62

Glucose S (g/L): 17.03

Lactic Acid P (g/L): 69.41

Maltose M (g/L): 71.48

Volume V (L): 6.72

t = 43.00 h

Biomass X (g/L): 80.26

Glucose S (g/L): 14.64

Lactic Acid P (g/L): 63.29

Maltose M (g/L): 70.39

Volume V (L): 6.78

t = 44.00 h

Biomass X (g/L): 83.66

Glucose S (g/L): 11.09

Lactic Acid P (g/L): 63.09

Maltose M (g/L): 60.49

Volume V (L): 6.57

t = 45.00 h

Biomass X (g/L): 86.48

Glucose S (g/L): 21.14

Lactic Acid P (g/L): 69.92

Maltose M (g/L): 66.02

Volume V (L): 6.46

t = 46.00 h

Biomass X (g/L): 74.82

Glucose S (g/L): 14.05

Lactic Acid P (g/L): 64.37

Maltose M (g/L): 62.13

Volume V (L): 7.08

t = 47.00 h

Biomass X (g/L): 79.46

Glucose S (g/L): 19.46

Lactic Acid P (g/L): 72.38

Maltose M (g/L): 64.47

Volume V (L): 6.90

t = 48.00 h

Biomass X (g/L): 79.07

Glucose S (g/L): 20.66

Lactic Acid P (g/L): 80.30

Maltose M (g/L): 67.49

Volume V (L): 7.15

t = 49.00 h

Biomass X (g/L): 93.31

Glucose S (g/L): 21.17

Lactic Acid P (g/L): 76.64

Maltose M (g/L): 71.12

Volume V (L): 6.88

t = 50.00 h

Biomass X (g/L): 76.67

Glucose S (g/L): 19.85

Lactic Acid P (g/L): 78.94

Maltose M (g/L): 65.56

Volume V (L): 6.95

t = 51.00 h

Biomass X (g/L): 68.33

Glucose S (g/L): 22.07

Lactic Acid P (g/L): 81.24

Maltose M (g/L): 63.47

Volume V (L): 7.22

t = 52.00 h

Biomass X (g/L): 66.35

Glucose S (g/L): 23.21
Lactic Acid P (g/L): 74.13
Maltose M (g/L): 60.71
Volume V (L): 7.27

t = 53.00 h

Biomass X (g/L): 71.18
Glucose S (g/L): 24.56
Lactic Acid P (g/L): 70.38
Maltose M (g/L): 62.28
Volume V (L): 7.32

t = 54.00 h

Biomass X (g/L): 98.57
Glucose S (g/L): 28.14
Lactic Acid P (g/L): 65.01
Maltose M (g/L): 66.32
Volume V (L): 7.10

t = 55.00 h

Biomass X (g/L): 85.58
Glucose S (g/L): 24.13
Lactic Acid P (g/L): 68.90
Maltose M (g/L): 60.69
Volume V (L): 7.36

t = 56.00 h

Biomass X (g/L): 90.27
Glucose S (g/L): 24.17
Lactic Acid P (g/L): 68.00
Maltose M (g/L): 57.17
Volume V (L): 7.52

t = 57.00 h

Biomass X (g/L): 90.59
Glucose S (g/L): 21.77
Lactic Acid P (g/L): 80.02
Maltose M (g/L): 54.06
Volume V (L): 7.27

t = 58.00 h

Biomass X (g/L): 85.93
Glucose S (g/L): 26.41
Lactic Acid P (g/L): 76.74
Maltose M (g/L): 61.69
Volume V (L): 7.21

t = 59.00 h

Biomass X (g/L): 84.63
Glucose S (g/L): 27.67
Lactic Acid P (g/L): 80.31
Maltose M (g/L): 60.35
Volume V (L): 7.70

t = 60.00 h

Biomass X (g/L): 95.40
Glucose S (g/L): 25.61
Lactic Acid P (g/L): 63.29
Maltose M (g/L): 56.65
Volume V (L): 7.25

t = 61.00 h

Biomass X (g/L): 86.66
Glucose S (g/L): 28.41
Lactic Acid P (g/L): 87.74
Maltose M (g/L): 62.57
Volume V (L): 7.60

t = 62.00 h

Biomass X (g/L): 98.70
Glucose S (g/L): 26.42
Lactic Acid P (g/L): 70.69
Maltose M (g/L): 56.81
Volume V (L): 7.48

t = 63.00 h

Biomass X (g/L): 91.11
Glucose S (g/L): 29.91
Lactic Acid P (g/L): 64.69
Maltose M (g/L): 57.55
Volume V (L): 7.09

t = 64.00 h

Biomass X (g/L): 79.07
Glucose S (g/L): 31.20
Lactic Acid P (g/L): 71.11
Maltose M (g/L): 57.66
Volume V (L): 7.85

t = 65.00 h

Biomass X (g/L): 80.15
Glucose S (g/L): 31.72
Lactic Acid P (g/L): 98.61
Maltose M (g/L): 56.08
Volume V (L): 6.95

t = 66.00 h

Biomass X (g/L): 77.83
Glucose S (g/L): 29.90
Lactic Acid P (g/L): 72.09
Maltose M (g/L): 57.96
Volume V (L): 7.17

t = 67.00 h

Biomass X (g/L): 96.51
Glucose S (g/L): 26.94
Lactic Acid P (g/L): 72.92
Maltose M (g/L): 61.07
Volume V (L): 7.82

t = 68.00 h

Biomass X (g/L): 80.73
Glucose S (g/L): 28.79
Lactic Acid P (g/L): 74.39
Maltose M (g/L): 52.16
Volume V (L): 7.86

t = 69.00 h

Biomass X (g/L): 76.82
Glucose S (g/L): 33.02
Lactic Acid P (g/L): 75.03
Maltose M (g/L): 57.96
Volume V (L): 7.62

t = 70.00 h

Biomass X (g/L): 87.50
Glucose S (g/L): 39.20
Lactic Acid P (g/L): 69.35
Maltose M (g/L): 56.46
Volume V (L): 7.85

t = 71.00 h

Biomass X (g/L): 83.35
Glucose S (g/L): 31.28
Lactic Acid P (g/L): 78.19
Maltose M (g/L): 56.09
Volume V (L): 7.24

t = 72.00 h

Biomass X (g/L): 87.00
Glucose S (g/L): 34.24
Lactic Acid P (g/L): 87.09
Maltose M (g/L): 58.48
Volume V (L): 8.07

t = 73.00 h

Biomass X (g/L): 102.27
Glucose S (g/L): 31.72
Lactic Acid P (g/L): 80.21
Maltose M (g/L): 55.10
Volume V (L): 7.99

t = 74.00 h

Biomass X (g/L): 92.70
Glucose S (g/L): 30.99
Lactic Acid P (g/L): 67.12
Maltose M (g/L): 56.42
Volume V (L): 7.96

t = 75.00 h

Biomass X (g/L): 83.99
Glucose S (g/L): 32.44
Lactic Acid P (g/L): 74.47
Maltose M (g/L): 57.01

Volume V (L): 8.02

Batch 46

Initial Conditions:

Batch: 46.00

X0 (g/L): 6.77

S0 (g/L): 97.75

M0 (g/L): 41.36

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.42

F_vol (L/h): 0.08

S_feed (g/L): 210.80

M_feed (g/L): 89.20

V_final (L): 10.86

Batch 46 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 11.48
Glucose S (g/L): 96.90
Lactic Acid P (g/L): 5.52
Maltose M (g/L): 44.45
Volume V (L): 4.86

t = 1.00 h

Biomass X (g/L): 8.82
Glucose S (g/L): 89.70
Lactic Acid P (g/L): 4.85
Maltose M (g/L): 43.47
Volume V (L): 5.39

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 101.39
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 41.38
Volume V (L): 5.09

t = 3.00 h

Biomass X (g/L): 36.49
Glucose S (g/L): 95.12
Lactic Acid P (g/L): 12.51
Maltose M (g/L): 40.45
Volume V (L): 5.27

t = 4.00 h

Biomass X (g/L): 31.07
Glucose S (g/L): 88.14
Lactic Acid P (g/L): 1.56
Maltose M (g/L): 41.20
Volume V (L): 4.13

t = 5.00 h

Biomass X (g/L): 11.89
Glucose S (g/L): 90.92

Lactic Acid P (g/L): 12.73

Maltose M (g/L): 41.26

Volume V (L): 5.40

t = 6.00 h

Biomass X (g/L): 25.70

Glucose S (g/L): 90.99

Lactic Acid P (g/L): 22.43

Maltose M (g/L): 39.43

Volume V (L): 4.61

t = 7.00 h

Biomass X (g/L): 31.72

Glucose S (g/L): 87.66

Lactic Acid P (g/L): 24.44

Maltose M (g/L): 40.86

Volume V (L): 4.97

t = 8.00 h

Biomass X (g/L): 49.21

Glucose S (g/L): 89.34

Lactic Acid P (g/L): 17.85

Maltose M (g/L): 42.19

Volume V (L): 5.00

t = 9.00 h

Biomass X (g/L): 23.07

Glucose S (g/L): 77.59

Lactic Acid P (g/L): 38.39

Maltose M (g/L): 38.45

Volume V (L): 5.21

t = 10.00 h

Biomass X (g/L): 43.41

Glucose S (g/L): 77.91

Lactic Acid P (g/L): 31.78

Maltose M (g/L): 41.37

Volume V (L): 5.52

t = 11.00 h

Biomass X (g/L): 35.53

Glucose S (g/L): 72.97
Lactic Acid P (g/L): 40.92
Maltose M (g/L): 41.88
Volume V (L): 5.63

t = 12.00 h

Biomass X (g/L): 53.43
Glucose S (g/L): 67.99
Lactic Acid P (g/L): 51.01
Maltose M (g/L): 39.50
Volume V (L): 6.25

t = 13.00 h

Biomass X (g/L): 55.09
Glucose S (g/L): 74.23
Lactic Acid P (g/L): 50.03
Maltose M (g/L): 41.98
Volume V (L): 6.12

t = 14.00 h

Biomass X (g/L): 63.47
Glucose S (g/L): 71.23
Lactic Acid P (g/L): 52.55
Maltose M (g/L): 39.08
Volume V (L): 5.20

t = 15.00 h

Biomass X (g/L): 59.72
Glucose S (g/L): 69.29
Lactic Acid P (g/L): 46.45
Maltose M (g/L): 40.24
Volume V (L): 6.21

t = 16.00 h

Biomass X (g/L): 60.81
Glucose S (g/L): 68.68
Lactic Acid P (g/L): 53.97
Maltose M (g/L): 39.23
Volume V (L): 6.40

t = 17.00 h

Biomass X (g/L): 76.11
Glucose S (g/L): 64.08
Lactic Acid P (g/L): 50.63
Maltose M (g/L): 39.08
Volume V (L): 6.17

t = 18.00 h

Biomass X (g/L): 76.51
Glucose S (g/L): 62.32
Lactic Acid P (g/L): 39.49
Maltose M (g/L): 38.57
Volume V (L): 6.38

t = 19.00 h

Biomass X (g/L): 79.19
Glucose S (g/L): 65.29
Lactic Acid P (g/L): 54.75
Maltose M (g/L): 39.93
Volume V (L): 6.57

t = 20.00 h

Biomass X (g/L): 73.80
Glucose S (g/L): 68.68
Lactic Acid P (g/L): 55.79
Maltose M (g/L): 39.95
Volume V (L): 6.92

t = 21.00 h

Biomass X (g/L): 71.42
Glucose S (g/L): 69.94
Lactic Acid P (g/L): 60.08
Maltose M (g/L): 38.73
Volume V (L): 5.90

t = 22.00 h

Biomass X (g/L): 76.97
Glucose S (g/L): 67.11
Lactic Acid P (g/L): 54.87
Maltose M (g/L): 37.07
Volume V (L): 6.60

t = 23.00 h

Biomass X (g/L): 83.99
Glucose S (g/L): 64.53
Lactic Acid P (g/L): 50.27
Maltose M (g/L): 38.95
Volume V (L): 7.50

t = 24.00 h

Biomass X (g/L): 59.40
Glucose S (g/L): 65.41
Lactic Acid P (g/L): 64.14
Maltose M (g/L): 38.81
Volume V (L): 7.54

t = 25.00 h

Biomass X (g/L): 84.80
Glucose S (g/L): 64.70
Lactic Acid P (g/L): 47.29
Maltose M (g/L): 37.90
Volume V (L): 7.35

t = 26.00 h

Biomass X (g/L): 84.76
Glucose S (g/L): 61.45
Lactic Acid P (g/L): 73.26
Maltose M (g/L): 40.14
Volume V (L): 7.28

t = 27.00 h

Biomass X (g/L): 83.07
Glucose S (g/L): 65.24
Lactic Acid P (g/L): 58.80
Maltose M (g/L): 36.00
Volume V (L): 7.89

t = 28.00 h

Biomass X (g/L): 81.65
Glucose S (g/L): 61.02
Lactic Acid P (g/L): 63.90
Maltose M (g/L): 35.07
Volume V (L): 8.09

t = 29.00 h

Biomass X (g/L): 59.11
Glucose S (g/L): 65.44
Lactic Acid P (g/L): 67.45
Maltose M (g/L): 36.62
Volume V (L): 7.21

t = 30.00 h

Biomass X (g/L): 79.87
Glucose S (g/L): 71.74
Lactic Acid P (g/L): 68.42
Maltose M (g/L): 38.12
Volume V (L): 6.89

t = 31.00 h

Biomass X (g/L): 101.55
Glucose S (g/L): 69.47
Lactic Acid P (g/L): 50.98
Maltose M (g/L): 35.90
Volume V (L): 7.30

t = 32.00 h

Biomass X (g/L): 84.94
Glucose S (g/L): 64.11
Lactic Acid P (g/L): 71.77
Maltose M (g/L): 35.82
Volume V (L): 8.95

t = 33.00 h

Biomass X (g/L): 82.19
Glucose S (g/L): 67.46
Lactic Acid P (g/L): 67.62
Maltose M (g/L): 37.88
Volume V (L): 6.84

t = 34.00 h

Biomass X (g/L): 85.54
Glucose S (g/L): 65.43
Lactic Acid P (g/L): 59.67
Maltose M (g/L): 37.14

Volume V (L): 7.44

t = 35.00 h

Biomass X (g/L): 88.99

Glucose S (g/L): 68.56

Lactic Acid P (g/L): 59.14

Maltose M (g/L): 36.87

Volume V (L): 7.07

t = 36.00 h

Biomass X (g/L): 92.07

Glucose S (g/L): 72.81

Lactic Acid P (g/L): 63.37

Maltose M (g/L): 37.16

Volume V (L): 7.87

t = 37.00 h

Biomass X (g/L): 83.75

Glucose S (g/L): 72.09

Lactic Acid P (g/L): 55.09

Maltose M (g/L): 35.59

Volume V (L): 7.33

t = 38.00 h

Biomass X (g/L): 90.11

Glucose S (g/L): 78.80

Lactic Acid P (g/L): 82.84

Maltose M (g/L): 35.47

Volume V (L): 7.92

t = 39.00 h

Biomass X (g/L): 102.16

Glucose S (g/L): 75.07

Lactic Acid P (g/L): 81.11

Maltose M (g/L): 36.67

Volume V (L): 7.13

t = 40.00 h

Biomass X (g/L): 95.83

Glucose S (g/L): 72.37

Lactic Acid P (g/L): 72.14

Maltose M (g/L): 34.00

Volume V (L): 8.42

t = 41.00 h

Biomass X (g/L): 108.61

Glucose S (g/L): 74.57

Lactic Acid P (g/L): 66.41

Maltose M (g/L): 32.90

Volume V (L): 8.78

t = 42.00 h

Biomass X (g/L): 80.03

Glucose S (g/L): 72.24

Lactic Acid P (g/L): 66.99

Maltose M (g/L): 34.70

Volume V (L): 8.10

t = 43.00 h

Biomass X (g/L): 102.31

Glucose S (g/L): 69.77

Lactic Acid P (g/L): 75.97

Maltose M (g/L): 36.11

Volume V (L): 8.30

t = 44.00 h

Biomass X (g/L): 78.75

Glucose S (g/L): 78.53

Lactic Acid P (g/L): 85.33

Maltose M (g/L): 35.15

Volume V (L): 8.36

t = 45.00 h

Biomass X (g/L): 89.65

Glucose S (g/L): 78.16

Lactic Acid P (g/L): 64.58

Maltose M (g/L): 35.68

Volume V (L): 8.60

t = 46.00 h

Biomass X (g/L): 93.23

Glucose S (g/L): 74.61

Lactic Acid P (g/L): 59.93

Maltose M (g/L): 32.18

Volume V (L): 8.58

t = 47.00 h

Biomass X (g/L): 81.31

Glucose S (g/L): 76.96

Lactic Acid P (g/L): 84.31

Maltose M (g/L): 34.70

Volume V (L): 9.03

t = 48.00 h

Biomass X (g/L): 86.95

Glucose S (g/L): 80.82

Lactic Acid P (g/L): 54.42

Maltose M (g/L): 34.92

Volume V (L): 9.33

t = 49.00 h

Biomass X (g/L): 94.73

Glucose S (g/L): 80.13

Lactic Acid P (g/L): 61.72

Maltose M (g/L): 32.43

Volume V (L): 8.45

t = 50.00 h

Biomass X (g/L): 79.17

Glucose S (g/L): 73.04

Lactic Acid P (g/L): 68.86

Maltose M (g/L): 34.82

Volume V (L): 8.76

t = 51.00 h

Biomass X (g/L): 101.79

Glucose S (g/L): 79.37

Lactic Acid P (g/L): 57.23

Maltose M (g/L): 33.52

Volume V (L): 9.41

t = 52.00 h

Biomass X (g/L): 72.53

Glucose S (g/L): 78.61
Lactic Acid P (g/L): 81.63
Maltose M (g/L): 37.04
Volume V (L): 8.40

t = 53.00 h

Biomass X (g/L): 104.15
Glucose S (g/L): 83.51
Lactic Acid P (g/L): 79.54
Maltose M (g/L): 30.79
Volume V (L): 8.22

t = 54.00 h

Biomass X (g/L): 95.07
Glucose S (g/L): 83.85
Lactic Acid P (g/L): 66.81
Maltose M (g/L): 34.73
Volume V (L): 9.33

t = 55.00 h

Biomass X (g/L): 84.53
Glucose S (g/L): 85.43
Lactic Acid P (g/L): 83.79
Maltose M (g/L): 32.52
Volume V (L): 9.53

t = 56.00 h

Biomass X (g/L): 81.40
Glucose S (g/L): 77.89
Lactic Acid P (g/L): 85.53
Maltose M (g/L): 30.93
Volume V (L): 9.04

t = 57.00 h

Biomass X (g/L): 80.27
Glucose S (g/L): 79.61
Lactic Acid P (g/L): 71.60
Maltose M (g/L): 32.03
Volume V (L): 9.18

t = 58.00 h

Biomass X (g/L): 83.66
Glucose S (g/L): 86.56
Lactic Acid P (g/L): 68.74
Maltose M (g/L): 32.79
Volume V (L): 9.18

t = 59.00 h

Biomass X (g/L): 84.34
Glucose S (g/L): 81.46
Lactic Acid P (g/L): 93.43
Maltose M (g/L): 33.51
Volume V (L): 9.20

t = 60.00 h

Biomass X (g/L): 96.63
Glucose S (g/L): 78.94
Lactic Acid P (g/L): 79.70
Maltose M (g/L): 34.14
Volume V (L): 10.01

t = 61.00 h

Biomass X (g/L): 103.28
Glucose S (g/L): 86.39
Lactic Acid P (g/L): 75.67
Maltose M (g/L): 33.63
Volume V (L): 9.21

t = 62.00 h

Biomass X (g/L): 86.23
Glucose S (g/L): 83.37
Lactic Acid P (g/L): 85.85
Maltose M (g/L): 30.70
Volume V (L): 9.72

t = 63.00 h

Biomass X (g/L): 76.09
Glucose S (g/L): 91.76
Lactic Acid P (g/L): 76.41
Maltose M (g/L): 32.96
Volume V (L): 9.50

t = 64.00 h

Biomass X (g/L): 78.78
Glucose S (g/L): 87.34
Lactic Acid P (g/L): 69.27
Maltose M (g/L): 31.95
Volume V (L): 9.32

t = 65.00 h

Biomass X (g/L): 82.06
Glucose S (g/L): 87.46
Lactic Acid P (g/L): 74.72
Maltose M (g/L): 30.57
Volume V (L): 10.49

t = 66.00 h

Biomass X (g/L): 73.80
Glucose S (g/L): 86.25
Lactic Acid P (g/L): 83.54
Maltose M (g/L): 30.93
Volume V (L): 9.91

t = 67.00 h

Biomass X (g/L): 72.20
Glucose S (g/L): 93.36
Lactic Acid P (g/L): 56.28
Maltose M (g/L): 30.19
Volume V (L): 10.32

t = 68.00 h

Biomass X (g/L): 103.29
Glucose S (g/L): 89.80
Lactic Acid P (g/L): 71.84
Maltose M (g/L): 30.90
Volume V (L): 9.61

t = 69.00 h

Biomass X (g/L): 85.98
Glucose S (g/L): 87.78
Lactic Acid P (g/L): 87.09
Maltose M (g/L): 29.85
Volume V (L): 10.04

t = 70.00 h

Biomass X (g/L): 74.95
Glucose S (g/L): 87.04
Lactic Acid P (g/L): 70.09
Maltose M (g/L): 29.35
Volume V (L): 10.82

t = 71.00 h

Biomass X (g/L): 90.51
Glucose S (g/L): 89.99
Lactic Acid P (g/L): 88.84
Maltose M (g/L): 30.94
Volume V (L): 11.76

t = 72.00 h

Biomass X (g/L): 71.28
Glucose S (g/L): 87.50
Lactic Acid P (g/L): 87.78
Maltose M (g/L): 30.96
Volume V (L): 9.70

t = 73.00 h

Biomass X (g/L): 71.88
Glucose S (g/L): 94.74
Lactic Acid P (g/L): 74.52
Maltose M (g/L): 29.86
Volume V (L): 10.61

t = 74.00 h

Biomass X (g/L): 74.48
Glucose S (g/L): 85.77
Lactic Acid P (g/L): 70.10
Maltose M (g/L): 31.51
Volume V (L): 10.98

t = 75.00 h

Biomass X (g/L): 97.74
Glucose S (g/L): 91.87
Lactic Acid P (g/L): 72.77
Maltose M (g/L): 29.47

Volume V (L): 10.57

Batch 47

Initial Conditions:

Batch: 47.00

X0 (g/L): 3.00

S0 (g/L): 44.49

M0 (g/L): 85.85

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.93

F_vol (L/h): 0.05

S_feed (g/L): 102.41

M_feed (g/L): 197.59

V_final (L): 8.89

Batch 47 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 11.26
Glucose S (g/L): 45.64
Lactic Acid P (g/L): 6.67
Maltose M (g/L): 86.28
Volume V (L): 4.88

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 45.66
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 87.07
Volume V (L): 5.10

t = 2.00 h

Biomass X (g/L): 6.77
Glucose S (g/L): 48.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 89.07
Volume V (L): 5.19

t = 3.00 h

Biomass X (g/L): 1.83
Glucose S (g/L): 43.76
Lactic Acid P (g/L): 5.72
Maltose M (g/L): 86.33
Volume V (L): 5.26

t = 4.00 h

Biomass X (g/L): 20.80
Glucose S (g/L): 52.07
Lactic Acid P (g/L): 9.35
Maltose M (g/L): 87.45
Volume V (L): 4.84

t = 5.00 h

Biomass X (g/L): 4.85
Glucose S (g/L): 52.45

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 80.35
Volume V (L): 4.74

t = 6.00 h

Biomass X (g/L): 10.95
Glucose S (g/L): 46.06
Lactic Acid P (g/L): 2.74
Maltose M (g/L): 81.91
Volume V (L): 5.13

t = 7.00 h

Biomass X (g/L): 26.94
Glucose S (g/L): 51.63
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 83.62
Volume V (L): 5.49

t = 8.00 h

Biomass X (g/L): 11.51
Glucose S (g/L): 41.58
Lactic Acid P (g/L): 23.10
Maltose M (g/L): 85.26
Volume V (L): 5.19

t = 9.00 h

Biomass X (g/L): 26.20
Glucose S (g/L): 41.80
Lactic Acid P (g/L): 13.43
Maltose M (g/L): 80.99
Volume V (L): 5.36

t = 10.00 h

Biomass X (g/L): 12.00
Glucose S (g/L): 42.48
Lactic Acid P (g/L): 23.92
Maltose M (g/L): 80.69
Volume V (L): 6.18

t = 11.00 h

Biomass X (g/L): 23.37

Glucose S (g/L): 34.52
Lactic Acid P (g/L): 35.81
Maltose M (g/L): 82.46
Volume V (L): 5.00

t = 12.00 h

Biomass X (g/L): 41.83
Glucose S (g/L): 29.41
Lactic Acid P (g/L): 27.92
Maltose M (g/L): 76.72
Volume V (L): 5.39

t = 13.00 h

Biomass X (g/L): 32.48
Glucose S (g/L): 26.43
Lactic Acid P (g/L): 27.90
Maltose M (g/L): 84.19
Volume V (L): 6.27

t = 14.00 h

Biomass X (g/L): 40.61
Glucose S (g/L): 35.01
Lactic Acid P (g/L): 33.59
Maltose M (g/L): 76.51
Volume V (L): 5.71

t = 15.00 h

Biomass X (g/L): 54.23
Glucose S (g/L): 25.48
Lactic Acid P (g/L): 35.12
Maltose M (g/L): 73.03
Volume V (L): 5.56

t = 16.00 h

Biomass X (g/L): 47.06
Glucose S (g/L): 24.75
Lactic Acid P (g/L): 48.48
Maltose M (g/L): 78.17
Volume V (L): 6.19

t = 17.00 h

Biomass X (g/L): 63.67
Glucose S (g/L): 20.53
Lactic Acid P (g/L): 42.28
Maltose M (g/L): 82.63
Volume V (L): 5.90

t = 18.00 h

Biomass X (g/L): 60.49
Glucose S (g/L): 25.25
Lactic Acid P (g/L): 36.64
Maltose M (g/L): 78.95
Volume V (L): 5.83

t = 19.00 h

Biomass X (g/L): 65.10
Glucose S (g/L): 19.10
Lactic Acid P (g/L): 53.22
Maltose M (g/L): 78.98
Volume V (L): 6.05

t = 20.00 h

Biomass X (g/L): 75.17
Glucose S (g/L): 19.04
Lactic Acid P (g/L): 56.76
Maltose M (g/L): 76.47
Volume V (L): 5.83

t = 21.00 h

Biomass X (g/L): 68.24
Glucose S (g/L): 20.62
Lactic Acid P (g/L): 34.98
Maltose M (g/L): 75.47
Volume V (L): 5.77

t = 22.00 h

Biomass X (g/L): 81.93
Glucose S (g/L): 16.05
Lactic Acid P (g/L): 62.84
Maltose M (g/L): 70.99
Volume V (L): 6.11

t = 23.00 h

Biomass X (g/L): 77.50
Glucose S (g/L): 18.19
Lactic Acid P (g/L): 53.64
Maltose M (g/L): 68.48
Volume V (L): 5.88

t = 24.00 h

Biomass X (g/L): 65.06
Glucose S (g/L): 22.49
Lactic Acid P (g/L): 54.26
Maltose M (g/L): 75.19
Volume V (L): 6.25

t = 25.00 h

Biomass X (g/L): 84.09
Glucose S (g/L): 19.42
Lactic Acid P (g/L): 62.16
Maltose M (g/L): 78.01
Volume V (L): 6.41

t = 26.00 h

Biomass X (g/L): 75.33
Glucose S (g/L): 18.43
Lactic Acid P (g/L): 48.96
Maltose M (g/L): 74.77
Volume V (L): 6.62

t = 27.00 h

Biomass X (g/L): 70.06
Glucose S (g/L): 18.63
Lactic Acid P (g/L): 64.00
Maltose M (g/L): 75.16
Volume V (L): 6.75

t = 28.00 h

Biomass X (g/L): 77.64
Glucose S (g/L): 16.90
Lactic Acid P (g/L): 52.88
Maltose M (g/L): 77.54
Volume V (L): 6.33

t = 29.00 h

Biomass X (g/L): 86.69
Glucose S (g/L): 23.97
Lactic Acid P (g/L): 43.82
Maltose M (g/L): 73.48
Volume V (L): 6.66

t = 30.00 h

Biomass X (g/L): 81.16
Glucose S (g/L): 15.92
Lactic Acid P (g/L): 49.12
Maltose M (g/L): 71.76
Volume V (L): 6.71

t = 31.00 h

Biomass X (g/L): 95.23
Glucose S (g/L): 19.12
Lactic Acid P (g/L): 64.31
Maltose M (g/L): 69.33
Volume V (L): 6.71

t = 32.00 h

Biomass X (g/L): 88.79
Glucose S (g/L): 19.09
Lactic Acid P (g/L): 51.31
Maltose M (g/L): 77.18
Volume V (L): 6.51

t = 33.00 h

Biomass X (g/L): 83.97
Glucose S (g/L): 22.00
Lactic Acid P (g/L): 75.47
Maltose M (g/L): 72.44
Volume V (L): 6.39

t = 34.00 h

Biomass X (g/L): 69.50
Glucose S (g/L): 22.61
Lactic Acid P (g/L): 72.38
Maltose M (g/L): 67.77

Volume V (L): 6.92

t = 35.00 h

Biomass X (g/L): 78.27

Glucose S (g/L): 19.66

Lactic Acid P (g/L): 74.41

Maltose M (g/L): 66.71

Volume V (L): 6.29

t = 36.00 h

Biomass X (g/L): 81.21

Glucose S (g/L): 21.00

Lactic Acid P (g/L): 64.03

Maltose M (g/L): 73.37

Volume V (L): 6.79

t = 37.00 h

Biomass X (g/L): 87.73

Glucose S (g/L): 21.53

Lactic Acid P (g/L): 65.90

Maltose M (g/L): 66.33

Volume V (L): 6.13

t = 38.00 h

Biomass X (g/L): 87.56

Glucose S (g/L): 14.37

Lactic Acid P (g/L): 53.24

Maltose M (g/L): 74.34

Volume V (L): 6.56

t = 39.00 h

Biomass X (g/L): 83.38

Glucose S (g/L): 16.84

Lactic Acid P (g/L): 73.85

Maltose M (g/L): 60.57

Volume V (L): 6.75

t = 40.00 h

Biomass X (g/L): 83.23

Glucose S (g/L): 20.20

Lactic Acid P (g/L): 68.78

Maltose M (g/L): 72.33

Volume V (L): 7.42

t = 41.00 h

Biomass X (g/L): 89.37

Glucose S (g/L): 20.83

Lactic Acid P (g/L): 71.04

Maltose M (g/L): 69.24

Volume V (L): 7.34

t = 42.00 h

Biomass X (g/L): 74.28

Glucose S (g/L): 19.48

Lactic Acid P (g/L): 81.81

Maltose M (g/L): 67.80

Volume V (L): 7.01

t = 43.00 h

Biomass X (g/L): 99.35

Glucose S (g/L): 25.71

Lactic Acid P (g/L): 61.32

Maltose M (g/L): 66.58

Volume V (L): 7.46

t = 44.00 h

Biomass X (g/L): 63.17

Glucose S (g/L): 24.07

Lactic Acid P (g/L): 66.19

Maltose M (g/L): 72.22

Volume V (L): 7.56

t = 45.00 h

Biomass X (g/L): 97.41

Glucose S (g/L): 23.99

Lactic Acid P (g/L): 70.79

Maltose M (g/L): 68.42

Volume V (L): 7.29

t = 46.00 h

Biomass X (g/L): 82.98

Glucose S (g/L): 21.51

Lactic Acid P (g/L): 59.68

Maltose M (g/L): 63.38

Volume V (L): 6.96

t = 47.00 h

Biomass X (g/L): 93.31

Glucose S (g/L): 25.63

Lactic Acid P (g/L): 66.95

Maltose M (g/L): 60.65

Volume V (L): 7.38

t = 48.00 h

Biomass X (g/L): 87.22

Glucose S (g/L): 18.66

Lactic Acid P (g/L): 73.58

Maltose M (g/L): 65.81

Volume V (L): 7.46

t = 49.00 h

Biomass X (g/L): 87.15

Glucose S (g/L): 26.45

Lactic Acid P (g/L): 66.48

Maltose M (g/L): 64.94

Volume V (L): 7.16

t = 50.00 h

Biomass X (g/L): 89.25

Glucose S (g/L): 27.71

Lactic Acid P (g/L): 81.81

Maltose M (g/L): 60.18

Volume V (L): 7.71

t = 51.00 h

Biomass X (g/L): 91.79

Glucose S (g/L): 28.83

Lactic Acid P (g/L): 56.38

Maltose M (g/L): 64.09

Volume V (L): 7.74

t = 52.00 h

Biomass X (g/L): 80.78

Glucose S (g/L): 24.16
Lactic Acid P (g/L): 80.83
Maltose M (g/L): 62.24
Volume V (L): 7.84

t = 53.00 h

Biomass X (g/L): 94.07
Glucose S (g/L): 29.99
Lactic Acid P (g/L): 76.94
Maltose M (g/L): 63.60
Volume V (L): 7.84

t = 54.00 h

Biomass X (g/L): 73.48
Glucose S (g/L): 26.91
Lactic Acid P (g/L): 66.02
Maltose M (g/L): 65.34
Volume V (L): 8.03

t = 55.00 h

Biomass X (g/L): 83.98
Glucose S (g/L): 28.19
Lactic Acid P (g/L): 63.09
Maltose M (g/L): 67.24
Volume V (L): 7.80

t = 56.00 h

Biomass X (g/L): 92.15
Glucose S (g/L): 33.28
Lactic Acid P (g/L): 52.82
Maltose M (g/L): 65.21
Volume V (L): 7.97

t = 57.00 h

Biomass X (g/L): 93.68
Glucose S (g/L): 37.68
Lactic Acid P (g/L): 64.87
Maltose M (g/L): 60.15
Volume V (L): 7.84

t = 58.00 h

Biomass X (g/L): 88.45
Glucose S (g/L): 32.57
Lactic Acid P (g/L): 82.01
Maltose M (g/L): 62.99
Volume V (L): 7.83

t = 59.00 h

Biomass X (g/L): 80.11
Glucose S (g/L): 30.65
Lactic Acid P (g/L): 72.20
Maltose M (g/L): 61.78
Volume V (L): 8.48

t = 60.00 h

Biomass X (g/L): 92.60
Glucose S (g/L): 34.21
Lactic Acid P (g/L): 83.62
Maltose M (g/L): 57.41
Volume V (L): 7.48

t = 61.00 h

Biomass X (g/L): 77.31
Glucose S (g/L): 37.02
Lactic Acid P (g/L): 67.57
Maltose M (g/L): 62.12
Volume V (L): 8.44

t = 62.00 h

Biomass X (g/L): 100.78
Glucose S (g/L): 36.71
Lactic Acid P (g/L): 79.97
Maltose M (g/L): 66.04
Volume V (L): 8.03

t = 63.00 h

Biomass X (g/L): 82.85
Glucose S (g/L): 43.54
Lactic Acid P (g/L): 67.18
Maltose M (g/L): 60.43
Volume V (L): 8.04

t = 64.00 h

Biomass X (g/L): 68.62
Glucose S (g/L): 34.05
Lactic Acid P (g/L): 65.15
Maltose M (g/L): 55.39
Volume V (L): 8.40

t = 65.00 h

Biomass X (g/L): 71.88
Glucose S (g/L): 37.49
Lactic Acid P (g/L): 73.24
Maltose M (g/L): 59.81
Volume V (L): 7.91

t = 66.00 h

Biomass X (g/L): 83.83
Glucose S (g/L): 41.90
Lactic Acid P (g/L): 68.36
Maltose M (g/L): 55.89
Volume V (L): 8.17

t = 67.00 h

Biomass X (g/L): 84.66
Glucose S (g/L): 42.33
Lactic Acid P (g/L): 60.55
Maltose M (g/L): 61.62
Volume V (L): 8.21

t = 68.00 h

Biomass X (g/L): 102.34
Glucose S (g/L): 32.90
Lactic Acid P (g/L): 73.21
Maltose M (g/L): 60.34
Volume V (L): 8.46

t = 69.00 h

Biomass X (g/L): 72.97
Glucose S (g/L): 39.20
Lactic Acid P (g/L): 76.06
Maltose M (g/L): 58.02
Volume V (L): 8.33

t = 70.00 h

Biomass X (g/L): 83.33
Glucose S (g/L): 44.16
Lactic Acid P (g/L): 86.89
Maltose M (g/L): 57.26
Volume V (L): 9.13

t = 71.00 h

Biomass X (g/L): 72.52
Glucose S (g/L): 48.88
Lactic Acid P (g/L): 84.05
Maltose M (g/L): 57.31
Volume V (L): 9.25

t = 72.00 h

Biomass X (g/L): 77.90
Glucose S (g/L): 43.09
Lactic Acid P (g/L): 86.84
Maltose M (g/L): 57.67
Volume V (L): 8.11

t = 73.00 h

Biomass X (g/L): 98.92
Glucose S (g/L): 40.56
Lactic Acid P (g/L): 82.45
Maltose M (g/L): 55.11
Volume V (L): 8.96

t = 74.00 h

Biomass X (g/L): 71.36
Glucose S (g/L): 39.29
Lactic Acid P (g/L): 79.09
Maltose M (g/L): 53.04
Volume V (L): 9.11

t = 75.00 h

Biomass X (g/L): 87.97
Glucose S (g/L): 49.32
Lactic Acid P (g/L): 64.98
Maltose M (g/L): 60.10

Volume V (L): 9.06

Batch 48

Initial Conditions:

Batch: 48.00

X0 (g/L): 1.56

S0 (g/L): 63.35

M0 (g/L): 124.39

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.96

F_vol (L/h): 0.08

S_feed (g/L): 101.23

M_feed (g/L): 198.77

V_final (L): 10.69

Batch 48 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 67.31
Lactic Acid P (g/L): 14.80
Maltose M (g/L): 124.46
Volume V (L): 4.46

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 67.25
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 124.50
Volume V (L): 5.45

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 70.70
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 126.98
Volume V (L): 4.86

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 70.74
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 116.92
Volume V (L): 5.44

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 73.49
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 116.76
Volume V (L): 6.05

t = 5.00 h

Biomass X (g/L): 5.33
Glucose S (g/L): 70.09

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 114.58
Volume V (L): 4.88

t = 6.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 71.78
Lactic Acid P (g/L): 4.87
Maltose M (g/L): 115.28
Volume V (L): 5.67

t = 7.00 h

Biomass X (g/L): 8.76
Glucose S (g/L): 76.85
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 123.85
Volume V (L): 5.78

t = 8.00 h

Biomass X (g/L): 1.29
Glucose S (g/L): 77.08
Lactic Acid P (g/L): 8.46
Maltose M (g/L): 118.04
Volume V (L): 5.21

t = 9.00 h

Biomass X (g/L): 18.79
Glucose S (g/L): 69.02
Lactic Acid P (g/L): 30.08
Maltose M (g/L): 107.55
Volume V (L): 6.30

t = 10.00 h

Biomass X (g/L): 13.75
Glucose S (g/L): 71.44
Lactic Acid P (g/L): 9.24
Maltose M (g/L): 109.41
Volume V (L): 5.63

t = 11.00 h

Biomass X (g/L): 27.21

Glucose S (g/L): 67.95
Lactic Acid P (g/L): 18.45
Maltose M (g/L): 113.73
Volume V (L): 5.66

t = 12.00 h

Biomass X (g/L): 39.11
Glucose S (g/L): 65.28
Lactic Acid P (g/L): 27.88
Maltose M (g/L): 105.65
Volume V (L): 6.87

t = 13.00 h

Biomass X (g/L): 48.33
Glucose S (g/L): 65.79
Lactic Acid P (g/L): 25.41
Maltose M (g/L): 105.64
Volume V (L): 6.22

t = 14.00 h

Biomass X (g/L): 12.20
Glucose S (g/L): 65.91
Lactic Acid P (g/L): 28.66
Maltose M (g/L): 116.06
Volume V (L): 6.48

t = 15.00 h

Biomass X (g/L): 32.89
Glucose S (g/L): 61.97
Lactic Acid P (g/L): 25.87
Maltose M (g/L): 104.29
Volume V (L): 6.75

t = 16.00 h

Biomass X (g/L): 41.99
Glucose S (g/L): 57.92
Lactic Acid P (g/L): 29.99
Maltose M (g/L): 111.98
Volume V (L): 5.92

t = 17.00 h

Biomass X (g/L): 53.14
Glucose S (g/L): 58.31
Lactic Acid P (g/L): 26.27
Maltose M (g/L): 112.70
Volume V (L): 6.86

t = 18.00 h

Biomass X (g/L): 35.87
Glucose S (g/L): 54.35
Lactic Acid P (g/L): 45.07
Maltose M (g/L): 100.80
Volume V (L): 6.14

t = 19.00 h

Biomass X (g/L): 45.55
Glucose S (g/L): 54.98
Lactic Acid P (g/L): 24.36
Maltose M (g/L): 99.03
Volume V (L): 6.19

t = 20.00 h

Biomass X (g/L): 59.31
Glucose S (g/L): 56.37
Lactic Acid P (g/L): 50.92
Maltose M (g/L): 101.52
Volume V (L): 6.39

t = 21.00 h

Biomass X (g/L): 61.48
Glucose S (g/L): 46.09
Lactic Acid P (g/L): 31.76
Maltose M (g/L): 96.16
Volume V (L): 6.31

t = 22.00 h

Biomass X (g/L): 66.72
Glucose S (g/L): 52.91
Lactic Acid P (g/L): 54.89
Maltose M (g/L): 101.96
Volume V (L): 7.04

t = 23.00 h

Biomass X (g/L): 69.51
Glucose S (g/L): 46.01
Lactic Acid P (g/L): 44.90
Maltose M (g/L): 99.31
Volume V (L): 6.87

t = 24.00 h

Biomass X (g/L): 70.42
Glucose S (g/L): 52.87
Lactic Acid P (g/L): 38.69
Maltose M (g/L): 101.59
Volume V (L): 6.60

t = 25.00 h

Biomass X (g/L): 70.75
Glucose S (g/L): 47.53
Lactic Acid P (g/L): 56.43
Maltose M (g/L): 106.48
Volume V (L): 7.20

t = 26.00 h

Biomass X (g/L): 70.98
Glucose S (g/L): 45.74
Lactic Acid P (g/L): 60.97
Maltose M (g/L): 102.47
Volume V (L): 6.85

t = 27.00 h

Biomass X (g/L): 78.37
Glucose S (g/L): 50.71
Lactic Acid P (g/L): 50.04
Maltose M (g/L): 99.86
Volume V (L): 6.90

t = 28.00 h

Biomass X (g/L): 51.20
Glucose S (g/L): 49.79
Lactic Acid P (g/L): 61.62
Maltose M (g/L): 85.53
Volume V (L): 7.14

t = 29.00 h

Biomass X (g/L): 89.43
Glucose S (g/L): 46.22
Lactic Acid P (g/L): 35.84
Maltose M (g/L): 98.04
Volume V (L): 7.41

t = 30.00 h

Biomass X (g/L): 53.75
Glucose S (g/L): 45.68
Lactic Acid P (g/L): 45.52
Maltose M (g/L): 104.33
Volume V (L): 7.45

t = 31.00 h

Biomass X (g/L): 65.71
Glucose S (g/L): 48.73
Lactic Acid P (g/L): 69.88
Maltose M (g/L): 77.47
Volume V (L): 7.18

t = 32.00 h

Biomass X (g/L): 74.91
Glucose S (g/L): 46.29
Lactic Acid P (g/L): 57.22
Maltose M (g/L): 97.14
Volume V (L): 7.81

t = 33.00 h

Biomass X (g/L): 84.48
Glucose S (g/L): 46.99
Lactic Acid P (g/L): 43.40
Maltose M (g/L): 91.72
Volume V (L): 7.40

t = 34.00 h

Biomass X (g/L): 49.23
Glucose S (g/L): 50.50
Lactic Acid P (g/L): 54.61
Maltose M (g/L): 87.20

Volume V (L): 7.28

t = 35.00 h

Biomass X (g/L): 78.92

Glucose S (g/L): 48.03

Lactic Acid P (g/L): 53.35

Maltose M (g/L): 95.59

Volume V (L): 6.99

t = 36.00 h

Biomass X (g/L): 87.50

Glucose S (g/L): 52.19

Lactic Acid P (g/L): 60.84

Maltose M (g/L): 99.90

Volume V (L): 8.18

t = 37.00 h

Biomass X (g/L): 89.98

Glucose S (g/L): 55.51

Lactic Acid P (g/L): 59.66

Maltose M (g/L): 94.06

Volume V (L): 7.81

t = 38.00 h

Biomass X (g/L): 73.84

Glucose S (g/L): 51.66

Lactic Acid P (g/L): 65.21

Maltose M (g/L): 87.05

Volume V (L): 7.98

t = 39.00 h

Biomass X (g/L): 78.12

Glucose S (g/L): 52.94

Lactic Acid P (g/L): 63.58

Maltose M (g/L): 83.23

Volume V (L): 8.43

t = 40.00 h

Biomass X (g/L): 75.38

Glucose S (g/L): 58.53

Lactic Acid P (g/L): 76.93

Maltose M (g/L): 88.56

Volume V (L): 7.82

t = 41.00 h

Biomass X (g/L): 77.87

Glucose S (g/L): 50.56

Lactic Acid P (g/L): 69.08

Maltose M (g/L): 91.93

Volume V (L): 7.48

t = 42.00 h

Biomass X (g/L): 71.88

Glucose S (g/L): 57.50

Lactic Acid P (g/L): 69.88

Maltose M (g/L): 84.34

Volume V (L): 8.22

t = 43.00 h

Biomass X (g/L): 73.51

Glucose S (g/L): 57.79

Lactic Acid P (g/L): 76.69

Maltose M (g/L): 84.23

Volume V (L): 8.40

t = 44.00 h

Biomass X (g/L): 86.68

Glucose S (g/L): 57.83

Lactic Acid P (g/L): 63.23

Maltose M (g/L): 100.77

Volume V (L): 8.22

t = 45.00 h

Biomass X (g/L): 83.89

Glucose S (g/L): 59.63

Lactic Acid P (g/L): 55.85

Maltose M (g/L): 90.54

Volume V (L): 9.08

t = 46.00 h

Biomass X (g/L): 87.56

Glucose S (g/L): 57.57

Lactic Acid P (g/L): 78.36

Maltose M (g/L): 90.32

Volume V (L): 7.36

t = 47.00 h

Biomass X (g/L): 84.97

Glucose S (g/L): 55.92

Lactic Acid P (g/L): 69.17

Maltose M (g/L): 75.32

Volume V (L): 9.07

t = 48.00 h

Biomass X (g/L): 89.87

Glucose S (g/L): 61.04

Lactic Acid P (g/L): 69.57

Maltose M (g/L): 78.88

Volume V (L): 9.00

t = 49.00 h

Biomass X (g/L): 91.66

Glucose S (g/L): 61.80

Lactic Acid P (g/L): 63.01

Maltose M (g/L): 82.61

Volume V (L): 8.92

t = 50.00 h

Biomass X (g/L): 66.05

Glucose S (g/L): 62.39

Lactic Acid P (g/L): 63.51

Maltose M (g/L): 76.10

Volume V (L): 7.27

t = 51.00 h

Biomass X (g/L): 74.79

Glucose S (g/L): 60.22

Lactic Acid P (g/L): 76.94

Maltose M (g/L): 89.87

Volume V (L): 9.45

t = 52.00 h

Biomass X (g/L): 80.92

Glucose S (g/L): 58.86
Lactic Acid P (g/L): 78.49
Maltose M (g/L): 88.26
Volume V (L): 8.48

t = 53.00 h

Biomass X (g/L): 82.97
Glucose S (g/L): 61.95
Lactic Acid P (g/L): 76.99
Maltose M (g/L): 76.98
Volume V (L): 8.79

t = 54.00 h

Biomass X (g/L): 76.63
Glucose S (g/L): 64.61
Lactic Acid P (g/L): 77.37
Maltose M (g/L): 89.21
Volume V (L): 9.02

t = 55.00 h

Biomass X (g/L): 92.11
Glucose S (g/L): 65.50
Lactic Acid P (g/L): 75.64
Maltose M (g/L): 78.81
Volume V (L): 9.35

t = 56.00 h

Biomass X (g/L): 102.54
Glucose S (g/L): 62.29
Lactic Acid P (g/L): 72.00
Maltose M (g/L): 92.58
Volume V (L): 9.54

t = 57.00 h

Biomass X (g/L): 84.93
Glucose S (g/L): 64.94
Lactic Acid P (g/L): 69.51
Maltose M (g/L): 78.56
Volume V (L): 9.73

t = 58.00 h

Biomass X (g/L): 83.88
Glucose S (g/L): 61.27
Lactic Acid P (g/L): 82.17
Maltose M (g/L): 80.42
Volume V (L): 9.77

t = 59.00 h

Biomass X (g/L): 91.32
Glucose S (g/L): 68.04
Lactic Acid P (g/L): 68.23
Maltose M (g/L): 84.44
Volume V (L): 9.59

t = 60.00 h

Biomass X (g/L): 64.65
Glucose S (g/L): 63.99
Lactic Acid P (g/L): 75.12
Maltose M (g/L): 83.26
Volume V (L): 9.74

t = 61.00 h

Biomass X (g/L): 83.13
Glucose S (g/L): 68.82
Lactic Acid P (g/L): 64.84
Maltose M (g/L): 78.53
Volume V (L): 9.13

t = 62.00 h

Biomass X (g/L): 92.35
Glucose S (g/L): 64.32
Lactic Acid P (g/L): 66.36
Maltose M (g/L): 74.43
Volume V (L): 9.36

t = 63.00 h

Biomass X (g/L): 76.37
Glucose S (g/L): 70.78
Lactic Acid P (g/L): 70.36
Maltose M (g/L): 77.55
Volume V (L): 10.18

t = 64.00 h

Biomass X (g/L): 70.55
Glucose S (g/L): 69.53
Lactic Acid P (g/L): 65.48
Maltose M (g/L): 69.73
Volume V (L): 10.23

t = 65.00 h

Biomass X (g/L): 80.64
Glucose S (g/L): 67.23
Lactic Acid P (g/L): 80.02
Maltose M (g/L): 74.59
Volume V (L): 10.26

t = 66.00 h

Biomass X (g/L): 81.76
Glucose S (g/L): 71.79
Lactic Acid P (g/L): 61.01
Maltose M (g/L): 67.16
Volume V (L): 9.77

t = 67.00 h

Biomass X (g/L): 73.33
Glucose S (g/L): 69.01
Lactic Acid P (g/L): 74.15
Maltose M (g/L): 62.98
Volume V (L): 10.49

t = 68.00 h

Biomass X (g/L): 102.64
Glucose S (g/L): 73.21
Lactic Acid P (g/L): 74.63
Maltose M (g/L): 72.62
Volume V (L): 9.63

t = 69.00 h

Biomass X (g/L): 77.29
Glucose S (g/L): 71.10
Lactic Acid P (g/L): 57.41
Maltose M (g/L): 78.84
Volume V (L): 10.72

t = 70.00 h

Biomass X (g/L): 70.55
Glucose S (g/L): 67.87
Lactic Acid P (g/L): 69.85
Maltose M (g/L): 76.53
Volume V (L): 10.19

t = 71.00 h

Biomass X (g/L): 80.79
Glucose S (g/L): 70.45
Lactic Acid P (g/L): 76.25
Maltose M (g/L): 70.53
Volume V (L): 11.15

t = 72.00 h

Biomass X (g/L): 90.47
Glucose S (g/L): 72.53
Lactic Acid P (g/L): 77.97
Maltose M (g/L): 80.17
Volume V (L): 9.77

t = 73.00 h

Biomass X (g/L): 81.60
Glucose S (g/L): 77.93
Lactic Acid P (g/L): 72.46
Maltose M (g/L): 66.96
Volume V (L): 11.34

t = 74.00 h

Biomass X (g/L): 86.04
Glucose S (g/L): 77.28
Lactic Acid P (g/L): 80.50
Maltose M (g/L): 59.09
Volume V (L): 10.45

t = 75.00 h

Biomass X (g/L): 72.59
Glucose S (g/L): 79.80
Lactic Acid P (g/L): 73.62
Maltose M (g/L): 57.20

Volume V (L): 10.51

Batch 49

Initial Conditions:

Batch: 49.00

X0 (g/L): 1.17

S0 (g/L): 84.67

M0 (g/L): 90.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.07

F_vol (L/h): 0.03

S_feed (g/L): 145.01

M_feed (g/L): 154.99

V_final (L): 7.09

Batch 49 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.21
Glucose S (g/L): 88.94
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 94.14
Volume V (L): 5.05

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 95.95
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 89.92
Volume V (L): 5.40

t = 2.00 h

Biomass X (g/L): 1.41
Glucose S (g/L): 84.49
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 82.59
Volume V (L): 4.99

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 84.10
Lactic Acid P (g/L): 9.78
Maltose M (g/L): 81.45
Volume V (L): 5.35

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 90.69
Lactic Acid P (g/L): 9.38
Maltose M (g/L): 74.85
Volume V (L): 4.99

t = 5.00 h

Biomass X (g/L): 10.36
Glucose S (g/L): 94.20

Lactic Acid P (g/L): 7.94
Maltose M (g/L): 79.61
Volume V (L): 5.18

t = 6.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 90.37
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 86.76
Volume V (L): 5.30

t = 7.00 h

Biomass X (g/L): 1.04
Glucose S (g/L): 86.10
Lactic Acid P (g/L): 14.40
Maltose M (g/L): 88.29
Volume V (L): 5.19

t = 8.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 89.22
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 83.83
Volume V (L): 5.17

t = 9.00 h

Biomass X (g/L): 9.54
Glucose S (g/L): 90.82
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 82.00
Volume V (L): 5.37

t = 10.00 h

Biomass X (g/L): 32.15
Glucose S (g/L): 84.80
Lactic Acid P (g/L): 0.06
Maltose M (g/L): 77.37
Volume V (L): 5.59

t = 11.00 h

Biomass X (g/L): 13.55

Glucose S (g/L): 86.71
Lactic Acid P (g/L): 15.11
Maltose M (g/L): 80.92
Volume V (L): 5.36

t = 12.00 h

Biomass X (g/L): 19.05
Glucose S (g/L): 86.37
Lactic Acid P (g/L): 25.19
Maltose M (g/L): 77.33
Volume V (L): 5.25

t = 13.00 h

Biomass X (g/L): 33.22
Glucose S (g/L): 79.64
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 84.11
Volume V (L): 5.33

t = 14.00 h

Biomass X (g/L): 31.82
Glucose S (g/L): 75.55
Lactic Acid P (g/L): 38.16
Maltose M (g/L): 73.27
Volume V (L): 5.81

t = 15.00 h

Biomass X (g/L): 63.50
Glucose S (g/L): 76.13
Lactic Acid P (g/L): 29.84
Maltose M (g/L): 79.70
Volume V (L): 5.44

t = 16.00 h

Biomass X (g/L): 42.75
Glucose S (g/L): 66.26
Lactic Acid P (g/L): 16.46
Maltose M (g/L): 76.30
Volume V (L): 5.35

t = 17.00 h

Biomass X (g/L): 52.02
Glucose S (g/L): 63.15
Lactic Acid P (g/L): 35.84
Maltose M (g/L): 74.63
Volume V (L): 5.28

t = 18.00 h

Biomass X (g/L): 62.19
Glucose S (g/L): 63.52
Lactic Acid P (g/L): 48.44
Maltose M (g/L): 59.52
Volume V (L): 5.20

t = 19.00 h

Biomass X (g/L): 65.49
Glucose S (g/L): 60.32
Lactic Acid P (g/L): 46.35
Maltose M (g/L): 73.77
Volume V (L): 5.59

t = 20.00 h

Biomass X (g/L): 61.47
Glucose S (g/L): 56.99
Lactic Acid P (g/L): 53.77
Maltose M (g/L): 67.43
Volume V (L): 5.67

t = 21.00 h

Biomass X (g/L): 60.28
Glucose S (g/L): 66.14
Lactic Acid P (g/L): 32.03
Maltose M (g/L): 64.92
Volume V (L): 5.52

t = 22.00 h

Biomass X (g/L): 85.84
Glucose S (g/L): 65.97
Lactic Acid P (g/L): 43.51
Maltose M (g/L): 58.70
Volume V (L): 5.61

t = 23.00 h

Biomass X (g/L): 66.03
Glucose S (g/L): 54.10
Lactic Acid P (g/L): 40.53
Maltose M (g/L): 65.48
Volume V (L): 5.45

t = 24.00 h

Biomass X (g/L): 64.64
Glucose S (g/L): 54.75
Lactic Acid P (g/L): 68.21
Maltose M (g/L): 64.86
Volume V (L): 5.69

t = 25.00 h

Biomass X (g/L): 64.50
Glucose S (g/L): 52.56
Lactic Acid P (g/L): 59.39
Maltose M (g/L): 57.16
Volume V (L): 5.78

t = 26.00 h

Biomass X (g/L): 54.98
Glucose S (g/L): 57.06
Lactic Acid P (g/L): 82.56
Maltose M (g/L): 55.50
Volume V (L): 5.63

t = 27.00 h

Biomass X (g/L): 83.30
Glucose S (g/L): 57.14
Lactic Acid P (g/L): 51.65
Maltose M (g/L): 64.83
Volume V (L): 6.03

t = 28.00 h

Biomass X (g/L): 51.58
Glucose S (g/L): 43.71
Lactic Acid P (g/L): 50.52
Maltose M (g/L): 66.07
Volume V (L): 5.64

t = 29.00 h

Biomass X (g/L): 77.45
Glucose S (g/L): 52.56
Lactic Acid P (g/L): 57.49
Maltose M (g/L): 61.81
Volume V (L): 5.86

t = 30.00 h

Biomass X (g/L): 76.57
Glucose S (g/L): 44.74
Lactic Acid P (g/L): 73.55
Maltose M (g/L): 66.05
Volume V (L): 5.98

t = 31.00 h

Biomass X (g/L): 103.27
Glucose S (g/L): 57.15
Lactic Acid P (g/L): 57.63
Maltose M (g/L): 58.96
Volume V (L): 5.85

t = 32.00 h

Biomass X (g/L): 61.62
Glucose S (g/L): 51.42
Lactic Acid P (g/L): 63.47
Maltose M (g/L): 58.43
Volume V (L): 5.86

t = 33.00 h

Biomass X (g/L): 72.26
Glucose S (g/L): 45.99
Lactic Acid P (g/L): 64.19
Maltose M (g/L): 53.30
Volume V (L): 5.68

t = 34.00 h

Biomass X (g/L): 81.08
Glucose S (g/L): 48.28
Lactic Acid P (g/L): 64.56
Maltose M (g/L): 49.32

Volume V (L): 6.17

t = 35.00 h

Biomass X (g/L): 81.17

Glucose S (g/L): 48.02

Lactic Acid P (g/L): 58.67

Maltose M (g/L): 66.28

Volume V (L): 5.95

t = 36.00 h

Biomass X (g/L): 78.14

Glucose S (g/L): 51.38

Lactic Acid P (g/L): 64.53

Maltose M (g/L): 61.92

Volume V (L): 6.09

t = 37.00 h

Biomass X (g/L): 83.64

Glucose S (g/L): 53.56

Lactic Acid P (g/L): 64.90

Maltose M (g/L): 48.27

Volume V (L): 6.14

t = 38.00 h

Biomass X (g/L): 71.83

Glucose S (g/L): 44.52

Lactic Acid P (g/L): 67.32

Maltose M (g/L): 55.65

Volume V (L): 5.95

t = 39.00 h

Biomass X (g/L): 89.25

Glucose S (g/L): 57.03

Lactic Acid P (g/L): 77.32

Maltose M (g/L): 53.73

Volume V (L): 6.05

t = 40.00 h

Biomass X (g/L): 61.62

Glucose S (g/L): 56.20

Lactic Acid P (g/L): 85.98

Maltose M (g/L): 52.25

Volume V (L): 6.34

t = 41.00 h

Biomass X (g/L): 86.05

Glucose S (g/L): 46.05

Lactic Acid P (g/L): 61.20

Maltose M (g/L): 50.97

Volume V (L): 5.96

t = 42.00 h

Biomass X (g/L): 80.38

Glucose S (g/L): 50.17

Lactic Acid P (g/L): 66.19

Maltose M (g/L): 54.49

Volume V (L): 6.53

t = 43.00 h

Biomass X (g/L): 97.61

Glucose S (g/L): 57.18

Lactic Acid P (g/L): 62.54

Maltose M (g/L): 56.19

Volume V (L): 6.07

t = 44.00 h

Biomass X (g/L): 105.62

Glucose S (g/L): 55.42

Lactic Acid P (g/L): 72.98

Maltose M (g/L): 53.50

Volume V (L): 6.03

t = 45.00 h

Biomass X (g/L): 84.01

Glucose S (g/L): 49.15

Lactic Acid P (g/L): 66.43

Maltose M (g/L): 50.28

Volume V (L): 6.28

t = 46.00 h

Biomass X (g/L): 102.24

Glucose S (g/L): 54.19

Lactic Acid P (g/L): 74.54

Maltose M (g/L): 49.57

Volume V (L): 6.65

t = 47.00 h

Biomass X (g/L): 84.54

Glucose S (g/L): 63.53

Lactic Acid P (g/L): 83.47

Maltose M (g/L): 54.70

Volume V (L): 5.99

t = 48.00 h

Biomass X (g/L): 77.88

Glucose S (g/L): 51.81

Lactic Acid P (g/L): 48.47

Maltose M (g/L): 51.76

Volume V (L): 6.31

t = 49.00 h

Biomass X (g/L): 106.41

Glucose S (g/L): 49.04

Lactic Acid P (g/L): 78.41

Maltose M (g/L): 49.25

Volume V (L): 6.44

t = 50.00 h

Biomass X (g/L): 93.68

Glucose S (g/L): 52.69

Lactic Acid P (g/L): 69.54

Maltose M (g/L): 48.21

Volume V (L): 6.40

t = 51.00 h

Biomass X (g/L): 88.36

Glucose S (g/L): 53.32

Lactic Acid P (g/L): 78.96

Maltose M (g/L): 49.71

Volume V (L): 6.48

t = 52.00 h

Biomass X (g/L): 95.57

Glucose S (g/L): 50.44
Lactic Acid P (g/L): 66.10
Maltose M (g/L): 45.05
Volume V (L): 6.33

t = 53.00 h

Biomass X (g/L): 91.29
Glucose S (g/L): 54.79
Lactic Acid P (g/L): 86.94
Maltose M (g/L): 51.25
Volume V (L): 6.38

t = 54.00 h

Biomass X (g/L): 87.71
Glucose S (g/L): 48.08
Lactic Acid P (g/L): 84.55
Maltose M (g/L): 41.02
Volume V (L): 6.48

t = 55.00 h

Biomass X (g/L): 74.56
Glucose S (g/L): 50.87
Lactic Acid P (g/L): 81.55
Maltose M (g/L): 44.01
Volume V (L): 6.34

t = 56.00 h

Biomass X (g/L): 85.61
Glucose S (g/L): 57.13
Lactic Acid P (g/L): 84.10
Maltose M (g/L): 48.85
Volume V (L): 6.55

t = 57.00 h

Biomass X (g/L): 94.95
Glucose S (g/L): 47.04
Lactic Acid P (g/L): 69.40
Maltose M (g/L): 39.44
Volume V (L): 6.52

t = 58.00 h

Biomass X (g/L): 87.10
Glucose S (g/L): 58.44
Lactic Acid P (g/L): 70.96
Maltose M (g/L): 36.76
Volume V (L): 6.54

t = 59.00 h

Biomass X (g/L): 97.80
Glucose S (g/L): 46.69
Lactic Acid P (g/L): 84.69
Maltose M (g/L): 44.79
Volume V (L): 6.70

t = 60.00 h

Biomass X (g/L): 92.31
Glucose S (g/L): 48.22
Lactic Acid P (g/L): 70.15
Maltose M (g/L): 35.32
Volume V (L): 6.37

t = 61.00 h

Biomass X (g/L): 79.02
Glucose S (g/L): 53.70
Lactic Acid P (g/L): 92.08
Maltose M (g/L): 36.98
Volume V (L): 6.66

t = 62.00 h

Biomass X (g/L): 74.33
Glucose S (g/L): 57.33
Lactic Acid P (g/L): 75.08
Maltose M (g/L): 47.20
Volume V (L): 7.11

t = 63.00 h

Biomass X (g/L): 86.41
Glucose S (g/L): 62.01
Lactic Acid P (g/L): 83.43
Maltose M (g/L): 41.87
Volume V (L): 6.57

t = 64.00 h

Biomass X (g/L): 82.47
Glucose S (g/L): 57.11
Lactic Acid P (g/L): 99.32
Maltose M (g/L): 49.19
Volume V (L): 7.00

t = 65.00 h

Biomass X (g/L): 91.38
Glucose S (g/L): 56.21
Lactic Acid P (g/L): 72.17
Maltose M (g/L): 35.98
Volume V (L): 6.82

t = 66.00 h

Biomass X (g/L): 80.23
Glucose S (g/L): 60.01
Lactic Acid P (g/L): 80.01
Maltose M (g/L): 41.35
Volume V (L): 6.74

t = 67.00 h

Biomass X (g/L): 81.18
Glucose S (g/L): 54.35
Lactic Acid P (g/L): 89.89
Maltose M (g/L): 43.55
Volume V (L): 6.65

t = 68.00 h

Biomass X (g/L): 98.53
Glucose S (g/L): 58.92
Lactic Acid P (g/L): 89.14
Maltose M (g/L): 35.11
Volume V (L): 6.92

t = 69.00 h

Biomass X (g/L): 88.92
Glucose S (g/L): 53.72
Lactic Acid P (g/L): 76.88
Maltose M (g/L): 46.13
Volume V (L): 7.17

t = 70.00 h

Biomass X (g/L): 79.87
Glucose S (g/L): 59.41
Lactic Acid P (g/L): 75.27
Maltose M (g/L): 43.11
Volume V (L): 6.88

t = 71.00 h

Biomass X (g/L): 87.28
Glucose S (g/L): 58.89
Lactic Acid P (g/L): 78.28
Maltose M (g/L): 31.85
Volume V (L): 6.96

t = 72.00 h

Biomass X (g/L): 101.38
Glucose S (g/L): 56.45
Lactic Acid P (g/L): 94.93
Maltose M (g/L): 39.64
Volume V (L): 7.17

t = 73.00 h

Biomass X (g/L): 89.86
Glucose S (g/L): 60.84
Lactic Acid P (g/L): 78.27
Maltose M (g/L): 41.51
Volume V (L): 7.09

t = 74.00 h

Biomass X (g/L): 92.84
Glucose S (g/L): 61.17
Lactic Acid P (g/L): 81.98
Maltose M (g/L): 36.17
Volume V (L): 6.80

t = 75.00 h

Biomass X (g/L): 78.59
Glucose S (g/L): 65.76
Lactic Acid P (g/L): 77.26
Maltose M (g/L): 31.28

Volume V (L): 7.11

Batch 50

Initial Conditions:

Batch: 50.00

X0 (g/L): 6.53

S0 (g/L): 125.81

M0 (g/L): 56.32

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.45

F_vol (L/h): 0.03

S_feed (g/L): 207.23

M_feed (g/L): 92.77

V_final (L): 6.90

Batch 50 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 22.37
Glucose S (g/L): 123.74
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 49.87
Volume V (L): 5.01

t = 1.00 h

Biomass X (g/L): 8.51
Glucose S (g/L): 129.78
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 54.59
Volume V (L): 4.81

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 119.88
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 58.42
Volume V (L): 4.80

t = 3.00 h

Biomass X (g/L): 19.68
Glucose S (g/L): 114.91
Lactic Acid P (g/L): 11.52
Maltose M (g/L): 66.84
Volume V (L): 5.01

t = 4.00 h

Biomass X (g/L): 17.87
Glucose S (g/L): 108.77
Lactic Acid P (g/L): 3.02
Maltose M (g/L): 48.83
Volume V (L): 5.03

t = 5.00 h

Biomass X (g/L): 5.65
Glucose S (g/L): 122.33

Lactic Acid P (g/L): 23.17

Maltose M (g/L): 53.44

Volume V (L): 4.99

t = 6.00 h

Biomass X (g/L): 12.45

Glucose S (g/L): 104.03

Lactic Acid P (g/L): 14.91

Maltose M (g/L): 49.32

Volume V (L): 5.05

t = 7.00 h

Biomass X (g/L): 45.64

Glucose S (g/L): 110.96

Lactic Acid P (g/L): 10.44

Maltose M (g/L): 57.28

Volume V (L): 5.18

t = 8.00 h

Biomass X (g/L): 33.11

Glucose S (g/L): 96.54

Lactic Acid P (g/L): 22.73

Maltose M (g/L): 45.17

Volume V (L): 5.09

t = 9.00 h

Biomass X (g/L): 36.45

Glucose S (g/L): 106.11

Lactic Acid P (g/L): 24.85

Maltose M (g/L): 51.02

Volume V (L): 5.24

t = 10.00 h

Biomass X (g/L): 50.46

Glucose S (g/L): 107.53

Lactic Acid P (g/L): 42.92

Maltose M (g/L): 43.51

Volume V (L): 5.20

t = 11.00 h

Biomass X (g/L): 48.60

Glucose S (g/L): 81.17
Lactic Acid P (g/L): 32.41
Maltose M (g/L): 46.43
Volume V (L): 5.30

t = 12.00 h

Biomass X (g/L): 60.53
Glucose S (g/L): 81.53
Lactic Acid P (g/L): 36.54
Maltose M (g/L): 46.68
Volume V (L): 5.44

t = 13.00 h

Biomass X (g/L): 61.11
Glucose S (g/L): 89.59
Lactic Acid P (g/L): 56.59
Maltose M (g/L): 49.68
Volume V (L): 5.46

t = 14.00 h

Biomass X (g/L): 54.04
Glucose S (g/L): 85.53
Lactic Acid P (g/L): 50.14
Maltose M (g/L): 41.85
Volume V (L): 5.55

t = 15.00 h

Biomass X (g/L): 64.29
Glucose S (g/L): 84.85
Lactic Acid P (g/L): 64.30
Maltose M (g/L): 47.01
Volume V (L): 5.51

t = 16.00 h

Biomass X (g/L): 77.00
Glucose S (g/L): 74.96
Lactic Acid P (g/L): 62.47
Maltose M (g/L): 44.22
Volume V (L): 5.44

t = 17.00 h

Biomass X (g/L): 75.23
Glucose S (g/L): 88.28
Lactic Acid P (g/L): 42.79
Maltose M (g/L): 44.45
Volume V (L): 5.37

t = 18.00 h

Biomass X (g/L): 76.11
Glucose S (g/L): 75.81
Lactic Acid P (g/L): 63.71
Maltose M (g/L): 46.12
Volume V (L): 5.40

t = 19.00 h

Biomass X (g/L): 99.21
Glucose S (g/L): 66.67
Lactic Acid P (g/L): 70.87
Maltose M (g/L): 38.05
Volume V (L): 5.35

t = 20.00 h

Biomass X (g/L): 88.32
Glucose S (g/L): 71.20
Lactic Acid P (g/L): 72.55
Maltose M (g/L): 44.45
Volume V (L): 5.51

t = 21.00 h

Biomass X (g/L): 91.15
Glucose S (g/L): 64.64
Lactic Acid P (g/L): 68.51
Maltose M (g/L): 39.28
Volume V (L): 5.64

t = 22.00 h

Biomass X (g/L): 81.87
Glucose S (g/L): 80.20
Lactic Acid P (g/L): 69.67
Maltose M (g/L): 44.28
Volume V (L): 5.45

t = 23.00 h

Biomass X (g/L): 87.16
Glucose S (g/L): 66.26
Lactic Acid P (g/L): 58.07
Maltose M (g/L): 41.19
Volume V (L): 5.58

t = 24.00 h

Biomass X (g/L): 81.46
Glucose S (g/L): 72.41
Lactic Acid P (g/L): 53.84
Maltose M (g/L): 38.84
Volume V (L): 5.51

t = 25.00 h

Biomass X (g/L): 95.12
Glucose S (g/L): 66.23
Lactic Acid P (g/L): 59.86
Maltose M (g/L): 37.05
Volume V (L): 5.67

t = 26.00 h

Biomass X (g/L): 86.72
Glucose S (g/L): 64.94
Lactic Acid P (g/L): 47.59
Maltose M (g/L): 41.26
Volume V (L): 5.58

t = 27.00 h

Biomass X (g/L): 91.00
Glucose S (g/L): 56.97
Lactic Acid P (g/L): 79.03
Maltose M (g/L): 42.22
Volume V (L): 5.57

t = 28.00 h

Biomass X (g/L): 91.07
Glucose S (g/L): 71.39
Lactic Acid P (g/L): 63.46
Maltose M (g/L): 35.74
Volume V (L): 5.61

t = 29.00 h

Biomass X (g/L): 89.84
Glucose S (g/L): 75.26
Lactic Acid P (g/L): 80.27
Maltose M (g/L): 33.26
Volume V (L): 5.81

t = 30.00 h

Biomass X (g/L): 121.38
Glucose S (g/L): 62.07
Lactic Acid P (g/L): 55.52
Maltose M (g/L): 38.89
Volume V (L): 5.75

t = 31.00 h

Biomass X (g/L): 84.11
Glucose S (g/L): 58.23
Lactic Acid P (g/L): 79.73
Maltose M (g/L): 31.28
Volume V (L): 5.72

t = 32.00 h

Biomass X (g/L): 84.47
Glucose S (g/L): 71.92
Lactic Acid P (g/L): 64.47
Maltose M (g/L): 31.31
Volume V (L): 5.85

t = 33.00 h

Biomass X (g/L): 99.21
Glucose S (g/L): 57.47
Lactic Acid P (g/L): 57.15
Maltose M (g/L): 31.68
Volume V (L): 5.77

t = 34.00 h

Biomass X (g/L): 86.05
Glucose S (g/L): 74.26
Lactic Acid P (g/L): 87.31
Maltose M (g/L): 33.41

Volume V (L): 6.03

t = 35.00 h

Biomass X (g/L): 89.53

Glucose S (g/L): 61.79

Lactic Acid P (g/L): 82.66

Maltose M (g/L): 29.92

Volume V (L): 5.74

t = 36.00 h

Biomass X (g/L): 81.47

Glucose S (g/L): 60.65

Lactic Acid P (g/L): 74.18

Maltose M (g/L): 27.50

Volume V (L): 5.88

t = 37.00 h

Biomass X (g/L): 98.22

Glucose S (g/L): 68.64

Lactic Acid P (g/L): 72.94

Maltose M (g/L): 29.77

Volume V (L): 5.72

t = 38.00 h

Biomass X (g/L): 98.54

Glucose S (g/L): 61.16

Lactic Acid P (g/L): 77.59

Maltose M (g/L): 35.31

Volume V (L): 5.91

t = 39.00 h

Biomass X (g/L): 90.12

Glucose S (g/L): 63.19

Lactic Acid P (g/L): 75.63

Maltose M (g/L): 28.54

Volume V (L): 5.73

t = 40.00 h

Biomass X (g/L): 101.04

Glucose S (g/L): 69.53

Lactic Acid P (g/L): 66.12

Maltose M (g/L): 31.27

Volume V (L): 6.13

t = 41.00 h

Biomass X (g/L): 74.18

Glucose S (g/L): 53.32

Lactic Acid P (g/L): 70.45

Maltose M (g/L): 38.52

Volume V (L): 6.16

t = 42.00 h

Biomass X (g/L): 86.32

Glucose S (g/L): 77.36

Lactic Acid P (g/L): 94.95

Maltose M (g/L): 31.50

Volume V (L): 6.18

t = 43.00 h

Biomass X (g/L): 92.01

Glucose S (g/L): 68.13

Lactic Acid P (g/L): 102.60

Maltose M (g/L): 32.45

Volume V (L): 6.09

t = 44.00 h

Biomass X (g/L): 81.54

Glucose S (g/L): 60.45

Lactic Acid P (g/L): 91.78

Maltose M (g/L): 27.81

Volume V (L): 5.98

t = 45.00 h

Biomass X (g/L): 75.51

Glucose S (g/L): 62.24

Lactic Acid P (g/L): 81.21

Maltose M (g/L): 37.84

Volume V (L): 6.02

t = 46.00 h

Biomass X (g/L): 90.61

Glucose S (g/L): 74.22

Lactic Acid P (g/L): 73.33

Maltose M (g/L): 28.48

Volume V (L): 6.40

t = 47.00 h

Biomass X (g/L): 101.08

Glucose S (g/L): 72.12

Lactic Acid P (g/L): 69.05

Maltose M (g/L): 29.20

Volume V (L): 6.41

t = 48.00 h

Biomass X (g/L): 107.21

Glucose S (g/L): 76.94

Lactic Acid P (g/L): 70.09

Maltose M (g/L): 34.03

Volume V (L): 6.24

t = 49.00 h

Biomass X (g/L): 87.99

Glucose S (g/L): 65.00

Lactic Acid P (g/L): 72.26

Maltose M (g/L): 28.71

Volume V (L): 6.16

t = 50.00 h

Biomass X (g/L): 101.34

Glucose S (g/L): 70.96

Lactic Acid P (g/L): 100.00

Maltose M (g/L): 29.17

Volume V (L): 6.48

t = 51.00 h

Biomass X (g/L): 100.25

Glucose S (g/L): 71.95

Lactic Acid P (g/L): 96.84

Maltose M (g/L): 27.77

Volume V (L): 6.19

t = 52.00 h

Biomass X (g/L): 106.26

Glucose S (g/L): 73.75
Lactic Acid P (g/L): 86.43
Maltose M (g/L): 27.50
Volume V (L): 6.35

t = 53.00 h

Biomass X (g/L): 89.72
Glucose S (g/L): 76.51
Lactic Acid P (g/L): 69.45
Maltose M (g/L): 29.36
Volume V (L): 6.55

t = 54.00 h

Biomass X (g/L): 74.44
Glucose S (g/L): 65.92
Lactic Acid P (g/L): 88.54
Maltose M (g/L): 29.68
Volume V (L): 6.24

t = 55.00 h

Biomass X (g/L): 93.90
Glucose S (g/L): 69.69
Lactic Acid P (g/L): 77.37
Maltose M (g/L): 25.41
Volume V (L): 6.52

t = 56.00 h

Biomass X (g/L): 82.41
Glucose S (g/L): 67.02
Lactic Acid P (g/L): 58.26
Maltose M (g/L): 24.73
Volume V (L): 6.50

t = 57.00 h

Biomass X (g/L): 85.32
Glucose S (g/L): 70.54
Lactic Acid P (g/L): 83.27
Maltose M (g/L): 24.98
Volume V (L): 6.44

t = 58.00 h

Biomass X (g/L): 89.60
Glucose S (g/L): 75.19
Lactic Acid P (g/L): 72.10
Maltose M (g/L): 22.08
Volume V (L): 6.32

t = 59.00 h

Biomass X (g/L): 89.05
Glucose S (g/L): 61.63
Lactic Acid P (g/L): 72.71
Maltose M (g/L): 21.19
Volume V (L): 6.39

t = 60.00 h

Biomass X (g/L): 93.93
Glucose S (g/L): 78.10
Lactic Acid P (g/L): 78.62
Maltose M (g/L): 27.23
Volume V (L): 6.57

t = 61.00 h

Biomass X (g/L): 99.65
Glucose S (g/L): 65.14
Lactic Acid P (g/L): 76.37
Maltose M (g/L): 29.12
Volume V (L): 6.33

t = 62.00 h

Biomass X (g/L): 88.26
Glucose S (g/L): 70.28
Lactic Acid P (g/L): 81.32
Maltose M (g/L): 28.65
Volume V (L): 6.59

t = 63.00 h

Biomass X (g/L): 96.08
Glucose S (g/L): 79.63
Lactic Acid P (g/L): 81.07
Maltose M (g/L): 17.56
Volume V (L): 6.68

t = 64.00 h

Biomass X (g/L): 91.91
Glucose S (g/L): 74.51
Lactic Acid P (g/L): 63.14
Maltose M (g/L): 24.45
Volume V (L): 6.77

t = 65.00 h

Biomass X (g/L): 101.37
Glucose S (g/L): 74.83
Lactic Acid P (g/L): 97.78
Maltose M (g/L): 21.36
Volume V (L): 6.44

t = 66.00 h

Biomass X (g/L): 89.40
Glucose S (g/L): 66.15
Lactic Acid P (g/L): 72.64
Maltose M (g/L): 25.29
Volume V (L): 6.87

t = 67.00 h

Biomass X (g/L): 90.35
Glucose S (g/L): 67.21
Lactic Acid P (g/L): 73.17
Maltose M (g/L): 23.20
Volume V (L): 6.96

t = 68.00 h

Biomass X (g/L): 87.06
Glucose S (g/L): 79.11
Lactic Acid P (g/L): 79.71
Maltose M (g/L): 26.35
Volume V (L): 6.79

t = 69.00 h

Biomass X (g/L): 89.08
Glucose S (g/L): 70.38
Lactic Acid P (g/L): 82.15
Maltose M (g/L): 24.10
Volume V (L): 6.59

t = 70.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 71.18
Lactic Acid P (g/L): 98.24
Maltose M (g/L): 17.63
Volume V (L): 6.72

t = 71.00 h

Biomass X (g/L): 81.12
Glucose S (g/L): 80.00
Lactic Acid P (g/L): 86.39
Maltose M (g/L): 20.52
Volume V (L): 6.79

t = 72.00 h

Biomass X (g/L): 98.28
Glucose S (g/L): 68.42
Lactic Acid P (g/L): 89.60
Maltose M (g/L): 23.56
Volume V (L): 6.87

t = 73.00 h

Biomass X (g/L): 78.59
Glucose S (g/L): 76.42
Lactic Acid P (g/L): 86.84
Maltose M (g/L): 23.65
Volume V (L): 6.85

t = 74.00 h

Biomass X (g/L): 86.38
Glucose S (g/L): 75.34
Lactic Acid P (g/L): 86.59
Maltose M (g/L): 18.46
Volume V (L): 6.92

t = 75.00 h

Biomass X (g/L): 100.15
Glucose S (g/L): 75.31
Lactic Acid P (g/L): 87.57
Maltose M (g/L): 17.83

Volume V (L): 6.89

Batch 51

Initial Conditions:

Batch: 51.00

X0 (g/L): 5.34

S0 (g/L): 68.82

M0 (g/L): 45.21

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.66

F_vol (L/h): 0.08

S_feed (g/L): 181.05

M_feed (g/L): 118.95

V_final (L): 10.93

Batch 51 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 7.86
Glucose S (g/L): 72.71
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 45.81
Volume V (L): 5.40

t = 1.00 h

Biomass X (g/L): 2.75
Glucose S (g/L): 73.14
Lactic Acid P (g/L): 14.95
Maltose M (g/L): 45.02
Volume V (L): 5.27

t = 2.00 h

Biomass X (g/L): 6.19
Glucose S (g/L): 68.44
Lactic Acid P (g/L): 1.93
Maltose M (g/L): 44.49
Volume V (L): 5.27

t = 3.00 h

Biomass X (g/L): 6.72
Glucose S (g/L): 65.37
Lactic Acid P (g/L): 23.72
Maltose M (g/L): 45.51
Volume V (L): 5.73

t = 4.00 h

Biomass X (g/L): 33.78
Glucose S (g/L): 67.12
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 45.49
Volume V (L): 5.99

t = 5.00 h

Biomass X (g/L): 33.28
Glucose S (g/L): 65.50

Lactic Acid P (g/L): 5.73
Maltose M (g/L): 47.18
Volume V (L): 4.80

t = 6.00 h

Biomass X (g/L): 28.85
Glucose S (g/L): 66.43
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 46.61
Volume V (L): 4.98

t = 7.00 h

Biomass X (g/L): 40.68
Glucose S (g/L): 62.37
Lactic Acid P (g/L): 6.02
Maltose M (g/L): 47.76
Volume V (L): 5.18

t = 8.00 h

Biomass X (g/L): 40.29
Glucose S (g/L): 57.81
Lactic Acid P (g/L): 12.89
Maltose M (g/L): 46.18
Volume V (L): 4.92

t = 9.00 h

Biomass X (g/L): 39.01
Glucose S (g/L): 61.13
Lactic Acid P (g/L): 26.60
Maltose M (g/L): 47.55
Volume V (L): 5.68

t = 10.00 h

Biomass X (g/L): 27.42
Glucose S (g/L): 55.57
Lactic Acid P (g/L): 29.54
Maltose M (g/L): 49.26
Volume V (L): 6.08

t = 11.00 h

Biomass X (g/L): 43.50

Glucose S (g/L): 54.16
Lactic Acid P (g/L): 24.27
Maltose M (g/L): 47.27
Volume V (L): 5.86

t = 12.00 h

Biomass X (g/L): 46.77
Glucose S (g/L): 45.18
Lactic Acid P (g/L): 13.42
Maltose M (g/L): 47.74
Volume V (L): 5.70

t = 13.00 h

Biomass X (g/L): 42.82
Glucose S (g/L): 41.06
Lactic Acid P (g/L): 49.52
Maltose M (g/L): 47.63
Volume V (L): 6.56

t = 14.00 h

Biomass X (g/L): 42.79
Glucose S (g/L): 42.54
Lactic Acid P (g/L): 36.32
Maltose M (g/L): 45.90
Volume V (L): 6.17

t = 15.00 h

Biomass X (g/L): 64.65
Glucose S (g/L): 44.18
Lactic Acid P (g/L): 31.29
Maltose M (g/L): 47.58
Volume V (L): 6.35

t = 16.00 h

Biomass X (g/L): 65.78
Glucose S (g/L): 46.36
Lactic Acid P (g/L): 46.00
Maltose M (g/L): 47.34
Volume V (L): 6.81

t = 17.00 h

Biomass X (g/L): 54.03
Glucose S (g/L): 36.17
Lactic Acid P (g/L): 47.40
Maltose M (g/L): 46.86
Volume V (L): 6.29

t = 18.00 h

Biomass X (g/L): 60.24
Glucose S (g/L): 47.89
Lactic Acid P (g/L): 44.54
Maltose M (g/L): 46.32
Volume V (L): 7.40

t = 19.00 h

Biomass X (g/L): 80.37
Glucose S (g/L): 42.82
Lactic Acid P (g/L): 50.48
Maltose M (g/L): 47.59
Volume V (L): 6.29

t = 20.00 h

Biomass X (g/L): 62.73
Glucose S (g/L): 46.18
Lactic Acid P (g/L): 40.36
Maltose M (g/L): 46.17
Volume V (L): 7.53

t = 21.00 h

Biomass X (g/L): 74.30
Glucose S (g/L): 38.80
Lactic Acid P (g/L): 58.99
Maltose M (g/L): 48.69
Volume V (L): 6.20

t = 22.00 h

Biomass X (g/L): 72.29
Glucose S (g/L): 43.07
Lactic Acid P (g/L): 44.56
Maltose M (g/L): 46.96
Volume V (L): 6.99

t = 23.00 h

Biomass X (g/L): 80.19
Glucose S (g/L): 41.71
Lactic Acid P (g/L): 49.30
Maltose M (g/L): 45.78
Volume V (L): 6.51

t = 24.00 h

Biomass X (g/L): 68.36
Glucose S (g/L): 44.10
Lactic Acid P (g/L): 61.70
Maltose M (g/L): 46.29
Volume V (L): 7.43

t = 25.00 h

Biomass X (g/L): 78.38
Glucose S (g/L): 40.93
Lactic Acid P (g/L): 67.74
Maltose M (g/L): 46.40
Volume V (L): 6.75

t = 26.00 h

Biomass X (g/L): 80.05
Glucose S (g/L): 43.39
Lactic Acid P (g/L): 72.15
Maltose M (g/L): 48.55
Volume V (L): 7.19

t = 27.00 h

Biomass X (g/L): 81.45
Glucose S (g/L): 45.68
Lactic Acid P (g/L): 68.31
Maltose M (g/L): 45.72
Volume V (L): 6.83

t = 28.00 h

Biomass X (g/L): 80.74
Glucose S (g/L): 40.48
Lactic Acid P (g/L): 56.78
Maltose M (g/L): 46.25
Volume V (L): 7.66

t = 29.00 h

Biomass X (g/L): 83.92
Glucose S (g/L): 43.72
Lactic Acid P (g/L): 46.65
Maltose M (g/L): 45.79
Volume V (L): 6.98

t = 30.00 h

Biomass X (g/L): 102.08
Glucose S (g/L): 37.42
Lactic Acid P (g/L): 73.84
Maltose M (g/L): 46.54
Volume V (L): 8.17

t = 31.00 h

Biomass X (g/L): 106.11
Glucose S (g/L): 46.11
Lactic Acid P (g/L): 74.24
Maltose M (g/L): 45.58
Volume V (L): 6.98

t = 32.00 h

Biomass X (g/L): 76.10
Glucose S (g/L): 50.60
Lactic Acid P (g/L): 56.56
Maltose M (g/L): 44.94
Volume V (L): 6.89

t = 33.00 h

Biomass X (g/L): 60.52
Glucose S (g/L): 41.02
Lactic Acid P (g/L): 71.48
Maltose M (g/L): 45.60
Volume V (L): 8.06

t = 34.00 h

Biomass X (g/L): 88.62
Glucose S (g/L): 45.26
Lactic Acid P (g/L): 60.82
Maltose M (g/L): 45.38

Volume V (L): 7.41

t = 35.00 h

Biomass X (g/L): 85.03

Glucose S (g/L): 48.93

Lactic Acid P (g/L): 77.83

Maltose M (g/L): 45.01

Volume V (L): 7.76

t = 36.00 h

Biomass X (g/L): 93.63

Glucose S (g/L): 46.87

Lactic Acid P (g/L): 60.26

Maltose M (g/L): 44.91

Volume V (L): 7.69

t = 37.00 h

Biomass X (g/L): 86.48

Glucose S (g/L): 47.93

Lactic Acid P (g/L): 83.14

Maltose M (g/L): 46.20

Volume V (L): 7.30

t = 38.00 h

Biomass X (g/L): 84.53

Glucose S (g/L): 46.82

Lactic Acid P (g/L): 72.91

Maltose M (g/L): 44.81

Volume V (L): 7.97

t = 39.00 h

Biomass X (g/L): 94.89

Glucose S (g/L): 47.37

Lactic Acid P (g/L): 82.43

Maltose M (g/L): 43.99

Volume V (L): 8.68

t = 40.00 h

Biomass X (g/L): 96.36

Glucose S (g/L): 49.13

Lactic Acid P (g/L): 62.56

Maltose M (g/L): 44.39

Volume V (L): 8.64

t = 41.00 h

Biomass X (g/L): 71.20

Glucose S (g/L): 50.22

Lactic Acid P (g/L): 70.77

Maltose M (g/L): 44.67

Volume V (L): 8.35

t = 42.00 h

Biomass X (g/L): 76.72

Glucose S (g/L): 47.04

Lactic Acid P (g/L): 58.82

Maltose M (g/L): 43.36

Volume V (L): 8.85

t = 43.00 h

Biomass X (g/L): 83.56

Glucose S (g/L): 51.27

Lactic Acid P (g/L): 75.47

Maltose M (g/L): 45.66

Volume V (L): 7.90

t = 44.00 h

Biomass X (g/L): 85.56

Glucose S (g/L): 57.15

Lactic Acid P (g/L): 76.32

Maltose M (g/L): 44.87

Volume V (L): 8.33

t = 45.00 h

Biomass X (g/L): 82.50

Glucose S (g/L): 51.56

Lactic Acid P (g/L): 90.10

Maltose M (g/L): 45.18

Volume V (L): 8.75

t = 46.00 h

Biomass X (g/L): 88.81

Glucose S (g/L): 49.68

Lactic Acid P (g/L): 73.24

Maltose M (g/L): 44.15

Volume V (L): 8.58

t = 47.00 h

Biomass X (g/L): 94.15

Glucose S (g/L): 53.83

Lactic Acid P (g/L): 75.72

Maltose M (g/L): 41.93

Volume V (L): 8.60

t = 48.00 h

Biomass X (g/L): 101.06

Glucose S (g/L): 52.75

Lactic Acid P (g/L): 65.30

Maltose M (g/L): 44.77

Volume V (L): 8.23

t = 49.00 h

Biomass X (g/L): 90.14

Glucose S (g/L): 51.49

Lactic Acid P (g/L): 74.67

Maltose M (g/L): 42.61

Volume V (L): 7.89

t = 50.00 h

Biomass X (g/L): 76.48

Glucose S (g/L): 56.88

Lactic Acid P (g/L): 66.68

Maltose M (g/L): 42.51

Volume V (L): 9.34

t = 51.00 h

Biomass X (g/L): 83.14

Glucose S (g/L): 54.90

Lactic Acid P (g/L): 71.40

Maltose M (g/L): 43.12

Volume V (L): 9.15

t = 52.00 h

Biomass X (g/L): 83.76

Glucose S (g/L): 55.88
Lactic Acid P (g/L): 72.18
Maltose M (g/L): 43.97
Volume V (L): 9.54

t = 53.00 h

Biomass X (g/L): 89.30
Glucose S (g/L): 59.05
Lactic Acid P (g/L): 67.70
Maltose M (g/L): 43.52
Volume V (L): 8.98

t = 54.00 h

Biomass X (g/L): 88.02
Glucose S (g/L): 54.42
Lactic Acid P (g/L): 71.67
Maltose M (g/L): 43.23
Volume V (L): 8.93

t = 55.00 h

Biomass X (g/L): 103.75
Glucose S (g/L): 54.46
Lactic Acid P (g/L): 67.97
Maltose M (g/L): 42.05
Volume V (L): 9.35

t = 56.00 h

Biomass X (g/L): 79.74
Glucose S (g/L): 62.80
Lactic Acid P (g/L): 68.15
Maltose M (g/L): 41.86
Volume V (L): 9.83

t = 57.00 h

Biomass X (g/L): 87.69
Glucose S (g/L): 55.25
Lactic Acid P (g/L): 81.44
Maltose M (g/L): 42.24
Volume V (L): 9.40

t = 58.00 h

Biomass X (g/L): 106.34
Glucose S (g/L): 58.97
Lactic Acid P (g/L): 73.56
Maltose M (g/L): 41.67
Volume V (L): 9.97

t = 59.00 h

Biomass X (g/L): 78.23
Glucose S (g/L): 57.05
Lactic Acid P (g/L): 75.31
Maltose M (g/L): 42.34
Volume V (L): 9.19

t = 60.00 h

Biomass X (g/L): 85.06
Glucose S (g/L): 61.89
Lactic Acid P (g/L): 59.31
Maltose M (g/L): 42.12
Volume V (L): 9.67

t = 61.00 h

Biomass X (g/L): 74.55
Glucose S (g/L): 65.47
Lactic Acid P (g/L): 76.38
Maltose M (g/L): 41.74
Volume V (L): 9.79

t = 62.00 h

Biomass X (g/L): 84.20
Glucose S (g/L): 65.14
Lactic Acid P (g/L): 77.53
Maltose M (g/L): 38.98
Volume V (L): 10.49

t = 63.00 h

Biomass X (g/L): 75.85
Glucose S (g/L): 73.03
Lactic Acid P (g/L): 69.37
Maltose M (g/L): 41.01
Volume V (L): 10.08

t = 64.00 h

Biomass X (g/L): 93.10
Glucose S (g/L): 69.57
Lactic Acid P (g/L): 91.84
Maltose M (g/L): 40.64
Volume V (L): 10.89

t = 65.00 h

Biomass X (g/L): 84.39
Glucose S (g/L): 63.84
Lactic Acid P (g/L): 52.75
Maltose M (g/L): 40.69
Volume V (L): 9.50

t = 66.00 h

Biomass X (g/L): 73.93
Glucose S (g/L): 63.65
Lactic Acid P (g/L): 80.50
Maltose M (g/L): 39.75
Volume V (L): 10.28

t = 67.00 h

Biomass X (g/L): 90.68
Glucose S (g/L): 76.67
Lactic Acid P (g/L): 80.02
Maltose M (g/L): 39.10
Volume V (L): 10.38

t = 68.00 h

Biomass X (g/L): 84.23
Glucose S (g/L): 72.10
Lactic Acid P (g/L): 78.64
Maltose M (g/L): 40.40
Volume V (L): 10.67

t = 69.00 h

Biomass X (g/L): 91.43
Glucose S (g/L): 71.25
Lactic Acid P (g/L): 75.52
Maltose M (g/L): 39.55
Volume V (L): 10.07

t = 70.00 h

Biomass X (g/L): 71.27
Glucose S (g/L): 70.25
Lactic Acid P (g/L): 79.03
Maltose M (g/L): 40.15
Volume V (L): 11.34

t = 71.00 h

Biomass X (g/L): 85.55
Glucose S (g/L): 71.51
Lactic Acid P (g/L): 82.03
Maltose M (g/L): 38.94
Volume V (L): 10.62

t = 72.00 h

Biomass X (g/L): 89.23
Glucose S (g/L): 66.58
Lactic Acid P (g/L): 67.87
Maltose M (g/L): 40.49
Volume V (L): 10.18

t = 73.00 h

Biomass X (g/L): 77.67
Glucose S (g/L): 72.39
Lactic Acid P (g/L): 77.21
Maltose M (g/L): 39.97
Volume V (L): 10.77

t = 74.00 h

Biomass X (g/L): 69.54
Glucose S (g/L): 72.73
Lactic Acid P (g/L): 73.01
Maltose M (g/L): 39.22
Volume V (L): 10.70

t = 75.00 h

Biomass X (g/L): 80.96
Glucose S (g/L): 70.25
Lactic Acid P (g/L): 64.98
Maltose M (g/L): 38.62

Volume V (L): 12.25

Batch 52

Initial Conditions:

Batch: 52.00

X0 (g/L): 8.83

S0 (g/L): 139.52

M0 (g/L): 50.25

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.36

F_vol (L/h): 0.04

S_feed (g/L): 220.56

M_feed (g/L): 79.44

V_final (L): 7.78

Batch 52 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 19.52
Glucose S (g/L): 134.74
Lactic Acid P (g/L): 14.70
Maltose M (g/L): 49.22
Volume V (L): 4.67

t = 1.00 h

Biomass X (g/L): 12.50
Glucose S (g/L): 140.33
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 47.16
Volume V (L): 5.08

t = 2.00 h

Biomass X (g/L): 14.78
Glucose S (g/L): 142.26
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 47.79
Volume V (L): 5.04

t = 3.00 h

Biomass X (g/L): 30.22
Glucose S (g/L): 140.01
Lactic Acid P (g/L): 4.52
Maltose M (g/L): 48.73
Volume V (L): 5.03

t = 4.00 h

Biomass X (g/L): 35.90
Glucose S (g/L): 126.54
Lactic Acid P (g/L): 4.59
Maltose M (g/L): 47.59
Volume V (L): 5.23

t = 5.00 h

Biomass X (g/L): 1.65
Glucose S (g/L): 125.97

Lactic Acid P (g/L): 9.70
Maltose M (g/L): 47.85
Volume V (L): 5.23

t = 6.00 h

Biomass X (g/L): 43.54
Glucose S (g/L): 122.46
Lactic Acid P (g/L): 18.55
Maltose M (g/L): 48.51
Volume V (L): 5.32

t = 7.00 h

Biomass X (g/L): 42.90
Glucose S (g/L): 119.09
Lactic Acid P (g/L): 41.92
Maltose M (g/L): 48.99
Volume V (L): 5.29

t = 8.00 h

Biomass X (g/L): 34.64
Glucose S (g/L): 106.27
Lactic Acid P (g/L): 22.41
Maltose M (g/L): 45.52
Volume V (L): 5.39

t = 9.00 h

Biomass X (g/L): 61.96
Glucose S (g/L): 117.88
Lactic Acid P (g/L): 49.07
Maltose M (g/L): 41.14
Volume V (L): 5.35

t = 10.00 h

Biomass X (g/L): 80.47
Glucose S (g/L): 109.19
Lactic Acid P (g/L): 44.64
Maltose M (g/L): 42.76
Volume V (L): 5.22

t = 11.00 h

Biomass X (g/L): 67.44

Glucose S (g/L): 106.15
Lactic Acid P (g/L): 43.45
Maltose M (g/L): 40.13
Volume V (L): 5.41

t = 12.00 h

Biomass X (g/L): 76.72
Glucose S (g/L): 104.78
Lactic Acid P (g/L): 49.87
Maltose M (g/L): 47.06
Volume V (L): 5.29

t = 13.00 h

Biomass X (g/L): 87.23
Glucose S (g/L): 98.46
Lactic Acid P (g/L): 45.36
Maltose M (g/L): 47.30
Volume V (L): 5.31

t = 14.00 h

Biomass X (g/L): 63.39
Glucose S (g/L): 87.33
Lactic Acid P (g/L): 60.31
Maltose M (g/L): 43.03
Volume V (L): 5.44

t = 15.00 h

Biomass X (g/L): 55.58
Glucose S (g/L): 90.56
Lactic Acid P (g/L): 55.48
Maltose M (g/L): 43.07
Volume V (L): 5.53

t = 16.00 h

Biomass X (g/L): 83.53
Glucose S (g/L): 80.26
Lactic Acid P (g/L): 35.59
Maltose M (g/L): 40.86
Volume V (L): 5.60

t = 17.00 h

Biomass X (g/L): 76.41
Glucose S (g/L): 90.74
Lactic Acid P (g/L): 56.70
Maltose M (g/L): 40.67
Volume V (L): 5.83

t = 18.00 h

Biomass X (g/L): 66.79
Glucose S (g/L): 86.19
Lactic Acid P (g/L): 63.19
Maltose M (g/L): 34.71
Volume V (L): 5.56

t = 19.00 h

Biomass X (g/L): 81.63
Glucose S (g/L): 85.92
Lactic Acid P (g/L): 54.20
Maltose M (g/L): 38.45
Volume V (L): 5.75

t = 20.00 h

Biomass X (g/L): 91.30
Glucose S (g/L): 92.21
Lactic Acid P (g/L): 59.70
Maltose M (g/L): 35.73
Volume V (L): 5.77

t = 21.00 h

Biomass X (g/L): 83.67
Glucose S (g/L): 81.94
Lactic Acid P (g/L): 53.76
Maltose M (g/L): 38.49
Volume V (L): 5.64

t = 22.00 h

Biomass X (g/L): 78.33
Glucose S (g/L): 82.51
Lactic Acid P (g/L): 47.46
Maltose M (g/L): 37.08
Volume V (L): 5.83

t = 23.00 h

Biomass X (g/L): 85.08
Glucose S (g/L): 83.59
Lactic Acid P (g/L): 46.89
Maltose M (g/L): 42.20
Volume V (L): 6.08

t = 24.00 h

Biomass X (g/L): 88.06
Glucose S (g/L): 72.47
Lactic Acid P (g/L): 48.07
Maltose M (g/L): 38.90
Volume V (L): 5.93

t = 25.00 h

Biomass X (g/L): 69.42
Glucose S (g/L): 81.20
Lactic Acid P (g/L): 70.71
Maltose M (g/L): 36.71
Volume V (L): 5.98

t = 26.00 h

Biomass X (g/L): 88.32
Glucose S (g/L): 68.18
Lactic Acid P (g/L): 40.59
Maltose M (g/L): 34.76
Volume V (L): 5.81

t = 27.00 h

Biomass X (g/L): 76.71
Glucose S (g/L): 78.92
Lactic Acid P (g/L): 59.35
Maltose M (g/L): 39.75
Volume V (L): 6.44

t = 28.00 h

Biomass X (g/L): 97.65
Glucose S (g/L): 87.07
Lactic Acid P (g/L): 59.18
Maltose M (g/L): 34.60
Volume V (L): 5.89

t = 29.00 h

Biomass X (g/L): 75.53
Glucose S (g/L): 84.64
Lactic Acid P (g/L): 64.68
Maltose M (g/L): 35.21
Volume V (L): 5.70

t = 30.00 h

Biomass X (g/L): 102.15
Glucose S (g/L): 78.61
Lactic Acid P (g/L): 49.92
Maltose M (g/L): 34.02
Volume V (L): 6.30

t = 31.00 h

Biomass X (g/L): 88.04
Glucose S (g/L): 82.27
Lactic Acid P (g/L): 72.79
Maltose M (g/L): 32.40
Volume V (L): 6.12

t = 32.00 h

Biomass X (g/L): 86.99
Glucose S (g/L): 75.90
Lactic Acid P (g/L): 56.94
Maltose M (g/L): 33.26
Volume V (L): 6.12

t = 33.00 h

Biomass X (g/L): 95.78
Glucose S (g/L): 82.14
Lactic Acid P (g/L): 81.17
Maltose M (g/L): 27.34
Volume V (L): 6.10

t = 34.00 h

Biomass X (g/L): 84.19
Glucose S (g/L): 99.09
Lactic Acid P (g/L): 64.50
Maltose M (g/L): 28.14

Volume V (L): 6.68

t = 35.00 h

Biomass X (g/L): 96.03

Glucose S (g/L): 92.99

Lactic Acid P (g/L): 70.21

Maltose M (g/L): 33.06

Volume V (L): 6.38

t = 36.00 h

Biomass X (g/L): 85.89

Glucose S (g/L): 82.83

Lactic Acid P (g/L): 71.70

Maltose M (g/L): 36.11

Volume V (L): 6.46

t = 37.00 h

Biomass X (g/L): 87.80

Glucose S (g/L): 83.62

Lactic Acid P (g/L): 74.55

Maltose M (g/L): 32.38

Volume V (L): 6.41

t = 38.00 h

Biomass X (g/L): 97.68

Glucose S (g/L): 77.25

Lactic Acid P (g/L): 72.97

Maltose M (g/L): 29.08

Volume V (L): 6.47

t = 39.00 h

Biomass X (g/L): 82.02

Glucose S (g/L): 78.48

Lactic Acid P (g/L): 68.86

Maltose M (g/L): 26.76

Volume V (L): 6.76

t = 40.00 h

Biomass X (g/L): 80.78

Glucose S (g/L): 85.39

Lactic Acid P (g/L): 76.85

Maltose M (g/L): 32.63

Volume V (L): 6.43

t = 41.00 h

Biomass X (g/L): 96.19

Glucose S (g/L): 80.53

Lactic Acid P (g/L): 78.46

Maltose M (g/L): 28.57

Volume V (L): 6.20

t = 42.00 h

Biomass X (g/L): 93.39

Glucose S (g/L): 88.98

Lactic Acid P (g/L): 66.76

Maltose M (g/L): 31.63

Volume V (L): 6.75

t = 43.00 h

Biomass X (g/L): 95.35

Glucose S (g/L): 93.07

Lactic Acid P (g/L): 78.83

Maltose M (g/L): 30.63

Volume V (L): 6.70

t = 44.00 h

Biomass X (g/L): 96.92

Glucose S (g/L): 86.22

Lactic Acid P (g/L): 86.71

Maltose M (g/L): 30.58

Volume V (L): 6.65

t = 45.00 h

Biomass X (g/L): 94.11

Glucose S (g/L): 85.16

Lactic Acid P (g/L): 82.28

Maltose M (g/L): 29.06

Volume V (L): 6.77

t = 46.00 h

Biomass X (g/L): 99.64

Glucose S (g/L): 75.20

Lactic Acid P (g/L): 84.84

Maltose M (g/L): 29.75

Volume V (L): 6.62

t = 47.00 h

Biomass X (g/L): 94.35

Glucose S (g/L): 89.44

Lactic Acid P (g/L): 77.69

Maltose M (g/L): 28.08

Volume V (L): 6.75

t = 48.00 h

Biomass X (g/L): 72.82

Glucose S (g/L): 78.86

Lactic Acid P (g/L): 67.11

Maltose M (g/L): 26.38

Volume V (L): 6.75

t = 49.00 h

Biomass X (g/L): 75.35

Glucose S (g/L): 83.33

Lactic Acid P (g/L): 68.65

Maltose M (g/L): 25.20

Volume V (L): 6.68

t = 50.00 h

Biomass X (g/L): 112.64

Glucose S (g/L): 91.55

Lactic Acid P (g/L): 80.03

Maltose M (g/L): 21.00

Volume V (L): 7.12

t = 51.00 h

Biomass X (g/L): 92.88

Glucose S (g/L): 92.26

Lactic Acid P (g/L): 86.43

Maltose M (g/L): 26.87

Volume V (L): 6.58

t = 52.00 h

Biomass X (g/L): 102.40

Glucose S (g/L): 83.97
Lactic Acid P (g/L): 104.80
Maltose M (g/L): 31.34
Volume V (L): 7.45

t = 53.00 h

Biomass X (g/L): 89.05
Glucose S (g/L): 84.61
Lactic Acid P (g/L): 87.34
Maltose M (g/L): 28.90
Volume V (L): 7.16

t = 54.00 h

Biomass X (g/L): 91.82
Glucose S (g/L): 83.25
Lactic Acid P (g/L): 66.45
Maltose M (g/L): 27.27
Volume V (L): 6.61

t = 55.00 h

Biomass X (g/L): 87.62
Glucose S (g/L): 80.70
Lactic Acid P (g/L): 70.65
Maltose M (g/L): 24.36
Volume V (L): 6.99

t = 56.00 h

Biomass X (g/L): 117.16
Glucose S (g/L): 84.26
Lactic Acid P (g/L): 100.80
Maltose M (g/L): 28.94
Volume V (L): 7.04

t = 57.00 h

Biomass X (g/L): 94.45
Glucose S (g/L): 78.95
Lactic Acid P (g/L): 76.54
Maltose M (g/L): 25.61
Volume V (L): 7.20

t = 58.00 h

Biomass X (g/L): 86.23
Glucose S (g/L): 92.47
Lactic Acid P (g/L): 79.34
Maltose M (g/L): 30.09
Volume V (L): 6.93

t = 59.00 h

Biomass X (g/L): 74.66
Glucose S (g/L): 91.13
Lactic Acid P (g/L): 69.65
Maltose M (g/L): 22.77
Volume V (L): 7.19

t = 60.00 h

Biomass X (g/L): 111.37
Glucose S (g/L): 92.41
Lactic Acid P (g/L): 94.67
Maltose M (g/L): 23.20
Volume V (L): 7.20

t = 61.00 h

Biomass X (g/L): 89.40
Glucose S (g/L): 84.02
Lactic Acid P (g/L): 84.41
Maltose M (g/L): 22.72
Volume V (L): 7.10

t = 62.00 h

Biomass X (g/L): 98.43
Glucose S (g/L): 82.96
Lactic Acid P (g/L): 72.09
Maltose M (g/L): 20.75
Volume V (L): 7.10

t = 63.00 h

Biomass X (g/L): 99.35
Glucose S (g/L): 90.45
Lactic Acid P (g/L): 70.13
Maltose M (g/L): 22.06
Volume V (L): 7.54

t = 64.00 h

Biomass X (g/L): 85.70
Glucose S (g/L): 90.83
Lactic Acid P (g/L): 83.10
Maltose M (g/L): 24.75
Volume V (L): 6.99

t = 65.00 h

Biomass X (g/L): 97.04
Glucose S (g/L): 89.94
Lactic Acid P (g/L): 80.70
Maltose M (g/L): 22.28
Volume V (L): 7.52

t = 66.00 h

Biomass X (g/L): 83.81
Glucose S (g/L): 89.44
Lactic Acid P (g/L): 88.61
Maltose M (g/L): 13.35
Volume V (L): 7.62

t = 67.00 h

Biomass X (g/L): 79.65
Glucose S (g/L): 101.26
Lactic Acid P (g/L): 85.99
Maltose M (g/L): 26.86
Volume V (L): 7.43

t = 68.00 h

Biomass X (g/L): 101.36
Glucose S (g/L): 94.19
Lactic Acid P (g/L): 78.18
Maltose M (g/L): 19.01
Volume V (L): 7.53

t = 69.00 h

Biomass X (g/L): 87.66
Glucose S (g/L): 103.38
Lactic Acid P (g/L): 75.61
Maltose M (g/L): 25.14
Volume V (L): 7.82

t = 70.00 h

Biomass X (g/L): 94.31
Glucose S (g/L): 99.66
Lactic Acid P (g/L): 79.54
Maltose M (g/L): 23.41
Volume V (L): 7.44

t = 71.00 h

Biomass X (g/L): 92.97
Glucose S (g/L): 102.47
Lactic Acid P (g/L): 75.86
Maltose M (g/L): 29.03
Volume V (L): 7.50

t = 72.00 h

Biomass X (g/L): 95.04
Glucose S (g/L): 86.82
Lactic Acid P (g/L): 80.14
Maltose M (g/L): 25.74
Volume V (L): 7.56

t = 73.00 h

Biomass X (g/L): 72.46
Glucose S (g/L): 96.71
Lactic Acid P (g/L): 81.73
Maltose M (g/L): 21.40
Volume V (L): 7.96

t = 74.00 h

Biomass X (g/L): 82.10
Glucose S (g/L): 91.13
Lactic Acid P (g/L): 72.90
Maltose M (g/L): 20.68
Volume V (L): 8.03

t = 75.00 h

Biomass X (g/L): 84.51
Glucose S (g/L): 84.36
Lactic Acid P (g/L): 88.06
Maltose M (g/L): 26.64

Volume V (L): 7.47

Batch 53

Initial Conditions:

Batch: 53.00

X0 (g/L): 5.13

S0 (g/L): 88.43

M0 (g/L): 111.07

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.26

F_vol (L/h): 0.03

S_feed (g/L): 132.97

M_feed (g/L): 167.03

V_final (L): 7.43

Batch 53 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 15.49
Glucose S (g/L): 88.09
Lactic Acid P (g/L): 13.53
Maltose M (g/L): 124.29
Volume V (L): 4.66

t = 1.00 h

Biomass X (g/L): 15.38
Glucose S (g/L): 94.47
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 106.61
Volume V (L): 4.69

t = 2.00 h

Biomass X (g/L): 25.29
Glucose S (g/L): 90.98
Lactic Acid P (g/L): 8.38
Maltose M (g/L): 115.57
Volume V (L): 5.19

t = 3.00 h

Biomass X (g/L): 16.19
Glucose S (g/L): 79.76
Lactic Acid P (g/L): 4.85
Maltose M (g/L): 102.83
Volume V (L): 5.19

t = 4.00 h

Biomass X (g/L): 15.30
Glucose S (g/L): 91.41
Lactic Acid P (g/L): 20.44
Maltose M (g/L): 110.00
Volume V (L): 5.16

t = 5.00 h

Biomass X (g/L): 31.63
Glucose S (g/L): 88.55

Lactic Acid P (g/L): 2.98
Maltose M (g/L): 108.48
Volume V (L): 4.99

t = 6.00 h

Biomass X (g/L): 31.59
Glucose S (g/L): 86.27
Lactic Acid P (g/L): 14.70
Maltose M (g/L): 107.23
Volume V (L): 5.34

t = 7.00 h

Biomass X (g/L): 52.37
Glucose S (g/L): 73.00
Lactic Acid P (g/L): 0.99
Maltose M (g/L): 102.03
Volume V (L): 5.36

t = 8.00 h

Biomass X (g/L): 44.14
Glucose S (g/L): 74.55
Lactic Acid P (g/L): 2.33
Maltose M (g/L): 106.96
Volume V (L): 5.28

t = 9.00 h

Biomass X (g/L): 40.85
Glucose S (g/L): 71.76
Lactic Acid P (g/L): 30.95
Maltose M (g/L): 99.01
Volume V (L): 5.06

t = 10.00 h

Biomass X (g/L): 32.03
Glucose S (g/L): 67.84
Lactic Acid P (g/L): 11.18
Maltose M (g/L): 106.25
Volume V (L): 5.24

t = 11.00 h

Biomass X (g/L): 42.72

Glucose S (g/L): 74.48
Lactic Acid P (g/L): 29.95
Maltose M (g/L): 83.21
Volume V (L): 5.52

t = 12.00 h

Biomass X (g/L): 38.20
Glucose S (g/L): 65.48
Lactic Acid P (g/L): 45.39
Maltose M (g/L): 94.63
Volume V (L): 5.65

t = 13.00 h

Biomass X (g/L): 54.71
Glucose S (g/L): 63.08
Lactic Acid P (g/L): 45.84
Maltose M (g/L): 92.87
Volume V (L): 5.36

t = 14.00 h

Biomass X (g/L): 70.04
Glucose S (g/L): 63.20
Lactic Acid P (g/L): 34.80
Maltose M (g/L): 83.65
Volume V (L): 5.40

t = 15.00 h

Biomass X (g/L): 63.73
Glucose S (g/L): 61.50
Lactic Acid P (g/L): 28.32
Maltose M (g/L): 92.23
Volume V (L): 5.45

t = 16.00 h

Biomass X (g/L): 77.23
Glucose S (g/L): 45.70
Lactic Acid P (g/L): 51.31
Maltose M (g/L): 77.55
Volume V (L): 5.57

t = 17.00 h

Biomass X (g/L): 72.83
Glucose S (g/L): 54.26
Lactic Acid P (g/L): 61.69
Maltose M (g/L): 81.76
Volume V (L): 5.44

t = 18.00 h

Biomass X (g/L): 67.59
Glucose S (g/L): 50.40
Lactic Acid P (g/L): 44.00
Maltose M (g/L): 71.85
Volume V (L): 5.02

t = 19.00 h

Biomass X (g/L): 74.78
Glucose S (g/L): 56.16
Lactic Acid P (g/L): 44.47
Maltose M (g/L): 93.79
Volume V (L): 5.85

t = 20.00 h

Biomass X (g/L): 80.65
Glucose S (g/L): 52.94
Lactic Acid P (g/L): 48.91
Maltose M (g/L): 81.03
Volume V (L): 5.99

t = 21.00 h

Biomass X (g/L): 78.95
Glucose S (g/L): 53.50
Lactic Acid P (g/L): 61.25
Maltose M (g/L): 95.50
Volume V (L): 5.56

t = 22.00 h

Biomass X (g/L): 80.61
Glucose S (g/L): 45.42
Lactic Acid P (g/L): 69.16
Maltose M (g/L): 85.56
Volume V (L): 5.50

t = 23.00 h

Biomass X (g/L): 72.54
Glucose S (g/L): 51.60
Lactic Acid P (g/L): 46.25
Maltose M (g/L): 82.20
Volume V (L): 5.72

t = 24.00 h

Biomass X (g/L): 77.24
Glucose S (g/L): 56.61
Lactic Acid P (g/L): 44.96
Maltose M (g/L): 74.45
Volume V (L): 5.74

t = 25.00 h

Biomass X (g/L): 106.06
Glucose S (g/L): 57.58
Lactic Acid P (g/L): 83.83
Maltose M (g/L): 70.45
Volume V (L): 6.00

t = 26.00 h

Biomass X (g/L): 96.47
Glucose S (g/L): 55.03
Lactic Acid P (g/L): 60.19
Maltose M (g/L): 79.65
Volume V (L): 5.85

t = 27.00 h

Biomass X (g/L): 69.56
Glucose S (g/L): 47.55
Lactic Acid P (g/L): 65.43
Maltose M (g/L): 74.82
Volume V (L): 5.72

t = 28.00 h

Biomass X (g/L): 61.18
Glucose S (g/L): 57.16
Lactic Acid P (g/L): 57.10
Maltose M (g/L): 65.64
Volume V (L): 5.65

t = 29.00 h

Biomass X (g/L): 87.04
Glucose S (g/L): 55.35
Lactic Acid P (g/L): 69.61
Maltose M (g/L): 63.56
Volume V (L): 6.11

t = 30.00 h

Biomass X (g/L): 96.10
Glucose S (g/L): 52.48
Lactic Acid P (g/L): 75.06
Maltose M (g/L): 66.28
Volume V (L): 6.09

t = 31.00 h

Biomass X (g/L): 82.14
Glucose S (g/L): 45.75
Lactic Acid P (g/L): 69.37
Maltose M (g/L): 83.44
Volume V (L): 6.21

t = 32.00 h

Biomass X (g/L): 98.52
Glucose S (g/L): 56.93
Lactic Acid P (g/L): 84.28
Maltose M (g/L): 85.12
Volume V (L): 5.85

t = 33.00 h

Biomass X (g/L): 89.08
Glucose S (g/L): 51.59
Lactic Acid P (g/L): 64.67
Maltose M (g/L): 71.23
Volume V (L): 6.42

t = 34.00 h

Biomass X (g/L): 87.19
Glucose S (g/L): 55.37
Lactic Acid P (g/L): 65.97
Maltose M (g/L): 70.03

Volume V (L): 5.76

t = 35.00 h

Biomass X (g/L): 97.01

Glucose S (g/L): 54.73

Lactic Acid P (g/L): 71.47

Maltose M (g/L): 66.07

Volume V (L): 6.27

t = 36.00 h

Biomass X (g/L): 84.06

Glucose S (g/L): 60.50

Lactic Acid P (g/L): 74.37

Maltose M (g/L): 61.39

Volume V (L): 5.79

t = 37.00 h

Biomass X (g/L): 86.86

Glucose S (g/L): 54.82

Lactic Acid P (g/L): 72.62

Maltose M (g/L): 75.61

Volume V (L): 6.04

t = 38.00 h

Biomass X (g/L): 96.94

Glucose S (g/L): 55.65

Lactic Acid P (g/L): 58.31

Maltose M (g/L): 78.08

Volume V (L): 6.53

t = 39.00 h

Biomass X (g/L): 89.92

Glucose S (g/L): 61.92

Lactic Acid P (g/L): 74.52

Maltose M (g/L): 59.36

Volume V (L): 6.42

t = 40.00 h

Biomass X (g/L): 110.06

Glucose S (g/L): 54.19

Lactic Acid P (g/L): 68.28

Maltose M (g/L): 63.90

Volume V (L): 6.19

t = 41.00 h

Biomass X (g/L): 97.70

Glucose S (g/L): 60.04

Lactic Acid P (g/L): 66.73

Maltose M (g/L): 60.80

Volume V (L): 6.23

t = 42.00 h

Biomass X (g/L): 97.14

Glucose S (g/L): 56.81

Lactic Acid P (g/L): 73.21

Maltose M (g/L): 60.29

Volume V (L): 6.38

t = 43.00 h

Biomass X (g/L): 90.25

Glucose S (g/L): 54.35

Lactic Acid P (g/L): 63.89

Maltose M (g/L): 70.03

Volume V (L): 6.49

t = 44.00 h

Biomass X (g/L): 99.82

Glucose S (g/L): 58.51

Lactic Acid P (g/L): 62.60

Maltose M (g/L): 53.13

Volume V (L): 6.67

t = 45.00 h

Biomass X (g/L): 87.55

Glucose S (g/L): 61.52

Lactic Acid P (g/L): 48.47

Maltose M (g/L): 64.02

Volume V (L): 6.37

t = 46.00 h

Biomass X (g/L): 93.80

Glucose S (g/L): 64.44

Lactic Acid P (g/L): 91.72

Maltose M (g/L): 65.07

Volume V (L): 6.17

t = 47.00 h

Biomass X (g/L): 81.46

Glucose S (g/L): 65.38

Lactic Acid P (g/L): 80.40

Maltose M (g/L): 58.79

Volume V (L): 6.55

t = 48.00 h

Biomass X (g/L): 101.58

Glucose S (g/L): 59.41

Lactic Acid P (g/L): 74.15

Maltose M (g/L): 64.89

Volume V (L): 6.46

t = 49.00 h

Biomass X (g/L): 93.85

Glucose S (g/L): 57.38

Lactic Acid P (g/L): 69.09

Maltose M (g/L): 59.44

Volume V (L): 6.33

t = 50.00 h

Biomass X (g/L): 111.22

Glucose S (g/L): 60.33

Lactic Acid P (g/L): 47.73

Maltose M (g/L): 65.78

Volume V (L): 6.67

t = 51.00 h

Biomass X (g/L): 115.38

Glucose S (g/L): 57.67

Lactic Acid P (g/L): 69.16

Maltose M (g/L): 47.41

Volume V (L): 6.61

t = 52.00 h

Biomass X (g/L): 98.54

Glucose S (g/L): 61.71
Lactic Acid P (g/L): 83.66
Maltose M (g/L): 54.46
Volume V (L): 6.74

t = 53.00 h

Biomass X (g/L): 91.31
Glucose S (g/L): 61.75
Lactic Acid P (g/L): 86.19
Maltose M (g/L): 55.69
Volume V (L): 6.54

t = 54.00 h

Biomass X (g/L): 80.78
Glucose S (g/L): 66.17
Lactic Acid P (g/L): 83.54
Maltose M (g/L): 38.26
Volume V (L): 6.84

t = 55.00 h

Biomass X (g/L): 99.73
Glucose S (g/L): 55.93
Lactic Acid P (g/L): 79.08
Maltose M (g/L): 48.02
Volume V (L): 6.52

t = 56.00 h

Biomass X (g/L): 92.47
Glucose S (g/L): 61.82
Lactic Acid P (g/L): 90.61
Maltose M (g/L): 49.25
Volume V (L): 6.65

t = 57.00 h

Biomass X (g/L): 95.72
Glucose S (g/L): 62.35
Lactic Acid P (g/L): 74.84
Maltose M (g/L): 60.17
Volume V (L): 6.80

t = 58.00 h

Biomass X (g/L): 107.85
Glucose S (g/L): 69.84
Lactic Acid P (g/L): 88.99
Maltose M (g/L): 48.92
Volume V (L): 7.21

t = 59.00 h

Biomass X (g/L): 81.43
Glucose S (g/L): 61.91
Lactic Acid P (g/L): 74.81
Maltose M (g/L): 47.69
Volume V (L): 7.06

t = 60.00 h

Biomass X (g/L): 80.26
Glucose S (g/L): 67.31
Lactic Acid P (g/L): 70.18
Maltose M (g/L): 57.82
Volume V (L): 7.15

t = 61.00 h

Biomass X (g/L): 87.18
Glucose S (g/L): 66.50
Lactic Acid P (g/L): 69.19
Maltose M (g/L): 41.53
Volume V (L): 7.25

t = 62.00 h

Biomass X (g/L): 97.91
Glucose S (g/L): 71.27
Lactic Acid P (g/L): 68.85
Maltose M (g/L): 38.11
Volume V (L): 7.70

t = 63.00 h

Biomass X (g/L): 95.00
Glucose S (g/L): 70.11
Lactic Acid P (g/L): 86.01
Maltose M (g/L): 48.75
Volume V (L): 7.05

t = 64.00 h

Biomass X (g/L): 79.91
Glucose S (g/L): 68.85
Lactic Acid P (g/L): 72.46
Maltose M (g/L): 41.24
Volume V (L): 7.35

t = 65.00 h

Biomass X (g/L): 96.74
Glucose S (g/L): 66.85
Lactic Acid P (g/L): 69.83
Maltose M (g/L): 41.04
Volume V (L): 7.30

t = 66.00 h

Biomass X (g/L): 76.41
Glucose S (g/L): 62.57
Lactic Acid P (g/L): 87.20
Maltose M (g/L): 53.48
Volume V (L): 7.33

t = 67.00 h

Biomass X (g/L): 79.96
Glucose S (g/L): 70.72
Lactic Acid P (g/L): 83.61
Maltose M (g/L): 55.01
Volume V (L): 7.51

t = 68.00 h

Biomass X (g/L): 87.21
Glucose S (g/L): 72.22
Lactic Acid P (g/L): 90.05
Maltose M (g/L): 54.02
Volume V (L): 7.12

t = 69.00 h

Biomass X (g/L): 82.51
Glucose S (g/L): 64.97
Lactic Acid P (g/L): 85.82
Maltose M (g/L): 55.68
Volume V (L): 7.13

t = 70.00 h

Biomass X (g/L): 97.16
Glucose S (g/L): 66.68
Lactic Acid P (g/L): 73.58
Maltose M (g/L): 56.13
Volume V (L): 7.33

t = 71.00 h

Biomass X (g/L): 90.75
Glucose S (g/L): 72.56
Lactic Acid P (g/L): 81.91
Maltose M (g/L): 33.85
Volume V (L): 7.26

t = 72.00 h

Biomass X (g/L): 105.77
Glucose S (g/L): 73.33
Lactic Acid P (g/L): 82.58
Maltose M (g/L): 50.20
Volume V (L): 7.39

t = 73.00 h

Biomass X (g/L): 84.85
Glucose S (g/L): 68.90
Lactic Acid P (g/L): 72.49
Maltose M (g/L): 43.46
Volume V (L): 7.66

t = 74.00 h

Biomass X (g/L): 85.93
Glucose S (g/L): 70.87
Lactic Acid P (g/L): 70.85
Maltose M (g/L): 41.66
Volume V (L): 7.30

t = 75.00 h

Biomass X (g/L): 99.44
Glucose S (g/L): 66.68
Lactic Acid P (g/L): 75.18
Maltose M (g/L): 49.78

Volume V (L): 7.37

Batch 54

Initial Conditions:

Batch: 54.00

X0 (g/L): 4.22

S0 (g/L): 14.25

M0 (g/L): 139.94

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 9.82

F_vol (L/h): 0.07

S_feed (g/L): 27.73

M_feed (g/L): 272.27

V_final (L): 10.54

Batch 54 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 5.83

Glucose S (g/L): 17.13

Lactic Acid P (g/L): 9.05

Maltose M (g/L): 134.96

Volume V (L): 4.44

t = 1.00 h

Biomass X (g/L): 3.17

Glucose S (g/L): 16.03

Lactic Acid P (g/L): 3.82

Maltose M (g/L): 144.98

Volume V (L): 5.14

t = 2.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 17.28

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 139.01

Volume V (L): 4.54

t = 3.00 h

Biomass X (g/L): 2.88

Glucose S (g/L): 22.20

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 143.57

Volume V (L): 4.33

t = 4.00 h

Biomass X (g/L): 16.54

Glucose S (g/L): 23.78

Lactic Acid P (g/L): 14.24

Maltose M (g/L): 144.65

Volume V (L): 5.53

t = 5.00 h

Biomass X (g/L): 25.77

Glucose S (g/L): 18.80

Lactic Acid P (g/L): 24.34

Maltose M (g/L): 136.51

Volume V (L): 5.51

t = 6.00 h

Biomass X (g/L): 4.52

Glucose S (g/L): 21.92

Lactic Acid P (g/L): 9.38

Maltose M (g/L): 133.77

Volume V (L): 5.64

t = 7.00 h

Biomass X (g/L): 7.43

Glucose S (g/L): 20.10

Lactic Acid P (g/L): 18.61

Maltose M (g/L): 134.88

Volume V (L): 5.63

t = 8.00 h

Biomass X (g/L): 21.61

Glucose S (g/L): 18.23

Lactic Acid P (g/L): 15.79

Maltose M (g/L): 133.31

Volume V (L): 5.56

t = 9.00 h

Biomass X (g/L): 5.56

Glucose S (g/L): 17.84

Lactic Acid P (g/L): 21.91

Maltose M (g/L): 124.86

Volume V (L): 5.37

t = 10.00 h

Biomass X (g/L): 24.05

Glucose S (g/L): 13.94

Lactic Acid P (g/L): 7.71

Maltose M (g/L): 139.87

Volume V (L): 5.97

t = 11.00 h

Biomass X (g/L): 15.59

Glucose S (g/L): 13.93
Lactic Acid P (g/L): 16.21
Maltose M (g/L): 132.68
Volume V (L): 6.49

t = 12.00 h

Biomass X (g/L): 47.58
Glucose S (g/L): 11.91
Lactic Acid P (g/L): 26.49
Maltose M (g/L): 124.99
Volume V (L): 5.66

t = 13.00 h

Biomass X (g/L): 31.07
Glucose S (g/L): 15.41
Lactic Acid P (g/L): 36.09
Maltose M (g/L): 119.78
Volume V (L): 5.51

t = 14.00 h

Biomass X (g/L): 56.44
Glucose S (g/L): 11.35
Lactic Acid P (g/L): 20.68
Maltose M (g/L): 117.74
Volume V (L): 5.53

t = 15.00 h

Biomass X (g/L): 54.30
Glucose S (g/L): 6.80
Lactic Acid P (g/L): 12.26
Maltose M (g/L): 132.21
Volume V (L): 5.72

t = 16.00 h

Biomass X (g/L): 40.70
Glucose S (g/L): 7.90
Lactic Acid P (g/L): 27.76
Maltose M (g/L): 124.63
Volume V (L): 6.27

t = 17.00 h

Biomass X (g/L): 39.20
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 42.90
Maltose M (g/L): 124.68
Volume V (L): 5.31

t = 18.00 h

Biomass X (g/L): 49.54
Glucose S (g/L): 7.67
Lactic Acid P (g/L): 22.04
Maltose M (g/L): 128.57
Volume V (L): 6.52

t = 19.00 h

Biomass X (g/L): 52.43
Glucose S (g/L): 9.04
Lactic Acid P (g/L): 27.31
Maltose M (g/L): 122.89
Volume V (L): 6.69

t = 20.00 h

Biomass X (g/L): 58.51
Glucose S (g/L): 8.32
Lactic Acid P (g/L): 40.57
Maltose M (g/L): 127.37
Volume V (L): 5.68

t = 21.00 h

Biomass X (g/L): 49.94
Glucose S (g/L): 14.80
Lactic Acid P (g/L): 39.71
Maltose M (g/L): 124.41
Volume V (L): 6.12

t = 22.00 h

Biomass X (g/L): 65.64
Glucose S (g/L): 10.44
Lactic Acid P (g/L): 47.36
Maltose M (g/L): 122.80
Volume V (L): 6.66

t = 23.00 h

Biomass X (g/L): 62.76
Glucose S (g/L): 10.36
Lactic Acid P (g/L): 32.78
Maltose M (g/L): 114.75
Volume V (L): 6.63

t = 24.00 h

Biomass X (g/L): 58.62
Glucose S (g/L): 3.75
Lactic Acid P (g/L): 41.98
Maltose M (g/L): 117.72
Volume V (L): 7.00

t = 25.00 h

Biomass X (g/L): 56.56
Glucose S (g/L): 7.89
Lactic Acid P (g/L): 56.51
Maltose M (g/L): 123.99
Volume V (L): 6.78

t = 26.00 h

Biomass X (g/L): 62.70
Glucose S (g/L): 7.73
Lactic Acid P (g/L): 36.82
Maltose M (g/L): 116.96
Volume V (L): 7.48

t = 27.00 h

Biomass X (g/L): 58.08
Glucose S (g/L): 7.06
Lactic Acid P (g/L): 61.30
Maltose M (g/L): 125.52
Volume V (L): 6.85

t = 28.00 h

Biomass X (g/L): 67.29
Glucose S (g/L): 7.32
Lactic Acid P (g/L): 40.14
Maltose M (g/L): 130.42
Volume V (L): 6.63

t = 29.00 h

Biomass X (g/L): 68.61
Glucose S (g/L): 4.53
Lactic Acid P (g/L): 57.94
Maltose M (g/L): 121.41
Volume V (L): 6.67

t = 30.00 h

Biomass X (g/L): 71.24
Glucose S (g/L): 6.54
Lactic Acid P (g/L): 48.26
Maltose M (g/L): 124.82
Volume V (L): 7.54

t = 31.00 h

Biomass X (g/L): 86.07
Glucose S (g/L): 13.60
Lactic Acid P (g/L): 42.47
Maltose M (g/L): 120.33
Volume V (L): 7.30

t = 32.00 h

Biomass X (g/L): 69.46
Glucose S (g/L): 10.86
Lactic Acid P (g/L): 49.07
Maltose M (g/L): 113.47
Volume V (L): 7.64

t = 33.00 h

Biomass X (g/L): 80.78
Glucose S (g/L): 10.18
Lactic Acid P (g/L): 59.07
Maltose M (g/L): 116.39
Volume V (L): 7.71

t = 34.00 h

Biomass X (g/L): 74.43
Glucose S (g/L): 12.20
Lactic Acid P (g/L): 55.10
Maltose M (g/L): 124.62

Volume V (L): 6.62

t = 35.00 h

Biomass X (g/L): 70.86

Glucose S (g/L): 16.32

Lactic Acid P (g/L): 65.76

Maltose M (g/L): 111.47

Volume V (L): 7.08

t = 36.00 h

Biomass X (g/L): 72.65

Glucose S (g/L): 13.22

Lactic Acid P (g/L): 69.33

Maltose M (g/L): 108.26

Volume V (L): 7.21

t = 37.00 h

Biomass X (g/L): 73.86

Glucose S (g/L): 15.41

Lactic Acid P (g/L): 47.19

Maltose M (g/L): 107.13

Volume V (L): 7.37

t = 38.00 h

Biomass X (g/L): 77.60

Glucose S (g/L): 11.66

Lactic Acid P (g/L): 47.29

Maltose M (g/L): 110.97

Volume V (L): 7.62

t = 39.00 h

Biomass X (g/L): 66.74

Glucose S (g/L): 12.93

Lactic Acid P (g/L): 47.57

Maltose M (g/L): 106.45

Volume V (L): 8.32

t = 40.00 h

Biomass X (g/L): 62.61

Glucose S (g/L): 16.53

Lactic Acid P (g/L): 51.79

Maltose M (g/L): 118.96

Volume V (L): 8.48

t = 41.00 h

Biomass X (g/L): 82.65

Glucose S (g/L): 21.89

Lactic Acid P (g/L): 69.96

Maltose M (g/L): 111.14

Volume V (L): 7.90

t = 42.00 h

Biomass X (g/L): 85.36

Glucose S (g/L): 21.06

Lactic Acid P (g/L): 65.83

Maltose M (g/L): 114.21

Volume V (L): 7.99

t = 43.00 h

Biomass X (g/L): 84.38

Glucose S (g/L): 19.06

Lactic Acid P (g/L): 56.77

Maltose M (g/L): 107.32

Volume V (L): 7.86

t = 44.00 h

Biomass X (g/L): 79.33

Glucose S (g/L): 18.36

Lactic Acid P (g/L): 62.02

Maltose M (g/L): 102.55

Volume V (L): 8.29

t = 45.00 h

Biomass X (g/L): 85.34

Glucose S (g/L): 22.87

Lactic Acid P (g/L): 74.95

Maltose M (g/L): 101.09

Volume V (L): 8.41

t = 46.00 h

Biomass X (g/L): 85.18

Glucose S (g/L): 21.65

Lactic Acid P (g/L): 79.89
Maltose M (g/L): 102.44
Volume V (L): 8.75

t = 47.00 h

Biomass X (g/L): 86.18
Glucose S (g/L): 12.74
Lactic Acid P (g/L): 66.00
Maltose M (g/L): 107.99
Volume V (L): 8.82

t = 48.00 h

Biomass X (g/L): 71.27
Glucose S (g/L): 17.25
Lactic Acid P (g/L): 40.98
Maltose M (g/L): 108.22
Volume V (L): 7.52

t = 49.00 h

Biomass X (g/L): 69.34
Glucose S (g/L): 22.25
Lactic Acid P (g/L): 67.31
Maltose M (g/L): 108.02
Volume V (L): 8.55

t = 50.00 h

Biomass X (g/L): 76.11
Glucose S (g/L): 16.35
Lactic Acid P (g/L): 73.35
Maltose M (g/L): 97.13
Volume V (L): 9.30

t = 51.00 h

Biomass X (g/L): 79.74
Glucose S (g/L): 19.80
Lactic Acid P (g/L): 67.58
Maltose M (g/L): 103.85
Volume V (L): 8.57

t = 52.00 h

Biomass X (g/L): 77.86

Glucose S (g/L): 21.11
Lactic Acid P (g/L): 72.40
Maltose M (g/L): 98.21
Volume V (L): 9.34

t = 53.00 h

Biomass X (g/L): 62.96
Glucose S (g/L): 23.19
Lactic Acid P (g/L): 69.59
Maltose M (g/L): 104.63
Volume V (L): 8.78

t = 54.00 h

Biomass X (g/L): 96.29
Glucose S (g/L): 23.80
Lactic Acid P (g/L): 56.07
Maltose M (g/L): 104.43
Volume V (L): 9.06

t = 55.00 h

Biomass X (g/L): 78.02
Glucose S (g/L): 24.50
Lactic Acid P (g/L): 59.51
Maltose M (g/L): 95.73
Volume V (L): 8.56

t = 56.00 h

Biomass X (g/L): 82.25
Glucose S (g/L): 28.67
Lactic Acid P (g/L): 64.82
Maltose M (g/L): 105.67
Volume V (L): 8.65

t = 57.00 h

Biomass X (g/L): 92.24
Glucose S (g/L): 26.00
Lactic Acid P (g/L): 77.30
Maltose M (g/L): 111.84
Volume V (L): 9.53

t = 58.00 h

Biomass X (g/L): 59.22
Glucose S (g/L): 26.35
Lactic Acid P (g/L): 61.64
Maltose M (g/L): 103.87
Volume V (L): 9.67

t = 59.00 h

Biomass X (g/L): 76.20
Glucose S (g/L): 24.08
Lactic Acid P (g/L): 72.66
Maltose M (g/L): 98.40
Volume V (L): 9.37

t = 60.00 h

Biomass X (g/L): 77.40
Glucose S (g/L): 32.16
Lactic Acid P (g/L): 71.66
Maltose M (g/L): 97.09
Volume V (L): 8.88

t = 61.00 h

Biomass X (g/L): 77.91
Glucose S (g/L): 27.62
Lactic Acid P (g/L): 83.19
Maltose M (g/L): 97.06
Volume V (L): 9.52

t = 62.00 h

Biomass X (g/L): 79.44
Glucose S (g/L): 27.05
Lactic Acid P (g/L): 69.26
Maltose M (g/L): 93.80
Volume V (L): 10.13

t = 63.00 h

Biomass X (g/L): 94.40
Glucose S (g/L): 31.80
Lactic Acid P (g/L): 54.58
Maltose M (g/L): 98.23
Volume V (L): 10.42

t = 64.00 h

Biomass X (g/L): 87.28
Glucose S (g/L): 29.48
Lactic Acid P (g/L): 75.86
Maltose M (g/L): 90.86
Volume V (L): 9.58

t = 65.00 h

Biomass X (g/L): 89.90
Glucose S (g/L): 30.95
Lactic Acid P (g/L): 89.18
Maltose M (g/L): 86.63
Volume V (L): 9.06

t = 66.00 h

Biomass X (g/L): 81.48
Glucose S (g/L): 34.10
Lactic Acid P (g/L): 76.98
Maltose M (g/L): 96.46
Volume V (L): 10.24

t = 67.00 h

Biomass X (g/L): 68.12
Glucose S (g/L): 33.00
Lactic Acid P (g/L): 64.13
Maltose M (g/L): 94.96
Volume V (L): 9.82

t = 68.00 h

Biomass X (g/L): 61.82
Glucose S (g/L): 41.89
Lactic Acid P (g/L): 61.81
Maltose M (g/L): 91.52
Volume V (L): 10.44

t = 69.00 h

Biomass X (g/L): 66.85
Glucose S (g/L): 35.21
Lactic Acid P (g/L): 63.19
Maltose M (g/L): 87.51
Volume V (L): 10.22

t = 70.00 h

Biomass X (g/L): 84.23
Glucose S (g/L): 38.06
Lactic Acid P (g/L): 89.27
Maltose M (g/L): 94.72
Volume V (L): 9.78

t = 71.00 h

Biomass X (g/L): 81.14
Glucose S (g/L): 30.84
Lactic Acid P (g/L): 69.13
Maltose M (g/L): 87.92
Volume V (L): 10.27

t = 72.00 h

Biomass X (g/L): 78.85
Glucose S (g/L): 34.89
Lactic Acid P (g/L): 66.42
Maltose M (g/L): 91.59
Volume V (L): 10.89

t = 73.00 h

Biomass X (g/L): 77.32
Glucose S (g/L): 34.89
Lactic Acid P (g/L): 71.05
Maltose M (g/L): 100.45
Volume V (L): 10.32

t = 74.00 h

Biomass X (g/L): 99.20
Glucose S (g/L): 36.00
Lactic Acid P (g/L): 69.88
Maltose M (g/L): 94.48
Volume V (L): 10.00

t = 75.00 h

Biomass X (g/L): 72.97
Glucose S (g/L): 40.94
Lactic Acid P (g/L): 86.57
Maltose M (g/L): 84.05

Volume V (L): 9.83

Batch 55

Initial Conditions:

Batch: 55.00

X0 (g/L): 1.35

S0 (g/L): 51.60

M0 (g/L): 77.28

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.50

F_vol (L/h): 0.09

S_feed (g/L): 120.10

M_feed (g/L): 179.90

V_final (L): 11.88

Batch 55 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 54.75
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 76.89
Volume V (L): 5.03

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 58.48
Lactic Acid P (g/L): 12.29
Maltose M (g/L): 79.44
Volume V (L): 5.04

t = 2.00 h

Biomass X (g/L): 10.27
Glucose S (g/L): 55.29
Lactic Acid P (g/L): 11.70
Maltose M (g/L): 79.72
Volume V (L): 5.92

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 58.03
Lactic Acid P (g/L): 9.42
Maltose M (g/L): 75.46
Volume V (L): 5.09

t = 4.00 h

Biomass X (g/L): 19.10
Glucose S (g/L): 55.13
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 77.29
Volume V (L): 4.85

t = 5.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 59.34

Lactic Acid P (g/L): 0.49
Maltose M (g/L): 77.43
Volume V (L): 4.98

t = 6.00 h

Biomass X (g/L): 29.46
Glucose S (g/L): 62.69
Lactic Acid P (g/L): 6.91
Maltose M (g/L): 74.56
Volume V (L): 5.91

t = 7.00 h

Biomass X (g/L): 25.60
Glucose S (g/L): 62.66
Lactic Acid P (g/L): 11.71
Maltose M (g/L): 79.86
Volume V (L): 4.87

t = 8.00 h

Biomass X (g/L): 12.73
Glucose S (g/L): 60.21
Lactic Acid P (g/L): 2.91
Maltose M (g/L): 78.60
Volume V (L): 5.62

t = 9.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 60.30
Lactic Acid P (g/L): 10.93
Maltose M (g/L): 78.88
Volume V (L): 6.56

t = 10.00 h

Biomass X (g/L): 16.88
Glucose S (g/L): 58.82
Lactic Acid P (g/L): 14.13
Maltose M (g/L): 81.36
Volume V (L): 6.08

t = 11.00 h

Biomass X (g/L): 17.26

Glucose S (g/L): 57.87
Lactic Acid P (g/L): 25.69
Maltose M (g/L): 79.27
Volume V (L): 5.97

t = 12.00 h

Biomass X (g/L): 12.55
Glucose S (g/L): 58.06
Lactic Acid P (g/L): 23.84
Maltose M (g/L): 78.39
Volume V (L): 6.08

t = 13.00 h

Biomass X (g/L): 31.64
Glucose S (g/L): 49.54
Lactic Acid P (g/L): 6.97
Maltose M (g/L): 80.03
Volume V (L): 5.63

t = 14.00 h

Biomass X (g/L): 24.96
Glucose S (g/L): 52.65
Lactic Acid P (g/L): 21.32
Maltose M (g/L): 80.79
Volume V (L): 5.81

t = 15.00 h

Biomass X (g/L): 30.02
Glucose S (g/L): 51.72
Lactic Acid P (g/L): 18.05
Maltose M (g/L): 79.45
Volume V (L): 6.27

t = 16.00 h

Biomass X (g/L): 38.84
Glucose S (g/L): 47.16
Lactic Acid P (g/L): 34.10
Maltose M (g/L): 80.01
Volume V (L): 6.62

t = 17.00 h

Biomass X (g/L): 38.97
Glucose S (g/L): 45.50
Lactic Acid P (g/L): 40.02
Maltose M (g/L): 78.51
Volume V (L): 7.08

t = 18.00 h

Biomass X (g/L): 68.00
Glucose S (g/L): 43.83
Lactic Acid P (g/L): 35.48
Maltose M (g/L): 78.49
Volume V (L): 6.70

t = 19.00 h

Biomass X (g/L): 46.06
Glucose S (g/L): 42.20
Lactic Acid P (g/L): 34.74
Maltose M (g/L): 78.68
Volume V (L): 7.49

t = 20.00 h

Biomass X (g/L): 58.44
Glucose S (g/L): 42.17
Lactic Acid P (g/L): 40.03
Maltose M (g/L): 77.08
Volume V (L): 7.72

t = 21.00 h

Biomass X (g/L): 58.32
Glucose S (g/L): 41.34
Lactic Acid P (g/L): 43.42
Maltose M (g/L): 77.22
Volume V (L): 7.36

t = 22.00 h

Biomass X (g/L): 62.80
Glucose S (g/L): 34.26
Lactic Acid P (g/L): 37.98
Maltose M (g/L): 80.30
Volume V (L): 7.69

t = 23.00 h

Biomass X (g/L): 65.80
Glucose S (g/L): 41.02
Lactic Acid P (g/L): 34.76
Maltose M (g/L): 81.16
Volume V (L): 6.56

t = 24.00 h

Biomass X (g/L): 72.21
Glucose S (g/L): 37.80
Lactic Acid P (g/L): 29.18
Maltose M (g/L): 80.26
Volume V (L): 6.57

t = 25.00 h

Biomass X (g/L): 68.14
Glucose S (g/L): 43.42
Lactic Acid P (g/L): 59.86
Maltose M (g/L): 75.18
Volume V (L): 7.58

t = 26.00 h

Biomass X (g/L): 69.42
Glucose S (g/L): 34.51
Lactic Acid P (g/L): 54.57
Maltose M (g/L): 74.80
Volume V (L): 6.65

t = 27.00 h

Biomass X (g/L): 69.34
Glucose S (g/L): 35.18
Lactic Acid P (g/L): 54.75
Maltose M (g/L): 74.96
Volume V (L): 7.32

t = 28.00 h

Biomass X (g/L): 68.82
Glucose S (g/L): 36.89
Lactic Acid P (g/L): 63.92
Maltose M (g/L): 76.94
Volume V (L): 7.99

t = 29.00 h

Biomass X (g/L): 62.61
Glucose S (g/L): 38.57
Lactic Acid P (g/L): 42.98
Maltose M (g/L): 71.77
Volume V (L): 8.44

t = 30.00 h

Biomass X (g/L): 71.93
Glucose S (g/L): 32.16
Lactic Acid P (g/L): 39.86
Maltose M (g/L): 75.23
Volume V (L): 7.61

t = 31.00 h

Biomass X (g/L): 75.86
Glucose S (g/L): 35.39
Lactic Acid P (g/L): 59.46
Maltose M (g/L): 78.31
Volume V (L): 7.48

t = 32.00 h

Biomass X (g/L): 78.33
Glucose S (g/L): 39.51
Lactic Acid P (g/L): 59.61
Maltose M (g/L): 77.34
Volume V (L): 8.11

t = 33.00 h

Biomass X (g/L): 77.03
Glucose S (g/L): 35.57
Lactic Acid P (g/L): 42.74
Maltose M (g/L): 74.58
Volume V (L): 7.43

t = 34.00 h

Biomass X (g/L): 73.80
Glucose S (g/L): 42.27
Lactic Acid P (g/L): 61.91
Maltose M (g/L): 74.43

Volume V (L): 8.02

t = 35.00 h

Biomass X (g/L): 76.17

Glucose S (g/L): 37.65

Lactic Acid P (g/L): 56.02

Maltose M (g/L): 75.82

Volume V (L): 8.20

t = 36.00 h

Biomass X (g/L): 72.00

Glucose S (g/L): 36.85

Lactic Acid P (g/L): 53.18

Maltose M (g/L): 76.15

Volume V (L): 8.29

t = 37.00 h

Biomass X (g/L): 80.90

Glucose S (g/L): 41.46

Lactic Acid P (g/L): 58.03

Maltose M (g/L): 73.41

Volume V (L): 8.44

t = 38.00 h

Biomass X (g/L): 76.25

Glucose S (g/L): 44.69

Lactic Acid P (g/L): 40.19

Maltose M (g/L): 73.62

Volume V (L): 8.02

t = 39.00 h

Biomass X (g/L): 80.56

Glucose S (g/L): 41.95

Lactic Acid P (g/L): 57.03

Maltose M (g/L): 72.77

Volume V (L): 8.18

t = 40.00 h

Biomass X (g/L): 88.86

Glucose S (g/L): 44.82

Lactic Acid P (g/L): 56.62

Maltose M (g/L): 71.19

Volume V (L): 8.26

t = 41.00 h

Biomass X (g/L): 72.75

Glucose S (g/L): 46.54

Lactic Acid P (g/L): 68.23

Maltose M (g/L): 72.75

Volume V (L): 7.87

t = 42.00 h

Biomass X (g/L): 85.65

Glucose S (g/L): 40.01

Lactic Acid P (g/L): 62.48

Maltose M (g/L): 72.32

Volume V (L): 9.04

t = 43.00 h

Biomass X (g/L): 80.50

Glucose S (g/L): 45.73

Lactic Acid P (g/L): 69.96

Maltose M (g/L): 70.40

Volume V (L): 9.43

t = 44.00 h

Biomass X (g/L): 99.90

Glucose S (g/L): 39.60

Lactic Acid P (g/L): 42.47

Maltose M (g/L): 73.27

Volume V (L): 9.47

t = 45.00 h

Biomass X (g/L): 78.57

Glucose S (g/L): 45.59

Lactic Acid P (g/L): 57.92

Maltose M (g/L): 72.69

Volume V (L): 8.98

t = 46.00 h

Biomass X (g/L): 87.54

Glucose S (g/L): 44.89

Lactic Acid P (g/L): 70.59

Maltose M (g/L): 72.73

Volume V (L): 8.77

t = 47.00 h

Biomass X (g/L): 81.16

Glucose S (g/L): 49.15

Lactic Acid P (g/L): 63.41

Maltose M (g/L): 69.98

Volume V (L): 9.72

t = 48.00 h

Biomass X (g/L): 67.78

Glucose S (g/L): 49.63

Lactic Acid P (g/L): 73.93

Maltose M (g/L): 67.63

Volume V (L): 9.75

t = 49.00 h

Biomass X (g/L): 76.25

Glucose S (g/L): 48.66

Lactic Acid P (g/L): 51.06

Maltose M (g/L): 71.41

Volume V (L): 9.61

t = 50.00 h

Biomass X (g/L): 89.85

Glucose S (g/L): 48.60

Lactic Acid P (g/L): 73.66

Maltose M (g/L): 69.20

Volume V (L): 10.54

t = 51.00 h

Biomass X (g/L): 77.28

Glucose S (g/L): 49.87

Lactic Acid P (g/L): 61.99

Maltose M (g/L): 68.50

Volume V (L): 9.66

t = 52.00 h

Biomass X (g/L): 60.13

Glucose S (g/L): 49.34
Lactic Acid P (g/L): 61.46
Maltose M (g/L): 67.57
Volume V (L): 9.63

t = 53.00 h

Biomass X (g/L): 81.26
Glucose S (g/L): 49.69
Lactic Acid P (g/L): 76.93
Maltose M (g/L): 68.89
Volume V (L): 9.64

t = 54.00 h

Biomass X (g/L): 71.07
Glucose S (g/L): 54.52
Lactic Acid P (g/L): 78.73
Maltose M (g/L): 71.67
Volume V (L): 9.09

t = 55.00 h

Biomass X (g/L): 82.88
Glucose S (g/L): 55.55
Lactic Acid P (g/L): 77.91
Maltose M (g/L): 67.87
Volume V (L): 10.41

t = 56.00 h

Biomass X (g/L): 96.56
Glucose S (g/L): 51.57
Lactic Acid P (g/L): 87.00
Maltose M (g/L): 66.57
Volume V (L): 10.08

t = 57.00 h

Biomass X (g/L): 89.03
Glucose S (g/L): 53.31
Lactic Acid P (g/L): 80.50
Maltose M (g/L): 69.16
Volume V (L): 10.51

t = 58.00 h

Biomass X (g/L): 96.40
Glucose S (g/L): 53.59
Lactic Acid P (g/L): 62.68
Maltose M (g/L): 68.04
Volume V (L): 11.33

t = 59.00 h

Biomass X (g/L): 76.16
Glucose S (g/L): 49.64
Lactic Acid P (g/L): 71.15
Maltose M (g/L): 63.89
Volume V (L): 10.19

t = 60.00 h

Biomass X (g/L): 80.15
Glucose S (g/L): 53.84
Lactic Acid P (g/L): 67.00
Maltose M (g/L): 68.22
Volume V (L): 9.21

t = 61.00 h

Biomass X (g/L): 71.42
Glucose S (g/L): 59.85
Lactic Acid P (g/L): 88.19
Maltose M (g/L): 70.35
Volume V (L): 10.86

t = 62.00 h

Biomass X (g/L): 74.84
Glucose S (g/L): 54.09
Lactic Acid P (g/L): 80.32
Maltose M (g/L): 62.13
Volume V (L): 10.78

t = 63.00 h

Biomass X (g/L): 72.44
Glucose S (g/L): 62.64
Lactic Acid P (g/L): 69.19
Maltose M (g/L): 64.10
Volume V (L): 10.37

t = 64.00 h

Biomass X (g/L): 68.71
Glucose S (g/L): 60.75
Lactic Acid P (g/L): 78.76
Maltose M (g/L): 61.08
Volume V (L): 11.09

t = 65.00 h

Biomass X (g/L): 94.37
Glucose S (g/L): 59.32
Lactic Acid P (g/L): 60.47
Maltose M (g/L): 65.40
Volume V (L): 11.39

t = 66.00 h

Biomass X (g/L): 60.98
Glucose S (g/L): 60.44
Lactic Acid P (g/L): 73.67
Maltose M (g/L): 66.87
Volume V (L): 10.53

t = 67.00 h

Biomass X (g/L): 94.24
Glucose S (g/L): 57.06
Lactic Acid P (g/L): 63.45
Maltose M (g/L): 65.07
Volume V (L): 10.98

t = 68.00 h

Biomass X (g/L): 92.95
Glucose S (g/L): 59.53
Lactic Acid P (g/L): 57.06
Maltose M (g/L): 66.91
Volume V (L): 12.05

t = 69.00 h

Biomass X (g/L): 85.66
Glucose S (g/L): 64.88
Lactic Acid P (g/L): 78.94
Maltose M (g/L): 65.66
Volume V (L): 10.93

t = 70.00 h

Biomass X (g/L): 70.00
Glucose S (g/L): 59.91
Lactic Acid P (g/L): 80.21
Maltose M (g/L): 66.80
Volume V (L): 11.45

t = 71.00 h

Biomass X (g/L): 65.39
Glucose S (g/L): 65.17
Lactic Acid P (g/L): 76.43
Maltose M (g/L): 65.58
Volume V (L): 10.07

t = 72.00 h

Biomass X (g/L): 75.23
Glucose S (g/L): 59.14
Lactic Acid P (g/L): 70.99
Maltose M (g/L): 62.66
Volume V (L): 10.79

t = 73.00 h

Biomass X (g/L): 86.42
Glucose S (g/L): 60.29
Lactic Acid P (g/L): 61.65
Maltose M (g/L): 63.14
Volume V (L): 11.78

t = 74.00 h

Biomass X (g/L): 80.30
Glucose S (g/L): 69.44
Lactic Acid P (g/L): 62.74
Maltose M (g/L): 61.27
Volume V (L): 12.46

t = 75.00 h

Biomass X (g/L): 70.10
Glucose S (g/L): 64.84
Lactic Acid P (g/L): 64.58
Maltose M (g/L): 62.03

Volume V (L): 11.13

Batch 56

Initial Conditions:

Batch: 56.00

X0 (g/L): 8.53

S0 (g/L): 77.42

M0 (g/L): 63.09

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.81

F_vol (L/h): 0.08

S_feed (g/L): 165.29

M_feed (g/L): 134.71

V_final (L): 11.30

Batch 56 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 9.23
Glucose S (g/L): 73.75
Lactic Acid P (g/L): 5.48
Maltose M (g/L): 62.34
Volume V (L): 4.65

t = 1.00 h

Biomass X (g/L): 7.87
Glucose S (g/L): 78.00
Lactic Acid P (g/L): 3.61
Maltose M (g/L): 61.99
Volume V (L): 5.35

t = 2.00 h

Biomass X (g/L): 23.76
Glucose S (g/L): 79.19
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 61.35
Volume V (L): 4.96

t = 3.00 h

Biomass X (g/L): 7.56
Glucose S (g/L): 69.45
Lactic Acid P (g/L): 23.90
Maltose M (g/L): 62.77
Volume V (L): 4.47

t = 4.00 h

Biomass X (g/L): 27.52
Glucose S (g/L): 80.79
Lactic Acid P (g/L): 17.36
Maltose M (g/L): 63.84
Volume V (L): 5.79

t = 5.00 h

Biomass X (g/L): 22.31
Glucose S (g/L): 73.64

Lactic Acid P (g/L): 8.00
Maltose M (g/L): 63.60
Volume V (L): 4.53

t = 6.00 h

Biomass X (g/L): 35.57
Glucose S (g/L): 66.72
Lactic Acid P (g/L): 18.44
Maltose M (g/L): 62.64
Volume V (L): 4.96

t = 7.00 h

Biomass X (g/L): 39.17
Glucose S (g/L): 60.96
Lactic Acid P (g/L): 12.17
Maltose M (g/L): 62.63
Volume V (L): 4.77

t = 8.00 h

Biomass X (g/L): 41.05
Glucose S (g/L): 60.34
Lactic Acid P (g/L): 42.59
Maltose M (g/L): 62.26
Volume V (L): 5.78

t = 9.00 h

Biomass X (g/L): 52.45
Glucose S (g/L): 58.58
Lactic Acid P (g/L): 21.55
Maltose M (g/L): 60.04
Volume V (L): 5.85

t = 10.00 h

Biomass X (g/L): 52.64
Glucose S (g/L): 57.41
Lactic Acid P (g/L): 28.30
Maltose M (g/L): 62.37
Volume V (L): 6.81

t = 11.00 h

Biomass X (g/L): 54.87

Glucose S (g/L): 58.93
Lactic Acid P (g/L): 29.48
Maltose M (g/L): 60.93
Volume V (L): 6.39

t = 12.00 h

Biomass X (g/L): 56.28
Glucose S (g/L): 50.42
Lactic Acid P (g/L): 39.55
Maltose M (g/L): 59.46
Volume V (L): 6.74

t = 13.00 h

Biomass X (g/L): 63.48
Glucose S (g/L): 52.46
Lactic Acid P (g/L): 53.68
Maltose M (g/L): 62.20
Volume V (L): 6.24

t = 14.00 h

Biomass X (g/L): 67.64
Glucose S (g/L): 44.93
Lactic Acid P (g/L): 59.76
Maltose M (g/L): 60.34
Volume V (L): 5.68

t = 15.00 h

Biomass X (g/L): 58.83
Glucose S (g/L): 46.94
Lactic Acid P (g/L): 42.22
Maltose M (g/L): 61.57
Volume V (L): 6.33

t = 16.00 h

Biomass X (g/L): 67.03
Glucose S (g/L): 48.87
Lactic Acid P (g/L): 51.13
Maltose M (g/L): 62.42
Volume V (L): 6.64

t = 17.00 h

Biomass X (g/L): 62.42
Glucose S (g/L): 49.38
Lactic Acid P (g/L): 50.22
Maltose M (g/L): 59.15
Volume V (L): 6.78

t = 18.00 h

Biomass X (g/L): 73.66
Glucose S (g/L): 44.07
Lactic Acid P (g/L): 42.27
Maltose M (g/L): 60.71
Volume V (L): 6.27

t = 19.00 h

Biomass X (g/L): 70.06
Glucose S (g/L): 44.90
Lactic Acid P (g/L): 74.30
Maltose M (g/L): 57.24
Volume V (L): 6.84

t = 20.00 h

Biomass X (g/L): 84.18
Glucose S (g/L): 51.27
Lactic Acid P (g/L): 66.13
Maltose M (g/L): 62.71
Volume V (L): 6.21

t = 21.00 h

Biomass X (g/L): 84.49
Glucose S (g/L): 50.26
Lactic Acid P (g/L): 56.83
Maltose M (g/L): 57.55
Volume V (L): 6.24

t = 22.00 h

Biomass X (g/L): 70.68
Glucose S (g/L): 47.13
Lactic Acid P (g/L): 70.47
Maltose M (g/L): 58.76
Volume V (L): 6.29

t = 23.00 h

Biomass X (g/L): 80.25
Glucose S (g/L): 39.02
Lactic Acid P (g/L): 68.20
Maltose M (g/L): 61.68
Volume V (L): 6.95

t = 24.00 h

Biomass X (g/L): 87.09
Glucose S (g/L): 55.45
Lactic Acid P (g/L): 59.02
Maltose M (g/L): 58.75
Volume V (L): 6.42

t = 25.00 h

Biomass X (g/L): 80.25
Glucose S (g/L): 47.19
Lactic Acid P (g/L): 62.12
Maltose M (g/L): 56.86
Volume V (L): 6.89

t = 26.00 h

Biomass X (g/L): 75.76
Glucose S (g/L): 46.59
Lactic Acid P (g/L): 73.61
Maltose M (g/L): 59.87
Volume V (L): 7.64

t = 27.00 h

Biomass X (g/L): 72.95
Glucose S (g/L): 46.34
Lactic Acid P (g/L): 52.32
Maltose M (g/L): 59.58
Volume V (L): 6.75

t = 28.00 h

Biomass X (g/L): 91.65
Glucose S (g/L): 49.89
Lactic Acid P (g/L): 76.58
Maltose M (g/L): 57.07
Volume V (L): 7.10

t = 29.00 h

Biomass X (g/L): 92.56
Glucose S (g/L): 50.46
Lactic Acid P (g/L): 60.21
Maltose M (g/L): 57.00
Volume V (L): 7.47

t = 30.00 h

Biomass X (g/L): 82.59
Glucose S (g/L): 52.95
Lactic Acid P (g/L): 76.57
Maltose M (g/L): 56.90
Volume V (L): 7.65

t = 31.00 h

Biomass X (g/L): 92.03
Glucose S (g/L): 47.17
Lactic Acid P (g/L): 63.44
Maltose M (g/L): 57.03
Volume V (L): 7.35

t = 32.00 h

Biomass X (g/L): 57.48
Glucose S (g/L): 49.88
Lactic Acid P (g/L): 67.08
Maltose M (g/L): 53.89
Volume V (L): 8.09

t = 33.00 h

Biomass X (g/L): 97.83
Glucose S (g/L): 58.49
Lactic Acid P (g/L): 64.28
Maltose M (g/L): 56.55
Volume V (L): 8.07

t = 34.00 h

Biomass X (g/L): 84.19
Glucose S (g/L): 53.01
Lactic Acid P (g/L): 60.32
Maltose M (g/L): 59.63

Volume V (L): 8.43

t = 35.00 h

Biomass X (g/L): 87.16

Glucose S (g/L): 52.34

Lactic Acid P (g/L): 63.15

Maltose M (g/L): 54.80

Volume V (L): 7.90

t = 36.00 h

Biomass X (g/L): 82.99

Glucose S (g/L): 50.41

Lactic Acid P (g/L): 78.42

Maltose M (g/L): 55.07

Volume V (L): 7.77

t = 37.00 h

Biomass X (g/L): 99.14

Glucose S (g/L): 56.03

Lactic Acid P (g/L): 45.68

Maltose M (g/L): 53.78

Volume V (L): 7.77

t = 38.00 h

Biomass X (g/L): 76.30

Glucose S (g/L): 58.53

Lactic Acid P (g/L): 55.75

Maltose M (g/L): 56.57

Volume V (L): 8.35

t = 39.00 h

Biomass X (g/L): 100.04

Glucose S (g/L): 58.71

Lactic Acid P (g/L): 60.75

Maltose M (g/L): 57.31

Volume V (L): 8.09

t = 40.00 h

Biomass X (g/L): 90.02

Glucose S (g/L): 55.58

Lactic Acid P (g/L): 57.95

Maltose M (g/L): 57.01

Volume V (L): 8.07

t = 41.00 h

Biomass X (g/L): 92.16

Glucose S (g/L): 58.42

Lactic Acid P (g/L): 81.55

Maltose M (g/L): 52.60

Volume V (L): 8.22

t = 42.00 h

Biomass X (g/L): 83.28

Glucose S (g/L): 63.60

Lactic Acid P (g/L): 79.63

Maltose M (g/L): 55.31

Volume V (L): 8.18

t = 43.00 h

Biomass X (g/L): 74.13

Glucose S (g/L): 62.29

Lactic Acid P (g/L): 73.97

Maltose M (g/L): 56.64

Volume V (L): 8.71

t = 44.00 h

Biomass X (g/L): 95.02

Glucose S (g/L): 56.72

Lactic Acid P (g/L): 63.24

Maltose M (g/L): 54.80

Volume V (L): 7.99

t = 45.00 h

Biomass X (g/L): 89.20

Glucose S (g/L): 51.46

Lactic Acid P (g/L): 81.25

Maltose M (g/L): 54.48

Volume V (L): 8.93

t = 46.00 h

Biomass X (g/L): 98.95

Glucose S (g/L): 61.86

Lactic Acid P (g/L): 55.29
Maltose M (g/L): 52.59
Volume V (L): 9.21

t = 47.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 64.30
Lactic Acid P (g/L): 89.06
Maltose M (g/L): 53.50
Volume V (L): 8.66

t = 48.00 h

Biomass X (g/L): 82.18
Glucose S (g/L): 58.61
Lactic Acid P (g/L): 63.94
Maltose M (g/L): 54.97
Volume V (L): 10.00

t = 49.00 h

Biomass X (g/L): 98.97
Glucose S (g/L): 63.01
Lactic Acid P (g/L): 67.99
Maltose M (g/L): 53.16
Volume V (L): 9.13

t = 50.00 h

Biomass X (g/L): 103.77
Glucose S (g/L): 62.87
Lactic Acid P (g/L): 66.74
Maltose M (g/L): 50.20
Volume V (L): 8.41

t = 51.00 h

Biomass X (g/L): 85.90
Glucose S (g/L): 64.84
Lactic Acid P (g/L): 61.19
Maltose M (g/L): 51.29
Volume V (L): 9.14

t = 52.00 h

Biomass X (g/L): 78.67

Glucose S (g/L): 63.35
Lactic Acid P (g/L): 88.35
Maltose M (g/L): 53.87
Volume V (L): 9.40

t = 53.00 h

Biomass X (g/L): 93.15
Glucose S (g/L): 70.31
Lactic Acid P (g/L): 78.37
Maltose M (g/L): 52.10
Volume V (L): 9.68

t = 54.00 h

Biomass X (g/L): 83.08
Glucose S (g/L): 67.13
Lactic Acid P (g/L): 82.41
Maltose M (g/L): 51.48
Volume V (L): 9.41

t = 55.00 h

Biomass X (g/L): 90.92
Glucose S (g/L): 64.34
Lactic Acid P (g/L): 60.71
Maltose M (g/L): 50.08
Volume V (L): 9.11

t = 56.00 h

Biomass X (g/L): 82.76
Glucose S (g/L): 71.15
Lactic Acid P (g/L): 73.53
Maltose M (g/L): 52.79
Volume V (L): 9.15

t = 57.00 h

Biomass X (g/L): 99.64
Glucose S (g/L): 64.43
Lactic Acid P (g/L): 76.78
Maltose M (g/L): 53.33
Volume V (L): 9.46

t = 58.00 h

Biomass X (g/L): 78.36
Glucose S (g/L): 73.79
Lactic Acid P (g/L): 74.72
Maltose M (g/L): 52.40
Volume V (L): 9.59

t = 59.00 h

Biomass X (g/L): 89.22
Glucose S (g/L): 73.48
Lactic Acid P (g/L): 62.05
Maltose M (g/L): 52.45
Volume V (L): 9.39

t = 60.00 h

Biomass X (g/L): 80.01
Glucose S (g/L): 75.51
Lactic Acid P (g/L): 82.30
Maltose M (g/L): 49.12
Volume V (L): 9.67

t = 61.00 h

Biomass X (g/L): 81.79
Glucose S (g/L): 70.92
Lactic Acid P (g/L): 73.91
Maltose M (g/L): 48.62
Volume V (L): 10.46

t = 62.00 h

Biomass X (g/L): 80.40
Glucose S (g/L): 70.44
Lactic Acid P (g/L): 64.41
Maltose M (g/L): 48.96
Volume V (L): 11.04

t = 63.00 h

Biomass X (g/L): 90.98
Glucose S (g/L): 73.08
Lactic Acid P (g/L): 56.49
Maltose M (g/L): 49.04
Volume V (L): 10.68

t = 64.00 h

Biomass X (g/L): 92.49
Glucose S (g/L): 73.17
Lactic Acid P (g/L): 78.76
Maltose M (g/L): 51.16
Volume V (L): 11.04

t = 65.00 h

Biomass X (g/L): 88.95
Glucose S (g/L): 71.89
Lactic Acid P (g/L): 95.47
Maltose M (g/L): 50.46
Volume V (L): 10.95

t = 66.00 h

Biomass X (g/L): 82.55
Glucose S (g/L): 72.87
Lactic Acid P (g/L): 64.50
Maltose M (g/L): 49.14
Volume V (L): 10.97

t = 67.00 h

Biomass X (g/L): 84.69
Glucose S (g/L): 74.15
Lactic Acid P (g/L): 75.30
Maltose M (g/L): 45.54
Volume V (L): 10.46

t = 68.00 h

Biomass X (g/L): 91.59
Glucose S (g/L): 72.05
Lactic Acid P (g/L): 83.88
Maltose M (g/L): 44.85
Volume V (L): 10.79

t = 69.00 h

Biomass X (g/L): 91.70
Glucose S (g/L): 78.97
Lactic Acid P (g/L): 62.65
Maltose M (g/L): 47.90
Volume V (L): 10.59

t = 70.00 h

Biomass X (g/L): 57.71
Glucose S (g/L): 77.10
Lactic Acid P (g/L): 69.88
Maltose M (g/L): 48.74
Volume V (L): 10.77

t = 71.00 h

Biomass X (g/L): 80.11
Glucose S (g/L): 76.85
Lactic Acid P (g/L): 66.60
Maltose M (g/L): 46.87
Volume V (L): 11.00

t = 72.00 h

Biomass X (g/L): 80.37
Glucose S (g/L): 74.73
Lactic Acid P (g/L): 83.30
Maltose M (g/L): 50.02
Volume V (L): 11.41

t = 73.00 h

Biomass X (g/L): 88.53
Glucose S (g/L): 75.17
Lactic Acid P (g/L): 86.49
Maltose M (g/L): 44.68
Volume V (L): 11.35

t = 74.00 h

Biomass X (g/L): 81.46
Glucose S (g/L): 86.30
Lactic Acid P (g/L): 76.14
Maltose M (g/L): 44.22
Volume V (L): 11.14

t = 75.00 h

Biomass X (g/L): 88.61
Glucose S (g/L): 78.18
Lactic Acid P (g/L): 67.28
Maltose M (g/L): 44.92

Volume V (L): 11.33

Batch 57

Initial Conditions:

Batch: 57.00

X0 (g/L): 2.77

S0 (g/L): 124.60

M0 (g/L): 33.48

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.27

F_vol (L/h): 0.06

S_feed (g/L): 236.47

M_feed (g/L): 63.53

V_final (L): 9.17

Batch 57 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 7.12
Glucose S (g/L): 126.35
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 34.90
Volume V (L): 5.04

t = 1.00 h

Biomass X (g/L): 10.47
Glucose S (g/L): 128.00
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 32.25
Volume V (L): 5.56

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 126.13
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 31.24
Volume V (L): 5.23

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 121.37
Lactic Acid P (g/L): 15.18
Maltose M (g/L): 32.50
Volume V (L): 5.13

t = 4.00 h

Biomass X (g/L): 20.12
Glucose S (g/L): 117.38
Lactic Acid P (g/L): 0.33
Maltose M (g/L): 31.79
Volume V (L): 5.27

t = 5.00 h

Biomass X (g/L): 11.10
Glucose S (g/L): 122.14

Lactic Acid P (g/L): 9.70
Maltose M (g/L): 33.07
Volume V (L): 4.87

t = 6.00 h

Biomass X (g/L): 10.69
Glucose S (g/L): 122.39
Lactic Acid P (g/L): 25.48
Maltose M (g/L): 30.17
Volume V (L): 5.44

t = 7.00 h

Biomass X (g/L): 16.04
Glucose S (g/L): 126.40
Lactic Acid P (g/L): 6.42
Maltose M (g/L): 32.83
Volume V (L): 4.72

t = 8.00 h

Biomass X (g/L): 20.20
Glucose S (g/L): 117.84
Lactic Acid P (g/L): 8.87
Maltose M (g/L): 33.02
Volume V (L): 5.16

t = 9.00 h

Biomass X (g/L): 32.92
Glucose S (g/L): 114.48
Lactic Acid P (g/L): 16.76
Maltose M (g/L): 30.85
Volume V (L): 5.54

t = 10.00 h

Biomass X (g/L): 20.85
Glucose S (g/L): 113.88
Lactic Acid P (g/L): 21.69
Maltose M (g/L): 30.67
Volume V (L): 5.95

t = 11.00 h

Biomass X (g/L): 36.88

Glucose S (g/L): 107.75
Lactic Acid P (g/L): 42.42
Maltose M (g/L): 29.24
Volume V (L): 6.09

t = 12.00 h

Biomass X (g/L): 43.66
Glucose S (g/L): 109.31
Lactic Acid P (g/L): 42.84
Maltose M (g/L): 29.30
Volume V (L): 5.59

t = 13.00 h

Biomass X (g/L): 49.53
Glucose S (g/L): 111.40
Lactic Acid P (g/L): 37.35
Maltose M (g/L): 29.24
Volume V (L): 5.77

t = 14.00 h

Biomass X (g/L): 51.89
Glucose S (g/L): 109.48
Lactic Acid P (g/L): 48.16
Maltose M (g/L): 31.58
Volume V (L): 6.11

t = 15.00 h

Biomass X (g/L): 58.63
Glucose S (g/L): 97.64
Lactic Acid P (g/L): 35.97
Maltose M (g/L): 29.21
Volume V (L): 5.39

t = 16.00 h

Biomass X (g/L): 51.19
Glucose S (g/L): 96.34
Lactic Acid P (g/L): 43.12
Maltose M (g/L): 27.22
Volume V (L): 6.02

t = 17.00 h

Biomass X (g/L): 50.66
Glucose S (g/L): 99.72
Lactic Acid P (g/L): 40.05
Maltose M (g/L): 29.19
Volume V (L): 5.67

t = 18.00 h

Biomass X (g/L): 87.64
Glucose S (g/L): 90.25
Lactic Acid P (g/L): 33.18
Maltose M (g/L): 29.76
Volume V (L): 5.94

t = 19.00 h

Biomass X (g/L): 62.56
Glucose S (g/L): 92.29
Lactic Acid P (g/L): 58.47
Maltose M (g/L): 28.33
Volume V (L): 5.90

t = 20.00 h

Biomass X (g/L): 66.07
Glucose S (g/L): 82.90
Lactic Acid P (g/L): 54.18
Maltose M (g/L): 30.20
Volume V (L): 6.26

t = 21.00 h

Biomass X (g/L): 76.82
Glucose S (g/L): 85.34
Lactic Acid P (g/L): 51.20
Maltose M (g/L): 29.01
Volume V (L): 6.35

t = 22.00 h

Biomass X (g/L): 87.92
Glucose S (g/L): 90.78
Lactic Acid P (g/L): 52.66
Maltose M (g/L): 26.23
Volume V (L): 6.52

t = 23.00 h

Biomass X (g/L): 68.13
Glucose S (g/L): 81.66
Lactic Acid P (g/L): 40.20
Maltose M (g/L): 26.71
Volume V (L): 6.39

t = 24.00 h

Biomass X (g/L): 65.09
Glucose S (g/L): 83.54
Lactic Acid P (g/L): 46.51
Maltose M (g/L): 27.59
Volume V (L): 6.39

t = 25.00 h

Biomass X (g/L): 64.01
Glucose S (g/L): 76.57
Lactic Acid P (g/L): 60.00
Maltose M (g/L): 24.72
Volume V (L): 7.01

t = 26.00 h

Biomass X (g/L): 93.67
Glucose S (g/L): 91.87
Lactic Acid P (g/L): 53.16
Maltose M (g/L): 27.32
Volume V (L): 5.94

t = 27.00 h

Biomass X (g/L): 68.36
Glucose S (g/L): 88.22
Lactic Acid P (g/L): 57.16
Maltose M (g/L): 24.18
Volume V (L): 7.07

t = 28.00 h

Biomass X (g/L): 79.48
Glucose S (g/L): 83.19
Lactic Acid P (g/L): 59.24
Maltose M (g/L): 27.88
Volume V (L): 6.46

t = 29.00 h

Biomass X (g/L): 70.82
Glucose S (g/L): 84.39
Lactic Acid P (g/L): 54.83
Maltose M (g/L): 27.52
Volume V (L): 6.80

t = 30.00 h

Biomass X (g/L): 66.88
Glucose S (g/L): 83.76
Lactic Acid P (g/L): 53.07
Maltose M (g/L): 26.76
Volume V (L): 6.56

t = 31.00 h

Biomass X (g/L): 79.17
Glucose S (g/L): 88.03
Lactic Acid P (g/L): 47.92
Maltose M (g/L): 24.91
Volume V (L): 6.49

t = 32.00 h

Biomass X (g/L): 73.31
Glucose S (g/L): 79.62
Lactic Acid P (g/L): 62.32
Maltose M (g/L): 25.91
Volume V (L): 6.94

t = 33.00 h

Biomass X (g/L): 76.34
Glucose S (g/L): 85.17
Lactic Acid P (g/L): 70.16
Maltose M (g/L): 24.77
Volume V (L): 7.18

t = 34.00 h

Biomass X (g/L): 103.89
Glucose S (g/L): 86.76
Lactic Acid P (g/L): 49.83
Maltose M (g/L): 23.86

Volume V (L): 7.05

t = 35.00 h

Biomass X (g/L): 88.46

Glucose S (g/L): 87.48

Lactic Acid P (g/L): 65.53

Maltose M (g/L): 24.64

Volume V (L): 6.76

t = 36.00 h

Biomass X (g/L): 93.92

Glucose S (g/L): 72.31

Lactic Acid P (g/L): 68.97

Maltose M (g/L): 25.92

Volume V (L): 8.20

t = 37.00 h

Biomass X (g/L): 93.18

Glucose S (g/L): 79.81

Lactic Acid P (g/L): 57.75

Maltose M (g/L): 24.82

Volume V (L): 7.46

t = 38.00 h

Biomass X (g/L): 84.19

Glucose S (g/L): 78.16

Lactic Acid P (g/L): 80.70

Maltose M (g/L): 26.62

Volume V (L): 7.42

t = 39.00 h

Biomass X (g/L): 81.04

Glucose S (g/L): 83.55

Lactic Acid P (g/L): 64.42

Maltose M (g/L): 25.49

Volume V (L): 7.05

t = 40.00 h

Biomass X (g/L): 88.16

Glucose S (g/L): 74.26

Lactic Acid P (g/L): 67.56

Maltose M (g/L): 25.17

Volume V (L): 7.01

t = 41.00 h

Biomass X (g/L): 87.53

Glucose S (g/L): 74.40

Lactic Acid P (g/L): 67.47

Maltose M (g/L): 21.82

Volume V (L): 7.18

t = 42.00 h

Biomass X (g/L): 80.29

Glucose S (g/L): 93.28

Lactic Acid P (g/L): 52.96

Maltose M (g/L): 23.51

Volume V (L): 7.69

t = 43.00 h

Biomass X (g/L): 87.42

Glucose S (g/L): 86.74

Lactic Acid P (g/L): 79.20

Maltose M (g/L): 22.26

Volume V (L): 7.06

t = 44.00 h

Biomass X (g/L): 79.13

Glucose S (g/L): 84.87

Lactic Acid P (g/L): 61.36

Maltose M (g/L): 23.33

Volume V (L): 7.09

t = 45.00 h

Biomass X (g/L): 84.57

Glucose S (g/L): 86.28

Lactic Acid P (g/L): 68.42

Maltose M (g/L): 22.46

Volume V (L): 7.68

t = 46.00 h

Biomass X (g/L): 82.98

Glucose S (g/L): 85.67

Lactic Acid P (g/L): 76.54

Maltose M (g/L): 22.74

Volume V (L): 7.66

t = 47.00 h

Biomass X (g/L): 73.64

Glucose S (g/L): 89.77

Lactic Acid P (g/L): 84.91

Maltose M (g/L): 24.60

Volume V (L): 7.35

t = 48.00 h

Biomass X (g/L): 62.79

Glucose S (g/L): 82.95

Lactic Acid P (g/L): 61.64

Maltose M (g/L): 23.63

Volume V (L): 7.30

t = 49.00 h

Biomass X (g/L): 76.96

Glucose S (g/L): 86.93

Lactic Acid P (g/L): 67.85

Maltose M (g/L): 24.22

Volume V (L): 8.20

t = 50.00 h

Biomass X (g/L): 104.46

Glucose S (g/L): 87.49

Lactic Acid P (g/L): 85.15

Maltose M (g/L): 20.42

Volume V (L): 7.90

t = 51.00 h

Biomass X (g/L): 83.22

Glucose S (g/L): 82.26

Lactic Acid P (g/L): 76.92

Maltose M (g/L): 22.60

Volume V (L): 7.38

t = 52.00 h

Biomass X (g/L): 94.06

Glucose S (g/L): 92.89
Lactic Acid P (g/L): 79.13
Maltose M (g/L): 23.73
Volume V (L): 7.46

t = 53.00 h

Biomass X (g/L): 64.90
Glucose S (g/L): 85.69
Lactic Acid P (g/L): 63.41
Maltose M (g/L): 24.34
Volume V (L): 7.82

t = 54.00 h

Biomass X (g/L): 76.03
Glucose S (g/L): 92.54
Lactic Acid P (g/L): 69.91
Maltose M (g/L): 19.91
Volume V (L): 8.24

t = 55.00 h

Biomass X (g/L): 72.47
Glucose S (g/L): 88.96
Lactic Acid P (g/L): 72.13
Maltose M (g/L): 21.45
Volume V (L): 8.25

t = 56.00 h

Biomass X (g/L): 83.94
Glucose S (g/L): 96.05
Lactic Acid P (g/L): 65.97
Maltose M (g/L): 23.26
Volume V (L): 8.49

t = 57.00 h

Biomass X (g/L): 110.16
Glucose S (g/L): 84.67
Lactic Acid P (g/L): 58.62
Maltose M (g/L): 21.40
Volume V (L): 7.67

t = 58.00 h

Biomass X (g/L): 90.37
Glucose S (g/L): 84.88
Lactic Acid P (g/L): 69.67
Maltose M (g/L): 20.26
Volume V (L): 8.30

t = 59.00 h

Biomass X (g/L): 102.64
Glucose S (g/L): 86.08
Lactic Acid P (g/L): 87.21
Maltose M (g/L): 19.23
Volume V (L): 8.75

t = 60.00 h

Biomass X (g/L): 82.95
Glucose S (g/L): 91.06
Lactic Acid P (g/L): 75.40
Maltose M (g/L): 19.94
Volume V (L): 8.78

t = 61.00 h

Biomass X (g/L): 99.06
Glucose S (g/L): 97.68
Lactic Acid P (g/L): 87.82
Maltose M (g/L): 18.77
Volume V (L): 8.51

t = 62.00 h

Biomass X (g/L): 106.03
Glucose S (g/L): 98.31
Lactic Acid P (g/L): 76.92
Maltose M (g/L): 21.09
Volume V (L): 9.23

t = 63.00 h

Biomass X (g/L): 83.99
Glucose S (g/L): 81.52
Lactic Acid P (g/L): 68.35
Maltose M (g/L): 22.14
Volume V (L): 8.70

t = 64.00 h

Biomass X (g/L): 79.52
Glucose S (g/L): 93.20
Lactic Acid P (g/L): 87.51
Maltose M (g/L): 18.99
Volume V (L): 8.90

t = 65.00 h

Biomass X (g/L): 71.55
Glucose S (g/L): 78.35
Lactic Acid P (g/L): 77.80
Maltose M (g/L): 20.99
Volume V (L): 8.63

t = 66.00 h

Biomass X (g/L): 81.51
Glucose S (g/L): 96.42
Lactic Acid P (g/L): 84.94
Maltose M (g/L): 21.69
Volume V (L): 8.41

t = 67.00 h

Biomass X (g/L): 85.27
Glucose S (g/L): 98.25
Lactic Acid P (g/L): 66.26
Maltose M (g/L): 20.38
Volume V (L): 8.43

t = 68.00 h

Biomass X (g/L): 104.65
Glucose S (g/L): 93.54
Lactic Acid P (g/L): 86.00
Maltose M (g/L): 18.01
Volume V (L): 8.84

t = 69.00 h

Biomass X (g/L): 71.11
Glucose S (g/L): 94.49
Lactic Acid P (g/L): 86.44
Maltose M (g/L): 20.69
Volume V (L): 8.76

t = 70.00 h

Biomass X (g/L): 84.52
Glucose S (g/L): 94.85
Lactic Acid P (g/L): 79.37
Maltose M (g/L): 19.95
Volume V (L): 8.30

t = 71.00 h

Biomass X (g/L): 65.54
Glucose S (g/L): 91.46
Lactic Acid P (g/L): 69.33
Maltose M (g/L): 19.69
Volume V (L): 8.60

t = 72.00 h

Biomass X (g/L): 79.28
Glucose S (g/L): 97.22
Lactic Acid P (g/L): 76.55
Maltose M (g/L): 19.03
Volume V (L): 8.89

t = 73.00 h

Biomass X (g/L): 86.50
Glucose S (g/L): 94.69
Lactic Acid P (g/L): 82.81
Maltose M (g/L): 19.80
Volume V (L): 9.30

t = 74.00 h

Biomass X (g/L): 88.33
Glucose S (g/L): 100.20
Lactic Acid P (g/L): 79.45
Maltose M (g/L): 20.28
Volume V (L): 9.11

t = 75.00 h

Biomass X (g/L): 88.90
Glucose S (g/L): 95.99
Lactic Acid P (g/L): 83.00
Maltose M (g/L): 22.81

Volume V (L): 9.14

Batch 58

Initial Conditions:

Batch: 58.00

X0 (g/L): 8.32

S0 (g/L): 49.95

M0 (g/L): 136.94

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.74

F_vol (L/h): 0.06

S_feed (g/L): 80.18

M_feed (g/L): 219.82

V_final (L): 9.53

Batch 58 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 56.85
Lactic Acid P (g/L): 1.72
Maltose M (g/L): 140.31
Volume V (L): 5.36

t = 1.00 h

Biomass X (g/L): 11.25
Glucose S (g/L): 52.03
Lactic Acid P (g/L): 7.43
Maltose M (g/L): 140.90
Volume V (L): 5.36

t = 2.00 h

Biomass X (g/L): 26.89
Glucose S (g/L): 51.64
Lactic Acid P (g/L): 8.58
Maltose M (g/L): 134.76
Volume V (L): 5.06

t = 3.00 h

Biomass X (g/L): 19.69
Glucose S (g/L): 42.32
Lactic Acid P (g/L): 6.90
Maltose M (g/L): 130.49
Volume V (L): 5.01

t = 4.00 h

Biomass X (g/L): 8.23
Glucose S (g/L): 49.88
Lactic Acid P (g/L): 6.69
Maltose M (g/L): 127.04
Volume V (L): 5.03

t = 5.00 h

Biomass X (g/L): 15.39
Glucose S (g/L): 46.68

Lactic Acid P (g/L): 17.92
Maltose M (g/L): 124.78
Volume V (L): 5.19

t = 6.00 h

Biomass X (g/L): 37.83
Glucose S (g/L): 48.50
Lactic Acid P (g/L): 16.69
Maltose M (g/L): 117.21
Volume V (L): 4.85

t = 7.00 h

Biomass X (g/L): 35.01
Glucose S (g/L): 38.73
Lactic Acid P (g/L): 38.11
Maltose M (g/L): 120.30
Volume V (L): 5.08

t = 8.00 h

Biomass X (g/L): 41.23
Glucose S (g/L): 33.74
Lactic Acid P (g/L): 30.49
Maltose M (g/L): 118.72
Volume V (L): 5.80

t = 9.00 h

Biomass X (g/L): 40.65
Glucose S (g/L): 32.61
Lactic Acid P (g/L): 20.04
Maltose M (g/L): 120.18
Volume V (L): 5.65

t = 10.00 h

Biomass X (g/L): 55.51
Glucose S (g/L): 31.23
Lactic Acid P (g/L): 27.41
Maltose M (g/L): 132.33
Volume V (L): 6.20

t = 11.00 h

Biomass X (g/L): 51.73

Glucose S (g/L): 27.12
Lactic Acid P (g/L): 35.28
Maltose M (g/L): 119.82
Volume V (L): 5.56

t = 12.00 h

Biomass X (g/L): 56.36
Glucose S (g/L): 34.67
Lactic Acid P (g/L): 49.61
Maltose M (g/L): 121.15
Volume V (L): 5.59

t = 13.00 h

Biomass X (g/L): 53.22
Glucose S (g/L): 33.56
Lactic Acid P (g/L): 53.21
Maltose M (g/L): 120.10
Volume V (L): 5.64

t = 14.00 h

Biomass X (g/L): 72.98
Glucose S (g/L): 26.82
Lactic Acid P (g/L): 38.90
Maltose M (g/L): 127.44
Volume V (L): 5.73

t = 15.00 h

Biomass X (g/L): 64.67
Glucose S (g/L): 25.30
Lactic Acid P (g/L): 36.11
Maltose M (g/L): 112.14
Volume V (L): 5.67

t = 16.00 h

Biomass X (g/L): 74.88
Glucose S (g/L): 19.37
Lactic Acid P (g/L): 36.90
Maltose M (g/L): 113.27
Volume V (L): 6.07

t = 17.00 h

Biomass X (g/L): 59.80
Glucose S (g/L): 28.44
Lactic Acid P (g/L): 50.82
Maltose M (g/L): 112.87
Volume V (L): 6.04

t = 18.00 h

Biomass X (g/L): 89.54
Glucose S (g/L): 31.81
Lactic Acid P (g/L): 71.39
Maltose M (g/L): 119.84
Volume V (L): 5.90

t = 19.00 h

Biomass X (g/L): 86.86
Glucose S (g/L): 23.09
Lactic Acid P (g/L): 56.66
Maltose M (g/L): 120.09
Volume V (L): 6.60

t = 20.00 h

Biomass X (g/L): 58.00
Glucose S (g/L): 25.08
Lactic Acid P (g/L): 38.42
Maltose M (g/L): 113.34
Volume V (L): 6.90

t = 21.00 h

Biomass X (g/L): 69.89
Glucose S (g/L): 27.29
Lactic Acid P (g/L): 47.73
Maltose M (g/L): 117.22
Volume V (L): 6.69

t = 22.00 h

Biomass X (g/L): 83.29
Glucose S (g/L): 32.37
Lactic Acid P (g/L): 49.89
Maltose M (g/L): 109.29
Volume V (L): 6.53

t = 23.00 h

Biomass X (g/L): 67.25
Glucose S (g/L): 26.94
Lactic Acid P (g/L): 49.20
Maltose M (g/L): 105.99
Volume V (L): 6.67

t = 24.00 h

Biomass X (g/L): 79.54
Glucose S (g/L): 27.30
Lactic Acid P (g/L): 54.26
Maltose M (g/L): 113.79
Volume V (L): 7.12

t = 25.00 h

Biomass X (g/L): 85.73
Glucose S (g/L): 27.68
Lactic Acid P (g/L): 62.22
Maltose M (g/L): 108.30
Volume V (L): 6.79

t = 26.00 h

Biomass X (g/L): 80.69
Glucose S (g/L): 22.13
Lactic Acid P (g/L): 66.03
Maltose M (g/L): 99.36
Volume V (L): 6.36

t = 27.00 h

Biomass X (g/L): 88.29
Glucose S (g/L): 22.59
Lactic Acid P (g/L): 70.68
Maltose M (g/L): 109.70
Volume V (L): 6.77

t = 28.00 h

Biomass X (g/L): 77.18
Glucose S (g/L): 25.57
Lactic Acid P (g/L): 66.26
Maltose M (g/L): 100.91
Volume V (L): 6.92

t = 29.00 h

Biomass X (g/L): 62.98
Glucose S (g/L): 33.04
Lactic Acid P (g/L): 44.19
Maltose M (g/L): 99.20
Volume V (L): 6.39

t = 30.00 h

Biomass X (g/L): 94.21
Glucose S (g/L): 28.61
Lactic Acid P (g/L): 53.18
Maltose M (g/L): 106.27
Volume V (L): 6.71

t = 31.00 h

Biomass X (g/L): 94.59
Glucose S (g/L): 28.54
Lactic Acid P (g/L): 62.16
Maltose M (g/L): 98.99
Volume V (L): 7.27

t = 32.00 h

Biomass X (g/L): 109.54
Glucose S (g/L): 29.17
Lactic Acid P (g/L): 65.62
Maltose M (g/L): 101.23
Volume V (L): 6.27

t = 33.00 h

Biomass X (g/L): 65.02
Glucose S (g/L): 31.42
Lactic Acid P (g/L): 72.81
Maltose M (g/L): 102.16
Volume V (L): 6.54

t = 34.00 h

Biomass X (g/L): 93.50
Glucose S (g/L): 39.08
Lactic Acid P (g/L): 65.86
Maltose M (g/L): 91.90

Volume V (L): 6.99

t = 35.00 h

Biomass X (g/L): 67.87

Glucose S (g/L): 32.85

Lactic Acid P (g/L): 66.48

Maltose M (g/L): 98.69

Volume V (L): 7.24

t = 36.00 h

Biomass X (g/L): 103.58

Glucose S (g/L): 31.21

Lactic Acid P (g/L): 68.73

Maltose M (g/L): 104.16

Volume V (L): 6.68

t = 37.00 h

Biomass X (g/L): 91.69

Glucose S (g/L): 35.03

Lactic Acid P (g/L): 64.69

Maltose M (g/L): 88.34

Volume V (L): 7.19

t = 38.00 h

Biomass X (g/L): 78.15

Glucose S (g/L): 42.32

Lactic Acid P (g/L): 68.32

Maltose M (g/L): 96.72

Volume V (L): 7.72

t = 39.00 h

Biomass X (g/L): 94.78

Glucose S (g/L): 34.60

Lactic Acid P (g/L): 78.35

Maltose M (g/L): 87.61

Volume V (L): 7.38

t = 40.00 h

Biomass X (g/L): 82.19

Glucose S (g/L): 32.83

Lactic Acid P (g/L): 67.64

Maltose M (g/L): 98.91

Volume V (L): 7.76

t = 41.00 h

Biomass X (g/L): 90.41

Glucose S (g/L): 36.18

Lactic Acid P (g/L): 68.55

Maltose M (g/L): 98.11

Volume V (L): 7.26

t = 42.00 h

Biomass X (g/L): 89.58

Glucose S (g/L): 39.34

Lactic Acid P (g/L): 68.91

Maltose M (g/L): 88.47

Volume V (L): 7.25

t = 43.00 h

Biomass X (g/L): 104.27

Glucose S (g/L): 42.24

Lactic Acid P (g/L): 81.56

Maltose M (g/L): 104.46

Volume V (L): 7.91

t = 44.00 h

Biomass X (g/L): 104.97

Glucose S (g/L): 44.24

Lactic Acid P (g/L): 59.76

Maltose M (g/L): 96.55

Volume V (L): 7.79

t = 45.00 h

Biomass X (g/L): 82.00

Glucose S (g/L): 37.30

Lactic Acid P (g/L): 86.69

Maltose M (g/L): 93.25

Volume V (L): 7.29

t = 46.00 h

Biomass X (g/L): 97.76

Glucose S (g/L): 38.46

Lactic Acid P (g/L): 59.82

Maltose M (g/L): 92.19

Volume V (L): 7.49

t = 47.00 h

Biomass X (g/L): 91.04

Glucose S (g/L): 41.61

Lactic Acid P (g/L): 64.95

Maltose M (g/L): 93.69

Volume V (L): 8.41

t = 48.00 h

Biomass X (g/L): 74.16

Glucose S (g/L): 46.20

Lactic Acid P (g/L): 78.16

Maltose M (g/L): 80.12

Volume V (L): 7.57

t = 49.00 h

Biomass X (g/L): 80.93

Glucose S (g/L): 39.66

Lactic Acid P (g/L): 81.09

Maltose M (g/L): 87.22

Volume V (L): 8.30

t = 50.00 h

Biomass X (g/L): 109.75

Glucose S (g/L): 42.91

Lactic Acid P (g/L): 58.95

Maltose M (g/L): 87.80

Volume V (L): 7.85

t = 51.00 h

Biomass X (g/L): 90.83

Glucose S (g/L): 43.74

Lactic Acid P (g/L): 62.09

Maltose M (g/L): 82.05

Volume V (L): 8.08

t = 52.00 h

Biomass X (g/L): 96.86

Glucose S (g/L): 50.30
Lactic Acid P (g/L): 74.99
Maltose M (g/L): 87.06
Volume V (L): 8.25

t = 53.00 h

Biomass X (g/L): 85.22
Glucose S (g/L): 49.51
Lactic Acid P (g/L): 94.30
Maltose M (g/L): 91.44
Volume V (L): 7.44

t = 54.00 h

Biomass X (g/L): 81.99
Glucose S (g/L): 51.70
Lactic Acid P (g/L): 84.43
Maltose M (g/L): 80.05
Volume V (L): 7.73

t = 55.00 h

Biomass X (g/L): 87.28
Glucose S (g/L): 54.10
Lactic Acid P (g/L): 70.90
Maltose M (g/L): 81.84
Volume V (L): 8.36

t = 56.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 48.67
Lactic Acid P (g/L): 51.13
Maltose M (g/L): 76.26
Volume V (L): 8.12

t = 57.00 h

Biomass X (g/L): 72.54
Glucose S (g/L): 46.10
Lactic Acid P (g/L): 74.40
Maltose M (g/L): 69.37
Volume V (L): 8.81

t = 58.00 h

Biomass X (g/L): 77.07
Glucose S (g/L): 46.88
Lactic Acid P (g/L): 77.21
Maltose M (g/L): 85.52
Volume V (L): 7.92

t = 59.00 h

Biomass X (g/L): 93.23
Glucose S (g/L): 45.15
Lactic Acid P (g/L): 70.55
Maltose M (g/L): 79.71
Volume V (L): 8.77

t = 60.00 h

Biomass X (g/L): 82.71
Glucose S (g/L): 51.08
Lactic Acid P (g/L): 66.45
Maltose M (g/L): 79.63
Volume V (L): 8.13

t = 61.00 h

Biomass X (g/L): 86.43
Glucose S (g/L): 52.09
Lactic Acid P (g/L): 75.96
Maltose M (g/L): 72.33
Volume V (L): 8.56

t = 62.00 h

Biomass X (g/L): 88.26
Glucose S (g/L): 57.23
Lactic Acid P (g/L): 66.00
Maltose M (g/L): 77.01
Volume V (L): 8.36

t = 63.00 h

Biomass X (g/L): 88.84
Glucose S (g/L): 55.05
Lactic Acid P (g/L): 92.13
Maltose M (g/L): 84.83
Volume V (L): 9.50

t = 64.00 h

Biomass X (g/L): 96.61
Glucose S (g/L): 53.85
Lactic Acid P (g/L): 89.03
Maltose M (g/L): 77.16
Volume V (L): 8.38

t = 65.00 h

Biomass X (g/L): 75.05
Glucose S (g/L): 55.97
Lactic Acid P (g/L): 81.30
Maltose M (g/L): 76.82
Volume V (L): 8.89

t = 66.00 h

Biomass X (g/L): 90.77
Glucose S (g/L): 53.19
Lactic Acid P (g/L): 76.24
Maltose M (g/L): 77.47
Volume V (L): 8.36

t = 67.00 h

Biomass X (g/L): 92.35
Glucose S (g/L): 61.66
Lactic Acid P (g/L): 69.50
Maltose M (g/L): 67.55
Volume V (L): 9.63

t = 68.00 h

Biomass X (g/L): 94.36
Glucose S (g/L): 55.53
Lactic Acid P (g/L): 75.55
Maltose M (g/L): 79.18
Volume V (L): 9.26

t = 69.00 h

Biomass X (g/L): 71.33
Glucose S (g/L): 54.88
Lactic Acid P (g/L): 91.01
Maltose M (g/L): 66.43
Volume V (L): 9.22

t = 70.00 h

Biomass X (g/L): 87.02
Glucose S (g/L): 62.93
Lactic Acid P (g/L): 84.07
Maltose M (g/L): 67.99
Volume V (L): 9.77

t = 71.00 h

Biomass X (g/L): 100.75
Glucose S (g/L): 62.68
Lactic Acid P (g/L): 78.09
Maltose M (g/L): 66.80
Volume V (L): 9.35

t = 72.00 h

Biomass X (g/L): 83.27
Glucose S (g/L): 54.69
Lactic Acid P (g/L): 84.88
Maltose M (g/L): 72.02
Volume V (L): 9.43

t = 73.00 h

Biomass X (g/L): 88.62
Glucose S (g/L): 65.06
Lactic Acid P (g/L): 81.54
Maltose M (g/L): 70.39
Volume V (L): 9.46

t = 74.00 h

Biomass X (g/L): 91.25
Glucose S (g/L): 59.70
Lactic Acid P (g/L): 80.30
Maltose M (g/L): 72.22
Volume V (L): 9.68

t = 75.00 h

Biomass X (g/L): 73.85
Glucose S (g/L): 61.73
Lactic Acid P (g/L): 85.83
Maltose M (g/L): 65.87

Volume V (L): 9.36

Batch 59

Initial Conditions:

Batch: 59.00

X0 (g/L): 7.39

S0 (g/L): 156.69

M0 (g/L): 35.71

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.23

F_vol (L/h): 0.07

S_feed (g/L): 244.32

M_feed (g/L): 55.68

V_final (L): 10.23

Batch 59 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 5.32
Glucose S (g/L): 149.38
Lactic Acid P (g/L): 2.78
Maltose M (g/L): 32.21
Volume V (L): 4.50

t = 1.00 h

Biomass X (g/L): 17.87
Glucose S (g/L): 157.10
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 35.53
Volume V (L): 5.77

t = 2.00 h

Biomass X (g/L): 15.98
Glucose S (g/L): 154.61
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 32.82
Volume V (L): 5.32

t = 3.00 h

Biomass X (g/L): 16.11
Glucose S (g/L): 147.64
Lactic Acid P (g/L): 3.72
Maltose M (g/L): 34.23
Volume V (L): 5.10

t = 4.00 h

Biomass X (g/L): 8.16
Glucose S (g/L): 156.88
Lactic Acid P (g/L): 23.66
Maltose M (g/L): 33.27
Volume V (L): 5.35

t = 5.00 h

Biomass X (g/L): 41.22
Glucose S (g/L): 148.87

Lactic Acid P (g/L): 14.30

Maltose M (g/L): 33.41

Volume V (L): 5.55

t = 6.00 h

Biomass X (g/L): 25.51

Glucose S (g/L): 145.60

Lactic Acid P (g/L): 9.31

Maltose M (g/L): 33.14

Volume V (L): 4.82

t = 7.00 h

Biomass X (g/L): 46.76

Glucose S (g/L): 136.10

Lactic Acid P (g/L): 13.34

Maltose M (g/L): 33.35

Volume V (L): 5.55

t = 8.00 h

Biomass X (g/L): 44.05

Glucose S (g/L): 134.69

Lactic Acid P (g/L): 23.71

Maltose M (g/L): 31.47

Volume V (L): 5.37

t = 9.00 h

Biomass X (g/L): 34.70

Glucose S (g/L): 134.16

Lactic Acid P (g/L): 31.91

Maltose M (g/L): 34.41

Volume V (L): 5.84

t = 10.00 h

Biomass X (g/L): 63.94

Glucose S (g/L): 128.56

Lactic Acid P (g/L): 37.66

Maltose M (g/L): 28.92

Volume V (L): 5.39

t = 11.00 h

Biomass X (g/L): 58.45

Glucose S (g/L): 132.57
Lactic Acid P (g/L): 37.60
Maltose M (g/L): 31.30
Volume V (L): 5.51

t = 12.00 h

Biomass X (g/L): 72.71
Glucose S (g/L): 121.18
Lactic Acid P (g/L): 31.70
Maltose M (g/L): 32.74
Volume V (L): 6.53

t = 13.00 h

Biomass X (g/L): 61.40
Glucose S (g/L): 117.38
Lactic Acid P (g/L): 38.26
Maltose M (g/L): 31.83
Volume V (L): 6.42

t = 14.00 h

Biomass X (g/L): 73.03
Glucose S (g/L): 120.85
Lactic Acid P (g/L): 51.37
Maltose M (g/L): 29.10
Volume V (L): 4.60

t = 15.00 h

Biomass X (g/L): 51.83
Glucose S (g/L): 109.92
Lactic Acid P (g/L): 36.35
Maltose M (g/L): 29.08
Volume V (L): 6.53

t = 16.00 h

Biomass X (g/L): 66.47
Glucose S (g/L): 118.99
Lactic Acid P (g/L): 44.64
Maltose M (g/L): 28.45
Volume V (L): 6.09

t = 17.00 h

Biomass X (g/L): 61.11
Glucose S (g/L): 106.96
Lactic Acid P (g/L): 56.26
Maltose M (g/L): 30.83
Volume V (L): 5.89

t = 18.00 h

Biomass X (g/L): 71.70
Glucose S (g/L): 115.26
Lactic Acid P (g/L): 60.51
Maltose M (g/L): 28.32
Volume V (L): 5.83

t = 19.00 h

Biomass X (g/L): 81.58
Glucose S (g/L): 108.14
Lactic Acid P (g/L): 52.24
Maltose M (g/L): 29.70
Volume V (L): 5.93

t = 20.00 h

Biomass X (g/L): 72.63
Glucose S (g/L): 112.84
Lactic Acid P (g/L): 51.65
Maltose M (g/L): 29.44
Volume V (L): 6.11

t = 21.00 h

Biomass X (g/L): 88.18
Glucose S (g/L): 114.31
Lactic Acid P (g/L): 44.57
Maltose M (g/L): 32.67
Volume V (L): 5.99

t = 22.00 h

Biomass X (g/L): 72.66
Glucose S (g/L): 110.18
Lactic Acid P (g/L): 62.74
Maltose M (g/L): 28.98
Volume V (L): 5.97

t = 23.00 h

Biomass X (g/L): 90.57
Glucose S (g/L): 103.26
Lactic Acid P (g/L): 71.12
Maltose M (g/L): 26.85
Volume V (L): 6.17

t = 24.00 h

Biomass X (g/L): 95.59
Glucose S (g/L): 104.44
Lactic Acid P (g/L): 83.57
Maltose M (g/L): 25.51
Volume V (L): 6.81

t = 25.00 h

Biomass X (g/L): 93.51
Glucose S (g/L): 105.05
Lactic Acid P (g/L): 64.50
Maltose M (g/L): 27.24
Volume V (L): 6.76

t = 26.00 h

Biomass X (g/L): 81.06
Glucose S (g/L): 99.28
Lactic Acid P (g/L): 61.35
Maltose M (g/L): 27.25
Volume V (L): 7.05

t = 27.00 h

Biomass X (g/L): 64.41
Glucose S (g/L): 113.82
Lactic Acid P (g/L): 54.18
Maltose M (g/L): 23.52
Volume V (L): 6.86

t = 28.00 h

Biomass X (g/L): 91.19
Glucose S (g/L): 108.89
Lactic Acid P (g/L): 46.12
Maltose M (g/L): 25.78
Volume V (L): 6.41

t = 29.00 h

Biomass X (g/L): 84.81
Glucose S (g/L): 102.90
Lactic Acid P (g/L): 65.70
Maltose M (g/L): 27.23
Volume V (L): 7.23

t = 30.00 h

Biomass X (g/L): 76.94
Glucose S (g/L): 115.51
Lactic Acid P (g/L): 86.32
Maltose M (g/L): 25.62
Volume V (L): 6.67

t = 31.00 h

Biomass X (g/L): 81.39
Glucose S (g/L): 105.20
Lactic Acid P (g/L): 77.02
Maltose M (g/L): 27.89
Volume V (L): 7.43

t = 32.00 h

Biomass X (g/L): 100.98
Glucose S (g/L): 109.95
Lactic Acid P (g/L): 76.19
Maltose M (g/L): 23.46
Volume V (L): 6.74

t = 33.00 h

Biomass X (g/L): 76.31
Glucose S (g/L): 104.93
Lactic Acid P (g/L): 68.73
Maltose M (g/L): 26.97
Volume V (L): 7.01

t = 34.00 h

Biomass X (g/L): 92.09
Glucose S (g/L): 99.37
Lactic Acid P (g/L): 70.29
Maltose M (g/L): 27.15

Volume V (L): 7.22

t = 35.00 h

Biomass X (g/L): 89.84

Glucose S (g/L): 102.75

Lactic Acid P (g/L): 45.21

Maltose M (g/L): 27.00

Volume V (L): 7.72

t = 36.00 h

Biomass X (g/L): 84.40

Glucose S (g/L): 107.43

Lactic Acid P (g/L): 74.93

Maltose M (g/L): 22.60

Volume V (L): 7.58

t = 37.00 h

Biomass X (g/L): 94.23

Glucose S (g/L): 114.21

Lactic Acid P (g/L): 63.04

Maltose M (g/L): 25.52

Volume V (L): 7.59

t = 38.00 h

Biomass X (g/L): 103.88

Glucose S (g/L): 111.82

Lactic Acid P (g/L): 62.33

Maltose M (g/L): 23.68

Volume V (L): 8.44

t = 39.00 h

Biomass X (g/L): 85.25

Glucose S (g/L): 111.12

Lactic Acid P (g/L): 66.20

Maltose M (g/L): 22.26

Volume V (L): 6.95

t = 40.00 h

Biomass X (g/L): 85.79

Glucose S (g/L): 101.03

Lactic Acid P (g/L): 66.26

Maltose M (g/L): 23.71

Volume V (L): 7.36

t = 41.00 h

Biomass X (g/L): 94.19

Glucose S (g/L): 117.20

Lactic Acid P (g/L): 63.30

Maltose M (g/L): 22.70

Volume V (L): 7.41

t = 42.00 h

Biomass X (g/L): 97.05

Glucose S (g/L): 112.47

Lactic Acid P (g/L): 70.36

Maltose M (g/L): 25.55

Volume V (L): 7.19

t = 43.00 h

Biomass X (g/L): 86.89

Glucose S (g/L): 111.15

Lactic Acid P (g/L): 65.43

Maltose M (g/L): 24.69

Volume V (L): 7.70

t = 44.00 h

Biomass X (g/L): 85.12

Glucose S (g/L): 117.54

Lactic Acid P (g/L): 71.78

Maltose M (g/L): 21.73

Volume V (L): 8.36

t = 45.00 h

Biomass X (g/L): 66.53

Glucose S (g/L): 118.44

Lactic Acid P (g/L): 57.50

Maltose M (g/L): 22.26

Volume V (L): 7.89

t = 46.00 h

Biomass X (g/L): 80.41

Glucose S (g/L): 118.34

Lactic Acid P (g/L): 69.47

Maltose M (g/L): 22.66

Volume V (L): 8.07

t = 47.00 h

Biomass X (g/L): 76.82

Glucose S (g/L): 111.92

Lactic Acid P (g/L): 61.64

Maltose M (g/L): 22.99

Volume V (L): 7.56

t = 48.00 h

Biomass X (g/L): 94.41

Glucose S (g/L): 108.62

Lactic Acid P (g/L): 66.08

Maltose M (g/L): 24.62

Volume V (L): 8.01

t = 49.00 h

Biomass X (g/L): 85.30

Glucose S (g/L): 118.62

Lactic Acid P (g/L): 73.54

Maltose M (g/L): 23.92

Volume V (L): 8.22

t = 50.00 h

Biomass X (g/L): 84.80

Glucose S (g/L): 107.60

Lactic Acid P (g/L): 57.73

Maltose M (g/L): 22.69

Volume V (L): 8.76

t = 51.00 h

Biomass X (g/L): 87.33

Glucose S (g/L): 123.97

Lactic Acid P (g/L): 65.14

Maltose M (g/L): 22.00

Volume V (L): 7.86

t = 52.00 h

Biomass X (g/L): 78.85

Glucose S (g/L): 115.06
Lactic Acid P (g/L): 78.14
Maltose M (g/L): 19.32
Volume V (L): 8.44

t = 53.00 h

Biomass X (g/L): 102.19
Glucose S (g/L): 110.16
Lactic Acid P (g/L): 60.15
Maltose M (g/L): 24.26
Volume V (L): 9.15

t = 54.00 h

Biomass X (g/L): 99.57
Glucose S (g/L): 109.70
Lactic Acid P (g/L): 89.67
Maltose M (g/L): 21.21
Volume V (L): 8.58

t = 55.00 h

Biomass X (g/L): 102.52
Glucose S (g/L): 109.84
Lactic Acid P (g/L): 84.61
Maltose M (g/L): 24.38
Volume V (L): 7.95

t = 56.00 h

Biomass X (g/L): 93.30
Glucose S (g/L): 117.06
Lactic Acid P (g/L): 69.20
Maltose M (g/L): 21.97
Volume V (L): 9.44

t = 57.00 h

Biomass X (g/L): 86.22
Glucose S (g/L): 116.71
Lactic Acid P (g/L): 66.70
Maltose M (g/L): 24.86
Volume V (L): 9.28

t = 58.00 h

Biomass X (g/L): 101.98
Glucose S (g/L): 111.13
Lactic Acid P (g/L): 75.96
Maltose M (g/L): 22.35
Volume V (L): 9.41

t = 59.00 h

Biomass X (g/L): 79.55
Glucose S (g/L): 115.74
Lactic Acid P (g/L): 60.99
Maltose M (g/L): 18.80
Volume V (L): 8.39

t = 60.00 h

Biomass X (g/L): 77.53
Glucose S (g/L): 111.21
Lactic Acid P (g/L): 56.07
Maltose M (g/L): 20.02
Volume V (L): 9.82

t = 61.00 h

Biomass X (g/L): 79.96
Glucose S (g/L): 121.03
Lactic Acid P (g/L): 70.40
Maltose M (g/L): 20.61
Volume V (L): 8.93

t = 62.00 h

Biomass X (g/L): 90.67
Glucose S (g/L): 118.27
Lactic Acid P (g/L): 73.55
Maltose M (g/L): 19.30
Volume V (L): 9.11

t = 63.00 h

Biomass X (g/L): 78.02
Glucose S (g/L): 117.06
Lactic Acid P (g/L): 75.81
Maltose M (g/L): 19.98
Volume V (L): 9.10

t = 64.00 h

Biomass X (g/L): 83.71
Glucose S (g/L): 113.01
Lactic Acid P (g/L): 69.45
Maltose M (g/L): 18.20
Volume V (L): 9.86

t = 65.00 h

Biomass X (g/L): 69.49
Glucose S (g/L): 118.14
Lactic Acid P (g/L): 86.94
Maltose M (g/L): 20.35
Volume V (L): 9.46

t = 66.00 h

Biomass X (g/L): 89.48
Glucose S (g/L): 117.93
Lactic Acid P (g/L): 61.65
Maltose M (g/L): 20.60
Volume V (L): 9.81

t = 67.00 h

Biomass X (g/L): 69.91
Glucose S (g/L): 121.81
Lactic Acid P (g/L): 65.07
Maltose M (g/L): 18.39
Volume V (L): 9.71

t = 68.00 h

Biomass X (g/L): 75.37
Glucose S (g/L): 121.43
Lactic Acid P (g/L): 80.14
Maltose M (g/L): 23.17
Volume V (L): 9.70

t = 69.00 h

Biomass X (g/L): 96.00
Glucose S (g/L): 124.79
Lactic Acid P (g/L): 65.13
Maltose M (g/L): 19.91
Volume V (L): 9.42

t = 70.00 h

Biomass X (g/L): 95.44
Glucose S (g/L): 118.95
Lactic Acid P (g/L): 64.91
Maltose M (g/L): 18.31
Volume V (L): 10.64

t = 71.00 h

Biomass X (g/L): 87.41
Glucose S (g/L): 112.63
Lactic Acid P (g/L): 77.19
Maltose M (g/L): 16.83
Volume V (L): 9.34

t = 72.00 h

Biomass X (g/L): 86.25
Glucose S (g/L): 126.75
Lactic Acid P (g/L): 66.51
Maltose M (g/L): 19.52
Volume V (L): 10.20

t = 73.00 h

Biomass X (g/L): 78.22
Glucose S (g/L): 117.76
Lactic Acid P (g/L): 75.08
Maltose M (g/L): 18.86
Volume V (L): 10.25

t = 74.00 h

Biomass X (g/L): 75.66
Glucose S (g/L): 117.26
Lactic Acid P (g/L): 70.00
Maltose M (g/L): 20.48
Volume V (L): 10.65

t = 75.00 h

Biomass X (g/L): 79.43
Glucose S (g/L): 122.72
Lactic Acid P (g/L): 68.63
Maltose M (g/L): 18.81

Volume V (L): 10.81

Batch 60

Initial Conditions:

Batch: 60.00

X0 (g/L): 3.84

S0 (g/L): 87.66

M0 (g/L): 17.55

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.20

F_vol (L/h): 0.02

S_feed (g/L): 249.95

M_feed (g/L): 50.05

V_final (L): 6.79

Batch 60 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 2.41
Glucose S (g/L): 79.11
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 16.89
Volume V (L): 4.93

t = 1.00 h

Biomass X (g/L): 10.72
Glucose S (g/L): 82.44
Lactic Acid P (g/L): 3.86
Maltose M (g/L): 18.41
Volume V (L): 4.75

t = 2.00 h

Biomass X (g/L): 25.31
Glucose S (g/L): 85.78
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 18.14
Volume V (L): 5.16

t = 3.00 h

Biomass X (g/L): 8.65
Glucose S (g/L): 87.64
Lactic Acid P (g/L): 23.78
Maltose M (g/L): 16.47
Volume V (L): 5.04

t = 4.00 h

Biomass X (g/L): 1.12
Glucose S (g/L): 85.29
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 16.56
Volume V (L): 5.16

t = 5.00 h

Biomass X (g/L): 11.60
Glucose S (g/L): 87.97

Lactic Acid P (g/L): 6.03
Maltose M (g/L): 16.32
Volume V (L): 4.91

t = 6.00 h

Biomass X (g/L): 6.00
Glucose S (g/L): 82.34
Lactic Acid P (g/L): 12.41
Maltose M (g/L): 15.49
Volume V (L): 5.34

t = 7.00 h

Biomass X (g/L): 27.75
Glucose S (g/L): 78.59
Lactic Acid P (g/L): 13.56
Maltose M (g/L): 15.62
Volume V (L): 5.00

t = 8.00 h

Biomass X (g/L): 39.61
Glucose S (g/L): 72.57
Lactic Acid P (g/L): 11.53
Maltose M (g/L): 16.93
Volume V (L): 5.30

t = 9.00 h

Biomass X (g/L): 32.16
Glucose S (g/L): 75.38
Lactic Acid P (g/L): 35.49
Maltose M (g/L): 16.41
Volume V (L): 5.23

t = 10.00 h

Biomass X (g/L): 21.44
Glucose S (g/L): 61.55
Lactic Acid P (g/L): 34.66
Maltose M (g/L): 16.08
Volume V (L): 5.21

t = 11.00 h

Biomass X (g/L): 53.53

Glucose S (g/L): 57.87
Lactic Acid P (g/L): 31.92
Maltose M (g/L): 14.71
Volume V (L): 5.22

t = 12.00 h

Biomass X (g/L): 29.00
Glucose S (g/L): 44.23
Lactic Acid P (g/L): 28.12
Maltose M (g/L): 15.65
Volume V (L): 5.20

t = 13.00 h

Biomass X (g/L): 35.57
Glucose S (g/L): 59.11
Lactic Acid P (g/L): 45.86
Maltose M (g/L): 13.68
Volume V (L): 5.50

t = 14.00 h

Biomass X (g/L): 53.61
Glucose S (g/L): 48.97
Lactic Acid P (g/L): 33.65
Maltose M (g/L): 15.92
Volume V (L): 5.34

t = 15.00 h

Biomass X (g/L): 61.60
Glucose S (g/L): 64.71
Lactic Acid P (g/L): 56.65
Maltose M (g/L): 15.34
Volume V (L): 5.18

t = 16.00 h

Biomass X (g/L): 66.58
Glucose S (g/L): 41.51
Lactic Acid P (g/L): 44.69
Maltose M (g/L): 13.99
Volume V (L): 5.33

t = 17.00 h

Biomass X (g/L): 64.43
Glucose S (g/L): 42.25
Lactic Acid P (g/L): 58.17
Maltose M (g/L): 16.10
Volume V (L): 5.42

t = 18.00 h

Biomass X (g/L): 85.98
Glucose S (g/L): 43.02
Lactic Acid P (g/L): 46.97
Maltose M (g/L): 13.38
Volume V (L): 5.56

t = 19.00 h

Biomass X (g/L): 70.79
Glucose S (g/L): 31.01
Lactic Acid P (g/L): 50.81
Maltose M (g/L): 15.01
Volume V (L): 5.37

t = 20.00 h

Biomass X (g/L): 82.97
Glucose S (g/L): 38.41
Lactic Acid P (g/L): 64.12
Maltose M (g/L): 14.91
Volume V (L): 5.40

t = 21.00 h

Biomass X (g/L): 82.32
Glucose S (g/L): 34.97
Lactic Acid P (g/L): 75.44
Maltose M (g/L): 13.42
Volume V (L): 5.51

t = 22.00 h

Biomass X (g/L): 62.50
Glucose S (g/L): 31.62
Lactic Acid P (g/L): 70.71
Maltose M (g/L): 14.21
Volume V (L): 5.65

t = 23.00 h

Biomass X (g/L): 75.69
Glucose S (g/L): 39.62
Lactic Acid P (g/L): 65.37
Maltose M (g/L): 12.87
Volume V (L): 5.35

t = 24.00 h

Biomass X (g/L): 100.86
Glucose S (g/L): 25.81
Lactic Acid P (g/L): 63.85
Maltose M (g/L): 11.84
Volume V (L): 5.65

t = 25.00 h

Biomass X (g/L): 74.89
Glucose S (g/L): 32.22
Lactic Acid P (g/L): 54.21
Maltose M (g/L): 13.07
Volume V (L): 5.53

t = 26.00 h

Biomass X (g/L): 74.27
Glucose S (g/L): 38.75
Lactic Acid P (g/L): 63.35
Maltose M (g/L): 13.30
Volume V (L): 5.58

t = 27.00 h

Biomass X (g/L): 84.06
Glucose S (g/L): 24.74
Lactic Acid P (g/L): 52.43
Maltose M (g/L): 14.14
Volume V (L): 5.93

t = 28.00 h

Biomass X (g/L): 78.36
Glucose S (g/L): 26.69
Lactic Acid P (g/L): 66.27
Maltose M (g/L): 13.60
Volume V (L): 5.64

t = 29.00 h

Biomass X (g/L): 75.83
Glucose S (g/L): 42.51
Lactic Acid P (g/L): 67.88
Maltose M (g/L): 13.51
Volume V (L): 5.80

t = 30.00 h

Biomass X (g/L): 79.80
Glucose S (g/L): 30.53
Lactic Acid P (g/L): 44.92
Maltose M (g/L): 12.56
Volume V (L): 5.82

t = 31.00 h

Biomass X (g/L): 100.33
Glucose S (g/L): 24.10
Lactic Acid P (g/L): 65.66
Maltose M (g/L): 12.82
Volume V (L): 5.57

t = 32.00 h

Biomass X (g/L): 84.68
Glucose S (g/L): 31.87
Lactic Acid P (g/L): 70.21
Maltose M (g/L): 12.42
Volume V (L): 5.50

t = 33.00 h

Biomass X (g/L): 72.88
Glucose S (g/L): 23.14
Lactic Acid P (g/L): 66.02
Maltose M (g/L): 12.09
Volume V (L): 5.91

t = 34.00 h

Biomass X (g/L): 71.47
Glucose S (g/L): 16.77
Lactic Acid P (g/L): 71.15
Maltose M (g/L): 12.47

Volume V (L): 5.66

t = 35.00 h

Biomass X (g/L): 110.45

Glucose S (g/L): 34.74

Lactic Acid P (g/L): 54.94

Maltose M (g/L): 12.88

Volume V (L): 5.60

t = 36.00 h

Biomass X (g/L): 85.66

Glucose S (g/L): 19.31

Lactic Acid P (g/L): 61.23

Maltose M (g/L): 12.30

Volume V (L): 6.01

t = 37.00 h

Biomass X (g/L): 92.83

Glucose S (g/L): 41.17

Lactic Acid P (g/L): 79.87

Maltose M (g/L): 12.39

Volume V (L): 5.74

t = 38.00 h

Biomass X (g/L): 80.65

Glucose S (g/L): 22.93

Lactic Acid P (g/L): 70.19

Maltose M (g/L): 13.07

Volume V (L): 5.80

t = 39.00 h

Biomass X (g/L): 92.24

Glucose S (g/L): 39.09

Lactic Acid P (g/L): 74.14

Maltose M (g/L): 12.07

Volume V (L): 6.22

t = 40.00 h

Biomass X (g/L): 104.28

Glucose S (g/L): 17.44

Lactic Acid P (g/L): 65.88

Maltose M (g/L): 11.35

Volume V (L): 5.72

t = 41.00 h

Biomass X (g/L): 82.61

Glucose S (g/L): 32.23

Lactic Acid P (g/L): 72.04

Maltose M (g/L): 12.70

Volume V (L): 5.80

t = 42.00 h

Biomass X (g/L): 94.31

Glucose S (g/L): 29.02

Lactic Acid P (g/L): 49.47

Maltose M (g/L): 12.39

Volume V (L): 5.94

t = 43.00 h

Biomass X (g/L): 75.14

Glucose S (g/L): 31.15

Lactic Acid P (g/L): 81.27

Maltose M (g/L): 12.35

Volume V (L): 5.99

t = 44.00 h

Biomass X (g/L): 110.80

Glucose S (g/L): 32.03

Lactic Acid P (g/L): 91.31

Maltose M (g/L): 12.49

Volume V (L): 5.92

t = 45.00 h

Biomass X (g/L): 92.73

Glucose S (g/L): 23.06

Lactic Acid P (g/L): 63.20

Maltose M (g/L): 12.28

Volume V (L): 5.91

t = 46.00 h

Biomass X (g/L): 85.76

Glucose S (g/L): 18.15

Lactic Acid P (g/L): 64.88

Maltose M (g/L): 11.88

Volume V (L): 6.32

t = 47.00 h

Biomass X (g/L): 87.02

Glucose S (g/L): 26.03

Lactic Acid P (g/L): 79.04

Maltose M (g/L): 13.59

Volume V (L): 6.16

t = 48.00 h

Biomass X (g/L): 84.87

Glucose S (g/L): 27.85

Lactic Acid P (g/L): 89.72

Maltose M (g/L): 10.72

Volume V (L): 6.23

t = 49.00 h

Biomass X (g/L): 85.43

Glucose S (g/L): 23.70

Lactic Acid P (g/L): 75.45

Maltose M (g/L): 11.17

Volume V (L): 6.15

t = 50.00 h

Biomass X (g/L): 64.83

Glucose S (g/L): 28.48

Lactic Acid P (g/L): 63.63

Maltose M (g/L): 12.28

Volume V (L): 6.24

t = 51.00 h

Biomass X (g/L): 89.18

Glucose S (g/L): 28.97

Lactic Acid P (g/L): 74.16

Maltose M (g/L): 11.58

Volume V (L): 6.35

t = 52.00 h

Biomass X (g/L): 88.74

Glucose S (g/L): 21.48
Lactic Acid P (g/L): 66.36
Maltose M (g/L): 11.97
Volume V (L): 6.10

t = 53.00 h

Biomass X (g/L): 95.22
Glucose S (g/L): 29.72
Lactic Acid P (g/L): 70.30
Maltose M (g/L): 11.45
Volume V (L): 6.31

t = 54.00 h

Biomass X (g/L): 84.70
Glucose S (g/L): 25.82
Lactic Acid P (g/L): 74.27
Maltose M (g/L): 12.29
Volume V (L): 6.39

t = 55.00 h

Biomass X (g/L): 82.08
Glucose S (g/L): 23.34
Lactic Acid P (g/L): 68.39
Maltose M (g/L): 10.41
Volume V (L): 6.54

t = 56.00 h

Biomass X (g/L): 89.93
Glucose S (g/L): 32.73
Lactic Acid P (g/L): 70.22
Maltose M (g/L): 12.13
Volume V (L): 6.62

t = 57.00 h

Biomass X (g/L): 73.23
Glucose S (g/L): 23.28
Lactic Acid P (g/L): 79.01
Maltose M (g/L): 11.93
Volume V (L): 6.37

t = 58.00 h

Biomass X (g/L): 91.34
Glucose S (g/L): 25.48
Lactic Acid P (g/L): 80.72
Maltose M (g/L): 10.06
Volume V (L): 6.55

t = 59.00 h

Biomass X (g/L): 70.47
Glucose S (g/L): 28.96
Lactic Acid P (g/L): 64.72
Maltose M (g/L): 11.39
Volume V (L): 6.03

t = 60.00 h

Biomass X (g/L): 101.28
Glucose S (g/L): 37.23
Lactic Acid P (g/L): 83.11
Maltose M (g/L): 10.81
Volume V (L): 6.66

t = 61.00 h

Biomass X (g/L): 96.65
Glucose S (g/L): 30.17
Lactic Acid P (g/L): 75.66
Maltose M (g/L): 10.32
Volume V (L): 6.35

t = 62.00 h

Biomass X (g/L): 79.85
Glucose S (g/L): 32.47
Lactic Acid P (g/L): 86.90
Maltose M (g/L): 9.28
Volume V (L): 6.45

t = 63.00 h

Biomass X (g/L): 96.18
Glucose S (g/L): 27.61
Lactic Acid P (g/L): 87.77
Maltose M (g/L): 9.88
Volume V (L): 6.57

t = 64.00 h

Biomass X (g/L): 90.36
Glucose S (g/L): 29.21
Lactic Acid P (g/L): 74.86
Maltose M (g/L): 11.13
Volume V (L): 6.51

t = 65.00 h

Biomass X (g/L): 78.61
Glucose S (g/L): 29.64
Lactic Acid P (g/L): 86.64
Maltose M (g/L): 11.15
Volume V (L): 6.58

t = 66.00 h

Biomass X (g/L): 67.97
Glucose S (g/L): 32.00
Lactic Acid P (g/L): 72.47
Maltose M (g/L): 10.31
Volume V (L): 6.81

t = 67.00 h

Biomass X (g/L): 92.07
Glucose S (g/L): 28.62
Lactic Acid P (g/L): 94.20
Maltose M (g/L): 11.03
Volume V (L): 6.55

t = 68.00 h

Biomass X (g/L): 80.41
Glucose S (g/L): 38.98
Lactic Acid P (g/L): 73.34
Maltose M (g/L): 11.07
Volume V (L): 6.47

t = 69.00 h

Biomass X (g/L): 99.83
Glucose S (g/L): 28.65
Lactic Acid P (g/L): 85.03
Maltose M (g/L): 10.41
Volume V (L): 6.91

t = 70.00 h

Biomass X (g/L): 88.47
Glucose S (g/L): 20.67
Lactic Acid P (g/L): 79.89
Maltose M (g/L): 9.69
Volume V (L): 6.75

t = 71.00 h

Biomass X (g/L): 80.91
Glucose S (g/L): 48.42
Lactic Acid P (g/L): 71.87
Maltose M (g/L): 10.52
Volume V (L): 6.97

t = 72.00 h

Biomass X (g/L): 106.63
Glucose S (g/L): 29.72
Lactic Acid P (g/L): 91.16
Maltose M (g/L): 10.50
Volume V (L): 6.85

t = 73.00 h

Biomass X (g/L): 93.80
Glucose S (g/L): 40.12
Lactic Acid P (g/L): 76.32
Maltose M (g/L): 10.58
Volume V (L): 6.56

t = 74.00 h

Biomass X (g/L): 49.77
Glucose S (g/L): 40.95
Lactic Acid P (g/L): 65.74
Maltose M (g/L): 10.68
Volume V (L): 6.86

t = 75.00 h

Biomass X (g/L): 82.43
Glucose S (g/L): 38.38
Lactic Acid P (g/L): 54.20
Maltose M (g/L): 8.93

Volume V (L): 6.65

Batch 61

Initial Conditions:

Batch: 61.00

X0 (g/L): 6.06

S0 (g/L): 28.95

M0 (g/L): 142.25

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 4.91

F_vol (L/h): 0.04

S_feed (g/L): 50.73

M_feed (g/L): 249.27

V_final (L): 8.27

Batch 61 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.13
Glucose S (g/L): 31.27
Lactic Acid P (g/L): 6.89
Maltose M (g/L): 134.46
Volume V (L): 4.76

t = 1.00 h

Biomass X (g/L): 7.17
Glucose S (g/L): 28.13
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 141.18
Volume V (L): 4.80

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 27.17
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 153.05
Volume V (L): 5.19

t = 3.00 h

Biomass X (g/L): 5.15
Glucose S (g/L): 28.77
Lactic Acid P (g/L): 13.58
Maltose M (g/L): 134.51
Volume V (L): 4.82

t = 4.00 h

Biomass X (g/L): 37.44
Glucose S (g/L): 36.08
Lactic Acid P (g/L): 3.43
Maltose M (g/L): 128.50
Volume V (L): 5.05

t = 5.00 h

Biomass X (g/L): 22.19
Glucose S (g/L): 29.77

Lactic Acid P (g/L): 3.53
Maltose M (g/L): 136.30
Volume V (L): 5.51

t = 6.00 h

Biomass X (g/L): 22.21
Glucose S (g/L): 34.13
Lactic Acid P (g/L): 9.31
Maltose M (g/L): 123.38
Volume V (L): 5.45

t = 7.00 h

Biomass X (g/L): 45.56
Glucose S (g/L): 25.33
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 142.53
Volume V (L): 5.82

t = 8.00 h

Biomass X (g/L): 28.61
Glucose S (g/L): 23.01
Lactic Acid P (g/L): 26.18
Maltose M (g/L): 138.76
Volume V (L): 5.39

t = 9.00 h

Biomass X (g/L): 35.79
Glucose S (g/L): 21.68
Lactic Acid P (g/L): 21.56
Maltose M (g/L): 126.23
Volume V (L): 5.23

t = 10.00 h

Biomass X (g/L): 50.31
Glucose S (g/L): 19.78
Lactic Acid P (g/L): 23.27
Maltose M (g/L): 119.60
Volume V (L): 5.45

t = 11.00 h

Biomass X (g/L): 40.80

Glucose S (g/L): 15.94
Lactic Acid P (g/L): 26.26
Maltose M (g/L): 121.47
Volume V (L): 5.44

t = 12.00 h

Biomass X (g/L): 41.09
Glucose S (g/L): 13.26
Lactic Acid P (g/L): 31.69
Maltose M (g/L): 120.98
Volume V (L): 5.49

t = 13.00 h

Biomass X (g/L): 29.91
Glucose S (g/L): 17.16
Lactic Acid P (g/L): 40.24
Maltose M (g/L): 124.37
Volume V (L): 5.56

t = 14.00 h

Biomass X (g/L): 45.63
Glucose S (g/L): 17.83
Lactic Acid P (g/L): 36.08
Maltose M (g/L): 127.64
Volume V (L): 5.37

t = 15.00 h

Biomass X (g/L): 48.31
Glucose S (g/L): 21.34
Lactic Acid P (g/L): 26.11
Maltose M (g/L): 111.71
Volume V (L): 5.73

t = 16.00 h

Biomass X (g/L): 61.45
Glucose S (g/L): 11.60
Lactic Acid P (g/L): 46.23
Maltose M (g/L): 124.82
Volume V (L): 5.33

t = 17.00 h

Biomass X (g/L): 56.34
Glucose S (g/L): 11.35
Lactic Acid P (g/L): 47.42
Maltose M (g/L): 120.79
Volume V (L): 5.55

t = 18.00 h

Biomass X (g/L): 85.05
Glucose S (g/L): 9.67
Lactic Acid P (g/L): 46.46
Maltose M (g/L): 111.63
Volume V (L): 5.75

t = 19.00 h

Biomass X (g/L): 56.02
Glucose S (g/L): 15.42
Lactic Acid P (g/L): 56.22
Maltose M (g/L): 106.34
Volume V (L): 5.24

t = 20.00 h

Biomass X (g/L): 81.41
Glucose S (g/L): 14.33
Lactic Acid P (g/L): 48.02
Maltose M (g/L): 113.61
Volume V (L): 6.07

t = 21.00 h

Biomass X (g/L): 71.64
Glucose S (g/L): 8.98
Lactic Acid P (g/L): 59.54
Maltose M (g/L): 104.18
Volume V (L): 5.38

t = 22.00 h

Biomass X (g/L): 78.75
Glucose S (g/L): 8.97
Lactic Acid P (g/L): 63.66
Maltose M (g/L): 117.69
Volume V (L): 6.13

t = 23.00 h

Biomass X (g/L): 79.51
Glucose S (g/L): 14.95
Lactic Acid P (g/L): 57.26
Maltose M (g/L): 101.46
Volume V (L): 5.66

t = 24.00 h

Biomass X (g/L): 52.24
Glucose S (g/L): 13.36
Lactic Acid P (g/L): 46.07
Maltose M (g/L): 112.12
Volume V (L): 5.94

t = 25.00 h

Biomass X (g/L): 74.61
Glucose S (g/L): 14.80
Lactic Acid P (g/L): 52.97
Maltose M (g/L): 106.62
Volume V (L): 6.23

t = 26.00 h

Biomass X (g/L): 73.26
Glucose S (g/L): 9.94
Lactic Acid P (g/L): 66.82
Maltose M (g/L): 102.19
Volume V (L): 6.71

t = 27.00 h

Biomass X (g/L): 65.86
Glucose S (g/L): 7.15
Lactic Acid P (g/L): 42.36
Maltose M (g/L): 104.38
Volume V (L): 6.21

t = 28.00 h

Biomass X (g/L): 82.09
Glucose S (g/L): 10.70
Lactic Acid P (g/L): 53.65
Maltose M (g/L): 111.68
Volume V (L): 5.70

t = 29.00 h

Biomass X (g/L): 95.02
Glucose S (g/L): 16.04
Lactic Acid P (g/L): 55.04
Maltose M (g/L): 94.70
Volume V (L): 6.51

t = 30.00 h

Biomass X (g/L): 53.57
Glucose S (g/L): 20.97
Lactic Acid P (g/L): 51.07
Maltose M (g/L): 99.22
Volume V (L): 6.25

t = 31.00 h

Biomass X (g/L): 74.86
Glucose S (g/L): 13.45
Lactic Acid P (g/L): 68.12
Maltose M (g/L): 95.87
Volume V (L): 6.37

t = 32.00 h

Biomass X (g/L): 68.68
Glucose S (g/L): 22.48
Lactic Acid P (g/L): 64.32
Maltose M (g/L): 99.95
Volume V (L): 6.54

t = 33.00 h

Biomass X (g/L): 81.96
Glucose S (g/L): 18.45
Lactic Acid P (g/L): 54.96
Maltose M (g/L): 94.48
Volume V (L): 5.86

t = 34.00 h

Biomass X (g/L): 78.70
Glucose S (g/L): 12.00
Lactic Acid P (g/L): 67.98
Maltose M (g/L): 85.95

Volume V (L): 6.48

t = 35.00 h

Biomass X (g/L): 76.52

Glucose S (g/L): 17.63

Lactic Acid P (g/L): 61.57

Maltose M (g/L): 90.08

Volume V (L): 6.42

t = 36.00 h

Biomass X (g/L): 89.45

Glucose S (g/L): 20.26

Lactic Acid P (g/L): 55.44

Maltose M (g/L): 108.15

Volume V (L): 6.89

t = 37.00 h

Biomass X (g/L): 98.98

Glucose S (g/L): 10.50

Lactic Acid P (g/L): 57.75

Maltose M (g/L): 97.51

Volume V (L): 6.80

t = 38.00 h

Biomass X (g/L): 109.24

Glucose S (g/L): 20.77

Lactic Acid P (g/L): 69.09

Maltose M (g/L): 96.42

Volume V (L): 6.23

t = 39.00 h

Biomass X (g/L): 76.88

Glucose S (g/L): 23.06

Lactic Acid P (g/L): 74.38

Maltose M (g/L): 92.12

Volume V (L): 7.03

t = 40.00 h

Biomass X (g/L): 66.80

Glucose S (g/L): 21.26

Lactic Acid P (g/L): 61.67

Maltose M (g/L): 92.93

Volume V (L): 6.86

t = 41.00 h

Biomass X (g/L): 89.99

Glucose S (g/L): 20.45

Lactic Acid P (g/L): 71.84

Maltose M (g/L): 89.47

Volume V (L): 6.38

t = 42.00 h

Biomass X (g/L): 100.37

Glucose S (g/L): 20.75

Lactic Acid P (g/L): 82.04

Maltose M (g/L): 95.42

Volume V (L): 6.84

t = 43.00 h

Biomass X (g/L): 92.93

Glucose S (g/L): 21.22

Lactic Acid P (g/L): 56.08

Maltose M (g/L): 91.65

Volume V (L): 7.14

t = 44.00 h

Biomass X (g/L): 100.77

Glucose S (g/L): 18.91

Lactic Acid P (g/L): 71.63

Maltose M (g/L): 81.95

Volume V (L): 6.60

t = 45.00 h

Biomass X (g/L): 83.33

Glucose S (g/L): 21.12

Lactic Acid P (g/L): 76.53

Maltose M (g/L): 91.23

Volume V (L): 7.32

t = 46.00 h

Biomass X (g/L): 79.79

Glucose S (g/L): 22.65

Lactic Acid P (g/L): 68.36

Maltose M (g/L): 71.41

Volume V (L): 6.61

t = 47.00 h

Biomass X (g/L): 86.90

Glucose S (g/L): 24.99

Lactic Acid P (g/L): 81.23

Maltose M (g/L): 90.90

Volume V (L): 6.84

t = 48.00 h

Biomass X (g/L): 82.73

Glucose S (g/L): 23.56

Lactic Acid P (g/L): 90.61

Maltose M (g/L): 77.37

Volume V (L): 6.85

t = 49.00 h

Biomass X (g/L): 92.38

Glucose S (g/L): 23.26

Lactic Acid P (g/L): 73.73

Maltose M (g/L): 85.92

Volume V (L): 7.50

t = 50.00 h

Biomass X (g/L): 95.41

Glucose S (g/L): 29.07

Lactic Acid P (g/L): 68.75

Maltose M (g/L): 77.50

Volume V (L): 7.05

t = 51.00 h

Biomass X (g/L): 96.22

Glucose S (g/L): 28.54

Lactic Acid P (g/L): 80.41

Maltose M (g/L): 85.26

Volume V (L): 6.92

t = 52.00 h

Biomass X (g/L): 91.09

Glucose S (g/L): 26.36
Lactic Acid P (g/L): 79.53
Maltose M (g/L): 88.64
Volume V (L): 7.38

t = 53.00 h

Biomass X (g/L): 88.57
Glucose S (g/L): 30.33
Lactic Acid P (g/L): 48.56
Maltose M (g/L): 94.50
Volume V (L): 7.48

t = 54.00 h

Biomass X (g/L): 93.86
Glucose S (g/L): 28.00
Lactic Acid P (g/L): 72.36
Maltose M (g/L): 72.44
Volume V (L): 7.53

t = 55.00 h

Biomass X (g/L): 67.34
Glucose S (g/L): 30.31
Lactic Acid P (g/L): 69.04
Maltose M (g/L): 89.52
Volume V (L): 7.33

t = 56.00 h

Biomass X (g/L): 88.23
Glucose S (g/L): 33.86
Lactic Acid P (g/L): 76.74
Maltose M (g/L): 82.96
Volume V (L): 7.51

t = 57.00 h

Biomass X (g/L): 86.68
Glucose S (g/L): 29.58
Lactic Acid P (g/L): 71.55
Maltose M (g/L): 78.26
Volume V (L): 7.31

t = 58.00 h

Biomass X (g/L): 88.19
Glucose S (g/L): 28.47
Lactic Acid P (g/L): 72.86
Maltose M (g/L): 86.17
Volume V (L): 7.97

t = 59.00 h

Biomass X (g/L): 85.25
Glucose S (g/L): 32.38
Lactic Acid P (g/L): 65.91
Maltose M (g/L): 78.58
Volume V (L): 7.87

t = 60.00 h

Biomass X (g/L): 90.19
Glucose S (g/L): 29.22
Lactic Acid P (g/L): 81.08
Maltose M (g/L): 84.70
Volume V (L): 7.90

t = 61.00 h

Biomass X (g/L): 80.44
Glucose S (g/L): 33.31
Lactic Acid P (g/L): 89.94
Maltose M (g/L): 72.10
Volume V (L): 7.87

t = 62.00 h

Biomass X (g/L): 96.75
Glucose S (g/L): 33.97
Lactic Acid P (g/L): 66.88
Maltose M (g/L): 74.07
Volume V (L): 7.84

t = 63.00 h

Biomass X (g/L): 90.36
Glucose S (g/L): 27.76
Lactic Acid P (g/L): 62.02
Maltose M (g/L): 84.60
Volume V (L): 7.42

t = 64.00 h

Biomass X (g/L): 87.16
Glucose S (g/L): 37.85
Lactic Acid P (g/L): 75.48
Maltose M (g/L): 87.87
Volume V (L): 7.95

t = 65.00 h

Biomass X (g/L): 78.89
Glucose S (g/L): 36.27
Lactic Acid P (g/L): 75.27
Maltose M (g/L): 76.63
Volume V (L): 7.86

t = 66.00 h

Biomass X (g/L): 93.34
Glucose S (g/L): 41.22
Lactic Acid P (g/L): 86.69
Maltose M (g/L): 56.74
Volume V (L): 8.21

t = 67.00 h

Biomass X (g/L): 87.37
Glucose S (g/L): 38.83
Lactic Acid P (g/L): 64.56
Maltose M (g/L): 74.43
Volume V (L): 8.13

t = 68.00 h

Biomass X (g/L): 77.99
Glucose S (g/L): 41.11
Lactic Acid P (g/L): 71.22
Maltose M (g/L): 75.54
Volume V (L): 8.11

t = 69.00 h

Biomass X (g/L): 84.58
Glucose S (g/L): 43.19
Lactic Acid P (g/L): 56.44
Maltose M (g/L): 73.17
Volume V (L): 7.57

t = 70.00 h

Biomass X (g/L): 91.04
Glucose S (g/L): 39.67
Lactic Acid P (g/L): 89.49
Maltose M (g/L): 76.67
Volume V (L): 8.03

t = 71.00 h

Biomass X (g/L): 90.16
Glucose S (g/L): 40.72
Lactic Acid P (g/L): 79.21
Maltose M (g/L): 70.46
Volume V (L): 7.95

t = 72.00 h

Biomass X (g/L): 89.39
Glucose S (g/L): 36.40
Lactic Acid P (g/L): 72.76
Maltose M (g/L): 79.23
Volume V (L): 8.20

t = 73.00 h

Biomass X (g/L): 85.45
Glucose S (g/L): 45.88
Lactic Acid P (g/L): 64.96
Maltose M (g/L): 64.64
Volume V (L): 8.07

t = 74.00 h

Biomass X (g/L): 94.27
Glucose S (g/L): 42.14
Lactic Acid P (g/L): 66.01
Maltose M (g/L): 81.08
Volume V (L): 8.15

t = 75.00 h

Biomass X (g/L): 96.15
Glucose S (g/L): 40.06
Lactic Acid P (g/L): 81.52
Maltose M (g/L): 82.81

Volume V (L): 8.39

Batch 62

Initial Conditions:

Batch: 62.00

X0 (g/L): 7.61

S0 (g/L): 44.76

M0 (g/L): 65.37

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.46

F_vol (L/h): 0.03

S_feed (g/L): 121.93

M_feed (g/L): 178.07

V_final (L): 7.62

Batch 62 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.53
Glucose S (g/L): 47.89
Lactic Acid P (g/L): 0.96
Maltose M (g/L): 64.86
Volume V (L): 5.08

t = 1.00 h

Biomass X (g/L): 11.24
Glucose S (g/L): 40.43
Lactic Acid P (g/L): 6.77
Maltose M (g/L): 66.42
Volume V (L): 5.21

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 39.77
Lactic Acid P (g/L): 3.97
Maltose M (g/L): 66.11
Volume V (L): 5.43

t = 3.00 h

Biomass X (g/L): 8.94
Glucose S (g/L): 38.17
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 62.72
Volume V (L): 5.21

t = 4.00 h

Biomass X (g/L): 27.09
Glucose S (g/L): 45.58
Lactic Acid P (g/L): 1.40
Maltose M (g/L): 61.52
Volume V (L): 4.99

t = 5.00 h

Biomass X (g/L): 13.84
Glucose S (g/L): 37.34

Lactic Acid P (g/L): 15.12

Maltose M (g/L): 60.11

Volume V (L): 5.30

t = 6.00 h

Biomass X (g/L): 41.89

Glucose S (g/L): 34.03

Lactic Acid P (g/L): 6.63

Maltose M (g/L): 62.67

Volume V (L): 5.20

t = 7.00 h

Biomass X (g/L): 44.11

Glucose S (g/L): 30.42

Lactic Acid P (g/L): 9.92

Maltose M (g/L): 58.85

Volume V (L): 4.75

t = 8.00 h

Biomass X (g/L): 37.08

Glucose S (g/L): 20.87

Lactic Acid P (g/L): 27.35

Maltose M (g/L): 61.95

Volume V (L): 5.38

t = 9.00 h

Biomass X (g/L): 72.28

Glucose S (g/L): 30.14

Lactic Acid P (g/L): 26.13

Maltose M (g/L): 64.11

Volume V (L): 5.34

t = 10.00 h

Biomass X (g/L): 59.87

Glucose S (g/L): 21.89

Lactic Acid P (g/L): 38.94

Maltose M (g/L): 60.05

Volume V (L): 5.59

t = 11.00 h

Biomass X (g/L): 42.99

Glucose S (g/L): 20.35
Lactic Acid P (g/L): 29.11
Maltose M (g/L): 59.21
Volume V (L): 5.23

t = 12.00 h

Biomass X (g/L): 62.35
Glucose S (g/L): 8.84
Lactic Acid P (g/L): 30.31
Maltose M (g/L): 64.68
Volume V (L): 5.10

t = 13.00 h

Biomass X (g/L): 58.98
Glucose S (g/L): 16.62
Lactic Acid P (g/L): 42.15
Maltose M (g/L): 63.14
Volume V (L): 5.49

t = 14.00 h

Biomass X (g/L): 53.73
Glucose S (g/L): 10.81
Lactic Acid P (g/L): 32.78
Maltose M (g/L): 57.47
Volume V (L): 5.57

t = 15.00 h

Biomass X (g/L): 53.35
Glucose S (g/L): 5.91
Lactic Acid P (g/L): 36.21
Maltose M (g/L): 59.17
Volume V (L): 5.54

t = 16.00 h

Biomass X (g/L): 74.91
Glucose S (g/L): 6.29
Lactic Acid P (g/L): 38.38
Maltose M (g/L): 59.22
Volume V (L): 5.56

t = 17.00 h

Biomass X (g/L): 61.76
Glucose S (g/L): 4.30
Lactic Acid P (g/L): 43.85
Maltose M (g/L): 58.55
Volume V (L): 5.37

t = 18.00 h

Biomass X (g/L): 68.09
Glucose S (g/L): 3.61
Lactic Acid P (g/L): 63.90
Maltose M (g/L): 61.91
Volume V (L): 5.60

t = 19.00 h

Biomass X (g/L): 61.85
Glucose S (g/L): 8.19
Lactic Acid P (g/L): 35.79
Maltose M (g/L): 57.53
Volume V (L): 5.59

t = 20.00 h

Biomass X (g/L): 64.80
Glucose S (g/L): 6.14
Lactic Acid P (g/L): 62.08
Maltose M (g/L): 57.51
Volume V (L): 5.92

t = 21.00 h

Biomass X (g/L): 71.84
Glucose S (g/L): 8.61
Lactic Acid P (g/L): 58.04
Maltose M (g/L): 58.19
Volume V (L): 5.81

t = 22.00 h

Biomass X (g/L): 77.78
Glucose S (g/L): 13.50
Lactic Acid P (g/L): 66.92
Maltose M (g/L): 59.00
Volume V (L): 5.71

t = 23.00 h

Biomass X (g/L): 50.73
Glucose S (g/L): 6.59
Lactic Acid P (g/L): 82.66
Maltose M (g/L): 58.06
Volume V (L): 6.16

t = 24.00 h

Biomass X (g/L): 76.17
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 67.68
Maltose M (g/L): 56.78
Volume V (L): 5.78

t = 25.00 h

Biomass X (g/L): 69.81
Glucose S (g/L): 6.15
Lactic Acid P (g/L): 63.89
Maltose M (g/L): 51.58
Volume V (L): 5.99

t = 26.00 h

Biomass X (g/L): 90.82
Glucose S (g/L): 11.62
Lactic Acid P (g/L): 43.83
Maltose M (g/L): 53.72
Volume V (L): 5.87

t = 27.00 h

Biomass X (g/L): 106.04
Glucose S (g/L): 6.89
Lactic Acid P (g/L): 45.87
Maltose M (g/L): 56.41
Volume V (L): 6.28

t = 28.00 h

Biomass X (g/L): 75.60
Glucose S (g/L): 13.11
Lactic Acid P (g/L): 61.18
Maltose M (g/L): 54.76
Volume V (L): 5.99

t = 29.00 h

Biomass X (g/L): 63.38
Glucose S (g/L): 9.20
Lactic Acid P (g/L): 42.70
Maltose M (g/L): 55.77
Volume V (L): 6.04

t = 30.00 h

Biomass X (g/L): 81.28
Glucose S (g/L): 0.74
Lactic Acid P (g/L): 65.13
Maltose M (g/L): 57.91
Volume V (L): 6.02

t = 31.00 h

Biomass X (g/L): 78.78
Glucose S (g/L): 8.29
Lactic Acid P (g/L): 40.12
Maltose M (g/L): 50.48
Volume V (L): 6.12

t = 32.00 h

Biomass X (g/L): 99.79
Glucose S (g/L): 9.22
Lactic Acid P (g/L): 71.61
Maltose M (g/L): 55.53
Volume V (L): 6.37

t = 33.00 h

Biomass X (g/L): 97.39
Glucose S (g/L): 12.41
Lactic Acid P (g/L): 58.50
Maltose M (g/L): 51.82
Volume V (L): 6.33

t = 34.00 h

Biomass X (g/L): 90.28
Glucose S (g/L): 15.46
Lactic Acid P (g/L): 63.36
Maltose M (g/L): 54.44

Volume V (L): 6.19

t = 35.00 h

Biomass X (g/L): 77.00

Glucose S (g/L): 12.29

Lactic Acid P (g/L): 55.82

Maltose M (g/L): 51.88

Volume V (L): 6.58

t = 36.00 h

Biomass X (g/L): 89.29

Glucose S (g/L): 10.81

Lactic Acid P (g/L): 45.68

Maltose M (g/L): 56.60

Volume V (L): 6.44

t = 37.00 h

Biomass X (g/L): 85.76

Glucose S (g/L): 12.00

Lactic Acid P (g/L): 75.62

Maltose M (g/L): 51.67

Volume V (L): 6.20

t = 38.00 h

Biomass X (g/L): 85.86

Glucose S (g/L): 9.13

Lactic Acid P (g/L): 77.16

Maltose M (g/L): 49.59

Volume V (L): 6.04

t = 39.00 h

Biomass X (g/L): 69.21

Glucose S (g/L): 4.57

Lactic Acid P (g/L): 61.94

Maltose M (g/L): 51.76

Volume V (L): 6.27

t = 40.00 h

Biomass X (g/L): 94.17

Glucose S (g/L): 10.58

Lactic Acid P (g/L): 71.98

Maltose M (g/L): 51.25

Volume V (L): 6.48

t = 41.00 h

Biomass X (g/L): 85.44

Glucose S (g/L): 12.15

Lactic Acid P (g/L): 62.66

Maltose M (g/L): 52.86

Volume V (L): 6.55

t = 42.00 h

Biomass X (g/L): 83.71

Glucose S (g/L): 6.59

Lactic Acid P (g/L): 69.55

Maltose M (g/L): 50.98

Volume V (L): 6.75

t = 43.00 h

Biomass X (g/L): 79.22

Glucose S (g/L): 16.49

Lactic Acid P (g/L): 74.46

Maltose M (g/L): 55.40

Volume V (L): 6.27

t = 44.00 h

Biomass X (g/L): 77.59

Glucose S (g/L): 15.66

Lactic Acid P (g/L): 73.05

Maltose M (g/L): 49.13

Volume V (L): 6.44

t = 45.00 h

Biomass X (g/L): 88.22

Glucose S (g/L): 7.07

Lactic Acid P (g/L): 80.88

Maltose M (g/L): 48.88

Volume V (L): 6.32

t = 46.00 h

Biomass X (g/L): 68.92

Glucose S (g/L): 17.79

Lactic Acid P (g/L): 73.65

Maltose M (g/L): 47.45

Volume V (L): 6.32

t = 47.00 h

Biomass X (g/L): 92.97

Glucose S (g/L): 11.87

Lactic Acid P (g/L): 69.51

Maltose M (g/L): 48.78

Volume V (L): 6.51

t = 48.00 h

Biomass X (g/L): 75.72

Glucose S (g/L): 10.50

Lactic Acid P (g/L): 62.56

Maltose M (g/L): 48.85

Volume V (L): 6.75

t = 49.00 h

Biomass X (g/L): 66.37

Glucose S (g/L): 17.92

Lactic Acid P (g/L): 65.64

Maltose M (g/L): 48.71

Volume V (L): 6.87

t = 50.00 h

Biomass X (g/L): 103.01

Glucose S (g/L): 14.07

Lactic Acid P (g/L): 78.11

Maltose M (g/L): 47.96

Volume V (L): 6.55

t = 51.00 h

Biomass X (g/L): 77.84

Glucose S (g/L): 21.85

Lactic Acid P (g/L): 56.88

Maltose M (g/L): 46.14

Volume V (L): 6.61

t = 52.00 h

Biomass X (g/L): 79.18

Glucose S (g/L): 11.89
Lactic Acid P (g/L): 81.68
Maltose M (g/L): 45.25
Volume V (L): 6.44

t = 53.00 h

Biomass X (g/L): 96.72
Glucose S (g/L): 9.01
Lactic Acid P (g/L): 69.08
Maltose M (g/L): 46.18
Volume V (L): 6.89

t = 54.00 h

Biomass X (g/L): 103.54
Glucose S (g/L): 14.92
Lactic Acid P (g/L): 64.42
Maltose M (g/L): 49.45
Volume V (L): 6.73

t = 55.00 h

Biomass X (g/L): 71.65
Glucose S (g/L): 19.39
Lactic Acid P (g/L): 62.16
Maltose M (g/L): 49.71
Volume V (L): 7.27

t = 56.00 h

Biomass X (g/L): 77.10
Glucose S (g/L): 13.37
Lactic Acid P (g/L): 79.11
Maltose M (g/L): 46.14
Volume V (L): 7.04

t = 57.00 h

Biomass X (g/L): 77.79
Glucose S (g/L): 12.29
Lactic Acid P (g/L): 73.00
Maltose M (g/L): 43.90
Volume V (L): 6.95

t = 58.00 h

Biomass X (g/L): 94.90
Glucose S (g/L): 18.62
Lactic Acid P (g/L): 88.34
Maltose M (g/L): 45.66
Volume V (L): 6.89

t = 59.00 h

Biomass X (g/L): 101.51
Glucose S (g/L): 12.70
Lactic Acid P (g/L): 61.97
Maltose M (g/L): 50.46
Volume V (L): 6.74

t = 60.00 h

Biomass X (g/L): 75.86
Glucose S (g/L): 19.83
Lactic Acid P (g/L): 68.70
Maltose M (g/L): 47.70
Volume V (L): 6.81

t = 61.00 h

Biomass X (g/L): 99.32
Glucose S (g/L): 17.85
Lactic Acid P (g/L): 73.55
Maltose M (g/L): 46.43
Volume V (L): 7.00

t = 62.00 h

Biomass X (g/L): 79.27
Glucose S (g/L): 21.46
Lactic Acid P (g/L): 67.68
Maltose M (g/L): 47.16
Volume V (L): 7.02

t = 63.00 h

Biomass X (g/L): 89.05
Glucose S (g/L): 14.84
Lactic Acid P (g/L): 75.07
Maltose M (g/L): 48.28
Volume V (L): 6.99

t = 64.00 h

Biomass X (g/L): 73.84
Glucose S (g/L): 25.42
Lactic Acid P (g/L): 78.28
Maltose M (g/L): 41.81
Volume V (L): 7.31

t = 65.00 h

Biomass X (g/L): 84.60
Glucose S (g/L): 20.30
Lactic Acid P (g/L): 80.11
Maltose M (g/L): 43.36
Volume V (L): 7.37

t = 66.00 h

Biomass X (g/L): 84.62
Glucose S (g/L): 24.33
Lactic Acid P (g/L): 89.94
Maltose M (g/L): 42.82
Volume V (L): 7.21

t = 67.00 h

Biomass X (g/L): 82.61
Glucose S (g/L): 25.54
Lactic Acid P (g/L): 63.91
Maltose M (g/L): 45.20
Volume V (L): 7.74

t = 68.00 h

Biomass X (g/L): 89.11
Glucose S (g/L): 18.03
Lactic Acid P (g/L): 79.53
Maltose M (g/L): 43.99
Volume V (L): 7.01

t = 69.00 h

Biomass X (g/L): 75.89
Glucose S (g/L): 26.19
Lactic Acid P (g/L): 70.56
Maltose M (g/L): 45.17
Volume V (L): 7.42

t = 70.00 h

Biomass X (g/L): 76.94
Glucose S (g/L): 17.96
Lactic Acid P (g/L): 80.89
Maltose M (g/L): 42.03
Volume V (L): 7.40

t = 71.00 h

Biomass X (g/L): 89.28
Glucose S (g/L): 18.63
Lactic Acid P (g/L): 81.14
Maltose M (g/L): 46.18
Volume V (L): 7.36

t = 72.00 h

Biomass X (g/L): 97.68
Glucose S (g/L): 31.68
Lactic Acid P (g/L): 68.10
Maltose M (g/L): 42.63
Volume V (L): 7.36

t = 73.00 h

Biomass X (g/L): 84.70
Glucose S (g/L): 28.94
Lactic Acid P (g/L): 69.13
Maltose M (g/L): 44.20
Volume V (L): 7.54

t = 74.00 h

Biomass X (g/L): 89.53
Glucose S (g/L): 33.03
Lactic Acid P (g/L): 82.58
Maltose M (g/L): 45.16
Volume V (L): 7.53

t = 75.00 h

Biomass X (g/L): 91.79
Glucose S (g/L): 24.29
Lactic Acid P (g/L): 78.23
Maltose M (g/L): 44.78

Volume V (L): 7.53

Batch 63

Initial Conditions:

Batch: 63.00

X0 (g/L): 9.16

S0 (g/L): 109.33

M0 (g/L): 7.24

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.07

F_vol (L/h): 0.05

S_feed (g/L): 281.36

M_feed (g/L): 18.64

V_final (L): 8.73

Batch 63 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 110.40
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 7.21
Volume V (L): 4.91

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 110.72
Lactic Acid P (g/L): 1.70
Maltose M (g/L): 7.23
Volume V (L): 5.22

t = 2.00 h

Biomass X (g/L): 13.45
Glucose S (g/L): 103.74
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 6.97
Volume V (L): 5.22

t = 3.00 h

Biomass X (g/L): 16.92
Glucose S (g/L): 96.72
Lactic Acid P (g/L): 10.05
Maltose M (g/L): 7.17
Volume V (L): 4.85

t = 4.00 h

Biomass X (g/L): 24.08
Glucose S (g/L): 100.65
Lactic Acid P (g/L): 23.63
Maltose M (g/L): 6.92
Volume V (L): 4.92

t = 5.00 h

Biomass X (g/L): 30.41
Glucose S (g/L): 101.27

Lactic Acid P (g/L): 2.97

Maltose M (g/L): 6.92

Volume V (L): 5.24

t = 6.00 h

Biomass X (g/L): 31.89

Glucose S (g/L): 92.86

Lactic Acid P (g/L): 25.23

Maltose M (g/L): 6.99

Volume V (L): 5.49

t = 7.00 h

Biomass X (g/L): 40.42

Glucose S (g/L): 88.96

Lactic Acid P (g/L): 18.93

Maltose M (g/L): 7.14

Volume V (L): 5.90

t = 8.00 h

Biomass X (g/L): 39.30

Glucose S (g/L): 83.37

Lactic Acid P (g/L): 36.29

Maltose M (g/L): 7.13

Volume V (L): 5.49

t = 9.00 h

Biomass X (g/L): 67.62

Glucose S (g/L): 83.69

Lactic Acid P (g/L): 33.88

Maltose M (g/L): 7.11

Volume V (L): 5.75

t = 10.00 h

Biomass X (g/L): 49.13

Glucose S (g/L): 75.96

Lactic Acid P (g/L): 39.66

Maltose M (g/L): 6.63

Volume V (L): 5.02

t = 11.00 h

Biomass X (g/L): 57.39

Glucose S (g/L): 71.82
Lactic Acid P (g/L): 39.58
Maltose M (g/L): 6.93
Volume V (L): 6.15

t = 12.00 h

Biomass X (g/L): 57.52
Glucose S (g/L): 77.02
Lactic Acid P (g/L): 54.46
Maltose M (g/L): 6.86
Volume V (L): 5.78

t = 13.00 h

Biomass X (g/L): 60.58
Glucose S (g/L): 66.84
Lactic Acid P (g/L): 47.76
Maltose M (g/L): 6.55
Volume V (L): 5.63

t = 14.00 h

Biomass X (g/L): 71.00
Glucose S (g/L): 71.99
Lactic Acid P (g/L): 40.14
Maltose M (g/L): 6.84
Volume V (L): 5.65

t = 15.00 h

Biomass X (g/L): 72.29
Glucose S (g/L): 71.50
Lactic Acid P (g/L): 49.94
Maltose M (g/L): 6.76
Volume V (L): 5.42

t = 16.00 h

Biomass X (g/L): 71.03
Glucose S (g/L): 67.54
Lactic Acid P (g/L): 47.46
Maltose M (g/L): 6.88
Volume V (L): 5.55

t = 17.00 h

Biomass X (g/L): 74.77
Glucose S (g/L): 57.53
Lactic Acid P (g/L): 54.30
Maltose M (g/L): 6.77
Volume V (L): 6.41

t = 18.00 h

Biomass X (g/L): 72.24
Glucose S (g/L): 62.90
Lactic Acid P (g/L): 50.41
Maltose M (g/L): 6.60
Volume V (L): 5.60

t = 19.00 h

Biomass X (g/L): 84.88
Glucose S (g/L): 61.04
Lactic Acid P (g/L): 49.46
Maltose M (g/L): 6.88
Volume V (L): 5.64

t = 20.00 h

Biomass X (g/L): 102.59
Glucose S (g/L): 59.01
Lactic Acid P (g/L): 59.16
Maltose M (g/L): 6.20
Volume V (L): 5.78

t = 21.00 h

Biomass X (g/L): 74.91
Glucose S (g/L): 61.69
Lactic Acid P (g/L): 59.06
Maltose M (g/L): 6.61
Volume V (L): 5.69

t = 22.00 h

Biomass X (g/L): 88.53
Glucose S (g/L): 64.43
Lactic Acid P (g/L): 69.81
Maltose M (g/L): 6.23
Volume V (L): 6.10

t = 23.00 h

Biomass X (g/L): 86.05
Glucose S (g/L): 58.16
Lactic Acid P (g/L): 56.84
Maltose M (g/L): 6.38
Volume V (L): 6.47

t = 24.00 h

Biomass X (g/L): 93.06
Glucose S (g/L): 59.41
Lactic Acid P (g/L): 42.97
Maltose M (g/L): 6.65
Volume V (L): 6.01

t = 25.00 h

Biomass X (g/L): 94.10
Glucose S (g/L): 59.00
Lactic Acid P (g/L): 59.26
Maltose M (g/L): 6.24
Volume V (L): 5.83

t = 26.00 h

Biomass X (g/L): 89.91
Glucose S (g/L): 65.89
Lactic Acid P (g/L): 48.14
Maltose M (g/L): 6.18
Volume V (L): 6.38

t = 27.00 h

Biomass X (g/L): 88.08
Glucose S (g/L): 58.96
Lactic Acid P (g/L): 68.35
Maltose M (g/L): 6.95
Volume V (L): 6.33

t = 28.00 h

Biomass X (g/L): 84.69
Glucose S (g/L): 54.27
Lactic Acid P (g/L): 49.26
Maltose M (g/L): 6.14
Volume V (L): 6.69

t = 29.00 h

Biomass X (g/L): 81.88
Glucose S (g/L): 52.75
Lactic Acid P (g/L): 56.49
Maltose M (g/L): 6.73
Volume V (L): 6.31

t = 30.00 h

Biomass X (g/L): 88.86
Glucose S (g/L): 62.47
Lactic Acid P (g/L): 63.10
Maltose M (g/L): 6.32
Volume V (L): 6.41

t = 31.00 h

Biomass X (g/L): 77.47
Glucose S (g/L): 57.18
Lactic Acid P (g/L): 81.84
Maltose M (g/L): 6.22
Volume V (L): 6.50

t = 32.00 h

Biomass X (g/L): 84.57
Glucose S (g/L): 64.47
Lactic Acid P (g/L): 57.58
Maltose M (g/L): 6.42
Volume V (L): 6.62

t = 33.00 h

Biomass X (g/L): 101.60
Glucose S (g/L): 54.94
Lactic Acid P (g/L): 68.45
Maltose M (g/L): 6.55
Volume V (L): 6.52

t = 34.00 h

Biomass X (g/L): 91.85
Glucose S (g/L): 59.15
Lactic Acid P (g/L): 74.38
Maltose M (g/L): 6.00

Volume V (L): 6.52

t = 35.00 h

Biomass X (g/L): 99.85

Glucose S (g/L): 55.60

Lactic Acid P (g/L): 83.47

Maltose M (g/L): 6.11

Volume V (L): 6.19

t = 36.00 h

Biomass X (g/L): 105.67

Glucose S (g/L): 64.65

Lactic Acid P (g/L): 77.24

Maltose M (g/L): 6.36

Volume V (L): 6.62

t = 37.00 h

Biomass X (g/L): 89.52

Glucose S (g/L): 66.08

Lactic Acid P (g/L): 80.70

Maltose M (g/L): 6.22

Volume V (L): 7.05

t = 38.00 h

Biomass X (g/L): 93.38

Glucose S (g/L): 64.96

Lactic Acid P (g/L): 78.95

Maltose M (g/L): 6.36

Volume V (L): 7.14

t = 39.00 h

Biomass X (g/L): 88.40

Glucose S (g/L): 58.37

Lactic Acid P (g/L): 59.67

Maltose M (g/L): 6.21

Volume V (L): 6.80

t = 40.00 h

Biomass X (g/L): 83.38

Glucose S (g/L): 60.46

Lactic Acid P (g/L): 74.97

Maltose M (g/L): 6.45

Volume V (L): 7.38

t = 41.00 h

Biomass X (g/L): 82.00

Glucose S (g/L): 65.72

Lactic Acid P (g/L): 79.99

Maltose M (g/L): 6.08

Volume V (L): 7.03

t = 42.00 h

Biomass X (g/L): 99.71

Glucose S (g/L): 56.71

Lactic Acid P (g/L): 63.20

Maltose M (g/L): 6.06

Volume V (L): 6.69

t = 43.00 h

Biomass X (g/L): 93.40

Glucose S (g/L): 64.28

Lactic Acid P (g/L): 67.13

Maltose M (g/L): 6.05

Volume V (L): 6.64

t = 44.00 h

Biomass X (g/L): 109.52

Glucose S (g/L): 73.38

Lactic Acid P (g/L): 65.33

Maltose M (g/L): 5.93

Volume V (L): 7.55

t = 45.00 h

Biomass X (g/L): 100.81

Glucose S (g/L): 65.51

Lactic Acid P (g/L): 80.62

Maltose M (g/L): 6.11

Volume V (L): 7.13

t = 46.00 h

Biomass X (g/L): 83.92

Glucose S (g/L): 59.55

Lactic Acid P (g/L): 72.93

Maltose M (g/L): 5.95

Volume V (L): 6.92

t = 47.00 h

Biomass X (g/L): 99.33

Glucose S (g/L): 64.35

Lactic Acid P (g/L): 70.83

Maltose M (g/L): 6.04

Volume V (L): 6.66

t = 48.00 h

Biomass X (g/L): 87.91

Glucose S (g/L): 64.72

Lactic Acid P (g/L): 61.99

Maltose M (g/L): 6.04

Volume V (L): 7.27

t = 49.00 h

Biomass X (g/L): 102.81

Glucose S (g/L): 78.83

Lactic Acid P (g/L): 73.79

Maltose M (g/L): 5.91

Volume V (L): 7.56

t = 50.00 h

Biomass X (g/L): 85.87

Glucose S (g/L): 76.98

Lactic Acid P (g/L): 69.30

Maltose M (g/L): 5.90

Volume V (L): 7.28

t = 51.00 h

Biomass X (g/L): 80.14

Glucose S (g/L): 72.87

Lactic Acid P (g/L): 77.26

Maltose M (g/L): 5.55

Volume V (L): 7.22

t = 52.00 h

Biomass X (g/L): 82.52

Glucose S (g/L): 68.08
Lactic Acid P (g/L): 64.33
Maltose M (g/L): 5.62
Volume V (L): 7.42

t = 53.00 h

Biomass X (g/L): 82.87
Glucose S (g/L): 76.16
Lactic Acid P (g/L): 77.12
Maltose M (g/L): 5.89
Volume V (L): 8.01

t = 54.00 h

Biomass X (g/L): 81.76
Glucose S (g/L): 65.17
Lactic Acid P (g/L): 71.49
Maltose M (g/L): 5.98
Volume V (L): 7.27

t = 55.00 h

Biomass X (g/L): 84.23
Glucose S (g/L): 76.53
Lactic Acid P (g/L): 64.01
Maltose M (g/L): 5.25
Volume V (L): 7.54

t = 56.00 h

Biomass X (g/L): 98.15
Glucose S (g/L): 76.29
Lactic Acid P (g/L): 63.48
Maltose M (g/L): 5.51
Volume V (L): 8.14

t = 57.00 h

Biomass X (g/L): 107.24
Glucose S (g/L): 71.47
Lactic Acid P (g/L): 65.11
Maltose M (g/L): 5.61
Volume V (L): 7.98

t = 58.00 h

Biomass X (g/L): 76.19
Glucose S (g/L): 78.30
Lactic Acid P (g/L): 74.53
Maltose M (g/L): 5.39
Volume V (L): 7.94

t = 59.00 h

Biomass X (g/L): 101.60
Glucose S (g/L): 70.99
Lactic Acid P (g/L): 77.84
Maltose M (g/L): 5.38
Volume V (L): 7.74

t = 60.00 h

Biomass X (g/L): 84.21
Glucose S (g/L): 67.46
Lactic Acid P (g/L): 79.44
Maltose M (g/L): 5.63
Volume V (L): 7.99

t = 61.00 h

Biomass X (g/L): 92.15
Glucose S (g/L): 78.97
Lactic Acid P (g/L): 79.98
Maltose M (g/L): 5.83
Volume V (L): 7.87

t = 62.00 h

Biomass X (g/L): 72.18
Glucose S (g/L): 74.23
Lactic Acid P (g/L): 70.03
Maltose M (g/L): 5.05
Volume V (L): 7.83

t = 63.00 h

Biomass X (g/L): 93.47
Glucose S (g/L): 78.09
Lactic Acid P (g/L): 87.47
Maltose M (g/L): 5.41
Volume V (L): 7.81

t = 64.00 h

Biomass X (g/L): 68.29
Glucose S (g/L): 84.56
Lactic Acid P (g/L): 73.38
Maltose M (g/L): 5.52
Volume V (L): 8.42

t = 65.00 h

Biomass X (g/L): 90.96
Glucose S (g/L): 72.53
Lactic Acid P (g/L): 82.02
Maltose M (g/L): 5.56
Volume V (L): 8.21

t = 66.00 h

Biomass X (g/L): 90.18
Glucose S (g/L): 75.85
Lactic Acid P (g/L): 76.34
Maltose M (g/L): 5.44
Volume V (L): 8.37

t = 67.00 h

Biomass X (g/L): 89.30
Glucose S (g/L): 84.75
Lactic Acid P (g/L): 81.62
Maltose M (g/L): 5.72
Volume V (L): 8.38

t = 68.00 h

Biomass X (g/L): 79.94
Glucose S (g/L): 80.79
Lactic Acid P (g/L): 76.17
Maltose M (g/L): 5.33
Volume V (L): 8.51

t = 69.00 h

Biomass X (g/L): 92.38
Glucose S (g/L): 74.53
Lactic Acid P (g/L): 82.09
Maltose M (g/L): 5.57
Volume V (L): 8.35

t = 70.00 h

Biomass X (g/L): 90.59
Glucose S (g/L): 73.74
Lactic Acid P (g/L): 78.43
Maltose M (g/L): 5.14
Volume V (L): 8.34

t = 71.00 h

Biomass X (g/L): 84.42
Glucose S (g/L): 76.30
Lactic Acid P (g/L): 86.37
Maltose M (g/L): 5.26
Volume V (L): 8.29

t = 72.00 h

Biomass X (g/L): 92.89
Glucose S (g/L): 75.72
Lactic Acid P (g/L): 71.00
Maltose M (g/L): 4.98
Volume V (L): 8.16

t = 73.00 h

Biomass X (g/L): 85.08
Glucose S (g/L): 80.97
Lactic Acid P (g/L): 62.09
Maltose M (g/L): 5.39
Volume V (L): 8.29

t = 74.00 h

Biomass X (g/L): 91.01
Glucose S (g/L): 80.65
Lactic Acid P (g/L): 69.79
Maltose M (g/L): 5.16
Volume V (L): 8.57

t = 75.00 h

Biomass X (g/L): 84.31
Glucose S (g/L): 82.08
Lactic Acid P (g/L): 89.32
Maltose M (g/L): 5.23

Volume V (L): 8.25

Batch 64

Initial Conditions:

Batch: 64.00

X0 (g/L): 5.55

S0 (g/L): 152.92

M0 (g/L): 20.83

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.14

F_vol (L/h): 0.03

S_feed (g/L): 264.04

M_feed (g/L): 35.96

V_final (L): 7.30

Batch 64 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 150.66
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 21.56
Volume V (L): 4.89

t = 1.00 h

Biomass X (g/L): 18.43
Glucose S (g/L): 152.98
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 19.88
Volume V (L): 4.77

t = 2.00 h

Biomass X (g/L): 12.54
Glucose S (g/L): 151.76
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 18.49
Volume V (L): 4.80

t = 3.00 h

Biomass X (g/L): 8.78
Glucose S (g/L): 146.20
Lactic Acid P (g/L): 9.56
Maltose M (g/L): 21.12
Volume V (L): 5.16

t = 4.00 h

Biomass X (g/L): 41.01
Glucose S (g/L): 149.81
Lactic Acid P (g/L): 12.67
Maltose M (g/L): 18.34
Volume V (L): 5.39

t = 5.00 h

Biomass X (g/L): 4.73
Glucose S (g/L): 137.12

Lactic Acid P (g/L): 26.43

Maltose M (g/L): 19.67

Volume V (L): 5.08

t = 6.00 h

Biomass X (g/L): 36.94

Glucose S (g/L): 146.78

Lactic Acid P (g/L): 15.62

Maltose M (g/L): 18.93

Volume V (L): 5.08

t = 7.00 h

Biomass X (g/L): 30.83

Glucose S (g/L): 137.71

Lactic Acid P (g/L): 1.88

Maltose M (g/L): 19.00

Volume V (L): 5.34

t = 8.00 h

Biomass X (g/L): 45.42

Glucose S (g/L): 122.72

Lactic Acid P (g/L): 20.88

Maltose M (g/L): 20.46

Volume V (L): 5.12

t = 9.00 h

Biomass X (g/L): 43.56

Glucose S (g/L): 136.22

Lactic Acid P (g/L): 13.32

Maltose M (g/L): 21.80

Volume V (L): 5.33

t = 10.00 h

Biomass X (g/L): 27.90

Glucose S (g/L): 121.67

Lactic Acid P (g/L): 18.83

Maltose M (g/L): 16.16

Volume V (L): 5.31

t = 11.00 h

Biomass X (g/L): 53.00

Glucose S (g/L): 111.71
Lactic Acid P (g/L): 40.11
Maltose M (g/L): 18.94
Volume V (L): 5.26

t = 12.00 h

Biomass X (g/L): 59.77
Glucose S (g/L): 120.49
Lactic Acid P (g/L): 39.55
Maltose M (g/L): 17.38
Volume V (L): 5.36

t = 13.00 h

Biomass X (g/L): 37.39
Glucose S (g/L): 113.07
Lactic Acid P (g/L): 46.50
Maltose M (g/L): 18.54
Volume V (L): 5.72

t = 14.00 h

Biomass X (g/L): 83.93
Glucose S (g/L): 94.76
Lactic Acid P (g/L): 39.90
Maltose M (g/L): 18.21
Volume V (L): 5.54

t = 15.00 h

Biomass X (g/L): 70.13
Glucose S (g/L): 100.55
Lactic Acid P (g/L): 53.86
Maltose M (g/L): 18.50
Volume V (L): 5.66

t = 16.00 h

Biomass X (g/L): 76.74
Glucose S (g/L): 103.64
Lactic Acid P (g/L): 44.24
Maltose M (g/L): 16.99
Volume V (L): 5.32

t = 17.00 h

Biomass X (g/L): 60.19
Glucose S (g/L): 85.02
Lactic Acid P (g/L): 61.59
Maltose M (g/L): 15.42
Volume V (L): 5.72

t = 18.00 h

Biomass X (g/L): 81.78
Glucose S (g/L): 98.37
Lactic Acid P (g/L): 55.88
Maltose M (g/L): 17.31
Volume V (L): 5.67

t = 19.00 h

Biomass X (g/L): 87.98
Glucose S (g/L): 97.21
Lactic Acid P (g/L): 52.19
Maltose M (g/L): 16.17
Volume V (L): 5.31

t = 20.00 h

Biomass X (g/L): 81.11
Glucose S (g/L): 86.66
Lactic Acid P (g/L): 58.01
Maltose M (g/L): 15.56
Volume V (L): 5.71

t = 21.00 h

Biomass X (g/L): 80.02
Glucose S (g/L): 95.22
Lactic Acid P (g/L): 57.25
Maltose M (g/L): 15.09
Volume V (L): 5.55

t = 22.00 h

Biomass X (g/L): 72.28
Glucose S (g/L): 92.29
Lactic Acid P (g/L): 74.70
Maltose M (g/L): 16.61
Volume V (L): 5.39

t = 23.00 h

Biomass X (g/L): 80.64
Glucose S (g/L): 83.16
Lactic Acid P (g/L): 62.58
Maltose M (g/L): 17.11
Volume V (L): 5.73

t = 24.00 h

Biomass X (g/L): 78.73
Glucose S (g/L): 96.58
Lactic Acid P (g/L): 61.33
Maltose M (g/L): 13.69
Volume V (L): 5.83

t = 25.00 h

Biomass X (g/L): 91.05
Glucose S (g/L): 95.24
Lactic Acid P (g/L): 60.10
Maltose M (g/L): 15.61
Volume V (L): 5.79

t = 26.00 h

Biomass X (g/L): 77.20
Glucose S (g/L): 86.91
Lactic Acid P (g/L): 71.90
Maltose M (g/L): 14.29
Volume V (L): 5.78

t = 27.00 h

Biomass X (g/L): 92.75
Glucose S (g/L): 86.92
Lactic Acid P (g/L): 71.53
Maltose M (g/L): 12.85
Volume V (L): 5.77

t = 28.00 h

Biomass X (g/L): 78.90
Glucose S (g/L): 92.66
Lactic Acid P (g/L): 62.69
Maltose M (g/L): 14.55
Volume V (L): 6.31

t = 29.00 h

Biomass X (g/L): 85.56
Glucose S (g/L): 85.92
Lactic Acid P (g/L): 54.79
Maltose M (g/L): 13.25
Volume V (L): 5.67

t = 30.00 h

Biomass X (g/L): 95.28
Glucose S (g/L): 73.79
Lactic Acid P (g/L): 58.12
Maltose M (g/L): 12.23
Volume V (L): 6.08

t = 31.00 h

Biomass X (g/L): 82.08
Glucose S (g/L): 89.83
Lactic Acid P (g/L): 66.02
Maltose M (g/L): 12.99
Volume V (L): 5.95

t = 32.00 h

Biomass X (g/L): 96.63
Glucose S (g/L): 80.72
Lactic Acid P (g/L): 81.98
Maltose M (g/L): 12.55
Volume V (L): 5.82

t = 33.00 h

Biomass X (g/L): 96.02
Glucose S (g/L): 83.34
Lactic Acid P (g/L): 66.77
Maltose M (g/L): 12.84
Volume V (L): 6.00

t = 34.00 h

Biomass X (g/L): 87.81
Glucose S (g/L): 95.65
Lactic Acid P (g/L): 63.99
Maltose M (g/L): 11.91

Volume V (L): 5.78

t = 35.00 h

Biomass X (g/L): 103.02

Glucose S (g/L): 84.69

Lactic Acid P (g/L): 53.01

Maltose M (g/L): 12.84

Volume V (L): 5.96

t = 36.00 h

Biomass X (g/L): 92.02

Glucose S (g/L): 85.47

Lactic Acid P (g/L): 86.14

Maltose M (g/L): 12.33

Volume V (L): 6.09

t = 37.00 h

Biomass X (g/L): 102.90

Glucose S (g/L): 88.80

Lactic Acid P (g/L): 77.31

Maltose M (g/L): 13.45

Volume V (L): 5.85

t = 38.00 h

Biomass X (g/L): 110.29

Glucose S (g/L): 88.69

Lactic Acid P (g/L): 80.09

Maltose M (g/L): 13.56

Volume V (L): 6.26

t = 39.00 h

Biomass X (g/L): 89.29

Glucose S (g/L): 78.98

Lactic Acid P (g/L): 82.25

Maltose M (g/L): 13.04

Volume V (L): 6.02

t = 40.00 h

Biomass X (g/L): 98.64

Glucose S (g/L): 84.77

Lactic Acid P (g/L): 71.54

Maltose M (g/L): 12.34

Volume V (L): 5.96

t = 41.00 h

Biomass X (g/L): 85.40

Glucose S (g/L): 85.41

Lactic Acid P (g/L): 74.48

Maltose M (g/L): 12.18

Volume V (L): 6.06

t = 42.00 h

Biomass X (g/L): 81.37

Glucose S (g/L): 87.01

Lactic Acid P (g/L): 88.87

Maltose M (g/L): 11.39

Volume V (L): 6.68

t = 43.00 h

Biomass X (g/L): 90.13

Glucose S (g/L): 87.15

Lactic Acid P (g/L): 76.29

Maltose M (g/L): 11.29

Volume V (L): 5.95

t = 44.00 h

Biomass X (g/L): 78.03

Glucose S (g/L): 102.76

Lactic Acid P (g/L): 80.76

Maltose M (g/L): 12.92

Volume V (L): 6.43

t = 45.00 h

Biomass X (g/L): 82.46

Glucose S (g/L): 77.32

Lactic Acid P (g/L): 83.96

Maltose M (g/L): 9.96

Volume V (L): 6.24

t = 46.00 h

Biomass X (g/L): 98.36

Glucose S (g/L): 91.99

Lactic Acid P (g/L): 82.08

Maltose M (g/L): 11.78

Volume V (L): 6.34

t = 47.00 h

Biomass X (g/L): 86.85

Glucose S (g/L): 93.45

Lactic Acid P (g/L): 74.78

Maltose M (g/L): 11.38

Volume V (L): 6.55

t = 48.00 h

Biomass X (g/L): 99.54

Glucose S (g/L): 88.86

Lactic Acid P (g/L): 76.07

Maltose M (g/L): 12.34

Volume V (L): 6.57

t = 49.00 h

Biomass X (g/L): 95.21

Glucose S (g/L): 81.22

Lactic Acid P (g/L): 84.47

Maltose M (g/L): 9.89

Volume V (L): 6.37

t = 50.00 h

Biomass X (g/L): 111.28

Glucose S (g/L): 97.02

Lactic Acid P (g/L): 84.79

Maltose M (g/L): 11.44

Volume V (L): 6.28

t = 51.00 h

Biomass X (g/L): 97.40

Glucose S (g/L): 71.00

Lactic Acid P (g/L): 89.00

Maltose M (g/L): 11.72

Volume V (L): 6.67

t = 52.00 h

Biomass X (g/L): 103.44

Glucose S (g/L): 81.35
Lactic Acid P (g/L): 83.17
Maltose M (g/L): 9.02
Volume V (L): 6.69

t = 53.00 h

Biomass X (g/L): 95.93
Glucose S (g/L): 94.06
Lactic Acid P (g/L): 83.92
Maltose M (g/L): 9.88
Volume V (L): 6.57

t = 54.00 h

Biomass X (g/L): 94.88
Glucose S (g/L): 83.75
Lactic Acid P (g/L): 73.85
Maltose M (g/L): 10.91
Volume V (L): 6.73

t = 55.00 h

Biomass X (g/L): 103.63
Glucose S (g/L): 97.23
Lactic Acid P (g/L): 72.79
Maltose M (g/L): 11.51
Volume V (L): 6.60

t = 56.00 h

Biomass X (g/L): 98.34
Glucose S (g/L): 76.32
Lactic Acid P (g/L): 86.59
Maltose M (g/L): 11.03
Volume V (L): 6.56

t = 57.00 h

Biomass X (g/L): 93.43
Glucose S (g/L): 79.19
Lactic Acid P (g/L): 73.39
Maltose M (g/L): 11.37
Volume V (L): 6.75

t = 58.00 h

Biomass X (g/L): 93.24
Glucose S (g/L): 83.28
Lactic Acid P (g/L): 79.15
Maltose M (g/L): 9.39
Volume V (L): 6.87

t = 59.00 h

Biomass X (g/L): 96.54
Glucose S (g/L): 89.88
Lactic Acid P (g/L): 93.30
Maltose M (g/L): 8.71
Volume V (L): 6.86

t = 60.00 h

Biomass X (g/L): 83.80
Glucose S (g/L): 85.72
Lactic Acid P (g/L): 91.97
Maltose M (g/L): 10.55
Volume V (L): 6.93

t = 61.00 h

Biomass X (g/L): 83.19
Glucose S (g/L): 84.08
Lactic Acid P (g/L): 83.08
Maltose M (g/L): 8.94
Volume V (L): 6.73

t = 62.00 h

Biomass X (g/L): 105.26
Glucose S (g/L): 97.21
Lactic Acid P (g/L): 77.36
Maltose M (g/L): 9.28
Volume V (L): 7.19

t = 63.00 h

Biomass X (g/L): 109.65
Glucose S (g/L): 98.02
Lactic Acid P (g/L): 79.30
Maltose M (g/L): 11.27
Volume V (L): 6.87

t = 64.00 h

Biomass X (g/L): 80.41
Glucose S (g/L): 85.14
Lactic Acid P (g/L): 89.37
Maltose M (g/L): 10.33
Volume V (L): 6.94

t = 65.00 h

Biomass X (g/L): 82.92
Glucose S (g/L): 76.55
Lactic Acid P (g/L): 80.59
Maltose M (g/L): 9.37
Volume V (L): 6.75

t = 66.00 h

Biomass X (g/L): 101.57
Glucose S (g/L): 73.71
Lactic Acid P (g/L): 75.61
Maltose M (g/L): 9.14
Volume V (L): 6.98

t = 67.00 h

Biomass X (g/L): 95.04
Glucose S (g/L): 90.14
Lactic Acid P (g/L): 93.45
Maltose M (g/L): 10.58
Volume V (L): 7.15

t = 68.00 h

Biomass X (g/L): 96.32
Glucose S (g/L): 93.16
Lactic Acid P (g/L): 83.37
Maltose M (g/L): 10.03
Volume V (L): 7.22

t = 69.00 h

Biomass X (g/L): 98.41
Glucose S (g/L): 89.33
Lactic Acid P (g/L): 76.02
Maltose M (g/L): 8.31
Volume V (L): 7.40

t = 70.00 h

Biomass X (g/L): 86.37
Glucose S (g/L): 88.62
Lactic Acid P (g/L): 86.70
Maltose M (g/L): 7.39
Volume V (L): 7.24

t = 71.00 h

Biomass X (g/L): 92.09
Glucose S (g/L): 83.85
Lactic Acid P (g/L): 91.78
Maltose M (g/L): 11.74
Volume V (L): 7.22

t = 72.00 h

Biomass X (g/L): 91.07
Glucose S (g/L): 93.94
Lactic Acid P (g/L): 72.29
Maltose M (g/L): 8.08
Volume V (L): 7.12

t = 73.00 h

Biomass X (g/L): 79.30
Glucose S (g/L): 101.07
Lactic Acid P (g/L): 80.49
Maltose M (g/L): 10.03
Volume V (L): 7.11

t = 74.00 h

Biomass X (g/L): 91.28
Glucose S (g/L): 74.88
Lactic Acid P (g/L): 80.20
Maltose M (g/L): 11.09
Volume V (L): 7.44

t = 75.00 h

Biomass X (g/L): 100.66
Glucose S (g/L): 89.58
Lactic Acid P (g/L): 87.32
Maltose M (g/L): 9.45

Volume V (L): 7.47

Batch 65

Initial Conditions:

Batch: 65.00

X0 (g/L): 2.28

S0 (g/L): 136.89

M0 (g/L): 45.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.33

F_vol (L/h): 0.07

S_feed (g/L): 225.78

M_feed (g/L): 74.22

V_final (L): 10.48

Batch 65 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 142.23
Lactic Acid P (g/L): 3.78
Maltose M (g/L): 43.63
Volume V (L): 5.54

t = 1.00 h

Biomass X (g/L): 10.24
Glucose S (g/L): 140.32
Lactic Acid P (g/L): 14.20
Maltose M (g/L): 44.94
Volume V (L): 5.32

t = 2.00 h

Biomass X (g/L): 10.61
Glucose S (g/L): 136.42
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 46.29
Volume V (L): 5.33

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 140.08
Lactic Acid P (g/L): 8.44
Maltose M (g/L): 42.59
Volume V (L): 5.38

t = 4.00 h

Biomass X (g/L): 12.76
Glucose S (g/L): 136.49
Lactic Acid P (g/L): 14.01
Maltose M (g/L): 45.07
Volume V (L): 5.56

t = 5.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 145.53

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 42.77
Volume V (L): 5.20

t = 6.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 145.08
Lactic Acid P (g/L): 5.87
Maltose M (g/L): 43.91
Volume V (L): 4.87

t = 7.00 h

Biomass X (g/L): 14.22
Glucose S (g/L): 139.89
Lactic Acid P (g/L): 0.28
Maltose M (g/L): 42.33
Volume V (L): 5.93

t = 8.00 h

Biomass X (g/L): 22.11
Glucose S (g/L): 141.32
Lactic Acid P (g/L): 16.58
Maltose M (g/L): 41.62
Volume V (L): 6.09

t = 9.00 h

Biomass X (g/L): 22.58
Glucose S (g/L): 133.27
Lactic Acid P (g/L): 12.48
Maltose M (g/L): 41.99
Volume V (L): 5.77

t = 10.00 h

Biomass X (g/L): 56.44
Glucose S (g/L): 132.95
Lactic Acid P (g/L): 17.79
Maltose M (g/L): 43.31
Volume V (L): 5.49

t = 11.00 h

Biomass X (g/L): 36.39

Glucose S (g/L): 131.84
Lactic Acid P (g/L): 18.59
Maltose M (g/L): 41.03
Volume V (L): 5.12

t = 12.00 h

Biomass X (g/L): 38.63
Glucose S (g/L): 123.92
Lactic Acid P (g/L): 32.20
Maltose M (g/L): 41.14
Volume V (L): 5.92

t = 13.00 h

Biomass X (g/L): 45.82
Glucose S (g/L): 121.28
Lactic Acid P (g/L): 31.05
Maltose M (g/L): 39.13
Volume V (L): 5.36

t = 14.00 h

Biomass X (g/L): 41.41
Glucose S (g/L): 121.91
Lactic Acid P (g/L): 34.23
Maltose M (g/L): 42.58
Volume V (L): 5.37

t = 15.00 h

Biomass X (g/L): 58.88
Glucose S (g/L): 118.16
Lactic Acid P (g/L): 36.59
Maltose M (g/L): 40.41
Volume V (L): 6.09

t = 16.00 h

Biomass X (g/L): 56.90
Glucose S (g/L): 110.19
Lactic Acid P (g/L): 43.13
Maltose M (g/L): 38.97
Volume V (L): 5.83

t = 17.00 h

Biomass X (g/L): 59.78
Glucose S (g/L): 102.34
Lactic Acid P (g/L): 38.17
Maltose M (g/L): 38.25
Volume V (L): 6.39

t = 18.00 h

Biomass X (g/L): 53.53
Glucose S (g/L): 108.48
Lactic Acid P (g/L): 58.35
Maltose M (g/L): 40.08
Volume V (L): 6.25

t = 19.00 h

Biomass X (g/L): 46.99
Glucose S (g/L): 99.16
Lactic Acid P (g/L): 33.98
Maltose M (g/L): 36.66
Volume V (L): 6.12

t = 20.00 h

Biomass X (g/L): 59.96
Glucose S (g/L): 102.14
Lactic Acid P (g/L): 31.56
Maltose M (g/L): 34.04
Volume V (L): 5.71

t = 21.00 h

Biomass X (g/L): 69.93
Glucose S (g/L): 101.37
Lactic Acid P (g/L): 32.67
Maltose M (g/L): 37.36
Volume V (L): 6.16

t = 22.00 h

Biomass X (g/L): 64.74
Glucose S (g/L): 101.54
Lactic Acid P (g/L): 31.93
Maltose M (g/L): 39.24
Volume V (L): 6.25

t = 23.00 h

Biomass X (g/L): 73.81
Glucose S (g/L): 106.20
Lactic Acid P (g/L): 58.43
Maltose M (g/L): 39.72
Volume V (L): 6.14

t = 24.00 h

Biomass X (g/L): 70.24
Glucose S (g/L): 106.33
Lactic Acid P (g/L): 58.77
Maltose M (g/L): 36.10
Volume V (L): 6.61

t = 25.00 h

Biomass X (g/L): 78.23
Glucose S (g/L): 103.62
Lactic Acid P (g/L): 66.05
Maltose M (g/L): 36.10
Volume V (L): 6.32

t = 26.00 h

Biomass X (g/L): 65.62
Glucose S (g/L): 103.62
Lactic Acid P (g/L): 46.77
Maltose M (g/L): 36.74
Volume V (L): 6.83

t = 27.00 h

Biomass X (g/L): 78.33
Glucose S (g/L): 103.34
Lactic Acid P (g/L): 64.22
Maltose M (g/L): 34.93
Volume V (L): 7.08

t = 28.00 h

Biomass X (g/L): 71.13
Glucose S (g/L): 90.47
Lactic Acid P (g/L): 59.04
Maltose M (g/L): 36.17
Volume V (L): 6.69

t = 29.00 h

Biomass X (g/L): 63.19
Glucose S (g/L): 100.26
Lactic Acid P (g/L): 61.01
Maltose M (g/L): 34.72
Volume V (L): 7.31

t = 30.00 h

Biomass X (g/L): 77.58
Glucose S (g/L): 92.92
Lactic Acid P (g/L): 66.07
Maltose M (g/L): 36.59
Volume V (L): 6.60

t = 31.00 h

Biomass X (g/L): 95.12
Glucose S (g/L): 102.05
Lactic Acid P (g/L): 54.40
Maltose M (g/L): 35.76
Volume V (L): 6.78

t = 32.00 h

Biomass X (g/L): 66.19
Glucose S (g/L): 91.65
Lactic Acid P (g/L): 85.47
Maltose M (g/L): 35.67
Volume V (L): 7.29

t = 33.00 h

Biomass X (g/L): 82.36
Glucose S (g/L): 99.32
Lactic Acid P (g/L): 74.84
Maltose M (g/L): 33.42
Volume V (L): 7.71

t = 34.00 h

Biomass X (g/L): 89.58
Glucose S (g/L): 102.31
Lactic Acid P (g/L): 58.65
Maltose M (g/L): 37.33

Volume V (L): 7.87

t = 35.00 h

Biomass X (g/L): 104.87

Glucose S (g/L): 95.78

Lactic Acid P (g/L): 54.24

Maltose M (g/L): 34.26

Volume V (L): 8.27

t = 36.00 h

Biomass X (g/L): 77.69

Glucose S (g/L): 104.24

Lactic Acid P (g/L): 63.93

Maltose M (g/L): 33.55

Volume V (L): 7.47

t = 37.00 h

Biomass X (g/L): 76.68

Glucose S (g/L): 102.61

Lactic Acid P (g/L): 66.96

Maltose M (g/L): 31.33

Volume V (L): 7.65

t = 38.00 h

Biomass X (g/L): 71.71

Glucose S (g/L): 102.32

Lactic Acid P (g/L): 75.50

Maltose M (g/L): 32.38

Volume V (L): 7.33

t = 39.00 h

Biomass X (g/L): 94.76

Glucose S (g/L): 103.83

Lactic Acid P (g/L): 55.77

Maltose M (g/L): 32.56

Volume V (L): 7.30

t = 40.00 h

Biomass X (g/L): 75.16

Glucose S (g/L): 99.87

Lactic Acid P (g/L): 69.43

Maltose M (g/L): 30.56

Volume V (L): 8.03

t = 41.00 h

Biomass X (g/L): 80.27

Glucose S (g/L): 102.19

Lactic Acid P (g/L): 53.46

Maltose M (g/L): 31.29

Volume V (L): 7.66

t = 42.00 h

Biomass X (g/L): 65.83

Glucose S (g/L): 101.93

Lactic Acid P (g/L): 71.69

Maltose M (g/L): 31.31

Volume V (L): 8.61

t = 43.00 h

Biomass X (g/L): 92.96

Glucose S (g/L): 102.15

Lactic Acid P (g/L): 67.18

Maltose M (g/L): 31.52

Volume V (L): 7.50

t = 44.00 h

Biomass X (g/L): 95.78

Glucose S (g/L): 99.84

Lactic Acid P (g/L): 86.09

Maltose M (g/L): 27.73

Volume V (L): 8.43

t = 45.00 h

Biomass X (g/L): 92.78

Glucose S (g/L): 97.90

Lactic Acid P (g/L): 54.92

Maltose M (g/L): 30.70

Volume V (L): 7.85

t = 46.00 h

Biomass X (g/L): 68.64

Glucose S (g/L): 101.87

Lactic Acid P (g/L): 81.70
Maltose M (g/L): 30.00
Volume V (L): 8.00

t = 47.00 h

Biomass X (g/L): 90.54
Glucose S (g/L): 102.00
Lactic Acid P (g/L): 54.20
Maltose M (g/L): 30.49
Volume V (L): 8.67

t = 48.00 h

Biomass X (g/L): 68.05
Glucose S (g/L): 100.99
Lactic Acid P (g/L): 57.93
Maltose M (g/L): 32.34
Volume V (L): 8.21

t = 49.00 h

Biomass X (g/L): 85.15
Glucose S (g/L): 107.70
Lactic Acid P (g/L): 55.08
Maltose M (g/L): 34.12
Volume V (L): 8.36

t = 50.00 h

Biomass X (g/L): 86.59
Glucose S (g/L): 100.00
Lactic Acid P (g/L): 72.84
Maltose M (g/L): 31.22
Volume V (L): 8.30

t = 51.00 h

Biomass X (g/L): 87.61
Glucose S (g/L): 104.37
Lactic Acid P (g/L): 60.23
Maltose M (g/L): 27.28
Volume V (L): 9.51

t = 52.00 h

Biomass X (g/L): 89.79

Glucose S (g/L): 110.09
Lactic Acid P (g/L): 58.92
Maltose M (g/L): 28.81
Volume V (L): 8.53

t = 53.00 h

Biomass X (g/L): 84.38
Glucose S (g/L): 109.45
Lactic Acid P (g/L): 73.24
Maltose M (g/L): 30.45
Volume V (L): 8.90

t = 54.00 h

Biomass X (g/L): 65.02
Glucose S (g/L): 111.06
Lactic Acid P (g/L): 75.75
Maltose M (g/L): 30.04
Volume V (L): 8.86

t = 55.00 h

Biomass X (g/L): 89.62
Glucose S (g/L): 104.24
Lactic Acid P (g/L): 72.26
Maltose M (g/L): 30.83
Volume V (L): 9.12

t = 56.00 h

Biomass X (g/L): 85.22
Glucose S (g/L): 110.12
Lactic Acid P (g/L): 70.67
Maltose M (g/L): 28.51
Volume V (L): 8.83

t = 57.00 h

Biomass X (g/L): 75.25
Glucose S (g/L): 103.14
Lactic Acid P (g/L): 88.94
Maltose M (g/L): 27.80
Volume V (L): 9.08

t = 58.00 h

Biomass X (g/L): 68.57
Glucose S (g/L): 106.68
Lactic Acid P (g/L): 73.91
Maltose M (g/L): 31.75
Volume V (L): 9.41

t = 59.00 h

Biomass X (g/L): 99.17
Glucose S (g/L): 107.45
Lactic Acid P (g/L): 80.10
Maltose M (g/L): 28.25
Volume V (L): 8.94

t = 60.00 h

Biomass X (g/L): 96.01
Glucose S (g/L): 106.27
Lactic Acid P (g/L): 74.31
Maltose M (g/L): 25.97
Volume V (L): 9.87

t = 61.00 h

Biomass X (g/L): 90.70
Glucose S (g/L): 107.51
Lactic Acid P (g/L): 63.27
Maltose M (g/L): 30.66
Volume V (L): 9.98

t = 62.00 h

Biomass X (g/L): 76.93
Glucose S (g/L): 107.38
Lactic Acid P (g/L): 78.39
Maltose M (g/L): 24.67
Volume V (L): 9.80

t = 63.00 h

Biomass X (g/L): 58.62
Glucose S (g/L): 105.82
Lactic Acid P (g/L): 67.65
Maltose M (g/L): 20.58
Volume V (L): 9.64

t = 64.00 h

Biomass X (g/L): 90.89
Glucose S (g/L): 113.54
Lactic Acid P (g/L): 72.95
Maltose M (g/L): 28.75
Volume V (L): 8.37

t = 65.00 h

Biomass X (g/L): 86.82
Glucose S (g/L): 109.79
Lactic Acid P (g/L): 76.72
Maltose M (g/L): 28.85
Volume V (L): 9.94

t = 66.00 h

Biomass X (g/L): 79.35
Glucose S (g/L): 111.70
Lactic Acid P (g/L): 56.54
Maltose M (g/L): 27.71
Volume V (L): 9.65

t = 67.00 h

Biomass X (g/L): 94.01
Glucose S (g/L): 111.94
Lactic Acid P (g/L): 59.71
Maltose M (g/L): 24.18
Volume V (L): 9.79

t = 68.00 h

Biomass X (g/L): 91.12
Glucose S (g/L): 111.88
Lactic Acid P (g/L): 82.69
Maltose M (g/L): 26.44
Volume V (L): 9.14

t = 69.00 h

Biomass X (g/L): 84.33
Glucose S (g/L): 112.41
Lactic Acid P (g/L): 68.60
Maltose M (g/L): 25.85
Volume V (L): 10.26

t = 70.00 h

Biomass X (g/L): 82.77
Glucose S (g/L): 111.20
Lactic Acid P (g/L): 69.18
Maltose M (g/L): 25.18
Volume V (L): 9.48

t = 71.00 h

Biomass X (g/L): 82.77
Glucose S (g/L): 118.45
Lactic Acid P (g/L): 76.42
Maltose M (g/L): 28.83
Volume V (L): 10.24

t = 72.00 h

Biomass X (g/L): 88.81
Glucose S (g/L): 116.20
Lactic Acid P (g/L): 71.57
Maltose M (g/L): 23.86
Volume V (L): 10.66

t = 73.00 h

Biomass X (g/L): 73.46
Glucose S (g/L): 113.61
Lactic Acid P (g/L): 78.99
Maltose M (g/L): 26.37
Volume V (L): 10.13

t = 74.00 h

Biomass X (g/L): 85.96
Glucose S (g/L): 112.12
Lactic Acid P (g/L): 74.88
Maltose M (g/L): 27.93
Volume V (L): 10.94

t = 75.00 h

Biomass X (g/L): 72.49
Glucose S (g/L): 113.79
Lactic Acid P (g/L): 77.28
Maltose M (g/L): 26.18

Volume V (L): 10.16

Batch 66

Initial Conditions:

Batch: 66.00

X0 (g/L): 2.97

S0 (g/L): 56.89

M0 (g/L): 45.28

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.80

F_vol (L/h): 0.09

S_feed (g/L): 167.04

M_feed (g/L): 132.96

V_final (L): 11.77

Batch 66 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 8.69
Glucose S (g/L): 58.32
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 45.25
Volume V (L): 5.27

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 60.21
Lactic Acid P (g/L): 7.87
Maltose M (g/L): 46.27
Volume V (L): 5.02

t = 2.00 h

Biomass X (g/L): 15.05
Glucose S (g/L): 58.86
Lactic Acid P (g/L): 6.23
Maltose M (g/L): 46.55
Volume V (L): 5.41

t = 3.00 h

Biomass X (g/L): 15.19
Glucose S (g/L): 62.18
Lactic Acid P (g/L): 2.31
Maltose M (g/L): 47.60
Volume V (L): 4.82

t = 4.00 h

Biomass X (g/L): 13.32
Glucose S (g/L): 62.34
Lactic Acid P (g/L): 8.88
Maltose M (g/L): 48.70
Volume V (L): 5.03

t = 5.00 h

Biomass X (g/L): 7.57
Glucose S (g/L): 61.70

Lactic Acid P (g/L): 7.47
Maltose M (g/L): 48.28
Volume V (L): 5.05

t = 6.00 h

Biomass X (g/L): 0.87
Glucose S (g/L): 61.40
Lactic Acid P (g/L): 16.85
Maltose M (g/L): 49.19
Volume V (L): 5.24

t = 7.00 h

Biomass X (g/L): 13.88
Glucose S (g/L): 60.60
Lactic Acid P (g/L): 19.24
Maltose M (g/L): 50.09
Volume V (L): 6.33

t = 8.00 h

Biomass X (g/L): 31.92
Glucose S (g/L): 59.42
Lactic Acid P (g/L): 4.18
Maltose M (g/L): 50.40
Volume V (L): 6.12

t = 9.00 h

Biomass X (g/L): 26.59
Glucose S (g/L): 57.46
Lactic Acid P (g/L): 18.97
Maltose M (g/L): 49.48
Volume V (L): 5.23

t = 10.00 h

Biomass X (g/L): 14.73
Glucose S (g/L): 51.83
Lactic Acid P (g/L): 4.93
Maltose M (g/L): 50.82
Volume V (L): 6.14

t = 11.00 h

Biomass X (g/L): 23.05

Glucose S (g/L): 52.71
Lactic Acid P (g/L): 23.98
Maltose M (g/L): 50.53
Volume V (L): 5.78

t = 12.00 h

Biomass X (g/L): 49.90
Glucose S (g/L): 51.91
Lactic Acid P (g/L): 31.27
Maltose M (g/L): 51.41
Volume V (L): 5.71

t = 13.00 h

Biomass X (g/L): 46.63
Glucose S (g/L): 49.35
Lactic Acid P (g/L): 17.38
Maltose M (g/L): 51.24
Volume V (L): 5.37

t = 14.00 h

Biomass X (g/L): 34.02
Glucose S (g/L): 50.14
Lactic Acid P (g/L): 28.20
Maltose M (g/L): 51.23
Volume V (L): 6.79

t = 15.00 h

Biomass X (g/L): 32.52
Glucose S (g/L): 47.38
Lactic Acid P (g/L): 34.70
Maltose M (g/L): 51.52
Volume V (L): 5.62

t = 16.00 h

Biomass X (g/L): 46.69
Glucose S (g/L): 41.21
Lactic Acid P (g/L): 39.25
Maltose M (g/L): 51.55
Volume V (L): 6.42

t = 17.00 h

Biomass X (g/L): 38.01
Glucose S (g/L): 42.11
Lactic Acid P (g/L): 37.11
Maltose M (g/L): 51.71
Volume V (L): 6.02

t = 18.00 h

Biomass X (g/L): 56.93
Glucose S (g/L): 40.57
Lactic Acid P (g/L): 31.43
Maltose M (g/L): 51.50
Volume V (L): 7.05

t = 19.00 h

Biomass X (g/L): 84.29
Glucose S (g/L): 40.64
Lactic Acid P (g/L): 43.33
Maltose M (g/L): 51.53
Volume V (L): 6.53

t = 20.00 h

Biomass X (g/L): 47.95
Glucose S (g/L): 39.94
Lactic Acid P (g/L): 31.10
Maltose M (g/L): 53.06
Volume V (L): 6.73

t = 21.00 h

Biomass X (g/L): 63.45
Glucose S (g/L): 41.32
Lactic Acid P (g/L): 45.36
Maltose M (g/L): 52.20
Volume V (L): 6.84

t = 22.00 h

Biomass X (g/L): 79.58
Glucose S (g/L): 39.57
Lactic Acid P (g/L): 45.11
Maltose M (g/L): 51.49
Volume V (L): 7.15

t = 23.00 h

Biomass X (g/L): 75.88
Glucose S (g/L): 42.62
Lactic Acid P (g/L): 49.29
Maltose M (g/L): 51.73
Volume V (L): 7.08

t = 24.00 h

Biomass X (g/L): 68.82
Glucose S (g/L): 39.39
Lactic Acid P (g/L): 46.89
Maltose M (g/L): 52.71
Volume V (L): 7.14

t = 25.00 h

Biomass X (g/L): 70.20
Glucose S (g/L): 36.02
Lactic Acid P (g/L): 39.78
Maltose M (g/L): 51.42
Volume V (L): 6.98

t = 26.00 h

Biomass X (g/L): 61.79
Glucose S (g/L): 37.23
Lactic Acid P (g/L): 51.02
Maltose M (g/L): 50.88
Volume V (L): 6.49

t = 27.00 h

Biomass X (g/L): 71.36
Glucose S (g/L): 40.75
Lactic Acid P (g/L): 63.55
Maltose M (g/L): 50.91
Volume V (L): 6.83

t = 28.00 h

Biomass X (g/L): 77.22
Glucose S (g/L): 34.04
Lactic Acid P (g/L): 67.47
Maltose M (g/L): 50.65
Volume V (L): 7.92

t = 29.00 h

Biomass X (g/L): 86.33
Glucose S (g/L): 35.18
Lactic Acid P (g/L): 78.81
Maltose M (g/L): 50.92
Volume V (L): 7.45

t = 30.00 h

Biomass X (g/L): 70.69
Glucose S (g/L): 39.79
Lactic Acid P (g/L): 73.09
Maltose M (g/L): 50.95
Volume V (L): 7.35

t = 31.00 h

Biomass X (g/L): 75.51
Glucose S (g/L): 43.15
Lactic Acid P (g/L): 48.42
Maltose M (g/L): 52.74
Volume V (L): 7.04

t = 32.00 h

Biomass X (g/L): 68.92
Glucose S (g/L): 44.73
Lactic Acid P (g/L): 56.15
Maltose M (g/L): 51.64
Volume V (L): 6.96

t = 33.00 h

Biomass X (g/L): 80.71
Glucose S (g/L): 40.45
Lactic Acid P (g/L): 60.28
Maltose M (g/L): 51.34
Volume V (L): 7.60

t = 34.00 h

Biomass X (g/L): 86.56
Glucose S (g/L): 44.99
Lactic Acid P (g/L): 60.40
Maltose M (g/L): 50.47

Volume V (L): 8.55

t = 35.00 h

Biomass X (g/L): 85.56

Glucose S (g/L): 42.46

Lactic Acid P (g/L): 62.42

Maltose M (g/L): 51.32

Volume V (L): 7.86

t = 36.00 h

Biomass X (g/L): 68.82

Glucose S (g/L): 38.03

Lactic Acid P (g/L): 83.65

Maltose M (g/L): 50.07

Volume V (L): 8.47

t = 37.00 h

Biomass X (g/L): 81.75

Glucose S (g/L): 43.41

Lactic Acid P (g/L): 53.43

Maltose M (g/L): 50.95

Volume V (L): 8.01

t = 38.00 h

Biomass X (g/L): 74.22

Glucose S (g/L): 42.14

Lactic Acid P (g/L): 57.23

Maltose M (g/L): 52.59

Volume V (L): 8.78

t = 39.00 h

Biomass X (g/L): 90.61

Glucose S (g/L): 47.60

Lactic Acid P (g/L): 78.54

Maltose M (g/L): 51.46

Volume V (L): 8.33

t = 40.00 h

Biomass X (g/L): 79.69

Glucose S (g/L): 45.32

Lactic Acid P (g/L): 59.09

Maltose M (g/L): 50.70

Volume V (L): 7.94

t = 41.00 h

Biomass X (g/L): 66.97

Glucose S (g/L): 42.08

Lactic Acid P (g/L): 68.46

Maltose M (g/L): 51.23

Volume V (L): 8.56

t = 42.00 h

Biomass X (g/L): 79.56

Glucose S (g/L): 42.08

Lactic Acid P (g/L): 74.56

Maltose M (g/L): 50.21

Volume V (L): 9.79

t = 43.00 h

Biomass X (g/L): 86.20

Glucose S (g/L): 47.41

Lactic Acid P (g/L): 70.15

Maltose M (g/L): 49.87

Volume V (L): 9.85

t = 44.00 h

Biomass X (g/L): 84.30

Glucose S (g/L): 40.60

Lactic Acid P (g/L): 67.95

Maltose M (g/L): 51.54

Volume V (L): 9.24

t = 45.00 h

Biomass X (g/L): 77.21

Glucose S (g/L): 47.28

Lactic Acid P (g/L): 74.92

Maltose M (g/L): 49.34

Volume V (L): 8.99

t = 46.00 h

Biomass X (g/L): 76.61

Glucose S (g/L): 50.34

Lactic Acid P (g/L): 59.13

Maltose M (g/L): 49.75

Volume V (L): 9.32

t = 47.00 h

Biomass X (g/L): 72.85

Glucose S (g/L): 47.04

Lactic Acid P (g/L): 62.98

Maltose M (g/L): 50.07

Volume V (L): 9.42

t = 48.00 h

Biomass X (g/L): 78.57

Glucose S (g/L): 54.19

Lactic Acid P (g/L): 72.42

Maltose M (g/L): 49.02

Volume V (L): 9.31

t = 49.00 h

Biomass X (g/L): 74.22

Glucose S (g/L): 55.53

Lactic Acid P (g/L): 49.76

Maltose M (g/L): 50.00

Volume V (L): 9.66

t = 50.00 h

Biomass X (g/L): 91.66

Glucose S (g/L): 49.35

Lactic Acid P (g/L): 80.83

Maltose M (g/L): 49.15

Volume V (L): 9.63

t = 51.00 h

Biomass X (g/L): 73.83

Glucose S (g/L): 52.34

Lactic Acid P (g/L): 71.31

Maltose M (g/L): 49.68

Volume V (L): 9.43

t = 52.00 h

Biomass X (g/L): 83.59

Glucose S (g/L): 57.12
Lactic Acid P (g/L): 70.27
Maltose M (g/L): 49.45
Volume V (L): 9.05

t = 53.00 h

Biomass X (g/L): 82.93
Glucose S (g/L): 54.73
Lactic Acid P (g/L): 54.81
Maltose M (g/L): 48.69
Volume V (L): 9.54

t = 54.00 h

Biomass X (g/L): 91.41
Glucose S (g/L): 51.10
Lactic Acid P (g/L): 58.11
Maltose M (g/L): 49.92
Volume V (L): 10.31

t = 55.00 h

Biomass X (g/L): 92.74
Glucose S (g/L): 55.74
Lactic Acid P (g/L): 66.21
Maltose M (g/L): 48.03
Volume V (L): 9.95

t = 56.00 h

Biomass X (g/L): 88.95
Glucose S (g/L): 54.25
Lactic Acid P (g/L): 77.10
Maltose M (g/L): 49.30
Volume V (L): 10.22

t = 57.00 h

Biomass X (g/L): 85.62
Glucose S (g/L): 61.11
Lactic Acid P (g/L): 51.89
Maltose M (g/L): 49.00
Volume V (L): 9.99

t = 58.00 h

Biomass X (g/L): 72.20
Glucose S (g/L): 60.26
Lactic Acid P (g/L): 77.40
Maltose M (g/L): 49.25
Volume V (L): 9.66

t = 59.00 h

Biomass X (g/L): 71.94
Glucose S (g/L): 58.09
Lactic Acid P (g/L): 67.50
Maltose M (g/L): 47.91
Volume V (L): 10.20

t = 60.00 h

Biomass X (g/L): 73.99
Glucose S (g/L): 60.23
Lactic Acid P (g/L): 68.92
Maltose M (g/L): 47.03
Volume V (L): 9.86

t = 61.00 h

Biomass X (g/L): 89.14
Glucose S (g/L): 56.40
Lactic Acid P (g/L): 73.34
Maltose M (g/L): 48.91
Volume V (L): 9.92

t = 62.00 h

Biomass X (g/L): 82.58
Glucose S (g/L): 58.84
Lactic Acid P (g/L): 73.69
Maltose M (g/L): 46.99
Volume V (L): 11.25

t = 63.00 h

Biomass X (g/L): 90.06
Glucose S (g/L): 66.77
Lactic Acid P (g/L): 65.22
Maltose M (g/L): 46.46
Volume V (L): 10.46

t = 64.00 h

Biomass X (g/L): 91.24
Glucose S (g/L): 62.77
Lactic Acid P (g/L): 77.22
Maltose M (g/L): 46.72
Volume V (L): 10.38

t = 65.00 h

Biomass X (g/L): 74.65
Glucose S (g/L): 66.21
Lactic Acid P (g/L): 51.43
Maltose M (g/L): 45.96
Volume V (L): 11.74

t = 66.00 h

Biomass X (g/L): 81.77
Glucose S (g/L): 64.71
Lactic Acid P (g/L): 68.10
Maltose M (g/L): 46.85
Volume V (L): 11.40

t = 67.00 h

Biomass X (g/L): 81.22
Glucose S (g/L): 68.78
Lactic Acid P (g/L): 73.61
Maltose M (g/L): 45.85
Volume V (L): 10.85

t = 68.00 h

Biomass X (g/L): 78.10
Glucose S (g/L): 59.08
Lactic Acid P (g/L): 69.38
Maltose M (g/L): 45.89
Volume V (L): 11.37

t = 69.00 h

Biomass X (g/L): 76.00
Glucose S (g/L): 65.42
Lactic Acid P (g/L): 56.48
Maltose M (g/L): 46.60
Volume V (L): 11.10

t = 70.00 h

Biomass X (g/L): 92.27
Glucose S (g/L): 71.83
Lactic Acid P (g/L): 79.45
Maltose M (g/L): 46.16
Volume V (L): 10.94

t = 71.00 h

Biomass X (g/L): 81.33
Glucose S (g/L): 70.34
Lactic Acid P (g/L): 61.91
Maltose M (g/L): 45.53
Volume V (L): 11.67

t = 72.00 h

Biomass X (g/L): 80.83
Glucose S (g/L): 72.15
Lactic Acid P (g/L): 72.49
Maltose M (g/L): 45.45
Volume V (L): 11.32

t = 73.00 h

Biomass X (g/L): 72.91
Glucose S (g/L): 68.63
Lactic Acid P (g/L): 74.59
Maltose M (g/L): 45.14
Volume V (L): 11.63

t = 74.00 h

Biomass X (g/L): 86.86
Glucose S (g/L): 67.61
Lactic Acid P (g/L): 73.56
Maltose M (g/L): 44.63
Volume V (L): 12.23

t = 75.00 h

Biomass X (g/L): 70.70
Glucose S (g/L): 66.72
Lactic Acid P (g/L): 83.85
Maltose M (g/L): 44.74

Volume V (L): 12.26

Batch 67

Initial Conditions:

Batch: 67.00

X0 (g/L): 1.73

S0 (g/L): 37.71

M0 (g/L): 154.13

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 4.09

F_vol (L/h): 0.04

S_feed (g/L): 58.97

M_feed (g/L): 241.03

V_final (L): 8.11

Batch 67 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 32.44
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 152.61
Volume V (L): 5.15

t = 1.00 h

Biomass X (g/L): 7.19
Glucose S (g/L): 38.75
Lactic Acid P (g/L): 3.08
Maltose M (g/L): 140.32
Volume V (L): 4.81

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 43.50
Lactic Acid P (g/L): 2.40
Maltose M (g/L): 141.46
Volume V (L): 5.20

t = 3.00 h

Biomass X (g/L): 8.55
Glucose S (g/L): 39.79
Lactic Acid P (g/L): 3.25
Maltose M (g/L): 151.71
Volume V (L): 5.12

t = 4.00 h

Biomass X (g/L): 6.76
Glucose S (g/L): 39.29
Lactic Acid P (g/L): 0.55
Maltose M (g/L): 145.72
Volume V (L): 4.82

t = 5.00 h

Biomass X (g/L): 10.67
Glucose S (g/L): 41.19

Lactic Acid P (g/L): 18.97
Maltose M (g/L): 137.68
Volume V (L): 5.10

t = 6.00 h

Biomass X (g/L): 5.84
Glucose S (g/L): 51.96
Lactic Acid P (g/L): 3.94
Maltose M (g/L): 152.49
Volume V (L): 5.79

t = 7.00 h

Biomass X (g/L): 32.26
Glucose S (g/L): 48.47
Lactic Acid P (g/L): 22.14
Maltose M (g/L): 129.35
Volume V (L): 5.34

t = 8.00 h

Biomass X (g/L): 5.03
Glucose S (g/L): 47.04
Lactic Acid P (g/L): 14.34
Maltose M (g/L): 134.61
Volume V (L): 5.90

t = 9.00 h

Biomass X (g/L): 27.21
Glucose S (g/L): 50.36
Lactic Acid P (g/L): 14.75
Maltose M (g/L): 131.56
Volume V (L): 5.01

t = 10.00 h

Biomass X (g/L): 10.69
Glucose S (g/L): 45.56
Lactic Acid P (g/L): 13.08
Maltose M (g/L): 139.10
Volume V (L): 5.60

t = 11.00 h

Biomass X (g/L): 30.15

Glucose S (g/L): 45.90
Lactic Acid P (g/L): 26.19
Maltose M (g/L): 132.61
Volume V (L): 4.83

t = 12.00 h

Biomass X (g/L): 35.02
Glucose S (g/L): 43.49
Lactic Acid P (g/L): 21.18
Maltose M (g/L): 129.59
Volume V (L): 5.50

t = 13.00 h

Biomass X (g/L): 34.00
Glucose S (g/L): 40.24
Lactic Acid P (g/L): 14.51
Maltose M (g/L): 113.44
Volume V (L): 5.70

t = 14.00 h

Biomass X (g/L): 32.64
Glucose S (g/L): 40.39
Lactic Acid P (g/L): 17.32
Maltose M (g/L): 134.37
Volume V (L): 5.79

t = 15.00 h

Biomass X (g/L): 43.11
Glucose S (g/L): 39.81
Lactic Acid P (g/L): 33.39
Maltose M (g/L): 123.74
Volume V (L): 5.42

t = 16.00 h

Biomass X (g/L): 48.25
Glucose S (g/L): 35.44
Lactic Acid P (g/L): 23.85
Maltose M (g/L): 124.95
Volume V (L): 5.92

t = 17.00 h

Biomass X (g/L): 67.24
Glucose S (g/L): 33.08
Lactic Acid P (g/L): 34.56
Maltose M (g/L): 116.22
Volume V (L): 5.57

t = 18.00 h

Biomass X (g/L): 43.78
Glucose S (g/L): 31.98
Lactic Acid P (g/L): 47.68
Maltose M (g/L): 113.41
Volume V (L): 5.71

t = 19.00 h

Biomass X (g/L): 50.09
Glucose S (g/L): 25.43
Lactic Acid P (g/L): 43.07
Maltose M (g/L): 131.98
Volume V (L): 5.68

t = 20.00 h

Biomass X (g/L): 61.43
Glucose S (g/L): 31.40
Lactic Acid P (g/L): 35.73
Maltose M (g/L): 103.28
Volume V (L): 5.55

t = 21.00 h

Biomass X (g/L): 58.60
Glucose S (g/L): 26.48
Lactic Acid P (g/L): 57.84
Maltose M (g/L): 107.87
Volume V (L): 6.26

t = 22.00 h

Biomass X (g/L): 62.56
Glucose S (g/L): 30.79
Lactic Acid P (g/L): 51.72
Maltose M (g/L): 127.62
Volume V (L): 5.92

t = 23.00 h

Biomass X (g/L): 76.53
Glucose S (g/L): 28.30
Lactic Acid P (g/L): 57.95
Maltose M (g/L): 119.28
Volume V (L): 6.22

t = 24.00 h

Biomass X (g/L): 39.29
Glucose S (g/L): 28.44
Lactic Acid P (g/L): 28.83
Maltose M (g/L): 109.91
Volume V (L): 6.26

t = 25.00 h

Biomass X (g/L): 63.49
Glucose S (g/L): 29.80
Lactic Acid P (g/L): 54.78
Maltose M (g/L): 121.72
Volume V (L): 5.80

t = 26.00 h

Biomass X (g/L): 53.59
Glucose S (g/L): 25.23
Lactic Acid P (g/L): 61.04
Maltose M (g/L): 113.80
Volume V (L): 6.38

t = 27.00 h

Biomass X (g/L): 60.58
Glucose S (g/L): 28.70
Lactic Acid P (g/L): 70.45
Maltose M (g/L): 110.79
Volume V (L): 6.25

t = 28.00 h

Biomass X (g/L): 98.23
Glucose S (g/L): 28.47
Lactic Acid P (g/L): 42.05
Maltose M (g/L): 109.52
Volume V (L): 5.90

t = 29.00 h

Biomass X (g/L): 81.25
Glucose S (g/L): 27.37
Lactic Acid P (g/L): 68.28
Maltose M (g/L): 110.34
Volume V (L): 6.31

t = 30.00 h

Biomass X (g/L): 85.93
Glucose S (g/L): 25.54
Lactic Acid P (g/L): 73.51
Maltose M (g/L): 109.52
Volume V (L): 5.99

t = 31.00 h

Biomass X (g/L): 76.55
Glucose S (g/L): 29.94
Lactic Acid P (g/L): 40.22
Maltose M (g/L): 122.47
Volume V (L): 6.69

t = 32.00 h

Biomass X (g/L): 85.93
Glucose S (g/L): 30.55
Lactic Acid P (g/L): 64.62
Maltose M (g/L): 101.44
Volume V (L): 6.53

t = 33.00 h

Biomass X (g/L): 61.32
Glucose S (g/L): 29.31
Lactic Acid P (g/L): 66.85
Maltose M (g/L): 100.61
Volume V (L): 6.35

t = 34.00 h

Biomass X (g/L): 90.36
Glucose S (g/L): 26.49
Lactic Acid P (g/L): 58.87
Maltose M (g/L): 90.60

Volume V (L): 6.72

t = 35.00 h

Biomass X (g/L): 72.32

Glucose S (g/L): 28.41

Lactic Acid P (g/L): 58.17

Maltose M (g/L): 99.44

Volume V (L): 6.06

t = 36.00 h

Biomass X (g/L): 75.86

Glucose S (g/L): 30.16

Lactic Acid P (g/L): 69.81

Maltose M (g/L): 114.37

Volume V (L): 6.31

t = 37.00 h

Biomass X (g/L): 86.88

Glucose S (g/L): 32.69

Lactic Acid P (g/L): 58.53

Maltose M (g/L): 106.05

Volume V (L): 6.47

t = 38.00 h

Biomass X (g/L): 85.50

Glucose S (g/L): 27.06

Lactic Acid P (g/L): 62.96

Maltose M (g/L): 92.33

Volume V (L): 7.13

t = 39.00 h

Biomass X (g/L): 83.62

Glucose S (g/L): 30.38

Lactic Acid P (g/L): 73.29

Maltose M (g/L): 96.15

Volume V (L): 6.35

t = 40.00 h

Biomass X (g/L): 69.45

Glucose S (g/L): 31.39

Lactic Acid P (g/L): 82.76

Maltose M (g/L): 84.15

Volume V (L): 6.23

t = 41.00 h

Biomass X (g/L): 84.58

Glucose S (g/L): 32.85

Lactic Acid P (g/L): 71.97

Maltose M (g/L): 85.95

Volume V (L): 6.34

t = 42.00 h

Biomass X (g/L): 93.59

Glucose S (g/L): 30.14

Lactic Acid P (g/L): 68.75

Maltose M (g/L): 94.37

Volume V (L): 6.97

t = 43.00 h

Biomass X (g/L): 78.84

Glucose S (g/L): 33.70

Lactic Acid P (g/L): 63.21

Maltose M (g/L): 88.11

Volume V (L): 6.64

t = 44.00 h

Biomass X (g/L): 97.73

Glucose S (g/L): 34.08

Lactic Acid P (g/L): 59.11

Maltose M (g/L): 82.65

Volume V (L): 6.79

t = 45.00 h

Biomass X (g/L): 75.91

Glucose S (g/L): 32.74

Lactic Acid P (g/L): 79.50

Maltose M (g/L): 94.58

Volume V (L): 7.06

t = 46.00 h

Biomass X (g/L): 84.98

Glucose S (g/L): 35.88

Lactic Acid P (g/L): 68.14
Maltose M (g/L): 88.97
Volume V (L): 6.87

t = 47.00 h

Biomass X (g/L): 100.57
Glucose S (g/L): 35.38
Lactic Acid P (g/L): 80.84
Maltose M (g/L): 75.77
Volume V (L): 7.00

t = 48.00 h

Biomass X (g/L): 78.38
Glucose S (g/L): 38.86
Lactic Acid P (g/L): 67.40
Maltose M (g/L): 70.73
Volume V (L): 7.12

t = 49.00 h

Biomass X (g/L): 96.00
Glucose S (g/L): 35.77
Lactic Acid P (g/L): 64.83
Maltose M (g/L): 82.48
Volume V (L): 6.93

t = 50.00 h

Biomass X (g/L): 105.39
Glucose S (g/L): 39.08
Lactic Acid P (g/L): 75.28
Maltose M (g/L): 73.21
Volume V (L): 7.03

t = 51.00 h

Biomass X (g/L): 87.29
Glucose S (g/L): 37.30
Lactic Acid P (g/L): 71.01
Maltose M (g/L): 70.11
Volume V (L): 7.18

t = 52.00 h

Biomass X (g/L): 80.32

Glucose S (g/L): 38.54
Lactic Acid P (g/L): 60.80
Maltose M (g/L): 94.34
Volume V (L): 7.37

t = 53.00 h

Biomass X (g/L): 77.90
Glucose S (g/L): 41.64
Lactic Acid P (g/L): 89.52
Maltose M (g/L): 83.38
Volume V (L): 7.00

t = 54.00 h

Biomass X (g/L): 98.87
Glucose S (g/L): 42.14
Lactic Acid P (g/L): 66.95
Maltose M (g/L): 81.25
Volume V (L): 6.49

t = 55.00 h

Biomass X (g/L): 98.60
Glucose S (g/L): 41.04
Lactic Acid P (g/L): 81.88
Maltose M (g/L): 81.18
Volume V (L): 7.45

t = 56.00 h

Biomass X (g/L): 100.35
Glucose S (g/L): 42.10
Lactic Acid P (g/L): 81.40
Maltose M (g/L): 72.57
Volume V (L): 7.55

t = 57.00 h

Biomass X (g/L): 85.39
Glucose S (g/L): 39.55
Lactic Acid P (g/L): 62.01
Maltose M (g/L): 84.79
Volume V (L): 7.41

t = 58.00 h

Biomass X (g/L): 96.16
Glucose S (g/L): 47.97
Lactic Acid P (g/L): 64.73
Maltose M (g/L): 79.60
Volume V (L): 7.47

t = 59.00 h

Biomass X (g/L): 85.17
Glucose S (g/L): 46.69
Lactic Acid P (g/L): 71.60
Maltose M (g/L): 73.72
Volume V (L): 7.31

t = 60.00 h

Biomass X (g/L): 76.71
Glucose S (g/L): 47.25
Lactic Acid P (g/L): 66.60
Maltose M (g/L): 74.31
Volume V (L): 7.61

t = 61.00 h

Biomass X (g/L): 96.88
Glucose S (g/L): 41.80
Lactic Acid P (g/L): 91.19
Maltose M (g/L): 68.51
Volume V (L): 7.74

t = 62.00 h

Biomass X (g/L): 101.95
Glucose S (g/L): 45.17
Lactic Acid P (g/L): 73.39
Maltose M (g/L): 74.72
Volume V (L): 7.59

t = 63.00 h

Biomass X (g/L): 75.12
Glucose S (g/L): 46.52
Lactic Acid P (g/L): 83.88
Maltose M (g/L): 75.49
Volume V (L): 8.23

t = 64.00 h

Biomass X (g/L): 74.84
Glucose S (g/L): 42.83
Lactic Acid P (g/L): 80.60
Maltose M (g/L): 73.27
Volume V (L): 7.27

t = 65.00 h

Biomass X (g/L): 85.62
Glucose S (g/L): 51.13
Lactic Acid P (g/L): 78.51
Maltose M (g/L): 76.74
Volume V (L): 7.74

t = 66.00 h

Biomass X (g/L): 94.76
Glucose S (g/L): 53.30
Lactic Acid P (g/L): 76.65
Maltose M (g/L): 75.03
Volume V (L): 7.45

t = 67.00 h

Biomass X (g/L): 73.58
Glucose S (g/L): 55.58
Lactic Acid P (g/L): 88.11
Maltose M (g/L): 77.03
Volume V (L): 8.00

t = 68.00 h

Biomass X (g/L): 87.18
Glucose S (g/L): 47.76
Lactic Acid P (g/L): 68.99
Maltose M (g/L): 82.40
Volume V (L): 7.43

t = 69.00 h

Biomass X (g/L): 78.81
Glucose S (g/L): 51.55
Lactic Acid P (g/L): 93.14
Maltose M (g/L): 78.92
Volume V (L): 7.53

t = 70.00 h

Biomass X (g/L): 91.70
Glucose S (g/L): 55.44
Lactic Acid P (g/L): 65.09
Maltose M (g/L): 81.38
Volume V (L): 7.75

t = 71.00 h

Biomass X (g/L): 76.45
Glucose S (g/L): 51.77
Lactic Acid P (g/L): 84.71
Maltose M (g/L): 65.82
Volume V (L): 8.01

t = 72.00 h

Biomass X (g/L): 92.14
Glucose S (g/L): 51.52
Lactic Acid P (g/L): 64.40
Maltose M (g/L): 68.11
Volume V (L): 7.84

t = 73.00 h

Biomass X (g/L): 82.62
Glucose S (g/L): 48.60
Lactic Acid P (g/L): 74.82
Maltose M (g/L): 77.27
Volume V (L): 7.88

t = 74.00 h

Biomass X (g/L): 74.61
Glucose S (g/L): 57.34
Lactic Acid P (g/L): 78.84
Maltose M (g/L): 76.90
Volume V (L): 8.02

t = 75.00 h

Biomass X (g/L): 84.55
Glucose S (g/L): 54.80
Lactic Acid P (g/L): 73.59
Maltose M (g/L): 65.14

Volume V (L): 8.05

Batch 68

Initial Conditions:

Batch: 68.00

X0 (g/L): 6.66

S0 (g/L): 60.24

M0 (g/L): 108.74

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.81

F_vol (L/h): 0.06

S_feed (g/L): 106.95

M_feed (g/L): 193.05

V_final (L): 9.43

Batch 68 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 21.02
Glucose S (g/L): 55.50
Lactic Acid P (g/L): 3.31
Maltose M (g/L): 109.68
Volume V (L): 5.27

t = 1.00 h

Biomass X (g/L): 15.51
Glucose S (g/L): 60.08
Lactic Acid P (g/L): 2.38
Maltose M (g/L): 107.56
Volume V (L): 5.16

t = 2.00 h

Biomass X (g/L): 8.54
Glucose S (g/L): 61.12
Lactic Acid P (g/L): 14.29
Maltose M (g/L): 108.37
Volume V (L): 5.04

t = 3.00 h

Biomass X (g/L): 35.27
Glucose S (g/L): 60.90
Lactic Acid P (g/L): 3.02
Maltose M (g/L): 113.88
Volume V (L): 5.17

t = 4.00 h

Biomass X (g/L): 23.68
Glucose S (g/L): 54.93
Lactic Acid P (g/L): 7.86
Maltose M (g/L): 101.62
Volume V (L): 5.75

t = 5.00 h

Biomass X (g/L): 13.94
Glucose S (g/L): 51.80

Lactic Acid P (g/L): 14.26
Maltose M (g/L): 103.63
Volume V (L): 5.68

t = 6.00 h

Biomass X (g/L): 30.44
Glucose S (g/L): 52.73
Lactic Acid P (g/L): 7.83
Maltose M (g/L): 99.35
Volume V (L): 6.05

t = 7.00 h

Biomass X (g/L): 37.96
Glucose S (g/L): 57.20
Lactic Acid P (g/L): 12.70
Maltose M (g/L): 102.85
Volume V (L): 5.49

t = 8.00 h

Biomass X (g/L): 30.97
Glucose S (g/L): 45.52
Lactic Acid P (g/L): 33.27
Maltose M (g/L): 97.78
Volume V (L): 5.16

t = 9.00 h

Biomass X (g/L): 32.13
Glucose S (g/L): 49.31
Lactic Acid P (g/L): 21.44
Maltose M (g/L): 97.04
Volume V (L): 5.38

t = 10.00 h

Biomass X (g/L): 47.57
Glucose S (g/L): 48.16
Lactic Acid P (g/L): 40.78
Maltose M (g/L): 91.67
Volume V (L): 5.17

t = 11.00 h

Biomass X (g/L): 47.39

Glucose S (g/L): 40.95
Lactic Acid P (g/L): 38.66
Maltose M (g/L): 98.17
Volume V (L): 5.92

t = 12.00 h

Biomass X (g/L): 68.95
Glucose S (g/L): 36.62
Lactic Acid P (g/L): 21.75
Maltose M (g/L): 98.86
Volume V (L): 5.50

t = 13.00 h

Biomass X (g/L): 65.30
Glucose S (g/L): 38.75
Lactic Acid P (g/L): 43.65
Maltose M (g/L): 104.78
Volume V (L): 4.69

t = 14.00 h

Biomass X (g/L): 55.10
Glucose S (g/L): 36.27
Lactic Acid P (g/L): 46.09
Maltose M (g/L): 97.31
Volume V (L): 6.70

t = 15.00 h

Biomass X (g/L): 68.23
Glucose S (g/L): 31.66
Lactic Acid P (g/L): 47.49
Maltose M (g/L): 92.09
Volume V (L): 5.68

t = 16.00 h

Biomass X (g/L): 78.89
Glucose S (g/L): 31.33
Lactic Acid P (g/L): 44.39
Maltose M (g/L): 86.93
Volume V (L): 5.74

t = 17.00 h

Biomass X (g/L): 55.49
Glucose S (g/L): 34.15
Lactic Acid P (g/L): 38.97
Maltose M (g/L): 86.77
Volume V (L): 6.60

t = 18.00 h

Biomass X (g/L): 61.11
Glucose S (g/L): 35.11
Lactic Acid P (g/L): 61.84
Maltose M (g/L): 85.61
Volume V (L): 5.63

t = 19.00 h

Biomass X (g/L): 72.71
Glucose S (g/L): 30.64
Lactic Acid P (g/L): 50.12
Maltose M (g/L): 99.92
Volume V (L): 6.52

t = 20.00 h

Biomass X (g/L): 74.81
Glucose S (g/L): 31.18
Lactic Acid P (g/L): 44.41
Maltose M (g/L): 91.58
Volume V (L): 5.97

t = 21.00 h

Biomass X (g/L): 75.43
Glucose S (g/L): 33.41
Lactic Acid P (g/L): 58.17
Maltose M (g/L): 86.67
Volume V (L): 5.77

t = 22.00 h

Biomass X (g/L): 74.26
Glucose S (g/L): 30.66
Lactic Acid P (g/L): 48.39
Maltose M (g/L): 84.13
Volume V (L): 6.90

t = 23.00 h

Biomass X (g/L): 75.30
Glucose S (g/L): 30.31
Lactic Acid P (g/L): 47.68
Maltose M (g/L): 83.88
Volume V (L): 6.69

t = 24.00 h

Biomass X (g/L): 83.22
Glucose S (g/L): 28.99
Lactic Acid P (g/L): 54.08
Maltose M (g/L): 84.33
Volume V (L): 6.74

t = 25.00 h

Biomass X (g/L): 91.41
Glucose S (g/L): 29.15
Lactic Acid P (g/L): 52.44
Maltose M (g/L): 92.78
Volume V (L): 6.61

t = 26.00 h

Biomass X (g/L): 76.67
Glucose S (g/L): 40.54
Lactic Acid P (g/L): 51.12
Maltose M (g/L): 85.67
Volume V (L): 6.75

t = 27.00 h

Biomass X (g/L): 105.62
Glucose S (g/L): 31.65
Lactic Acid P (g/L): 67.88
Maltose M (g/L): 78.23
Volume V (L): 6.16

t = 28.00 h

Biomass X (g/L): 101.69
Glucose S (g/L): 37.08
Lactic Acid P (g/L): 60.24
Maltose M (g/L): 95.43
Volume V (L): 6.69

t = 29.00 h

Biomass X (g/L): 96.21
Glucose S (g/L): 38.56
Lactic Acid P (g/L): 73.63
Maltose M (g/L): 78.34
Volume V (L): 6.38

t = 30.00 h

Biomass X (g/L): 89.93
Glucose S (g/L): 33.02
Lactic Acid P (g/L): 54.95
Maltose M (g/L): 80.29
Volume V (L): 6.28

t = 31.00 h

Biomass X (g/L): 103.79
Glucose S (g/L): 35.31
Lactic Acid P (g/L): 47.60
Maltose M (g/L): 88.69
Volume V (L): 6.98

t = 32.00 h

Biomass X (g/L): 82.04
Glucose S (g/L): 35.00
Lactic Acid P (g/L): 65.57
Maltose M (g/L): 83.79
Volume V (L): 7.30

t = 33.00 h

Biomass X (g/L): 77.19
Glucose S (g/L): 30.04
Lactic Acid P (g/L): 81.95
Maltose M (g/L): 78.95
Volume V (L): 6.81

t = 34.00 h

Biomass X (g/L): 80.48
Glucose S (g/L): 30.83
Lactic Acid P (g/L): 70.32
Maltose M (g/L): 74.70

Volume V (L): 6.93

t = 35.00 h

Biomass X (g/L): 76.17

Glucose S (g/L): 36.61

Lactic Acid P (g/L): 95.64

Maltose M (g/L): 75.35

Volume V (L): 6.66

t = 36.00 h

Biomass X (g/L): 87.28

Glucose S (g/L): 35.72

Lactic Acid P (g/L): 69.59

Maltose M (g/L): 75.03

Volume V (L): 7.13

t = 37.00 h

Biomass X (g/L): 89.22

Glucose S (g/L): 43.84

Lactic Acid P (g/L): 69.21

Maltose M (g/L): 78.78

Volume V (L): 7.45

t = 38.00 h

Biomass X (g/L): 88.45

Glucose S (g/L): 37.64

Lactic Acid P (g/L): 47.64

Maltose M (g/L): 79.07

Volume V (L): 7.52

t = 39.00 h

Biomass X (g/L): 86.55

Glucose S (g/L): 45.36

Lactic Acid P (g/L): 71.93

Maltose M (g/L): 76.88

Volume V (L): 7.50

t = 40.00 h

Biomass X (g/L): 88.08

Glucose S (g/L): 38.21

Lactic Acid P (g/L): 69.98

Maltose M (g/L): 76.83

Volume V (L): 7.14

t = 41.00 h

Biomass X (g/L): 103.66

Glucose S (g/L): 46.61

Lactic Acid P (g/L): 65.09

Maltose M (g/L): 71.46

Volume V (L): 7.19

t = 42.00 h

Biomass X (g/L): 92.96

Glucose S (g/L): 42.77

Lactic Acid P (g/L): 67.98

Maltose M (g/L): 77.30

Volume V (L): 7.76

t = 43.00 h

Biomass X (g/L): 95.94

Glucose S (g/L): 43.65

Lactic Acid P (g/L): 69.00

Maltose M (g/L): 81.45

Volume V (L): 7.83

t = 44.00 h

Biomass X (g/L): 88.92

Glucose S (g/L): 40.89

Lactic Acid P (g/L): 68.34

Maltose M (g/L): 74.04

Volume V (L): 7.54

t = 45.00 h

Biomass X (g/L): 71.50

Glucose S (g/L): 45.11

Lactic Acid P (g/L): 89.63

Maltose M (g/L): 73.42

Volume V (L): 7.51

t = 46.00 h

Biomass X (g/L): 69.59

Glucose S (g/L): 49.42

Lactic Acid P (g/L): 67.28

Maltose M (g/L): 74.16

Volume V (L): 8.13

t = 47.00 h

Biomass X (g/L): 107.19

Glucose S (g/L): 44.20

Lactic Acid P (g/L): 72.58

Maltose M (g/L): 78.57

Volume V (L): 7.65

t = 48.00 h

Biomass X (g/L): 81.92

Glucose S (g/L): 44.29

Lactic Acid P (g/L): 67.61

Maltose M (g/L): 68.48

Volume V (L): 8.10

t = 49.00 h

Biomass X (g/L): 81.06

Glucose S (g/L): 45.30

Lactic Acid P (g/L): 78.87

Maltose M (g/L): 79.14

Volume V (L): 7.85

t = 50.00 h

Biomass X (g/L): 89.33

Glucose S (g/L): 52.03

Lactic Acid P (g/L): 73.56

Maltose M (g/L): 73.74

Volume V (L): 7.58

t = 51.00 h

Biomass X (g/L): 92.20

Glucose S (g/L): 50.27

Lactic Acid P (g/L): 69.63

Maltose M (g/L): 71.39

Volume V (L): 8.58

t = 52.00 h

Biomass X (g/L): 93.35

Glucose S (g/L): 52.89
Lactic Acid P (g/L): 54.72
Maltose M (g/L): 76.54
Volume V (L): 8.11

t = 53.00 h

Biomass X (g/L): 103.85
Glucose S (g/L): 49.94
Lactic Acid P (g/L): 74.99
Maltose M (g/L): 64.67
Volume V (L): 7.88

t = 54.00 h

Biomass X (g/L): 101.11
Glucose S (g/L): 49.76
Lactic Acid P (g/L): 70.14
Maltose M (g/L): 72.30
Volume V (L): 7.70

t = 55.00 h

Biomass X (g/L): 78.69
Glucose S (g/L): 51.70
Lactic Acid P (g/L): 72.68
Maltose M (g/L): 71.40
Volume V (L): 7.79

t = 56.00 h

Biomass X (g/L): 95.76
Glucose S (g/L): 54.48
Lactic Acid P (g/L): 62.57
Maltose M (g/L): 64.45
Volume V (L): 8.18

t = 57.00 h

Biomass X (g/L): 82.57
Glucose S (g/L): 50.93
Lactic Acid P (g/L): 86.93
Maltose M (g/L): 70.63
Volume V (L): 9.05

t = 58.00 h

Biomass X (g/L): 84.82
Glucose S (g/L): 56.83
Lactic Acid P (g/L): 81.83
Maltose M (g/L): 69.34
Volume V (L): 8.22

t = 59.00 h

Biomass X (g/L): 75.46
Glucose S (g/L): 54.28
Lactic Acid P (g/L): 87.34
Maltose M (g/L): 66.70
Volume V (L): 9.00

t = 60.00 h

Biomass X (g/L): 69.53
Glucose S (g/L): 46.03
Lactic Acid P (g/L): 84.93
Maltose M (g/L): 71.22
Volume V (L): 8.50

t = 61.00 h

Biomass X (g/L): 85.42
Glucose S (g/L): 57.32
Lactic Acid P (g/L): 80.02
Maltose M (g/L): 63.97
Volume V (L): 8.18

t = 62.00 h

Biomass X (g/L): 76.34
Glucose S (g/L): 51.04
Lactic Acid P (g/L): 64.21
Maltose M (g/L): 67.55
Volume V (L): 8.49

t = 63.00 h

Biomass X (g/L): 94.37
Glucose S (g/L): 58.58
Lactic Acid P (g/L): 91.94
Maltose M (g/L): 65.21
Volume V (L): 8.27

t = 64.00 h

Biomass X (g/L): 82.54
Glucose S (g/L): 59.38
Lactic Acid P (g/L): 66.57
Maltose M (g/L): 62.50
Volume V (L): 8.48

t = 65.00 h

Biomass X (g/L): 88.06
Glucose S (g/L): 58.20
Lactic Acid P (g/L): 72.83
Maltose M (g/L): 67.41
Volume V (L): 8.53

t = 66.00 h

Biomass X (g/L): 98.11
Glucose S (g/L): 59.55
Lactic Acid P (g/L): 82.00
Maltose M (g/L): 63.62
Volume V (L): 8.22

t = 67.00 h

Biomass X (g/L): 70.40
Glucose S (g/L): 61.99
Lactic Acid P (g/L): 82.63
Maltose M (g/L): 57.20
Volume V (L): 9.49

t = 68.00 h

Biomass X (g/L): 77.56
Glucose S (g/L): 57.49
Lactic Acid P (g/L): 81.09
Maltose M (g/L): 55.25
Volume V (L): 9.09

t = 69.00 h

Biomass X (g/L): 86.12
Glucose S (g/L): 62.29
Lactic Acid P (g/L): 72.28
Maltose M (g/L): 67.18
Volume V (L): 8.83

t = 70.00 h

Biomass X (g/L): 87.21
Glucose S (g/L): 64.12
Lactic Acid P (g/L): 70.53
Maltose M (g/L): 57.23
Volume V (L): 9.21

t = 71.00 h

Biomass X (g/L): 94.66
Glucose S (g/L): 66.68
Lactic Acid P (g/L): 99.66
Maltose M (g/L): 63.98
Volume V (L): 8.90

t = 72.00 h

Biomass X (g/L): 92.19
Glucose S (g/L): 58.06
Lactic Acid P (g/L): 76.07
Maltose M (g/L): 62.54
Volume V (L): 8.31

t = 73.00 h

Biomass X (g/L): 84.56
Glucose S (g/L): 63.56
Lactic Acid P (g/L): 75.09
Maltose M (g/L): 60.15
Volume V (L): 10.36

t = 74.00 h

Biomass X (g/L): 86.36
Glucose S (g/L): 59.76
Lactic Acid P (g/L): 70.14
Maltose M (g/L): 70.84
Volume V (L): 9.44

t = 75.00 h

Biomass X (g/L): 86.43
Glucose S (g/L): 60.23
Lactic Acid P (g/L): 79.12
Maltose M (g/L): 62.03

Volume V (L): 9.22

Batch 69

Initial Conditions:

Batch: 69.00

X0 (g/L): 1.09

S0 (g/L): 157.40

M0 (g/L): 12.01

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.08

F_vol (L/h): 0.09

S_feed (g/L): 278.74

M_feed (g/L): 21.26

V_final (L): 12.06

Batch 69 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 157.43

Lactic Acid P (g/L): 1.34

Maltose M (g/L): 12.48

Volume V (L): 4.95

t = 1.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 159.55

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 11.36

Volume V (L): 5.00

t = 2.00 h

Biomass X (g/L): 18.22

Glucose S (g/L): 155.65

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 11.59

Volume V (L): 5.47

t = 3.00 h

Biomass X (g/L): 17.42

Glucose S (g/L): 162.63

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 11.68

Volume V (L): 4.25

t = 4.00 h

Biomass X (g/L): 11.01

Glucose S (g/L): 165.44

Lactic Acid P (g/L): 6.74

Maltose M (g/L): 12.07

Volume V (L): 4.75

t = 5.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 170.30

Lactic Acid P (g/L): 11.23

Maltose M (g/L): 11.79

Volume V (L): 5.02

t = 6.00 h

Biomass X (g/L): 21.18

Glucose S (g/L): 162.74

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 12.56

Volume V (L): 5.31

t = 7.00 h

Biomass X (g/L): 12.74

Glucose S (g/L): 174.53

Lactic Acid P (g/L): 2.63

Maltose M (g/L): 11.23

Volume V (L): 5.16

t = 8.00 h

Biomass X (g/L): 17.16

Glucose S (g/L): 167.40

Lactic Acid P (g/L): 5.82

Maltose M (g/L): 10.91

Volume V (L): 5.94

t = 9.00 h

Biomass X (g/L): 18.00

Glucose S (g/L): 158.35

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 12.21

Volume V (L): 5.89

t = 10.00 h

Biomass X (g/L): 2.20

Glucose S (g/L): 159.44

Lactic Acid P (g/L): 10.55

Maltose M (g/L): 10.77

Volume V (L): 6.73

t = 11.00 h

Biomass X (g/L): 8.98

Glucose S (g/L): 166.22
Lactic Acid P (g/L): 15.67
Maltose M (g/L): 11.61
Volume V (L): 7.43

t = 12.00 h

Biomass X (g/L): 19.81
Glucose S (g/L): 163.14
Lactic Acid P (g/L): 21.94
Maltose M (g/L): 10.83
Volume V (L): 5.42

t = 13.00 h

Biomass X (g/L): 35.40
Glucose S (g/L): 155.73
Lactic Acid P (g/L): 21.18
Maltose M (g/L): 10.30
Volume V (L): 6.31

t = 14.00 h

Biomass X (g/L): 19.91
Glucose S (g/L): 153.43
Lactic Acid P (g/L): 15.91
Maltose M (g/L): 10.96
Volume V (L): 6.40

t = 15.00 h

Biomass X (g/L): 36.04
Glucose S (g/L): 145.50
Lactic Acid P (g/L): 44.73
Maltose M (g/L): 11.44
Volume V (L): 6.40

t = 16.00 h

Biomass X (g/L): 40.20
Glucose S (g/L): 152.37
Lactic Acid P (g/L): 37.39
Maltose M (g/L): 10.47
Volume V (L): 5.89

t = 17.00 h

Biomass X (g/L): 53.67
Glucose S (g/L): 140.79
Lactic Acid P (g/L): 37.94
Maltose M (g/L): 11.17
Volume V (L): 6.08

t = 18.00 h

Biomass X (g/L): 56.22
Glucose S (g/L): 145.45
Lactic Acid P (g/L): 44.77
Maltose M (g/L): 11.19
Volume V (L): 6.83

t = 19.00 h

Biomass X (g/L): 43.13
Glucose S (g/L): 145.18
Lactic Acid P (g/L): 40.68
Maltose M (g/L): 10.25
Volume V (L): 6.25

t = 20.00 h

Biomass X (g/L): 70.36
Glucose S (g/L): 138.13
Lactic Acid P (g/L): 52.36
Maltose M (g/L): 10.63
Volume V (L): 6.97

t = 21.00 h

Biomass X (g/L): 47.63
Glucose S (g/L): 139.91
Lactic Acid P (g/L): 43.74
Maltose M (g/L): 11.35
Volume V (L): 6.84

t = 22.00 h

Biomass X (g/L): 72.33
Glucose S (g/L): 141.76
Lactic Acid P (g/L): 45.34
Maltose M (g/L): 10.88
Volume V (L): 7.15

t = 23.00 h

Biomass X (g/L): 58.57
Glucose S (g/L): 134.27
Lactic Acid P (g/L): 52.87
Maltose M (g/L): 10.43
Volume V (L): 6.63

t = 24.00 h

Biomass X (g/L): 67.68
Glucose S (g/L): 131.37
Lactic Acid P (g/L): 62.97
Maltose M (g/L): 10.65
Volume V (L): 7.35

t = 25.00 h

Biomass X (g/L): 70.18
Glucose S (g/L): 130.77
Lactic Acid P (g/L): 64.01
Maltose M (g/L): 10.12
Volume V (L): 6.85

t = 26.00 h

Biomass X (g/L): 71.61
Glucose S (g/L): 127.29
Lactic Acid P (g/L): 57.37
Maltose M (g/L): 10.83
Volume V (L): 7.54

t = 27.00 h

Biomass X (g/L): 77.81
Glucose S (g/L): 126.47
Lactic Acid P (g/L): 58.06
Maltose M (g/L): 10.76
Volume V (L): 8.30

t = 28.00 h

Biomass X (g/L): 76.40
Glucose S (g/L): 130.54
Lactic Acid P (g/L): 60.91
Maltose M (g/L): 10.28
Volume V (L): 7.78

t = 29.00 h

Biomass X (g/L): 81.48
Glucose S (g/L): 127.69
Lactic Acid P (g/L): 64.64
Maltose M (g/L): 10.26
Volume V (L): 7.91

t = 30.00 h

Biomass X (g/L): 79.45
Glucose S (g/L): 129.19
Lactic Acid P (g/L): 68.49
Maltose M (g/L): 9.67
Volume V (L): 8.28

t = 31.00 h

Biomass X (g/L): 63.48
Glucose S (g/L): 121.83
Lactic Acid P (g/L): 55.90
Maltose M (g/L): 9.46
Volume V (L): 7.12

t = 32.00 h

Biomass X (g/L): 79.41
Glucose S (g/L): 127.00
Lactic Acid P (g/L): 70.59
Maltose M (g/L): 9.81
Volume V (L): 8.98

t = 33.00 h

Biomass X (g/L): 72.03
Glucose S (g/L): 131.17
Lactic Acid P (g/L): 62.81
Maltose M (g/L): 9.88
Volume V (L): 7.51

t = 34.00 h

Biomass X (g/L): 67.38
Glucose S (g/L): 122.72
Lactic Acid P (g/L): 66.34
Maltose M (g/L): 10.31

Volume V (L): 8.70

t = 35.00 h

Biomass X (g/L): 76.05

Glucose S (g/L): 127.05

Lactic Acid P (g/L): 73.57

Maltose M (g/L): 10.19

Volume V (L): 8.71

t = 36.00 h

Biomass X (g/L): 81.25

Glucose S (g/L): 119.78

Lactic Acid P (g/L): 68.65

Maltose M (g/L): 9.81

Volume V (L): 7.88

t = 37.00 h

Biomass X (g/L): 82.40

Glucose S (g/L): 121.92

Lactic Acid P (g/L): 70.80

Maltose M (g/L): 10.19

Volume V (L): 9.05

t = 38.00 h

Biomass X (g/L): 90.75

Glucose S (g/L): 130.16

Lactic Acid P (g/L): 59.86

Maltose M (g/L): 9.21

Volume V (L): 8.86

t = 39.00 h

Biomass X (g/L): 93.60

Glucose S (g/L): 125.92

Lactic Acid P (g/L): 77.15

Maltose M (g/L): 10.02

Volume V (L): 9.74

t = 40.00 h

Biomass X (g/L): 84.42

Glucose S (g/L): 134.29

Lactic Acid P (g/L): 70.59

Maltose M (g/L): 9.16

Volume V (L): 8.87

t = 41.00 h

Biomass X (g/L): 86.50

Glucose S (g/L): 129.05

Lactic Acid P (g/L): 66.06

Maltose M (g/L): 9.97

Volume V (L): 8.80

t = 42.00 h

Biomass X (g/L): 82.63

Glucose S (g/L): 129.59

Lactic Acid P (g/L): 75.18

Maltose M (g/L): 9.38

Volume V (L): 8.66

t = 43.00 h

Biomass X (g/L): 89.44

Glucose S (g/L): 123.55

Lactic Acid P (g/L): 88.81

Maltose M (g/L): 8.47

Volume V (L): 8.06

t = 44.00 h

Biomass X (g/L): 86.89

Glucose S (g/L): 126.42

Lactic Acid P (g/L): 55.99

Maltose M (g/L): 9.58

Volume V (L): 8.96

t = 45.00 h

Biomass X (g/L): 81.75

Glucose S (g/L): 125.29

Lactic Acid P (g/L): 59.63

Maltose M (g/L): 9.56

Volume V (L): 9.08

t = 46.00 h

Biomass X (g/L): 103.52

Glucose S (g/L): 124.23

Lactic Acid P (g/L): 90.91

Maltose M (g/L): 8.96

Volume V (L): 9.48

t = 47.00 h

Biomass X (g/L): 77.18

Glucose S (g/L): 131.16

Lactic Acid P (g/L): 71.30

Maltose M (g/L): 8.94

Volume V (L): 8.62

t = 48.00 h

Biomass X (g/L): 96.89

Glucose S (g/L): 132.19

Lactic Acid P (g/L): 69.57

Maltose M (g/L): 8.98

Volume V (L): 9.38

t = 49.00 h

Biomass X (g/L): 69.41

Glucose S (g/L): 130.64

Lactic Acid P (g/L): 67.20

Maltose M (g/L): 8.16

Volume V (L): 9.94

t = 50.00 h

Biomass X (g/L): 90.72

Glucose S (g/L): 126.15

Lactic Acid P (g/L): 79.24

Maltose M (g/L): 7.92

Volume V (L): 10.09

t = 51.00 h

Biomass X (g/L): 96.86

Glucose S (g/L): 130.91

Lactic Acid P (g/L): 91.97

Maltose M (g/L): 9.44

Volume V (L): 8.75

t = 52.00 h

Biomass X (g/L): 64.09

Glucose S (g/L): 122.49
Lactic Acid P (g/L): 63.27
Maltose M (g/L): 8.47
Volume V (L): 10.29

t = 53.00 h

Biomass X (g/L): 84.16
Glucose S (g/L): 135.31
Lactic Acid P (g/L): 68.58
Maltose M (g/L): 9.57
Volume V (L): 9.85

t = 54.00 h

Biomass X (g/L): 89.80
Glucose S (g/L): 125.44
Lactic Acid P (g/L): 66.61
Maltose M (g/L): 9.31
Volume V (L): 9.98

t = 55.00 h

Biomass X (g/L): 85.23
Glucose S (g/L): 126.41
Lactic Acid P (g/L): 76.47
Maltose M (g/L): 9.34
Volume V (L): 8.84

t = 56.00 h

Biomass X (g/L): 86.43
Glucose S (g/L): 131.49
Lactic Acid P (g/L): 81.06
Maltose M (g/L): 8.34
Volume V (L): 11.11

t = 57.00 h

Biomass X (g/L): 84.01
Glucose S (g/L): 132.13
Lactic Acid P (g/L): 84.83
Maltose M (g/L): 7.88
Volume V (L): 10.46

t = 58.00 h

Biomass X (g/L): 71.89
Glucose S (g/L): 131.34
Lactic Acid P (g/L): 77.50
Maltose M (g/L): 8.05
Volume V (L): 9.59

t = 59.00 h

Biomass X (g/L): 105.77
Glucose S (g/L): 127.62
Lactic Acid P (g/L): 93.25
Maltose M (g/L): 8.27
Volume V (L): 10.76

t = 60.00 h

Biomass X (g/L): 77.54
Glucose S (g/L): 136.61
Lactic Acid P (g/L): 66.08
Maltose M (g/L): 7.40
Volume V (L): 10.84

t = 61.00 h

Biomass X (g/L): 86.65
Glucose S (g/L): 125.98
Lactic Acid P (g/L): 63.99
Maltose M (g/L): 8.16
Volume V (L): 11.43

t = 62.00 h

Biomass X (g/L): 75.96
Glucose S (g/L): 127.06
Lactic Acid P (g/L): 84.59
Maltose M (g/L): 8.23
Volume V (L): 10.66

t = 63.00 h

Biomass X (g/L): 59.89
Glucose S (g/L): 132.68
Lactic Acid P (g/L): 72.87
Maltose M (g/L): 7.71
Volume V (L): 10.93

t = 64.00 h

Biomass X (g/L): 73.04
Glucose S (g/L): 128.49
Lactic Acid P (g/L): 59.80
Maltose M (g/L): 7.77
Volume V (L): 10.66

t = 65.00 h

Biomass X (g/L): 67.80
Glucose S (g/L): 138.42
Lactic Acid P (g/L): 69.36
Maltose M (g/L): 8.08
Volume V (L): 11.45

t = 66.00 h

Biomass X (g/L): 92.58
Glucose S (g/L): 129.41
Lactic Acid P (g/L): 74.61
Maltose M (g/L): 8.28
Volume V (L): 10.77

t = 67.00 h

Biomass X (g/L): 75.81
Glucose S (g/L): 140.29
Lactic Acid P (g/L): 66.80
Maltose M (g/L): 7.69
Volume V (L): 10.80

t = 68.00 h

Biomass X (g/L): 76.60
Glucose S (g/L): 138.56
Lactic Acid P (g/L): 63.71
Maltose M (g/L): 8.15
Volume V (L): 11.36

t = 69.00 h

Biomass X (g/L): 71.07
Glucose S (g/L): 139.62
Lactic Acid P (g/L): 71.02
Maltose M (g/L): 8.04
Volume V (L): 10.80

t = 70.00 h

Biomass X (g/L): 82.30
Glucose S (g/L): 142.43
Lactic Acid P (g/L): 63.60
Maltose M (g/L): 7.96
Volume V (L): 12.49

t = 71.00 h

Biomass X (g/L): 74.21
Glucose S (g/L): 135.72
Lactic Acid P (g/L): 78.80
Maltose M (g/L): 7.61
Volume V (L): 12.77

t = 72.00 h

Biomass X (g/L): 90.83
Glucose S (g/L): 131.76
Lactic Acid P (g/L): 56.12
Maltose M (g/L): 7.27
Volume V (L): 11.52

t = 73.00 h

Biomass X (g/L): 83.92
Glucose S (g/L): 135.56
Lactic Acid P (g/L): 74.28
Maltose M (g/L): 8.53
Volume V (L): 11.09

t = 74.00 h

Biomass X (g/L): 75.21
Glucose S (g/L): 134.04
Lactic Acid P (g/L): 78.23
Maltose M (g/L): 8.03
Volume V (L): 12.15

t = 75.00 h

Biomass X (g/L): 89.00
Glucose S (g/L): 133.76
Lactic Acid P (g/L): 72.97
Maltose M (g/L): 8.14

Volume V (L): 11.78

Batch 70

Initial Conditions:

Batch: 70.00

X0 (g/L): 7.90

S0 (g/L): 76.18

M0 (g/L): 42.32

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.56

F_vol (L/h): 0.04

S_feed (g/L): 192.86

M_feed (g/L): 107.14

V_final (L): 7.89

Batch 70 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 19.58
Glucose S (g/L): 75.88
Lactic Acid P (g/L): 9.30
Maltose M (g/L): 42.27
Volume V (L): 5.03

t = 1.00 h

Biomass X (g/L): 2.81
Glucose S (g/L): 78.62
Lactic Acid P (g/L): 11.58
Maltose M (g/L): 40.31
Volume V (L): 5.02

t = 2.00 h

Biomass X (g/L): 19.43
Glucose S (g/L): 71.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 41.08
Volume V (L): 4.77

t = 3.00 h

Biomass X (g/L): 19.18
Glucose S (g/L): 69.07
Lactic Acid P (g/L): 12.59
Maltose M (g/L): 41.30
Volume V (L): 5.14

t = 4.00 h

Biomass X (g/L): 15.92
Glucose S (g/L): 70.85
Lactic Acid P (g/L): 19.26
Maltose M (g/L): 42.07
Volume V (L): 5.18

t = 5.00 h

Biomass X (g/L): 30.03
Glucose S (g/L): 62.86

Lactic Acid P (g/L): 9.99
Maltose M (g/L): 42.68
Volume V (L): 4.97

t = 6.00 h

Biomass X (g/L): 41.60
Glucose S (g/L): 71.42
Lactic Acid P (g/L): 15.53
Maltose M (g/L): 40.06
Volume V (L): 4.96

t = 7.00 h

Biomass X (g/L): 38.98
Glucose S (g/L): 59.65
Lactic Acid P (g/L): 27.43
Maltose M (g/L): 43.08
Volume V (L): 5.16

t = 8.00 h

Biomass X (g/L): 44.19
Glucose S (g/L): 53.29
Lactic Acid P (g/L): 11.57
Maltose M (g/L): 39.60
Volume V (L): 5.32

t = 9.00 h

Biomass X (g/L): 39.39
Glucose S (g/L): 53.61
Lactic Acid P (g/L): 19.30
Maltose M (g/L): 40.88
Volume V (L): 5.24

t = 10.00 h

Biomass X (g/L): 63.95
Glucose S (g/L): 44.06
Lactic Acid P (g/L): 23.69
Maltose M (g/L): 42.30
Volume V (L): 5.14

t = 11.00 h

Biomass X (g/L): 67.09

Glucose S (g/L): 47.93
Lactic Acid P (g/L): 19.12
Maltose M (g/L): 39.26
Volume V (L): 5.49

t = 12.00 h

Biomass X (g/L): 56.38
Glucose S (g/L): 45.19
Lactic Acid P (g/L): 49.82
Maltose M (g/L): 36.65
Volume V (L): 5.59

t = 13.00 h

Biomass X (g/L): 62.60
Glucose S (g/L): 44.65
Lactic Acid P (g/L): 22.91
Maltose M (g/L): 38.64
Volume V (L): 5.53

t = 14.00 h

Biomass X (g/L): 80.92
Glucose S (g/L): 34.46
Lactic Acid P (g/L): 47.83
Maltose M (g/L): 37.29
Volume V (L): 5.41

t = 15.00 h

Biomass X (g/L): 58.30
Glucose S (g/L): 28.96
Lactic Acid P (g/L): 54.59
Maltose M (g/L): 37.55
Volume V (L): 5.72

t = 16.00 h

Biomass X (g/L): 77.99
Glucose S (g/L): 30.48
Lactic Acid P (g/L): 46.00
Maltose M (g/L): 40.77
Volume V (L): 5.57

t = 17.00 h

Biomass X (g/L): 68.59
Glucose S (g/L): 30.84
Lactic Acid P (g/L): 44.27
Maltose M (g/L): 35.57
Volume V (L): 5.38

t = 18.00 h

Biomass X (g/L): 93.74
Glucose S (g/L): 39.26
Lactic Acid P (g/L): 54.67
Maltose M (g/L): 37.67
Volume V (L): 5.93

t = 19.00 h

Biomass X (g/L): 78.42
Glucose S (g/L): 40.96
Lactic Acid P (g/L): 54.02
Maltose M (g/L): 35.61
Volume V (L): 5.81

t = 20.00 h

Biomass X (g/L): 91.85
Glucose S (g/L): 31.28
Lactic Acid P (g/L): 65.13
Maltose M (g/L): 36.24
Volume V (L): 5.74

t = 21.00 h

Biomass X (g/L): 73.13
Glucose S (g/L): 29.09
Lactic Acid P (g/L): 31.49
Maltose M (g/L): 37.62
Volume V (L): 5.45

t = 22.00 h

Biomass X (g/L): 69.51
Glucose S (g/L): 26.90
Lactic Acid P (g/L): 49.21
Maltose M (g/L): 36.73
Volume V (L): 5.68

t = 23.00 h

Biomass X (g/L): 80.69
Glucose S (g/L): 32.25
Lactic Acid P (g/L): 64.62
Maltose M (g/L): 35.65
Volume V (L): 6.35

t = 24.00 h

Biomass X (g/L): 86.61
Glucose S (g/L): 38.61
Lactic Acid P (g/L): 60.78
Maltose M (g/L): 34.77
Volume V (L): 5.58

t = 25.00 h

Biomass X (g/L): 79.80
Glucose S (g/L): 36.49
Lactic Acid P (g/L): 68.54
Maltose M (g/L): 36.02
Volume V (L): 5.90

t = 26.00 h

Biomass X (g/L): 93.20
Glucose S (g/L): 33.25
Lactic Acid P (g/L): 64.77
Maltose M (g/L): 37.94
Volume V (L): 6.00

t = 27.00 h

Biomass X (g/L): 84.28
Glucose S (g/L): 33.15
Lactic Acid P (g/L): 61.83
Maltose M (g/L): 33.70
Volume V (L): 6.11

t = 28.00 h

Biomass X (g/L): 66.30
Glucose S (g/L): 31.76
Lactic Acid P (g/L): 61.05
Maltose M (g/L): 37.81
Volume V (L): 5.73

t = 29.00 h

Biomass X (g/L): 93.04
Glucose S (g/L): 27.58
Lactic Acid P (g/L): 87.14
Maltose M (g/L): 36.50
Volume V (L): 6.16

t = 30.00 h

Biomass X (g/L): 79.27
Glucose S (g/L): 24.36
Lactic Acid P (g/L): 80.27
Maltose M (g/L): 36.54
Volume V (L): 5.77

t = 31.00 h

Biomass X (g/L): 84.88
Glucose S (g/L): 27.99
Lactic Acid P (g/L): 70.96
Maltose M (g/L): 34.70
Volume V (L): 6.22

t = 32.00 h

Biomass X (g/L): 104.30
Glucose S (g/L): 30.34
Lactic Acid P (g/L): 67.61
Maltose M (g/L): 35.82
Volume V (L): 6.08

t = 33.00 h

Biomass X (g/L): 106.83
Glucose S (g/L): 21.48
Lactic Acid P (g/L): 72.66
Maltose M (g/L): 32.25
Volume V (L): 6.07

t = 34.00 h

Biomass X (g/L): 83.78
Glucose S (g/L): 28.05
Lactic Acid P (g/L): 76.04
Maltose M (g/L): 32.57

Volume V (L): 6.50

t = 35.00 h

Biomass X (g/L): 95.50

Glucose S (g/L): 25.08

Lactic Acid P (g/L): 72.68

Maltose M (g/L): 34.58

Volume V (L): 6.29

t = 36.00 h

Biomass X (g/L): 84.49

Glucose S (g/L): 34.31

Lactic Acid P (g/L): 61.95

Maltose M (g/L): 33.65

Volume V (L): 6.50

t = 37.00 h

Biomass X (g/L): 87.31

Glucose S (g/L): 27.06

Lactic Acid P (g/L): 86.90

Maltose M (g/L): 32.77

Volume V (L): 6.54

t = 38.00 h

Biomass X (g/L): 84.30

Glucose S (g/L): 31.67

Lactic Acid P (g/L): 72.58

Maltose M (g/L): 28.94

Volume V (L): 6.76

t = 39.00 h

Biomass X (g/L): 73.31

Glucose S (g/L): 27.06

Lactic Acid P (g/L): 90.09

Maltose M (g/L): 31.94

Volume V (L): 6.22

t = 40.00 h

Biomass X (g/L): 91.50

Glucose S (g/L): 38.58

Lactic Acid P (g/L): 72.94

Maltose M (g/L): 30.61

Volume V (L): 6.75

t = 41.00 h

Biomass X (g/L): 93.50

Glucose S (g/L): 22.99

Lactic Acid P (g/L): 69.21

Maltose M (g/L): 32.99

Volume V (L): 6.58

t = 42.00 h

Biomass X (g/L): 84.02

Glucose S (g/L): 33.96

Lactic Acid P (g/L): 85.33

Maltose M (g/L): 31.75

Volume V (L): 6.53

t = 43.00 h

Biomass X (g/L): 101.54

Glucose S (g/L): 25.88

Lactic Acid P (g/L): 75.30

Maltose M (g/L): 30.89

Volume V (L): 6.39

t = 44.00 h

Biomass X (g/L): 86.41

Glucose S (g/L): 30.04

Lactic Acid P (g/L): 74.07

Maltose M (g/L): 35.75

Volume V (L): 6.84

t = 45.00 h

Biomass X (g/L): 84.41

Glucose S (g/L): 33.95

Lactic Acid P (g/L): 61.59

Maltose M (g/L): 33.10

Volume V (L): 6.52

t = 46.00 h

Biomass X (g/L): 95.96

Glucose S (g/L): 34.63

Lactic Acid P (g/L): 79.88

Maltose M (g/L): 32.95

Volume V (L): 6.89

t = 47.00 h

Biomass X (g/L): 75.35

Glucose S (g/L): 34.96

Lactic Acid P (g/L): 69.11

Maltose M (g/L): 34.54

Volume V (L): 7.26

t = 48.00 h

Biomass X (g/L): 99.65

Glucose S (g/L): 35.05

Lactic Acid P (g/L): 69.88

Maltose M (g/L): 32.20

Volume V (L): 7.08

t = 49.00 h

Biomass X (g/L): 86.58

Glucose S (g/L): 31.67

Lactic Acid P (g/L): 69.80

Maltose M (g/L): 29.61

Volume V (L): 7.21

t = 50.00 h

Biomass X (g/L): 100.34

Glucose S (g/L): 34.46

Lactic Acid P (g/L): 79.06

Maltose M (g/L): 29.26

Volume V (L): 6.76

t = 51.00 h

Biomass X (g/L): 103.22

Glucose S (g/L): 35.89

Lactic Acid P (g/L): 60.34

Maltose M (g/L): 31.94

Volume V (L): 7.14

t = 52.00 h

Biomass X (g/L): 96.69

Glucose S (g/L): 37.16
Lactic Acid P (g/L): 73.40
Maltose M (g/L): 27.12
Volume V (L): 6.74

t = 53.00 h

Biomass X (g/L): 82.14
Glucose S (g/L): 36.38
Lactic Acid P (g/L): 85.15
Maltose M (g/L): 30.01
Volume V (L): 6.99

t = 54.00 h

Biomass X (g/L): 77.66
Glucose S (g/L): 40.72
Lactic Acid P (g/L): 63.98
Maltose M (g/L): 26.62
Volume V (L): 7.48

t = 55.00 h

Biomass X (g/L): 75.72
Glucose S (g/L): 38.13
Lactic Acid P (g/L): 77.92
Maltose M (g/L): 29.95
Volume V (L): 6.98

t = 56.00 h

Biomass X (g/L): 100.80
Glucose S (g/L): 39.14
Lactic Acid P (g/L): 70.42
Maltose M (g/L): 30.96
Volume V (L): 7.41

t = 57.00 h

Biomass X (g/L): 88.36
Glucose S (g/L): 36.80
Lactic Acid P (g/L): 72.87
Maltose M (g/L): 29.91
Volume V (L): 7.43

t = 58.00 h

Biomass X (g/L): 83.60
Glucose S (g/L): 34.05
Lactic Acid P (g/L): 92.95
Maltose M (g/L): 30.77
Volume V (L): 7.48

t = 59.00 h

Biomass X (g/L): 88.08
Glucose S (g/L): 40.39
Lactic Acid P (g/L): 61.34
Maltose M (g/L): 32.20
Volume V (L): 6.98

t = 60.00 h

Biomass X (g/L): 99.98
Glucose S (g/L): 45.99
Lactic Acid P (g/L): 84.84
Maltose M (g/L): 28.80
Volume V (L): 7.43

t = 61.00 h

Biomass X (g/L): 96.81
Glucose S (g/L): 42.79
Lactic Acid P (g/L): 73.08
Maltose M (g/L): 33.20
Volume V (L): 7.31

t = 62.00 h

Biomass X (g/L): 93.26
Glucose S (g/L): 37.17
Lactic Acid P (g/L): 72.82
Maltose M (g/L): 29.05
Volume V (L): 7.64

t = 63.00 h

Biomass X (g/L): 95.06
Glucose S (g/L): 45.57
Lactic Acid P (g/L): 67.41
Maltose M (g/L): 30.03
Volume V (L): 7.04

t = 64.00 h

Biomass X (g/L): 90.33
Glucose S (g/L): 40.49
Lactic Acid P (g/L): 85.59
Maltose M (g/L): 27.48
Volume V (L): 7.55

t = 65.00 h

Biomass X (g/L): 91.35
Glucose S (g/L): 46.08
Lactic Acid P (g/L): 76.68
Maltose M (g/L): 30.77
Volume V (L): 7.56

t = 66.00 h

Biomass X (g/L): 90.36
Glucose S (g/L): 41.32
Lactic Acid P (g/L): 75.66
Maltose M (g/L): 29.74
Volume V (L): 7.08

t = 67.00 h

Biomass X (g/L): 87.96
Glucose S (g/L): 45.31
Lactic Acid P (g/L): 79.97
Maltose M (g/L): 30.26
Volume V (L): 7.38

t = 68.00 h

Biomass X (g/L): 95.25
Glucose S (g/L): 41.57
Lactic Acid P (g/L): 77.45
Maltose M (g/L): 27.93
Volume V (L): 7.20

t = 69.00 h

Biomass X (g/L): 91.16
Glucose S (g/L): 42.19
Lactic Acid P (g/L): 60.00
Maltose M (g/L): 25.78
Volume V (L): 7.49

t = 70.00 h

Biomass X (g/L): 78.60
Glucose S (g/L): 37.47
Lactic Acid P (g/L): 74.72
Maltose M (g/L): 28.28
Volume V (L): 7.82

t = 71.00 h

Biomass X (g/L): 92.41
Glucose S (g/L): 48.03
Lactic Acid P (g/L): 88.25
Maltose M (g/L): 24.82
Volume V (L): 7.78

t = 72.00 h

Biomass X (g/L): 89.40
Glucose S (g/L): 47.83
Lactic Acid P (g/L): 65.28
Maltose M (g/L): 26.78
Volume V (L): 7.81

t = 73.00 h

Biomass X (g/L): 92.81
Glucose S (g/L): 48.58
Lactic Acid P (g/L): 69.46
Maltose M (g/L): 27.46
Volume V (L): 7.57

t = 74.00 h

Biomass X (g/L): 85.48
Glucose S (g/L): 48.33
Lactic Acid P (g/L): 76.01
Maltose M (g/L): 30.70
Volume V (L): 8.11

t = 75.00 h

Biomass X (g/L): 94.37
Glucose S (g/L): 43.78
Lactic Acid P (g/L): 73.29
Maltose M (g/L): 27.39

Volume V (L): 7.88

Batch 71

Initial Conditions:

Batch: 71.00

X0 (g/L): 7.84

S0 (g/L): 15.52

M0 (g/L): 119.30

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 7.69

F_vol (L/h): 0.07

S_feed (g/L): 34.54

M_feed (g/L): 265.46

V_final (L): 10.39

Batch 71 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 8.13
Glucose S (g/L): 20.60
Lactic Acid P (g/L): 8.17
Maltose M (g/L): 121.09
Volume V (L): 5.33

t = 1.00 h

Biomass X (g/L): 18.14
Glucose S (g/L): 15.80
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 110.86
Volume V (L): 5.07

t = 2.00 h

Biomass X (g/L): 3.35
Glucose S (g/L): 14.83
Lactic Acid P (g/L): 18.66
Maltose M (g/L): 115.09
Volume V (L): 5.78

t = 3.00 h

Biomass X (g/L): 16.87
Glucose S (g/L): 19.09
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 118.70
Volume V (L): 5.23

t = 4.00 h

Biomass X (g/L): 21.61
Glucose S (g/L): 15.01
Lactic Acid P (g/L): 4.94
Maltose M (g/L): 117.44
Volume V (L): 5.29

t = 5.00 h

Biomass X (g/L): 16.27
Glucose S (g/L): 10.70

Lactic Acid P (g/L): 10.53
Maltose M (g/L): 116.85
Volume V (L): 5.65

t = 6.00 h

Biomass X (g/L): 30.29
Glucose S (g/L): 13.04
Lactic Acid P (g/L): 29.39
Maltose M (g/L): 114.34
Volume V (L): 5.96

t = 7.00 h

Biomass X (g/L): 32.16
Glucose S (g/L): 9.38
Lactic Acid P (g/L): 23.18
Maltose M (g/L): 112.59
Volume V (L): 4.92

t = 8.00 h

Biomass X (g/L): 30.39
Glucose S (g/L): 1.65
Lactic Acid P (g/L): 13.74
Maltose M (g/L): 123.65
Volume V (L): 5.09

t = 9.00 h

Biomass X (g/L): 33.34
Glucose S (g/L): 6.68
Lactic Acid P (g/L): 12.02
Maltose M (g/L): 116.66
Volume V (L): 5.51

t = 10.00 h

Biomass X (g/L): 42.16
Glucose S (g/L): 6.69
Lactic Acid P (g/L): 31.35
Maltose M (g/L): 118.34
Volume V (L): 7.08

t = 11.00 h

Biomass X (g/L): 47.30

Glucose S (g/L): 4.02
Lactic Acid P (g/L): 20.15
Maltose M (g/L): 115.83
Volume V (L): 5.68

t = 12.00 h

Biomass X (g/L): 43.42
Glucose S (g/L): 8.39
Lactic Acid P (g/L): 28.73
Maltose M (g/L): 115.56
Volume V (L): 5.81

t = 13.00 h

Biomass X (g/L): 55.13
Glucose S (g/L): 3.95
Lactic Acid P (g/L): 16.79
Maltose M (g/L): 114.79
Volume V (L): 6.20

t = 14.00 h

Biomass X (g/L): 53.19
Glucose S (g/L): 7.25
Lactic Acid P (g/L): 35.47
Maltose M (g/L): 110.83
Volume V (L): 6.43

t = 15.00 h

Biomass X (g/L): 33.47
Glucose S (g/L): 2.04
Lactic Acid P (g/L): 36.43
Maltose M (g/L): 109.09
Volume V (L): 6.15

t = 16.00 h

Biomass X (g/L): 38.72
Glucose S (g/L): 8.13
Lactic Acid P (g/L): 22.91
Maltose M (g/L): 115.01
Volume V (L): 6.65

t = 17.00 h

Biomass X (g/L): 63.39
Glucose S (g/L): 1.71
Lactic Acid P (g/L): 44.22
Maltose M (g/L): 113.91
Volume V (L): 5.71

t = 18.00 h

Biomass X (g/L): 52.34
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 41.49
Maltose M (g/L): 115.97
Volume V (L): 5.96

t = 19.00 h

Biomass X (g/L): 57.53
Glucose S (g/L): 9.08
Lactic Acid P (g/L): 44.47
Maltose M (g/L): 114.84
Volume V (L): 6.87

t = 20.00 h

Biomass X (g/L): 67.64
Glucose S (g/L): 3.08
Lactic Acid P (g/L): 36.03
Maltose M (g/L): 109.22
Volume V (L): 6.44

t = 21.00 h

Biomass X (g/L): 58.88
Glucose S (g/L): 10.81
Lactic Acid P (g/L): 42.16
Maltose M (g/L): 115.24
Volume V (L): 5.96

t = 22.00 h

Biomass X (g/L): 60.60
Glucose S (g/L): 7.50
Lactic Acid P (g/L): 42.39
Maltose M (g/L): 113.55
Volume V (L): 6.01

t = 23.00 h

Biomass X (g/L): 53.85
Glucose S (g/L): 8.34
Lactic Acid P (g/L): 41.74
Maltose M (g/L): 112.83
Volume V (L): 6.30

t = 24.00 h

Biomass X (g/L): 66.06
Glucose S (g/L): 9.29
Lactic Acid P (g/L): 45.67
Maltose M (g/L): 117.56
Volume V (L): 6.72

t = 25.00 h

Biomass X (g/L): 69.15
Glucose S (g/L): 6.81
Lactic Acid P (g/L): 56.27
Maltose M (g/L): 116.40
Volume V (L): 7.00

t = 26.00 h

Biomass X (g/L): 59.95
Glucose S (g/L): 7.09
Lactic Acid P (g/L): 55.78
Maltose M (g/L): 111.38
Volume V (L): 6.73

t = 27.00 h

Biomass X (g/L): 57.97
Glucose S (g/L): 10.73
Lactic Acid P (g/L): 37.15
Maltose M (g/L): 110.48
Volume V (L): 7.10

t = 28.00 h

Biomass X (g/L): 90.46
Glucose S (g/L): 7.61
Lactic Acid P (g/L): 57.08
Maltose M (g/L): 111.72
Volume V (L): 6.82

t = 29.00 h

Biomass X (g/L): 67.33
Glucose S (g/L): 5.48
Lactic Acid P (g/L): 59.87
Maltose M (g/L): 105.66
Volume V (L): 7.61

t = 30.00 h

Biomass X (g/L): 63.03
Glucose S (g/L): 9.63
Lactic Acid P (g/L): 49.14
Maltose M (g/L): 105.22
Volume V (L): 8.25

t = 31.00 h

Biomass X (g/L): 76.26
Glucose S (g/L): 2.12
Lactic Acid P (g/L): 55.83
Maltose M (g/L): 108.88
Volume V (L): 7.06

t = 32.00 h

Biomass X (g/L): 70.39
Glucose S (g/L): 12.48
Lactic Acid P (g/L): 56.61
Maltose M (g/L): 103.27
Volume V (L): 7.50

t = 33.00 h

Biomass X (g/L): 62.02
Glucose S (g/L): 12.92
Lactic Acid P (g/L): 45.77
Maltose M (g/L): 107.62
Volume V (L): 7.54

t = 34.00 h

Biomass X (g/L): 76.71
Glucose S (g/L): 12.91
Lactic Acid P (g/L): 50.71
Maltose M (g/L): 105.67

Volume V (L): 7.50

t = 35.00 h

Biomass X (g/L): 65.59

Glucose S (g/L): 13.21

Lactic Acid P (g/L): 61.42

Maltose M (g/L): 101.79

Volume V (L): 7.79

t = 36.00 h

Biomass X (g/L): 87.82

Glucose S (g/L): 10.03

Lactic Acid P (g/L): 51.34

Maltose M (g/L): 104.24

Volume V (L): 7.79

t = 37.00 h

Biomass X (g/L): 78.88

Glucose S (g/L): 10.41

Lactic Acid P (g/L): 64.95

Maltose M (g/L): 108.02

Volume V (L): 7.45

t = 38.00 h

Biomass X (g/L): 72.66

Glucose S (g/L): 7.61

Lactic Acid P (g/L): 68.57

Maltose M (g/L): 103.44

Volume V (L): 8.30

t = 39.00 h

Biomass X (g/L): 76.67

Glucose S (g/L): 16.76

Lactic Acid P (g/L): 67.89

Maltose M (g/L): 102.95

Volume V (L): 7.89

t = 40.00 h

Biomass X (g/L): 84.25

Glucose S (g/L): 10.76

Lactic Acid P (g/L): 75.88

Maltose M (g/L): 103.27
Volume V (L): 7.71

t = 41.00 h

Biomass X (g/L): 78.41
Glucose S (g/L): 13.67
Lactic Acid P (g/L): 58.50
Maltose M (g/L): 103.21
Volume V (L): 9.00

t = 42.00 h

Biomass X (g/L): 96.06
Glucose S (g/L): 13.95
Lactic Acid P (g/L): 61.49
Maltose M (g/L): 100.07
Volume V (L): 7.97

t = 43.00 h

Biomass X (g/L): 87.76
Glucose S (g/L): 9.85
Lactic Acid P (g/L): 63.37
Maltose M (g/L): 101.86
Volume V (L): 8.20

t = 44.00 h

Biomass X (g/L): 82.43
Glucose S (g/L): 9.23
Lactic Acid P (g/L): 60.65
Maltose M (g/L): 96.68
Volume V (L): 8.23

t = 45.00 h

Biomass X (g/L): 70.19
Glucose S (g/L): 14.89
Lactic Acid P (g/L): 66.50
Maltose M (g/L): 97.03
Volume V (L): 8.18

t = 46.00 h

Biomass X (g/L): 77.56
Glucose S (g/L): 14.13

Lactic Acid P (g/L): 47.41
Maltose M (g/L): 96.97
Volume V (L): 8.50

t = 47.00 h

Biomass X (g/L): 83.44
Glucose S (g/L): 15.77
Lactic Acid P (g/L): 64.67
Maltose M (g/L): 100.29
Volume V (L): 8.01

t = 48.00 h

Biomass X (g/L): 72.20
Glucose S (g/L): 19.73
Lactic Acid P (g/L): 51.40
Maltose M (g/L): 98.77
Volume V (L): 8.70

t = 49.00 h

Biomass X (g/L): 91.26
Glucose S (g/L): 17.89
Lactic Acid P (g/L): 55.53
Maltose M (g/L): 97.30
Volume V (L): 7.76

t = 50.00 h

Biomass X (g/L): 71.41
Glucose S (g/L): 20.06
Lactic Acid P (g/L): 70.81
Maltose M (g/L): 99.57
Volume V (L): 8.03

t = 51.00 h

Biomass X (g/L): 74.63
Glucose S (g/L): 19.77
Lactic Acid P (g/L): 63.16
Maltose M (g/L): 98.05
Volume V (L): 9.45

t = 52.00 h

Biomass X (g/L): 75.51

Glucose S (g/L): 18.19
Lactic Acid P (g/L): 71.44
Maltose M (g/L): 95.33
Volume V (L): 8.19

t = 53.00 h

Biomass X (g/L): 87.29
Glucose S (g/L): 16.83
Lactic Acid P (g/L): 63.87
Maltose M (g/L): 92.36
Volume V (L): 7.96

t = 54.00 h

Biomass X (g/L): 73.73
Glucose S (g/L): 21.16
Lactic Acid P (g/L): 75.87
Maltose M (g/L): 92.60
Volume V (L): 8.35

t = 55.00 h

Biomass X (g/L): 65.06
Glucose S (g/L): 21.29
Lactic Acid P (g/L): 64.56
Maltose M (g/L): 89.96
Volume V (L): 9.70

t = 56.00 h

Biomass X (g/L): 90.81
Glucose S (g/L): 24.36
Lactic Acid P (g/L): 49.99
Maltose M (g/L): 94.87
Volume V (L): 8.25

t = 57.00 h

Biomass X (g/L): 77.35
Glucose S (g/L): 21.92
Lactic Acid P (g/L): 69.40
Maltose M (g/L): 91.04
Volume V (L): 8.68

t = 58.00 h

Biomass X (g/L): 71.92
Glucose S (g/L): 22.16
Lactic Acid P (g/L): 48.89
Maltose M (g/L): 91.80
Volume V (L): 8.54

t = 59.00 h

Biomass X (g/L): 74.79
Glucose S (g/L): 23.02
Lactic Acid P (g/L): 55.50
Maltose M (g/L): 92.56
Volume V (L): 9.12

t = 60.00 h

Biomass X (g/L): 96.78
Glucose S (g/L): 21.69
Lactic Acid P (g/L): 60.82
Maltose M (g/L): 96.02
Volume V (L): 9.12

t = 61.00 h

Biomass X (g/L): 75.79
Glucose S (g/L): 29.24
Lactic Acid P (g/L): 60.07
Maltose M (g/L): 95.17
Volume V (L): 9.17

t = 62.00 h

Biomass X (g/L): 80.57
Glucose S (g/L): 29.02
Lactic Acid P (g/L): 64.44
Maltose M (g/L): 91.42
Volume V (L): 10.25

t = 63.00 h

Biomass X (g/L): 77.74
Glucose S (g/L): 22.86
Lactic Acid P (g/L): 60.42
Maltose M (g/L): 95.38
Volume V (L): 9.01

t = 64.00 h

Biomass X (g/L): 70.79
Glucose S (g/L): 29.40
Lactic Acid P (g/L): 57.56
Maltose M (g/L): 83.96
Volume V (L): 9.55

t = 65.00 h

Biomass X (g/L): 75.15
Glucose S (g/L): 22.38
Lactic Acid P (g/L): 83.56
Maltose M (g/L): 96.51
Volume V (L): 9.81

t = 66.00 h

Biomass X (g/L): 70.20
Glucose S (g/L): 31.29
Lactic Acid P (g/L): 64.49
Maltose M (g/L): 92.10
Volume V (L): 9.32

t = 67.00 h

Biomass X (g/L): 90.20
Glucose S (g/L): 27.08
Lactic Acid P (g/L): 51.23
Maltose M (g/L): 89.68
Volume V (L): 10.02

t = 68.00 h

Biomass X (g/L): 82.60
Glucose S (g/L): 30.83
Lactic Acid P (g/L): 67.81
Maltose M (g/L): 85.91
Volume V (L): 9.71

t = 69.00 h

Biomass X (g/L): 103.69
Glucose S (g/L): 24.85
Lactic Acid P (g/L): 61.42
Maltose M (g/L): 86.98
Volume V (L): 10.12

t = 70.00 h

Biomass X (g/L): 75.35
Glucose S (g/L): 27.66
Lactic Acid P (g/L): 63.77
Maltose M (g/L): 90.05
Volume V (L): 9.98

t = 71.00 h

Biomass X (g/L): 73.98
Glucose S (g/L): 33.18
Lactic Acid P (g/L): 77.93
Maltose M (g/L): 89.69
Volume V (L): 9.21

t = 72.00 h

Biomass X (g/L): 95.14
Glucose S (g/L): 30.46
Lactic Acid P (g/L): 68.07
Maltose M (g/L): 89.37
Volume V (L): 10.73

t = 73.00 h

Biomass X (g/L): 90.86
Glucose S (g/L): 30.73
Lactic Acid P (g/L): 62.16
Maltose M (g/L): 84.27
Volume V (L): 11.13

t = 74.00 h

Biomass X (g/L): 84.82
Glucose S (g/L): 37.91
Lactic Acid P (g/L): 76.34
Maltose M (g/L): 83.16
Volume V (L): 10.24

t = 75.00 h

Biomass X (g/L): 71.46
Glucose S (g/L): 40.95
Lactic Acid P (g/L): 82.70
Maltose M (g/L): 86.50

Volume V (L): 10.00

Batch 72

Initial Conditions:

Batch: 72.00

X0 (g/L): 7.34

S0 (g/L): 103.34

M0 (g/L): 29.24

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.28

F_vol (L/h): 0.06

S_feed (g/L): 233.83

M_feed (g/L): 66.17

V_final (L): 9.36

Batch 72 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 22.07
Glucose S (g/L): 98.77
Lactic Acid P (g/L): 1.43
Maltose M (g/L): 30.53
Volume V (L): 5.08

t = 1.00 h

Biomass X (g/L): 28.40
Glucose S (g/L): 104.78
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 29.08
Volume V (L): 4.77

t = 2.00 h

Biomass X (g/L): 11.46
Glucose S (g/L): 106.64
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 29.32
Volume V (L): 5.49

t = 3.00 h

Biomass X (g/L): 14.81
Glucose S (g/L): 98.04
Lactic Acid P (g/L): 14.06
Maltose M (g/L): 27.97
Volume V (L): 4.98

t = 4.00 h

Biomass X (g/L): 20.59
Glucose S (g/L): 94.49
Lactic Acid P (g/L): 16.48
Maltose M (g/L): 30.26
Volume V (L): 4.63

t = 5.00 h

Biomass X (g/L): 9.86
Glucose S (g/L): 92.21

Lactic Acid P (g/L): 5.12
Maltose M (g/L): 28.38
Volume V (L): 5.78

t = 6.00 h

Biomass X (g/L): 28.53
Glucose S (g/L): 85.86
Lactic Acid P (g/L): 3.09
Maltose M (g/L): 28.47
Volume V (L): 5.41

t = 7.00 h

Biomass X (g/L): 53.37
Glucose S (g/L): 81.56
Lactic Acid P (g/L): 22.85
Maltose M (g/L): 27.88
Volume V (L): 5.42

t = 8.00 h

Biomass X (g/L): 40.05
Glucose S (g/L): 76.55
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 29.24
Volume V (L): 5.85

t = 9.00 h

Biomass X (g/L): 36.40
Glucose S (g/L): 74.57
Lactic Acid P (g/L): 18.49
Maltose M (g/L): 28.16
Volume V (L): 5.83

t = 10.00 h

Biomass X (g/L): 56.00
Glucose S (g/L): 84.14
Lactic Acid P (g/L): 49.07
Maltose M (g/L): 28.55
Volume V (L): 5.32

t = 11.00 h

Biomass X (g/L): 48.81

Glucose S (g/L): 71.44
Lactic Acid P (g/L): 49.38
Maltose M (g/L): 28.02
Volume V (L): 5.48

t = 12.00 h

Biomass X (g/L): 58.50
Glucose S (g/L): 59.36
Lactic Acid P (g/L): 43.45
Maltose M (g/L): 27.34
Volume V (L): 5.31

t = 13.00 h

Biomass X (g/L): 50.17
Glucose S (g/L): 67.26
Lactic Acid P (g/L): 44.51
Maltose M (g/L): 28.28
Volume V (L): 5.24

t = 14.00 h

Biomass X (g/L): 55.53
Glucose S (g/L): 61.13
Lactic Acid P (g/L): 25.99
Maltose M (g/L): 27.59
Volume V (L): 6.02

t = 15.00 h

Biomass X (g/L): 75.65
Glucose S (g/L): 66.60
Lactic Acid P (g/L): 39.53
Maltose M (g/L): 26.23
Volume V (L): 6.01

t = 16.00 h

Biomass X (g/L): 72.02
Glucose S (g/L): 64.81
Lactic Acid P (g/L): 45.76
Maltose M (g/L): 26.06
Volume V (L): 6.19

t = 17.00 h

Biomass X (g/L): 65.49
Glucose S (g/L): 64.16
Lactic Acid P (g/L): 43.71
Maltose M (g/L): 26.99
Volume V (L): 6.43

t = 18.00 h

Biomass X (g/L): 76.20
Glucose S (g/L): 62.84
Lactic Acid P (g/L): 48.88
Maltose M (g/L): 27.56
Volume V (L): 6.32

t = 19.00 h

Biomass X (g/L): 70.82
Glucose S (g/L): 60.70
Lactic Acid P (g/L): 65.83
Maltose M (g/L): 28.10
Volume V (L): 6.40

t = 20.00 h

Biomass X (g/L): 70.63
Glucose S (g/L): 55.50
Lactic Acid P (g/L): 50.13
Maltose M (g/L): 24.76
Volume V (L): 5.44

t = 21.00 h

Biomass X (g/L): 91.06
Glucose S (g/L): 60.97
Lactic Acid P (g/L): 53.33
Maltose M (g/L): 26.31
Volume V (L): 6.11

t = 22.00 h

Biomass X (g/L): 88.54
Glucose S (g/L): 65.33
Lactic Acid P (g/L): 64.63
Maltose M (g/L): 26.61
Volume V (L): 6.61

t = 23.00 h

Biomass X (g/L): 81.76
Glucose S (g/L): 68.52
Lactic Acid P (g/L): 45.66
Maltose M (g/L): 26.29
Volume V (L): 5.93

t = 24.00 h

Biomass X (g/L): 84.34
Glucose S (g/L): 59.64
Lactic Acid P (g/L): 52.10
Maltose M (g/L): 26.82
Volume V (L): 6.43

t = 25.00 h

Biomass X (g/L): 83.69
Glucose S (g/L): 60.08
Lactic Acid P (g/L): 54.03
Maltose M (g/L): 26.36
Volume V (L): 6.63

t = 26.00 h

Biomass X (g/L): 89.29
Glucose S (g/L): 62.81
Lactic Acid P (g/L): 62.33
Maltose M (g/L): 24.68
Volume V (L): 6.67

t = 27.00 h

Biomass X (g/L): 77.37
Glucose S (g/L): 67.16
Lactic Acid P (g/L): 59.45
Maltose M (g/L): 24.52
Volume V (L): 6.70

t = 28.00 h

Biomass X (g/L): 94.22
Glucose S (g/L): 58.86
Lactic Acid P (g/L): 60.53
Maltose M (g/L): 25.63
Volume V (L): 6.84

t = 29.00 h

Biomass X (g/L): 89.76
Glucose S (g/L): 64.11
Lactic Acid P (g/L): 58.00
Maltose M (g/L): 26.13
Volume V (L): 6.75

t = 30.00 h

Biomass X (g/L): 84.18
Glucose S (g/L): 57.89
Lactic Acid P (g/L): 54.91
Maltose M (g/L): 25.20
Volume V (L): 6.23

t = 31.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 63.11
Lactic Acid P (g/L): 71.21
Maltose M (g/L): 23.46
Volume V (L): 6.98

t = 32.00 h

Biomass X (g/L): 95.71
Glucose S (g/L): 66.18
Lactic Acid P (g/L): 63.01
Maltose M (g/L): 23.82
Volume V (L): 6.96

t = 33.00 h

Biomass X (g/L): 86.31
Glucose S (g/L): 56.87
Lactic Acid P (g/L): 72.50
Maltose M (g/L): 25.32
Volume V (L): 6.41

t = 34.00 h

Biomass X (g/L): 81.19
Glucose S (g/L): 60.53
Lactic Acid P (g/L): 56.93
Maltose M (g/L): 24.22

Volume V (L): 7.46

t = 35.00 h

Biomass X (g/L): 84.48

Glucose S (g/L): 67.26

Lactic Acid P (g/L): 61.60

Maltose M (g/L): 24.52

Volume V (L): 7.22

t = 36.00 h

Biomass X (g/L): 95.94

Glucose S (g/L): 66.24

Lactic Acid P (g/L): 66.23

Maltose M (g/L): 25.28

Volume V (L): 6.61

t = 37.00 h

Biomass X (g/L): 72.80

Glucose S (g/L): 68.48

Lactic Acid P (g/L): 67.56

Maltose M (g/L): 23.03

Volume V (L): 6.86

t = 38.00 h

Biomass X (g/L): 85.19

Glucose S (g/L): 68.14

Lactic Acid P (g/L): 76.22

Maltose M (g/L): 23.64

Volume V (L): 7.01

t = 39.00 h

Biomass X (g/L): 99.23

Glucose S (g/L): 63.54

Lactic Acid P (g/L): 65.61

Maltose M (g/L): 23.41

Volume V (L): 7.07

t = 40.00 h

Biomass X (g/L): 87.17

Glucose S (g/L): 74.87

Lactic Acid P (g/L): 76.97

Maltose M (g/L): 24.11

Volume V (L): 7.43

t = 41.00 h

Biomass X (g/L): 91.52

Glucose S (g/L): 66.56

Lactic Acid P (g/L): 73.89

Maltose M (g/L): 24.37

Volume V (L): 7.04

t = 42.00 h

Biomass X (g/L): 99.50

Glucose S (g/L): 53.43

Lactic Acid P (g/L): 69.77

Maltose M (g/L): 22.14

Volume V (L): 7.83

t = 43.00 h

Biomass X (g/L): 97.59

Glucose S (g/L): 65.79

Lactic Acid P (g/L): 62.25

Maltose M (g/L): 23.40

Volume V (L): 7.86

t = 44.00 h

Biomass X (g/L): 84.53

Glucose S (g/L): 65.52

Lactic Acid P (g/L): 78.50

Maltose M (g/L): 24.09

Volume V (L): 7.77

t = 45.00 h

Biomass X (g/L): 87.52

Glucose S (g/L): 71.44

Lactic Acid P (g/L): 63.14

Maltose M (g/L): 21.42

Volume V (L): 7.63

t = 46.00 h

Biomass X (g/L): 97.59

Glucose S (g/L): 66.65

Lactic Acid P (g/L): 71.88

Maltose M (g/L): 22.98

Volume V (L): 8.00

t = 47.00 h

Biomass X (g/L): 97.86

Glucose S (g/L): 73.81

Lactic Acid P (g/L): 71.87

Maltose M (g/L): 21.77

Volume V (L): 7.63

t = 48.00 h

Biomass X (g/L): 87.73

Glucose S (g/L): 72.03

Lactic Acid P (g/L): 70.73

Maltose M (g/L): 22.19

Volume V (L): 7.54

t = 49.00 h

Biomass X (g/L): 87.97

Glucose S (g/L): 65.92

Lactic Acid P (g/L): 69.30

Maltose M (g/L): 20.33

Volume V (L): 8.26

t = 50.00 h

Biomass X (g/L): 91.01

Glucose S (g/L): 70.13

Lactic Acid P (g/L): 65.53

Maltose M (g/L): 23.86

Volume V (L): 8.02

t = 51.00 h

Biomass X (g/L): 101.65

Glucose S (g/L): 67.96

Lactic Acid P (g/L): 58.38

Maltose M (g/L): 22.11

Volume V (L): 7.72

t = 52.00 h

Biomass X (g/L): 98.60

Glucose S (g/L): 72.60
Lactic Acid P (g/L): 86.76
Maltose M (g/L): 21.57
Volume V (L): 7.88

t = 53.00 h

Biomass X (g/L): 94.80
Glucose S (g/L): 69.02
Lactic Acid P (g/L): 69.34
Maltose M (g/L): 22.52
Volume V (L): 7.82

t = 54.00 h

Biomass X (g/L): 77.33
Glucose S (g/L): 73.22
Lactic Acid P (g/L): 72.95
Maltose M (g/L): 21.12
Volume V (L): 8.84

t = 55.00 h

Biomass X (g/L): 85.04
Glucose S (g/L): 72.28
Lactic Acid P (g/L): 73.12
Maltose M (g/L): 20.91
Volume V (L): 8.20

t = 56.00 h

Biomass X (g/L): 85.40
Glucose S (g/L): 73.33
Lactic Acid P (g/L): 84.78
Maltose M (g/L): 23.28
Volume V (L): 8.31

t = 57.00 h

Biomass X (g/L): 101.29
Glucose S (g/L): 83.28
Lactic Acid P (g/L): 92.79
Maltose M (g/L): 23.45
Volume V (L): 8.56

t = 58.00 h

Biomass X (g/L): 84.91
Glucose S (g/L): 73.52
Lactic Acid P (g/L): 76.47
Maltose M (g/L): 21.09
Volume V (L): 7.95

t = 59.00 h

Biomass X (g/L): 99.34
Glucose S (g/L): 67.33
Lactic Acid P (g/L): 73.71
Maltose M (g/L): 22.53
Volume V (L): 8.81

t = 60.00 h

Biomass X (g/L): 81.31
Glucose S (g/L): 80.46
Lactic Acid P (g/L): 64.90
Maltose M (g/L): 21.75
Volume V (L): 8.27

t = 61.00 h

Biomass X (g/L): 81.91
Glucose S (g/L): 80.96
Lactic Acid P (g/L): 62.10
Maltose M (g/L): 19.98
Volume V (L): 9.18

t = 62.00 h

Biomass X (g/L): 91.24
Glucose S (g/L): 68.73
Lactic Acid P (g/L): 59.20
Maltose M (g/L): 20.33
Volume V (L): 8.49

t = 63.00 h

Biomass X (g/L): 101.68
Glucose S (g/L): 74.71
Lactic Acid P (g/L): 68.48
Maltose M (g/L): 22.55
Volume V (L): 9.02

t = 64.00 h

Biomass X (g/L): 84.07
Glucose S (g/L): 80.09
Lactic Acid P (g/L): 56.58
Maltose M (g/L): 19.65
Volume V (L): 7.85

t = 65.00 h

Biomass X (g/L): 60.87
Glucose S (g/L): 80.99
Lactic Acid P (g/L): 86.27
Maltose M (g/L): 23.02
Volume V (L): 8.37

t = 66.00 h

Biomass X (g/L): 70.44
Glucose S (g/L): 79.24
Lactic Acid P (g/L): 80.32
Maltose M (g/L): 20.24
Volume V (L): 8.73

t = 67.00 h

Biomass X (g/L): 88.46
Glucose S (g/L): 79.85
Lactic Acid P (g/L): 74.19
Maltose M (g/L): 22.15
Volume V (L): 8.18

t = 68.00 h

Biomass X (g/L): 70.91
Glucose S (g/L): 76.28
Lactic Acid P (g/L): 87.01
Maltose M (g/L): 20.28
Volume V (L): 9.16

t = 69.00 h

Biomass X (g/L): 76.66
Glucose S (g/L): 70.54
Lactic Acid P (g/L): 66.85
Maltose M (g/L): 21.21
Volume V (L): 9.12

t = 70.00 h

Biomass X (g/L): 85.04
Glucose S (g/L): 90.05
Lactic Acid P (g/L): 69.33
Maltose M (g/L): 21.67
Volume V (L): 8.72

t = 71.00 h

Biomass X (g/L): 60.75
Glucose S (g/L): 77.66
Lactic Acid P (g/L): 59.09
Maltose M (g/L): 21.56
Volume V (L): 9.12

t = 72.00 h

Biomass X (g/L): 80.78
Glucose S (g/L): 83.04
Lactic Acid P (g/L): 84.45
Maltose M (g/L): 19.61
Volume V (L): 9.46

t = 73.00 h

Biomass X (g/L): 72.88
Glucose S (g/L): 82.10
Lactic Acid P (g/L): 69.82
Maltose M (g/L): 19.25
Volume V (L): 8.51

t = 74.00 h

Biomass X (g/L): 69.40
Glucose S (g/L): 87.10
Lactic Acid P (g/L): 77.65
Maltose M (g/L): 19.88
Volume V (L): 8.86

t = 75.00 h

Biomass X (g/L): 76.00
Glucose S (g/L): 84.24
Lactic Acid P (g/L): 86.97
Maltose M (g/L): 18.64

Volume V (L): 9.24

Batch 73

Initial Conditions:

Batch: 73.00

X0 (g/L): 6.44

S0 (g/L): 31.74

M0 (g/L): 77.15

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.43

F_vol (L/h): 0.03

S_feed (g/L): 87.45

M_feed (g/L): 212.55

V_final (L): 7.58

Batch 73 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.66
Glucose S (g/L): 31.66
Lactic Acid P (g/L): 1.95
Maltose M (g/L): 77.79
Volume V (L): 4.84

t = 1.00 h

Biomass X (g/L): 7.91
Glucose S (g/L): 28.81
Lactic Acid P (g/L): 19.86
Maltose M (g/L): 71.67
Volume V (L): 5.35

t = 2.00 h

Biomass X (g/L): 19.08
Glucose S (g/L): 29.48
Lactic Acid P (g/L): 4.90
Maltose M (g/L): 73.94
Volume V (L): 4.84

t = 3.00 h

Biomass X (g/L): 18.26
Glucose S (g/L): 30.42
Lactic Acid P (g/L): 14.47
Maltose M (g/L): 76.50
Volume V (L): 4.62

t = 4.00 h

Biomass X (g/L): 23.70
Glucose S (g/L): 26.29
Lactic Acid P (g/L): 5.06
Maltose M (g/L): 78.05
Volume V (L): 4.87

t = 5.00 h

Biomass X (g/L): 8.37
Glucose S (g/L): 27.80

Lactic Acid P (g/L): 2.56
Maltose M (g/L): 72.92
Volume V (L): 5.25

t = 6.00 h

Biomass X (g/L): 26.53
Glucose S (g/L): 25.50
Lactic Acid P (g/L): 20.33
Maltose M (g/L): 73.78
Volume V (L): 5.18

t = 7.00 h

Biomass X (g/L): 26.59
Glucose S (g/L): 20.97
Lactic Acid P (g/L): 16.13
Maltose M (g/L): 69.01
Volume V (L): 5.40

t = 8.00 h

Biomass X (g/L): 47.02
Glucose S (g/L): 13.31
Lactic Acid P (g/L): 27.62
Maltose M (g/L): 76.84
Volume V (L): 5.38

t = 9.00 h

Biomass X (g/L): 39.25
Glucose S (g/L): 20.55
Lactic Acid P (g/L): 26.32
Maltose M (g/L): 75.40
Volume V (L): 5.18

t = 10.00 h

Biomass X (g/L): 35.31
Glucose S (g/L): 12.91
Lactic Acid P (g/L): 18.81
Maltose M (g/L): 73.64
Volume V (L): 5.24

t = 11.00 h

Biomass X (g/L): 52.75

Glucose S (g/L): 18.90
Lactic Acid P (g/L): 38.43
Maltose M (g/L): 72.49
Volume V (L): 5.30

t = 12.00 h

Biomass X (g/L): 53.22
Glucose S (g/L): 11.48
Lactic Acid P (g/L): 23.23
Maltose M (g/L): 75.13
Volume V (L): 5.80

t = 13.00 h

Biomass X (g/L): 43.29
Glucose S (g/L): 7.65
Lactic Acid P (g/L): 39.01
Maltose M (g/L): 68.23
Volume V (L): 5.59

t = 14.00 h

Biomass X (g/L): 50.07
Glucose S (g/L): 8.30
Lactic Acid P (g/L): 46.94
Maltose M (g/L): 67.47
Volume V (L): 5.87

t = 15.00 h

Biomass X (g/L): 52.37
Glucose S (g/L): 0.60
Lactic Acid P (g/L): 33.60
Maltose M (g/L): 72.38
Volume V (L): 5.69

t = 16.00 h

Biomass X (g/L): 68.54
Glucose S (g/L): 8.32
Lactic Acid P (g/L): 49.17
Maltose M (g/L): 71.75
Volume V (L): 5.54

t = 17.00 h

Biomass X (g/L): 60.80
Glucose S (g/L): 12.08
Lactic Acid P (g/L): 43.75
Maltose M (g/L): 65.34
Volume V (L): 5.37

t = 18.00 h

Biomass X (g/L): 67.08
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 49.20
Maltose M (g/L): 66.70
Volume V (L): 5.66

t = 19.00 h

Biomass X (g/L): 57.23
Glucose S (g/L): 3.64
Lactic Acid P (g/L): 35.12
Maltose M (g/L): 72.31
Volume V (L): 5.64

t = 20.00 h

Biomass X (g/L): 66.05
Glucose S (g/L): 4.00
Lactic Acid P (g/L): 45.81
Maltose M (g/L): 68.39
Volume V (L): 5.67

t = 21.00 h

Biomass X (g/L): 76.06
Glucose S (g/L): 6.62
Lactic Acid P (g/L): 32.19
Maltose M (g/L): 68.99
Volume V (L): 5.68

t = 22.00 h

Biomass X (g/L): 61.85
Glucose S (g/L): 9.99
Lactic Acid P (g/L): 61.53
Maltose M (g/L): 63.27
Volume V (L): 5.33

t = 23.00 h

Biomass X (g/L): 67.21
Glucose S (g/L): 5.26
Lactic Acid P (g/L): 28.75
Maltose M (g/L): 67.80
Volume V (L): 6.05

t = 24.00 h

Biomass X (g/L): 54.58
Glucose S (g/L): 10.15
Lactic Acid P (g/L): 44.77
Maltose M (g/L): 61.73
Volume V (L): 5.81

t = 25.00 h

Biomass X (g/L): 60.96
Glucose S (g/L): 4.79
Lactic Acid P (g/L): 40.48
Maltose M (g/L): 59.80
Volume V (L): 5.57

t = 26.00 h

Biomass X (g/L): 65.76
Glucose S (g/L): 5.65
Lactic Acid P (g/L): 55.96
Maltose M (g/L): 66.78
Volume V (L): 5.99

t = 27.00 h

Biomass X (g/L): 59.91
Glucose S (g/L): 5.42
Lactic Acid P (g/L): 68.18
Maltose M (g/L): 68.31
Volume V (L): 5.75

t = 28.00 h

Biomass X (g/L): 82.55
Glucose S (g/L): 8.00
Lactic Acid P (g/L): 57.54
Maltose M (g/L): 63.32
Volume V (L): 6.08

t = 29.00 h

Biomass X (g/L): 63.48
Glucose S (g/L): 3.62
Lactic Acid P (g/L): 79.31
Maltose M (g/L): 62.14
Volume V (L): 6.18

t = 30.00 h

Biomass X (g/L): 79.10
Glucose S (g/L): 7.18
Lactic Acid P (g/L): 40.78
Maltose M (g/L): 67.91
Volume V (L): 6.24

t = 31.00 h

Biomass X (g/L): 74.28
Glucose S (g/L): 1.70
Lactic Acid P (g/L): 42.28
Maltose M (g/L): 62.34
Volume V (L): 6.03

t = 32.00 h

Biomass X (g/L): 73.51
Glucose S (g/L): 4.48
Lactic Acid P (g/L): 46.77
Maltose M (g/L): 62.32
Volume V (L): 6.01

t = 33.00 h

Biomass X (g/L): 87.49
Glucose S (g/L): 3.69
Lactic Acid P (g/L): 56.74
Maltose M (g/L): 66.75
Volume V (L): 6.00

t = 34.00 h

Biomass X (g/L): 53.72
Glucose S (g/L): 5.88
Lactic Acid P (g/L): 47.08
Maltose M (g/L): 60.81

Volume V (L): 6.02

t = 35.00 h

Biomass X (g/L): 73.24

Glucose S (g/L): 4.97

Lactic Acid P (g/L): 50.31

Maltose M (g/L): 59.63

Volume V (L): 6.20

t = 36.00 h

Biomass X (g/L): 65.68

Glucose S (g/L): 9.17

Lactic Acid P (g/L): 63.48

Maltose M (g/L): 63.28

Volume V (L): 6.19

t = 37.00 h

Biomass X (g/L): 58.12

Glucose S (g/L): 3.85

Lactic Acid P (g/L): 47.31

Maltose M (g/L): 58.77

Volume V (L): 6.10

t = 38.00 h

Biomass X (g/L): 65.66

Glucose S (g/L): 1.73

Lactic Acid P (g/L): 48.96

Maltose M (g/L): 59.97

Volume V (L): 6.07

t = 39.00 h

Biomass X (g/L): 77.26

Glucose S (g/L): 3.00

Lactic Acid P (g/L): 56.29

Maltose M (g/L): 56.43

Volume V (L): 6.32

t = 40.00 h

Biomass X (g/L): 70.90

Glucose S (g/L): 9.73

Lactic Acid P (g/L): 57.80

Maltose M (g/L): 60.19

Volume V (L): 6.28

t = 41.00 h

Biomass X (g/L): 86.22

Glucose S (g/L): 4.58

Lactic Acid P (g/L): 70.36

Maltose M (g/L): 59.54

Volume V (L): 6.17

t = 42.00 h

Biomass X (g/L): 76.44

Glucose S (g/L): 8.61

Lactic Acid P (g/L): 53.66

Maltose M (g/L): 61.66

Volume V (L): 6.61

t = 43.00 h

Biomass X (g/L): 75.81

Glucose S (g/L): 6.98

Lactic Acid P (g/L): 55.90

Maltose M (g/L): 61.03

Volume V (L): 6.84

t = 44.00 h

Biomass X (g/L): 76.22

Glucose S (g/L): 10.35

Lactic Acid P (g/L): 68.30

Maltose M (g/L): 59.87

Volume V (L): 6.71

t = 45.00 h

Biomass X (g/L): 78.61

Glucose S (g/L): 7.73

Lactic Acid P (g/L): 70.87

Maltose M (g/L): 59.13

Volume V (L): 6.40

t = 46.00 h

Biomass X (g/L): 76.00

Glucose S (g/L): 16.28

Lactic Acid P (g/L): 77.13

Maltose M (g/L): 57.54

Volume V (L): 6.57

t = 47.00 h

Biomass X (g/L): 72.34

Glucose S (g/L): 14.20

Lactic Acid P (g/L): 68.97

Maltose M (g/L): 59.09

Volume V (L): 6.93

t = 48.00 h

Biomass X (g/L): 89.51

Glucose S (g/L): 8.42

Lactic Acid P (g/L): 66.44

Maltose M (g/L): 54.09

Volume V (L): 6.41

t = 49.00 h

Biomass X (g/L): 74.98

Glucose S (g/L): 8.37

Lactic Acid P (g/L): 70.99

Maltose M (g/L): 50.42

Volume V (L): 6.84

t = 50.00 h

Biomass X (g/L): 77.43

Glucose S (g/L): 18.57

Lactic Acid P (g/L): 63.86

Maltose M (g/L): 59.68

Volume V (L): 6.42

t = 51.00 h

Biomass X (g/L): 85.12

Glucose S (g/L): 7.31

Lactic Acid P (g/L): 64.29

Maltose M (g/L): 58.05

Volume V (L): 7.10

t = 52.00 h

Biomass X (g/L): 84.32

Glucose S (g/L): 10.47
Lactic Acid P (g/L): 51.18
Maltose M (g/L): 55.57
Volume V (L): 6.70

t = 53.00 h

Biomass X (g/L): 73.40
Glucose S (g/L): 11.04
Lactic Acid P (g/L): 75.84
Maltose M (g/L): 58.55
Volume V (L): 7.28

t = 54.00 h

Biomass X (g/L): 88.71
Glucose S (g/L): 11.33
Lactic Acid P (g/L): 70.12
Maltose M (g/L): 53.35
Volume V (L): 7.21

t = 55.00 h

Biomass X (g/L): 86.88
Glucose S (g/L): 6.91
Lactic Acid P (g/L): 75.70
Maltose M (g/L): 53.13
Volume V (L): 6.87

t = 56.00 h

Biomass X (g/L): 83.73
Glucose S (g/L): 13.44
Lactic Acid P (g/L): 43.55
Maltose M (g/L): 52.47
Volume V (L): 6.78

t = 57.00 h

Biomass X (g/L): 78.28
Glucose S (g/L): 9.77
Lactic Acid P (g/L): 69.08
Maltose M (g/L): 56.04
Volume V (L): 7.00

t = 58.00 h

Biomass X (g/L): 78.62
Glucose S (g/L): 16.78
Lactic Acid P (g/L): 62.16
Maltose M (g/L): 49.90
Volume V (L): 6.96

t = 59.00 h

Biomass X (g/L): 80.61
Glucose S (g/L): 9.20
Lactic Acid P (g/L): 83.94
Maltose M (g/L): 51.85
Volume V (L): 6.94

t = 60.00 h

Biomass X (g/L): 71.26
Glucose S (g/L): 13.76
Lactic Acid P (g/L): 76.29
Maltose M (g/L): 54.46
Volume V (L): 6.72

t = 61.00 h

Biomass X (g/L): 82.28
Glucose S (g/L): 7.89
Lactic Acid P (g/L): 73.23
Maltose M (g/L): 50.12
Volume V (L): 6.99

t = 62.00 h

Biomass X (g/L): 90.89
Glucose S (g/L): 8.46
Lactic Acid P (g/L): 67.56
Maltose M (g/L): 51.34
Volume V (L): 6.90

t = 63.00 h

Biomass X (g/L): 90.81
Glucose S (g/L): 12.48
Lactic Acid P (g/L): 75.45
Maltose M (g/L): 46.85
Volume V (L): 7.28

t = 64.00 h

Biomass X (g/L): 79.32
Glucose S (g/L): 13.28
Lactic Acid P (g/L): 63.78
Maltose M (g/L): 54.26
Volume V (L): 7.40

t = 65.00 h

Biomass X (g/L): 92.61
Glucose S (g/L): 19.65
Lactic Acid P (g/L): 71.64
Maltose M (g/L): 50.48
Volume V (L): 7.39

t = 66.00 h

Biomass X (g/L): 100.35
Glucose S (g/L): 15.92
Lactic Acid P (g/L): 67.64
Maltose M (g/L): 49.11
Volume V (L): 7.34

t = 67.00 h

Biomass X (g/L): 79.98
Glucose S (g/L): 20.01
Lactic Acid P (g/L): 93.77
Maltose M (g/L): 48.32
Volume V (L): 7.13

t = 68.00 h

Biomass X (g/L): 89.23
Glucose S (g/L): 15.82
Lactic Acid P (g/L): 86.87
Maltose M (g/L): 50.46
Volume V (L): 7.38

t = 69.00 h

Biomass X (g/L): 62.13
Glucose S (g/L): 18.96
Lactic Acid P (g/L): 83.49
Maltose M (g/L): 47.05
Volume V (L): 7.53

t = 70.00 h

Biomass X (g/L): 71.33
Glucose S (g/L): 15.71
Lactic Acid P (g/L): 60.64
Maltose M (g/L): 56.40
Volume V (L): 7.28

t = 71.00 h

Biomass X (g/L): 83.68
Glucose S (g/L): 15.95
Lactic Acid P (g/L): 82.29
Maltose M (g/L): 44.37
Volume V (L): 7.46

t = 72.00 h

Biomass X (g/L): 73.58
Glucose S (g/L): 13.55
Lactic Acid P (g/L): 76.45
Maltose M (g/L): 49.40
Volume V (L): 7.44

t = 73.00 h

Biomass X (g/L): 101.48
Glucose S (g/L): 19.92
Lactic Acid P (g/L): 86.12
Maltose M (g/L): 50.68
Volume V (L): 7.85

t = 74.00 h

Biomass X (g/L): 88.21
Glucose S (g/L): 20.15
Lactic Acid P (g/L): 78.23
Maltose M (g/L): 48.04
Volume V (L): 7.72

t = 75.00 h

Biomass X (g/L): 91.53
Glucose S (g/L): 22.01
Lactic Acid P (g/L): 75.52
Maltose M (g/L): 50.49

Volume V (L): 7.70

Batch 74

Initial Conditions:

Batch: 74.00

X0 (g/L): 3.98

S0 (g/L): 142.12

M0 (g/L): 28.78

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.20

F_vol (L/h): 0.05

S_feed (g/L): 249.48

M_feed (g/L): 50.52

V_final (L): 8.43

Batch 74 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 143.93

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 29.78

Volume V (L): 5.08

t = 1.00 h

Biomass X (g/L): 7.18

Glucose S (g/L): 140.05

Lactic Acid P (g/L): 3.17

Maltose M (g/L): 28.39

Volume V (L): 4.82

t = 2.00 h

Biomass X (g/L): 5.75

Glucose S (g/L): 147.30

Lactic Acid P (g/L): 2.93

Maltose M (g/L): 26.28

Volume V (L): 5.21

t = 3.00 h

Biomass X (g/L): 8.08

Glucose S (g/L): 136.85

Lactic Acid P (g/L): 12.83

Maltose M (g/L): 29.09

Volume V (L): 4.92

t = 4.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 132.29

Lactic Acid P (g/L): 1.85

Maltose M (g/L): 26.99

Volume V (L): 5.22

t = 5.00 h

Biomass X (g/L): 5.25

Glucose S (g/L): 141.37

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 25.37
Volume V (L): 5.12

t = 6.00 h

Biomass X (g/L): 19.69
Glucose S (g/L): 132.13
Lactic Acid P (g/L): 15.43
Maltose M (g/L): 29.85
Volume V (L): 5.75

t = 7.00 h

Biomass X (g/L): 36.21
Glucose S (g/L): 129.40
Lactic Acid P (g/L): 10.79
Maltose M (g/L): 24.98
Volume V (L): 5.58

t = 8.00 h

Biomass X (g/L): 23.71
Glucose S (g/L): 124.22
Lactic Acid P (g/L): 26.32
Maltose M (g/L): 27.38
Volume V (L): 5.73

t = 9.00 h

Biomass X (g/L): 30.57
Glucose S (g/L): 121.34
Lactic Acid P (g/L): 22.58
Maltose M (g/L): 24.77
Volume V (L): 5.42

t = 10.00 h

Biomass X (g/L): 57.58
Glucose S (g/L): 127.62
Lactic Acid P (g/L): 32.36
Maltose M (g/L): 25.75
Volume V (L): 5.68

t = 11.00 h

Biomass X (g/L): 41.21

Glucose S (g/L): 125.40
Lactic Acid P (g/L): 20.03
Maltose M (g/L): 24.01
Volume V (L): 5.53

t = 12.00 h

Biomass X (g/L): 47.38
Glucose S (g/L): 122.70
Lactic Acid P (g/L): 34.42
Maltose M (g/L): 23.26
Volume V (L): 5.95

t = 13.00 h

Biomass X (g/L): 51.66
Glucose S (g/L): 115.79
Lactic Acid P (g/L): 29.58
Maltose M (g/L): 25.45
Volume V (L): 5.57

t = 14.00 h

Biomass X (g/L): 42.99
Glucose S (g/L): 109.59
Lactic Acid P (g/L): 27.58
Maltose M (g/L): 25.02
Volume V (L): 5.26

t = 15.00 h

Biomass X (g/L): 65.92
Glucose S (g/L): 96.89
Lactic Acid P (g/L): 33.73
Maltose M (g/L): 25.80
Volume V (L): 5.14

t = 16.00 h

Biomass X (g/L): 56.64
Glucose S (g/L): 101.23
Lactic Acid P (g/L): 39.25
Maltose M (g/L): 23.89
Volume V (L): 6.19

t = 17.00 h

Biomass X (g/L): 65.79
Glucose S (g/L): 111.69
Lactic Acid P (g/L): 58.56
Maltose M (g/L): 26.80
Volume V (L): 5.91

t = 18.00 h

Biomass X (g/L): 73.16
Glucose S (g/L): 96.09
Lactic Acid P (g/L): 43.41
Maltose M (g/L): 23.82
Volume V (L): 5.97

t = 19.00 h

Biomass X (g/L): 90.64
Glucose S (g/L): 104.92
Lactic Acid P (g/L): 38.69
Maltose M (g/L): 24.00
Volume V (L): 6.28

t = 20.00 h

Biomass X (g/L): 59.61
Glucose S (g/L): 95.78
Lactic Acid P (g/L): 53.80
Maltose M (g/L): 21.07
Volume V (L): 5.46

t = 21.00 h

Biomass X (g/L): 83.59
Glucose S (g/L): 80.03
Lactic Acid P (g/L): 54.46
Maltose M (g/L): 21.88
Volume V (L): 5.86

t = 22.00 h

Biomass X (g/L): 85.27
Glucose S (g/L): 96.71
Lactic Acid P (g/L): 51.00
Maltose M (g/L): 23.24
Volume V (L): 5.94

t = 23.00 h

Biomass X (g/L): 71.44
Glucose S (g/L): 99.17
Lactic Acid P (g/L): 55.53
Maltose M (g/L): 22.17
Volume V (L): 6.25

t = 24.00 h

Biomass X (g/L): 85.32
Glucose S (g/L): 91.42
Lactic Acid P (g/L): 67.62
Maltose M (g/L): 22.33
Volume V (L): 5.91

t = 25.00 h

Biomass X (g/L): 97.08
Glucose S (g/L): 97.62
Lactic Acid P (g/L): 65.90
Maltose M (g/L): 20.71
Volume V (L): 5.81

t = 26.00 h

Biomass X (g/L): 73.69
Glucose S (g/L): 81.63
Lactic Acid P (g/L): 66.18
Maltose M (g/L): 21.82
Volume V (L): 5.86

t = 27.00 h

Biomass X (g/L): 84.84
Glucose S (g/L): 82.70
Lactic Acid P (g/L): 50.39
Maltose M (g/L): 21.61
Volume V (L): 6.55

t = 28.00 h

Biomass X (g/L): 87.34
Glucose S (g/L): 90.82
Lactic Acid P (g/L): 90.04
Maltose M (g/L): 19.59
Volume V (L): 6.08

t = 29.00 h

Biomass X (g/L): 89.40
Glucose S (g/L): 91.99
Lactic Acid P (g/L): 73.48
Maltose M (g/L): 21.94
Volume V (L): 6.17

t = 30.00 h

Biomass X (g/L): 92.45
Glucose S (g/L): 96.45
Lactic Acid P (g/L): 69.23
Maltose M (g/L): 18.47
Volume V (L): 6.54

t = 31.00 h

Biomass X (g/L): 86.92
Glucose S (g/L): 92.47
Lactic Acid P (g/L): 54.03
Maltose M (g/L): 20.92
Volume V (L): 6.22

t = 32.00 h

Biomass X (g/L): 90.96
Glucose S (g/L): 82.59
Lactic Acid P (g/L): 74.84
Maltose M (g/L): 22.94
Volume V (L): 5.99

t = 33.00 h

Biomass X (g/L): 76.96
Glucose S (g/L): 90.02
Lactic Acid P (g/L): 61.76
Maltose M (g/L): 21.24
Volume V (L): 6.07

t = 34.00 h

Biomass X (g/L): 88.28
Glucose S (g/L): 92.06
Lactic Acid P (g/L): 65.32
Maltose M (g/L): 21.05

Volume V (L): 6.54

t = 35.00 h

Biomass X (g/L): 82.14

Glucose S (g/L): 78.88

Lactic Acid P (g/L): 51.81

Maltose M (g/L): 18.67

Volume V (L): 6.74

t = 36.00 h

Biomass X (g/L): 74.80

Glucose S (g/L): 88.69

Lactic Acid P (g/L): 56.56

Maltose M (g/L): 22.23

Volume V (L): 6.59

t = 37.00 h

Biomass X (g/L): 82.38

Glucose S (g/L): 87.21

Lactic Acid P (g/L): 73.09

Maltose M (g/L): 21.51

Volume V (L): 6.81

t = 38.00 h

Biomass X (g/L): 90.51

Glucose S (g/L): 85.61

Lactic Acid P (g/L): 79.00

Maltose M (g/L): 18.24

Volume V (L): 6.29

t = 39.00 h

Biomass X (g/L): 92.39

Glucose S (g/L): 91.88

Lactic Acid P (g/L): 76.45

Maltose M (g/L): 18.59

Volume V (L): 6.36

t = 40.00 h

Biomass X (g/L): 75.26

Glucose S (g/L): 83.24

Lactic Acid P (g/L): 76.59

Maltose M (g/L): 20.85

Volume V (L): 7.44

t = 41.00 h

Biomass X (g/L): 78.75

Glucose S (g/L): 91.76

Lactic Acid P (g/L): 70.24

Maltose M (g/L): 21.44

Volume V (L): 7.21

t = 42.00 h

Biomass X (g/L): 95.72

Glucose S (g/L): 92.71

Lactic Acid P (g/L): 67.42

Maltose M (g/L): 17.01

Volume V (L): 7.13

t = 43.00 h

Biomass X (g/L): 93.60

Glucose S (g/L): 95.71

Lactic Acid P (g/L): 74.44

Maltose M (g/L): 17.22

Volume V (L): 6.78

t = 44.00 h

Biomass X (g/L): 97.50

Glucose S (g/L): 84.16

Lactic Acid P (g/L): 79.89

Maltose M (g/L): 18.27

Volume V (L): 6.94

t = 45.00 h

Biomass X (g/L): 102.41

Glucose S (g/L): 82.35

Lactic Acid P (g/L): 72.51

Maltose M (g/L): 15.28

Volume V (L): 7.06

t = 46.00 h

Biomass X (g/L): 94.00

Glucose S (g/L): 92.10

Lactic Acid P (g/L): 84.32

Maltose M (g/L): 18.20

Volume V (L): 7.46

t = 47.00 h

Biomass X (g/L): 84.33

Glucose S (g/L): 89.34

Lactic Acid P (g/L): 71.12

Maltose M (g/L): 19.69

Volume V (L): 7.26

t = 48.00 h

Biomass X (g/L): 89.68

Glucose S (g/L): 106.60

Lactic Acid P (g/L): 64.17

Maltose M (g/L): 15.88

Volume V (L): 7.34

t = 49.00 h

Biomass X (g/L): 105.32

Glucose S (g/L): 93.84

Lactic Acid P (g/L): 73.71

Maltose M (g/L): 17.93

Volume V (L): 7.19

t = 50.00 h

Biomass X (g/L): 116.48

Glucose S (g/L): 91.98

Lactic Acid P (g/L): 75.12

Maltose M (g/L): 16.80

Volume V (L): 7.50

t = 51.00 h

Biomass X (g/L): 93.06

Glucose S (g/L): 91.03

Lactic Acid P (g/L): 73.63

Maltose M (g/L): 19.77

Volume V (L): 7.41

t = 52.00 h

Biomass X (g/L): 93.71

Glucose S (g/L): 85.75
Lactic Acid P (g/L): 86.87
Maltose M (g/L): 16.55
Volume V (L): 7.39

t = 53.00 h

Biomass X (g/L): 95.55
Glucose S (g/L): 95.54
Lactic Acid P (g/L): 82.62
Maltose M (g/L): 19.34
Volume V (L): 7.22

t = 54.00 h

Biomass X (g/L): 88.33
Glucose S (g/L): 92.41
Lactic Acid P (g/L): 61.62
Maltose M (g/L): 16.25
Volume V (L): 7.44

t = 55.00 h

Biomass X (g/L): 99.79
Glucose S (g/L): 92.72
Lactic Acid P (g/L): 90.07
Maltose M (g/L): 18.04
Volume V (L): 7.22

t = 56.00 h

Biomass X (g/L): 98.09
Glucose S (g/L): 93.94
Lactic Acid P (g/L): 49.67
Maltose M (g/L): 17.19
Volume V (L): 7.55

t = 57.00 h

Biomass X (g/L): 91.33
Glucose S (g/L): 94.93
Lactic Acid P (g/L): 96.41
Maltose M (g/L): 16.52
Volume V (L): 7.63

t = 58.00 h

Biomass X (g/L): 84.37
Glucose S (g/L): 84.79
Lactic Acid P (g/L): 73.46
Maltose M (g/L): 15.49
Volume V (L): 6.99

t = 59.00 h

Biomass X (g/L): 72.02
Glucose S (g/L): 98.84
Lactic Acid P (g/L): 74.61
Maltose M (g/L): 16.93
Volume V (L): 7.63

t = 60.00 h

Biomass X (g/L): 94.73
Glucose S (g/L): 90.18
Lactic Acid P (g/L): 76.06
Maltose M (g/L): 15.74
Volume V (L): 7.42

t = 61.00 h

Biomass X (g/L): 89.30
Glucose S (g/L): 94.27
Lactic Acid P (g/L): 80.50
Maltose M (g/L): 15.28
Volume V (L): 7.92

t = 62.00 h

Biomass X (g/L): 79.49
Glucose S (g/L): 102.18
Lactic Acid P (g/L): 77.04
Maltose M (g/L): 18.19
Volume V (L): 8.01

t = 63.00 h

Biomass X (g/L): 75.35
Glucose S (g/L): 105.13
Lactic Acid P (g/L): 75.97
Maltose M (g/L): 16.78
Volume V (L): 7.76

t = 64.00 h

Biomass X (g/L): 93.35
Glucose S (g/L): 86.90
Lactic Acid P (g/L): 89.87
Maltose M (g/L): 16.81
Volume V (L): 8.07

t = 65.00 h

Biomass X (g/L): 70.21
Glucose S (g/L): 96.44
Lactic Acid P (g/L): 82.99
Maltose M (g/L): 15.25
Volume V (L): 8.13

t = 66.00 h

Biomass X (g/L): 89.32
Glucose S (g/L): 98.06
Lactic Acid P (g/L): 68.68
Maltose M (g/L): 15.43
Volume V (L): 7.95

t = 67.00 h

Biomass X (g/L): 94.65
Glucose S (g/L): 88.89
Lactic Acid P (g/L): 85.01
Maltose M (g/L): 15.24
Volume V (L): 8.48

t = 68.00 h

Biomass X (g/L): 91.48
Glucose S (g/L): 101.52
Lactic Acid P (g/L): 83.13
Maltose M (g/L): 15.95
Volume V (L): 7.91

t = 69.00 h

Biomass X (g/L): 81.74
Glucose S (g/L): 98.18
Lactic Acid P (g/L): 73.13
Maltose M (g/L): 16.28
Volume V (L): 8.56

t = 70.00 h

Biomass X (g/L): 72.83
Glucose S (g/L): 98.92
Lactic Acid P (g/L): 86.00
Maltose M (g/L): 15.24
Volume V (L): 8.14

t = 71.00 h

Biomass X (g/L): 91.27
Glucose S (g/L): 93.31
Lactic Acid P (g/L): 77.13
Maltose M (g/L): 14.07
Volume V (L): 8.01

t = 72.00 h

Biomass X (g/L): 97.29
Glucose S (g/L): 100.72
Lactic Acid P (g/L): 76.67
Maltose M (g/L): 16.14
Volume V (L): 8.19

t = 73.00 h

Biomass X (g/L): 79.09
Glucose S (g/L): 103.31
Lactic Acid P (g/L): 76.57
Maltose M (g/L): 15.03
Volume V (L): 8.03

t = 74.00 h

Biomass X (g/L): 89.84
Glucose S (g/L): 99.83
Lactic Acid P (g/L): 56.00
Maltose M (g/L): 12.87
Volume V (L): 8.60

t = 75.00 h

Biomass X (g/L): 100.53
Glucose S (g/L): 96.73
Lactic Acid P (g/L): 74.21
Maltose M (g/L): 11.81

Volume V (L): 7.99

Batch 75

Initial Conditions:

Batch: 75.00

X0 (g/L): 1.45

S0 (g/L): 50.80

M0 (g/L): 49.40

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.97

F_vol (L/h): 0.04

S_feed (g/L): 152.10

M_feed (g/L): 147.90

V_final (L): 8.14

Batch 75 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 15.23
Glucose S (g/L): 47.85
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 51.42
Volume V (L): 5.14

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 56.96
Lactic Acid P (g/L): 3.62
Maltose M (g/L): 49.09
Volume V (L): 4.69

t = 2.00 h

Biomass X (g/L): 10.89
Glucose S (g/L): 50.46
Lactic Acid P (g/L): 3.75
Maltose M (g/L): 48.68
Volume V (L): 5.12

t = 3.00 h

Biomass X (g/L): 3.70
Glucose S (g/L): 53.87
Lactic Acid P (g/L): 19.01
Maltose M (g/L): 48.95
Volume V (L): 5.44

t = 4.00 h

Biomass X (g/L): 7.26
Glucose S (g/L): 48.30
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 51.51
Volume V (L): 5.26

t = 5.00 h

Biomass X (g/L): 0.16
Glucose S (g/L): 59.27

Lactic Acid P (g/L): 1.63
Maltose M (g/L): 49.61
Volume V (L): 5.29

t = 6.00 h

Biomass X (g/L): 11.31
Glucose S (g/L): 47.66
Lactic Acid P (g/L): 6.85
Maltose M (g/L): 47.39
Volume V (L): 4.74

t = 7.00 h

Biomass X (g/L): 11.37
Glucose S (g/L): 48.10
Lactic Acid P (g/L): 15.17
Maltose M (g/L): 48.42
Volume V (L): 5.09

t = 8.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 57.16
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 48.64
Volume V (L): 5.85

t = 9.00 h

Biomass X (g/L): 6.25
Glucose S (g/L): 58.92
Lactic Acid P (g/L): 11.41
Maltose M (g/L): 46.78
Volume V (L): 5.23

t = 10.00 h

Biomass X (g/L): 13.94
Glucose S (g/L): 48.70
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 48.85
Volume V (L): 5.74

t = 11.00 h

Biomass X (g/L): 15.66

Glucose S (g/L): 47.31
Lactic Acid P (g/L): 13.32
Maltose M (g/L): 46.76
Volume V (L): 5.31

t = 12.00 h

Biomass X (g/L): 39.62
Glucose S (g/L): 48.33
Lactic Acid P (g/L): 28.73
Maltose M (g/L): 45.74
Volume V (L): 5.43

t = 13.00 h

Biomass X (g/L): 31.45
Glucose S (g/L): 43.29
Lactic Acid P (g/L): 15.20
Maltose M (g/L): 47.53
Volume V (L): 5.67

t = 14.00 h

Biomass X (g/L): 27.83
Glucose S (g/L): 38.05
Lactic Acid P (g/L): 26.64
Maltose M (g/L): 47.17
Volume V (L): 5.61

t = 15.00 h

Biomass X (g/L): 34.76
Glucose S (g/L): 34.92
Lactic Acid P (g/L): 23.17
Maltose M (g/L): 47.11
Volume V (L): 5.72

t = 16.00 h

Biomass X (g/L): 38.53
Glucose S (g/L): 28.28
Lactic Acid P (g/L): 23.07
Maltose M (g/L): 46.01
Volume V (L): 5.46

t = 17.00 h

Biomass X (g/L): 44.35
Glucose S (g/L): 32.62
Lactic Acid P (g/L): 42.79
Maltose M (g/L): 46.01
Volume V (L): 5.43

t = 18.00 h

Biomass X (g/L): 39.97
Glucose S (g/L): 29.21
Lactic Acid P (g/L): 26.79
Maltose M (g/L): 48.00
Volume V (L): 5.53

t = 19.00 h

Biomass X (g/L): 45.71
Glucose S (g/L): 25.95
Lactic Acid P (g/L): 41.62
Maltose M (g/L): 46.98
Volume V (L): 5.66

t = 20.00 h

Biomass X (g/L): 64.27
Glucose S (g/L): 22.08
Lactic Acid P (g/L): 44.05
Maltose M (g/L): 45.10
Volume V (L): 5.78

t = 21.00 h

Biomass X (g/L): 61.14
Glucose S (g/L): 20.93
Lactic Acid P (g/L): 35.46
Maltose M (g/L): 43.71
Volume V (L): 6.14

t = 22.00 h

Biomass X (g/L): 60.19
Glucose S (g/L): 25.30
Lactic Acid P (g/L): 46.72
Maltose M (g/L): 44.81
Volume V (L): 5.94

t = 23.00 h

Biomass X (g/L): 63.89
Glucose S (g/L): 25.14
Lactic Acid P (g/L): 38.46
Maltose M (g/L): 45.95
Volume V (L): 6.09

t = 24.00 h

Biomass X (g/L): 65.38
Glucose S (g/L): 19.65
Lactic Acid P (g/L): 53.59
Maltose M (g/L): 44.81
Volume V (L): 6.04

t = 25.00 h

Biomass X (g/L): 65.54
Glucose S (g/L): 18.44
Lactic Acid P (g/L): 38.18
Maltose M (g/L): 44.54
Volume V (L): 5.77

t = 26.00 h

Biomass X (g/L): 74.35
Glucose S (g/L): 16.19
Lactic Acid P (g/L): 48.62
Maltose M (g/L): 44.52
Volume V (L): 6.09

t = 27.00 h

Biomass X (g/L): 75.08
Glucose S (g/L): 22.35
Lactic Acid P (g/L): 50.76
Maltose M (g/L): 44.98
Volume V (L): 6.16

t = 28.00 h

Biomass X (g/L): 56.43
Glucose S (g/L): 18.51
Lactic Acid P (g/L): 64.63
Maltose M (g/L): 45.86
Volume V (L): 6.36

t = 29.00 h

Biomass X (g/L): 71.91
Glucose S (g/L): 19.44
Lactic Acid P (g/L): 56.81
Maltose M (g/L): 44.72
Volume V (L): 5.95

t = 30.00 h

Biomass X (g/L): 59.38
Glucose S (g/L): 19.15
Lactic Acid P (g/L): 65.43
Maltose M (g/L): 44.63
Volume V (L): 5.74

t = 31.00 h

Biomass X (g/L): 81.60
Glucose S (g/L): 16.48
Lactic Acid P (g/L): 62.75
Maltose M (g/L): 44.10
Volume V (L): 6.34

t = 32.00 h

Biomass X (g/L): 83.82
Glucose S (g/L): 13.16
Lactic Acid P (g/L): 63.66
Maltose M (g/L): 43.59
Volume V (L): 6.43

t = 33.00 h

Biomass X (g/L): 70.98
Glucose S (g/L): 13.75
Lactic Acid P (g/L): 56.43
Maltose M (g/L): 45.13
Volume V (L): 6.34

t = 34.00 h

Biomass X (g/L): 76.63
Glucose S (g/L): 21.72
Lactic Acid P (g/L): 69.79
Maltose M (g/L): 44.77

Volume V (L): 6.43

t = 35.00 h

Biomass X (g/L): 57.54

Glucose S (g/L): 14.83

Lactic Acid P (g/L): 57.88

Maltose M (g/L): 46.19

Volume V (L): 6.69

t = 36.00 h

Biomass X (g/L): 88.74

Glucose S (g/L): 16.12

Lactic Acid P (g/L): 45.96

Maltose M (g/L): 44.26

Volume V (L): 6.39

t = 37.00 h

Biomass X (g/L): 86.70

Glucose S (g/L): 13.55

Lactic Acid P (g/L): 60.69

Maltose M (g/L): 42.72

Volume V (L): 6.48

t = 38.00 h

Biomass X (g/L): 91.58

Glucose S (g/L): 13.43

Lactic Acid P (g/L): 67.56

Maltose M (g/L): 42.05

Volume V (L): 6.61

t = 39.00 h

Biomass X (g/L): 95.39

Glucose S (g/L): 15.31

Lactic Acid P (g/L): 49.20

Maltose M (g/L): 44.16

Volume V (L): 6.33

t = 40.00 h

Biomass X (g/L): 68.95

Glucose S (g/L): 24.76

Lactic Acid P (g/L): 56.50

Maltose M (g/L): 41.82

Volume V (L): 6.64

t = 41.00 h

Biomass X (g/L): 90.27

Glucose S (g/L): 21.17

Lactic Acid P (g/L): 53.92

Maltose M (g/L): 43.64

Volume V (L): 6.70

t = 42.00 h

Biomass X (g/L): 96.49

Glucose S (g/L): 17.02

Lactic Acid P (g/L): 73.13

Maltose M (g/L): 41.54

Volume V (L): 7.17

t = 43.00 h

Biomass X (g/L): 67.33

Glucose S (g/L): 23.37

Lactic Acid P (g/L): 81.88

Maltose M (g/L): 42.73

Volume V (L): 6.73

t = 44.00 h

Biomass X (g/L): 80.73

Glucose S (g/L): 19.07

Lactic Acid P (g/L): 65.02

Maltose M (g/L): 40.83

Volume V (L): 6.89

t = 45.00 h

Biomass X (g/L): 79.08

Glucose S (g/L): 25.07

Lactic Acid P (g/L): 72.38

Maltose M (g/L): 42.92

Volume V (L): 7.12

t = 46.00 h

Biomass X (g/L): 91.39

Glucose S (g/L): 28.65

Lactic Acid P (g/L): 72.64

Maltose M (g/L): 40.27

Volume V (L): 6.72

t = 47.00 h

Biomass X (g/L): 59.39

Glucose S (g/L): 24.60

Lactic Acid P (g/L): 82.79

Maltose M (g/L): 41.95

Volume V (L): 7.31

t = 48.00 h

Biomass X (g/L): 101.30

Glucose S (g/L): 21.50

Lactic Acid P (g/L): 62.87

Maltose M (g/L): 41.21

Volume V (L): 7.45

t = 49.00 h

Biomass X (g/L): 82.34

Glucose S (g/L): 24.17

Lactic Acid P (g/L): 67.31

Maltose M (g/L): 41.40

Volume V (L): 7.05

t = 50.00 h

Biomass X (g/L): 92.67

Glucose S (g/L): 22.03

Lactic Acid P (g/L): 66.61

Maltose M (g/L): 40.62

Volume V (L): 6.96

t = 51.00 h

Biomass X (g/L): 83.11

Glucose S (g/L): 15.52

Lactic Acid P (g/L): 70.87

Maltose M (g/L): 40.43

Volume V (L): 7.55

t = 52.00 h

Biomass X (g/L): 88.68

Glucose S (g/L): 20.39
Lactic Acid P (g/L): 93.70
Maltose M (g/L): 40.97
Volume V (L): 7.38

t = 53.00 h

Biomass X (g/L): 60.87
Glucose S (g/L): 19.72
Lactic Acid P (g/L): 71.24
Maltose M (g/L): 39.38
Volume V (L): 7.24

t = 54.00 h

Biomass X (g/L): 80.13
Glucose S (g/L): 21.21
Lactic Acid P (g/L): 63.42
Maltose M (g/L): 39.20
Volume V (L): 7.62

t = 55.00 h

Biomass X (g/L): 92.40
Glucose S (g/L): 30.92
Lactic Acid P (g/L): 61.01
Maltose M (g/L): 39.84
Volume V (L): 7.64

t = 56.00 h

Biomass X (g/L): 86.89
Glucose S (g/L): 28.88
Lactic Acid P (g/L): 83.60
Maltose M (g/L): 39.62
Volume V (L): 7.26

t = 57.00 h

Biomass X (g/L): 67.83
Glucose S (g/L): 26.80
Lactic Acid P (g/L): 64.42
Maltose M (g/L): 38.71
Volume V (L): 7.45

t = 58.00 h

Biomass X (g/L): 84.67
Glucose S (g/L): 28.67
Lactic Acid P (g/L): 78.81
Maltose M (g/L): 40.92
Volume V (L): 7.57

t = 59.00 h

Biomass X (g/L): 75.94
Glucose S (g/L): 19.02
Lactic Acid P (g/L): 66.72
Maltose M (g/L): 39.21
Volume V (L): 7.48

t = 60.00 h

Biomass X (g/L): 75.06
Glucose S (g/L): 29.32
Lactic Acid P (g/L): 61.60
Maltose M (g/L): 39.18
Volume V (L): 7.16

t = 61.00 h

Biomass X (g/L): 68.94
Glucose S (g/L): 30.61
Lactic Acid P (g/L): 79.19
Maltose M (g/L): 37.67
Volume V (L): 7.32

t = 62.00 h

Biomass X (g/L): 90.11
Glucose S (g/L): 19.19
Lactic Acid P (g/L): 68.06
Maltose M (g/L): 41.68
Volume V (L): 7.26

t = 63.00 h

Biomass X (g/L): 94.96
Glucose S (g/L): 30.94
Lactic Acid P (g/L): 78.10
Maltose M (g/L): 41.96
Volume V (L): 7.76

t = 64.00 h

Biomass X (g/L): 104.89
Glucose S (g/L): 23.24
Lactic Acid P (g/L): 79.05
Maltose M (g/L): 37.86
Volume V (L): 7.38

t = 65.00 h

Biomass X (g/L): 69.20
Glucose S (g/L): 21.69
Lactic Acid P (g/L): 79.75
Maltose M (g/L): 40.16
Volume V (L): 7.77

t = 66.00 h

Biomass X (g/L): 91.34
Glucose S (g/L): 34.55
Lactic Acid P (g/L): 72.48
Maltose M (g/L): 38.39
Volume V (L): 8.22

t = 67.00 h

Biomass X (g/L): 91.91
Glucose S (g/L): 26.31
Lactic Acid P (g/L): 81.55
Maltose M (g/L): 40.31
Volume V (L): 8.17

t = 68.00 h

Biomass X (g/L): 77.81
Glucose S (g/L): 27.91
Lactic Acid P (g/L): 66.93
Maltose M (g/L): 38.60
Volume V (L): 7.39

t = 69.00 h

Biomass X (g/L): 74.38
Glucose S (g/L): 34.80
Lactic Acid P (g/L): 53.02
Maltose M (g/L): 38.15
Volume V (L): 7.57

t = 70.00 h

Biomass X (g/L): 72.61
Glucose S (g/L): 30.46
Lactic Acid P (g/L): 82.45
Maltose M (g/L): 34.80
Volume V (L): 7.74

t = 71.00 h

Biomass X (g/L): 78.04
Glucose S (g/L): 31.43
Lactic Acid P (g/L): 77.67
Maltose M (g/L): 36.64
Volume V (L): 7.68

t = 72.00 h

Biomass X (g/L): 85.16
Glucose S (g/L): 31.86
Lactic Acid P (g/L): 70.72
Maltose M (g/L): 37.59
Volume V (L): 7.60

t = 73.00 h

Biomass X (g/L): 96.52
Glucose S (g/L): 34.43
Lactic Acid P (g/L): 62.98
Maltose M (g/L): 37.66
Volume V (L): 8.05

t = 74.00 h

Biomass X (g/L): 93.87
Glucose S (g/L): 32.78
Lactic Acid P (g/L): 57.97
Maltose M (g/L): 35.77
Volume V (L): 8.28

t = 75.00 h

Biomass X (g/L): 66.77
Glucose S (g/L): 34.56
Lactic Acid P (g/L): 77.46
Maltose M (g/L): 36.21

Volume V (L): 8.38

Batch 76

Initial Conditions:

Batch: 76.00

X0 (g/L): 4.80

S0 (g/L): 81.84

M0 (g/L): 25.03

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.31

F_vol (L/h): 0.02

S_feed (g/L): 229.75

M_feed (g/L): 70.25

V_final (L): 6.71

Batch 76 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 85.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 25.57
Volume V (L): 5.14

t = 1.00 h

Biomass X (g/L): 3.71
Glucose S (g/L): 87.75
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 23.58
Volume V (L): 4.89

t = 2.00 h

Biomass X (g/L): 12.39
Glucose S (g/L): 83.52
Lactic Acid P (g/L): 12.05
Maltose M (g/L): 24.08
Volume V (L): 5.18

t = 3.00 h

Biomass X (g/L): 10.37
Glucose S (g/L): 81.22
Lactic Acid P (g/L): 22.14
Maltose M (g/L): 22.75
Volume V (L): 4.83

t = 4.00 h

Biomass X (g/L): 23.20
Glucose S (g/L): 73.09
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 23.76
Volume V (L): 5.06

t = 5.00 h

Biomass X (g/L): 23.19
Glucose S (g/L): 63.16

Lactic Acid P (g/L): 12.00
Maltose M (g/L): 23.86
Volume V (L): 5.18

t = 6.00 h

Biomass X (g/L): 19.27
Glucose S (g/L): 61.63
Lactic Acid P (g/L): 24.34
Maltose M (g/L): 23.07
Volume V (L): 4.85

t = 7.00 h

Biomass X (g/L): 8.76
Glucose S (g/L): 80.96
Lactic Acid P (g/L): 11.67
Maltose M (g/L): 22.69
Volume V (L): 5.45

t = 8.00 h

Biomass X (g/L): 48.26
Glucose S (g/L): 64.45
Lactic Acid P (g/L): 27.23
Maltose M (g/L): 21.14
Volume V (L): 5.36

t = 9.00 h

Biomass X (g/L): 30.68
Glucose S (g/L): 56.99
Lactic Acid P (g/L): 23.81
Maltose M (g/L): 23.10
Volume V (L): 5.29

t = 10.00 h

Biomass X (g/L): 46.25
Glucose S (g/L): 62.29
Lactic Acid P (g/L): 39.62
Maltose M (g/L): 20.32
Volume V (L): 5.10

t = 11.00 h

Biomass X (g/L): 27.41

Glucose S (g/L): 46.26
Lactic Acid P (g/L): 18.41
Maltose M (g/L): 22.30
Volume V (L): 5.24

t = 12.00 h

Biomass X (g/L): 51.34
Glucose S (g/L): 50.91
Lactic Acid P (g/L): 23.46
Maltose M (g/L): 20.02
Volume V (L): 5.38

t = 13.00 h

Biomass X (g/L): 56.45
Glucose S (g/L): 44.61
Lactic Acid P (g/L): 40.32
Maltose M (g/L): 21.88
Volume V (L): 5.14

t = 14.00 h

Biomass X (g/L): 60.58
Glucose S (g/L): 44.04
Lactic Acid P (g/L): 41.32
Maltose M (g/L): 21.60
Volume V (L): 5.29

t = 15.00 h

Biomass X (g/L): 93.40
Glucose S (g/L): 36.57
Lactic Acid P (g/L): 42.41
Maltose M (g/L): 21.86
Volume V (L): 5.47

t = 16.00 h

Biomass X (g/L): 58.80
Glucose S (g/L): 30.39
Lactic Acid P (g/L): 51.12
Maltose M (g/L): 22.17
Volume V (L): 5.16

t = 17.00 h

Biomass X (g/L): 76.44
Glucose S (g/L): 29.19
Lactic Acid P (g/L): 38.30
Maltose M (g/L): 22.17
Volume V (L): 5.43

t = 18.00 h

Biomass X (g/L): 75.79
Glucose S (g/L): 27.59
Lactic Acid P (g/L): 35.95
Maltose M (g/L): 18.99
Volume V (L): 5.17

t = 19.00 h

Biomass X (g/L): 65.57
Glucose S (g/L): 30.73
Lactic Acid P (g/L): 46.48
Maltose M (g/L): 20.54
Volume V (L): 5.30

t = 20.00 h

Biomass X (g/L): 82.81
Glucose S (g/L): 19.81
Lactic Acid P (g/L): 49.31
Maltose M (g/L): 22.68
Volume V (L): 5.39

t = 21.00 h

Biomass X (g/L): 74.56
Glucose S (g/L): 20.44
Lactic Acid P (g/L): 51.94
Maltose M (g/L): 20.36
Volume V (L): 5.55

t = 22.00 h

Biomass X (g/L): 86.85
Glucose S (g/L): 27.78
Lactic Acid P (g/L): 68.90
Maltose M (g/L): 19.11
Volume V (L): 5.41

t = 23.00 h

Biomass X (g/L): 99.45
Glucose S (g/L): 25.20
Lactic Acid P (g/L): 65.28
Maltose M (g/L): 19.77
Volume V (L): 5.54

t = 24.00 h

Biomass X (g/L): 82.07
Glucose S (g/L): 26.30
Lactic Acid P (g/L): 64.18
Maltose M (g/L): 19.26
Volume V (L): 5.45

t = 25.00 h

Biomass X (g/L): 78.77
Glucose S (g/L): 24.12
Lactic Acid P (g/L): 62.26
Maltose M (g/L): 18.19
Volume V (L): 5.50

t = 26.00 h

Biomass X (g/L): 93.53
Glucose S (g/L): 39.14
Lactic Acid P (g/L): 56.82
Maltose M (g/L): 20.32
Volume V (L): 5.76

t = 27.00 h

Biomass X (g/L): 98.02
Glucose S (g/L): 23.97
Lactic Acid P (g/L): 53.66
Maltose M (g/L): 19.84
Volume V (L): 5.46

t = 28.00 h

Biomass X (g/L): 75.36
Glucose S (g/L): 20.40
Lactic Acid P (g/L): 51.21
Maltose M (g/L): 17.51
Volume V (L): 5.74

t = 29.00 h

Biomass X (g/L): 80.55
Glucose S (g/L): 22.92
Lactic Acid P (g/L): 57.14
Maltose M (g/L): 18.75
Volume V (L): 5.66

t = 30.00 h

Biomass X (g/L): 92.86
Glucose S (g/L): 28.18
Lactic Acid P (g/L): 64.51
Maltose M (g/L): 18.06
Volume V (L): 5.60

t = 31.00 h

Biomass X (g/L): 85.16
Glucose S (g/L): 21.35
Lactic Acid P (g/L): 54.23
Maltose M (g/L): 16.74
Volume V (L): 5.54

t = 32.00 h

Biomass X (g/L): 77.23
Glucose S (g/L): 25.60
Lactic Acid P (g/L): 68.22
Maltose M (g/L): 16.80
Volume V (L): 5.57

t = 33.00 h

Biomass X (g/L): 93.14
Glucose S (g/L): 23.63
Lactic Acid P (g/L): 65.33
Maltose M (g/L): 16.81
Volume V (L): 5.65

t = 34.00 h

Biomass X (g/L): 94.93
Glucose S (g/L): 19.28
Lactic Acid P (g/L): 64.01
Maltose M (g/L): 18.03

Volume V (L): 5.75

t = 35.00 h

Biomass X (g/L): 84.96

Glucose S (g/L): 21.61

Lactic Acid P (g/L): 80.23

Maltose M (g/L): 17.43

Volume V (L): 5.88

t = 36.00 h

Biomass X (g/L): 89.39

Glucose S (g/L): 30.61

Lactic Acid P (g/L): 44.74

Maltose M (g/L): 16.73

Volume V (L): 5.78

t = 37.00 h

Biomass X (g/L): 83.13

Glucose S (g/L): 22.27

Lactic Acid P (g/L): 66.07

Maltose M (g/L): 17.22

Volume V (L): 5.81

t = 38.00 h

Biomass X (g/L): 78.06

Glucose S (g/L): 14.74

Lactic Acid P (g/L): 47.01

Maltose M (g/L): 18.37

Volume V (L): 5.85

t = 39.00 h

Biomass X (g/L): 84.47

Glucose S (g/L): 18.80

Lactic Acid P (g/L): 73.42

Maltose M (g/L): 17.80

Volume V (L): 5.95

t = 40.00 h

Biomass X (g/L): 90.57

Glucose S (g/L): 7.86

Lactic Acid P (g/L): 79.19

Maltose M (g/L): 18.39

Volume V (L): 6.00

t = 41.00 h

Biomass X (g/L): 74.80

Glucose S (g/L): 21.65

Lactic Acid P (g/L): 58.63

Maltose M (g/L): 16.88

Volume V (L): 5.74

t = 42.00 h

Biomass X (g/L): 85.79

Glucose S (g/L): 28.89

Lactic Acid P (g/L): 60.41

Maltose M (g/L): 18.81

Volume V (L): 5.92

t = 43.00 h

Biomass X (g/L): 91.20

Glucose S (g/L): 12.76

Lactic Acid P (g/L): 56.15

Maltose M (g/L): 17.27

Volume V (L): 6.15

t = 44.00 h

Biomass X (g/L): 102.72

Glucose S (g/L): 21.96

Lactic Acid P (g/L): 74.19

Maltose M (g/L): 16.21

Volume V (L): 5.82

t = 45.00 h

Biomass X (g/L): 87.28

Glucose S (g/L): 27.17

Lactic Acid P (g/L): 70.61

Maltose M (g/L): 16.74

Volume V (L): 6.04

t = 46.00 h

Biomass X (g/L): 77.73

Glucose S (g/L): 27.94

Lactic Acid P (g/L): 71.66

Maltose M (g/L): 15.11

Volume V (L): 6.02

t = 47.00 h

Biomass X (g/L): 88.46

Glucose S (g/L): 29.23

Lactic Acid P (g/L): 71.20

Maltose M (g/L): 16.05

Volume V (L): 5.99

t = 48.00 h

Biomass X (g/L): 84.82

Glucose S (g/L): 10.15

Lactic Acid P (g/L): 65.56

Maltose M (g/L): 16.77

Volume V (L): 6.31

t = 49.00 h

Biomass X (g/L): 96.37

Glucose S (g/L): 24.80

Lactic Acid P (g/L): 86.96

Maltose M (g/L): 16.86

Volume V (L): 6.12

t = 50.00 h

Biomass X (g/L): 87.20

Glucose S (g/L): 12.69

Lactic Acid P (g/L): 78.77

Maltose M (g/L): 15.60

Volume V (L): 6.35

t = 51.00 h

Biomass X (g/L): 62.74

Glucose S (g/L): 15.96

Lactic Acid P (g/L): 64.61

Maltose M (g/L): 19.03

Volume V (L): 6.16

t = 52.00 h

Biomass X (g/L): 128.70

Glucose S (g/L): 27.26
Lactic Acid P (g/L): 69.81
Maltose M (g/L): 15.64
Volume V (L): 6.25

t = 53.00 h

Biomass X (g/L): 92.09
Glucose S (g/L): 23.40
Lactic Acid P (g/L): 68.45
Maltose M (g/L): 13.84
Volume V (L): 6.13

t = 54.00 h

Biomass X (g/L): 91.50
Glucose S (g/L): 27.07
Lactic Acid P (g/L): 91.51
Maltose M (g/L): 16.99
Volume V (L): 6.15

t = 55.00 h

Biomass X (g/L): 74.64
Glucose S (g/L): 17.17
Lactic Acid P (g/L): 84.92
Maltose M (g/L): 16.21
Volume V (L): 6.31

t = 56.00 h

Biomass X (g/L): 75.36
Glucose S (g/L): 32.06
Lactic Acid P (g/L): 76.89
Maltose M (g/L): 15.68
Volume V (L): 6.33

t = 57.00 h

Biomass X (g/L): 89.43
Glucose S (g/L): 27.48
Lactic Acid P (g/L): 95.50
Maltose M (g/L): 14.80
Volume V (L): 6.38

t = 58.00 h

Biomass X (g/L): 93.76
Glucose S (g/L): 28.93
Lactic Acid P (g/L): 93.26
Maltose M (g/L): 15.67
Volume V (L): 6.37

t = 59.00 h

Biomass X (g/L): 97.44
Glucose S (g/L): 17.13
Lactic Acid P (g/L): 76.78
Maltose M (g/L): 16.95
Volume V (L): 6.16

t = 60.00 h

Biomass X (g/L): 92.55
Glucose S (g/L): 31.12
Lactic Acid P (g/L): 75.28
Maltose M (g/L): 15.26
Volume V (L): 6.26

t = 61.00 h

Biomass X (g/L): 77.77
Glucose S (g/L): 22.27
Lactic Acid P (g/L): 76.05
Maltose M (g/L): 13.28
Volume V (L): 6.51

t = 62.00 h

Biomass X (g/L): 64.07
Glucose S (g/L): 20.71
Lactic Acid P (g/L): 70.44
Maltose M (g/L): 10.96
Volume V (L): 6.15

t = 63.00 h

Biomass X (g/L): 88.72
Glucose S (g/L): 28.50
Lactic Acid P (g/L): 97.52
Maltose M (g/L): 15.19
Volume V (L): 6.48

t = 64.00 h

Biomass X (g/L): 84.12
Glucose S (g/L): 28.49
Lactic Acid P (g/L): 70.46
Maltose M (g/L): 13.56
Volume V (L): 6.44

t = 65.00 h

Biomass X (g/L): 76.48
Glucose S (g/L): 20.01
Lactic Acid P (g/L): 89.42
Maltose M (g/L): 14.47
Volume V (L): 6.60

t = 66.00 h

Biomass X (g/L): 97.89
Glucose S (g/L): 37.99
Lactic Acid P (g/L): 87.90
Maltose M (g/L): 13.58
Volume V (L): 6.56

t = 67.00 h

Biomass X (g/L): 72.97
Glucose S (g/L): 36.77
Lactic Acid P (g/L): 78.23
Maltose M (g/L): 13.42
Volume V (L): 6.44

t = 68.00 h

Biomass X (g/L): 95.42
Glucose S (g/L): 29.64
Lactic Acid P (g/L): 99.57
Maltose M (g/L): 13.46
Volume V (L): 6.45

t = 69.00 h

Biomass X (g/L): 74.37
Glucose S (g/L): 28.84
Lactic Acid P (g/L): 66.21
Maltose M (g/L): 13.48
Volume V (L): 6.50

t = 70.00 h

Biomass X (g/L): 90.13
Glucose S (g/L): 28.48
Lactic Acid P (g/L): 81.20
Maltose M (g/L): 15.03
Volume V (L): 6.67

t = 71.00 h

Biomass X (g/L): 98.21
Glucose S (g/L): 17.26
Lactic Acid P (g/L): 77.35
Maltose M (g/L): 14.72
Volume V (L): 6.30

t = 72.00 h

Biomass X (g/L): 84.93
Glucose S (g/L): 27.69
Lactic Acid P (g/L): 92.04
Maltose M (g/L): 12.64
Volume V (L): 6.66

t = 73.00 h

Biomass X (g/L): 105.66
Glucose S (g/L): 27.87
Lactic Acid P (g/L): 86.21
Maltose M (g/L): 12.71
Volume V (L): 6.54

t = 74.00 h

Biomass X (g/L): 107.84
Glucose S (g/L): 24.01
Lactic Acid P (g/L): 65.68
Maltose M (g/L): 14.27
Volume V (L): 6.71

t = 75.00 h

Biomass X (g/L): 90.34
Glucose S (g/L): 25.91
Lactic Acid P (g/L): 95.49
Maltose M (g/L): 14.95

Volume V (L): 6.76

Batch 77

Initial Conditions:

Batch: 77.00

X0 (g/L): 3.35

S0 (g/L): 95.27

M0 (g/L): 65.26

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.69

F_vol (L/h): 0.08

S_feed (g/L): 178.04

M_feed (g/L): 121.96

V_final (L): 11.08

Batch 77 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 101.01

Lactic Acid P (g/L): 5.18

Maltose M (g/L): 69.68

Volume V (L): 5.20

t = 1.00 h

Biomass X (g/L): 2.32

Glucose S (g/L): 93.58

Lactic Acid P (g/L): 5.27

Maltose M (g/L): 60.68

Volume V (L): 4.94

t = 2.00 h

Biomass X (g/L): 7.55

Glucose S (g/L): 95.33

Lactic Acid P (g/L): 10.12

Maltose M (g/L): 61.48

Volume V (L): 6.33

t = 3.00 h

Biomass X (g/L): 8.83

Glucose S (g/L): 96.70

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 65.95

Volume V (L): 5.65

t = 4.00 h

Biomass X (g/L): 9.21

Glucose S (g/L): 96.30

Lactic Acid P (g/L): 5.52

Maltose M (g/L): 62.37

Volume V (L): 5.10

t = 5.00 h

Biomass X (g/L): 1.49

Glucose S (g/L): 103.25

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 64.92
Volume V (L): 5.45

t = 6.00 h

Biomass X (g/L): 6.38
Glucose S (g/L): 94.52
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 61.20
Volume V (L): 5.80

t = 7.00 h

Biomass X (g/L): 7.26
Glucose S (g/L): 93.24
Lactic Acid P (g/L): 16.04
Maltose M (g/L): 63.11
Volume V (L): 5.49

t = 8.00 h

Biomass X (g/L): 30.19
Glucose S (g/L): 97.24
Lactic Acid P (g/L): 24.33
Maltose M (g/L): 66.47
Volume V (L): 5.71

t = 9.00 h

Biomass X (g/L): 28.15
Glucose S (g/L): 91.65
Lactic Acid P (g/L): 17.39
Maltose M (g/L): 59.30
Volume V (L): 5.33

t = 10.00 h

Biomass X (g/L): 33.29
Glucose S (g/L): 90.32
Lactic Acid P (g/L): 37.27
Maltose M (g/L): 64.08
Volume V (L): 4.99

t = 11.00 h

Biomass X (g/L): 21.75

Glucose S (g/L): 81.94
Lactic Acid P (g/L): 27.74
Maltose M (g/L): 62.93
Volume V (L): 5.94

t = 12.00 h

Biomass X (g/L): 42.28
Glucose S (g/L): 80.56
Lactic Acid P (g/L): 22.13
Maltose M (g/L): 57.54
Volume V (L): 6.04

t = 13.00 h

Biomass X (g/L): 52.84
Glucose S (g/L): 76.63
Lactic Acid P (g/L): 22.47
Maltose M (g/L): 61.93
Volume V (L): 6.31

t = 14.00 h

Biomass X (g/L): 32.68
Glucose S (g/L): 80.10
Lactic Acid P (g/L): 44.27
Maltose M (g/L): 60.45
Volume V (L): 5.85

t = 15.00 h

Biomass X (g/L): 48.87
Glucose S (g/L): 79.46
Lactic Acid P (g/L): 39.04
Maltose M (g/L): 60.14
Volume V (L): 5.56

t = 16.00 h

Biomass X (g/L): 42.06
Glucose S (g/L): 72.19
Lactic Acid P (g/L): 46.48
Maltose M (g/L): 60.35
Volume V (L): 5.83

t = 17.00 h

Biomass X (g/L): 56.12
Glucose S (g/L): 71.45
Lactic Acid P (g/L): 18.93
Maltose M (g/L): 58.41
Volume V (L): 6.89

t = 18.00 h

Biomass X (g/L): 54.11
Glucose S (g/L): 70.35
Lactic Acid P (g/L): 47.10
Maltose M (g/L): 59.03
Volume V (L): 6.14

t = 19.00 h

Biomass X (g/L): 68.91
Glucose S (g/L): 65.57
Lactic Acid P (g/L): 42.29
Maltose M (g/L): 59.07
Volume V (L): 6.33

t = 20.00 h

Biomass X (g/L): 69.00
Glucose S (g/L): 67.41
Lactic Acid P (g/L): 36.19
Maltose M (g/L): 59.89
Volume V (L): 6.47

t = 21.00 h

Biomass X (g/L): 81.79
Glucose S (g/L): 71.17
Lactic Acid P (g/L): 49.78
Maltose M (g/L): 54.28
Volume V (L): 6.40

t = 22.00 h

Biomass X (g/L): 70.48
Glucose S (g/L): 65.62
Lactic Acid P (g/L): 61.65
Maltose M (g/L): 60.46
Volume V (L): 6.45

t = 23.00 h

Biomass X (g/L): 75.82
Glucose S (g/L): 68.20
Lactic Acid P (g/L): 73.59
Maltose M (g/L): 54.45
Volume V (L): 6.44

t = 24.00 h

Biomass X (g/L): 68.52
Glucose S (g/L): 63.67
Lactic Acid P (g/L): 56.73
Maltose M (g/L): 58.55
Volume V (L): 7.63

t = 25.00 h

Biomass X (g/L): 64.48
Glucose S (g/L): 67.08
Lactic Acid P (g/L): 49.40
Maltose M (g/L): 56.62
Volume V (L): 7.36

t = 26.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 65.75
Lactic Acid P (g/L): 49.96
Maltose M (g/L): 53.13
Volume V (L): 7.72

t = 27.00 h

Biomass X (g/L): 89.31
Glucose S (g/L): 63.78
Lactic Acid P (g/L): 66.52
Maltose M (g/L): 56.99
Volume V (L): 7.89

t = 28.00 h

Biomass X (g/L): 84.99
Glucose S (g/L): 70.04
Lactic Acid P (g/L): 49.89
Maltose M (g/L): 53.24
Volume V (L): 6.13

t = 29.00 h

Biomass X (g/L): 80.21
Glucose S (g/L): 70.38
Lactic Acid P (g/L): 65.30
Maltose M (g/L): 57.57
Volume V (L): 7.52

t = 30.00 h

Biomass X (g/L): 70.21
Glucose S (g/L): 65.06
Lactic Acid P (g/L): 63.17
Maltose M (g/L): 53.12
Volume V (L): 7.15

t = 31.00 h

Biomass X (g/L): 91.34
Glucose S (g/L): 67.65
Lactic Acid P (g/L): 62.95
Maltose M (g/L): 50.97
Volume V (L): 8.06

t = 32.00 h

Biomass X (g/L): 86.45
Glucose S (g/L): 68.77
Lactic Acid P (g/L): 47.97
Maltose M (g/L): 57.66
Volume V (L): 7.23

t = 33.00 h

Biomass X (g/L): 100.95
Glucose S (g/L): 63.45
Lactic Acid P (g/L): 59.51
Maltose M (g/L): 54.20
Volume V (L): 7.16

t = 34.00 h

Biomass X (g/L): 70.59
Glucose S (g/L): 65.12
Lactic Acid P (g/L): 50.27
Maltose M (g/L): 55.47

Volume V (L): 7.06

t = 35.00 h

Biomass X (g/L): 90.51

Glucose S (g/L): 73.76

Lactic Acid P (g/L): 58.66

Maltose M (g/L): 51.16

Volume V (L): 7.69

t = 36.00 h

Biomass X (g/L): 87.82

Glucose S (g/L): 67.02

Lactic Acid P (g/L): 72.72

Maltose M (g/L): 48.43

Volume V (L): 8.36

t = 37.00 h

Biomass X (g/L): 79.48

Glucose S (g/L): 74.60

Lactic Acid P (g/L): 76.07

Maltose M (g/L): 56.38

Volume V (L): 8.12

t = 38.00 h

Biomass X (g/L): 96.27

Glucose S (g/L): 67.14

Lactic Acid P (g/L): 67.45

Maltose M (g/L): 52.17

Volume V (L): 7.50

t = 39.00 h

Biomass X (g/L): 87.17

Glucose S (g/L): 73.78

Lactic Acid P (g/L): 60.90

Maltose M (g/L): 53.01

Volume V (L): 9.21

t = 40.00 h

Biomass X (g/L): 74.39

Glucose S (g/L): 71.35

Lactic Acid P (g/L): 80.70

Maltose M (g/L): 53.08

Volume V (L): 8.56

t = 41.00 h

Biomass X (g/L): 82.48

Glucose S (g/L): 72.32

Lactic Acid P (g/L): 61.66

Maltose M (g/L): 50.48

Volume V (L): 8.70

t = 42.00 h

Biomass X (g/L): 74.56

Glucose S (g/L): 79.07

Lactic Acid P (g/L): 57.12

Maltose M (g/L): 51.47

Volume V (L): 9.03

t = 43.00 h

Biomass X (g/L): 79.32

Glucose S (g/L): 70.12

Lactic Acid P (g/L): 73.93

Maltose M (g/L): 54.34

Volume V (L): 7.69

t = 44.00 h

Biomass X (g/L): 90.77

Glucose S (g/L): 76.34

Lactic Acid P (g/L): 72.16

Maltose M (g/L): 50.27

Volume V (L): 8.12

t = 45.00 h

Biomass X (g/L): 93.59

Glucose S (g/L): 76.79

Lactic Acid P (g/L): 63.55

Maltose M (g/L): 52.23

Volume V (L): 9.05

t = 46.00 h

Biomass X (g/L): 86.20

Glucose S (g/L): 85.93

Lactic Acid P (g/L): 69.65

Maltose M (g/L): 45.13

Volume V (L): 8.34

t = 47.00 h

Biomass X (g/L): 76.81

Glucose S (g/L): 72.97

Lactic Acid P (g/L): 73.04

Maltose M (g/L): 52.06

Volume V (L): 8.68

t = 48.00 h

Biomass X (g/L): 87.78

Glucose S (g/L): 74.93

Lactic Acid P (g/L): 71.21

Maltose M (g/L): 45.63

Volume V (L): 8.85

t = 49.00 h

Biomass X (g/L): 91.09

Glucose S (g/L): 79.93

Lactic Acid P (g/L): 73.79

Maltose M (g/L): 51.21

Volume V (L): 8.35

t = 50.00 h

Biomass X (g/L): 64.95

Glucose S (g/L): 77.42

Lactic Acid P (g/L): 65.43

Maltose M (g/L): 49.49

Volume V (L): 8.88

t = 51.00 h

Biomass X (g/L): 85.15

Glucose S (g/L): 77.21

Lactic Acid P (g/L): 74.95

Maltose M (g/L): 48.07

Volume V (L): 9.58

t = 52.00 h

Biomass X (g/L): 82.14

Glucose S (g/L): 82.14
Lactic Acid P (g/L): 70.66
Maltose M (g/L): 47.78
Volume V (L): 9.57

t = 53.00 h

Biomass X (g/L): 93.16
Glucose S (g/L): 78.99
Lactic Acid P (g/L): 66.52
Maltose M (g/L): 47.15
Volume V (L): 9.84

t = 54.00 h

Biomass X (g/L): 87.59
Glucose S (g/L): 77.67
Lactic Acid P (g/L): 68.79
Maltose M (g/L): 45.55
Volume V (L): 9.87

t = 55.00 h

Biomass X (g/L): 88.47
Glucose S (g/L): 83.20
Lactic Acid P (g/L): 70.36
Maltose M (g/L): 46.70
Volume V (L): 9.66

t = 56.00 h

Biomass X (g/L): 79.03
Glucose S (g/L): 82.72
Lactic Acid P (g/L): 58.50
Maltose M (g/L): 52.02
Volume V (L): 9.70

t = 57.00 h

Biomass X (g/L): 83.62
Glucose S (g/L): 81.20
Lactic Acid P (g/L): 65.98
Maltose M (g/L): 49.65
Volume V (L): 9.75

t = 58.00 h

Biomass X (g/L): 92.64
Glucose S (g/L): 81.64
Lactic Acid P (g/L): 72.48
Maltose M (g/L): 44.34
Volume V (L): 9.40

t = 59.00 h

Biomass X (g/L): 82.77
Glucose S (g/L): 80.52
Lactic Acid P (g/L): 59.99
Maltose M (g/L): 46.71
Volume V (L): 9.28

t = 60.00 h

Biomass X (g/L): 84.97
Glucose S (g/L): 86.11
Lactic Acid P (g/L): 66.74
Maltose M (g/L): 46.05
Volume V (L): 9.53

t = 61.00 h

Biomass X (g/L): 68.28
Glucose S (g/L): 83.36
Lactic Acid P (g/L): 93.87
Maltose M (g/L): 45.23
Volume V (L): 9.79

t = 62.00 h

Biomass X (g/L): 82.37
Glucose S (g/L): 82.68
Lactic Acid P (g/L): 62.56
Maltose M (g/L): 42.98
Volume V (L): 9.71

t = 63.00 h

Biomass X (g/L): 63.41
Glucose S (g/L): 81.67
Lactic Acid P (g/L): 69.42
Maltose M (g/L): 46.08
Volume V (L): 10.31

t = 64.00 h

Biomass X (g/L): 77.77
Glucose S (g/L): 87.36
Lactic Acid P (g/L): 65.57
Maltose M (g/L): 46.75
Volume V (L): 10.29

t = 65.00 h

Biomass X (g/L): 83.24
Glucose S (g/L): 85.17
Lactic Acid P (g/L): 72.17
Maltose M (g/L): 42.60
Volume V (L): 10.42

t = 66.00 h

Biomass X (g/L): 89.92
Glucose S (g/L): 81.12
Lactic Acid P (g/L): 66.08
Maltose M (g/L): 48.81
Volume V (L): 10.52

t = 67.00 h

Biomass X (g/L): 72.66
Glucose S (g/L): 84.15
Lactic Acid P (g/L): 71.36
Maltose M (g/L): 47.55
Volume V (L): 11.01

t = 68.00 h

Biomass X (g/L): 85.81
Glucose S (g/L): 87.66
Lactic Acid P (g/L): 67.26
Maltose M (g/L): 45.05
Volume V (L): 10.51

t = 69.00 h

Biomass X (g/L): 96.26
Glucose S (g/L): 88.28
Lactic Acid P (g/L): 83.32
Maltose M (g/L): 40.68
Volume V (L): 11.09

t = 70.00 h

Biomass X (g/L): 75.67
Glucose S (g/L): 88.70
Lactic Acid P (g/L): 74.74
Maltose M (g/L): 44.22
Volume V (L): 10.93

t = 71.00 h

Biomass X (g/L): 66.73
Glucose S (g/L): 88.81
Lactic Acid P (g/L): 65.16
Maltose M (g/L): 43.70
Volume V (L): 11.31

t = 72.00 h

Biomass X (g/L): 84.36
Glucose S (g/L): 86.00
Lactic Acid P (g/L): 81.84
Maltose M (g/L): 43.99
Volume V (L): 11.62

t = 73.00 h

Biomass X (g/L): 69.84
Glucose S (g/L): 88.36
Lactic Acid P (g/L): 65.85
Maltose M (g/L): 46.25
Volume V (L): 11.20

t = 74.00 h

Biomass X (g/L): 73.29
Glucose S (g/L): 95.17
Lactic Acid P (g/L): 60.55
Maltose M (g/L): 41.92
Volume V (L): 11.05

t = 75.00 h

Biomass X (g/L): 90.00
Glucose S (g/L): 95.23
Lactic Acid P (g/L): 74.68
Maltose M (g/L): 46.84

Volume V (L): 11.13

Batch 78

Initial Conditions:

Batch: 78.00

X0 (g/L): 3.14

S0 (g/L): 116.73

M0 (g/L): 56.13

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.48

F_vol (L/h): 0.05

S_feed (g/L): 202.59

M_feed (g/L): 97.41

V_final (L): 9.05

Batch 78 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 120.66
Lactic Acid P (g/L): 16.87
Maltose M (g/L): 55.75
Volume V (L): 5.13

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 124.25
Lactic Acid P (g/L): 14.75
Maltose M (g/L): 57.90
Volume V (L): 4.73

t = 2.00 h

Biomass X (g/L): 14.80
Glucose S (g/L): 117.14
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 56.64
Volume V (L): 5.07

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 122.54
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 52.83
Volume V (L): 5.01

t = 4.00 h

Biomass X (g/L): 26.27
Glucose S (g/L): 117.87
Lactic Acid P (g/L): 6.29
Maltose M (g/L): 52.06
Volume V (L): 5.02

t = 5.00 h

Biomass X (g/L): 11.50
Glucose S (g/L): 113.76

Lactic Acid P (g/L): 7.69
Maltose M (g/L): 48.63
Volume V (L): 5.57

t = 6.00 h

Biomass X (g/L): 45.20
Glucose S (g/L): 111.36
Lactic Acid P (g/L): 12.75
Maltose M (g/L): 51.40
Volume V (L): 6.14

t = 7.00 h

Biomass X (g/L): 13.51
Glucose S (g/L): 107.78
Lactic Acid P (g/L): 19.60
Maltose M (g/L): 47.45
Volume V (L): 5.74

t = 8.00 h

Biomass X (g/L): 27.32
Glucose S (g/L): 106.59
Lactic Acid P (g/L): 20.09
Maltose M (g/L): 56.77
Volume V (L): 5.42

t = 9.00 h

Biomass X (g/L): 25.88
Glucose S (g/L): 111.38
Lactic Acid P (g/L): 8.87
Maltose M (g/L): 48.65
Volume V (L): 5.90

t = 10.00 h

Biomass X (g/L): 26.16
Glucose S (g/L): 106.53
Lactic Acid P (g/L): 25.50
Maltose M (g/L): 53.22
Volume V (L): 6.10

t = 11.00 h

Biomass X (g/L): 49.45

Glucose S (g/L): 95.21
Lactic Acid P (g/L): 34.19
Maltose M (g/L): 54.55
Volume V (L): 5.14

t = 12.00 h

Biomass X (g/L): 32.22
Glucose S (g/L): 90.01
Lactic Acid P (g/L): 22.88
Maltose M (g/L): 49.89
Volume V (L): 6.05

t = 13.00 h

Biomass X (g/L): 40.81
Glucose S (g/L): 89.76
Lactic Acid P (g/L): 35.66
Maltose M (g/L): 48.58
Volume V (L): 5.95

t = 14.00 h

Biomass X (g/L): 57.46
Glucose S (g/L): 88.48
Lactic Acid P (g/L): 30.78
Maltose M (g/L): 44.92
Volume V (L): 6.14

t = 15.00 h

Biomass X (g/L): 62.74
Glucose S (g/L): 88.03
Lactic Acid P (g/L): 57.38
Maltose M (g/L): 46.57
Volume V (L): 6.07

t = 16.00 h

Biomass X (g/L): 44.67
Glucose S (g/L): 92.04
Lactic Acid P (g/L): 29.06
Maltose M (g/L): 51.58
Volume V (L): 5.79

t = 17.00 h

Biomass X (g/L): 67.52
Glucose S (g/L): 82.94
Lactic Acid P (g/L): 54.63
Maltose M (g/L): 47.87
Volume V (L): 6.58

t = 18.00 h

Biomass X (g/L): 65.74
Glucose S (g/L): 90.59
Lactic Acid P (g/L): 30.53
Maltose M (g/L): 46.58
Volume V (L): 6.08

t = 19.00 h

Biomass X (g/L): 57.73
Glucose S (g/L): 75.09
Lactic Acid P (g/L): 64.86
Maltose M (g/L): 49.48
Volume V (L): 5.89

t = 20.00 h

Biomass X (g/L): 61.26
Glucose S (g/L): 79.43
Lactic Acid P (g/L): 39.41
Maltose M (g/L): 47.45
Volume V (L): 6.25

t = 21.00 h

Biomass X (g/L): 50.78
Glucose S (g/L): 83.13
Lactic Acid P (g/L): 53.54
Maltose M (g/L): 44.84
Volume V (L): 6.45

t = 22.00 h

Biomass X (g/L): 90.20
Glucose S (g/L): 82.51
Lactic Acid P (g/L): 38.19
Maltose M (g/L): 45.37
Volume V (L): 6.22

t = 23.00 h

Biomass X (g/L): 84.78
Glucose S (g/L): 76.84
Lactic Acid P (g/L): 42.26
Maltose M (g/L): 42.41
Volume V (L): 6.37

t = 24.00 h

Biomass X (g/L): 63.12
Glucose S (g/L): 81.84
Lactic Acid P (g/L): 64.56
Maltose M (g/L): 42.54
Volume V (L): 6.33

t = 25.00 h

Biomass X (g/L): 85.35
Glucose S (g/L): 80.58
Lactic Acid P (g/L): 67.03
Maltose M (g/L): 43.07
Volume V (L): 6.14

t = 26.00 h

Biomass X (g/L): 72.75
Glucose S (g/L): 73.96
Lactic Acid P (g/L): 60.84
Maltose M (g/L): 44.67
Volume V (L): 6.14

t = 27.00 h

Biomass X (g/L): 88.72
Glucose S (g/L): 77.85
Lactic Acid P (g/L): 66.59
Maltose M (g/L): 39.60
Volume V (L): 6.00

t = 28.00 h

Biomass X (g/L): 75.50
Glucose S (g/L): 79.29
Lactic Acid P (g/L): 70.42
Maltose M (g/L): 42.67
Volume V (L): 7.01

t = 29.00 h

Biomass X (g/L): 77.67
Glucose S (g/L): 73.67
Lactic Acid P (g/L): 55.24
Maltose M (g/L): 41.09
Volume V (L): 6.50

t = 30.00 h

Biomass X (g/L): 79.16
Glucose S (g/L): 83.27
Lactic Acid P (g/L): 49.45
Maltose M (g/L): 39.41
Volume V (L): 5.51

t = 31.00 h

Biomass X (g/L): 76.12
Glucose S (g/L): 72.69
Lactic Acid P (g/L): 63.18
Maltose M (g/L): 40.39
Volume V (L): 6.80

t = 32.00 h

Biomass X (g/L): 97.34
Glucose S (g/L): 76.85
Lactic Acid P (g/L): 58.69
Maltose M (g/L): 35.02
Volume V (L): 6.85

t = 33.00 h

Biomass X (g/L): 92.84
Glucose S (g/L): 71.12
Lactic Acid P (g/L): 68.41
Maltose M (g/L): 38.74
Volume V (L): 6.18

t = 34.00 h

Biomass X (g/L): 85.40
Glucose S (g/L): 76.93
Lactic Acid P (g/L): 70.57
Maltose M (g/L): 37.51

Volume V (L): 6.57

t = 35.00 h

Biomass X (g/L): 94.98

Glucose S (g/L): 72.22

Lactic Acid P (g/L): 59.75

Maltose M (g/L): 39.31

Volume V (L): 6.61

t = 36.00 h

Biomass X (g/L): 68.74

Glucose S (g/L): 75.67

Lactic Acid P (g/L): 73.26

Maltose M (g/L): 33.79

Volume V (L): 6.71

t = 37.00 h

Biomass X (g/L): 73.05

Glucose S (g/L): 70.50

Lactic Acid P (g/L): 73.36

Maltose M (g/L): 39.88

Volume V (L): 7.39

t = 38.00 h

Biomass X (g/L): 90.41

Glucose S (g/L): 80.76

Lactic Acid P (g/L): 74.54

Maltose M (g/L): 41.43

Volume V (L): 7.01

t = 39.00 h

Biomass X (g/L): 93.26

Glucose S (g/L): 79.20

Lactic Acid P (g/L): 64.51

Maltose M (g/L): 38.46

Volume V (L): 6.97

t = 40.00 h

Biomass X (g/L): 88.88

Glucose S (g/L): 82.57

Lactic Acid P (g/L): 76.06

Maltose M (g/L): 39.75

Volume V (L): 7.53

t = 41.00 h

Biomass X (g/L): 67.37

Glucose S (g/L): 76.89

Lactic Acid P (g/L): 74.10

Maltose M (g/L): 36.83

Volume V (L): 7.38

t = 42.00 h

Biomass X (g/L): 75.73

Glucose S (g/L): 83.44

Lactic Acid P (g/L): 75.66

Maltose M (g/L): 33.83

Volume V (L): 6.71

t = 43.00 h

Biomass X (g/L): 90.67

Glucose S (g/L): 79.22

Lactic Acid P (g/L): 85.74

Maltose M (g/L): 37.58

Volume V (L): 7.12

t = 44.00 h

Biomass X (g/L): 110.92

Glucose S (g/L): 79.90

Lactic Acid P (g/L): 73.56

Maltose M (g/L): 39.05

Volume V (L): 7.48

t = 45.00 h

Biomass X (g/L): 98.86

Glucose S (g/L): 88.66

Lactic Acid P (g/L): 65.24

Maltose M (g/L): 33.57

Volume V (L): 7.13

t = 46.00 h

Biomass X (g/L): 102.50

Glucose S (g/L): 78.18

Lactic Acid P (g/L): 60.91
Maltose M (g/L): 35.52
Volume V (L): 8.15

t = 47.00 h

Biomass X (g/L): 90.11
Glucose S (g/L): 85.75
Lactic Acid P (g/L): 82.11
Maltose M (g/L): 35.82
Volume V (L): 7.58

t = 48.00 h

Biomass X (g/L): 90.05
Glucose S (g/L): 83.03
Lactic Acid P (g/L): 75.27
Maltose M (g/L): 43.58
Volume V (L): 7.35

t = 49.00 h

Biomass X (g/L): 76.33
Glucose S (g/L): 79.04
Lactic Acid P (g/L): 87.82
Maltose M (g/L): 36.50
Volume V (L): 7.28

t = 50.00 h

Biomass X (g/L): 96.22
Glucose S (g/L): 76.35
Lactic Acid P (g/L): 53.65
Maltose M (g/L): 30.72
Volume V (L): 8.00

t = 51.00 h

Biomass X (g/L): 85.90
Glucose S (g/L): 77.96
Lactic Acid P (g/L): 81.34
Maltose M (g/L): 29.82
Volume V (L): 7.55

t = 52.00 h

Biomass X (g/L): 93.52

Glucose S (g/L): 87.44
Lactic Acid P (g/L): 70.71
Maltose M (g/L): 37.04
Volume V (L): 7.93

t = 53.00 h

Biomass X (g/L): 86.81
Glucose S (g/L): 90.40
Lactic Acid P (g/L): 70.12
Maltose M (g/L): 41.35
Volume V (L): 7.49

t = 54.00 h

Biomass X (g/L): 89.23
Glucose S (g/L): 83.96
Lactic Acid P (g/L): 66.56
Maltose M (g/L): 33.92
Volume V (L): 7.94

t = 55.00 h

Biomass X (g/L): 85.46
Glucose S (g/L): 84.62
Lactic Acid P (g/L): 82.32
Maltose M (g/L): 33.53
Volume V (L): 7.77

t = 56.00 h

Biomass X (g/L): 88.49
Glucose S (g/L): 87.70
Lactic Acid P (g/L): 79.90
Maltose M (g/L): 35.29
Volume V (L): 7.93

t = 57.00 h

Biomass X (g/L): 96.16
Glucose S (g/L): 85.20
Lactic Acid P (g/L): 63.77
Maltose M (g/L): 31.86
Volume V (L): 7.82

t = 58.00 h

Biomass X (g/L): 73.31
Glucose S (g/L): 89.94
Lactic Acid P (g/L): 76.18
Maltose M (g/L): 27.10
Volume V (L): 8.47

t = 59.00 h

Biomass X (g/L): 70.92
Glucose S (g/L): 87.00
Lactic Acid P (g/L): 81.45
Maltose M (g/L): 36.16
Volume V (L): 8.03

t = 60.00 h

Biomass X (g/L): 90.03
Glucose S (g/L): 86.78
Lactic Acid P (g/L): 76.74
Maltose M (g/L): 31.65
Volume V (L): 8.20

t = 61.00 h

Biomass X (g/L): 100.07
Glucose S (g/L): 83.73
Lactic Acid P (g/L): 85.61
Maltose M (g/L): 35.75
Volume V (L): 8.27

t = 62.00 h

Biomass X (g/L): 102.45
Glucose S (g/L): 83.14
Lactic Acid P (g/L): 72.79
Maltose M (g/L): 30.83
Volume V (L): 8.88

t = 63.00 h

Biomass X (g/L): 73.04
Glucose S (g/L): 74.31
Lactic Acid P (g/L): 54.68
Maltose M (g/L): 32.90
Volume V (L): 8.70

t = 64.00 h

Biomass X (g/L): 81.13
Glucose S (g/L): 93.13
Lactic Acid P (g/L): 77.62
Maltose M (g/L): 34.59
Volume V (L): 8.72

t = 65.00 h

Biomass X (g/L): 70.17
Glucose S (g/L): 87.79
Lactic Acid P (g/L): 77.74
Maltose M (g/L): 34.93
Volume V (L): 8.00

t = 66.00 h

Biomass X (g/L): 79.32
Glucose S (g/L): 87.16
Lactic Acid P (g/L): 79.14
Maltose M (g/L): 32.55
Volume V (L): 8.36

t = 67.00 h

Biomass X (g/L): 95.93
Glucose S (g/L): 81.11
Lactic Acid P (g/L): 86.73
Maltose M (g/L): 32.22
Volume V (L): 9.01

t = 68.00 h

Biomass X (g/L): 84.37
Glucose S (g/L): 87.02
Lactic Acid P (g/L): 58.39
Maltose M (g/L): 33.15
Volume V (L): 8.30

t = 69.00 h

Biomass X (g/L): 80.57
Glucose S (g/L): 86.90
Lactic Acid P (g/L): 74.52
Maltose M (g/L): 32.54
Volume V (L): 8.76

t = 70.00 h

Biomass X (g/L): 89.60
Glucose S (g/L): 87.23
Lactic Acid P (g/L): 68.31
Maltose M (g/L): 27.79
Volume V (L): 8.44

t = 71.00 h

Biomass X (g/L): 75.98
Glucose S (g/L): 92.37
Lactic Acid P (g/L): 65.81
Maltose M (g/L): 27.64
Volume V (L): 8.53

t = 72.00 h

Biomass X (g/L): 99.74
Glucose S (g/L): 91.91
Lactic Acid P (g/L): 76.54
Maltose M (g/L): 25.28
Volume V (L): 9.44

t = 73.00 h

Biomass X (g/L): 83.65
Glucose S (g/L): 86.13
Lactic Acid P (g/L): 83.90
Maltose M (g/L): 32.55
Volume V (L): 9.18

t = 74.00 h

Biomass X (g/L): 79.36
Glucose S (g/L): 86.81
Lactic Acid P (g/L): 80.31
Maltose M (g/L): 29.34
Volume V (L): 9.32

t = 75.00 h

Biomass X (g/L): 89.08
Glucose S (g/L): 87.21
Lactic Acid P (g/L): 71.24
Maltose M (g/L): 32.31

Volume V (L): 9.31

Batch 79

Initial Conditions:

Batch: 79.00

X0 (g/L): 3.59

S0 (g/L): 23.24

M0 (g/L): 80.39

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 3.46

F_vol (L/h): 0.03

S_feed (g/L): 67.29

M_feed (g/L): 232.71

V_final (L): 7.26

Batch 79 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 2.32

Glucose S (g/L): 26.40

Lactic Acid P (g/L): 8.86

Maltose M (g/L): 82.33

Volume V (L): 5.09

t = 1.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 23.96

Lactic Acid P (g/L): 2.17

Maltose M (g/L): 78.86

Volume V (L): 4.91

t = 2.00 h

Biomass X (g/L): 2.33

Glucose S (g/L): 24.53

Lactic Acid P (g/L): 16.21

Maltose M (g/L): 80.23

Volume V (L): 5.00

t = 3.00 h

Biomass X (g/L): 5.92

Glucose S (g/L): 20.27

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 72.55

Volume V (L): 5.17

t = 4.00 h

Biomass X (g/L): 25.93

Glucose S (g/L): 24.47

Lactic Acid P (g/L): 8.75

Maltose M (g/L): 79.23

Volume V (L): 5.29

t = 5.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 23.27

Lactic Acid P (g/L): 4.06
Maltose M (g/L): 80.52
Volume V (L): 4.83

t = 6.00 h

Biomass X (g/L): 9.87
Glucose S (g/L): 19.56
Lactic Acid P (g/L): 20.40
Maltose M (g/L): 71.50
Volume V (L): 5.29

t = 7.00 h

Biomass X (g/L): 9.72
Glucose S (g/L): 21.01
Lactic Acid P (g/L): 8.83
Maltose M (g/L): 77.23
Volume V (L): 5.13

t = 8.00 h

Biomass X (g/L): 31.09
Glucose S (g/L): 18.20
Lactic Acid P (g/L): 20.58
Maltose M (g/L): 79.51
Volume V (L): 5.44

t = 9.00 h

Biomass X (g/L): 41.91
Glucose S (g/L): 16.61
Lactic Acid P (g/L): 6.31
Maltose M (g/L): 79.94
Volume V (L): 4.94

t = 10.00 h

Biomass X (g/L): 31.39
Glucose S (g/L): 15.06
Lactic Acid P (g/L): 14.76
Maltose M (g/L): 72.98
Volume V (L): 5.74

t = 11.00 h

Biomass X (g/L): 35.66

Glucose S (g/L): 11.96
Lactic Acid P (g/L): 20.30
Maltose M (g/L): 76.20
Volume V (L): 5.17

t = 12.00 h

Biomass X (g/L): 36.28
Glucose S (g/L): 11.56
Lactic Acid P (g/L): 34.61
Maltose M (g/L): 77.57
Volume V (L): 5.62

t = 13.00 h

Biomass X (g/L): 42.56
Glucose S (g/L): 9.12
Lactic Acid P (g/L): 6.74
Maltose M (g/L): 74.03
Volume V (L): 5.38

t = 14.00 h

Biomass X (g/L): 48.46
Glucose S (g/L): 8.82
Lactic Acid P (g/L): 25.23
Maltose M (g/L): 76.11
Volume V (L): 5.27

t = 15.00 h

Biomass X (g/L): 47.82
Glucose S (g/L): 5.46
Lactic Acid P (g/L): 26.93
Maltose M (g/L): 71.72
Volume V (L): 5.50

t = 16.00 h

Biomass X (g/L): 50.73
Glucose S (g/L): 3.34
Lactic Acid P (g/L): 13.77
Maltose M (g/L): 72.48
Volume V (L): 5.67

t = 17.00 h

Biomass X (g/L): 55.22
Glucose S (g/L): 3.34
Lactic Acid P (g/L): 29.49
Maltose M (g/L): 69.00
Volume V (L): 5.23

t = 18.00 h

Biomass X (g/L): 33.64
Glucose S (g/L): 7.36
Lactic Acid P (g/L): 21.53
Maltose M (g/L): 69.86
Volume V (L): 5.67

t = 19.00 h

Biomass X (g/L): 51.45
Glucose S (g/L): 5.93
Lactic Acid P (g/L): 15.96
Maltose M (g/L): 71.13
Volume V (L): 5.57

t = 20.00 h

Biomass X (g/L): 55.61
Glucose S (g/L): 6.20
Lactic Acid P (g/L): 35.27
Maltose M (g/L): 67.34
Volume V (L): 5.71

t = 21.00 h

Biomass X (g/L): 57.90
Glucose S (g/L): 6.17
Lactic Acid P (g/L): 36.12
Maltose M (g/L): 68.27
Volume V (L): 5.63

t = 22.00 h

Biomass X (g/L): 53.73
Glucose S (g/L): 2.34
Lactic Acid P (g/L): 37.66
Maltose M (g/L): 70.70
Volume V (L): 5.56

t = 23.00 h

Biomass X (g/L): 57.87
Glucose S (g/L): 3.90
Lactic Acid P (g/L): 43.05
Maltose M (g/L): 74.03
Volume V (L): 5.49

t = 24.00 h

Biomass X (g/L): 50.87
Glucose S (g/L): 3.41
Lactic Acid P (g/L): 34.15
Maltose M (g/L): 61.55
Volume V (L): 5.84

t = 25.00 h

Biomass X (g/L): 42.24
Glucose S (g/L): 4.65
Lactic Acid P (g/L): 58.47
Maltose M (g/L): 72.93
Volume V (L): 5.87

t = 26.00 h

Biomass X (g/L): 68.38
Glucose S (g/L): 2.66
Lactic Acid P (g/L): 43.46
Maltose M (g/L): 65.93
Volume V (L): 5.90

t = 27.00 h

Biomass X (g/L): 49.96
Glucose S (g/L): 3.52
Lactic Acid P (g/L): 55.92
Maltose M (g/L): 69.09
Volume V (L): 5.89

t = 28.00 h

Biomass X (g/L): 59.74
Glucose S (g/L): 3.53
Lactic Acid P (g/L): 30.88
Maltose M (g/L): 69.25
Volume V (L): 6.01

t = 29.00 h

Biomass X (g/L): 55.22
Glucose S (g/L): 4.57
Lactic Acid P (g/L): 39.15
Maltose M (g/L): 70.02
Volume V (L): 6.13

t = 30.00 h

Biomass X (g/L): 77.38
Glucose S (g/L): 4.91
Lactic Acid P (g/L): 53.31
Maltose M (g/L): 62.88
Volume V (L): 5.56

t = 31.00 h

Biomass X (g/L): 62.96
Glucose S (g/L): 3.79
Lactic Acid P (g/L): 46.18
Maltose M (g/L): 63.95
Volume V (L): 5.51

t = 32.00 h

Biomass X (g/L): 61.63
Glucose S (g/L): 3.95
Lactic Acid P (g/L): 39.61
Maltose M (g/L): 64.06
Volume V (L): 6.28

t = 33.00 h

Biomass X (g/L): 72.05
Glucose S (g/L): 4.26
Lactic Acid P (g/L): 33.73
Maltose M (g/L): 62.77
Volume V (L): 5.75

t = 34.00 h

Biomass X (g/L): 85.64
Glucose S (g/L): 7.38
Lactic Acid P (g/L): 57.02
Maltose M (g/L): 67.30

Volume V (L): 5.83

t = 35.00 h

Biomass X (g/L): 70.36

Glucose S (g/L): 0.32

Lactic Acid P (g/L): 57.09

Maltose M (g/L): 62.82

Volume V (L): 6.00

t = 36.00 h

Biomass X (g/L): 63.89

Glucose S (g/L): 2.06

Lactic Acid P (g/L): 54.29

Maltose M (g/L): 57.92

Volume V (L): 5.79

t = 37.00 h

Biomass X (g/L): 74.35

Glucose S (g/L): 5.67

Lactic Acid P (g/L): 48.59

Maltose M (g/L): 60.44

Volume V (L): 6.05

t = 38.00 h

Biomass X (g/L): 54.27

Glucose S (g/L): 4.94

Lactic Acid P (g/L): 43.94

Maltose M (g/L): 57.30

Volume V (L): 5.97

t = 39.00 h

Biomass X (g/L): 69.67

Glucose S (g/L): 5.06

Lactic Acid P (g/L): 53.94

Maltose M (g/L): 60.90

Volume V (L): 6.13

t = 40.00 h

Biomass X (g/L): 88.05

Glucose S (g/L): 6.06

Lactic Acid P (g/L): 43.62

Maltose M (g/L): 63.18

Volume V (L): 6.26

t = 41.00 h

Biomass X (g/L): 71.86

Glucose S (g/L): 5.24

Lactic Acid P (g/L): 49.26

Maltose M (g/L): 58.93

Volume V (L): 6.17

t = 42.00 h

Biomass X (g/L): 71.95

Glucose S (g/L): 3.75

Lactic Acid P (g/L): 67.54

Maltose M (g/L): 59.32

Volume V (L): 6.19

t = 43.00 h

Biomass X (g/L): 71.69

Glucose S (g/L): 6.44

Lactic Acid P (g/L): 41.02

Maltose M (g/L): 59.35

Volume V (L): 6.34

t = 44.00 h

Biomass X (g/L): 68.41

Glucose S (g/L): 5.31

Lactic Acid P (g/L): 63.79

Maltose M (g/L): 60.23

Volume V (L): 6.51

t = 45.00 h

Biomass X (g/L): 75.38

Glucose S (g/L): 4.49

Lactic Acid P (g/L): 71.58

Maltose M (g/L): 61.00

Volume V (L): 6.65

t = 46.00 h

Biomass X (g/L): 80.06

Glucose S (g/L): 7.09

Lactic Acid P (g/L): 66.29

Maltose M (g/L): 61.52

Volume V (L): 6.35

t = 47.00 h

Biomass X (g/L): 70.41

Glucose S (g/L): 7.19

Lactic Acid P (g/L): 63.27

Maltose M (g/L): 65.44

Volume V (L): 6.75

t = 48.00 h

Biomass X (g/L): 51.14

Glucose S (g/L): 6.45

Lactic Acid P (g/L): 56.35

Maltose M (g/L): 58.86

Volume V (L): 6.61

t = 49.00 h

Biomass X (g/L): 78.55

Glucose S (g/L): 9.84

Lactic Acid P (g/L): 55.36

Maltose M (g/L): 60.34

Volume V (L): 6.25

t = 50.00 h

Biomass X (g/L): 94.17

Glucose S (g/L): 9.22

Lactic Acid P (g/L): 41.94

Maltose M (g/L): 53.20

Volume V (L): 6.28

t = 51.00 h

Biomass X (g/L): 71.64

Glucose S (g/L): 8.74

Lactic Acid P (g/L): 62.50

Maltose M (g/L): 56.44

Volume V (L): 6.43

t = 52.00 h

Biomass X (g/L): 62.01

Glucose S (g/L): 12.28
Lactic Acid P (g/L): 59.22
Maltose M (g/L): 56.45
Volume V (L): 6.83

t = 53.00 h

Biomass X (g/L): 81.97
Glucose S (g/L): 6.74
Lactic Acid P (g/L): 57.19
Maltose M (g/L): 53.95
Volume V (L): 6.51

t = 54.00 h

Biomass X (g/L): 92.04
Glucose S (g/L): 6.30
Lactic Acid P (g/L): 63.66
Maltose M (g/L): 55.06
Volume V (L): 6.29

t = 55.00 h

Biomass X (g/L): 81.26
Glucose S (g/L): 9.35
Lactic Acid P (g/L): 64.91
Maltose M (g/L): 56.02
Volume V (L): 6.62

t = 56.00 h

Biomass X (g/L): 92.47
Glucose S (g/L): 7.23
Lactic Acid P (g/L): 59.33
Maltose M (g/L): 54.56
Volume V (L): 6.33

t = 57.00 h

Biomass X (g/L): 84.25
Glucose S (g/L): 8.04
Lactic Acid P (g/L): 79.50
Maltose M (g/L): 63.39
Volume V (L): 6.53

t = 58.00 h

Biomass X (g/L): 83.72
Glucose S (g/L): 6.17
Lactic Acid P (g/L): 72.28
Maltose M (g/L): 56.97
Volume V (L): 6.64

t = 59.00 h

Biomass X (g/L): 72.79
Glucose S (g/L): 9.81
Lactic Acid P (g/L): 74.21
Maltose M (g/L): 61.05
Volume V (L): 6.81

t = 60.00 h

Biomass X (g/L): 80.35
Glucose S (g/L): 5.67
Lactic Acid P (g/L): 65.95
Maltose M (g/L): 59.28
Volume V (L): 6.99

t = 61.00 h

Biomass X (g/L): 77.70
Glucose S (g/L): 8.86
Lactic Acid P (g/L): 71.79
Maltose M (g/L): 52.75
Volume V (L): 6.96

t = 62.00 h

Biomass X (g/L): 77.59
Glucose S (g/L): 9.31
Lactic Acid P (g/L): 59.95
Maltose M (g/L): 54.46
Volume V (L): 6.66

t = 63.00 h

Biomass X (g/L): 76.04
Glucose S (g/L): 8.70
Lactic Acid P (g/L): 74.05
Maltose M (g/L): 52.80
Volume V (L): 6.83

t = 64.00 h

Biomass X (g/L): 80.60
Glucose S (g/L): 6.18
Lactic Acid P (g/L): 66.87
Maltose M (g/L): 53.33
Volume V (L): 6.94

t = 65.00 h

Biomass X (g/L): 66.86
Glucose S (g/L): 6.46
Lactic Acid P (g/L): 60.15
Maltose M (g/L): 50.67
Volume V (L): 6.91

t = 66.00 h

Biomass X (g/L): 86.26
Glucose S (g/L): 8.92
Lactic Acid P (g/L): 69.30
Maltose M (g/L): 51.98
Volume V (L): 6.94

t = 67.00 h

Biomass X (g/L): 70.97
Glucose S (g/L): 11.95
Lactic Acid P (g/L): 67.35
Maltose M (g/L): 54.07
Volume V (L): 6.71

t = 68.00 h

Biomass X (g/L): 76.83
Glucose S (g/L): 11.73
Lactic Acid P (g/L): 90.61
Maltose M (g/L): 53.74
Volume V (L): 6.84

t = 69.00 h

Biomass X (g/L): 82.25
Glucose S (g/L): 11.39
Lactic Acid P (g/L): 74.50
Maltose M (g/L): 53.25
Volume V (L): 7.01

t = 70.00 h

Biomass X (g/L): 70.86
Glucose S (g/L): 9.86
Lactic Acid P (g/L): 84.88
Maltose M (g/L): 52.08
Volume V (L): 6.55

t = 71.00 h

Biomass X (g/L): 61.46
Glucose S (g/L): 12.79
Lactic Acid P (g/L): 81.33
Maltose M (g/L): 54.53
Volume V (L): 7.26

t = 72.00 h

Biomass X (g/L): 76.00
Glucose S (g/L): 8.57
Lactic Acid P (g/L): 72.14
Maltose M (g/L): 51.01
Volume V (L): 6.92

t = 73.00 h

Biomass X (g/L): 66.50
Glucose S (g/L): 7.63
Lactic Acid P (g/L): 68.28
Maltose M (g/L): 55.60
Volume V (L): 7.32

t = 74.00 h

Biomass X (g/L): 86.13
Glucose S (g/L): 12.53
Lactic Acid P (g/L): 57.30
Maltose M (g/L): 45.85
Volume V (L): 7.34

t = 75.00 h

Biomass X (g/L): 69.15
Glucose S (g/L): 12.22
Lactic Acid P (g/L): 65.24
Maltose M (g/L): 52.96

Volume V (L): 7.16

Batch 80

Initial Conditions:

Batch: 80.00

X0 (g/L): 6.28

S0 (g/L): 5.92

M0 (g/L): 106.40

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 17.99

F_vol (L/h): 0.03

S_feed (g/L): 15.80

M_feed (g/L): 284.20

V_final (L): 7.34

Batch 80 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 15.04
Glucose S (g/L): 4.12
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 106.12
Volume V (L): 4.92

t = 1.00 h

Biomass X (g/L): 13.90
Glucose S (g/L): 5.84
Lactic Acid P (g/L): 4.88
Maltose M (g/L): 98.69
Volume V (L): 4.98

t = 2.00 h

Biomass X (g/L): 6.05
Glucose S (g/L): 7.39
Lactic Acid P (g/L): 2.44
Maltose M (g/L): 98.12
Volume V (L): 5.20

t = 3.00 h

Biomass X (g/L): 6.67
Glucose S (g/L): 7.77
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 105.44
Volume V (L): 5.07

t = 4.00 h

Biomass X (g/L): 21.83
Glucose S (g/L): 8.45
Lactic Acid P (g/L): 6.07
Maltose M (g/L): 97.38
Volume V (L): 5.04

t = 5.00 h

Biomass X (g/L): 16.32
Glucose S (g/L): 8.27

Lactic Acid P (g/L): 28.78

Maltose M (g/L): 109.60

Volume V (L): 5.06

t = 6.00 h

Biomass X (g/L): 25.69

Glucose S (g/L): 6.05

Lactic Acid P (g/L): 1.93

Maltose M (g/L): 97.05

Volume V (L): 4.82

t = 7.00 h

Biomass X (g/L): 0.00

Glucose S (g/L): 7.44

Lactic Acid P (g/L): 7.38

Maltose M (g/L): 100.11

Volume V (L): 5.36

t = 8.00 h

Biomass X (g/L): 22.21

Glucose S (g/L): 6.53

Lactic Acid P (g/L): 5.70

Maltose M (g/L): 99.94

Volume V (L): 5.06

t = 9.00 h

Biomass X (g/L): 9.96

Glucose S (g/L): 6.25

Lactic Acid P (g/L): 8.42

Maltose M (g/L): 95.41

Volume V (L): 5.63

t = 10.00 h

Biomass X (g/L): 25.91

Glucose S (g/L): 5.99

Lactic Acid P (g/L): 21.82

Maltose M (g/L): 88.29

Volume V (L): 5.26

t = 11.00 h

Biomass X (g/L): 34.64

Glucose S (g/L): 4.41
Lactic Acid P (g/L): 22.72
Maltose M (g/L): 97.65
Volume V (L): 5.59

t = 12.00 h

Biomass X (g/L): 45.74
Glucose S (g/L): 5.09
Lactic Acid P (g/L): 18.30
Maltose M (g/L): 92.93
Volume V (L): 5.04

t = 13.00 h

Biomass X (g/L): 24.28
Glucose S (g/L): 2.93
Lactic Acid P (g/L): 28.55
Maltose M (g/L): 95.80
Volume V (L): 5.54

t = 14.00 h

Biomass X (g/L): 29.75
Glucose S (g/L): 2.58
Lactic Acid P (g/L): 28.08
Maltose M (g/L): 94.05
Volume V (L): 5.62

t = 15.00 h

Biomass X (g/L): 35.26
Glucose S (g/L): 3.51
Lactic Acid P (g/L): 21.45
Maltose M (g/L): 94.92
Volume V (L): 5.71

t = 16.00 h

Biomass X (g/L): 29.22
Glucose S (g/L): 2.68
Lactic Acid P (g/L): 11.27
Maltose M (g/L): 99.61
Volume V (L): 5.21

t = 17.00 h

Biomass X (g/L): 47.89
Glucose S (g/L): 3.10
Lactic Acid P (g/L): 32.45
Maltose M (g/L): 90.09
Volume V (L): 5.40

t = 18.00 h

Biomass X (g/L): 40.21
Glucose S (g/L): 4.46
Lactic Acid P (g/L): 28.16
Maltose M (g/L): 89.71
Volume V (L): 5.67

t = 19.00 h

Biomass X (g/L): 43.33
Glucose S (g/L): 3.08
Lactic Acid P (g/L): 28.14
Maltose M (g/L): 84.01
Volume V (L): 5.62

t = 20.00 h

Biomass X (g/L): 49.09
Glucose S (g/L): 2.90
Lactic Acid P (g/L): 25.23
Maltose M (g/L): 89.47
Volume V (L): 5.68

t = 21.00 h

Biomass X (g/L): 44.38
Glucose S (g/L): 1.98
Lactic Acid P (g/L): 36.58
Maltose M (g/L): 91.33
Volume V (L): 5.59

t = 22.00 h

Biomass X (g/L): 39.15
Glucose S (g/L): 1.29
Lactic Acid P (g/L): 34.39
Maltose M (g/L): 86.94
Volume V (L): 5.98

t = 23.00 h

Biomass X (g/L): 47.03
Glucose S (g/L): 2.65
Lactic Acid P (g/L): 30.59
Maltose M (g/L): 90.04
Volume V (L): 6.21

t = 24.00 h

Biomass X (g/L): 41.32
Glucose S (g/L): 3.62
Lactic Acid P (g/L): 23.38
Maltose M (g/L): 86.92
Volume V (L): 5.77

t = 25.00 h

Biomass X (g/L): 50.96
Glucose S (g/L): 2.48
Lactic Acid P (g/L): 43.75
Maltose M (g/L): 81.59
Volume V (L): 5.87

t = 26.00 h

Biomass X (g/L): 59.91
Glucose S (g/L): 3.11
Lactic Acid P (g/L): 40.91
Maltose M (g/L): 87.72
Volume V (L): 5.82

t = 27.00 h

Biomass X (g/L): 42.02
Glucose S (g/L): 4.00
Lactic Acid P (g/L): 39.02
Maltose M (g/L): 84.94
Volume V (L): 6.08

t = 28.00 h

Biomass X (g/L): 40.00
Glucose S (g/L): 4.02
Lactic Acid P (g/L): 15.55
Maltose M (g/L): 85.71
Volume V (L): 6.03

t = 29.00 h

Biomass X (g/L): 54.93
Glucose S (g/L): 3.26
Lactic Acid P (g/L): 38.59
Maltose M (g/L): 82.24
Volume V (L): 5.71

t = 30.00 h

Biomass X (g/L): 48.82
Glucose S (g/L): 3.09
Lactic Acid P (g/L): 37.87
Maltose M (g/L): 81.29
Volume V (L): 5.82

t = 31.00 h

Biomass X (g/L): 56.84
Glucose S (g/L): 1.74
Lactic Acid P (g/L): 35.68
Maltose M (g/L): 84.11
Volume V (L): 5.78

t = 32.00 h

Biomass X (g/L): 60.36
Glucose S (g/L): 2.54
Lactic Acid P (g/L): 48.18
Maltose M (g/L): 81.92
Volume V (L): 6.50

t = 33.00 h

Biomass X (g/L): 39.74
Glucose S (g/L): 2.90
Lactic Acid P (g/L): 48.80
Maltose M (g/L): 89.53
Volume V (L): 6.15

t = 34.00 h

Biomass X (g/L): 65.50
Glucose S (g/L): 2.49
Lactic Acid P (g/L): 47.39
Maltose M (g/L): 85.11

Volume V (L): 5.95

t = 35.00 h

Biomass X (g/L): 50.12

Glucose S (g/L): 3.48

Lactic Acid P (g/L): 53.36

Maltose M (g/L): 83.51

Volume V (L): 5.97

t = 36.00 h

Biomass X (g/L): 61.06

Glucose S (g/L): 4.95

Lactic Acid P (g/L): 36.93

Maltose M (g/L): 79.96

Volume V (L): 6.28

t = 37.00 h

Biomass X (g/L): 56.34

Glucose S (g/L): 4.71

Lactic Acid P (g/L): 42.60

Maltose M (g/L): 71.71

Volume V (L): 6.33

t = 38.00 h

Biomass X (g/L): 58.12

Glucose S (g/L): 3.74

Lactic Acid P (g/L): 47.91

Maltose M (g/L): 76.28

Volume V (L): 6.34

t = 39.00 h

Biomass X (g/L): 78.60

Glucose S (g/L): 4.28

Lactic Acid P (g/L): 42.91

Maltose M (g/L): 76.44

Volume V (L): 6.14

t = 40.00 h

Biomass X (g/L): 67.35

Glucose S (g/L): 3.12

Lactic Acid P (g/L): 55.10

Maltose M (g/L): 77.49

Volume V (L): 6.31

t = 41.00 h

Biomass X (g/L): 73.24

Glucose S (g/L): 5.05

Lactic Acid P (g/L): 45.41

Maltose M (g/L): 78.01

Volume V (L): 6.30

t = 42.00 h

Biomass X (g/L): 67.38

Glucose S (g/L): 2.64

Lactic Acid P (g/L): 61.54

Maltose M (g/L): 80.40

Volume V (L): 6.14

t = 43.00 h

Biomass X (g/L): 72.21

Glucose S (g/L): 4.82

Lactic Acid P (g/L): 43.58

Maltose M (g/L): 78.31

Volume V (L): 6.17

t = 44.00 h

Biomass X (g/L): 71.13

Glucose S (g/L): 4.60

Lactic Acid P (g/L): 70.13

Maltose M (g/L): 81.54

Volume V (L): 6.45

t = 45.00 h

Biomass X (g/L): 70.37

Glucose S (g/L): 3.68

Lactic Acid P (g/L): 70.04

Maltose M (g/L): 74.92

Volume V (L): 6.44

t = 46.00 h

Biomass X (g/L): 78.49

Glucose S (g/L): 4.27

Lactic Acid P (g/L): 52.93

Maltose M (g/L): 76.84

Volume V (L): 6.54

t = 47.00 h

Biomass X (g/L): 70.79

Glucose S (g/L): 3.64

Lactic Acid P (g/L): 64.11

Maltose M (g/L): 76.74

Volume V (L): 6.62

t = 48.00 h

Biomass X (g/L): 90.04

Glucose S (g/L): 3.82

Lactic Acid P (g/L): 51.83

Maltose M (g/L): 75.88

Volume V (L): 6.78

t = 49.00 h

Biomass X (g/L): 79.03

Glucose S (g/L): 3.56

Lactic Acid P (g/L): 64.43

Maltose M (g/L): 79.77

Volume V (L): 6.39

t = 50.00 h

Biomass X (g/L): 61.20

Glucose S (g/L): 5.68

Lactic Acid P (g/L): 55.13

Maltose M (g/L): 77.33

Volume V (L): 6.68

t = 51.00 h

Biomass X (g/L): 70.17

Glucose S (g/L): 6.21

Lactic Acid P (g/L): 46.48

Maltose M (g/L): 68.02

Volume V (L): 6.73

t = 52.00 h

Biomass X (g/L): 75.72

Glucose S (g/L): 3.58
Lactic Acid P (g/L): 28.25
Maltose M (g/L): 72.63
Volume V (L): 6.57

t = 53.00 h

Biomass X (g/L): 73.13
Glucose S (g/L): 7.09
Lactic Acid P (g/L): 56.37
Maltose M (g/L): 72.63
Volume V (L): 6.68

t = 54.00 h

Biomass X (g/L): 79.24
Glucose S (g/L): 5.73
Lactic Acid P (g/L): 61.62
Maltose M (g/L): 81.26
Volume V (L): 6.63

t = 55.00 h

Biomass X (g/L): 80.67
Glucose S (g/L): 6.17
Lactic Acid P (g/L): 71.10
Maltose M (g/L): 70.48
Volume V (L): 7.11

t = 56.00 h

Biomass X (g/L): 62.22
Glucose S (g/L): 4.90
Lactic Acid P (g/L): 60.84
Maltose M (g/L): 67.35
Volume V (L): 6.77

t = 57.00 h

Biomass X (g/L): 85.77
Glucose S (g/L): 7.43
Lactic Acid P (g/L): 69.57
Maltose M (g/L): 72.88
Volume V (L): 6.87

t = 58.00 h

Biomass X (g/L): 70.61
Glucose S (g/L): 6.71
Lactic Acid P (g/L): 57.92
Maltose M (g/L): 72.73
Volume V (L): 6.76

t = 59.00 h

Biomass X (g/L): 70.12
Glucose S (g/L): 7.70
Lactic Acid P (g/L): 59.48
Maltose M (g/L): 69.96
Volume V (L): 6.75

t = 60.00 h

Biomass X (g/L): 70.46
Glucose S (g/L): 8.00
Lactic Acid P (g/L): 71.75
Maltose M (g/L): 68.96
Volume V (L): 6.63

t = 61.00 h

Biomass X (g/L): 83.14
Glucose S (g/L): 8.32
Lactic Acid P (g/L): 42.72
Maltose M (g/L): 73.46
Volume V (L): 7.11

t = 62.00 h

Biomass X (g/L): 74.05
Glucose S (g/L): 6.90
Lactic Acid P (g/L): 64.10
Maltose M (g/L): 65.56
Volume V (L): 7.08

t = 63.00 h

Biomass X (g/L): 70.89
Glucose S (g/L): 7.51
Lactic Acid P (g/L): 61.17
Maltose M (g/L): 67.33
Volume V (L): 6.91

t = 64.00 h

Biomass X (g/L): 73.29
Glucose S (g/L): 7.42
Lactic Acid P (g/L): 76.50
Maltose M (g/L): 66.20
Volume V (L): 6.83

t = 65.00 h

Biomass X (g/L): 61.57
Glucose S (g/L): 8.93
Lactic Acid P (g/L): 61.12
Maltose M (g/L): 72.59
Volume V (L): 7.06

t = 66.00 h

Biomass X (g/L): 70.17
Glucose S (g/L): 8.73
Lactic Acid P (g/L): 70.41
Maltose M (g/L): 68.85
Volume V (L): 6.97

t = 67.00 h

Biomass X (g/L): 66.94
Glucose S (g/L): 9.60
Lactic Acid P (g/L): 74.30
Maltose M (g/L): 62.86
Volume V (L): 7.05

t = 68.00 h

Biomass X (g/L): 75.67
Glucose S (g/L): 9.91
Lactic Acid P (g/L): 54.48
Maltose M (g/L): 62.93
Volume V (L): 6.89

t = 69.00 h

Biomass X (g/L): 84.07
Glucose S (g/L): 7.82
Lactic Acid P (g/L): 66.33
Maltose M (g/L): 74.44
Volume V (L): 7.20

t = 70.00 h

Biomass X (g/L): 83.31
Glucose S (g/L): 10.67
Lactic Acid P (g/L): 64.33
Maltose M (g/L): 70.79
Volume V (L): 7.23

t = 71.00 h

Biomass X (g/L): 89.65
Glucose S (g/L): 10.63
Lactic Acid P (g/L): 76.36
Maltose M (g/L): 69.45
Volume V (L): 7.42

t = 72.00 h

Biomass X (g/L): 87.88
Glucose S (g/L): 9.82
Lactic Acid P (g/L): 74.52
Maltose M (g/L): 65.65
Volume V (L): 7.22

t = 73.00 h

Biomass X (g/L): 74.35
Glucose S (g/L): 10.72
Lactic Acid P (g/L): 67.30
Maltose M (g/L): 71.57
Volume V (L): 7.22

t = 74.00 h

Biomass X (g/L): 79.30
Glucose S (g/L): 10.62
Lactic Acid P (g/L): 60.87
Maltose M (g/L): 60.33
Volume V (L): 7.02

t = 75.00 h

Biomass X (g/L): 78.98
Glucose S (g/L): 11.55
Lactic Acid P (g/L): 78.82
Maltose M (g/L): 67.76

Volume V (L): 7.55

Batch 81

Initial Conditions:

Batch: 81.00

X0 (g/L): 9.29

S0 (g/L): 93.02

M0 (g/L): 11.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.12

F_vol (L/h): 0.06

S_feed (g/L): 266.99

M_feed (g/L): 33.01

V_final (L): 9.77

Batch 81 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 17.87
Glucose S (g/L): 90.84
Lactic Acid P (g/L): 8.20
Maltose M (g/L): 11.67
Volume V (L): 5.24

t = 1.00 h

Biomass X (g/L): 12.75
Glucose S (g/L): 92.02
Lactic Acid P (g/L): 14.74
Maltose M (g/L): 11.49
Volume V (L): 5.95

t = 2.00 h

Biomass X (g/L): 3.57
Glucose S (g/L): 95.54
Lactic Acid P (g/L): 1.93
Maltose M (g/L): 11.58
Volume V (L): 4.94

t = 3.00 h

Biomass X (g/L): 9.19
Glucose S (g/L): 85.05
Lactic Acid P (g/L): 2.13
Maltose M (g/L): 11.42
Volume V (L): 4.81

t = 4.00 h

Biomass X (g/L): 20.62
Glucose S (g/L): 83.62
Lactic Acid P (g/L): 17.66
Maltose M (g/L): 12.12
Volume V (L): 5.00

t = 5.00 h

Biomass X (g/L): 40.49
Glucose S (g/L): 80.89

Lactic Acid P (g/L): 19.57

Maltose M (g/L): 12.01

Volume V (L): 5.99

t = 6.00 h

Biomass X (g/L): 39.42

Glucose S (g/L): 90.66

Lactic Acid P (g/L): 26.78

Maltose M (g/L): 11.74

Volume V (L): 5.93

t = 7.00 h

Biomass X (g/L): 27.59

Glucose S (g/L): 77.53

Lactic Acid P (g/L): 43.54

Maltose M (g/L): 12.17

Volume V (L): 5.24

t = 8.00 h

Biomass X (g/L): 50.90

Glucose S (g/L): 76.28

Lactic Acid P (g/L): 29.74

Maltose M (g/L): 11.88

Volume V (L): 4.61

t = 9.00 h

Biomass X (g/L): 43.67

Glucose S (g/L): 71.92

Lactic Acid P (g/L): 20.78

Maltose M (g/L): 11.69

Volume V (L): 5.49

t = 10.00 h

Biomass X (g/L): 51.05

Glucose S (g/L): 69.52

Lactic Acid P (g/L): 35.79

Maltose M (g/L): 11.96

Volume V (L): 5.25

t = 11.00 h

Biomass X (g/L): 69.33

Glucose S (g/L): 65.81
Lactic Acid P (g/L): 40.49
Maltose M (g/L): 11.87
Volume V (L): 5.51

t = 12.00 h

Biomass X (g/L): 70.87
Glucose S (g/L): 63.41
Lactic Acid P (g/L): 52.80
Maltose M (g/L): 11.97
Volume V (L): 5.36

t = 13.00 h

Biomass X (g/L): 77.16
Glucose S (g/L): 54.81
Lactic Acid P (g/L): 41.81
Maltose M (g/L): 11.54
Volume V (L): 5.21

t = 14.00 h

Biomass X (g/L): 69.22
Glucose S (g/L): 60.31
Lactic Acid P (g/L): 42.17
Maltose M (g/L): 11.65
Volume V (L): 6.61

t = 15.00 h

Biomass X (g/L): 80.41
Glucose S (g/L): 57.39
Lactic Acid P (g/L): 35.38
Maltose M (g/L): 12.10
Volume V (L): 5.69

t = 16.00 h

Biomass X (g/L): 68.48
Glucose S (g/L): 58.98
Lactic Acid P (g/L): 55.06
Maltose M (g/L): 12.13
Volume V (L): 6.56

t = 17.00 h

Biomass X (g/L): 73.59
Glucose S (g/L): 55.92
Lactic Acid P (g/L): 82.73
Maltose M (g/L): 12.07
Volume V (L): 6.00

t = 18.00 h

Biomass X (g/L): 74.66
Glucose S (g/L): 58.57
Lactic Acid P (g/L): 51.28
Maltose M (g/L): 12.22
Volume V (L): 5.71

t = 19.00 h

Biomass X (g/L): 79.55
Glucose S (g/L): 58.87
Lactic Acid P (g/L): 58.98
Maltose M (g/L): 11.89
Volume V (L): 6.40

t = 20.00 h

Biomass X (g/L): 68.31
Glucose S (g/L): 43.95
Lactic Acid P (g/L): 51.07
Maltose M (g/L): 11.86
Volume V (L): 6.12

t = 21.00 h

Biomass X (g/L): 79.71
Glucose S (g/L): 53.28
Lactic Acid P (g/L): 45.52
Maltose M (g/L): 11.57
Volume V (L): 6.26

t = 22.00 h

Biomass X (g/L): 96.90
Glucose S (g/L): 47.84
Lactic Acid P (g/L): 62.53
Maltose M (g/L): 11.62
Volume V (L): 6.45

t = 23.00 h

Biomass X (g/L): 73.37
Glucose S (g/L): 55.90
Lactic Acid P (g/L): 59.82
Maltose M (g/L): 11.33
Volume V (L): 6.15

t = 24.00 h

Biomass X (g/L): 71.92
Glucose S (g/L): 53.99
Lactic Acid P (g/L): 38.46
Maltose M (g/L): 11.63
Volume V (L): 5.91

t = 25.00 h

Biomass X (g/L): 73.42
Glucose S (g/L): 60.52
Lactic Acid P (g/L): 76.34
Maltose M (g/L): 11.28
Volume V (L): 6.57

t = 26.00 h

Biomass X (g/L): 111.67
Glucose S (g/L): 56.43
Lactic Acid P (g/L): 66.13
Maltose M (g/L): 11.87
Volume V (L): 7.02

t = 27.00 h

Biomass X (g/L): 91.11
Glucose S (g/L): 54.12
Lactic Acid P (g/L): 63.75
Maltose M (g/L): 11.71
Volume V (L): 6.37

t = 28.00 h

Biomass X (g/L): 84.53
Glucose S (g/L): 59.92
Lactic Acid P (g/L): 59.90
Maltose M (g/L): 11.21
Volume V (L): 6.84

t = 29.00 h

Biomass X (g/L): 88.21
Glucose S (g/L): 55.55
Lactic Acid P (g/L): 68.24
Maltose M (g/L): 11.55
Volume V (L): 7.42

t = 30.00 h

Biomass X (g/L): 90.12
Glucose S (g/L): 55.31
Lactic Acid P (g/L): 69.31
Maltose M (g/L): 11.94
Volume V (L): 6.87

t = 31.00 h

Biomass X (g/L): 82.09
Glucose S (g/L): 59.19
Lactic Acid P (g/L): 69.46
Maltose M (g/L): 11.50
Volume V (L): 7.00

t = 32.00 h

Biomass X (g/L): 83.54
Glucose S (g/L): 64.35
Lactic Acid P (g/L): 76.20
Maltose M (g/L): 11.31
Volume V (L): 7.06

t = 33.00 h

Biomass X (g/L): 87.23
Glucose S (g/L): 61.98
Lactic Acid P (g/L): 78.84
Maltose M (g/L): 11.12
Volume V (L): 6.84

t = 34.00 h

Biomass X (g/L): 81.97
Glucose S (g/L): 61.44
Lactic Acid P (g/L): 85.93
Maltose M (g/L): 11.10

Volume V (L): 6.78

t = 35.00 h

Biomass X (g/L): 85.27

Glucose S (g/L): 70.33

Lactic Acid P (g/L): 60.11

Maltose M (g/L): 11.51

Volume V (L): 6.85

t = 36.00 h

Biomass X (g/L): 92.61

Glucose S (g/L): 57.63

Lactic Acid P (g/L): 63.00

Maltose M (g/L): 11.29

Volume V (L): 7.43

t = 37.00 h

Biomass X (g/L): 79.71

Glucose S (g/L): 58.14

Lactic Acid P (g/L): 72.09

Maltose M (g/L): 11.28

Volume V (L): 7.64

t = 38.00 h

Biomass X (g/L): 114.47

Glucose S (g/L): 62.18

Lactic Acid P (g/L): 61.95

Maltose M (g/L): 11.23

Volume V (L): 7.72

t = 39.00 h

Biomass X (g/L): 96.65

Glucose S (g/L): 68.00

Lactic Acid P (g/L): 63.83

Maltose M (g/L): 11.25

Volume V (L): 7.33

t = 40.00 h

Biomass X (g/L): 77.21

Glucose S (g/L): 59.26

Lactic Acid P (g/L): 74.33

Maltose M (g/L): 11.00

Volume V (L): 7.96

t = 41.00 h

Biomass X (g/L): 65.59

Glucose S (g/L): 70.16

Lactic Acid P (g/L): 78.62

Maltose M (g/L): 11.27

Volume V (L): 7.72

t = 42.00 h

Biomass X (g/L): 82.79

Glucose S (g/L): 58.62

Lactic Acid P (g/L): 82.44

Maltose M (g/L): 10.89

Volume V (L): 7.70

t = 43.00 h

Biomass X (g/L): 95.94

Glucose S (g/L): 64.27

Lactic Acid P (g/L): 69.17

Maltose M (g/L): 10.99

Volume V (L): 6.46

t = 44.00 h

Biomass X (g/L): 94.02

Glucose S (g/L): 66.03

Lactic Acid P (g/L): 75.56

Maltose M (g/L): 11.30

Volume V (L): 7.49

t = 45.00 h

Biomass X (g/L): 88.17

Glucose S (g/L): 62.64

Lactic Acid P (g/L): 73.82

Maltose M (g/L): 11.14

Volume V (L): 7.28

t = 46.00 h

Biomass X (g/L): 83.37

Glucose S (g/L): 64.28

Lactic Acid P (g/L): 72.02

Maltose M (g/L): 11.16

Volume V (L): 8.16

t = 47.00 h

Biomass X (g/L): 80.36

Glucose S (g/L): 70.32

Lactic Acid P (g/L): 64.59

Maltose M (g/L): 11.41

Volume V (L): 8.02

t = 48.00 h

Biomass X (g/L): 85.08

Glucose S (g/L): 73.41

Lactic Acid P (g/L): 63.68

Maltose M (g/L): 10.94

Volume V (L): 7.64

t = 49.00 h

Biomass X (g/L): 79.10

Glucose S (g/L): 60.50

Lactic Acid P (g/L): 78.83

Maltose M (g/L): 10.91

Volume V (L): 8.23

t = 50.00 h

Biomass X (g/L): 84.87

Glucose S (g/L): 69.88

Lactic Acid P (g/L): 65.20

Maltose M (g/L): 10.82

Volume V (L): 7.64

t = 51.00 h

Biomass X (g/L): 101.37

Glucose S (g/L): 69.39

Lactic Acid P (g/L): 56.70

Maltose M (g/L): 10.74

Volume V (L): 7.84

t = 52.00 h

Biomass X (g/L): 81.01

Glucose S (g/L): 70.86
Lactic Acid P (g/L): 73.08
Maltose M (g/L): 10.59
Volume V (L): 8.03

t = 53.00 h

Biomass X (g/L): 94.15
Glucose S (g/L): 69.87
Lactic Acid P (g/L): 87.07
Maltose M (g/L): 10.77
Volume V (L): 8.58

t = 54.00 h

Biomass X (g/L): 95.46
Glucose S (g/L): 66.79
Lactic Acid P (g/L): 47.95
Maltose M (g/L): 10.78
Volume V (L): 8.97

t = 55.00 h

Biomass X (g/L): 77.57
Glucose S (g/L): 78.20
Lactic Acid P (g/L): 71.35
Maltose M (g/L): 10.75
Volume V (L): 8.27

t = 56.00 h

Biomass X (g/L): 79.67
Glucose S (g/L): 77.36
Lactic Acid P (g/L): 80.47
Maltose M (g/L): 10.64
Volume V (L): 9.06

t = 57.00 h

Biomass X (g/L): 97.88
Glucose S (g/L): 73.16
Lactic Acid P (g/L): 57.54
Maltose M (g/L): 10.77
Volume V (L): 8.64

t = 58.00 h

Biomass X (g/L): 78.52
Glucose S (g/L): 72.31
Lactic Acid P (g/L): 72.63
Maltose M (g/L): 10.56
Volume V (L): 8.78

t = 59.00 h

Biomass X (g/L): 86.68
Glucose S (g/L): 72.45
Lactic Acid P (g/L): 64.24
Maltose M (g/L): 10.43
Volume V (L): 8.73

t = 60.00 h

Biomass X (g/L): 106.66
Glucose S (g/L): 65.52
Lactic Acid P (g/L): 65.64
Maltose M (g/L): 10.29
Volume V (L): 8.74

t = 61.00 h

Biomass X (g/L): 96.27
Glucose S (g/L): 77.32
Lactic Acid P (g/L): 71.41
Maltose M (g/L): 10.13
Volume V (L): 9.05

t = 62.00 h

Biomass X (g/L): 87.47
Glucose S (g/L): 79.54
Lactic Acid P (g/L): 64.91
Maltose M (g/L): 10.67
Volume V (L): 9.22

t = 63.00 h

Biomass X (g/L): 94.54
Glucose S (g/L): 77.68
Lactic Acid P (g/L): 64.30
Maltose M (g/L): 10.21
Volume V (L): 8.67

t = 64.00 h

Biomass X (g/L): 85.71
Glucose S (g/L): 77.41
Lactic Acid P (g/L): 75.08
Maltose M (g/L): 10.13
Volume V (L): 9.00

t = 65.00 h

Biomass X (g/L): 95.83
Glucose S (g/L): 78.83
Lactic Acid P (g/L): 96.39
Maltose M (g/L): 10.18
Volume V (L): 9.77

t = 66.00 h

Biomass X (g/L): 66.16
Glucose S (g/L): 75.64
Lactic Acid P (g/L): 82.94
Maltose M (g/L): 10.17
Volume V (L): 9.26

t = 67.00 h

Biomass X (g/L): 84.28
Glucose S (g/L): 77.56
Lactic Acid P (g/L): 85.43
Maltose M (g/L): 10.50
Volume V (L): 9.37

t = 68.00 h

Biomass X (g/L): 65.35
Glucose S (g/L): 79.39
Lactic Acid P (g/L): 83.60
Maltose M (g/L): 10.08
Volume V (L): 9.77

t = 69.00 h

Biomass X (g/L): 86.75
Glucose S (g/L): 83.24
Lactic Acid P (g/L): 69.90
Maltose M (g/L): 10.61
Volume V (L): 9.20

t = 70.00 h

Biomass X (g/L): 82.71
Glucose S (g/L): 86.40
Lactic Acid P (g/L): 79.37
Maltose M (g/L): 10.07
Volume V (L): 9.44

t = 71.00 h

Biomass X (g/L): 97.85
Glucose S (g/L): 80.82
Lactic Acid P (g/L): 86.78
Maltose M (g/L): 10.42
Volume V (L): 9.83

t = 72.00 h

Biomass X (g/L): 91.39
Glucose S (g/L): 78.26
Lactic Acid P (g/L): 84.42
Maltose M (g/L): 10.15
Volume V (L): 10.05

t = 73.00 h

Biomass X (g/L): 79.26
Glucose S (g/L): 82.91
Lactic Acid P (g/L): 78.68
Maltose M (g/L): 10.01
Volume V (L): 8.74

t = 74.00 h

Biomass X (g/L): 80.19
Glucose S (g/L): 90.98
Lactic Acid P (g/L): 81.82
Maltose M (g/L): 10.20
Volume V (L): 9.42

t = 75.00 h

Biomass X (g/L): 87.54
Glucose S (g/L): 80.19
Lactic Acid P (g/L): 99.92
Maltose M (g/L): 10.07

Volume V (L): 9.39

Batch 82

Initial Conditions:

Batch: 82.00

X0 (g/L): 2.81

S0 (g/L): 90.02

M0 (g/L): 46.33

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.51

F_vol (L/h): 0.02

S_feed (g/L): 198.07

M_feed (g/L): 101.93

V_final (L): 6.53

Batch 82 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 1.51
Glucose S (g/L): 94.54
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 45.40
Volume V (L): 4.97

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 89.74
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 47.98
Volume V (L): 5.23

t = 2.00 h

Biomass X (g/L): 12.19
Glucose S (g/L): 97.55
Lactic Acid P (g/L): 2.19
Maltose M (g/L): 41.23
Volume V (L): 5.08

t = 3.00 h

Biomass X (g/L): 8.76
Glucose S (g/L): 94.81
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 37.93
Volume V (L): 4.88

t = 4.00 h

Biomass X (g/L): 18.55
Glucose S (g/L): 88.49
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 46.70
Volume V (L): 4.91

t = 5.00 h

Biomass X (g/L): 21.98
Glucose S (g/L): 87.59

Lactic Acid P (g/L): 20.25

Maltose M (g/L): 41.00

Volume V (L): 5.21

t = 6.00 h

Biomass X (g/L): 22.40

Glucose S (g/L): 74.63

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 44.75

Volume V (L): 5.21

t = 7.00 h

Biomass X (g/L): 15.53

Glucose S (g/L): 89.64

Lactic Acid P (g/L): 18.66

Maltose M (g/L): 42.07

Volume V (L): 5.03

t = 8.00 h

Biomass X (g/L): 24.42

Glucose S (g/L): 77.67

Lactic Acid P (g/L): 5.97

Maltose M (g/L): 38.42

Volume V (L): 5.25

t = 9.00 h

Biomass X (g/L): 23.03

Glucose S (g/L): 73.83

Lactic Acid P (g/L): 6.96

Maltose M (g/L): 42.38

Volume V (L): 5.38

t = 10.00 h

Biomass X (g/L): 18.24

Glucose S (g/L): 69.33

Lactic Acid P (g/L): 15.67

Maltose M (g/L): 39.23

Volume V (L): 5.38

t = 11.00 h

Biomass X (g/L): 47.42

Glucose S (g/L): 71.44
Lactic Acid P (g/L): 31.52
Maltose M (g/L): 37.00
Volume V (L): 5.42

t = 12.00 h

Biomass X (g/L): 41.13
Glucose S (g/L): 58.64
Lactic Acid P (g/L): 14.93
Maltose M (g/L): 36.54
Volume V (L): 5.31

t = 13.00 h

Biomass X (g/L): 30.63
Glucose S (g/L): 55.98
Lactic Acid P (g/L): 34.99
Maltose M (g/L): 34.29
Volume V (L): 5.24

t = 14.00 h

Biomass X (g/L): 56.29
Glucose S (g/L): 58.09
Lactic Acid P (g/L): 25.59
Maltose M (g/L): 39.23
Volume V (L): 5.24

t = 15.00 h

Biomass X (g/L): 52.15
Glucose S (g/L): 51.75
Lactic Acid P (g/L): 30.39
Maltose M (g/L): 35.33
Volume V (L): 5.30

t = 16.00 h

Biomass X (g/L): 36.38
Glucose S (g/L): 54.54
Lactic Acid P (g/L): 45.07
Maltose M (g/L): 37.14
Volume V (L): 5.22

t = 17.00 h

Biomass X (g/L): 46.86
Glucose S (g/L): 46.71
Lactic Acid P (g/L): 65.50
Maltose M (g/L): 36.30
Volume V (L): 5.30

t = 18.00 h

Biomass X (g/L): 56.91
Glucose S (g/L): 57.09
Lactic Acid P (g/L): 36.81
Maltose M (g/L): 33.90
Volume V (L): 5.52

t = 19.00 h

Biomass X (g/L): 69.18
Glucose S (g/L): 49.89
Lactic Acid P (g/L): 55.43
Maltose M (g/L): 31.97
Volume V (L): 5.19

t = 20.00 h

Biomass X (g/L): 77.69
Glucose S (g/L): 48.04
Lactic Acid P (g/L): 47.06
Maltose M (g/L): 41.87
Volume V (L): 5.22

t = 21.00 h

Biomass X (g/L): 79.52
Glucose S (g/L): 39.04
Lactic Acid P (g/L): 47.29
Maltose M (g/L): 34.85
Volume V (L): 5.53

t = 22.00 h

Biomass X (g/L): 79.90
Glucose S (g/L): 55.03
Lactic Acid P (g/L): 64.60
Maltose M (g/L): 39.20
Volume V (L): 5.41

t = 23.00 h

Biomass X (g/L): 74.98
Glucose S (g/L): 37.38
Lactic Acid P (g/L): 46.91
Maltose M (g/L): 37.11
Volume V (L): 5.23

t = 24.00 h

Biomass X (g/L): 95.09
Glucose S (g/L): 34.70
Lactic Acid P (g/L): 45.26
Maltose M (g/L): 32.91
Volume V (L): 5.48

t = 25.00 h

Biomass X (g/L): 85.78
Glucose S (g/L): 41.95
Lactic Acid P (g/L): 62.91
Maltose M (g/L): 33.79
Volume V (L): 5.48

t = 26.00 h

Biomass X (g/L): 82.06
Glucose S (g/L): 38.16
Lactic Acid P (g/L): 75.63
Maltose M (g/L): 33.33
Volume V (L): 5.48

t = 27.00 h

Biomass X (g/L): 86.88
Glucose S (g/L): 31.57
Lactic Acid P (g/L): 68.32
Maltose M (g/L): 37.26
Volume V (L): 5.63

t = 28.00 h

Biomass X (g/L): 77.79
Glucose S (g/L): 48.34
Lactic Acid P (g/L): 47.72
Maltose M (g/L): 34.30
Volume V (L): 5.41

t = 29.00 h

Biomass X (g/L): 82.36
Glucose S (g/L): 40.74
Lactic Acid P (g/L): 56.75
Maltose M (g/L): 29.76
Volume V (L): 5.59

t = 30.00 h

Biomass X (g/L): 61.91
Glucose S (g/L): 34.65
Lactic Acid P (g/L): 73.82
Maltose M (g/L): 32.36
Volume V (L): 5.59

t = 31.00 h

Biomass X (g/L): 80.57
Glucose S (g/L): 48.07
Lactic Acid P (g/L): 80.82
Maltose M (g/L): 33.16
Volume V (L): 5.64

t = 32.00 h

Biomass X (g/L): 103.90
Glucose S (g/L): 38.54
Lactic Acid P (g/L): 68.08
Maltose M (g/L): 27.77
Volume V (L): 5.44

t = 33.00 h

Biomass X (g/L): 87.58
Glucose S (g/L): 33.48
Lactic Acid P (g/L): 60.78
Maltose M (g/L): 31.69
Volume V (L): 5.61

t = 34.00 h

Biomass X (g/L): 87.73
Glucose S (g/L): 46.48
Lactic Acid P (g/L): 84.00
Maltose M (g/L): 33.53

Volume V (L): 5.70

t = 35.00 h

Biomass X (g/L): 97.65

Glucose S (g/L): 36.90

Lactic Acid P (g/L): 67.42

Maltose M (g/L): 27.59

Volume V (L): 5.78

t = 36.00 h

Biomass X (g/L): 91.38

Glucose S (g/L): 36.48

Lactic Acid P (g/L): 62.79

Maltose M (g/L): 27.62

Volume V (L): 5.96

t = 37.00 h

Biomass X (g/L): 79.75

Glucose S (g/L): 30.57

Lactic Acid P (g/L): 61.77

Maltose M (g/L): 32.10

Volume V (L): 5.85

t = 38.00 h

Biomass X (g/L): 88.67

Glucose S (g/L): 32.99

Lactic Acid P (g/L): 58.20

Maltose M (g/L): 27.93

Volume V (L): 5.83

t = 39.00 h

Biomass X (g/L): 76.80

Glucose S (g/L): 30.26

Lactic Acid P (g/L): 99.41

Maltose M (g/L): 26.03

Volume V (L): 5.95

t = 40.00 h

Biomass X (g/L): 91.00

Glucose S (g/L): 40.91

Lactic Acid P (g/L): 85.78

Maltose M (g/L): 30.49

Volume V (L): 5.78

t = 41.00 h

Biomass X (g/L): 80.51

Glucose S (g/L): 23.28

Lactic Acid P (g/L): 65.43

Maltose M (g/L): 29.43

Volume V (L): 6.03

t = 42.00 h

Biomass X (g/L): 99.35

Glucose S (g/L): 40.47

Lactic Acid P (g/L): 90.05

Maltose M (g/L): 27.92

Volume V (L): 5.99

t = 43.00 h

Biomass X (g/L): 84.40

Glucose S (g/L): 27.54

Lactic Acid P (g/L): 78.50

Maltose M (g/L): 30.27

Volume V (L): 5.82

t = 44.00 h

Biomass X (g/L): 81.27

Glucose S (g/L): 32.32

Lactic Acid P (g/L): 78.42

Maltose M (g/L): 30.18

Volume V (L): 6.13

t = 45.00 h

Biomass X (g/L): 88.92

Glucose S (g/L): 31.01

Lactic Acid P (g/L): 66.91

Maltose M (g/L): 26.65

Volume V (L): 5.97

t = 46.00 h

Biomass X (g/L): 94.09

Glucose S (g/L): 29.79

Lactic Acid P (g/L): 69.44
Maltose M (g/L): 31.63
Volume V (L): 6.21

t = 47.00 h

Biomass X (g/L): 77.63
Glucose S (g/L): 36.87
Lactic Acid P (g/L): 74.65
Maltose M (g/L): 30.38
Volume V (L): 5.99

t = 48.00 h

Biomass X (g/L): 84.94
Glucose S (g/L): 32.96
Lactic Acid P (g/L): 55.72
Maltose M (g/L): 21.03
Volume V (L): 5.91

t = 49.00 h

Biomass X (g/L): 91.61
Glucose S (g/L): 29.25
Lactic Acid P (g/L): 71.05
Maltose M (g/L): 23.22
Volume V (L): 5.90

t = 50.00 h

Biomass X (g/L): 104.65
Glucose S (g/L): 34.12
Lactic Acid P (g/L): 76.42
Maltose M (g/L): 19.85
Volume V (L): 5.98

t = 51.00 h

Biomass X (g/L): 63.95
Glucose S (g/L): 29.35
Lactic Acid P (g/L): 89.97
Maltose M (g/L): 24.54
Volume V (L): 5.97

t = 52.00 h

Biomass X (g/L): 97.08

Glucose S (g/L): 30.41
Lactic Acid P (g/L): 70.96
Maltose M (g/L): 22.05
Volume V (L): 6.21

t = 53.00 h

Biomass X (g/L): 86.57
Glucose S (g/L): 25.03
Lactic Acid P (g/L): 85.74
Maltose M (g/L): 22.83
Volume V (L): 6.08

t = 54.00 h

Biomass X (g/L): 85.38
Glucose S (g/L): 31.23
Lactic Acid P (g/L): 57.17
Maltose M (g/L): 27.47
Volume V (L): 5.90

t = 55.00 h

Biomass X (g/L): 83.40
Glucose S (g/L): 36.86
Lactic Acid P (g/L): 90.04
Maltose M (g/L): 19.99
Volume V (L): 6.15

t = 56.00 h

Biomass X (g/L): 86.70
Glucose S (g/L): 34.39
Lactic Acid P (g/L): 69.57
Maltose M (g/L): 24.58
Volume V (L): 6.16

t = 57.00 h

Biomass X (g/L): 97.03
Glucose S (g/L): 44.84
Lactic Acid P (g/L): 80.29
Maltose M (g/L): 24.35
Volume V (L): 6.10

t = 58.00 h

Biomass X (g/L): 84.78
Glucose S (g/L): 49.64
Lactic Acid P (g/L): 83.40
Maltose M (g/L): 23.62
Volume V (L): 6.21

t = 59.00 h

Biomass X (g/L): 107.84
Glucose S (g/L): 32.06
Lactic Acid P (g/L): 81.75
Maltose M (g/L): 23.96
Volume V (L): 6.20

t = 60.00 h

Biomass X (g/L): 96.86
Glucose S (g/L): 30.12
Lactic Acid P (g/L): 78.88
Maltose M (g/L): 17.74
Volume V (L): 6.30

t = 61.00 h

Biomass X (g/L): 92.29
Glucose S (g/L): 41.66
Lactic Acid P (g/L): 67.75
Maltose M (g/L): 21.17
Volume V (L): 6.31

t = 62.00 h

Biomass X (g/L): 88.36
Glucose S (g/L): 37.27
Lactic Acid P (g/L): 71.60
Maltose M (g/L): 26.97
Volume V (L): 6.31

t = 63.00 h

Biomass X (g/L): 102.54
Glucose S (g/L): 42.10
Lactic Acid P (g/L): 90.90
Maltose M (g/L): 19.80
Volume V (L): 6.12

t = 64.00 h

Biomass X (g/L): 103.97
Glucose S (g/L): 38.06
Lactic Acid P (g/L): 75.13
Maltose M (g/L): 26.91
Volume V (L): 6.56

t = 65.00 h

Biomass X (g/L): 77.55
Glucose S (g/L): 34.84
Lactic Acid P (g/L): 80.39
Maltose M (g/L): 18.89
Volume V (L): 6.13

t = 66.00 h

Biomass X (g/L): 82.27
Glucose S (g/L): 35.09
Lactic Acid P (g/L): 66.67
Maltose M (g/L): 18.41
Volume V (L): 6.33

t = 67.00 h

Biomass X (g/L): 96.22
Glucose S (g/L): 38.90
Lactic Acid P (g/L): 90.62
Maltose M (g/L): 19.70
Volume V (L): 6.23

t = 68.00 h

Biomass X (g/L): 99.27
Glucose S (g/L): 35.40
Lactic Acid P (g/L): 68.95
Maltose M (g/L): 24.83
Volume V (L): 6.43

t = 69.00 h

Biomass X (g/L): 86.10
Glucose S (g/L): 43.88
Lactic Acid P (g/L): 91.73
Maltose M (g/L): 20.39
Volume V (L): 6.42

t = 70.00 h

Biomass X (g/L): 81.30
Glucose S (g/L): 29.54
Lactic Acid P (g/L): 72.05
Maltose M (g/L): 22.50
Volume V (L): 6.41

t = 71.00 h

Biomass X (g/L): 89.80
Glucose S (g/L): 44.29
Lactic Acid P (g/L): 93.22
Maltose M (g/L): 21.64
Volume V (L): 6.63

t = 72.00 h

Biomass X (g/L): 104.37
Glucose S (g/L): 26.68
Lactic Acid P (g/L): 83.35
Maltose M (g/L): 21.15
Volume V (L): 6.60

t = 73.00 h

Biomass X (g/L): 89.09
Glucose S (g/L): 38.57
Lactic Acid P (g/L): 77.86
Maltose M (g/L): 24.38
Volume V (L): 6.45

t = 74.00 h

Biomass X (g/L): 75.18
Glucose S (g/L): 36.04
Lactic Acid P (g/L): 99.85
Maltose M (g/L): 22.95
Volume V (L): 6.61

t = 75.00 h

Biomass X (g/L): 89.49
Glucose S (g/L): 29.42
Lactic Acid P (g/L): 83.26
Maltose M (g/L): 17.91

Volume V (L): 6.72

Batch 83

Initial Conditions:

Batch: 83.00

X0 (g/L): 3.46

S0 (g/L): 52.82

M0 (g/L): 108.89

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.06

F_vol (L/h): 0.09

S_feed (g/L): 97.98

M_feed (g/L): 202.02

V_final (L): 11.53

Batch 83 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 13.94
Glucose S (g/L): 51.34
Lactic Acid P (g/L): 6.04
Maltose M (g/L): 109.73
Volume V (L): 5.30

t = 1.00 h

Biomass X (g/L): 3.30
Glucose S (g/L): 55.63
Lactic Acid P (g/L): 9.77
Maltose M (g/L): 105.66
Volume V (L): 5.01

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 58.66
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 100.81
Volume V (L): 5.20

t = 3.00 h

Biomass X (g/L): 10.99
Glucose S (g/L): 51.48
Lactic Acid P (g/L): 7.23
Maltose M (g/L): 110.41
Volume V (L): 5.35

t = 4.00 h

Biomass X (g/L): 16.53
Glucose S (g/L): 59.10
Lactic Acid P (g/L): 1.88
Maltose M (g/L): 101.95
Volume V (L): 5.22

t = 5.00 h

Biomass X (g/L): 5.59
Glucose S (g/L): 56.40

Lactic Acid P (g/L): 14.86
Maltose M (g/L): 104.78
Volume V (L): 4.72

t = 6.00 h

Biomass X (g/L): 16.32
Glucose S (g/L): 56.09
Lactic Acid P (g/L): 6.61
Maltose M (g/L): 112.09
Volume V (L): 6.37

t = 7.00 h

Biomass X (g/L): 18.92
Glucose S (g/L): 56.83
Lactic Acid P (g/L): 6.91
Maltose M (g/L): 107.54
Volume V (L): 5.60

t = 8.00 h

Biomass X (g/L): 36.84
Glucose S (g/L): 59.04
Lactic Acid P (g/L): 14.60
Maltose M (g/L): 106.16
Volume V (L): 5.79

t = 9.00 h

Biomass X (g/L): 45.43
Glucose S (g/L): 52.46
Lactic Acid P (g/L): 6.15
Maltose M (g/L): 107.70
Volume V (L): 6.87

t = 10.00 h

Biomass X (g/L): 43.23
Glucose S (g/L): 44.99
Lactic Acid P (g/L): 19.97
Maltose M (g/L): 107.01
Volume V (L): 5.51

t = 11.00 h

Biomass X (g/L): 35.48

Glucose S (g/L): 46.73
Lactic Acid P (g/L): 27.56
Maltose M (g/L): 109.00
Volume V (L): 6.30

t = 12.00 h

Biomass X (g/L): 51.32
Glucose S (g/L): 48.08
Lactic Acid P (g/L): 35.16
Maltose M (g/L): 100.22
Volume V (L): 5.63

t = 13.00 h

Biomass X (g/L): 50.51
Glucose S (g/L): 46.26
Lactic Acid P (g/L): 32.65
Maltose M (g/L): 100.51
Volume V (L): 6.30

t = 14.00 h

Biomass X (g/L): 47.31
Glucose S (g/L): 43.79
Lactic Acid P (g/L): 20.60
Maltose M (g/L): 97.89
Volume V (L): 6.32

t = 15.00 h

Biomass X (g/L): 71.33
Glucose S (g/L): 38.39
Lactic Acid P (g/L): 47.89
Maltose M (g/L): 100.68
Volume V (L): 6.72

t = 16.00 h

Biomass X (g/L): 47.56
Glucose S (g/L): 34.23
Lactic Acid P (g/L): 51.10
Maltose M (g/L): 99.74
Volume V (L): 7.02

t = 17.00 h

Biomass X (g/L): 63.19
Glucose S (g/L): 39.50
Lactic Acid P (g/L): 41.67
Maltose M (g/L): 93.07
Volume V (L): 6.18

t = 18.00 h

Biomass X (g/L): 68.04
Glucose S (g/L): 32.67
Lactic Acid P (g/L): 36.14
Maltose M (g/L): 96.66
Volume V (L): 7.31

t = 19.00 h

Biomass X (g/L): 59.68
Glucose S (g/L): 38.17
Lactic Acid P (g/L): 40.03
Maltose M (g/L): 103.98
Volume V (L): 7.43

t = 20.00 h

Biomass X (g/L): 69.87
Glucose S (g/L): 37.60
Lactic Acid P (g/L): 44.73
Maltose M (g/L): 98.92
Volume V (L): 6.61

t = 21.00 h

Biomass X (g/L): 75.47
Glucose S (g/L): 33.39
Lactic Acid P (g/L): 50.65
Maltose M (g/L): 100.13
Volume V (L): 6.82

t = 22.00 h

Biomass X (g/L): 70.55
Glucose S (g/L): 34.28
Lactic Acid P (g/L): 47.02
Maltose M (g/L): 96.42
Volume V (L): 6.66

t = 23.00 h

Biomass X (g/L): 74.29
Glucose S (g/L): 35.16
Lactic Acid P (g/L): 43.53
Maltose M (g/L): 89.21
Volume V (L): 6.84

t = 24.00 h

Biomass X (g/L): 72.21
Glucose S (g/L): 32.47
Lactic Acid P (g/L): 61.48
Maltose M (g/L): 95.28
Volume V (L): 6.46

t = 25.00 h

Biomass X (g/L): 80.07
Glucose S (g/L): 30.03
Lactic Acid P (g/L): 55.91
Maltose M (g/L): 99.13
Volume V (L): 7.58

t = 26.00 h

Biomass X (g/L): 71.46
Glucose S (g/L): 34.58
Lactic Acid P (g/L): 45.45
Maltose M (g/L): 90.19
Volume V (L): 6.19

t = 27.00 h

Biomass X (g/L): 71.72
Glucose S (g/L): 35.16
Lactic Acid P (g/L): 58.38
Maltose M (g/L): 96.92
Volume V (L): 6.99

t = 28.00 h

Biomass X (g/L): 75.81
Glucose S (g/L): 39.71
Lactic Acid P (g/L): 58.21
Maltose M (g/L): 87.14
Volume V (L): 7.41

t = 29.00 h

Biomass X (g/L): 87.72
Glucose S (g/L): 33.06
Lactic Acid P (g/L): 51.34
Maltose M (g/L): 93.08
Volume V (L): 8.20

t = 30.00 h

Biomass X (g/L): 60.61
Glucose S (g/L): 39.54
Lactic Acid P (g/L): 69.12
Maltose M (g/L): 94.51
Volume V (L): 7.33

t = 31.00 h

Biomass X (g/L): 69.26
Glucose S (g/L): 32.76
Lactic Acid P (g/L): 60.54
Maltose M (g/L): 90.74
Volume V (L): 7.92

t = 32.00 h

Biomass X (g/L): 100.17
Glucose S (g/L): 32.62
Lactic Acid P (g/L): 58.18
Maltose M (g/L): 100.69
Volume V (L): 7.74

t = 33.00 h

Biomass X (g/L): 81.70
Glucose S (g/L): 36.58
Lactic Acid P (g/L): 60.07
Maltose M (g/L): 88.99
Volume V (L): 8.37

t = 34.00 h

Biomass X (g/L): 75.71
Glucose S (g/L): 35.06
Lactic Acid P (g/L): 65.40
Maltose M (g/L): 88.13

Volume V (L): 8.16

t = 35.00 h

Biomass X (g/L): 87.58

Glucose S (g/L): 34.30

Lactic Acid P (g/L): 61.46

Maltose M (g/L): 90.55

Volume V (L): 8.48

t = 36.00 h

Biomass X (g/L): 72.96

Glucose S (g/L): 45.74

Lactic Acid P (g/L): 61.98

Maltose M (g/L): 86.40

Volume V (L): 7.85

t = 37.00 h

Biomass X (g/L): 60.78

Glucose S (g/L): 38.55

Lactic Acid P (g/L): 72.69

Maltose M (g/L): 83.29

Volume V (L): 7.69

t = 38.00 h

Biomass X (g/L): 84.46

Glucose S (g/L): 48.26

Lactic Acid P (g/L): 51.24

Maltose M (g/L): 84.07

Volume V (L): 8.50

t = 39.00 h

Biomass X (g/L): 74.42

Glucose S (g/L): 38.12

Lactic Acid P (g/L): 53.22

Maltose M (g/L): 90.54

Volume V (L): 8.14

t = 40.00 h

Biomass X (g/L): 82.58

Glucose S (g/L): 36.49

Lactic Acid P (g/L): 71.65

Maltose M (g/L): 86.02

Volume V (L): 8.50

t = 41.00 h

Biomass X (g/L): 84.96

Glucose S (g/L): 47.01

Lactic Acid P (g/L): 86.61

Maltose M (g/L): 93.06

Volume V (L): 8.73

t = 42.00 h

Biomass X (g/L): 83.36

Glucose S (g/L): 41.13

Lactic Acid P (g/L): 73.43

Maltose M (g/L): 84.48

Volume V (L): 9.30

t = 43.00 h

Biomass X (g/L): 65.92

Glucose S (g/L): 44.69

Lactic Acid P (g/L): 82.59

Maltose M (g/L): 88.31

Volume V (L): 9.15

t = 44.00 h

Biomass X (g/L): 82.55

Glucose S (g/L): 40.53

Lactic Acid P (g/L): 62.65

Maltose M (g/L): 87.06

Volume V (L): 8.47

t = 45.00 h

Biomass X (g/L): 86.64

Glucose S (g/L): 45.58

Lactic Acid P (g/L): 61.23

Maltose M (g/L): 82.91

Volume V (L): 8.32

t = 46.00 h

Biomass X (g/L): 69.19

Glucose S (g/L): 41.96

Lactic Acid P (g/L): 71.28

Maltose M (g/L): 80.82

Volume V (L): 9.90

t = 47.00 h

Biomass X (g/L): 68.13

Glucose S (g/L): 46.96

Lactic Acid P (g/L): 76.02

Maltose M (g/L): 77.12

Volume V (L): 8.26

t = 48.00 h

Biomass X (g/L): 72.47

Glucose S (g/L): 46.40

Lactic Acid P (g/L): 56.31

Maltose M (g/L): 85.24

Volume V (L): 8.94

t = 49.00 h

Biomass X (g/L): 89.77

Glucose S (g/L): 49.45

Lactic Acid P (g/L): 66.36

Maltose M (g/L): 87.18

Volume V (L): 9.60

t = 50.00 h

Biomass X (g/L): 63.69

Glucose S (g/L): 51.50

Lactic Acid P (g/L): 68.31

Maltose M (g/L): 76.19

Volume V (L): 10.21

t = 51.00 h

Biomass X (g/L): 90.59

Glucose S (g/L): 50.99

Lactic Acid P (g/L): 62.84

Maltose M (g/L): 80.31

Volume V (L): 9.34

t = 52.00 h

Biomass X (g/L): 86.42

Glucose S (g/L): 52.43
Lactic Acid P (g/L): 64.16
Maltose M (g/L): 78.71
Volume V (L): 9.54

t = 53.00 h

Biomass X (g/L): 80.23
Glucose S (g/L): 51.34
Lactic Acid P (g/L): 76.87
Maltose M (g/L): 82.83
Volume V (L): 9.16

t = 54.00 h

Biomass X (g/L): 68.57
Glucose S (g/L): 51.02
Lactic Acid P (g/L): 73.84
Maltose M (g/L): 79.05
Volume V (L): 9.77

t = 55.00 h

Biomass X (g/L): 75.04
Glucose S (g/L): 54.73
Lactic Acid P (g/L): 77.00
Maltose M (g/L): 84.80
Volume V (L): 9.19

t = 56.00 h

Biomass X (g/L): 90.00
Glucose S (g/L): 53.57
Lactic Acid P (g/L): 73.17
Maltose M (g/L): 77.91
Volume V (L): 9.48

t = 57.00 h

Biomass X (g/L): 94.46
Glucose S (g/L): 54.43
Lactic Acid P (g/L): 62.70
Maltose M (g/L): 74.24
Volume V (L): 10.36

t = 58.00 h

Biomass X (g/L): 72.63
Glucose S (g/L): 56.71
Lactic Acid P (g/L): 74.40
Maltose M (g/L): 76.68
Volume V (L): 9.82

t = 59.00 h

Biomass X (g/L): 93.23
Glucose S (g/L): 52.76
Lactic Acid P (g/L): 62.46
Maltose M (g/L): 76.64
Volume V (L): 10.61

t = 60.00 h

Biomass X (g/L): 88.81
Glucose S (g/L): 61.55
Lactic Acid P (g/L): 68.34
Maltose M (g/L): 79.52
Volume V (L): 10.14

t = 61.00 h

Biomass X (g/L): 82.13
Glucose S (g/L): 56.78
Lactic Acid P (g/L): 62.15
Maltose M (g/L): 75.69
Volume V (L): 9.59

t = 62.00 h

Biomass X (g/L): 75.09
Glucose S (g/L): 58.72
Lactic Acid P (g/L): 66.11
Maltose M (g/L): 78.31
Volume V (L): 11.01

t = 63.00 h

Biomass X (g/L): 76.48
Glucose S (g/L): 58.88
Lactic Acid P (g/L): 72.52
Maltose M (g/L): 73.24
Volume V (L): 9.80

t = 64.00 h

Biomass X (g/L): 99.23
Glucose S (g/L): 61.88
Lactic Acid P (g/L): 67.87
Maltose M (g/L): 74.67
Volume V (L): 10.26

t = 65.00 h

Biomass X (g/L): 83.48
Glucose S (g/L): 61.56
Lactic Acid P (g/L): 64.29
Maltose M (g/L): 72.40
Volume V (L): 10.55

t = 66.00 h

Biomass X (g/L): 94.32
Glucose S (g/L): 60.98
Lactic Acid P (g/L): 82.35
Maltose M (g/L): 71.74
Volume V (L): 11.27

t = 67.00 h

Biomass X (g/L): 83.85
Glucose S (g/L): 66.59
Lactic Acid P (g/L): 71.41
Maltose M (g/L): 79.30
Volume V (L): 10.56

t = 68.00 h

Biomass X (g/L): 77.12
Glucose S (g/L): 52.05
Lactic Acid P (g/L): 67.97
Maltose M (g/L): 71.39
Volume V (L): 10.43

t = 69.00 h

Biomass X (g/L): 86.85
Glucose S (g/L): 67.29
Lactic Acid P (g/L): 63.23
Maltose M (g/L): 72.35
Volume V (L): 11.09

t = 70.00 h

Biomass X (g/L): 101.68
Glucose S (g/L): 66.91
Lactic Acid P (g/L): 56.41
Maltose M (g/L): 77.18
Volume V (L): 10.98

t = 71.00 h

Biomass X (g/L): 74.91
Glucose S (g/L): 58.56
Lactic Acid P (g/L): 71.17
Maltose M (g/L): 75.66
Volume V (L): 10.94

t = 72.00 h

Biomass X (g/L): 89.58
Glucose S (g/L): 68.28
Lactic Acid P (g/L): 61.10
Maltose M (g/L): 78.61
Volume V (L): 11.38

t = 73.00 h

Biomass X (g/L): 82.56
Glucose S (g/L): 69.37
Lactic Acid P (g/L): 79.75
Maltose M (g/L): 73.84
Volume V (L): 10.73

t = 74.00 h

Biomass X (g/L): 66.82
Glucose S (g/L): 68.40
Lactic Acid P (g/L): 73.86
Maltose M (g/L): 69.36
Volume V (L): 10.54

t = 75.00 h

Biomass X (g/L): 66.47
Glucose S (g/L): 62.11
Lactic Acid P (g/L): 87.53
Maltose M (g/L): 78.97

Volume V (L): 12.64

Batch 84

Initial Conditions:

Batch: 84.00

X0 (g/L): 9.67

S0 (g/L): 102.99

M0 (g/L): 19.15

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.19

F_vol (L/h): 0.04

S_feed (g/L): 252.96

M_feed (g/L): 47.04

V_final (L): 7.66

Batch 84 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 10.72
Glucose S (g/L): 107.00
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 18.83
Volume V (L): 5.01

t = 1.00 h

Biomass X (g/L): 14.83
Glucose S (g/L): 96.84
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 19.05
Volume V (L): 5.33

t = 2.00 h

Biomass X (g/L): 9.33
Glucose S (g/L): 99.03
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 17.45
Volume V (L): 5.27

t = 3.00 h

Biomass X (g/L): 31.12
Glucose S (g/L): 92.84
Lactic Acid P (g/L): 7.75
Maltose M (g/L): 19.12
Volume V (L): 5.04

t = 4.00 h

Biomass X (g/L): 22.02
Glucose S (g/L): 88.02
Lactic Acid P (g/L): 14.38
Maltose M (g/L): 18.55
Volume V (L): 5.35

t = 5.00 h

Biomass X (g/L): 4.92
Glucose S (g/L): 85.53

Lactic Acid P (g/L): 31.68
Maltose M (g/L): 19.09
Volume V (L): 5.10

t = 6.00 h

Biomass X (g/L): 33.05
Glucose S (g/L): 81.46
Lactic Acid P (g/L): 22.66
Maltose M (g/L): 18.63
Volume V (L): 4.90

t = 7.00 h

Biomass X (g/L): 41.50
Glucose S (g/L): 84.11
Lactic Acid P (g/L): 31.64
Maltose M (g/L): 18.10
Volume V (L): 5.25

t = 8.00 h

Biomass X (g/L): 47.90
Glucose S (g/L): 81.92
Lactic Acid P (g/L): 21.92
Maltose M (g/L): 17.59
Volume V (L): 4.96

t = 9.00 h

Biomass X (g/L): 52.50
Glucose S (g/L): 72.99
Lactic Acid P (g/L): 33.74
Maltose M (g/L): 18.87
Volume V (L): 5.49

t = 10.00 h

Biomass X (g/L): 52.47
Glucose S (g/L): 73.11
Lactic Acid P (g/L): 38.15
Maltose M (g/L): 19.22
Volume V (L): 5.53

t = 11.00 h

Biomass X (g/L): 55.52

Glucose S (g/L): 67.56
Lactic Acid P (g/L): 38.56
Maltose M (g/L): 17.16
Volume V (L): 5.20

t = 12.00 h

Biomass X (g/L): 63.61
Glucose S (g/L): 62.73
Lactic Acid P (g/L): 42.60
Maltose M (g/L): 18.07
Volume V (L): 5.46

t = 13.00 h

Biomass X (g/L): 53.22
Glucose S (g/L): 65.91
Lactic Acid P (g/L): 46.68
Maltose M (g/L): 16.59
Volume V (L): 5.49

t = 14.00 h

Biomass X (g/L): 52.94
Glucose S (g/L): 47.75
Lactic Acid P (g/L): 52.54
Maltose M (g/L): 16.55
Volume V (L): 5.68

t = 15.00 h

Biomass X (g/L): 76.24
Glucose S (g/L): 49.57
Lactic Acid P (g/L): 61.47
Maltose M (g/L): 16.57
Volume V (L): 5.27

t = 16.00 h

Biomass X (g/L): 78.67
Glucose S (g/L): 47.22
Lactic Acid P (g/L): 29.69
Maltose M (g/L): 17.25
Volume V (L): 5.70

t = 17.00 h

Biomass X (g/L): 88.40
Glucose S (g/L): 48.90
Lactic Acid P (g/L): 64.70
Maltose M (g/L): 15.78
Volume V (L): 5.32

t = 18.00 h

Biomass X (g/L): 78.72
Glucose S (g/L): 53.50
Lactic Acid P (g/L): 38.59
Maltose M (g/L): 17.65
Volume V (L): 5.80

t = 19.00 h

Biomass X (g/L): 70.83
Glucose S (g/L): 42.05
Lactic Acid P (g/L): 59.95
Maltose M (g/L): 17.06
Volume V (L): 5.90

t = 20.00 h

Biomass X (g/L): 64.70
Glucose S (g/L): 50.65
Lactic Acid P (g/L): 81.46
Maltose M (g/L): 16.72
Volume V (L): 5.57

t = 21.00 h

Biomass X (g/L): 76.55
Glucose S (g/L): 43.62
Lactic Acid P (g/L): 59.18
Maltose M (g/L): 15.87
Volume V (L): 6.01

t = 22.00 h

Biomass X (g/L): 90.92
Glucose S (g/L): 44.45
Lactic Acid P (g/L): 62.55
Maltose M (g/L): 16.61
Volume V (L): 5.45

t = 23.00 h

Biomass X (g/L): 102.19
Glucose S (g/L): 47.17
Lactic Acid P (g/L): 58.84
Maltose M (g/L): 13.63
Volume V (L): 6.05

t = 24.00 h

Biomass X (g/L): 89.06
Glucose S (g/L): 55.41
Lactic Acid P (g/L): 69.48
Maltose M (g/L): 16.08
Volume V (L): 6.16

t = 25.00 h

Biomass X (g/L): 77.03
Glucose S (g/L): 41.19
Lactic Acid P (g/L): 67.44
Maltose M (g/L): 15.75
Volume V (L): 5.96

t = 26.00 h

Biomass X (g/L): 81.09
Glucose S (g/L): 44.65
Lactic Acid P (g/L): 58.13
Maltose M (g/L): 14.07
Volume V (L): 5.88

t = 27.00 h

Biomass X (g/L): 82.01
Glucose S (g/L): 38.69
Lactic Acid P (g/L): 64.50
Maltose M (g/L): 15.71
Volume V (L): 6.03

t = 28.00 h

Biomass X (g/L): 87.02
Glucose S (g/L): 40.66
Lactic Acid P (g/L): 55.42
Maltose M (g/L): 15.74
Volume V (L): 5.87

t = 29.00 h

Biomass X (g/L): 102.91
Glucose S (g/L): 38.49
Lactic Acid P (g/L): 65.99
Maltose M (g/L): 13.77
Volume V (L): 6.59

t = 30.00 h

Biomass X (g/L): 104.10
Glucose S (g/L): 46.24
Lactic Acid P (g/L): 63.74
Maltose M (g/L): 14.17
Volume V (L): 5.94

t = 31.00 h

Biomass X (g/L): 90.61
Glucose S (g/L): 47.33
Lactic Acid P (g/L): 69.87
Maltose M (g/L): 16.55
Volume V (L): 6.38

t = 32.00 h

Biomass X (g/L): 94.51
Glucose S (g/L): 48.94
Lactic Acid P (g/L): 62.18
Maltose M (g/L): 14.73
Volume V (L): 5.98

t = 33.00 h

Biomass X (g/L): 84.87
Glucose S (g/L): 40.67
Lactic Acid P (g/L): 67.65
Maltose M (g/L): 15.51
Volume V (L): 6.51

t = 34.00 h

Biomass X (g/L): 75.88
Glucose S (g/L): 48.15
Lactic Acid P (g/L): 70.41
Maltose M (g/L): 15.53

Volume V (L): 6.64

t = 35.00 h

Biomass X (g/L): 111.39

Glucose S (g/L): 46.02

Lactic Acid P (g/L): 71.64

Maltose M (g/L): 14.87

Volume V (L): 6.33

t = 36.00 h

Biomass X (g/L): 70.03

Glucose S (g/L): 45.22

Lactic Acid P (g/L): 74.08

Maltose M (g/L): 15.25

Volume V (L): 6.34

t = 37.00 h

Biomass X (g/L): 75.06

Glucose S (g/L): 38.55

Lactic Acid P (g/L): 70.16

Maltose M (g/L): 12.76

Volume V (L): 6.11

t = 38.00 h

Biomass X (g/L): 111.83

Glucose S (g/L): 46.90

Lactic Acid P (g/L): 64.61

Maltose M (g/L): 14.10

Volume V (L): 6.12

t = 39.00 h

Biomass X (g/L): 93.93

Glucose S (g/L): 47.34

Lactic Acid P (g/L): 73.60

Maltose M (g/L): 11.83

Volume V (L): 6.45

t = 40.00 h

Biomass X (g/L): 82.43

Glucose S (g/L): 47.12

Lactic Acid P (g/L): 71.46

Maltose M (g/L): 12.76

Volume V (L): 6.73

t = 41.00 h

Biomass X (g/L): 100.71

Glucose S (g/L): 49.14

Lactic Acid P (g/L): 83.70

Maltose M (g/L): 13.58

Volume V (L): 6.34

t = 42.00 h

Biomass X (g/L): 101.62

Glucose S (g/L): 51.15

Lactic Acid P (g/L): 85.34

Maltose M (g/L): 13.40

Volume V (L): 6.96

t = 43.00 h

Biomass X (g/L): 80.34

Glucose S (g/L): 41.04

Lactic Acid P (g/L): 60.93

Maltose M (g/L): 13.86

Volume V (L): 6.40

t = 44.00 h

Biomass X (g/L): 88.37

Glucose S (g/L): 46.48

Lactic Acid P (g/L): 80.84

Maltose M (g/L): 13.72

Volume V (L): 6.61

t = 45.00 h

Biomass X (g/L): 102.55

Glucose S (g/L): 51.07

Lactic Acid P (g/L): 83.17

Maltose M (g/L): 13.75

Volume V (L): 6.65

t = 46.00 h

Biomass X (g/L): 85.38

Glucose S (g/L): 46.85

Lactic Acid P (g/L): 62.24
Maltose M (g/L): 13.70
Volume V (L): 6.61

t = 47.00 h

Biomass X (g/L): 96.39
Glucose S (g/L): 54.34
Lactic Acid P (g/L): 81.49
Maltose M (g/L): 13.74
Volume V (L): 6.23

t = 48.00 h

Biomass X (g/L): 107.83
Glucose S (g/L): 49.64
Lactic Acid P (g/L): 84.92
Maltose M (g/L): 12.67
Volume V (L): 6.57

t = 49.00 h

Biomass X (g/L): 69.17
Glucose S (g/L): 38.10
Lactic Acid P (g/L): 67.08
Maltose M (g/L): 12.76
Volume V (L): 6.78

t = 50.00 h

Biomass X (g/L): 89.77
Glucose S (g/L): 54.72
Lactic Acid P (g/L): 74.39
Maltose M (g/L): 13.21
Volume V (L): 6.75

t = 51.00 h

Biomass X (g/L): 95.72
Glucose S (g/L): 49.52
Lactic Acid P (g/L): 78.83
Maltose M (g/L): 13.14
Volume V (L): 6.95

t = 52.00 h

Biomass X (g/L): 74.84

Glucose S (g/L): 40.29
Lactic Acid P (g/L): 82.88
Maltose M (g/L): 12.25
Volume V (L): 7.01

t = 53.00 h

Biomass X (g/L): 87.44
Glucose S (g/L): 49.49
Lactic Acid P (g/L): 67.83
Maltose M (g/L): 13.83
Volume V (L): 6.83

t = 54.00 h

Biomass X (g/L): 103.18
Glucose S (g/L): 43.96
Lactic Acid P (g/L): 89.17
Maltose M (g/L): 13.48
Volume V (L): 6.71

t = 55.00 h

Biomass X (g/L): 85.35
Glucose S (g/L): 56.11
Lactic Acid P (g/L): 58.11
Maltose M (g/L): 12.19
Volume V (L): 7.28

t = 56.00 h

Biomass X (g/L): 77.30
Glucose S (g/L): 51.91
Lactic Acid P (g/L): 91.31
Maltose M (g/L): 12.62
Volume V (L): 6.70

t = 57.00 h

Biomass X (g/L): 103.53
Glucose S (g/L): 47.13
Lactic Acid P (g/L): 83.94
Maltose M (g/L): 14.24
Volume V (L): 6.90

t = 58.00 h

Biomass X (g/L): 96.41
Glucose S (g/L): 57.61
Lactic Acid P (g/L): 72.66
Maltose M (g/L): 12.75
Volume V (L): 6.88

t = 59.00 h

Biomass X (g/L): 91.04
Glucose S (g/L): 55.96
Lactic Acid P (g/L): 87.49
Maltose M (g/L): 12.86
Volume V (L): 7.25

t = 60.00 h

Biomass X (g/L): 90.54
Glucose S (g/L): 52.31
Lactic Acid P (g/L): 68.91
Maltose M (g/L): 12.02
Volume V (L): 7.41

t = 61.00 h

Biomass X (g/L): 101.72
Glucose S (g/L): 46.83
Lactic Acid P (g/L): 80.55
Maltose M (g/L): 12.18
Volume V (L): 7.13

t = 62.00 h

Biomass X (g/L): 94.10
Glucose S (g/L): 49.58
Lactic Acid P (g/L): 81.97
Maltose M (g/L): 13.50
Volume V (L): 7.25

t = 63.00 h

Biomass X (g/L): 97.43
Glucose S (g/L): 49.89
Lactic Acid P (g/L): 82.15
Maltose M (g/L): 12.37
Volume V (L): 7.33

t = 64.00 h

Biomass X (g/L): 83.70
Glucose S (g/L): 59.61
Lactic Acid P (g/L): 71.77
Maltose M (g/L): 12.16
Volume V (L): 7.66

t = 65.00 h

Biomass X (g/L): 71.12
Glucose S (g/L): 45.89
Lactic Acid P (g/L): 75.34
Maltose M (g/L): 12.15
Volume V (L): 6.93

t = 66.00 h

Biomass X (g/L): 96.54
Glucose S (g/L): 55.79
Lactic Acid P (g/L): 81.65
Maltose M (g/L): 14.02
Volume V (L): 7.50

t = 67.00 h

Biomass X (g/L): 101.15
Glucose S (g/L): 55.01
Lactic Acid P (g/L): 73.44
Maltose M (g/L): 12.04
Volume V (L): 7.46

t = 68.00 h

Biomass X (g/L): 87.53
Glucose S (g/L): 62.88
Lactic Acid P (g/L): 83.43
Maltose M (g/L): 11.31
Volume V (L): 7.34

t = 69.00 h

Biomass X (g/L): 98.54
Glucose S (g/L): 58.61
Lactic Acid P (g/L): 74.90
Maltose M (g/L): 11.38
Volume V (L): 7.52

t = 70.00 h

Biomass X (g/L): 85.95
Glucose S (g/L): 60.77
Lactic Acid P (g/L): 90.93
Maltose M (g/L): 12.42
Volume V (L): 7.64

t = 71.00 h

Biomass X (g/L): 88.19
Glucose S (g/L): 54.99
Lactic Acid P (g/L): 96.63
Maltose M (g/L): 10.67
Volume V (L): 7.65

t = 72.00 h

Biomass X (g/L): 106.56
Glucose S (g/L): 51.28
Lactic Acid P (g/L): 78.51
Maltose M (g/L): 12.04
Volume V (L): 8.02

t = 73.00 h

Biomass X (g/L): 85.68
Glucose S (g/L): 56.94
Lactic Acid P (g/L): 90.64
Maltose M (g/L): 11.08
Volume V (L): 7.62

t = 74.00 h

Biomass X (g/L): 84.84
Glucose S (g/L): 63.75
Lactic Acid P (g/L): 79.66
Maltose M (g/L): 10.16
Volume V (L): 7.63

t = 75.00 h

Biomass X (g/L): 69.26
Glucose S (g/L): 64.12
Lactic Acid P (g/L): 72.69
Maltose M (g/L): 10.77

Volume V (L): 7.62

Batch 85

Initial Conditions:

Batch: 85.00

X0 (g/L): 2.43

S0 (g/L): 105.56

M0 (g/L): 53.71

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.51

F_vol (L/h): 0.03

S_feed (g/L): 198.83

M_feed (g/L): 101.17

V_final (L): 7.16

Batch 85 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 8.20

Glucose S (g/L): 101.19

Lactic Acid P (g/L): 8.80

Maltose M (g/L): 51.79

Volume V (L): 5.26

t = 1.00 h

Biomass X (g/L): 2.99

Glucose S (g/L): 110.32

Lactic Acid P (g/L): 3.92

Maltose M (g/L): 56.83

Volume V (L): 5.02

t = 2.00 h

Biomass X (g/L): 4.64

Glucose S (g/L): 108.99

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 51.65

Volume V (L): 4.93

t = 3.00 h

Biomass X (g/L): 6.24

Glucose S (g/L): 109.99

Lactic Acid P (g/L): 0.00

Maltose M (g/L): 53.56

Volume V (L): 5.42

t = 4.00 h

Biomass X (g/L): 6.64

Glucose S (g/L): 109.83

Lactic Acid P (g/L): 0.10

Maltose M (g/L): 52.05

Volume V (L): 4.95

t = 5.00 h

Biomass X (g/L): 4.18

Glucose S (g/L): 96.69

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 48.34
Volume V (L): 4.98

t = 6.00 h

Biomass X (g/L): 17.92
Glucose S (g/L): 99.93
Lactic Acid P (g/L): 9.34
Maltose M (g/L): 43.68
Volume V (L): 5.39

t = 7.00 h

Biomass X (g/L): 14.14
Glucose S (g/L): 94.21
Lactic Acid P (g/L): 19.98
Maltose M (g/L): 52.16
Volume V (L): 5.32

t = 8.00 h

Biomass X (g/L): 27.55
Glucose S (g/L): 99.60
Lactic Acid P (g/L): 36.90
Maltose M (g/L): 45.26
Volume V (L): 5.21

t = 9.00 h

Biomass X (g/L): 26.71
Glucose S (g/L): 88.71
Lactic Acid P (g/L): 6.00
Maltose M (g/L): 46.29
Volume V (L): 5.40

t = 10.00 h

Biomass X (g/L): 26.15
Glucose S (g/L): 81.97
Lactic Acid P (g/L): 29.15
Maltose M (g/L): 46.40
Volume V (L): 5.34

t = 11.00 h

Biomass X (g/L): 51.26

Glucose S (g/L): 87.49
Lactic Acid P (g/L): 16.80
Maltose M (g/L): 46.08
Volume V (L): 5.11

t = 12.00 h

Biomass X (g/L): 38.54
Glucose S (g/L): 73.81
Lactic Acid P (g/L): 16.14
Maltose M (g/L): 48.41
Volume V (L): 4.93

t = 13.00 h

Biomass X (g/L): 50.64
Glucose S (g/L): 83.18
Lactic Acid P (g/L): 19.53
Maltose M (g/L): 45.07
Volume V (L): 5.28

t = 14.00 h

Biomass X (g/L): 41.12
Glucose S (g/L): 72.13
Lactic Acid P (g/L): 35.27
Maltose M (g/L): 44.12
Volume V (L): 5.59

t = 15.00 h

Biomass X (g/L): 31.29
Glucose S (g/L): 71.31
Lactic Acid P (g/L): 27.80
Maltose M (g/L): 46.87
Volume V (L): 5.36

t = 16.00 h

Biomass X (g/L): 50.22
Glucose S (g/L): 61.14
Lactic Acid P (g/L): 33.85
Maltose M (g/L): 45.12
Volume V (L): 5.54

t = 17.00 h

Biomass X (g/L): 37.21
Glucose S (g/L): 79.01
Lactic Acid P (g/L): 38.15
Maltose M (g/L): 40.73
Volume V (L): 5.52

t = 18.00 h

Biomass X (g/L): 60.14
Glucose S (g/L): 60.71
Lactic Acid P (g/L): 37.66
Maltose M (g/L): 39.02
Volume V (L): 5.23

t = 19.00 h

Biomass X (g/L): 80.74
Glucose S (g/L): 64.44
Lactic Acid P (g/L): 39.22
Maltose M (g/L): 39.44
Volume V (L): 5.48

t = 20.00 h

Biomass X (g/L): 58.49
Glucose S (g/L): 55.33
Lactic Acid P (g/L): 25.41
Maltose M (g/L): 41.18
Volume V (L): 5.64

t = 21.00 h

Biomass X (g/L): 48.24
Glucose S (g/L): 58.75
Lactic Acid P (g/L): 52.09
Maltose M (g/L): 37.24
Volume V (L): 5.47

t = 22.00 h

Biomass X (g/L): 87.95
Glucose S (g/L): 70.80
Lactic Acid P (g/L): 53.68
Maltose M (g/L): 40.62
Volume V (L): 5.57

t = 23.00 h

Biomass X (g/L): 69.19
Glucose S (g/L): 62.12
Lactic Acid P (g/L): 46.81
Maltose M (g/L): 39.52
Volume V (L): 5.73

t = 24.00 h

Biomass X (g/L): 75.66
Glucose S (g/L): 55.83
Lactic Acid P (g/L): 59.72
Maltose M (g/L): 32.09
Volume V (L): 5.71

t = 25.00 h

Biomass X (g/L): 85.57
Glucose S (g/L): 58.52
Lactic Acid P (g/L): 71.33
Maltose M (g/L): 34.92
Volume V (L): 5.90

t = 26.00 h

Biomass X (g/L): 84.16
Glucose S (g/L): 55.05
Lactic Acid P (g/L): 59.13
Maltose M (g/L): 30.21
Volume V (L): 5.82

t = 27.00 h

Biomass X (g/L): 87.77
Glucose S (g/L): 56.83
Lactic Acid P (g/L): 73.20
Maltose M (g/L): 37.12
Volume V (L): 5.61

t = 28.00 h

Biomass X (g/L): 90.44
Glucose S (g/L): 60.61
Lactic Acid P (g/L): 83.27
Maltose M (g/L): 40.80
Volume V (L): 5.73

t = 29.00 h

Biomass X (g/L): 63.81
Glucose S (g/L): 54.60
Lactic Acid P (g/L): 87.21
Maltose M (g/L): 31.82
Volume V (L): 5.84

t = 30.00 h

Biomass X (g/L): 68.17
Glucose S (g/L): 57.07
Lactic Acid P (g/L): 51.64
Maltose M (g/L): 33.06
Volume V (L): 6.10

t = 31.00 h

Biomass X (g/L): 95.97
Glucose S (g/L): 56.84
Lactic Acid P (g/L): 52.63
Maltose M (g/L): 36.01
Volume V (L): 5.81

t = 32.00 h

Biomass X (g/L): 76.32
Glucose S (g/L): 53.69
Lactic Acid P (g/L): 60.78
Maltose M (g/L): 33.02
Volume V (L): 5.62

t = 33.00 h

Biomass X (g/L): 78.11
Glucose S (g/L): 51.38
Lactic Acid P (g/L): 67.25
Maltose M (g/L): 32.63
Volume V (L): 6.03

t = 34.00 h

Biomass X (g/L): 72.54
Glucose S (g/L): 52.33
Lactic Acid P (g/L): 82.37
Maltose M (g/L): 36.91

Volume V (L): 5.85

t = 35.00 h

Biomass X (g/L): 82.02

Glucose S (g/L): 58.56

Lactic Acid P (g/L): 73.68

Maltose M (g/L): 33.55

Volume V (L): 5.79

t = 36.00 h

Biomass X (g/L): 83.07

Glucose S (g/L): 58.94

Lactic Acid P (g/L): 56.38

Maltose M (g/L): 30.32

Volume V (L): 6.04

t = 37.00 h

Biomass X (g/L): 77.35

Glucose S (g/L): 48.99

Lactic Acid P (g/L): 62.87

Maltose M (g/L): 28.23

Volume V (L): 6.24

t = 38.00 h

Biomass X (g/L): 76.52

Glucose S (g/L): 53.54

Lactic Acid P (g/L): 69.11

Maltose M (g/L): 40.30

Volume V (L): 6.00

t = 39.00 h

Biomass X (g/L): 91.17

Glucose S (g/L): 51.04

Lactic Acid P (g/L): 55.48

Maltose M (g/L): 36.84

Volume V (L): 6.21

t = 40.00 h

Biomass X (g/L): 79.70

Glucose S (g/L): 59.15

Lactic Acid P (g/L): 70.84

Maltose M (g/L): 31.95

Volume V (L): 6.27

t = 41.00 h

Biomass X (g/L): 82.06

Glucose S (g/L): 52.67

Lactic Acid P (g/L): 59.57

Maltose M (g/L): 33.18

Volume V (L): 6.32

t = 42.00 h

Biomass X (g/L): 65.78

Glucose S (g/L): 66.61

Lactic Acid P (g/L): 55.55

Maltose M (g/L): 31.20

Volume V (L): 6.08

t = 43.00 h

Biomass X (g/L): 109.03

Glucose S (g/L): 56.49

Lactic Acid P (g/L): 62.98

Maltose M (g/L): 33.77

Volume V (L): 6.05

t = 44.00 h

Biomass X (g/L): 98.79

Glucose S (g/L): 61.64

Lactic Acid P (g/L): 76.82

Maltose M (g/L): 29.52

Volume V (L): 6.10

t = 45.00 h

Biomass X (g/L): 83.92

Glucose S (g/L): 50.45

Lactic Acid P (g/L): 77.22

Maltose M (g/L): 30.12

Volume V (L): 6.44

t = 46.00 h

Biomass X (g/L): 79.81

Glucose S (g/L): 55.95

Lactic Acid P (g/L): 74.64

Maltose M (g/L): 32.71

Volume V (L): 6.44

t = 47.00 h

Biomass X (g/L): 89.40

Glucose S (g/L): 59.27

Lactic Acid P (g/L): 83.87

Maltose M (g/L): 35.31

Volume V (L): 6.19

t = 48.00 h

Biomass X (g/L): 89.18

Glucose S (g/L): 50.96

Lactic Acid P (g/L): 83.24

Maltose M (g/L): 32.97

Volume V (L): 6.75

t = 49.00 h

Biomass X (g/L): 76.07

Glucose S (g/L): 56.63

Lactic Acid P (g/L): 76.94

Maltose M (g/L): 34.11

Volume V (L): 6.23

t = 50.00 h

Biomass X (g/L): 62.80

Glucose S (g/L): 62.20

Lactic Acid P (g/L): 75.97

Maltose M (g/L): 29.37

Volume V (L): 6.62

t = 51.00 h

Biomass X (g/L): 83.94

Glucose S (g/L): 55.31

Lactic Acid P (g/L): 76.96

Maltose M (g/L): 28.21

Volume V (L): 6.73

t = 52.00 h

Biomass X (g/L): 73.11

Glucose S (g/L): 52.86
Lactic Acid P (g/L): 74.59
Maltose M (g/L): 24.14
Volume V (L): 6.52

t = 53.00 h

Biomass X (g/L): 91.20
Glucose S (g/L): 66.84
Lactic Acid P (g/L): 55.01
Maltose M (g/L): 29.69
Volume V (L): 6.40

t = 54.00 h

Biomass X (g/L): 91.23
Glucose S (g/L): 51.73
Lactic Acid P (g/L): 87.89
Maltose M (g/L): 30.33
Volume V (L): 6.43

t = 55.00 h

Biomass X (g/L): 79.16
Glucose S (g/L): 58.86
Lactic Acid P (g/L): 80.96
Maltose M (g/L): 28.11
Volume V (L): 6.54

t = 56.00 h

Biomass X (g/L): 83.68
Glucose S (g/L): 54.67
Lactic Acid P (g/L): 65.40
Maltose M (g/L): 26.49
Volume V (L): 6.80

t = 57.00 h

Biomass X (g/L): 107.60
Glucose S (g/L): 53.77
Lactic Acid P (g/L): 91.71
Maltose M (g/L): 32.09
Volume V (L): 6.44

t = 58.00 h

Biomass X (g/L): 97.20
Glucose S (g/L): 62.74
Lactic Acid P (g/L): 72.16
Maltose M (g/L): 30.09
Volume V (L): 6.36

t = 59.00 h

Biomass X (g/L): 84.30
Glucose S (g/L): 58.13
Lactic Acid P (g/L): 77.19
Maltose M (g/L): 31.82
Volume V (L): 6.76

t = 60.00 h

Biomass X (g/L): 112.85
Glucose S (g/L): 61.59
Lactic Acid P (g/L): 88.60
Maltose M (g/L): 25.05
Volume V (L): 6.83

t = 61.00 h

Biomass X (g/L): 76.31
Glucose S (g/L): 48.12
Lactic Acid P (g/L): 80.97
Maltose M (g/L): 25.53
Volume V (L): 7.10

t = 62.00 h

Biomass X (g/L): 80.21
Glucose S (g/L): 62.68
Lactic Acid P (g/L): 92.20
Maltose M (g/L): 30.83
Volume V (L): 6.55

t = 63.00 h

Biomass X (g/L): 87.41
Glucose S (g/L): 56.66
Lactic Acid P (g/L): 75.86
Maltose M (g/L): 27.01
Volume V (L): 6.79

t = 64.00 h

Biomass X (g/L): 72.45
Glucose S (g/L): 50.97
Lactic Acid P (g/L): 96.20
Maltose M (g/L): 25.29
Volume V (L): 7.05

t = 65.00 h

Biomass X (g/L): 106.99
Glucose S (g/L): 70.74
Lactic Acid P (g/L): 72.17
Maltose M (g/L): 25.28
Volume V (L): 6.90

t = 66.00 h

Biomass X (g/L): 90.06
Glucose S (g/L): 64.40
Lactic Acid P (g/L): 93.01
Maltose M (g/L): 26.48
Volume V (L): 6.87

t = 67.00 h

Biomass X (g/L): 79.75
Glucose S (g/L): 62.78
Lactic Acid P (g/L): 58.01
Maltose M (g/L): 29.21
Volume V (L): 7.07

t = 68.00 h

Biomass X (g/L): 92.52
Glucose S (g/L): 68.37
Lactic Acid P (g/L): 69.69
Maltose M (g/L): 25.65
Volume V (L): 6.86

t = 69.00 h

Biomass X (g/L): 89.21
Glucose S (g/L): 63.50
Lactic Acid P (g/L): 83.66
Maltose M (g/L): 28.21
Volume V (L): 7.14

t = 70.00 h

Biomass X (g/L): 84.34
Glucose S (g/L): 58.40
Lactic Acid P (g/L): 77.01
Maltose M (g/L): 27.08
Volume V (L): 6.80

t = 71.00 h

Biomass X (g/L): 74.73
Glucose S (g/L): 61.93
Lactic Acid P (g/L): 80.42
Maltose M (g/L): 23.00
Volume V (L): 6.79

t = 72.00 h

Biomass X (g/L): 72.18
Glucose S (g/L): 59.64
Lactic Acid P (g/L): 81.77
Maltose M (g/L): 22.82
Volume V (L): 7.03

t = 73.00 h

Biomass X (g/L): 109.82
Glucose S (g/L): 60.87
Lactic Acid P (g/L): 69.85
Maltose M (g/L): 23.87
Volume V (L): 6.66

t = 74.00 h

Biomass X (g/L): 72.69
Glucose S (g/L): 57.26
Lactic Acid P (g/L): 70.78
Maltose M (g/L): 21.13
Volume V (L): 7.15

t = 75.00 h

Biomass X (g/L): 104.35
Glucose S (g/L): 59.06
Lactic Acid P (g/L): 84.23
Maltose M (g/L): 23.33

Volume V (L): 6.86

Batch 86

Initial Conditions:

Batch: 86.00

X0 (g/L): 3.96

S0 (g/L): 75.49

M0 (g/L): 122.55

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.62

F_vol (L/h): 0.08

S_feed (g/L): 114.36

M_feed (g/L): 185.64

V_final (L): 11.26

Batch 86 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.67
Glucose S (g/L): 74.84
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 123.36
Volume V (L): 5.41

t = 1.00 h

Biomass X (g/L): 7.08
Glucose S (g/L): 75.96
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 123.09
Volume V (L): 4.26

t = 2.00 h

Biomass X (g/L): 6.38
Glucose S (g/L): 84.50
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 115.38
Volume V (L): 4.38

t = 3.00 h

Biomass X (g/L): 13.41
Glucose S (g/L): 74.76
Lactic Acid P (g/L): 11.25
Maltose M (g/L): 115.91
Volume V (L): 5.59

t = 4.00 h

Biomass X (g/L): 13.86
Glucose S (g/L): 78.68
Lactic Acid P (g/L): 3.86
Maltose M (g/L): 116.71
Volume V (L): 5.51

t = 5.00 h

Biomass X (g/L): 12.27
Glucose S (g/L): 71.93

Lactic Acid P (g/L): 25.00

Maltose M (g/L): 122.81

Volume V (L): 5.03

t = 6.00 h

Biomass X (g/L): 36.32

Glucose S (g/L): 78.16

Lactic Acid P (g/L): 5.94

Maltose M (g/L): 119.64

Volume V (L): 5.68

t = 7.00 h

Biomass X (g/L): 26.58

Glucose S (g/L): 71.45

Lactic Acid P (g/L): 15.21

Maltose M (g/L): 124.65

Volume V (L): 4.88

t = 8.00 h

Biomass X (g/L): 34.55

Glucose S (g/L): 76.99

Lactic Acid P (g/L): 14.58

Maltose M (g/L): 104.53

Volume V (L): 7.30

t = 9.00 h

Biomass X (g/L): 41.92

Glucose S (g/L): 70.55

Lactic Acid P (g/L): 24.42

Maltose M (g/L): 104.35

Volume V (L): 6.41

t = 10.00 h

Biomass X (g/L): 42.53

Glucose S (g/L): 64.88

Lactic Acid P (g/L): 33.37

Maltose M (g/L): 116.73

Volume V (L): 6.15

t = 11.00 h

Biomass X (g/L): 60.31

Glucose S (g/L): 60.48
Lactic Acid P (g/L): 32.58
Maltose M (g/L): 117.60
Volume V (L): 5.96

t = 12.00 h

Biomass X (g/L): 46.29
Glucose S (g/L): 61.79
Lactic Acid P (g/L): 12.17
Maltose M (g/L): 108.69
Volume V (L): 5.69

t = 13.00 h

Biomass X (g/L): 48.57
Glucose S (g/L): 63.57
Lactic Acid P (g/L): 37.80
Maltose M (g/L): 82.69
Volume V (L): 5.43

t = 14.00 h

Biomass X (g/L): 46.05
Glucose S (g/L): 62.38
Lactic Acid P (g/L): 27.12
Maltose M (g/L): 102.35
Volume V (L): 6.11

t = 15.00 h

Biomass X (g/L): 53.17
Glucose S (g/L): 58.04
Lactic Acid P (g/L): 36.01
Maltose M (g/L): 107.35
Volume V (L): 5.82

t = 16.00 h

Biomass X (g/L): 51.66
Glucose S (g/L): 61.11
Lactic Acid P (g/L): 47.36
Maltose M (g/L): 99.52
Volume V (L): 5.93

t = 17.00 h

Biomass X (g/L): 67.94
Glucose S (g/L): 53.72
Lactic Acid P (g/L): 33.42
Maltose M (g/L): 104.59
Volume V (L): 5.71

t = 18.00 h

Biomass X (g/L): 55.17
Glucose S (g/L): 47.37
Lactic Acid P (g/L): 55.81
Maltose M (g/L): 113.17
Volume V (L): 6.63

t = 19.00 h

Biomass X (g/L): 66.87
Glucose S (g/L): 56.53
Lactic Acid P (g/L): 44.20
Maltose M (g/L): 109.63
Volume V (L): 6.40

t = 20.00 h

Biomass X (g/L): 73.55
Glucose S (g/L): 59.11
Lactic Acid P (g/L): 68.48
Maltose M (g/L): 105.30
Volume V (L): 6.94

t = 21.00 h

Biomass X (g/L): 71.43
Glucose S (g/L): 58.79
Lactic Acid P (g/L): 51.28
Maltose M (g/L): 99.56
Volume V (L): 6.98

t = 22.00 h

Biomass X (g/L): 63.01
Glucose S (g/L): 50.61
Lactic Acid P (g/L): 67.36
Maltose M (g/L): 96.30
Volume V (L): 6.95

t = 23.00 h

Biomass X (g/L): 79.77
Glucose S (g/L): 52.83
Lactic Acid P (g/L): 64.17
Maltose M (g/L): 102.39
Volume V (L): 8.25

t = 24.00 h

Biomass X (g/L): 73.54
Glucose S (g/L): 56.74
Lactic Acid P (g/L): 49.25
Maltose M (g/L): 97.48
Volume V (L): 7.42

t = 25.00 h

Biomass X (g/L): 47.95
Glucose S (g/L): 52.71
Lactic Acid P (g/L): 44.22
Maltose M (g/L): 96.84
Volume V (L): 6.46

t = 26.00 h

Biomass X (g/L): 67.14
Glucose S (g/L): 55.14
Lactic Acid P (g/L): 57.93
Maltose M (g/L): 87.08
Volume V (L): 7.40

t = 27.00 h

Biomass X (g/L): 90.96
Glucose S (g/L): 52.89
Lactic Acid P (g/L): 52.70
Maltose M (g/L): 90.92
Volume V (L): 7.23

t = 28.00 h

Biomass X (g/L): 92.60
Glucose S (g/L): 52.07
Lactic Acid P (g/L): 72.94
Maltose M (g/L): 98.27
Volume V (L): 6.67

t = 29.00 h

Biomass X (g/L): 84.15
Glucose S (g/L): 53.05
Lactic Acid P (g/L): 57.47
Maltose M (g/L): 87.20
Volume V (L): 6.71

t = 30.00 h

Biomass X (g/L): 87.29
Glucose S (g/L): 59.16
Lactic Acid P (g/L): 57.79
Maltose M (g/L): 94.02
Volume V (L): 7.73

t = 31.00 h

Biomass X (g/L): 72.89
Glucose S (g/L): 56.74
Lactic Acid P (g/L): 48.75
Maltose M (g/L): 94.08
Volume V (L): 7.94

t = 32.00 h

Biomass X (g/L): 78.39
Glucose S (g/L): 61.11
Lactic Acid P (g/L): 65.67
Maltose M (g/L): 93.66
Volume V (L): 7.15

t = 33.00 h

Biomass X (g/L): 93.22
Glucose S (g/L): 59.35
Lactic Acid P (g/L): 52.09
Maltose M (g/L): 93.42
Volume V (L): 8.24

t = 34.00 h

Biomass X (g/L): 85.87
Glucose S (g/L): 63.25
Lactic Acid P (g/L): 67.16
Maltose M (g/L): 93.07

Volume V (L): 7.98

t = 35.00 h

Biomass X (g/L): 82.77

Glucose S (g/L): 61.43

Lactic Acid P (g/L): 68.97

Maltose M (g/L): 85.41

Volume V (L): 8.52

t = 36.00 h

Biomass X (g/L): 80.67

Glucose S (g/L): 61.86

Lactic Acid P (g/L): 57.64

Maltose M (g/L): 80.03

Volume V (L): 8.55

t = 37.00 h

Biomass X (g/L): 94.02

Glucose S (g/L): 59.80

Lactic Acid P (g/L): 73.40

Maltose M (g/L): 82.63

Volume V (L): 8.01

t = 38.00 h

Biomass X (g/L): 93.76

Glucose S (g/L): 70.03

Lactic Acid P (g/L): 62.89

Maltose M (g/L): 90.85

Volume V (L): 8.99

t = 39.00 h

Biomass X (g/L): 89.44

Glucose S (g/L): 61.81

Lactic Acid P (g/L): 52.67

Maltose M (g/L): 79.46

Volume V (L): 7.69

t = 40.00 h

Biomass X (g/L): 84.02

Glucose S (g/L): 56.09

Lactic Acid P (g/L): 80.07

Maltose M (g/L): 79.69

Volume V (L): 9.54

t = 41.00 h

Biomass X (g/L): 80.11

Glucose S (g/L): 63.64

Lactic Acid P (g/L): 58.43

Maltose M (g/L): 79.01

Volume V (L): 7.98

t = 42.00 h

Biomass X (g/L): 90.31

Glucose S (g/L): 60.49

Lactic Acid P (g/L): 71.61

Maltose M (g/L): 73.92

Volume V (L): 8.48

t = 43.00 h

Biomass X (g/L): 73.18

Glucose S (g/L): 67.31

Lactic Acid P (g/L): 71.27

Maltose M (g/L): 77.17

Volume V (L): 8.31

t = 44.00 h

Biomass X (g/L): 75.25

Glucose S (g/L): 67.11

Lactic Acid P (g/L): 71.31

Maltose M (g/L): 88.47

Volume V (L): 8.49

t = 45.00 h

Biomass X (g/L): 69.96

Glucose S (g/L): 64.99

Lactic Acid P (g/L): 71.50

Maltose M (g/L): 78.31

Volume V (L): 10.22

t = 46.00 h

Biomass X (g/L): 95.79

Glucose S (g/L): 63.56

Lactic Acid P (g/L): 59.85

Maltose M (g/L): 71.97

Volume V (L): 8.61

t = 47.00 h

Biomass X (g/L): 77.03

Glucose S (g/L): 67.43

Lactic Acid P (g/L): 83.95

Maltose M (g/L): 78.24

Volume V (L): 9.35

t = 48.00 h

Biomass X (g/L): 87.84

Glucose S (g/L): 65.71

Lactic Acid P (g/L): 77.69

Maltose M (g/L): 83.33

Volume V (L): 8.89

t = 49.00 h

Biomass X (g/L): 82.74

Glucose S (g/L): 70.79

Lactic Acid P (g/L): 73.68

Maltose M (g/L): 69.74

Volume V (L): 9.56

t = 50.00 h

Biomass X (g/L): 111.74

Glucose S (g/L): 70.78

Lactic Acid P (g/L): 57.81

Maltose M (g/L): 84.62

Volume V (L): 8.90

t = 51.00 h

Biomass X (g/L): 88.13

Glucose S (g/L): 78.77

Lactic Acid P (g/L): 61.25

Maltose M (g/L): 70.76

Volume V (L): 9.39

t = 52.00 h

Biomass X (g/L): 65.97

Glucose S (g/L): 69.10
Lactic Acid P (g/L): 76.13
Maltose M (g/L): 76.00
Volume V (L): 9.66

t = 53.00 h

Biomass X (g/L): 95.07
Glucose S (g/L): 70.23
Lactic Acid P (g/L): 52.60
Maltose M (g/L): 75.37
Volume V (L): 9.07

t = 54.00 h

Biomass X (g/L): 80.00
Glucose S (g/L): 71.37
Lactic Acid P (g/L): 73.78
Maltose M (g/L): 73.98
Volume V (L): 10.89

t = 55.00 h

Biomass X (g/L): 81.72
Glucose S (g/L): 72.19
Lactic Acid P (g/L): 59.09
Maltose M (g/L): 76.19
Volume V (L): 9.94

t = 56.00 h

Biomass X (g/L): 77.10
Glucose S (g/L): 76.35
Lactic Acid P (g/L): 63.46
Maltose M (g/L): 80.34
Volume V (L): 9.56

t = 57.00 h

Biomass X (g/L): 85.35
Glucose S (g/L): 73.03
Lactic Acid P (g/L): 58.82
Maltose M (g/L): 75.26
Volume V (L): 9.41

t = 58.00 h

Biomass X (g/L): 85.12
Glucose S (g/L): 80.60
Lactic Acid P (g/L): 73.84
Maltose M (g/L): 71.61
Volume V (L): 9.89

t = 59.00 h

Biomass X (g/L): 77.06
Glucose S (g/L): 77.00
Lactic Acid P (g/L): 81.83
Maltose M (g/L): 77.00
Volume V (L): 9.32

t = 60.00 h

Biomass X (g/L): 78.27
Glucose S (g/L): 74.39
Lactic Acid P (g/L): 75.16
Maltose M (g/L): 75.10
Volume V (L): 10.90

t = 61.00 h

Biomass X (g/L): 84.00
Glucose S (g/L): 74.27
Lactic Acid P (g/L): 76.33
Maltose M (g/L): 69.73
Volume V (L): 10.31

t = 62.00 h

Biomass X (g/L): 60.41
Glucose S (g/L): 80.38
Lactic Acid P (g/L): 73.93
Maltose M (g/L): 73.19
Volume V (L): 10.13

t = 63.00 h

Biomass X (g/L): 68.94
Glucose S (g/L): 77.92
Lactic Acid P (g/L): 78.71
Maltose M (g/L): 68.28
Volume V (L): 10.79

t = 64.00 h

Biomass X (g/L): 98.67
Glucose S (g/L): 78.50
Lactic Acid P (g/L): 76.96
Maltose M (g/L): 67.17
Volume V (L): 9.98

t = 65.00 h

Biomass X (g/L): 85.65
Glucose S (g/L): 77.28
Lactic Acid P (g/L): 48.70
Maltose M (g/L): 75.97
Volume V (L): 9.85

t = 66.00 h

Biomass X (g/L): 77.32
Glucose S (g/L): 79.55
Lactic Acid P (g/L): 81.23
Maltose M (g/L): 73.51
Volume V (L): 10.45

t = 67.00 h

Biomass X (g/L): 65.94
Glucose S (g/L): 78.40
Lactic Acid P (g/L): 80.45
Maltose M (g/L): 74.84
Volume V (L): 10.66

t = 68.00 h

Biomass X (g/L): 81.53
Glucose S (g/L): 85.15
Lactic Acid P (g/L): 70.22
Maltose M (g/L): 73.63
Volume V (L): 10.95

t = 69.00 h

Biomass X (g/L): 91.38
Glucose S (g/L): 83.79
Lactic Acid P (g/L): 55.97
Maltose M (g/L): 64.72
Volume V (L): 10.02

t = 70.00 h

Biomass X (g/L): 107.10
Glucose S (g/L): 81.66
Lactic Acid P (g/L): 75.93
Maltose M (g/L): 74.75
Volume V (L): 10.62

t = 71.00 h

Biomass X (g/L): 99.77
Glucose S (g/L): 82.22
Lactic Acid P (g/L): 76.83
Maltose M (g/L): 67.10
Volume V (L): 9.38

t = 72.00 h

Biomass X (g/L): 72.93
Glucose S (g/L): 83.68
Lactic Acid P (g/L): 70.12
Maltose M (g/L): 73.47
Volume V (L): 10.42

t = 73.00 h

Biomass X (g/L): 76.52
Glucose S (g/L): 83.97
Lactic Acid P (g/L): 75.86
Maltose M (g/L): 67.18
Volume V (L): 11.46

t = 74.00 h

Biomass X (g/L): 66.99
Glucose S (g/L): 81.64
Lactic Acid P (g/L): 83.91
Maltose M (g/L): 69.18
Volume V (L): 11.66

t = 75.00 h

Biomass X (g/L): 72.87
Glucose S (g/L): 79.32
Lactic Acid P (g/L): 95.89
Maltose M (g/L): 68.93

Volume V (L): 10.69

Batch 87

Initial Conditions:

Batch: 87.00

X0 (g/L): 8.94

S0 (g/L): 20.68

M0 (g/L): 127.53

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 6.17

F_vol (L/h): 0.10

S_feed (g/L): 41.85

M_feed (g/L): 258.15

V_final (L): 12.42

Batch 87 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.11
Glucose S (g/L): 19.68
Lactic Acid P (g/L): 4.98
Maltose M (g/L): 130.93
Volume V (L): 4.91

t = 1.00 h

Biomass X (g/L): 11.53
Glucose S (g/L): 23.53
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 129.01
Volume V (L): 4.66

t = 2.00 h

Biomass X (g/L): 23.52
Glucose S (g/L): 26.54
Lactic Acid P (g/L): 1.85
Maltose M (g/L): 130.81
Volume V (L): 4.91

t = 3.00 h

Biomass X (g/L): 30.64
Glucose S (g/L): 20.06
Lactic Acid P (g/L): 14.19
Maltose M (g/L): 125.88
Volume V (L): 5.83

t = 4.00 h

Biomass X (g/L): 22.13
Glucose S (g/L): 17.73
Lactic Acid P (g/L): 11.23
Maltose M (g/L): 129.06
Volume V (L): 4.26

t = 5.00 h

Biomass X (g/L): 22.24
Glucose S (g/L): 22.68

Lactic Acid P (g/L): 2.84
Maltose M (g/L): 115.85
Volume V (L): 3.71

t = 6.00 h

Biomass X (g/L): 15.73
Glucose S (g/L): 24.36
Lactic Acid P (g/L): 14.63
Maltose M (g/L): 129.62
Volume V (L): 6.13

t = 7.00 h

Biomass X (g/L): 33.07
Glucose S (g/L): 23.15
Lactic Acid P (g/L): 21.52
Maltose M (g/L): 122.95
Volume V (L): 5.86

t = 8.00 h

Biomass X (g/L): 53.92
Glucose S (g/L): 12.26
Lactic Acid P (g/L): 29.13
Maltose M (g/L): 125.81
Volume V (L): 5.80

t = 9.00 h

Biomass X (g/L): 50.54
Glucose S (g/L): 11.35
Lactic Acid P (g/L): 16.75
Maltose M (g/L): 129.27
Volume V (L): 5.12

t = 10.00 h

Biomass X (g/L): 44.04
Glucose S (g/L): 7.36
Lactic Acid P (g/L): 21.00
Maltose M (g/L): 121.14
Volume V (L): 4.97

t = 11.00 h

Biomass X (g/L): 48.14

Glucose S (g/L): 10.82
Lactic Acid P (g/L): 22.19
Maltose M (g/L): 129.90
Volume V (L): 5.71

t = 12.00 h

Biomass X (g/L): 42.20
Glucose S (g/L): 5.56
Lactic Acid P (g/L): 24.81
Maltose M (g/L): 125.39
Volume V (L): 6.40

t = 13.00 h

Biomass X (g/L): 47.57
Glucose S (g/L): 7.11
Lactic Acid P (g/L): 35.94
Maltose M (g/L): 130.49
Volume V (L): 5.94

t = 14.00 h

Biomass X (g/L): 54.24
Glucose S (g/L): 11.41
Lactic Acid P (g/L): 39.47
Maltose M (g/L): 122.01
Volume V (L): 6.92

t = 15.00 h

Biomass X (g/L): 56.34
Glucose S (g/L): 9.99
Lactic Acid P (g/L): 45.18
Maltose M (g/L): 118.50
Volume V (L): 5.92

t = 16.00 h

Biomass X (g/L): 65.76
Glucose S (g/L): 1.49
Lactic Acid P (g/L): 36.95
Maltose M (g/L): 124.25
Volume V (L): 6.28

t = 17.00 h

Biomass X (g/L): 67.10
Glucose S (g/L): 14.35
Lactic Acid P (g/L): 27.13
Maltose M (g/L): 125.96
Volume V (L): 7.10

t = 18.00 h

Biomass X (g/L): 64.89
Glucose S (g/L): 4.29
Lactic Acid P (g/L): 46.55
Maltose M (g/L): 125.83
Volume V (L): 6.56

t = 19.00 h

Biomass X (g/L): 53.15
Glucose S (g/L): 4.91
Lactic Acid P (g/L): 43.52
Maltose M (g/L): 119.16
Volume V (L): 6.69

t = 20.00 h

Biomass X (g/L): 54.91
Glucose S (g/L): 4.05
Lactic Acid P (g/L): 43.99
Maltose M (g/L): 117.02
Volume V (L): 6.60

t = 21.00 h

Biomass X (g/L): 56.90
Glucose S (g/L): 6.49
Lactic Acid P (g/L): 33.63
Maltose M (g/L): 121.08
Volume V (L): 6.59

t = 22.00 h

Biomass X (g/L): 67.97
Glucose S (g/L): 5.89
Lactic Acid P (g/L): 44.24
Maltose M (g/L): 123.46
Volume V (L): 7.39

t = 23.00 h

Biomass X (g/L): 49.41
Glucose S (g/L): 6.32
Lactic Acid P (g/L): 36.77
Maltose M (g/L): 122.80
Volume V (L): 7.56

t = 24.00 h

Biomass X (g/L): 68.16
Glucose S (g/L): 0.63
Lactic Acid P (g/L): 52.37
Maltose M (g/L): 126.11
Volume V (L): 7.51

t = 25.00 h

Biomass X (g/L): 81.56
Glucose S (g/L): 10.91
Lactic Acid P (g/L): 45.54
Maltose M (g/L): 118.90
Volume V (L): 7.11

t = 26.00 h

Biomass X (g/L): 76.07
Glucose S (g/L): 11.74
Lactic Acid P (g/L): 45.78
Maltose M (g/L): 113.32
Volume V (L): 7.45

t = 27.00 h

Biomass X (g/L): 68.12
Glucose S (g/L): 19.87
Lactic Acid P (g/L): 37.96
Maltose M (g/L): 123.29
Volume V (L): 8.62

t = 28.00 h

Biomass X (g/L): 65.96
Glucose S (g/L): 13.67
Lactic Acid P (g/L): 61.08
Maltose M (g/L): 119.31
Volume V (L): 6.93

t = 29.00 h

Biomass X (g/L): 79.91
Glucose S (g/L): 10.45
Lactic Acid P (g/L): 50.85
Maltose M (g/L): 119.08
Volume V (L): 8.99

t = 30.00 h

Biomass X (g/L): 76.32
Glucose S (g/L): 8.31
Lactic Acid P (g/L): 54.70
Maltose M (g/L): 120.21
Volume V (L): 8.03

t = 31.00 h

Biomass X (g/L): 72.21
Glucose S (g/L): 13.08
Lactic Acid P (g/L): 48.42
Maltose M (g/L): 121.39
Volume V (L): 7.96

t = 32.00 h

Biomass X (g/L): 69.89
Glucose S (g/L): 15.75
Lactic Acid P (g/L): 41.30
Maltose M (g/L): 116.41
Volume V (L): 8.73

t = 33.00 h

Biomass X (g/L): 76.33
Glucose S (g/L): 12.73
Lactic Acid P (g/L): 51.74
Maltose M (g/L): 119.99
Volume V (L): 7.04

t = 34.00 h

Biomass X (g/L): 73.60
Glucose S (g/L): 15.51
Lactic Acid P (g/L): 53.86
Maltose M (g/L): 117.29

Volume V (L): 9.19

t = 35.00 h

Biomass X (g/L): 83.64

Glucose S (g/L): 23.15

Lactic Acid P (g/L): 60.31

Maltose M (g/L): 112.57

Volume V (L): 9.57

t = 36.00 h

Biomass X (g/L): 88.14

Glucose S (g/L): 15.12

Lactic Acid P (g/L): 55.38

Maltose M (g/L): 113.47

Volume V (L): 9.21

t = 37.00 h

Biomass X (g/L): 65.83

Glucose S (g/L): 10.08

Lactic Acid P (g/L): 52.88

Maltose M (g/L): 115.85

Volume V (L): 7.41

t = 38.00 h

Biomass X (g/L): 67.50

Glucose S (g/L): 21.28

Lactic Acid P (g/L): 46.71

Maltose M (g/L): 110.72

Volume V (L): 9.22

t = 39.00 h

Biomass X (g/L): 90.37

Glucose S (g/L): 19.86

Lactic Acid P (g/L): 60.74

Maltose M (g/L): 115.65

Volume V (L): 9.84

t = 40.00 h

Biomass X (g/L): 80.42

Glucose S (g/L): 12.87

Lactic Acid P (g/L): 54.00

Maltose M (g/L): 112.58
Volume V (L): 9.88

t = 41.00 h

Biomass X (g/L): 69.43
Glucose S (g/L): 21.74
Lactic Acid P (g/L): 72.87
Maltose M (g/L): 112.79
Volume V (L): 8.22

t = 42.00 h

Biomass X (g/L): 88.20
Glucose S (g/L): 17.05
Lactic Acid P (g/L): 63.27
Maltose M (g/L): 115.88
Volume V (L): 9.76

t = 43.00 h

Biomass X (g/L): 82.89
Glucose S (g/L): 23.28
Lactic Acid P (g/L): 67.34
Maltose M (g/L): 111.58
Volume V (L): 10.32

t = 44.00 h

Biomass X (g/L): 99.39
Glucose S (g/L): 29.61
Lactic Acid P (g/L): 60.82
Maltose M (g/L): 110.03
Volume V (L): 9.37

t = 45.00 h

Biomass X (g/L): 78.34
Glucose S (g/L): 22.44
Lactic Acid P (g/L): 65.18
Maltose M (g/L): 108.94
Volume V (L): 8.59

t = 46.00 h

Biomass X (g/L): 90.28
Glucose S (g/L): 28.01

Lactic Acid P (g/L): 56.19
Maltose M (g/L): 108.20
Volume V (L): 10.05

t = 47.00 h

Biomass X (g/L): 68.14
Glucose S (g/L): 19.98
Lactic Acid P (g/L): 54.14
Maltose M (g/L): 109.17
Volume V (L): 10.28

t = 48.00 h

Biomass X (g/L): 93.14
Glucose S (g/L): 19.66
Lactic Acid P (g/L): 64.88
Maltose M (g/L): 112.24
Volume V (L): 9.67

t = 49.00 h

Biomass X (g/L): 89.34
Glucose S (g/L): 33.91
Lactic Acid P (g/L): 70.09
Maltose M (g/L): 104.87
Volume V (L): 9.14

t = 50.00 h

Biomass X (g/L): 95.23
Glucose S (g/L): 28.19
Lactic Acid P (g/L): 48.71
Maltose M (g/L): 104.05
Volume V (L): 10.53

t = 51.00 h

Biomass X (g/L): 91.03
Glucose S (g/L): 30.23
Lactic Acid P (g/L): 61.60
Maltose M (g/L): 110.29
Volume V (L): 9.59

t = 52.00 h

Biomass X (g/L): 80.56

Glucose S (g/L): 32.98
Lactic Acid P (g/L): 82.22
Maltose M (g/L): 103.36
Volume V (L): 10.38

t = 53.00 h

Biomass X (g/L): 71.28
Glucose S (g/L): 27.38
Lactic Acid P (g/L): 70.78
Maltose M (g/L): 105.50
Volume V (L): 11.17

t = 54.00 h

Biomass X (g/L): 76.31
Glucose S (g/L): 33.33
Lactic Acid P (g/L): 80.27
Maltose M (g/L): 102.16
Volume V (L): 9.36

t = 55.00 h

Biomass X (g/L): 66.24
Glucose S (g/L): 33.55
Lactic Acid P (g/L): 61.81
Maltose M (g/L): 97.86
Volume V (L): 10.98

t = 56.00 h

Biomass X (g/L): 73.62
Glucose S (g/L): 26.85
Lactic Acid P (g/L): 63.65
Maltose M (g/L): 103.79
Volume V (L): 10.42

t = 57.00 h

Biomass X (g/L): 85.40
Glucose S (g/L): 38.33
Lactic Acid P (g/L): 76.28
Maltose M (g/L): 106.40
Volume V (L): 10.54

t = 58.00 h

Biomass X (g/L): 81.28
Glucose S (g/L): 38.34
Lactic Acid P (g/L): 73.61
Maltose M (g/L): 101.80
Volume V (L): 11.05

t = 59.00 h

Biomass X (g/L): 79.74
Glucose S (g/L): 29.27
Lactic Acid P (g/L): 67.68
Maltose M (g/L): 96.36
Volume V (L): 10.38

t = 60.00 h

Biomass X (g/L): 74.76
Glucose S (g/L): 35.71
Lactic Acid P (g/L): 57.85
Maltose M (g/L): 101.31
Volume V (L): 10.24

t = 61.00 h

Biomass X (g/L): 66.20
Glucose S (g/L): 34.86
Lactic Acid P (g/L): 79.05
Maltose M (g/L): 97.79
Volume V (L): 9.98

t = 62.00 h

Biomass X (g/L): 83.23
Glucose S (g/L): 41.93
Lactic Acid P (g/L): 91.02
Maltose M (g/L): 90.83
Volume V (L): 11.87

t = 63.00 h

Biomass X (g/L): 84.92
Glucose S (g/L): 31.94
Lactic Acid P (g/L): 55.96
Maltose M (g/L): 99.05
Volume V (L): 11.02

t = 64.00 h

Biomass X (g/L): 72.90
Glucose S (g/L): 51.81
Lactic Acid P (g/L): 81.94
Maltose M (g/L): 100.58
Volume V (L): 11.67

t = 65.00 h

Biomass X (g/L): 76.85
Glucose S (g/L): 38.21
Lactic Acid P (g/L): 65.99
Maltose M (g/L): 102.33
Volume V (L): 11.15

t = 66.00 h

Biomass X (g/L): 94.91
Glucose S (g/L): 42.46
Lactic Acid P (g/L): 67.39
Maltose M (g/L): 103.80
Volume V (L): 10.84

t = 67.00 h

Biomass X (g/L): 91.51
Glucose S (g/L): 45.70
Lactic Acid P (g/L): 72.83
Maltose M (g/L): 99.47
Volume V (L): 12.02

t = 68.00 h

Biomass X (g/L): 68.35
Glucose S (g/L): 40.44
Lactic Acid P (g/L): 60.29
Maltose M (g/L): 99.93
Volume V (L): 11.93

t = 69.00 h

Biomass X (g/L): 80.49
Glucose S (g/L): 41.51
Lactic Acid P (g/L): 70.61
Maltose M (g/L): 91.77
Volume V (L): 11.69

t = 70.00 h

Biomass X (g/L): 93.85
Glucose S (g/L): 40.80
Lactic Acid P (g/L): 57.19
Maltose M (g/L): 101.27
Volume V (L): 12.11

t = 71.00 h

Biomass X (g/L): 79.17
Glucose S (g/L): 40.09
Lactic Acid P (g/L): 75.45
Maltose M (g/L): 99.54
Volume V (L): 11.46

t = 72.00 h

Biomass X (g/L): 78.62
Glucose S (g/L): 45.25
Lactic Acid P (g/L): 90.85
Maltose M (g/L): 93.40
Volume V (L): 12.76

t = 73.00 h

Biomass X (g/L): 90.41
Glucose S (g/L): 44.45
Lactic Acid P (g/L): 66.19
Maltose M (g/L): 96.51
Volume V (L): 12.99

t = 74.00 h

Biomass X (g/L): 89.38
Glucose S (g/L): 43.77
Lactic Acid P (g/L): 70.45
Maltose M (g/L): 95.67
Volume V (L): 12.78

t = 75.00 h

Biomass X (g/L): 88.82
Glucose S (g/L): 44.72
Lactic Acid P (g/L): 71.06
Maltose M (g/L): 98.24

Volume V (L): 12.20

Batch 88

Initial Conditions:

Batch: 88.00

X0 (g/L): 2.66

S0 (g/L): 123.33

M0 (g/L): 56.57

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.46

F_vol (L/h): 0.08

S_feed (g/L): 205.66

M_feed (g/L): 94.34

V_final (L): 10.81

Batch 88 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.24
Glucose S (g/L): 126.03
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 54.76
Volume V (L): 4.53

t = 1.00 h

Biomass X (g/L): 18.98
Glucose S (g/L): 126.37
Lactic Acid P (g/L): 6.26
Maltose M (g/L): 58.68
Volume V (L): 4.73

t = 2.00 h

Biomass X (g/L): 17.40
Glucose S (g/L): 129.91
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 52.32
Volume V (L): 5.74

t = 3.00 h

Biomass X (g/L): 26.00
Glucose S (g/L): 118.11
Lactic Acid P (g/L): 4.27
Maltose M (g/L): 51.28
Volume V (L): 6.07

t = 4.00 h

Biomass X (g/L): 0.87
Glucose S (g/L): 129.42
Lactic Acid P (g/L): 18.38
Maltose M (g/L): 53.64
Volume V (L): 4.73

t = 5.00 h

Biomass X (g/L): 2.53
Glucose S (g/L): 132.99

Lactic Acid P (g/L): 5.02
Maltose M (g/L): 57.70
Volume V (L): 5.57

t = 6.00 h

Biomass X (g/L): 15.42
Glucose S (g/L): 122.63
Lactic Acid P (g/L): 14.07
Maltose M (g/L): 51.28
Volume V (L): 5.32

t = 7.00 h

Biomass X (g/L): 21.64
Glucose S (g/L): 120.65
Lactic Acid P (g/L): 11.55
Maltose M (g/L): 53.14
Volume V (L): 6.19

t = 8.00 h

Biomass X (g/L): 20.23
Glucose S (g/L): 125.84
Lactic Acid P (g/L): 12.95
Maltose M (g/L): 55.55
Volume V (L): 5.68

t = 9.00 h

Biomass X (g/L): 22.10
Glucose S (g/L): 115.44
Lactic Acid P (g/L): 15.00
Maltose M (g/L): 54.29
Volume V (L): 5.69

t = 10.00 h

Biomass X (g/L): 42.82
Glucose S (g/L): 113.66
Lactic Acid P (g/L): 26.73
Maltose M (g/L): 57.11
Volume V (L): 5.43

t = 11.00 h

Biomass X (g/L): 34.05

Glucose S (g/L): 113.84
Lactic Acid P (g/L): 17.19
Maltose M (g/L): 51.41
Volume V (L): 5.14

t = 12.00 h

Biomass X (g/L): 27.50
Glucose S (g/L): 107.53
Lactic Acid P (g/L): 16.92
Maltose M (g/L): 49.92
Volume V (L): 6.28

t = 13.00 h

Biomass X (g/L): 32.93
Glucose S (g/L): 110.81
Lactic Acid P (g/L): 31.03
Maltose M (g/L): 47.95
Volume V (L): 6.17

t = 14.00 h

Biomass X (g/L): 50.19
Glucose S (g/L): 102.27
Lactic Acid P (g/L): 35.17
Maltose M (g/L): 49.26
Volume V (L): 6.26

t = 15.00 h

Biomass X (g/L): 80.96
Glucose S (g/L): 104.06
Lactic Acid P (g/L): 19.90
Maltose M (g/L): 47.11
Volume V (L): 5.88

t = 16.00 h

Biomass X (g/L): 45.44
Glucose S (g/L): 101.69
Lactic Acid P (g/L): 38.94
Maltose M (g/L): 45.06
Volume V (L): 6.17

t = 17.00 h

Biomass X (g/L): 55.09
Glucose S (g/L): 104.78
Lactic Acid P (g/L): 30.47
Maltose M (g/L): 53.60
Volume V (L): 6.68

t = 18.00 h

Biomass X (g/L): 84.61
Glucose S (g/L): 101.37
Lactic Acid P (g/L): 49.47
Maltose M (g/L): 48.66
Volume V (L): 6.85

t = 19.00 h

Biomass X (g/L): 58.89
Glucose S (g/L): 93.88
Lactic Acid P (g/L): 29.68
Maltose M (g/L): 54.32
Volume V (L): 5.92

t = 20.00 h

Biomass X (g/L): 70.43
Glucose S (g/L): 90.05
Lactic Acid P (g/L): 65.22
Maltose M (g/L): 45.78
Volume V (L): 6.25

t = 21.00 h

Biomass X (g/L): 86.45
Glucose S (g/L): 96.61
Lactic Acid P (g/L): 46.60
Maltose M (g/L): 46.38
Volume V (L): 5.98

t = 22.00 h

Biomass X (g/L): 60.62
Glucose S (g/L): 90.42
Lactic Acid P (g/L): 43.48
Maltose M (g/L): 45.19
Volume V (L): 6.80

t = 23.00 h

Biomass X (g/L): 68.60
Glucose S (g/L): 90.88
Lactic Acid P (g/L): 50.22
Maltose M (g/L): 45.14
Volume V (L): 7.19

t = 24.00 h

Biomass X (g/L): 68.40
Glucose S (g/L): 87.74
Lactic Acid P (g/L): 59.17
Maltose M (g/L): 48.38
Volume V (L): 6.73

t = 25.00 h

Biomass X (g/L): 59.62
Glucose S (g/L): 93.57
Lactic Acid P (g/L): 63.45
Maltose M (g/L): 47.30
Volume V (L): 6.63

t = 26.00 h

Biomass X (g/L): 91.09
Glucose S (g/L): 88.50
Lactic Acid P (g/L): 54.12
Maltose M (g/L): 47.68
Volume V (L): 6.86

t = 27.00 h

Biomass X (g/L): 71.05
Glucose S (g/L): 90.48
Lactic Acid P (g/L): 44.72
Maltose M (g/L): 48.45
Volume V (L): 8.16

t = 28.00 h

Biomass X (g/L): 69.17
Glucose S (g/L): 90.41
Lactic Acid P (g/L): 47.76
Maltose M (g/L): 47.55
Volume V (L): 7.72

t = 29.00 h

Biomass X (g/L): 74.27
Glucose S (g/L): 91.08
Lactic Acid P (g/L): 62.20
Maltose M (g/L): 42.18
Volume V (L): 7.59

t = 30.00 h

Biomass X (g/L): 88.50
Glucose S (g/L): 92.81
Lactic Acid P (g/L): 69.39
Maltose M (g/L): 43.00
Volume V (L): 7.47

t = 31.00 h

Biomass X (g/L): 81.19
Glucose S (g/L): 89.63
Lactic Acid P (g/L): 71.37
Maltose M (g/L): 44.21
Volume V (L): 7.40

t = 32.00 h

Biomass X (g/L): 72.94
Glucose S (g/L): 87.96
Lactic Acid P (g/L): 63.57
Maltose M (g/L): 52.46
Volume V (L): 7.06

t = 33.00 h

Biomass X (g/L): 85.25
Glucose S (g/L): 84.11
Lactic Acid P (g/L): 71.48
Maltose M (g/L): 45.11
Volume V (L): 7.06

t = 34.00 h

Biomass X (g/L): 89.68
Glucose S (g/L): 91.37
Lactic Acid P (g/L): 80.67
Maltose M (g/L): 43.22

Volume V (L): 7.37

t = 35.00 h

Biomass X (g/L): 81.91

Glucose S (g/L): 90.67

Lactic Acid P (g/L): 74.78

Maltose M (g/L): 44.25

Volume V (L): 7.27

t = 36.00 h

Biomass X (g/L): 80.22

Glucose S (g/L): 88.96

Lactic Acid P (g/L): 72.95

Maltose M (g/L): 40.91

Volume V (L): 7.51

t = 37.00 h

Biomass X (g/L): 92.80

Glucose S (g/L): 92.04

Lactic Acid P (g/L): 57.93

Maltose M (g/L): 42.66

Volume V (L): 7.82

t = 38.00 h

Biomass X (g/L): 81.38

Glucose S (g/L): 95.21

Lactic Acid P (g/L): 74.40

Maltose M (g/L): 44.01

Volume V (L): 7.93

t = 39.00 h

Biomass X (g/L): 99.14

Glucose S (g/L): 96.48

Lactic Acid P (g/L): 70.37

Maltose M (g/L): 41.37

Volume V (L): 7.73

t = 40.00 h

Biomass X (g/L): 76.17

Glucose S (g/L): 87.33

Lactic Acid P (g/L): 72.55

Maltose M (g/L): 39.35

Volume V (L): 8.33

t = 41.00 h

Biomass X (g/L): 76.12

Glucose S (g/L): 91.93

Lactic Acid P (g/L): 78.49

Maltose M (g/L): 40.68

Volume V (L): 8.09

t = 42.00 h

Biomass X (g/L): 75.06

Glucose S (g/L): 97.07

Lactic Acid P (g/L): 70.78

Maltose M (g/L): 39.82

Volume V (L): 8.35

t = 43.00 h

Biomass X (g/L): 81.70

Glucose S (g/L): 99.76

Lactic Acid P (g/L): 67.03

Maltose M (g/L): 37.07

Volume V (L): 7.98

t = 44.00 h

Biomass X (g/L): 84.16

Glucose S (g/L): 91.50

Lactic Acid P (g/L): 58.69

Maltose M (g/L): 38.66

Volume V (L): 8.73

t = 45.00 h

Biomass X (g/L): 71.22

Glucose S (g/L): 99.82

Lactic Acid P (g/L): 66.02

Maltose M (g/L): 36.43

Volume V (L): 8.24

t = 46.00 h

Biomass X (g/L): 85.22

Glucose S (g/L): 90.45

Lactic Acid P (g/L): 70.96

Maltose M (g/L): 36.53

Volume V (L): 9.31

t = 47.00 h

Biomass X (g/L): 89.84

Glucose S (g/L): 99.87

Lactic Acid P (g/L): 71.88

Maltose M (g/L): 36.49

Volume V (L): 8.54

t = 48.00 h

Biomass X (g/L): 69.96

Glucose S (g/L): 95.17

Lactic Acid P (g/L): 71.89

Maltose M (g/L): 41.20

Volume V (L): 8.67

t = 49.00 h

Biomass X (g/L): 79.21

Glucose S (g/L): 102.25

Lactic Acid P (g/L): 78.64

Maltose M (g/L): 42.21

Volume V (L): 9.07

t = 50.00 h

Biomass X (g/L): 89.87

Glucose S (g/L): 95.42

Lactic Acid P (g/L): 78.09

Maltose M (g/L): 34.76

Volume V (L): 9.25

t = 51.00 h

Biomass X (g/L): 73.60

Glucose S (g/L): 96.43

Lactic Acid P (g/L): 70.01

Maltose M (g/L): 41.91

Volume V (L): 9.38

t = 52.00 h

Biomass X (g/L): 81.14

Glucose S (g/L): 97.52
Lactic Acid P (g/L): 63.20
Maltose M (g/L): 35.22
Volume V (L): 9.85

t = 53.00 h

Biomass X (g/L): 94.85
Glucose S (g/L): 92.94
Lactic Acid P (g/L): 74.55
Maltose M (g/L): 39.43
Volume V (L): 9.41

t = 54.00 h

Biomass X (g/L): 101.71
Glucose S (g/L): 92.59
Lactic Acid P (g/L): 91.59
Maltose M (g/L): 38.02
Volume V (L): 9.15

t = 55.00 h

Biomass X (g/L): 88.36
Glucose S (g/L): 101.22
Lactic Acid P (g/L): 74.48
Maltose M (g/L): 38.67
Volume V (L): 9.01

t = 56.00 h

Biomass X (g/L): 82.31
Glucose S (g/L): 103.40
Lactic Acid P (g/L): 69.15
Maltose M (g/L): 34.14
Volume V (L): 9.28

t = 57.00 h

Biomass X (g/L): 78.96
Glucose S (g/L): 102.49
Lactic Acid P (g/L): 76.87
Maltose M (g/L): 32.37
Volume V (L): 10.11

t = 58.00 h

Biomass X (g/L): 92.70
Glucose S (g/L): 96.26
Lactic Acid P (g/L): 77.63
Maltose M (g/L): 34.53
Volume V (L): 8.85

t = 59.00 h

Biomass X (g/L): 94.77
Glucose S (g/L): 102.95
Lactic Acid P (g/L): 77.74
Maltose M (g/L): 37.22
Volume V (L): 9.19

t = 60.00 h

Biomass X (g/L): 72.22
Glucose S (g/L): 98.12
Lactic Acid P (g/L): 85.04
Maltose M (g/L): 40.63
Volume V (L): 9.55

t = 61.00 h

Biomass X (g/L): 92.35
Glucose S (g/L): 103.60
Lactic Acid P (g/L): 76.35
Maltose M (g/L): 33.30
Volume V (L): 9.34

t = 62.00 h

Biomass X (g/L): 77.36
Glucose S (g/L): 106.65
Lactic Acid P (g/L): 85.00
Maltose M (g/L): 35.28
Volume V (L): 9.61

t = 63.00 h

Biomass X (g/L): 74.75
Glucose S (g/L): 104.36
Lactic Acid P (g/L): 75.31
Maltose M (g/L): 33.51
Volume V (L): 10.72

t = 64.00 h

Biomass X (g/L): 81.53
Glucose S (g/L): 108.76
Lactic Acid P (g/L): 87.05
Maltose M (g/L): 35.03
Volume V (L): 9.82

t = 65.00 h

Biomass X (g/L): 92.62
Glucose S (g/L): 103.40
Lactic Acid P (g/L): 84.47
Maltose M (g/L): 30.26
Volume V (L): 10.23

t = 66.00 h

Biomass X (g/L): 85.78
Glucose S (g/L): 103.60
Lactic Acid P (g/L): 82.07
Maltose M (g/L): 38.20
Volume V (L): 9.34

t = 67.00 h

Biomass X (g/L): 85.37
Glucose S (g/L): 105.06
Lactic Acid P (g/L): 81.04
Maltose M (g/L): 34.48
Volume V (L): 9.68

t = 68.00 h

Biomass X (g/L): 85.90
Glucose S (g/L): 100.43
Lactic Acid P (g/L): 94.82
Maltose M (g/L): 36.39
Volume V (L): 10.70

t = 69.00 h

Biomass X (g/L): 86.88
Glucose S (g/L): 104.24
Lactic Acid P (g/L): 83.94
Maltose M (g/L): 33.45
Volume V (L): 11.12

t = 70.00 h

Biomass X (g/L): 81.02
Glucose S (g/L): 104.99
Lactic Acid P (g/L): 66.74
Maltose M (g/L): 34.25
Volume V (L): 10.53

t = 71.00 h

Biomass X (g/L): 88.16
Glucose S (g/L): 105.53
Lactic Acid P (g/L): 76.75
Maltose M (g/L): 36.27
Volume V (L): 10.36

t = 72.00 h

Biomass X (g/L): 75.34
Glucose S (g/L): 107.93
Lactic Acid P (g/L): 72.30
Maltose M (g/L): 33.70
Volume V (L): 10.81

t = 73.00 h

Biomass X (g/L): 88.02
Glucose S (g/L): 102.07
Lactic Acid P (g/L): 93.43
Maltose M (g/L): 33.58
Volume V (L): 9.45

t = 74.00 h

Biomass X (g/L): 89.87
Glucose S (g/L): 103.69
Lactic Acid P (g/L): 86.24
Maltose M (g/L): 31.10
Volume V (L): 10.84

t = 75.00 h

Biomass X (g/L): 88.15
Glucose S (g/L): 112.29
Lactic Acid P (g/L): 93.40
Maltose M (g/L): 31.52

Volume V (L): 10.32

Batch 89

Initial Conditions:

Batch: 89.00

X0 (g/L): 5.27

S0 (g/L): 26.93

M0 (g/L): 122.50

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 4.55

F_vol (L/h): 0.03

S_feed (g/L): 54.06

M_feed (g/L): 245.94

V_final (L): 7.01

Batch 89 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 12.96
Glucose S (g/L): 26.77
Lactic Acid P (g/L): 10.11
Maltose M (g/L): 119.63
Volume V (L): 4.85

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 27.59
Lactic Acid P (g/L): 17.66
Maltose M (g/L): 110.42
Volume V (L): 5.23

t = 2.00 h

Biomass X (g/L): 2.81
Glucose S (g/L): 32.25
Lactic Acid P (g/L): 1.11
Maltose M (g/L): 115.88
Volume V (L): 5.02

t = 3.00 h

Biomass X (g/L): 10.96
Glucose S (g/L): 26.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 117.03
Volume V (L): 5.00

t = 4.00 h

Biomass X (g/L): 6.91
Glucose S (g/L): 22.53
Lactic Acid P (g/L): 9.85
Maltose M (g/L): 108.15
Volume V (L): 5.19

t = 5.00 h

Biomass X (g/L): 14.01
Glucose S (g/L): 27.16

Lactic Acid P (g/L): 10.30
Maltose M (g/L): 96.33
Volume V (L): 5.04

t = 6.00 h

Biomass X (g/L): 30.14
Glucose S (g/L): 27.57
Lactic Acid P (g/L): 7.07
Maltose M (g/L): 114.20
Volume V (L): 5.16

t = 7.00 h

Biomass X (g/L): 25.13
Glucose S (g/L): 23.64
Lactic Acid P (g/L): 19.22
Maltose M (g/L): 124.94
Volume V (L): 5.48

t = 8.00 h

Biomass X (g/L): 20.47
Glucose S (g/L): 22.24
Lactic Acid P (g/L): 26.70
Maltose M (g/L): 107.63
Volume V (L): 5.50

t = 9.00 h

Biomass X (g/L): 42.74
Glucose S (g/L): 20.69
Lactic Acid P (g/L): 13.45
Maltose M (g/L): 114.74
Volume V (L): 5.18

t = 10.00 h

Biomass X (g/L): 40.08
Glucose S (g/L): 17.27
Lactic Acid P (g/L): 48.00
Maltose M (g/L): 116.89
Volume V (L): 5.24

t = 11.00 h

Biomass X (g/L): 44.32

Glucose S (g/L): 16.31
Lactic Acid P (g/L): 12.29
Maltose M (g/L): 93.39
Volume V (L): 5.20

t = 12.00 h

Biomass X (g/L): 36.41
Glucose S (g/L): 14.55
Lactic Acid P (g/L): 21.49
Maltose M (g/L): 101.44
Volume V (L): 5.01

t = 13.00 h

Biomass X (g/L): 23.67
Glucose S (g/L): 7.68
Lactic Acid P (g/L): 34.33
Maltose M (g/L): 94.81
Volume V (L): 5.51

t = 14.00 h

Biomass X (g/L): 42.57
Glucose S (g/L): 10.86
Lactic Acid P (g/L): 55.76
Maltose M (g/L): 96.60
Volume V (L): 5.76

t = 15.00 h

Biomass X (g/L): 44.29
Glucose S (g/L): 6.19
Lactic Acid P (g/L): 31.30
Maltose M (g/L): 111.66
Volume V (L): 5.33

t = 16.00 h

Biomass X (g/L): 50.56
Glucose S (g/L): 8.65
Lactic Acid P (g/L): 28.53
Maltose M (g/L): 100.83
Volume V (L): 5.30

t = 17.00 h

Biomass X (g/L): 50.56
Glucose S (g/L): 7.02
Lactic Acid P (g/L): 45.67
Maltose M (g/L): 103.31
Volume V (L): 5.41

t = 18.00 h

Biomass X (g/L): 61.81
Glucose S (g/L): 9.81
Lactic Acid P (g/L): 39.80
Maltose M (g/L): 96.72
Volume V (L): 5.50

t = 19.00 h

Biomass X (g/L): 71.07
Glucose S (g/L): 7.75
Lactic Acid P (g/L): 50.04
Maltose M (g/L): 87.99
Volume V (L): 5.33

t = 20.00 h

Biomass X (g/L): 74.63
Glucose S (g/L): 7.99
Lactic Acid P (g/L): 44.54
Maltose M (g/L): 99.15
Volume V (L): 5.46

t = 21.00 h

Biomass X (g/L): 73.85
Glucose S (g/L): 6.41
Lactic Acid P (g/L): 38.47
Maltose M (g/L): 101.79
Volume V (L): 5.73

t = 22.00 h

Biomass X (g/L): 58.46
Glucose S (g/L): 9.00
Lactic Acid P (g/L): 70.16
Maltose M (g/L): 94.64
Volume V (L): 5.61

t = 23.00 h

Biomass X (g/L): 69.04
Glucose S (g/L): 3.29
Lactic Acid P (g/L): 21.15
Maltose M (g/L): 94.70
Volume V (L): 5.82

t = 24.00 h

Biomass X (g/L): 84.87
Glucose S (g/L): 4.83
Lactic Acid P (g/L): 67.59
Maltose M (g/L): 82.46
Volume V (L): 5.70

t = 25.00 h

Biomass X (g/L): 55.50
Glucose S (g/L): 9.59
Lactic Acid P (g/L): 60.32
Maltose M (g/L): 94.08
Volume V (L): 5.49

t = 26.00 h

Biomass X (g/L): 85.00
Glucose S (g/L): 6.00
Lactic Acid P (g/L): 45.50
Maltose M (g/L): 97.24
Volume V (L): 5.60

t = 27.00 h

Biomass X (g/L): 63.30
Glucose S (g/L): 2.47
Lactic Acid P (g/L): 71.35
Maltose M (g/L): 90.15
Volume V (L): 5.76

t = 28.00 h

Biomass X (g/L): 76.75
Glucose S (g/L): 10.38
Lactic Acid P (g/L): 41.62
Maltose M (g/L): 74.89
Volume V (L): 5.57

t = 29.00 h

Biomass X (g/L): 78.81
Glucose S (g/L): 9.77
Lactic Acid P (g/L): 65.14
Maltose M (g/L): 96.28
Volume V (L): 5.58

t = 30.00 h

Biomass X (g/L): 77.50
Glucose S (g/L): 9.20
Lactic Acid P (g/L): 36.79
Maltose M (g/L): 78.59
Volume V (L): 5.61

t = 31.00 h

Biomass X (g/L): 98.05
Glucose S (g/L): 8.11
Lactic Acid P (g/L): 46.58
Maltose M (g/L): 86.36
Volume V (L): 5.91

t = 32.00 h

Biomass X (g/L): 74.31
Glucose S (g/L): 11.74
Lactic Acid P (g/L): 47.43
Maltose M (g/L): 89.16
Volume V (L): 5.80

t = 33.00 h

Biomass X (g/L): 62.05
Glucose S (g/L): 11.15
Lactic Acid P (g/L): 52.05
Maltose M (g/L): 89.13
Volume V (L): 6.07

t = 34.00 h

Biomass X (g/L): 82.06
Glucose S (g/L): 12.06
Lactic Acid P (g/L): 60.10
Maltose M (g/L): 69.56

Volume V (L): 5.77

t = 35.00 h

Biomass X (g/L): 77.59

Glucose S (g/L): 10.53

Lactic Acid P (g/L): 52.95

Maltose M (g/L): 81.20

Volume V (L): 5.54

t = 36.00 h

Biomass X (g/L): 85.21

Glucose S (g/L): 9.61

Lactic Acid P (g/L): 42.25

Maltose M (g/L): 79.96

Volume V (L): 5.94

t = 37.00 h

Biomass X (g/L): 88.94

Glucose S (g/L): 8.82

Lactic Acid P (g/L): 66.69

Maltose M (g/L): 88.20

Volume V (L): 6.26

t = 38.00 h

Biomass X (g/L): 80.59

Glucose S (g/L): 7.30

Lactic Acid P (g/L): 63.48

Maltose M (g/L): 81.04

Volume V (L): 6.02

t = 39.00 h

Biomass X (g/L): 85.63

Glucose S (g/L): 11.45

Lactic Acid P (g/L): 67.59

Maltose M (g/L): 86.72

Volume V (L): 5.82

t = 40.00 h

Biomass X (g/L): 83.26

Glucose S (g/L): 12.00

Lactic Acid P (g/L): 66.02

Maltose M (g/L): 80.06

Volume V (L): 5.81

t = 41.00 h

Biomass X (g/L): 99.65

Glucose S (g/L): 10.36

Lactic Acid P (g/L): 74.09

Maltose M (g/L): 75.34

Volume V (L): 6.29

t = 42.00 h

Biomass X (g/L): 79.10

Glucose S (g/L): 12.81

Lactic Acid P (g/L): 64.97

Maltose M (g/L): 76.36

Volume V (L): 6.00

t = 43.00 h

Biomass X (g/L): 76.40

Glucose S (g/L): 11.17

Lactic Acid P (g/L): 75.90

Maltose M (g/L): 84.30

Volume V (L): 6.40

t = 44.00 h

Biomass X (g/L): 91.93

Glucose S (g/L): 14.15

Lactic Acid P (g/L): 59.55

Maltose M (g/L): 65.27

Volume V (L): 6.28

t = 45.00 h

Biomass X (g/L): 74.71

Glucose S (g/L): 10.80

Lactic Acid P (g/L): 60.67

Maltose M (g/L): 78.92

Volume V (L): 6.08

t = 46.00 h

Biomass X (g/L): 87.80

Glucose S (g/L): 10.84

Lactic Acid P (g/L): 81.08

Maltose M (g/L): 68.28

Volume V (L): 6.42

t = 47.00 h

Biomass X (g/L): 82.63

Glucose S (g/L): 13.01

Lactic Acid P (g/L): 81.43

Maltose M (g/L): 66.23

Volume V (L): 6.11

t = 48.00 h

Biomass X (g/L): 83.53

Glucose S (g/L): 15.74

Lactic Acid P (g/L): 74.55

Maltose M (g/L): 77.72

Volume V (L): 6.05

t = 49.00 h

Biomass X (g/L): 108.35

Glucose S (g/L): 11.90

Lactic Acid P (g/L): 61.20

Maltose M (g/L): 64.69

Volume V (L): 6.17

t = 50.00 h

Biomass X (g/L): 96.89

Glucose S (g/L): 17.08

Lactic Acid P (g/L): 60.93

Maltose M (g/L): 72.27

Volume V (L): 6.46

t = 51.00 h

Biomass X (g/L): 95.02

Glucose S (g/L): 11.74

Lactic Acid P (g/L): 83.19

Maltose M (g/L): 66.03

Volume V (L): 6.55

t = 52.00 h

Biomass X (g/L): 90.42

Glucose S (g/L): 15.18
Lactic Acid P (g/L): 83.13
Maltose M (g/L): 53.29
Volume V (L): 6.33

t = 53.00 h

Biomass X (g/L): 78.72
Glucose S (g/L): 13.13
Lactic Acid P (g/L): 70.92
Maltose M (g/L): 71.14
Volume V (L): 6.12

t = 54.00 h

Biomass X (g/L): 88.08
Glucose S (g/L): 11.01
Lactic Acid P (g/L): 94.16
Maltose M (g/L): 68.28
Volume V (L): 6.20

t = 55.00 h

Biomass X (g/L): 91.31
Glucose S (g/L): 17.86
Lactic Acid P (g/L): 76.75
Maltose M (g/L): 54.47
Volume V (L): 6.30

t = 56.00 h

Biomass X (g/L): 87.68
Glucose S (g/L): 17.15
Lactic Acid P (g/L): 62.32
Maltose M (g/L): 61.79
Volume V (L): 6.29

t = 57.00 h

Biomass X (g/L): 86.74
Glucose S (g/L): 14.99
Lactic Acid P (g/L): 70.61
Maltose M (g/L): 54.04
Volume V (L): 6.57

t = 58.00 h

Biomass X (g/L): 82.04
Glucose S (g/L): 17.34
Lactic Acid P (g/L): 73.67
Maltose M (g/L): 70.02
Volume V (L): 6.86

t = 59.00 h

Biomass X (g/L): 88.59
Glucose S (g/L): 19.86
Lactic Acid P (g/L): 74.82
Maltose M (g/L): 73.76
Volume V (L): 6.71

t = 60.00 h

Biomass X (g/L): 80.88
Glucose S (g/L): 18.59
Lactic Acid P (g/L): 86.22
Maltose M (g/L): 69.74
Volume V (L): 6.43

t = 61.00 h

Biomass X (g/L): 92.81
Glucose S (g/L): 19.42
Lactic Acid P (g/L): 65.67
Maltose M (g/L): 70.15
Volume V (L): 6.52

t = 62.00 h

Biomass X (g/L): 66.48
Glucose S (g/L): 19.46
Lactic Acid P (g/L): 62.47
Maltose M (g/L): 59.85
Volume V (L): 6.58

t = 63.00 h

Biomass X (g/L): 99.17
Glucose S (g/L): 19.53
Lactic Acid P (g/L): 81.14
Maltose M (g/L): 66.28
Volume V (L): 6.75

t = 64.00 h

Biomass X (g/L): 73.29
Glucose S (g/L): 19.97
Lactic Acid P (g/L): 70.97
Maltose M (g/L): 62.19
Volume V (L): 6.97

t = 65.00 h

Biomass X (g/L): 75.64
Glucose S (g/L): 20.37
Lactic Acid P (g/L): 99.06
Maltose M (g/L): 62.98
Volume V (L): 6.67

t = 66.00 h

Biomass X (g/L): 89.91
Glucose S (g/L): 20.56
Lactic Acid P (g/L): 85.36
Maltose M (g/L): 67.54
Volume V (L): 7.01

t = 67.00 h

Biomass X (g/L): 91.14
Glucose S (g/L): 20.46
Lactic Acid P (g/L): 71.75
Maltose M (g/L): 68.50
Volume V (L): 7.13

t = 68.00 h

Biomass X (g/L): 105.99
Glucose S (g/L): 22.29
Lactic Acid P (g/L): 88.53
Maltose M (g/L): 57.87
Volume V (L): 6.76

t = 69.00 h

Biomass X (g/L): 88.67
Glucose S (g/L): 22.20
Lactic Acid P (g/L): 84.25
Maltose M (g/L): 50.81
Volume V (L): 6.79

t = 70.00 h

Biomass X (g/L): 85.34
Glucose S (g/L): 21.73
Lactic Acid P (g/L): 83.62
Maltose M (g/L): 58.31
Volume V (L): 6.82

t = 71.00 h

Biomass X (g/L): 87.51
Glucose S (g/L): 26.15
Lactic Acid P (g/L): 76.74
Maltose M (g/L): 57.68
Volume V (L): 6.91

t = 72.00 h

Biomass X (g/L): 100.99
Glucose S (g/L): 27.61
Lactic Acid P (g/L): 82.16
Maltose M (g/L): 60.53
Volume V (L): 6.87

t = 73.00 h

Biomass X (g/L): 108.98
Glucose S (g/L): 25.42
Lactic Acid P (g/L): 58.52
Maltose M (g/L): 63.69
Volume V (L): 7.10

t = 74.00 h

Biomass X (g/L): 83.61
Glucose S (g/L): 26.02
Lactic Acid P (g/L): 91.19
Maltose M (g/L): 45.41
Volume V (L): 6.73

t = 75.00 h

Biomass X (g/L): 90.56
Glucose S (g/L): 24.85
Lactic Acid P (g/L): 46.41
Maltose M (g/L): 53.10

Volume V (L): 6.92

Batch 90

Initial Conditions:

Batch: 90.00

X0 (g/L): 3.63

S0 (g/L): 86.20

M0 (g/L): 76.00

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.88

F_vol (L/h): 0.05

S_feed (g/L): 159.43

M_feed (g/L): 140.57

V_final (L): 8.38

Batch 90 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 6.95
Glucose S (g/L): 83.78
Lactic Acid P (g/L): 3.30
Maltose M (g/L): 77.56
Volume V (L): 4.92

t = 1.00 h

Biomass X (g/L): 13.68
Glucose S (g/L): 84.50
Lactic Acid P (g/L): 19.30
Maltose M (g/L): 77.70
Volume V (L): 5.46

t = 2.00 h

Biomass X (g/L): 28.71
Glucose S (g/L): 87.55
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 68.18
Volume V (L): 4.90

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 87.54
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 74.41
Volume V (L): 5.12

t = 4.00 h

Biomass X (g/L): 4.83
Glucose S (g/L): 82.85
Lactic Acid P (g/L): 3.20
Maltose M (g/L): 69.76
Volume V (L): 5.56

t = 5.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 84.35

Lactic Acid P (g/L): 17.78

Maltose M (g/L): 73.18

Volume V (L): 5.08

t = 6.00 h

Biomass X (g/L): 7.13

Glucose S (g/L): 77.86

Lactic Acid P (g/L): 11.24

Maltose M (g/L): 70.64

Volume V (L): 5.57

t = 7.00 h

Biomass X (g/L): 4.66

Glucose S (g/L): 86.09

Lactic Acid P (g/L): 2.75

Maltose M (g/L): 66.84

Volume V (L): 5.07

t = 8.00 h

Biomass X (g/L): 21.72

Glucose S (g/L): 71.65

Lactic Acid P (g/L): 17.86

Maltose M (g/L): 73.40

Volume V (L): 4.58

t = 9.00 h

Biomass X (g/L): 29.35

Glucose S (g/L): 68.67

Lactic Acid P (g/L): 4.18

Maltose M (g/L): 67.22

Volume V (L): 5.74

t = 10.00 h

Biomass X (g/L): 41.09

Glucose S (g/L): 74.17

Lactic Acid P (g/L): 30.00

Maltose M (g/L): 68.94

Volume V (L): 5.57

t = 11.00 h

Biomass X (g/L): 37.97

Glucose S (g/L): 64.69
Lactic Acid P (g/L): 24.59
Maltose M (g/L): 71.94
Volume V (L): 5.63

t = 12.00 h

Biomass X (g/L): 50.14
Glucose S (g/L): 64.66
Lactic Acid P (g/L): 30.07
Maltose M (g/L): 63.31
Volume V (L): 5.25

t = 13.00 h

Biomass X (g/L): 50.72
Glucose S (g/L): 64.17
Lactic Acid P (g/L): 38.06
Maltose M (g/L): 65.91
Volume V (L): 5.31

t = 14.00 h

Biomass X (g/L): 43.02
Glucose S (g/L): 64.10
Lactic Acid P (g/L): 38.94
Maltose M (g/L): 63.45
Volume V (L): 5.94

t = 15.00 h

Biomass X (g/L): 39.99
Glucose S (g/L): 58.45
Lactic Acid P (g/L): 34.64
Maltose M (g/L): 63.59
Volume V (L): 5.74

t = 16.00 h

Biomass X (g/L): 66.09
Glucose S (g/L): 64.26
Lactic Acid P (g/L): 52.09
Maltose M (g/L): 71.58
Volume V (L): 5.69

t = 17.00 h

Biomass X (g/L): 79.73
Glucose S (g/L): 55.70
Lactic Acid P (g/L): 52.21
Maltose M (g/L): 64.91
Volume V (L): 5.83

t = 18.00 h

Biomass X (g/L): 91.33
Glucose S (g/L): 58.87
Lactic Acid P (g/L): 30.07
Maltose M (g/L): 63.78
Volume V (L): 6.22

t = 19.00 h

Biomass X (g/L): 77.60
Glucose S (g/L): 54.11
Lactic Acid P (g/L): 46.66
Maltose M (g/L): 63.92
Volume V (L): 6.05

t = 20.00 h

Biomass X (g/L): 85.01
Glucose S (g/L): 60.25
Lactic Acid P (g/L): 40.46
Maltose M (g/L): 62.99
Volume V (L): 5.77

t = 21.00 h

Biomass X (g/L): 76.90
Glucose S (g/L): 55.24
Lactic Acid P (g/L): 40.37
Maltose M (g/L): 54.38
Volume V (L): 5.57

t = 22.00 h

Biomass X (g/L): 77.04
Glucose S (g/L): 50.62
Lactic Acid P (g/L): 58.98
Maltose M (g/L): 68.48
Volume V (L): 6.15

t = 23.00 h

Biomass X (g/L): 76.67
Glucose S (g/L): 52.32
Lactic Acid P (g/L): 51.39
Maltose M (g/L): 54.53
Volume V (L): 5.79

t = 24.00 h

Biomass X (g/L): 75.89
Glucose S (g/L): 48.44
Lactic Acid P (g/L): 55.34
Maltose M (g/L): 65.36
Volume V (L): 5.76

t = 25.00 h

Biomass X (g/L): 84.33
Glucose S (g/L): 48.08
Lactic Acid P (g/L): 51.59
Maltose M (g/L): 60.85
Volume V (L): 6.48

t = 26.00 h

Biomass X (g/L): 74.10
Glucose S (g/L): 43.86
Lactic Acid P (g/L): 62.57
Maltose M (g/L): 58.76
Volume V (L): 6.41

t = 27.00 h

Biomass X (g/L): 78.88
Glucose S (g/L): 45.64
Lactic Acid P (g/L): 66.55
Maltose M (g/L): 56.51
Volume V (L): 6.32

t = 28.00 h

Biomass X (g/L): 69.41
Glucose S (g/L): 56.67
Lactic Acid P (g/L): 81.65
Maltose M (g/L): 66.35
Volume V (L): 5.93

t = 29.00 h

Biomass X (g/L): 86.88
Glucose S (g/L): 44.05
Lactic Acid P (g/L): 60.35
Maltose M (g/L): 58.73
Volume V (L): 6.79

t = 30.00 h

Biomass X (g/L): 73.17
Glucose S (g/L): 46.24
Lactic Acid P (g/L): 49.25
Maltose M (g/L): 56.15
Volume V (L): 6.63

t = 31.00 h

Biomass X (g/L): 83.83
Glucose S (g/L): 48.53
Lactic Acid P (g/L): 58.99
Maltose M (g/L): 53.70
Volume V (L): 6.32

t = 32.00 h

Biomass X (g/L): 89.11
Glucose S (g/L): 48.83
Lactic Acid P (g/L): 72.54
Maltose M (g/L): 52.80
Volume V (L): 6.30

t = 33.00 h

Biomass X (g/L): 90.50
Glucose S (g/L): 50.76
Lactic Acid P (g/L): 57.95
Maltose M (g/L): 54.84
Volume V (L): 6.51

t = 34.00 h

Biomass X (g/L): 92.83
Glucose S (g/L): 47.71
Lactic Acid P (g/L): 51.93
Maltose M (g/L): 50.89

Volume V (L): 6.40

t = 35.00 h

Biomass X (g/L): 83.32

Glucose S (g/L): 51.53

Lactic Acid P (g/L): 59.48

Maltose M (g/L): 56.16

Volume V (L): 7.18

t = 36.00 h

Biomass X (g/L): 90.73

Glucose S (g/L): 54.54

Lactic Acid P (g/L): 62.16

Maltose M (g/L): 51.00

Volume V (L): 6.52

t = 37.00 h

Biomass X (g/L): 81.25

Glucose S (g/L): 55.36

Lactic Acid P (g/L): 57.55

Maltose M (g/L): 56.44

Volume V (L): 6.38

t = 38.00 h

Biomass X (g/L): 66.58

Glucose S (g/L): 49.78

Lactic Acid P (g/L): 70.39

Maltose M (g/L): 55.93

Volume V (L): 7.14

t = 39.00 h

Biomass X (g/L): 92.01

Glucose S (g/L): 53.57

Lactic Acid P (g/L): 68.12

Maltose M (g/L): 51.81

Volume V (L): 6.61

t = 40.00 h

Biomass X (g/L): 89.89

Glucose S (g/L): 47.17

Lactic Acid P (g/L): 76.83

Maltose M (g/L): 53.66

Volume V (L): 7.02

t = 41.00 h

Biomass X (g/L): 97.26

Glucose S (g/L): 49.23

Lactic Acid P (g/L): 65.63

Maltose M (g/L): 53.43

Volume V (L): 7.03

t = 42.00 h

Biomass X (g/L): 86.95

Glucose S (g/L): 47.92

Lactic Acid P (g/L): 75.57

Maltose M (g/L): 45.58

Volume V (L): 6.88

t = 43.00 h

Biomass X (g/L): 71.17

Glucose S (g/L): 52.73

Lactic Acid P (g/L): 66.08

Maltose M (g/L): 56.55

Volume V (L): 6.68

t = 44.00 h

Biomass X (g/L): 86.49

Glucose S (g/L): 50.04

Lactic Acid P (g/L): 60.47

Maltose M (g/L): 49.09

Volume V (L): 6.97

t = 45.00 h

Biomass X (g/L): 96.42

Glucose S (g/L): 47.95

Lactic Acid P (g/L): 66.41

Maltose M (g/L): 52.09

Volume V (L): 6.71

t = 46.00 h

Biomass X (g/L): 92.33

Glucose S (g/L): 52.81

Lactic Acid P (g/L): 56.63

Maltose M (g/L): 51.81

Volume V (L): 7.07

t = 47.00 h

Biomass X (g/L): 86.09

Glucose S (g/L): 48.10

Lactic Acid P (g/L): 89.20

Maltose M (g/L): 40.01

Volume V (L): 7.33

t = 48.00 h

Biomass X (g/L): 70.82

Glucose S (g/L): 60.41

Lactic Acid P (g/L): 73.99

Maltose M (g/L): 46.16

Volume V (L): 7.63

t = 49.00 h

Biomass X (g/L): 77.24

Glucose S (g/L): 53.62

Lactic Acid P (g/L): 65.46

Maltose M (g/L): 51.13

Volume V (L): 6.92

t = 50.00 h

Biomass X (g/L): 92.36

Glucose S (g/L): 60.08

Lactic Acid P (g/L): 69.74

Maltose M (g/L): 48.01

Volume V (L): 7.19

t = 51.00 h

Biomass X (g/L): 70.75

Glucose S (g/L): 54.87

Lactic Acid P (g/L): 50.46

Maltose M (g/L): 49.89

Volume V (L): 7.16

t = 52.00 h

Biomass X (g/L): 84.27

Glucose S (g/L): 52.66
Lactic Acid P (g/L): 87.48
Maltose M (g/L): 47.42
Volume V (L): 7.42

t = 53.00 h

Biomass X (g/L): 87.80
Glucose S (g/L): 62.34
Lactic Acid P (g/L): 86.85
Maltose M (g/L): 51.13
Volume V (L): 6.98

t = 54.00 h

Biomass X (g/L): 98.92
Glucose S (g/L): 58.65
Lactic Acid P (g/L): 75.51
Maltose M (g/L): 46.83
Volume V (L): 7.19

t = 55.00 h

Biomass X (g/L): 93.44
Glucose S (g/L): 66.12
Lactic Acid P (g/L): 77.79
Maltose M (g/L): 51.16
Volume V (L): 7.02

t = 56.00 h

Biomass X (g/L): 96.86
Glucose S (g/L): 57.53
Lactic Acid P (g/L): 63.05
Maltose M (g/L): 42.29
Volume V (L): 7.14

t = 57.00 h

Biomass X (g/L): 90.10
Glucose S (g/L): 58.13
Lactic Acid P (g/L): 63.99
Maltose M (g/L): 45.00
Volume V (L): 7.39

t = 58.00 h

Biomass X (g/L): 78.43
Glucose S (g/L): 64.25
Lactic Acid P (g/L): 59.49
Maltose M (g/L): 43.03
Volume V (L): 8.07

t = 59.00 h

Biomass X (g/L): 80.64
Glucose S (g/L): 64.79
Lactic Acid P (g/L): 74.24
Maltose M (g/L): 46.21
Volume V (L): 7.76

t = 60.00 h

Biomass X (g/L): 77.30
Glucose S (g/L): 61.98
Lactic Acid P (g/L): 86.03
Maltose M (g/L): 43.53
Volume V (L): 7.72

t = 61.00 h

Biomass X (g/L): 92.51
Glucose S (g/L): 74.80
Lactic Acid P (g/L): 72.60
Maltose M (g/L): 43.41
Volume V (L): 8.12

t = 62.00 h

Biomass X (g/L): 79.43
Glucose S (g/L): 61.33
Lactic Acid P (g/L): 79.72
Maltose M (g/L): 41.42
Volume V (L): 7.90

t = 63.00 h

Biomass X (g/L): 82.30
Glucose S (g/L): 58.01
Lactic Acid P (g/L): 85.90
Maltose M (g/L): 42.00
Volume V (L): 7.97

t = 64.00 h

Biomass X (g/L): 86.35
Glucose S (g/L): 59.56
Lactic Acid P (g/L): 61.50
Maltose M (g/L): 43.31
Volume V (L): 7.76

t = 65.00 h

Biomass X (g/L): 102.50
Glucose S (g/L): 62.88
Lactic Acid P (g/L): 94.50
Maltose M (g/L): 35.87
Volume V (L): 7.76

t = 66.00 h

Biomass X (g/L): 92.42
Glucose S (g/L): 70.32
Lactic Acid P (g/L): 88.18
Maltose M (g/L): 34.60
Volume V (L): 7.62

t = 67.00 h

Biomass X (g/L): 77.84
Glucose S (g/L): 64.23
Lactic Acid P (g/L): 82.29
Maltose M (g/L): 42.72
Volume V (L): 7.86

t = 68.00 h

Biomass X (g/L): 86.99
Glucose S (g/L): 71.51
Lactic Acid P (g/L): 86.35
Maltose M (g/L): 32.38
Volume V (L): 7.99

t = 69.00 h

Biomass X (g/L): 101.91
Glucose S (g/L): 67.63
Lactic Acid P (g/L): 86.97
Maltose M (g/L): 41.28
Volume V (L): 8.21

t = 70.00 h

Biomass X (g/L): 91.30
Glucose S (g/L): 70.64
Lactic Acid P (g/L): 80.73
Maltose M (g/L): 40.36
Volume V (L): 8.15

t = 71.00 h

Biomass X (g/L): 68.81
Glucose S (g/L): 68.44
Lactic Acid P (g/L): 91.64
Maltose M (g/L): 43.37
Volume V (L): 8.10

t = 72.00 h

Biomass X (g/L): 75.51
Glucose S (g/L): 68.90
Lactic Acid P (g/L): 74.02
Maltose M (g/L): 38.79
Volume V (L): 8.00

t = 73.00 h

Biomass X (g/L): 74.54
Glucose S (g/L): 70.83
Lactic Acid P (g/L): 82.83
Maltose M (g/L): 46.51
Volume V (L): 8.40

t = 74.00 h

Biomass X (g/L): 98.86
Glucose S (g/L): 66.21
Lactic Acid P (g/L): 91.11
Maltose M (g/L): 43.69
Volume V (L): 8.20

t = 75.00 h

Biomass X (g/L): 96.80
Glucose S (g/L): 66.73
Lactic Acid P (g/L): 53.33
Maltose M (g/L): 35.05

Volume V (L): 8.48

Batch 91

Initial Conditions:

Batch: 91.00

X0 (g/L): 5.96

S0 (g/L): 36.22

M0 (g/L): 114.66

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 3.17

F_vol (L/h): 0.05

S_feed (g/L): 72.02

M_feed (g/L): 227.98

V_final (L): 8.71

Batch 91 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 32.03
Lactic Acid P (g/L): 1.35
Maltose M (g/L): 126.81
Volume V (L): 4.96

t = 1.00 h

Biomass X (g/L): 11.04
Glucose S (g/L): 37.66
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 118.22
Volume V (L): 5.46

t = 2.00 h

Biomass X (g/L): 17.30
Glucose S (g/L): 35.15
Lactic Acid P (g/L): 2.95
Maltose M (g/L): 111.87
Volume V (L): 5.08

t = 3.00 h

Biomass X (g/L): 18.93
Glucose S (g/L): 35.12
Lactic Acid P (g/L): 21.17
Maltose M (g/L): 114.23
Volume V (L): 4.92

t = 4.00 h

Biomass X (g/L): 14.48
Glucose S (g/L): 33.07
Lactic Acid P (g/L): 9.72
Maltose M (g/L): 100.34
Volume V (L): 5.24

t = 5.00 h

Biomass X (g/L): 28.85
Glucose S (g/L): 33.73

Lactic Acid P (g/L): 7.01
Maltose M (g/L): 114.03
Volume V (L): 5.23

t = 6.00 h

Biomass X (g/L): 12.11
Glucose S (g/L): 26.29
Lactic Acid P (g/L): 11.41
Maltose M (g/L): 104.44
Volume V (L): 5.43

t = 7.00 h

Biomass X (g/L): 27.71
Glucose S (g/L): 32.12
Lactic Acid P (g/L): 15.26
Maltose M (g/L): 107.85
Volume V (L): 5.83

t = 8.00 h

Biomass X (g/L): 31.12
Glucose S (g/L): 27.71
Lactic Acid P (g/L): 12.63
Maltose M (g/L): 102.95
Volume V (L): 5.51

t = 9.00 h

Biomass X (g/L): 47.17
Glucose S (g/L): 24.16
Lactic Acid P (g/L): 22.87
Maltose M (g/L): 102.52
Volume V (L): 6.05

t = 10.00 h

Biomass X (g/L): 22.22
Glucose S (g/L): 18.77
Lactic Acid P (g/L): 39.70
Maltose M (g/L): 103.71
Volume V (L): 5.63

t = 11.00 h

Biomass X (g/L): 35.33

Glucose S (g/L): 22.62
Lactic Acid P (g/L): 41.67
Maltose M (g/L): 104.04
Volume V (L): 5.32

t = 12.00 h

Biomass X (g/L): 65.72
Glucose S (g/L): 17.65
Lactic Acid P (g/L): 25.30
Maltose M (g/L): 102.64
Volume V (L): 6.27

t = 13.00 h

Biomass X (g/L): 29.27
Glucose S (g/L): 21.83
Lactic Acid P (g/L): 46.62
Maltose M (g/L): 101.88
Volume V (L): 5.61

t = 14.00 h

Biomass X (g/L): 63.24
Glucose S (g/L): 18.07
Lactic Acid P (g/L): 44.31
Maltose M (g/L): 93.20
Volume V (L): 5.38

t = 15.00 h

Biomass X (g/L): 51.58
Glucose S (g/L): 20.08
Lactic Acid P (g/L): 45.37
Maltose M (g/L): 111.41
Volume V (L): 6.43

t = 16.00 h

Biomass X (g/L): 63.19
Glucose S (g/L): 15.05
Lactic Acid P (g/L): 45.57
Maltose M (g/L): 100.46
Volume V (L): 5.57

t = 17.00 h

Biomass X (g/L): 76.49
Glucose S (g/L): 17.14
Lactic Acid P (g/L): 34.92
Maltose M (g/L): 100.63
Volume V (L): 5.61

t = 18.00 h

Biomass X (g/L): 67.86
Glucose S (g/L): 14.79
Lactic Acid P (g/L): 41.50
Maltose M (g/L): 96.07
Volume V (L): 5.73

t = 19.00 h

Biomass X (g/L): 80.59
Glucose S (g/L): 11.70
Lactic Acid P (g/L): 54.53
Maltose M (g/L): 92.70
Volume V (L): 5.99

t = 20.00 h

Biomass X (g/L): 63.91
Glucose S (g/L): 13.34
Lactic Acid P (g/L): 37.25
Maltose M (g/L): 96.86
Volume V (L): 6.27

t = 21.00 h

Biomass X (g/L): 66.70
Glucose S (g/L): 10.17
Lactic Acid P (g/L): 40.18
Maltose M (g/L): 96.75
Volume V (L): 5.25

t = 22.00 h

Biomass X (g/L): 79.02
Glucose S (g/L): 18.73
Lactic Acid P (g/L): 53.90
Maltose M (g/L): 100.87
Volume V (L): 6.36

t = 23.00 h

Biomass X (g/L): 68.61
Glucose S (g/L): 12.33
Lactic Acid P (g/L): 45.04
Maltose M (g/L): 103.86
Volume V (L): 5.97

t = 24.00 h

Biomass X (g/L): 79.66
Glucose S (g/L): 12.82
Lactic Acid P (g/L): 61.07
Maltose M (g/L): 100.23
Volume V (L): 6.07

t = 25.00 h

Biomass X (g/L): 86.32
Glucose S (g/L): 14.07
Lactic Acid P (g/L): 47.63
Maltose M (g/L): 83.88
Volume V (L): 6.75

t = 26.00 h

Biomass X (g/L): 84.31
Glucose S (g/L): 9.92
Lactic Acid P (g/L): 57.86
Maltose M (g/L): 89.46
Volume V (L): 6.35

t = 27.00 h

Biomass X (g/L): 61.89
Glucose S (g/L): 17.33
Lactic Acid P (g/L): 59.61
Maltose M (g/L): 92.59
Volume V (L): 6.47

t = 28.00 h

Biomass X (g/L): 74.52
Glucose S (g/L): 10.70
Lactic Acid P (g/L): 53.08
Maltose M (g/L): 88.36
Volume V (L): 6.37

t = 29.00 h

Biomass X (g/L): 81.23
Glucose S (g/L): 10.38
Lactic Acid P (g/L): 56.20
Maltose M (g/L): 89.39
Volume V (L): 6.66

t = 30.00 h

Biomass X (g/L): 76.93
Glucose S (g/L): 13.99
Lactic Acid P (g/L): 52.99
Maltose M (g/L): 78.60
Volume V (L): 6.97

t = 31.00 h

Biomass X (g/L): 80.66
Glucose S (g/L): 11.11
Lactic Acid P (g/L): 59.89
Maltose M (g/L): 92.68
Volume V (L): 5.80

t = 32.00 h

Biomass X (g/L): 83.31
Glucose S (g/L): 19.98
Lactic Acid P (g/L): 51.98
Maltose M (g/L): 78.59
Volume V (L): 6.91

t = 33.00 h

Biomass X (g/L): 76.03
Glucose S (g/L): 15.14
Lactic Acid P (g/L): 59.51
Maltose M (g/L): 79.40
Volume V (L): 7.09

t = 34.00 h

Biomass X (g/L): 79.71
Glucose S (g/L): 6.78
Lactic Acid P (g/L): 69.17
Maltose M (g/L): 81.46

Volume V (L): 6.83

t = 35.00 h

Biomass X (g/L): 70.76

Glucose S (g/L): 17.43

Lactic Acid P (g/L): 52.48

Maltose M (g/L): 85.25

Volume V (L): 6.53

t = 36.00 h

Biomass X (g/L): 77.89

Glucose S (g/L): 19.01

Lactic Acid P (g/L): 59.10

Maltose M (g/L): 77.29

Volume V (L): 6.44

t = 37.00 h

Biomass X (g/L): 68.28

Glucose S (g/L): 16.68

Lactic Acid P (g/L): 68.98

Maltose M (g/L): 79.39

Volume V (L): 7.53

t = 38.00 h

Biomass X (g/L): 84.29

Glucose S (g/L): 14.47

Lactic Acid P (g/L): 45.67

Maltose M (g/L): 89.44

Volume V (L): 6.61

t = 39.00 h

Biomass X (g/L): 81.65

Glucose S (g/L): 18.76

Lactic Acid P (g/L): 82.46

Maltose M (g/L): 82.78

Volume V (L): 7.38

t = 40.00 h

Biomass X (g/L): 75.05

Glucose S (g/L): 18.23

Lactic Acid P (g/L): 75.54

Maltose M (g/L): 84.82

Volume V (L): 7.10

t = 41.00 h

Biomass X (g/L): 101.05

Glucose S (g/L): 17.37

Lactic Acid P (g/L): 52.27

Maltose M (g/L): 76.21

Volume V (L): 7.64

t = 42.00 h

Biomass X (g/L): 81.93

Glucose S (g/L): 17.72

Lactic Acid P (g/L): 47.57

Maltose M (g/L): 81.34

Volume V (L): 6.99

t = 43.00 h

Biomass X (g/L): 87.20

Glucose S (g/L): 21.96

Lactic Acid P (g/L): 67.71

Maltose M (g/L): 80.39

Volume V (L): 6.65

t = 44.00 h

Biomass X (g/L): 78.56

Glucose S (g/L): 24.80

Lactic Acid P (g/L): 57.58

Maltose M (g/L): 76.66

Volume V (L): 7.18

t = 45.00 h

Biomass X (g/L): 80.85

Glucose S (g/L): 24.12

Lactic Acid P (g/L): 77.21

Maltose M (g/L): 79.70

Volume V (L): 6.79

t = 46.00 h

Biomass X (g/L): 88.03

Glucose S (g/L): 18.28

Lactic Acid P (g/L): 72.41
Maltose M (g/L): 70.30
Volume V (L): 8.07

t = 47.00 h

Biomass X (g/L): 81.88
Glucose S (g/L): 25.87
Lactic Acid P (g/L): 70.34
Maltose M (g/L): 81.00
Volume V (L): 7.51

t = 48.00 h

Biomass X (g/L): 85.18
Glucose S (g/L): 26.98
Lactic Acid P (g/L): 64.35
Maltose M (g/L): 76.44
Volume V (L): 7.61

t = 49.00 h

Biomass X (g/L): 92.91
Glucose S (g/L): 29.47
Lactic Acid P (g/L): 84.73
Maltose M (g/L): 76.61
Volume V (L): 7.66

t = 50.00 h

Biomass X (g/L): 80.34
Glucose S (g/L): 23.19
Lactic Acid P (g/L): 77.31
Maltose M (g/L): 81.44
Volume V (L): 7.27

t = 51.00 h

Biomass X (g/L): 72.63
Glucose S (g/L): 26.49
Lactic Acid P (g/L): 76.28
Maltose M (g/L): 78.92
Volume V (L): 7.49

t = 52.00 h

Biomass X (g/L): 70.81

Glucose S (g/L): 29.00
Lactic Acid P (g/L): 82.13
Maltose M (g/L): 74.81
Volume V (L): 7.61

t = 53.00 h

Biomass X (g/L): 99.07
Glucose S (g/L): 26.42
Lactic Acid P (g/L): 62.03
Maltose M (g/L): 70.48
Volume V (L): 8.20

t = 54.00 h

Biomass X (g/L): 94.24
Glucose S (g/L): 30.28
Lactic Acid P (g/L): 83.58
Maltose M (g/L): 81.77
Volume V (L): 7.33

t = 55.00 h

Biomass X (g/L): 91.60
Glucose S (g/L): 29.24
Lactic Acid P (g/L): 69.62
Maltose M (g/L): 68.44
Volume V (L): 7.51

t = 56.00 h

Biomass X (g/L): 81.18
Glucose S (g/L): 27.46
Lactic Acid P (g/L): 75.45
Maltose M (g/L): 77.46
Volume V (L): 7.67

t = 57.00 h

Biomass X (g/L): 72.59
Glucose S (g/L): 28.38
Lactic Acid P (g/L): 71.22
Maltose M (g/L): 66.27
Volume V (L): 7.62

t = 58.00 h

Biomass X (g/L): 82.56
Glucose S (g/L): 35.00
Lactic Acid P (g/L): 55.66
Maltose M (g/L): 77.93
Volume V (L): 7.44

t = 59.00 h

Biomass X (g/L): 84.30
Glucose S (g/L): 29.01
Lactic Acid P (g/L): 82.05
Maltose M (g/L): 77.92
Volume V (L): 7.12

t = 60.00 h

Biomass X (g/L): 74.34
Glucose S (g/L): 29.16
Lactic Acid P (g/L): 56.70
Maltose M (g/L): 63.39
Volume V (L): 7.43

t = 61.00 h

Biomass X (g/L): 71.05
Glucose S (g/L): 34.30
Lactic Acid P (g/L): 64.86
Maltose M (g/L): 74.32
Volume V (L): 8.35

t = 62.00 h

Biomass X (g/L): 68.48
Glucose S (g/L): 33.84
Lactic Acid P (g/L): 69.85
Maltose M (g/L): 73.49
Volume V (L): 8.07

t = 63.00 h

Biomass X (g/L): 91.14
Glucose S (g/L): 30.70
Lactic Acid P (g/L): 56.27
Maltose M (g/L): 74.68
Volume V (L): 8.37

t = 64.00 h

Biomass X (g/L): 73.29
Glucose S (g/L): 33.70
Lactic Acid P (g/L): 64.37
Maltose M (g/L): 72.62
Volume V (L): 8.00

t = 65.00 h

Biomass X (g/L): 87.62
Glucose S (g/L): 33.47
Lactic Acid P (g/L): 76.99
Maltose M (g/L): 65.81
Volume V (L): 8.19

t = 66.00 h

Biomass X (g/L): 80.77
Glucose S (g/L): 34.64
Lactic Acid P (g/L): 71.77
Maltose M (g/L): 68.98
Volume V (L): 8.75

t = 67.00 h

Biomass X (g/L): 90.06
Glucose S (g/L): 44.67
Lactic Acid P (g/L): 72.75
Maltose M (g/L): 77.63
Volume V (L): 8.78

t = 68.00 h

Biomass X (g/L): 87.52
Glucose S (g/L): 36.93
Lactic Acid P (g/L): 66.97
Maltose M (g/L): 63.87
Volume V (L): 8.65

t = 69.00 h

Biomass X (g/L): 81.98
Glucose S (g/L): 35.85
Lactic Acid P (g/L): 76.03
Maltose M (g/L): 70.36
Volume V (L): 8.48

t = 70.00 h

Biomass X (g/L): 85.30
Glucose S (g/L): 45.09
Lactic Acid P (g/L): 72.82
Maltose M (g/L): 73.95
Volume V (L): 8.62

t = 71.00 h

Biomass X (g/L): 82.87
Glucose S (g/L): 42.37
Lactic Acid P (g/L): 60.90
Maltose M (g/L): 63.24
Volume V (L): 8.63

t = 72.00 h

Biomass X (g/L): 77.91
Glucose S (g/L): 42.51
Lactic Acid P (g/L): 74.35
Maltose M (g/L): 60.80
Volume V (L): 8.52

t = 73.00 h

Biomass X (g/L): 92.96
Glucose S (g/L): 38.52
Lactic Acid P (g/L): 64.95
Maltose M (g/L): 61.14
Volume V (L): 8.50

t = 74.00 h

Biomass X (g/L): 79.44
Glucose S (g/L): 45.14
Lactic Acid P (g/L): 84.84
Maltose M (g/L): 68.76
Volume V (L): 9.00

t = 75.00 h

Biomass X (g/L): 82.49
Glucose S (g/L): 44.65
Lactic Acid P (g/L): 73.15
Maltose M (g/L): 60.34

Volume V (L): 8.33

Batch 92

Initial Conditions:

Batch: 92.00

X0 (g/L): 2.11

S0 (g/L): 136.62

M0 (g/L): 20.96

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.15

F_vol (L/h): 0.06

S_feed (g/L): 260.09

M_feed (g/L): 39.91

V_final (L): 9.46

Batch 92 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 17.03
Glucose S (g/L): 136.79
Lactic Acid P (g/L): 15.02
Maltose M (g/L): 20.05
Volume V (L): 4.75

t = 1.00 h

Biomass X (g/L): 0.12
Glucose S (g/L): 142.69
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 20.68
Volume V (L): 5.24

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 128.03
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 18.15
Volume V (L): 4.76

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 140.15
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 20.15
Volume V (L): 5.36

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 137.56
Lactic Acid P (g/L): 6.44
Maltose M (g/L): 19.89
Volume V (L): 5.30

t = 5.00 h

Biomass X (g/L): 3.40
Glucose S (g/L): 134.97

Lactic Acid P (g/L): 5.70
Maltose M (g/L): 21.28
Volume V (L): 4.76

t = 6.00 h

Biomass X (g/L): 5.91
Glucose S (g/L): 133.91
Lactic Acid P (g/L): 8.99
Maltose M (g/L): 19.41
Volume V (L): 5.70

t = 7.00 h

Biomass X (g/L): 8.73
Glucose S (g/L): 123.87
Lactic Acid P (g/L): 1.37
Maltose M (g/L): 19.53
Volume V (L): 4.97

t = 8.00 h

Biomass X (g/L): 15.69
Glucose S (g/L): 129.98
Lactic Acid P (g/L): 25.89
Maltose M (g/L): 19.53
Volume V (L): 5.75

t = 9.00 h

Biomass X (g/L): 22.25
Glucose S (g/L): 136.76
Lactic Acid P (g/L): 11.01
Maltose M (g/L): 18.94
Volume V (L): 5.58

t = 10.00 h

Biomass X (g/L): 34.96
Glucose S (g/L): 131.77
Lactic Acid P (g/L): 29.39
Maltose M (g/L): 19.89
Volume V (L): 5.43

t = 11.00 h

Biomass X (g/L): 29.41

Glucose S (g/L): 124.03
Lactic Acid P (g/L): 50.76
Maltose M (g/L): 17.44
Volume V (L): 5.26

t = 12.00 h

Biomass X (g/L): 39.73
Glucose S (g/L): 115.33
Lactic Acid P (g/L): 17.16
Maltose M (g/L): 19.94
Volume V (L): 5.83

t = 13.00 h

Biomass X (g/L): 41.02
Glucose S (g/L): 122.17
Lactic Acid P (g/L): 31.72
Maltose M (g/L): 18.82
Volume V (L): 5.46

t = 14.00 h

Biomass X (g/L): 50.66
Glucose S (g/L): 121.31
Lactic Acid P (g/L): 30.58
Maltose M (g/L): 19.78
Volume V (L): 6.26

t = 15.00 h

Biomass X (g/L): 49.74
Glucose S (g/L): 117.22
Lactic Acid P (g/L): 53.60
Maltose M (g/L): 16.20
Volume V (L): 6.34

t = 16.00 h

Biomass X (g/L): 46.53
Glucose S (g/L): 105.62
Lactic Acid P (g/L): 35.16
Maltose M (g/L): 16.33
Volume V (L): 6.54

t = 17.00 h

Biomass X (g/L): 47.25
Glucose S (g/L): 103.72
Lactic Acid P (g/L): 42.72
Maltose M (g/L): 18.00
Volume V (L): 6.11

t = 18.00 h

Biomass X (g/L): 61.70
Glucose S (g/L): 112.91
Lactic Acid P (g/L): 45.00
Maltose M (g/L): 18.54
Volume V (L): 5.93

t = 19.00 h

Biomass X (g/L): 75.21
Glucose S (g/L): 98.19
Lactic Acid P (g/L): 44.26
Maltose M (g/L): 17.50
Volume V (L): 6.02

t = 20.00 h

Biomass X (g/L): 62.70
Glucose S (g/L): 104.12
Lactic Acid P (g/L): 54.25
Maltose M (g/L): 18.33
Volume V (L): 6.06

t = 21.00 h

Biomass X (g/L): 60.29
Glucose S (g/L): 92.57
Lactic Acid P (g/L): 50.07
Maltose M (g/L): 18.54
Volume V (L): 6.34

t = 22.00 h

Biomass X (g/L): 66.83
Glucose S (g/L): 97.97
Lactic Acid P (g/L): 55.43
Maltose M (g/L): 18.14
Volume V (L): 6.46

t = 23.00 h

Biomass X (g/L): 65.64
Glucose S (g/L): 100.88
Lactic Acid P (g/L): 54.58
Maltose M (g/L): 17.53
Volume V (L): 6.81

t = 24.00 h

Biomass X (g/L): 66.25
Glucose S (g/L): 97.22
Lactic Acid P (g/L): 48.53
Maltose M (g/L): 17.39
Volume V (L): 6.43

t = 25.00 h

Biomass X (g/L): 59.97
Glucose S (g/L): 98.94
Lactic Acid P (g/L): 53.01
Maltose M (g/L): 16.67
Volume V (L): 5.99

t = 26.00 h

Biomass X (g/L): 73.35
Glucose S (g/L): 90.45
Lactic Acid P (g/L): 61.89
Maltose M (g/L): 17.02
Volume V (L): 6.40

t = 27.00 h

Biomass X (g/L): 79.38
Glucose S (g/L): 91.75
Lactic Acid P (g/L): 63.04
Maltose M (g/L): 16.07
Volume V (L): 7.16

t = 28.00 h

Biomass X (g/L): 67.24
Glucose S (g/L): 97.17
Lactic Acid P (g/L): 70.53
Maltose M (g/L): 16.15
Volume V (L): 6.11

t = 29.00 h

Biomass X (g/L): 70.26
Glucose S (g/L): 97.79
Lactic Acid P (g/L): 77.21
Maltose M (g/L): 17.13
Volume V (L): 6.99

t = 30.00 h

Biomass X (g/L): 70.43
Glucose S (g/L): 95.40
Lactic Acid P (g/L): 56.20
Maltose M (g/L): 16.19
Volume V (L): 6.64

t = 31.00 h

Biomass X (g/L): 66.80
Glucose S (g/L): 98.47
Lactic Acid P (g/L): 63.14
Maltose M (g/L): 15.84
Volume V (L): 6.53

t = 32.00 h

Biomass X (g/L): 71.82
Glucose S (g/L): 86.66
Lactic Acid P (g/L): 60.93
Maltose M (g/L): 16.73
Volume V (L): 7.04

t = 33.00 h

Biomass X (g/L): 80.04
Glucose S (g/L): 91.94
Lactic Acid P (g/L): 53.28
Maltose M (g/L): 16.56
Volume V (L): 7.52

t = 34.00 h

Biomass X (g/L): 69.11
Glucose S (g/L): 102.19
Lactic Acid P (g/L): 69.42
Maltose M (g/L): 15.94

Volume V (L): 7.14

t = 35.00 h

Biomass X (g/L): 81.68

Glucose S (g/L): 99.79

Lactic Acid P (g/L): 72.07

Maltose M (g/L): 15.88

Volume V (L): 6.99

t = 36.00 h

Biomass X (g/L): 76.44

Glucose S (g/L): 96.94

Lactic Acid P (g/L): 55.60

Maltose M (g/L): 16.25

Volume V (L): 6.68

t = 37.00 h

Biomass X (g/L): 74.55

Glucose S (g/L): 93.50

Lactic Acid P (g/L): 62.69

Maltose M (g/L): 16.61

Volume V (L): 7.16

t = 38.00 h

Biomass X (g/L): 87.99

Glucose S (g/L): 98.97

Lactic Acid P (g/L): 62.36

Maltose M (g/L): 15.86

Volume V (L): 6.78

t = 39.00 h

Biomass X (g/L): 78.07

Glucose S (g/L): 96.42

Lactic Acid P (g/L): 53.34

Maltose M (g/L): 14.57

Volume V (L): 7.21

t = 40.00 h

Biomass X (g/L): 80.03

Glucose S (g/L): 87.87

Lactic Acid P (g/L): 67.42

Maltose M (g/L): 15.29

Volume V (L): 7.36

t = 41.00 h

Biomass X (g/L): 97.98

Glucose S (g/L): 99.35

Lactic Acid P (g/L): 57.68

Maltose M (g/L): 15.03

Volume V (L): 6.60

t = 42.00 h

Biomass X (g/L): 75.91

Glucose S (g/L): 91.23

Lactic Acid P (g/L): 70.17

Maltose M (g/L): 14.38

Volume V (L): 7.99

t = 43.00 h

Biomass X (g/L): 101.00

Glucose S (g/L): 98.76

Lactic Acid P (g/L): 71.62

Maltose M (g/L): 15.59

Volume V (L): 7.12

t = 44.00 h

Biomass X (g/L): 81.93

Glucose S (g/L): 88.37

Lactic Acid P (g/L): 73.18

Maltose M (g/L): 15.56

Volume V (L): 7.55

t = 45.00 h

Biomass X (g/L): 74.12

Glucose S (g/L): 103.07

Lactic Acid P (g/L): 74.02

Maltose M (g/L): 15.59

Volume V (L): 7.87

t = 46.00 h

Biomass X (g/L): 80.03

Glucose S (g/L): 94.95

Lactic Acid P (g/L): 84.45

Maltose M (g/L): 14.03

Volume V (L): 8.20

t = 47.00 h

Biomass X (g/L): 86.33

Glucose S (g/L): 91.51

Lactic Acid P (g/L): 63.24

Maltose M (g/L): 14.91

Volume V (L): 7.51

t = 48.00 h

Biomass X (g/L): 98.26

Glucose S (g/L): 93.70

Lactic Acid P (g/L): 64.04

Maltose M (g/L): 14.68

Volume V (L): 6.96

t = 49.00 h

Biomass X (g/L): 93.70

Glucose S (g/L): 95.60

Lactic Acid P (g/L): 77.42

Maltose M (g/L): 14.85

Volume V (L): 8.30

t = 50.00 h

Biomass X (g/L): 86.33

Glucose S (g/L): 97.41

Lactic Acid P (g/L): 70.47

Maltose M (g/L): 14.61

Volume V (L): 7.53

t = 51.00 h

Biomass X (g/L): 98.31

Glucose S (g/L): 103.68

Lactic Acid P (g/L): 82.37

Maltose M (g/L): 14.40

Volume V (L): 8.16

t = 52.00 h

Biomass X (g/L): 66.77

Glucose S (g/L): 98.42
Lactic Acid P (g/L): 77.47
Maltose M (g/L): 14.20
Volume V (L): 8.24

t = 53.00 h

Biomass X (g/L): 88.10
Glucose S (g/L): 88.71
Lactic Acid P (g/L): 90.26
Maltose M (g/L): 15.16
Volume V (L): 6.99

t = 54.00 h

Biomass X (g/L): 88.65
Glucose S (g/L): 93.28
Lactic Acid P (g/L): 72.00
Maltose M (g/L): 13.41
Volume V (L): 7.63

t = 55.00 h

Biomass X (g/L): 88.70
Glucose S (g/L): 111.52
Lactic Acid P (g/L): 59.43
Maltose M (g/L): 13.90
Volume V (L): 8.11

t = 56.00 h

Biomass X (g/L): 69.24
Glucose S (g/L): 100.12
Lactic Acid P (g/L): 76.49
Maltose M (g/L): 14.88
Volume V (L): 7.41

t = 57.00 h

Biomass X (g/L): 75.10
Glucose S (g/L): 104.13
Lactic Acid P (g/L): 72.93
Maltose M (g/L): 13.19
Volume V (L): 7.91

t = 58.00 h

Biomass X (g/L): 83.53
Glucose S (g/L): 98.84
Lactic Acid P (g/L): 78.76
Maltose M (g/L): 13.15
Volume V (L): 8.49

t = 59.00 h

Biomass X (g/L): 96.27
Glucose S (g/L): 96.08
Lactic Acid P (g/L): 80.16
Maltose M (g/L): 14.33
Volume V (L): 8.48

t = 60.00 h

Biomass X (g/L): 104.18
Glucose S (g/L): 98.79
Lactic Acid P (g/L): 69.34
Maltose M (g/L): 12.42
Volume V (L): 8.68

t = 61.00 h

Biomass X (g/L): 78.19
Glucose S (g/L): 101.89
Lactic Acid P (g/L): 65.18
Maltose M (g/L): 14.10
Volume V (L): 8.79

t = 62.00 h

Biomass X (g/L): 71.58
Glucose S (g/L): 99.61
Lactic Acid P (g/L): 73.34
Maltose M (g/L): 13.05
Volume V (L): 8.09

t = 63.00 h

Biomass X (g/L): 92.77
Glucose S (g/L): 110.65
Lactic Acid P (g/L): 52.92
Maltose M (g/L): 13.73
Volume V (L): 8.29

t = 64.00 h

Biomass X (g/L): 110.67
Glucose S (g/L): 99.64
Lactic Acid P (g/L): 75.82
Maltose M (g/L): 12.27
Volume V (L): 9.08

t = 65.00 h

Biomass X (g/L): 95.40
Glucose S (g/L): 98.04
Lactic Acid P (g/L): 71.51
Maltose M (g/L): 14.23
Volume V (L): 9.41

t = 66.00 h

Biomass X (g/L): 76.63
Glucose S (g/L): 106.76
Lactic Acid P (g/L): 83.39
Maltose M (g/L): 12.12
Volume V (L): 8.58

t = 67.00 h

Biomass X (g/L): 112.68
Glucose S (g/L): 107.68
Lactic Acid P (g/L): 57.30
Maltose M (g/L): 13.69
Volume V (L): 9.08

t = 68.00 h

Biomass X (g/L): 90.71
Glucose S (g/L): 104.00
Lactic Acid P (g/L): 75.85
Maltose M (g/L): 12.00
Volume V (L): 9.33

t = 69.00 h

Biomass X (g/L): 75.73
Glucose S (g/L): 101.52
Lactic Acid P (g/L): 90.49
Maltose M (g/L): 14.45
Volume V (L): 9.19

t = 70.00 h

Biomass X (g/L): 102.89
Glucose S (g/L): 96.05
Lactic Acid P (g/L): 82.35
Maltose M (g/L): 12.79
Volume V (L): 9.07

t = 71.00 h

Biomass X (g/L): 74.59
Glucose S (g/L): 110.42
Lactic Acid P (g/L): 85.32
Maltose M (g/L): 14.43
Volume V (L): 8.99

t = 72.00 h

Biomass X (g/L): 76.91
Glucose S (g/L): 109.19
Lactic Acid P (g/L): 79.56
Maltose M (g/L): 12.01
Volume V (L): 9.99

t = 73.00 h

Biomass X (g/L): 90.25
Glucose S (g/L): 103.68
Lactic Acid P (g/L): 61.25
Maltose M (g/L): 12.25
Volume V (L): 9.38

t = 74.00 h

Biomass X (g/L): 82.53
Glucose S (g/L): 106.27
Lactic Acid P (g/L): 78.26
Maltose M (g/L): 13.58
Volume V (L): 9.54

t = 75.00 h

Biomass X (g/L): 82.97
Glucose S (g/L): 105.34
Lactic Acid P (g/L): 80.63
Maltose M (g/L): 11.28

Volume V (L): 9.58

Batch 93

Initial Conditions:

Batch: 93.00

X0 (g/L): 8.26

S0 (g/L): 105.12

M0 (g/L): 60.98

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.58

F_vol (L/h): 0.08

S_feed (g/L): 189.86

M_feed (g/L): 110.14

V_final (L): 10.98

Batch 93 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.56
Glucose S (g/L): 106.88
Lactic Acid P (g/L): 6.72
Maltose M (g/L): 60.78
Volume V (L): 5.15

t = 1.00 h

Biomass X (g/L): 11.57
Glucose S (g/L): 101.89
Lactic Acid P (g/L): 5.36
Maltose M (g/L): 58.08
Volume V (L): 5.97

t = 2.00 h

Biomass X (g/L): 11.53
Glucose S (g/L): 101.77
Lactic Acid P (g/L): 4.16
Maltose M (g/L): 59.75
Volume V (L): 5.14

t = 3.00 h

Biomass X (g/L): 15.47
Glucose S (g/L): 100.71
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 60.57
Volume V (L): 4.46

t = 4.00 h

Biomass X (g/L): 26.34
Glucose S (g/L): 97.86
Lactic Acid P (g/L): 17.47
Maltose M (g/L): 54.10
Volume V (L): 5.22

t = 5.00 h

Biomass X (g/L): 19.68
Glucose S (g/L): 98.79

Lactic Acid P (g/L): 17.75

Maltose M (g/L): 61.51

Volume V (L): 6.00

t = 6.00 h

Biomass X (g/L): 22.14

Glucose S (g/L): 88.50

Lactic Acid P (g/L): 5.05

Maltose M (g/L): 60.49

Volume V (L): 5.15

t = 7.00 h

Biomass X (g/L): 43.49

Glucose S (g/L): 85.85

Lactic Acid P (g/L): 14.04

Maltose M (g/L): 53.98

Volume V (L): 6.09

t = 8.00 h

Biomass X (g/L): 38.70

Glucose S (g/L): 85.76

Lactic Acid P (g/L): 1.50

Maltose M (g/L): 53.23

Volume V (L): 5.72

t = 9.00 h

Biomass X (g/L): 45.19

Glucose S (g/L): 79.01

Lactic Acid P (g/L): 17.81

Maltose M (g/L): 59.39

Volume V (L): 6.03

t = 10.00 h

Biomass X (g/L): 34.55

Glucose S (g/L): 74.45

Lactic Acid P (g/L): 30.70

Maltose M (g/L): 58.83

Volume V (L): 6.69

t = 11.00 h

Biomass X (g/L): 65.92

Glucose S (g/L): 73.56
Lactic Acid P (g/L): 27.10
Maltose M (g/L): 58.13
Volume V (L): 5.00

t = 12.00 h

Biomass X (g/L): 65.36
Glucose S (g/L): 76.00
Lactic Acid P (g/L): 31.98
Maltose M (g/L): 53.06
Volume V (L): 6.39

t = 13.00 h

Biomass X (g/L): 70.32
Glucose S (g/L): 77.39
Lactic Acid P (g/L): 54.70
Maltose M (g/L): 60.15
Volume V (L): 5.99

t = 14.00 h

Biomass X (g/L): 70.16
Glucose S (g/L): 76.02
Lactic Acid P (g/L): 53.79
Maltose M (g/L): 53.85
Volume V (L): 5.66

t = 15.00 h

Biomass X (g/L): 54.69
Glucose S (g/L): 78.18
Lactic Acid P (g/L): 40.35
Maltose M (g/L): 55.77
Volume V (L): 6.15

t = 16.00 h

Biomass X (g/L): 73.49
Glucose S (g/L): 71.91
Lactic Acid P (g/L): 55.77
Maltose M (g/L): 55.43
Volume V (L): 7.09

t = 17.00 h

Biomass X (g/L): 67.56
Glucose S (g/L): 71.64
Lactic Acid P (g/L): 61.36
Maltose M (g/L): 58.14
Volume V (L): 7.00

t = 18.00 h

Biomass X (g/L): 65.99
Glucose S (g/L): 69.34
Lactic Acid P (g/L): 57.59
Maltose M (g/L): 54.42
Volume V (L): 5.48

t = 19.00 h

Biomass X (g/L): 81.43
Glucose S (g/L): 69.10
Lactic Acid P (g/L): 56.17
Maltose M (g/L): 54.73
Volume V (L): 6.15

t = 20.00 h

Biomass X (g/L): 93.80
Glucose S (g/L): 76.99
Lactic Acid P (g/L): 64.76
Maltose M (g/L): 53.25
Volume V (L): 7.05

t = 21.00 h

Biomass X (g/L): 93.47
Glucose S (g/L): 70.34
Lactic Acid P (g/L): 63.03
Maltose M (g/L): 50.33
Volume V (L): 6.67

t = 22.00 h

Biomass X (g/L): 86.31
Glucose S (g/L): 70.61
Lactic Acid P (g/L): 46.23
Maltose M (g/L): 47.41
Volume V (L): 6.75

t = 23.00 h

Biomass X (g/L): 81.10
Glucose S (g/L): 65.31
Lactic Acid P (g/L): 56.55
Maltose M (g/L): 52.46
Volume V (L): 6.08

t = 24.00 h

Biomass X (g/L): 80.20
Glucose S (g/L): 71.01
Lactic Acid P (g/L): 54.24
Maltose M (g/L): 48.80
Volume V (L): 6.56

t = 25.00 h

Biomass X (g/L): 74.42
Glucose S (g/L): 72.10
Lactic Acid P (g/L): 70.57
Maltose M (g/L): 52.50
Volume V (L): 6.34

t = 26.00 h

Biomass X (g/L): 81.10
Glucose S (g/L): 73.97
Lactic Acid P (g/L): 55.72
Maltose M (g/L): 51.71
Volume V (L): 6.28

t = 27.00 h

Biomass X (g/L): 92.84
Glucose S (g/L): 72.57
Lactic Acid P (g/L): 54.95
Maltose M (g/L): 51.32
Volume V (L): 6.55

t = 28.00 h

Biomass X (g/L): 102.01
Glucose S (g/L): 70.23
Lactic Acid P (g/L): 71.78
Maltose M (g/L): 48.29
Volume V (L): 6.79

t = 29.00 h

Biomass X (g/L): 77.45
Glucose S (g/L): 70.47
Lactic Acid P (g/L): 65.51
Maltose M (g/L): 52.75
Volume V (L): 6.95

t = 30.00 h

Biomass X (g/L): 66.67
Glucose S (g/L): 71.26
Lactic Acid P (g/L): 82.91
Maltose M (g/L): 51.14
Volume V (L): 6.99

t = 31.00 h

Biomass X (g/L): 80.87
Glucose S (g/L): 72.37
Lactic Acid P (g/L): 56.59
Maltose M (g/L): 47.02
Volume V (L): 7.96

t = 32.00 h

Biomass X (g/L): 94.75
Glucose S (g/L): 77.01
Lactic Acid P (g/L): 60.26
Maltose M (g/L): 54.15
Volume V (L): 7.12

t = 33.00 h

Biomass X (g/L): 106.05
Glucose S (g/L): 76.83
Lactic Acid P (g/L): 75.64
Maltose M (g/L): 48.60
Volume V (L): 8.32

t = 34.00 h

Biomass X (g/L): 99.04
Glucose S (g/L): 76.22
Lactic Acid P (g/L): 54.41
Maltose M (g/L): 45.71

Volume V (L): 7.48

t = 35.00 h

Biomass X (g/L): 98.97

Glucose S (g/L): 81.33

Lactic Acid P (g/L): 78.40

Maltose M (g/L): 46.72

Volume V (L): 8.00

t = 36.00 h

Biomass X (g/L): 78.88

Glucose S (g/L): 65.14

Lactic Acid P (g/L): 55.97

Maltose M (g/L): 44.69

Volume V (L): 8.10

t = 37.00 h

Biomass X (g/L): 79.25

Glucose S (g/L): 73.57

Lactic Acid P (g/L): 72.85

Maltose M (g/L): 50.96

Volume V (L): 7.52

t = 38.00 h

Biomass X (g/L): 74.57

Glucose S (g/L): 75.76

Lactic Acid P (g/L): 51.05

Maltose M (g/L): 50.02

Volume V (L): 7.87

t = 39.00 h

Biomass X (g/L): 80.85

Glucose S (g/L): 74.00

Lactic Acid P (g/L): 68.52

Maltose M (g/L): 48.94

Volume V (L): 7.87

t = 40.00 h

Biomass X (g/L): 95.85

Glucose S (g/L): 73.68

Lactic Acid P (g/L): 56.08

Maltose M (g/L): 48.21

Volume V (L): 8.37

t = 41.00 h

Biomass X (g/L): 102.58

Glucose S (g/L): 71.47

Lactic Acid P (g/L): 86.25

Maltose M (g/L): 49.51

Volume V (L): 8.21

t = 42.00 h

Biomass X (g/L): 90.46

Glucose S (g/L): 77.10

Lactic Acid P (g/L): 59.45

Maltose M (g/L): 45.80

Volume V (L): 7.99

t = 43.00 h

Biomass X (g/L): 84.74

Glucose S (g/L): 82.03

Lactic Acid P (g/L): 61.96

Maltose M (g/L): 47.04

Volume V (L): 8.49

t = 44.00 h

Biomass X (g/L): 77.01

Glucose S (g/L): 82.21

Lactic Acid P (g/L): 74.61

Maltose M (g/L): 43.92

Volume V (L): 8.67

t = 45.00 h

Biomass X (g/L): 86.00

Glucose S (g/L): 77.37

Lactic Acid P (g/L): 64.62

Maltose M (g/L): 45.14

Volume V (L): 9.48

t = 46.00 h

Biomass X (g/L): 88.92

Glucose S (g/L): 81.47

Lactic Acid P (g/L): 76.67

Maltose M (g/L): 46.25

Volume V (L): 8.70

t = 47.00 h

Biomass X (g/L): 73.05

Glucose S (g/L): 86.15

Lactic Acid P (g/L): 64.17

Maltose M (g/L): 37.87

Volume V (L): 8.75

t = 48.00 h

Biomass X (g/L): 95.55

Glucose S (g/L): 76.47

Lactic Acid P (g/L): 63.73

Maltose M (g/L): 43.24

Volume V (L): 9.03

t = 49.00 h

Biomass X (g/L): 82.70

Glucose S (g/L): 81.95

Lactic Acid P (g/L): 70.86

Maltose M (g/L): 44.76

Volume V (L): 8.45

t = 50.00 h

Biomass X (g/L): 94.77

Glucose S (g/L): 84.16

Lactic Acid P (g/L): 60.33

Maltose M (g/L): 43.93

Volume V (L): 8.16

t = 51.00 h

Biomass X (g/L): 86.38

Glucose S (g/L): 83.78

Lactic Acid P (g/L): 75.16

Maltose M (g/L): 42.80

Volume V (L): 8.92

t = 52.00 h

Biomass X (g/L): 89.17

Glucose S (g/L): 87.39
Lactic Acid P (g/L): 79.73
Maltose M (g/L): 44.92
Volume V (L): 9.66

t = 53.00 h

Biomass X (g/L): 79.57
Glucose S (g/L): 79.60
Lactic Acid P (g/L): 77.01
Maltose M (g/L): 41.69
Volume V (L): 9.82

t = 54.00 h

Biomass X (g/L): 82.82
Glucose S (g/L): 81.96
Lactic Acid P (g/L): 78.04
Maltose M (g/L): 40.25
Volume V (L): 8.41

t = 55.00 h

Biomass X (g/L): 92.02
Glucose S (g/L): 88.59
Lactic Acid P (g/L): 54.99
Maltose M (g/L): 45.50
Volume V (L): 9.71

t = 56.00 h

Biomass X (g/L): 84.53
Glucose S (g/L): 80.65
Lactic Acid P (g/L): 68.43
Maltose M (g/L): 44.92
Volume V (L): 8.42

t = 57.00 h

Biomass X (g/L): 75.15
Glucose S (g/L): 90.23
Lactic Acid P (g/L): 70.74
Maltose M (g/L): 40.90
Volume V (L): 9.77

t = 58.00 h

Biomass X (g/L): 86.52
Glucose S (g/L): 89.52
Lactic Acid P (g/L): 86.14
Maltose M (g/L): 41.86
Volume V (L): 9.82

t = 59.00 h

Biomass X (g/L): 88.94
Glucose S (g/L): 91.15
Lactic Acid P (g/L): 67.89
Maltose M (g/L): 43.71
Volume V (L): 8.88

t = 60.00 h

Biomass X (g/L): 98.12
Glucose S (g/L): 87.39
Lactic Acid P (g/L): 52.49
Maltose M (g/L): 41.62
Volume V (L): 10.27

t = 61.00 h

Biomass X (g/L): 95.86
Glucose S (g/L): 91.46
Lactic Acid P (g/L): 64.40
Maltose M (g/L): 42.49
Volume V (L): 10.70

t = 62.00 h

Biomass X (g/L): 84.15
Glucose S (g/L): 86.01
Lactic Acid P (g/L): 67.67
Maltose M (g/L): 42.12
Volume V (L): 9.92

t = 63.00 h

Biomass X (g/L): 79.40
Glucose S (g/L): 89.28
Lactic Acid P (g/L): 54.29
Maltose M (g/L): 40.06
Volume V (L): 10.09

t = 64.00 h

Biomass X (g/L): 82.24
Glucose S (g/L): 91.45
Lactic Acid P (g/L): 82.73
Maltose M (g/L): 41.73
Volume V (L): 10.81

t = 65.00 h

Biomass X (g/L): 79.25
Glucose S (g/L): 88.66
Lactic Acid P (g/L): 59.91
Maltose M (g/L): 38.10
Volume V (L): 9.65

t = 66.00 h

Biomass X (g/L): 88.82
Glucose S (g/L): 93.20
Lactic Acid P (g/L): 78.29
Maltose M (g/L): 40.98
Volume V (L): 9.74

t = 67.00 h

Biomass X (g/L): 93.70
Glucose S (g/L): 88.86
Lactic Acid P (g/L): 64.51
Maltose M (g/L): 39.68
Volume V (L): 9.94

t = 68.00 h

Biomass X (g/L): 95.39
Glucose S (g/L): 94.45
Lactic Acid P (g/L): 88.18
Maltose M (g/L): 38.79
Volume V (L): 10.44

t = 69.00 h

Biomass X (g/L): 79.54
Glucose S (g/L): 100.25
Lactic Acid P (g/L): 72.17
Maltose M (g/L): 36.53
Volume V (L): 10.52

t = 70.00 h

Biomass X (g/L): 78.40
Glucose S (g/L): 90.05
Lactic Acid P (g/L): 70.47
Maltose M (g/L): 39.46
Volume V (L): 10.32

t = 71.00 h

Biomass X (g/L): 79.07
Glucose S (g/L): 94.99
Lactic Acid P (g/L): 79.72
Maltose M (g/L): 39.02
Volume V (L): 10.52

t = 72.00 h

Biomass X (g/L): 64.56
Glucose S (g/L): 89.80
Lactic Acid P (g/L): 65.94
Maltose M (g/L): 36.79
Volume V (L): 10.49

t = 73.00 h

Biomass X (g/L): 71.26
Glucose S (g/L): 93.11
Lactic Acid P (g/L): 80.39
Maltose M (g/L): 34.83
Volume V (L): 10.59

t = 74.00 h

Biomass X (g/L): 82.13
Glucose S (g/L): 99.11
Lactic Acid P (g/L): 70.95
Maltose M (g/L): 34.50
Volume V (L): 10.54

t = 75.00 h

Biomass X (g/L): 85.85
Glucose S (g/L): 95.55
Lactic Acid P (g/L): 70.04
Maltose M (g/L): 37.47

Volume V (L): 10.62

Batch 94

Initial Conditions:

Batch: 94.00

X0 (g/L): 4.66

S0 (g/L): 51.82

M0 (g/L): 68.86

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 1.33

F_vol (L/h): 0.06

S_feed (g/L): 128.83

M_feed (g/L): 171.17

V_final (L): 9.73

Batch 94 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 50.56
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 66.68
Volume V (L): 4.54

t = 1.00 h

Biomass X (g/L): 4.06
Glucose S (g/L): 48.29
Lactic Acid P (g/L): 13.84
Maltose M (g/L): 68.98
Volume V (L): 5.19

t = 2.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 51.87
Lactic Acid P (g/L): 3.74
Maltose M (g/L): 65.24
Volume V (L): 4.99

t = 3.00 h

Biomass X (g/L): 17.78
Glucose S (g/L): 55.95
Lactic Acid P (g/L): 14.03
Maltose M (g/L): 69.75
Volume V (L): 5.67

t = 4.00 h

Biomass X (g/L): 8.16
Glucose S (g/L): 55.18
Lactic Acid P (g/L): 18.61
Maltose M (g/L): 68.75
Volume V (L): 4.91

t = 5.00 h

Biomass X (g/L): 15.79
Glucose S (g/L): 51.48

Lactic Acid P (g/L): 6.70
Maltose M (g/L): 69.08
Volume V (L): 6.16

t = 6.00 h

Biomass X (g/L): 17.21
Glucose S (g/L): 47.65
Lactic Acid P (g/L): 17.11
Maltose M (g/L): 67.83
Volume V (L): 5.70

t = 7.00 h

Biomass X (g/L): 22.63
Glucose S (g/L): 48.03
Lactic Acid P (g/L): 11.32
Maltose M (g/L): 68.64
Volume V (L): 5.93

t = 8.00 h

Biomass X (g/L): 41.45
Glucose S (g/L): 41.89
Lactic Acid P (g/L): 26.70
Maltose M (g/L): 68.87
Volume V (L): 5.12

t = 9.00 h

Biomass X (g/L): 20.85
Glucose S (g/L): 40.42
Lactic Acid P (g/L): 22.39
Maltose M (g/L): 68.27
Volume V (L): 6.24

t = 10.00 h

Biomass X (g/L): 22.63
Glucose S (g/L): 36.51
Lactic Acid P (g/L): 33.57
Maltose M (g/L): 65.52
Volume V (L): 5.17

t = 11.00 h

Biomass X (g/L): 43.55

Glucose S (g/L): 39.01
Lactic Acid P (g/L): 41.89
Maltose M (g/L): 67.91
Volume V (L): 5.61

t = 12.00 h

Biomass X (g/L): 41.65
Glucose S (g/L): 36.33
Lactic Acid P (g/L): 36.06
Maltose M (g/L): 70.56
Volume V (L): 6.25

t = 13.00 h

Biomass X (g/L): 53.56
Glucose S (g/L): 31.80
Lactic Acid P (g/L): 33.11
Maltose M (g/L): 67.67
Volume V (L): 6.01

t = 14.00 h

Biomass X (g/L): 60.40
Glucose S (g/L): 35.27
Lactic Acid P (g/L): 35.93
Maltose M (g/L): 67.32
Volume V (L): 5.41

t = 15.00 h

Biomass X (g/L): 46.06
Glucose S (g/L): 29.70
Lactic Acid P (g/L): 12.28
Maltose M (g/L): 69.52
Volume V (L): 6.38

t = 16.00 h

Biomass X (g/L): 65.27
Glucose S (g/L): 28.17
Lactic Acid P (g/L): 34.92
Maltose M (g/L): 65.45
Volume V (L): 6.14

t = 17.00 h

Biomass X (g/L): 52.27
Glucose S (g/L): 26.70
Lactic Acid P (g/L): 51.95
Maltose M (g/L): 67.99
Volume V (L): 5.91

t = 18.00 h

Biomass X (g/L): 65.09
Glucose S (g/L): 21.21
Lactic Acid P (g/L): 54.22
Maltose M (g/L): 65.40
Volume V (L): 5.55

t = 19.00 h

Biomass X (g/L): 54.61
Glucose S (g/L): 21.57
Lactic Acid P (g/L): 40.53
Maltose M (g/L): 67.90
Volume V (L): 6.36

t = 20.00 h

Biomass X (g/L): 69.61
Glucose S (g/L): 23.51
Lactic Acid P (g/L): 50.33
Maltose M (g/L): 67.20
Volume V (L): 6.20

t = 21.00 h

Biomass X (g/L): 69.84
Glucose S (g/L): 23.10
Lactic Acid P (g/L): 48.63
Maltose M (g/L): 64.95
Volume V (L): 6.38

t = 22.00 h

Biomass X (g/L): 61.01
Glucose S (g/L): 24.71
Lactic Acid P (g/L): 58.01
Maltose M (g/L): 64.97
Volume V (L): 5.79

t = 23.00 h

Biomass X (g/L): 59.94
Glucose S (g/L): 20.73
Lactic Acid P (g/L): 73.09
Maltose M (g/L): 66.83
Volume V (L): 6.29

t = 24.00 h

Biomass X (g/L): 81.40
Glucose S (g/L): 22.87
Lactic Acid P (g/L): 60.69
Maltose M (g/L): 61.84
Volume V (L): 6.70

t = 25.00 h

Biomass X (g/L): 70.99
Glucose S (g/L): 22.93
Lactic Acid P (g/L): 49.31
Maltose M (g/L): 63.98
Volume V (L): 7.29

t = 26.00 h

Biomass X (g/L): 86.05
Glucose S (g/L): 23.13
Lactic Acid P (g/L): 55.73
Maltose M (g/L): 65.53
Volume V (L): 6.56

t = 27.00 h

Biomass X (g/L): 72.07
Glucose S (g/L): 22.95
Lactic Acid P (g/L): 70.15
Maltose M (g/L): 63.73
Volume V (L): 7.25

t = 28.00 h

Biomass X (g/L): 90.59
Glucose S (g/L): 18.89
Lactic Acid P (g/L): 45.57
Maltose M (g/L): 64.23
Volume V (L): 7.06

t = 29.00 h

Biomass X (g/L): 69.78
Glucose S (g/L): 16.58
Lactic Acid P (g/L): 53.64
Maltose M (g/L): 60.32
Volume V (L): 6.92

t = 30.00 h

Biomass X (g/L): 93.66
Glucose S (g/L): 20.87
Lactic Acid P (g/L): 66.48
Maltose M (g/L): 62.07
Volume V (L): 6.51

t = 31.00 h

Biomass X (g/L): 81.84
Glucose S (g/L): 22.47
Lactic Acid P (g/L): 72.56
Maltose M (g/L): 64.40
Volume V (L): 6.59

t = 32.00 h

Biomass X (g/L): 84.55
Glucose S (g/L): 20.93
Lactic Acid P (g/L): 67.07
Maltose M (g/L): 61.75
Volume V (L): 7.25

t = 33.00 h

Biomass X (g/L): 71.61
Glucose S (g/L): 24.74
Lactic Acid P (g/L): 56.73
Maltose M (g/L): 60.57
Volume V (L): 6.88

t = 34.00 h

Biomass X (g/L): 88.66
Glucose S (g/L): 28.66
Lactic Acid P (g/L): 67.76
Maltose M (g/L): 64.28

Volume V (L): 6.91

t = 35.00 h

Biomass X (g/L): 89.40

Glucose S (g/L): 28.06

Lactic Acid P (g/L): 67.27

Maltose M (g/L): 63.11

Volume V (L): 7.14

t = 36.00 h

Biomass X (g/L): 79.34

Glucose S (g/L): 27.80

Lactic Acid P (g/L): 67.29

Maltose M (g/L): 60.16

Volume V (L): 7.21

t = 37.00 h

Biomass X (g/L): 62.33

Glucose S (g/L): 29.90

Lactic Acid P (g/L): 63.79

Maltose M (g/L): 61.96

Volume V (L): 7.67

t = 38.00 h

Biomass X (g/L): 72.39

Glucose S (g/L): 24.72

Lactic Acid P (g/L): 65.69

Maltose M (g/L): 59.70

Volume V (L): 7.21

t = 39.00 h

Biomass X (g/L): 72.65

Glucose S (g/L): 26.70

Lactic Acid P (g/L): 57.48

Maltose M (g/L): 60.96

Volume V (L): 7.76

t = 40.00 h

Biomass X (g/L): 87.84

Glucose S (g/L): 30.52

Lactic Acid P (g/L): 88.71

Maltose M (g/L): 61.25

Volume V (L): 7.69

t = 41.00 h

Biomass X (g/L): 79.28

Glucose S (g/L): 27.85

Lactic Acid P (g/L): 68.10

Maltose M (g/L): 59.34

Volume V (L): 6.95

t = 42.00 h

Biomass X (g/L): 79.63

Glucose S (g/L): 28.63

Lactic Acid P (g/L): 72.10

Maltose M (g/L): 61.68

Volume V (L): 7.76

t = 43.00 h

Biomass X (g/L): 78.62

Glucose S (g/L): 36.13

Lactic Acid P (g/L): 65.57

Maltose M (g/L): 61.68

Volume V (L): 6.70

t = 44.00 h

Biomass X (g/L): 98.24

Glucose S (g/L): 31.45

Lactic Acid P (g/L): 75.63

Maltose M (g/L): 60.35

Volume V (L): 7.44

t = 45.00 h

Biomass X (g/L): 84.70

Glucose S (g/L): 37.12

Lactic Acid P (g/L): 56.58

Maltose M (g/L): 60.35

Volume V (L): 8.01

t = 46.00 h

Biomass X (g/L): 86.24

Glucose S (g/L): 26.97

Lactic Acid P (g/L): 52.90

Maltose M (g/L): 56.95

Volume V (L): 7.90

t = 47.00 h

Biomass X (g/L): 78.47

Glucose S (g/L): 31.22

Lactic Acid P (g/L): 68.82

Maltose M (g/L): 61.42

Volume V (L): 7.30

t = 48.00 h

Biomass X (g/L): 84.17

Glucose S (g/L): 35.01

Lactic Acid P (g/L): 88.09

Maltose M (g/L): 58.71

Volume V (L): 7.60

t = 49.00 h

Biomass X (g/L): 75.13

Glucose S (g/L): 34.73

Lactic Acid P (g/L): 80.75

Maltose M (g/L): 57.03

Volume V (L): 7.10

t = 50.00 h

Biomass X (g/L): 87.78

Glucose S (g/L): 28.93

Lactic Acid P (g/L): 62.78

Maltose M (g/L): 58.59

Volume V (L): 7.65

t = 51.00 h

Biomass X (g/L): 73.51

Glucose S (g/L): 35.10

Lactic Acid P (g/L): 75.11

Maltose M (g/L): 59.34

Volume V (L): 8.44

t = 52.00 h

Biomass X (g/L): 91.60

Glucose S (g/L): 43.92
Lactic Acid P (g/L): 64.96
Maltose M (g/L): 58.68
Volume V (L): 8.07

t = 53.00 h

Biomass X (g/L): 77.41
Glucose S (g/L): 37.89
Lactic Acid P (g/L): 74.89
Maltose M (g/L): 55.74
Volume V (L): 8.41

t = 54.00 h

Biomass X (g/L): 93.50
Glucose S (g/L): 39.64
Lactic Acid P (g/L): 75.88
Maltose M (g/L): 58.58
Volume V (L): 8.41

t = 55.00 h

Biomass X (g/L): 102.63
Glucose S (g/L): 36.56
Lactic Acid P (g/L): 79.98
Maltose M (g/L): 56.94
Volume V (L): 8.48

t = 56.00 h

Biomass X (g/L): 84.31
Glucose S (g/L): 39.91
Lactic Acid P (g/L): 57.63
Maltose M (g/L): 57.63
Volume V (L): 8.92

t = 57.00 h

Biomass X (g/L): 96.80
Glucose S (g/L): 39.82
Lactic Acid P (g/L): 60.61
Maltose M (g/L): 55.18
Volume V (L): 8.52

t = 58.00 h

Biomass X (g/L): 91.36
Glucose S (g/L): 43.90
Lactic Acid P (g/L): 71.65
Maltose M (g/L): 58.03
Volume V (L): 9.00

t = 59.00 h

Biomass X (g/L): 70.35
Glucose S (g/L): 44.48
Lactic Acid P (g/L): 63.21
Maltose M (g/L): 56.33
Volume V (L): 7.78

t = 60.00 h

Biomass X (g/L): 85.04
Glucose S (g/L): 43.11
Lactic Acid P (g/L): 62.74
Maltose M (g/L): 57.76
Volume V (L): 9.21

t = 61.00 h

Biomass X (g/L): 88.68
Glucose S (g/L): 42.77
Lactic Acid P (g/L): 75.50
Maltose M (g/L): 52.93
Volume V (L): 8.36

t = 62.00 h

Biomass X (g/L): 89.50
Glucose S (g/L): 38.17
Lactic Acid P (g/L): 64.94
Maltose M (g/L): 52.90
Volume V (L): 8.50

t = 63.00 h

Biomass X (g/L): 85.37
Glucose S (g/L): 38.73
Lactic Acid P (g/L): 93.95
Maltose M (g/L): 54.60
Volume V (L): 8.71

t = 64.00 h

Biomass X (g/L): 85.12
Glucose S (g/L): 44.80
Lactic Acid P (g/L): 68.83
Maltose M (g/L): 57.22
Volume V (L): 9.45

t = 65.00 h

Biomass X (g/L): 83.47
Glucose S (g/L): 47.85
Lactic Acid P (g/L): 71.42
Maltose M (g/L): 57.80
Volume V (L): 8.76

t = 66.00 h

Biomass X (g/L): 84.56
Glucose S (g/L): 49.89
Lactic Acid P (g/L): 76.77
Maltose M (g/L): 52.76
Volume V (L): 8.92

t = 67.00 h

Biomass X (g/L): 71.43
Glucose S (g/L): 43.86
Lactic Acid P (g/L): 60.57
Maltose M (g/L): 54.09
Volume V (L): 8.86

t = 68.00 h

Biomass X (g/L): 80.75
Glucose S (g/L): 42.64
Lactic Acid P (g/L): 79.05
Maltose M (g/L): 54.02
Volume V (L): 9.39

t = 69.00 h

Biomass X (g/L): 68.95
Glucose S (g/L): 50.65
Lactic Acid P (g/L): 77.72
Maltose M (g/L): 54.72
Volume V (L): 9.63

t = 70.00 h

Biomass X (g/L): 77.96
Glucose S (g/L): 46.03
Lactic Acid P (g/L): 71.62
Maltose M (g/L): 54.75
Volume V (L): 10.06

t = 71.00 h

Biomass X (g/L): 76.91
Glucose S (g/L): 44.38
Lactic Acid P (g/L): 77.45
Maltose M (g/L): 51.15
Volume V (L): 10.39

t = 72.00 h

Biomass X (g/L): 76.15
Glucose S (g/L): 48.28
Lactic Acid P (g/L): 79.02
Maltose M (g/L): 52.89
Volume V (L): 9.79

t = 73.00 h

Biomass X (g/L): 72.18
Glucose S (g/L): 48.86
Lactic Acid P (g/L): 88.31
Maltose M (g/L): 49.16
Volume V (L): 9.74

t = 74.00 h

Biomass X (g/L): 84.68
Glucose S (g/L): 55.01
Lactic Acid P (g/L): 98.14
Maltose M (g/L): 53.66
Volume V (L): 9.21

t = 75.00 h

Biomass X (g/L): 78.89
Glucose S (g/L): 54.27
Lactic Acid P (g/L): 65.98
Maltose M (g/L): 51.49

Volume V (L): 10.45

Batch 95

Initial Conditions:

Batch: 95.00

X0 (g/L): 1.82

S0 (g/L): 18.76

M0 (g/L): 155.81

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 8.31

F_vol (L/h): 0.07

S_feed (g/L): 32.23

M_feed (g/L): 267.77

V_final (L): 10.00

Batch 95 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 3.89
Glucose S (g/L): 19.23
Lactic Acid P (g/L): 4.15
Maltose M (g/L): 163.40
Volume V (L): 5.08

t = 1.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 19.92
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 150.34
Volume V (L): 5.32

t = 2.00 h

Biomass X (g/L): 6.70
Glucose S (g/L): 18.43
Lactic Acid P (g/L): 4.13
Maltose M (g/L): 147.05
Volume V (L): 5.43

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 30.30
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 151.12
Volume V (L): 5.66

t = 4.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 30.39
Lactic Acid P (g/L): 9.66
Maltose M (g/L): 142.39
Volume V (L): 5.86

t = 5.00 h

Biomass X (g/L): 11.74
Glucose S (g/L): 28.99

Lactic Acid P (g/L): 0.00
Maltose M (g/L): 135.74
Volume V (L): 4.78

t = 6.00 h

Biomass X (g/L): 20.82
Glucose S (g/L): 27.82
Lactic Acid P (g/L): 14.49
Maltose M (g/L): 147.62
Volume V (L): 5.56

t = 7.00 h

Biomass X (g/L): 15.30
Glucose S (g/L): 31.51
Lactic Acid P (g/L): 23.60
Maltose M (g/L): 149.34
Volume V (L): 5.64

t = 8.00 h

Biomass X (g/L): 12.28
Glucose S (g/L): 32.62
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 152.58
Volume V (L): 5.50

t = 9.00 h

Biomass X (g/L): 15.27
Glucose S (g/L): 33.70
Lactic Acid P (g/L): 9.36
Maltose M (g/L): 147.19
Volume V (L): 5.27

t = 10.00 h

Biomass X (g/L): 19.94
Glucose S (g/L): 33.41
Lactic Acid P (g/L): 2.64
Maltose M (g/L): 155.03
Volume V (L): 6.24

t = 11.00 h

Biomass X (g/L): 23.07

Glucose S (g/L): 33.06
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 149.71
Volume V (L): 5.50

t = 12.00 h

Biomass X (g/L): 24.76
Glucose S (g/L): 23.43
Lactic Acid P (g/L): 16.37
Maltose M (g/L): 144.28
Volume V (L): 4.92

t = 13.00 h

Biomass X (g/L): 29.34
Glucose S (g/L): 26.38
Lactic Acid P (g/L): 25.74
Maltose M (g/L): 141.25
Volume V (L): 5.87

t = 14.00 h

Biomass X (g/L): 47.83
Glucose S (g/L): 28.83
Lactic Acid P (g/L): 29.62
Maltose M (g/L): 145.16
Volume V (L): 5.84

t = 15.00 h

Biomass X (g/L): 41.14
Glucose S (g/L): 21.55
Lactic Acid P (g/L): 27.07
Maltose M (g/L): 120.85
Volume V (L): 6.74

t = 16.00 h

Biomass X (g/L): 41.19
Glucose S (g/L): 26.60
Lactic Acid P (g/L): 13.61
Maltose M (g/L): 131.33
Volume V (L): 6.26

t = 17.00 h

Biomass X (g/L): 49.93
Glucose S (g/L): 27.06
Lactic Acid P (g/L): 32.11
Maltose M (g/L): 120.93
Volume V (L): 6.01

t = 18.00 h

Biomass X (g/L): 53.14
Glucose S (g/L): 24.01
Lactic Acid P (g/L): 20.21
Maltose M (g/L): 136.75
Volume V (L): 6.12

t = 19.00 h

Biomass X (g/L): 58.87
Glucose S (g/L): 21.39
Lactic Acid P (g/L): 23.12
Maltose M (g/L): 126.20
Volume V (L): 6.18

t = 20.00 h

Biomass X (g/L): 45.25
Glucose S (g/L): 17.19
Lactic Acid P (g/L): 29.12
Maltose M (g/L): 150.39
Volume V (L): 7.22

t = 21.00 h

Biomass X (g/L): 50.45
Glucose S (g/L): 18.69
Lactic Acid P (g/L): 49.35
Maltose M (g/L): 125.55
Volume V (L): 6.67

t = 22.00 h

Biomass X (g/L): 49.57
Glucose S (g/L): 13.23
Lactic Acid P (g/L): 46.75
Maltose M (g/L): 128.69
Volume V (L): 5.90

t = 23.00 h

Biomass X (g/L): 66.91
Glucose S (g/L): 14.53
Lactic Acid P (g/L): 46.54
Maltose M (g/L): 124.23
Volume V (L): 6.43

t = 24.00 h

Biomass X (g/L): 68.40
Glucose S (g/L): 14.46
Lactic Acid P (g/L): 37.65
Maltose M (g/L): 122.63
Volume V (L): 6.59

t = 25.00 h

Biomass X (g/L): 51.05
Glucose S (g/L): 16.63
Lactic Acid P (g/L): 43.41
Maltose M (g/L): 116.53
Volume V (L): 6.22

t = 26.00 h

Biomass X (g/L): 68.70
Glucose S (g/L): 15.20
Lactic Acid P (g/L): 36.12
Maltose M (g/L): 115.31
Volume V (L): 6.02

t = 27.00 h

Biomass X (g/L): 79.65
Glucose S (g/L): 17.20
Lactic Acid P (g/L): 61.78
Maltose M (g/L): 108.29
Volume V (L): 7.04

t = 28.00 h

Biomass X (g/L): 65.58
Glucose S (g/L): 19.17
Lactic Acid P (g/L): 70.45
Maltose M (g/L): 125.64
Volume V (L): 6.54

t = 29.00 h

Biomass X (g/L): 72.14
Glucose S (g/L): 14.56
Lactic Acid P (g/L): 57.06
Maltose M (g/L): 118.18
Volume V (L): 7.43

t = 30.00 h

Biomass X (g/L): 77.33
Glucose S (g/L): 17.40
Lactic Acid P (g/L): 59.74
Maltose M (g/L): 133.09
Volume V (L): 7.00

t = 31.00 h

Biomass X (g/L): 58.68
Glucose S (g/L): 16.00
Lactic Acid P (g/L): 52.88
Maltose M (g/L): 110.81
Volume V (L): 6.88

t = 32.00 h

Biomass X (g/L): 73.84
Glucose S (g/L): 18.19
Lactic Acid P (g/L): 40.90
Maltose M (g/L): 128.19
Volume V (L): 6.17

t = 33.00 h

Biomass X (g/L): 85.54
Glucose S (g/L): 20.08
Lactic Acid P (g/L): 48.75
Maltose M (g/L): 127.94
Volume V (L): 7.69

t = 34.00 h

Biomass X (g/L): 68.44
Glucose S (g/L): 21.24
Lactic Acid P (g/L): 53.01
Maltose M (g/L): 125.07

Volume V (L): 7.38

t = 35.00 h

Biomass X (g/L): 82.94

Glucose S (g/L): 20.95

Lactic Acid P (g/L): 54.58

Maltose M (g/L): 117.20

Volume V (L): 7.15

t = 36.00 h

Biomass X (g/L): 75.91

Glucose S (g/L): 20.69

Lactic Acid P (g/L): 53.35

Maltose M (g/L): 112.49

Volume V (L): 7.04

t = 37.00 h

Biomass X (g/L): 69.82

Glucose S (g/L): 15.69

Lactic Acid P (g/L): 60.02

Maltose M (g/L): 122.49

Volume V (L): 8.10

t = 38.00 h

Biomass X (g/L): 75.99

Glucose S (g/L): 17.27

Lactic Acid P (g/L): 52.77

Maltose M (g/L): 107.76

Volume V (L): 7.11

t = 39.00 h

Biomass X (g/L): 72.42

Glucose S (g/L): 18.64

Lactic Acid P (g/L): 80.30

Maltose M (g/L): 114.08

Volume V (L): 7.45

t = 40.00 h

Biomass X (g/L): 69.44

Glucose S (g/L): 22.82

Lactic Acid P (g/L): 49.56

Maltose M (g/L): 111.15
Volume V (L): 7.89

t = 41.00 h

Biomass X (g/L): 76.01
Glucose S (g/L): 18.46
Lactic Acid P (g/L): 70.22
Maltose M (g/L): 104.66
Volume V (L): 8.31

t = 42.00 h

Biomass X (g/L): 79.54
Glucose S (g/L): 24.02
Lactic Acid P (g/L): 60.62
Maltose M (g/L): 111.72
Volume V (L): 7.71

t = 43.00 h

Biomass X (g/L): 77.68
Glucose S (g/L): 24.46
Lactic Acid P (g/L): 67.64
Maltose M (g/L): 114.06
Volume V (L): 7.57

t = 44.00 h

Biomass X (g/L): 77.08
Glucose S (g/L): 18.48
Lactic Acid P (g/L): 68.60
Maltose M (g/L): 102.05
Volume V (L): 7.89

t = 45.00 h

Biomass X (g/L): 82.13
Glucose S (g/L): 21.73
Lactic Acid P (g/L): 66.02
Maltose M (g/L): 110.64
Volume V (L): 8.36

t = 46.00 h

Biomass X (g/L): 85.36
Glucose S (g/L): 28.86

Lactic Acid P (g/L): 55.06

Maltose M (g/L): 91.82

Volume V (L): 8.52

t = 47.00 h

Biomass X (g/L): 69.50

Glucose S (g/L): 28.52

Lactic Acid P (g/L): 62.67

Maltose M (g/L): 99.01

Volume V (L): 7.58

t = 48.00 h

Biomass X (g/L): 78.32

Glucose S (g/L): 24.55

Lactic Acid P (g/L): 67.60

Maltose M (g/L): 102.25

Volume V (L): 8.24

t = 49.00 h

Biomass X (g/L): 61.97

Glucose S (g/L): 24.71

Lactic Acid P (g/L): 67.49

Maltose M (g/L): 107.00

Volume V (L): 7.69

t = 50.00 h

Biomass X (g/L): 79.58

Glucose S (g/L): 21.15

Lactic Acid P (g/L): 57.89

Maltose M (g/L): 109.73

Volume V (L): 8.68

t = 51.00 h

Biomass X (g/L): 82.75

Glucose S (g/L): 33.72

Lactic Acid P (g/L): 78.88

Maltose M (g/L): 103.74

Volume V (L): 8.57

t = 52.00 h

Biomass X (g/L): 86.93

Glucose S (g/L): 30.13
Lactic Acid P (g/L): 45.63
Maltose M (g/L): 106.30
Volume V (L): 8.10

t = 53.00 h

Biomass X (g/L): 83.05
Glucose S (g/L): 36.43
Lactic Acid P (g/L): 65.97
Maltose M (g/L): 100.73
Volume V (L): 9.14

t = 54.00 h

Biomass X (g/L): 86.95
Glucose S (g/L): 30.89
Lactic Acid P (g/L): 60.37
Maltose M (g/L): 103.85
Volume V (L): 8.73

t = 55.00 h

Biomass X (g/L): 82.39
Glucose S (g/L): 33.06
Lactic Acid P (g/L): 70.54
Maltose M (g/L): 102.82
Volume V (L): 8.75

t = 56.00 h

Biomass X (g/L): 75.34
Glucose S (g/L): 35.64
Lactic Acid P (g/L): 78.31
Maltose M (g/L): 96.38
Volume V (L): 8.66

t = 57.00 h

Biomass X (g/L): 71.15
Glucose S (g/L): 38.63
Lactic Acid P (g/L): 57.09
Maltose M (g/L): 96.76
Volume V (L): 8.79

t = 58.00 h

Biomass X (g/L): 79.40
Glucose S (g/L): 35.93
Lactic Acid P (g/L): 90.85
Maltose M (g/L): 102.53
Volume V (L): 8.71

t = 59.00 h

Biomass X (g/L): 74.59
Glucose S (g/L): 35.16
Lactic Acid P (g/L): 72.03
Maltose M (g/L): 110.04
Volume V (L): 8.63

t = 60.00 h

Biomass X (g/L): 80.82
Glucose S (g/L): 35.98
Lactic Acid P (g/L): 73.93
Maltose M (g/L): 105.34
Volume V (L): 9.49

t = 61.00 h

Biomass X (g/L): 77.24
Glucose S (g/L): 33.47
Lactic Acid P (g/L): 58.03
Maltose M (g/L): 89.02
Volume V (L): 8.87

t = 62.00 h

Biomass X (g/L): 85.99
Glucose S (g/L): 36.20
Lactic Acid P (g/L): 70.69
Maltose M (g/L): 88.43
Volume V (L): 8.92

t = 63.00 h

Biomass X (g/L): 82.41
Glucose S (g/L): 38.24
Lactic Acid P (g/L): 80.02
Maltose M (g/L): 91.79
Volume V (L): 9.15

t = 64.00 h

Biomass X (g/L): 86.22
Glucose S (g/L): 42.99
Lactic Acid P (g/L): 73.51
Maltose M (g/L): 95.45
Volume V (L): 9.83

t = 65.00 h

Biomass X (g/L): 86.13
Glucose S (g/L): 38.24
Lactic Acid P (g/L): 73.15
Maltose M (g/L): 94.52
Volume V (L): 9.78

t = 66.00 h

Biomass X (g/L): 80.10
Glucose S (g/L): 43.47
Lactic Acid P (g/L): 71.69
Maltose M (g/L): 101.94
Volume V (L): 9.24

t = 67.00 h

Biomass X (g/L): 80.63
Glucose S (g/L): 40.73
Lactic Acid P (g/L): 60.14
Maltose M (g/L): 101.73
Volume V (L): 9.95

t = 68.00 h

Biomass X (g/L): 86.53
Glucose S (g/L): 45.45
Lactic Acid P (g/L): 60.32
Maltose M (g/L): 92.49
Volume V (L): 9.57

t = 69.00 h

Biomass X (g/L): 105.16
Glucose S (g/L): 44.45
Lactic Acid P (g/L): 83.25
Maltose M (g/L): 90.35
Volume V (L): 9.26

t = 70.00 h

Biomass X (g/L): 72.04
Glucose S (g/L): 44.96
Lactic Acid P (g/L): 80.91
Maltose M (g/L): 94.79
Volume V (L): 9.47

t = 71.00 h

Biomass X (g/L): 81.87
Glucose S (g/L): 46.78
Lactic Acid P (g/L): 82.44
Maltose M (g/L): 97.91
Volume V (L): 10.28

t = 72.00 h

Biomass X (g/L): 69.87
Glucose S (g/L): 47.04
Lactic Acid P (g/L): 66.18
Maltose M (g/L): 93.93
Volume V (L): 10.63

t = 73.00 h

Biomass X (g/L): 81.08
Glucose S (g/L): 48.43
Lactic Acid P (g/L): 72.29
Maltose M (g/L): 86.78
Volume V (L): 9.99

t = 74.00 h

Biomass X (g/L): 74.69
Glucose S (g/L): 52.35
Lactic Acid P (g/L): 84.84
Maltose M (g/L): 85.03
Volume V (L): 9.70

t = 75.00 h

Biomass X (g/L): 79.91
Glucose S (g/L): 51.48
Lactic Acid P (g/L): 63.16
Maltose M (g/L): 81.15

Volume V (L): 9.95

Batch 96

Initial Conditions:

Batch: 96.00

X0 (g/L): 7.70

S0 (g/L): 16.14

M0 (g/L): 108.93

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 6.75

F_vol (L/h): 0.05

S_feed (g/L): 38.71

M_feed (g/L): 261.29

V_final (L): 9.11

Batch 96 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 0.90
Glucose S (g/L): 17.49
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 112.18
Volume V (L): 4.97

t = 1.00 h

Biomass X (g/L): 4.19
Glucose S (g/L): 16.77
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 106.31
Volume V (L): 5.28

t = 2.00 h

Biomass X (g/L): 5.67
Glucose S (g/L): 16.13
Lactic Acid P (g/L): 14.19
Maltose M (g/L): 115.38
Volume V (L): 4.99

t = 3.00 h

Biomass X (g/L): 14.07
Glucose S (g/L): 16.86
Lactic Acid P (g/L): 9.72
Maltose M (g/L): 108.69
Volume V (L): 5.04

t = 4.00 h

Biomass X (g/L): 18.91
Glucose S (g/L): 10.68
Lactic Acid P (g/L): 0.82
Maltose M (g/L): 109.87
Volume V (L): 6.03

t = 5.00 h

Biomass X (g/L): 19.79
Glucose S (g/L): 13.42

Lactic Acid P (g/L): 12.21
Maltose M (g/L): 104.70
Volume V (L): 4.57

t = 6.00 h

Biomass X (g/L): 22.55
Glucose S (g/L): 9.55
Lactic Acid P (g/L): 21.47
Maltose M (g/L): 109.40
Volume V (L): 4.91

t = 7.00 h

Biomass X (g/L): 34.74
Glucose S (g/L): 10.92
Lactic Acid P (g/L): 16.20
Maltose M (g/L): 106.82
Volume V (L): 5.02

t = 8.00 h

Biomass X (g/L): 29.25
Glucose S (g/L): 7.81
Lactic Acid P (g/L): 19.85
Maltose M (g/L): 103.99
Volume V (L): 6.03

t = 9.00 h

Biomass X (g/L): 32.98
Glucose S (g/L): 6.70
Lactic Acid P (g/L): 29.25
Maltose M (g/L): 101.21
Volume V (L): 5.28

t = 10.00 h

Biomass X (g/L): 35.80
Glucose S (g/L): 4.07
Lactic Acid P (g/L): 35.96
Maltose M (g/L): 103.51
Volume V (L): 5.63

t = 11.00 h

Biomass X (g/L): 53.82

Glucose S (g/L): 8.79
Lactic Acid P (g/L): 23.51
Maltose M (g/L): 102.76
Volume V (L): 5.45

t = 12.00 h

Biomass X (g/L): 38.09
Glucose S (g/L): 4.01
Lactic Acid P (g/L): 14.74
Maltose M (g/L): 106.21
Volume V (L): 5.86

t = 13.00 h

Biomass X (g/L): 49.98
Glucose S (g/L): 3.64
Lactic Acid P (g/L): 16.83
Maltose M (g/L): 102.10
Volume V (L): 5.71

t = 14.00 h

Biomass X (g/L): 39.99
Glucose S (g/L): 8.22
Lactic Acid P (g/L): 23.86
Maltose M (g/L): 96.06
Volume V (L): 5.85

t = 15.00 h

Biomass X (g/L): 51.63
Glucose S (g/L): 5.79
Lactic Acid P (g/L): 23.27
Maltose M (g/L): 104.04
Volume V (L): 6.25

t = 16.00 h

Biomass X (g/L): 55.09
Glucose S (g/L): 0.00
Lactic Acid P (g/L): 18.17
Maltose M (g/L): 105.69
Volume V (L): 6.02

t = 17.00 h

Biomass X (g/L): 61.50
Glucose S (g/L): 7.45
Lactic Acid P (g/L): 50.58
Maltose M (g/L): 95.46
Volume V (L): 5.69

t = 18.00 h

Biomass X (g/L): 49.92
Glucose S (g/L): 8.04
Lactic Acid P (g/L): 40.33
Maltose M (g/L): 106.41
Volume V (L): 6.09

t = 19.00 h

Biomass X (g/L): 44.86
Glucose S (g/L): 7.85
Lactic Acid P (g/L): 32.37
Maltose M (g/L): 104.59
Volume V (L): 6.84

t = 20.00 h

Biomass X (g/L): 49.19
Glucose S (g/L): 5.88
Lactic Acid P (g/L): 37.94
Maltose M (g/L): 97.43
Volume V (L): 6.28

t = 21.00 h

Biomass X (g/L): 66.12
Glucose S (g/L): 1.91
Lactic Acid P (g/L): 23.75
Maltose M (g/L): 95.35
Volume V (L): 6.20

t = 22.00 h

Biomass X (g/L): 77.36
Glucose S (g/L): 4.43
Lactic Acid P (g/L): 50.80
Maltose M (g/L): 102.02
Volume V (L): 5.76

t = 23.00 h

Biomass X (g/L): 67.46
Glucose S (g/L): 1.35
Lactic Acid P (g/L): 24.65
Maltose M (g/L): 93.57
Volume V (L): 6.50

t = 24.00 h

Biomass X (g/L): 49.94
Glucose S (g/L): 2.82
Lactic Acid P (g/L): 47.39
Maltose M (g/L): 95.40
Volume V (L): 6.60

t = 25.00 h

Biomass X (g/L): 69.49
Glucose S (g/L): 2.42
Lactic Acid P (g/L): 27.85
Maltose M (g/L): 92.80
Volume V (L): 6.47

t = 26.00 h

Biomass X (g/L): 55.03
Glucose S (g/L): 3.69
Lactic Acid P (g/L): 41.43
Maltose M (g/L): 96.67
Volume V (L): 6.46

t = 27.00 h

Biomass X (g/L): 61.40
Glucose S (g/L): 5.68
Lactic Acid P (g/L): 57.04
Maltose M (g/L): 101.01
Volume V (L): 7.41

t = 28.00 h

Biomass X (g/L): 67.20
Glucose S (g/L): 4.49
Lactic Acid P (g/L): 32.78
Maltose M (g/L): 96.95
Volume V (L): 6.82

t = 29.00 h

Biomass X (g/L): 69.69
Glucose S (g/L): 4.60
Lactic Acid P (g/L): 46.83
Maltose M (g/L): 98.59
Volume V (L): 7.00

t = 30.00 h

Biomass X (g/L): 68.05
Glucose S (g/L): 5.28
Lactic Acid P (g/L): 54.62
Maltose M (g/L): 93.61
Volume V (L): 6.67

t = 31.00 h

Biomass X (g/L): 63.01
Glucose S (g/L): 9.61
Lactic Acid P (g/L): 54.13
Maltose M (g/L): 94.63
Volume V (L): 6.66

t = 32.00 h

Biomass X (g/L): 78.71
Glucose S (g/L): 11.38
Lactic Acid P (g/L): 51.59
Maltose M (g/L): 89.80
Volume V (L): 6.96

t = 33.00 h

Biomass X (g/L): 65.38
Glucose S (g/L): 5.67
Lactic Acid P (g/L): 56.23
Maltose M (g/L): 93.06
Volume V (L): 6.77

t = 34.00 h

Biomass X (g/L): 82.09
Glucose S (g/L): 7.75
Lactic Acid P (g/L): 47.39
Maltose M (g/L): 90.46

Volume V (L): 6.85

t = 35.00 h

Biomass X (g/L): 59.54

Glucose S (g/L): 5.35

Lactic Acid P (g/L): 70.59

Maltose M (g/L): 91.93

Volume V (L): 7.07

t = 36.00 h

Biomass X (g/L): 74.32

Glucose S (g/L): 9.33

Lactic Acid P (g/L): 40.29

Maltose M (g/L): 91.25

Volume V (L): 6.80

t = 37.00 h

Biomass X (g/L): 62.16

Glucose S (g/L): 6.81

Lactic Acid P (g/L): 65.46

Maltose M (g/L): 90.51

Volume V (L): 7.10

t = 38.00 h

Biomass X (g/L): 66.33

Glucose S (g/L): 6.61

Lactic Acid P (g/L): 57.70

Maltose M (g/L): 98.03

Volume V (L): 7.11

t = 39.00 h

Biomass X (g/L): 69.83

Glucose S (g/L): 13.16

Lactic Acid P (g/L): 62.22

Maltose M (g/L): 96.46

Volume V (L): 7.46

t = 40.00 h

Biomass X (g/L): 88.57

Glucose S (g/L): 5.84

Lactic Acid P (g/L): 64.52

Maltose M (g/L): 94.85

Volume V (L): 6.99

t = 41.00 h

Biomass X (g/L): 87.25

Glucose S (g/L): 3.71

Lactic Acid P (g/L): 63.14

Maltose M (g/L): 95.66

Volume V (L): 6.59

t = 42.00 h

Biomass X (g/L): 83.92

Glucose S (g/L): 11.38

Lactic Acid P (g/L): 51.83

Maltose M (g/L): 93.97

Volume V (L): 7.20

t = 43.00 h

Biomass X (g/L): 76.93

Glucose S (g/L): 9.26

Lactic Acid P (g/L): 57.12

Maltose M (g/L): 88.78

Volume V (L): 7.49

t = 44.00 h

Biomass X (g/L): 78.79

Glucose S (g/L): 9.03

Lactic Acid P (g/L): 47.54

Maltose M (g/L): 88.00

Volume V (L): 7.73

t = 45.00 h

Biomass X (g/L): 88.87

Glucose S (g/L): 17.18

Lactic Acid P (g/L): 58.31

Maltose M (g/L): 88.75

Volume V (L): 7.24

t = 46.00 h

Biomass X (g/L): 84.26

Glucose S (g/L): 13.04

Lactic Acid P (g/L): 64.39

Maltose M (g/L): 86.85

Volume V (L): 7.70

t = 47.00 h

Biomass X (g/L): 78.56

Glucose S (g/L): 11.68

Lactic Acid P (g/L): 67.49

Maltose M (g/L): 83.68

Volume V (L): 7.95

t = 48.00 h

Biomass X (g/L): 92.04

Glucose S (g/L): 10.66

Lactic Acid P (g/L): 66.18

Maltose M (g/L): 87.66

Volume V (L): 7.88

t = 49.00 h

Biomass X (g/L): 86.39

Glucose S (g/L): 13.76

Lactic Acid P (g/L): 47.88

Maltose M (g/L): 81.09

Volume V (L): 7.43

t = 50.00 h

Biomass X (g/L): 81.61

Glucose S (g/L): 9.26

Lactic Acid P (g/L): 58.91

Maltose M (g/L): 86.28

Volume V (L): 7.52

t = 51.00 h

Biomass X (g/L): 75.66

Glucose S (g/L): 12.53

Lactic Acid P (g/L): 58.56

Maltose M (g/L): 82.12

Volume V (L): 7.94

t = 52.00 h

Biomass X (g/L): 75.13

Glucose S (g/L): 14.12
Lactic Acid P (g/L): 53.38
Maltose M (g/L): 84.71
Volume V (L): 8.39

t = 53.00 h

Biomass X (g/L): 77.29
Glucose S (g/L): 15.20
Lactic Acid P (g/L): 58.05
Maltose M (g/L): 85.19
Volume V (L): 7.91

t = 54.00 h

Biomass X (g/L): 84.60
Glucose S (g/L): 14.90
Lactic Acid P (g/L): 65.89
Maltose M (g/L): 82.98
Volume V (L): 7.91

t = 55.00 h

Biomass X (g/L): 88.19
Glucose S (g/L): 14.17
Lactic Acid P (g/L): 78.77
Maltose M (g/L): 78.33
Volume V (L): 7.56

t = 56.00 h

Biomass X (g/L): 68.05
Glucose S (g/L): 18.38
Lactic Acid P (g/L): 58.42
Maltose M (g/L): 79.41
Volume V (L): 7.95

t = 57.00 h

Biomass X (g/L): 63.75
Glucose S (g/L): 13.04
Lactic Acid P (g/L): 57.31
Maltose M (g/L): 84.42
Volume V (L): 7.89

t = 58.00 h

Biomass X (g/L): 72.78
Glucose S (g/L): 16.72
Lactic Acid P (g/L): 62.39
Maltose M (g/L): 80.67
Volume V (L): 8.19

t = 59.00 h

Biomass X (g/L): 77.78
Glucose S (g/L): 16.72
Lactic Acid P (g/L): 71.04
Maltose M (g/L): 83.95
Volume V (L): 7.74

t = 60.00 h

Biomass X (g/L): 92.21
Glucose S (g/L): 20.77
Lactic Acid P (g/L): 79.22
Maltose M (g/L): 81.73
Volume V (L): 8.18

t = 61.00 h

Biomass X (g/L): 79.15
Glucose S (g/L): 17.97
Lactic Acid P (g/L): 68.10
Maltose M (g/L): 80.65
Volume V (L): 8.01

t = 62.00 h

Biomass X (g/L): 84.02
Glucose S (g/L): 20.58
Lactic Acid P (g/L): 56.53
Maltose M (g/L): 82.05
Volume V (L): 8.71

t = 63.00 h

Biomass X (g/L): 85.56
Glucose S (g/L): 20.91
Lactic Acid P (g/L): 58.76
Maltose M (g/L): 78.29
Volume V (L): 8.17

t = 64.00 h

Biomass X (g/L): 72.76
Glucose S (g/L): 16.52
Lactic Acid P (g/L): 71.60
Maltose M (g/L): 81.44
Volume V (L): 8.32

t = 65.00 h

Biomass X (g/L): 94.97
Glucose S (g/L): 21.14
Lactic Acid P (g/L): 79.04
Maltose M (g/L): 78.35
Volume V (L): 8.66

t = 66.00 h

Biomass X (g/L): 76.06
Glucose S (g/L): 17.88
Lactic Acid P (g/L): 86.58
Maltose M (g/L): 87.70
Volume V (L): 8.47

t = 67.00 h

Biomass X (g/L): 95.93
Glucose S (g/L): 23.25
Lactic Acid P (g/L): 60.56
Maltose M (g/L): 75.23
Volume V (L): 9.45

t = 68.00 h

Biomass X (g/L): 80.91
Glucose S (g/L): 21.36
Lactic Acid P (g/L): 72.49
Maltose M (g/L): 77.41
Volume V (L): 9.03

t = 69.00 h

Biomass X (g/L): 79.03
Glucose S (g/L): 26.32
Lactic Acid P (g/L): 62.52
Maltose M (g/L): 77.17
Volume V (L): 8.79

t = 70.00 h

Biomass X (g/L): 72.13
Glucose S (g/L): 23.78
Lactic Acid P (g/L): 63.97
Maltose M (g/L): 73.38
Volume V (L): 8.96

t = 71.00 h

Biomass X (g/L): 83.79
Glucose S (g/L): 23.02
Lactic Acid P (g/L): 63.51
Maltose M (g/L): 79.85
Volume V (L): 8.12

t = 72.00 h

Biomass X (g/L): 102.12
Glucose S (g/L): 28.11
Lactic Acid P (g/L): 69.50
Maltose M (g/L): 75.68
Volume V (L): 9.35

t = 73.00 h

Biomass X (g/L): 85.66
Glucose S (g/L): 23.96
Lactic Acid P (g/L): 64.95
Maltose M (g/L): 68.89
Volume V (L): 9.17

t = 74.00 h

Biomass X (g/L): 84.35
Glucose S (g/L): 28.59
Lactic Acid P (g/L): 94.75
Maltose M (g/L): 78.65
Volume V (L): 9.03

t = 75.00 h

Biomass X (g/L): 75.94
Glucose S (g/L): 28.20
Lactic Acid P (g/L): 69.19
Maltose M (g/L): 73.90

Volume V (L): 9.30

Batch 97

Initial Conditions:

Batch: 97.00

X0 (g/L): 6.91

S0 (g/L): 43.13

M0 (g/L): 100.89

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 2.34

F_vol (L/h): 0.09

S_feed (g/L): 89.84

M_feed (g/L): 210.16

V_final (L): 11.66

Batch 97 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 13.22
Glucose S (g/L): 43.66
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 101.91
Volume V (L): 4.84

t = 1.00 h

Biomass X (g/L): 5.25
Glucose S (g/L): 42.18
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 99.28
Volume V (L): 5.69

t = 2.00 h

Biomass X (g/L): 15.22
Glucose S (g/L): 41.92
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 104.18
Volume V (L): 5.91

t = 3.00 h

Biomass X (g/L): 24.52
Glucose S (g/L): 46.61
Lactic Acid P (g/L): 14.03
Maltose M (g/L): 97.39
Volume V (L): 4.61

t = 4.00 h

Biomass X (g/L): 22.71
Glucose S (g/L): 36.00
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 100.18
Volume V (L): 5.24

t = 5.00 h

Biomass X (g/L): 20.26
Glucose S (g/L): 36.13

Lactic Acid P (g/L): 17.91
Maltose M (g/L): 99.50
Volume V (L): 5.21

t = 6.00 h

Biomass X (g/L): 29.60
Glucose S (g/L): 40.68
Lactic Acid P (g/L): 11.05
Maltose M (g/L): 100.89
Volume V (L): 5.70

t = 7.00 h

Biomass X (g/L): 44.00
Glucose S (g/L): 35.71
Lactic Acid P (g/L): 18.15
Maltose M (g/L): 102.61
Volume V (L): 4.41

t = 8.00 h

Biomass X (g/L): 42.94
Glucose S (g/L): 37.48
Lactic Acid P (g/L): 15.59
Maltose M (g/L): 100.58
Volume V (L): 4.93

t = 9.00 h

Biomass X (g/L): 29.37
Glucose S (g/L): 32.93
Lactic Acid P (g/L): 14.89
Maltose M (g/L): 99.25
Volume V (L): 5.74

t = 10.00 h

Biomass X (g/L): 49.51
Glucose S (g/L): 24.55
Lactic Acid P (g/L): 40.71
Maltose M (g/L): 101.27
Volume V (L): 6.31

t = 11.00 h

Biomass X (g/L): 52.11

Glucose S (g/L): 27.96
Lactic Acid P (g/L): 38.20
Maltose M (g/L): 101.42
Volume V (L): 6.02

t = 12.00 h

Biomass X (g/L): 62.14
Glucose S (g/L): 28.13
Lactic Acid P (g/L): 39.44
Maltose M (g/L): 99.55
Volume V (L): 6.42

t = 13.00 h

Biomass X (g/L): 54.97
Glucose S (g/L): 25.64
Lactic Acid P (g/L): 44.32
Maltose M (g/L): 99.58
Volume V (L): 6.26

t = 14.00 h

Biomass X (g/L): 54.19
Glucose S (g/L): 21.88
Lactic Acid P (g/L): 43.09
Maltose M (g/L): 99.99
Volume V (L): 6.92

t = 15.00 h

Biomass X (g/L): 47.66
Glucose S (g/L): 21.71
Lactic Acid P (g/L): 49.15
Maltose M (g/L): 93.35
Volume V (L): 7.12

t = 16.00 h

Biomass X (g/L): 56.21
Glucose S (g/L): 19.94
Lactic Acid P (g/L): 36.93
Maltose M (g/L): 96.58
Volume V (L): 6.02

t = 17.00 h

Biomass X (g/L): 75.80
Glucose S (g/L): 16.56
Lactic Acid P (g/L): 56.78
Maltose M (g/L): 98.25
Volume V (L): 6.98

t = 18.00 h

Biomass X (g/L): 68.46
Glucose S (g/L): 15.69
Lactic Acid P (g/L): 57.67
Maltose M (g/L): 96.05
Volume V (L): 6.87

t = 19.00 h

Biomass X (g/L): 67.46
Glucose S (g/L): 22.45
Lactic Acid P (g/L): 51.20
Maltose M (g/L): 95.70
Volume V (L): 8.02

t = 20.00 h

Biomass X (g/L): 81.83
Glucose S (g/L): 17.63
Lactic Acid P (g/L): 46.73
Maltose M (g/L): 97.16
Volume V (L): 6.35

t = 21.00 h

Biomass X (g/L): 81.36
Glucose S (g/L): 23.11
Lactic Acid P (g/L): 39.79
Maltose M (g/L): 96.85
Volume V (L): 6.80

t = 22.00 h

Biomass X (g/L): 65.88
Glucose S (g/L): 18.84
Lactic Acid P (g/L): 53.85
Maltose M (g/L): 96.66
Volume V (L): 6.40

t = 23.00 h

Biomass X (g/L): 84.81
Glucose S (g/L): 16.37
Lactic Acid P (g/L): 67.86
Maltose M (g/L): 92.79
Volume V (L): 6.52

t = 24.00 h

Biomass X (g/L): 59.62
Glucose S (g/L): 20.07
Lactic Acid P (g/L): 43.80
Maltose M (g/L): 90.67
Volume V (L): 7.34

t = 25.00 h

Biomass X (g/L): 80.00
Glucose S (g/L): 29.83
Lactic Acid P (g/L): 64.05
Maltose M (g/L): 96.41
Volume V (L): 7.33

t = 26.00 h

Biomass X (g/L): 79.77
Glucose S (g/L): 22.02
Lactic Acid P (g/L): 51.84
Maltose M (g/L): 91.73
Volume V (L): 7.60

t = 27.00 h

Biomass X (g/L): 87.97
Glucose S (g/L): 25.28
Lactic Acid P (g/L): 53.47
Maltose M (g/L): 91.13
Volume V (L): 7.68

t = 28.00 h

Biomass X (g/L): 81.62
Glucose S (g/L): 24.01
Lactic Acid P (g/L): 47.34
Maltose M (g/L): 90.47
Volume V (L): 7.36

t = 29.00 h

Biomass X (g/L): 76.51
Glucose S (g/L): 18.65
Lactic Acid P (g/L): 51.86
Maltose M (g/L): 95.72
Volume V (L): 7.45

t = 30.00 h

Biomass X (g/L): 77.82
Glucose S (g/L): 29.96
Lactic Acid P (g/L): 64.96
Maltose M (g/L): 92.70
Volume V (L): 7.15

t = 31.00 h

Biomass X (g/L): 104.57
Glucose S (g/L): 24.21
Lactic Acid P (g/L): 66.75
Maltose M (g/L): 92.64
Volume V (L): 7.38

t = 32.00 h

Biomass X (g/L): 81.08
Glucose S (g/L): 20.87
Lactic Acid P (g/L): 62.69
Maltose M (g/L): 93.17
Volume V (L): 7.81

t = 33.00 h

Biomass X (g/L): 86.22
Glucose S (g/L): 24.55
Lactic Acid P (g/L): 56.81
Maltose M (g/L): 90.73
Volume V (L): 8.04

t = 34.00 h

Biomass X (g/L): 62.58
Glucose S (g/L): 26.39
Lactic Acid P (g/L): 52.09
Maltose M (g/L): 93.21

Volume V (L): 8.40

t = 35.00 h

Biomass X (g/L): 73.87

Glucose S (g/L): 30.69

Lactic Acid P (g/L): 62.95

Maltose M (g/L): 86.68

Volume V (L): 8.43

t = 36.00 h

Biomass X (g/L): 86.53

Glucose S (g/L): 26.05

Lactic Acid P (g/L): 59.72

Maltose M (g/L): 86.55

Volume V (L): 9.03

t = 37.00 h

Biomass X (g/L): 67.22

Glucose S (g/L): 39.19

Lactic Acid P (g/L): 65.33

Maltose M (g/L): 85.97

Volume V (L): 7.77

t = 38.00 h

Biomass X (g/L): 87.32

Glucose S (g/L): 26.59

Lactic Acid P (g/L): 61.11

Maltose M (g/L): 89.85

Volume V (L): 8.25

t = 39.00 h

Biomass X (g/L): 75.83

Glucose S (g/L): 33.17

Lactic Acid P (g/L): 61.81

Maltose M (g/L): 89.65

Volume V (L): 7.99

t = 40.00 h

Biomass X (g/L): 73.18

Glucose S (g/L): 24.23

Lactic Acid P (g/L): 66.27

Maltose M (g/L): 86.51

Volume V (L): 9.23

t = 41.00 h

Biomass X (g/L): 76.14

Glucose S (g/L): 31.96

Lactic Acid P (g/L): 68.41

Maltose M (g/L): 86.74

Volume V (L): 8.37

t = 42.00 h

Biomass X (g/L): 96.97

Glucose S (g/L): 34.59

Lactic Acid P (g/L): 76.52

Maltose M (g/L): 90.44

Volume V (L): 9.35

t = 43.00 h

Biomass X (g/L): 86.13

Glucose S (g/L): 29.80

Lactic Acid P (g/L): 63.13

Maltose M (g/L): 84.59

Volume V (L): 9.59

t = 44.00 h

Biomass X (g/L): 74.81

Glucose S (g/L): 40.43

Lactic Acid P (g/L): 71.30

Maltose M (g/L): 86.89

Volume V (L): 9.13

t = 45.00 h

Biomass X (g/L): 90.35

Glucose S (g/L): 37.23

Lactic Acid P (g/L): 82.78

Maltose M (g/L): 83.75

Volume V (L): 9.15

t = 46.00 h

Biomass X (g/L): 90.90

Glucose S (g/L): 35.21

Lactic Acid P (g/L): 72.25

Maltose M (g/L): 88.23

Volume V (L): 9.11

t = 47.00 h

Biomass X (g/L): 79.83

Glucose S (g/L): 38.79

Lactic Acid P (g/L): 77.81

Maltose M (g/L): 86.51

Volume V (L): 9.01

t = 48.00 h

Biomass X (g/L): 92.16

Glucose S (g/L): 35.78

Lactic Acid P (g/L): 62.37

Maltose M (g/L): 80.48

Volume V (L): 9.03

t = 49.00 h

Biomass X (g/L): 92.61

Glucose S (g/L): 43.50

Lactic Acid P (g/L): 61.80

Maltose M (g/L): 83.19

Volume V (L): 9.27

t = 50.00 h

Biomass X (g/L): 91.18

Glucose S (g/L): 43.42

Lactic Acid P (g/L): 67.06

Maltose M (g/L): 77.74

Volume V (L): 10.41

t = 51.00 h

Biomass X (g/L): 77.17

Glucose S (g/L): 41.81

Lactic Acid P (g/L): 68.29

Maltose M (g/L): 85.53

Volume V (L): 8.98

t = 52.00 h

Biomass X (g/L): 72.51

Glucose S (g/L): 43.50
Lactic Acid P (g/L): 78.49
Maltose M (g/L): 88.27
Volume V (L): 9.73

t = 53.00 h

Biomass X (g/L): 67.76
Glucose S (g/L): 36.46
Lactic Acid P (g/L): 62.86
Maltose M (g/L): 76.18
Volume V (L): 10.09

t = 54.00 h

Biomass X (g/L): 77.23
Glucose S (g/L): 48.31
Lactic Acid P (g/L): 68.78
Maltose M (g/L): 77.83
Volume V (L): 9.65

t = 55.00 h

Biomass X (g/L): 79.82
Glucose S (g/L): 43.22
Lactic Acid P (g/L): 79.31
Maltose M (g/L): 81.81
Volume V (L): 10.11

t = 56.00 h

Biomass X (g/L): 80.50
Glucose S (g/L): 48.23
Lactic Acid P (g/L): 71.97
Maltose M (g/L): 80.87
Volume V (L): 9.17

t = 57.00 h

Biomass X (g/L): 82.62
Glucose S (g/L): 47.98
Lactic Acid P (g/L): 70.20
Maltose M (g/L): 79.37
Volume V (L): 10.77

t = 58.00 h

Biomass X (g/L): 98.88
Glucose S (g/L): 49.73
Lactic Acid P (g/L): 72.63
Maltose M (g/L): 75.29
Volume V (L): 10.76

t = 59.00 h

Biomass X (g/L): 79.78
Glucose S (g/L): 42.78
Lactic Acid P (g/L): 60.79
Maltose M (g/L): 80.56
Volume V (L): 10.19

t = 60.00 h

Biomass X (g/L): 82.35
Glucose S (g/L): 44.03
Lactic Acid P (g/L): 60.87
Maltose M (g/L): 81.70
Volume V (L): 9.86

t = 61.00 h

Biomass X (g/L): 66.02
Glucose S (g/L): 56.70
Lactic Acid P (g/L): 57.15
Maltose M (g/L): 86.70
Volume V (L): 10.84

t = 62.00 h

Biomass X (g/L): 103.04
Glucose S (g/L): 43.31
Lactic Acid P (g/L): 73.21
Maltose M (g/L): 82.04
Volume V (L): 9.90

t = 63.00 h

Biomass X (g/L): 70.41
Glucose S (g/L): 45.66
Lactic Acid P (g/L): 79.69
Maltose M (g/L): 74.44
Volume V (L): 10.06

t = 64.00 h

Biomass X (g/L): 75.89
Glucose S (g/L): 50.69
Lactic Acid P (g/L): 78.53
Maltose M (g/L): 77.58
Volume V (L): 10.66

t = 65.00 h

Biomass X (g/L): 88.64
Glucose S (g/L): 51.45
Lactic Acid P (g/L): 82.06
Maltose M (g/L): 78.84
Volume V (L): 11.58

t = 66.00 h

Biomass X (g/L): 84.39
Glucose S (g/L): 49.99
Lactic Acid P (g/L): 84.99
Maltose M (g/L): 74.91
Volume V (L): 11.20

t = 67.00 h

Biomass X (g/L): 97.08
Glucose S (g/L): 55.07
Lactic Acid P (g/L): 73.60
Maltose M (g/L): 78.56
Volume V (L): 10.61

t = 68.00 h

Biomass X (g/L): 88.48
Glucose S (g/L): 56.88
Lactic Acid P (g/L): 71.92
Maltose M (g/L): 77.95
Volume V (L): 10.14

t = 69.00 h

Biomass X (g/L): 80.99
Glucose S (g/L): 54.15
Lactic Acid P (g/L): 85.70
Maltose M (g/L): 79.03
Volume V (L): 11.15

t = 70.00 h

Biomass X (g/L): 91.18
Glucose S (g/L): 53.93
Lactic Acid P (g/L): 66.64
Maltose M (g/L): 77.43
Volume V (L): 11.35

t = 71.00 h

Biomass X (g/L): 92.11
Glucose S (g/L): 55.57
Lactic Acid P (g/L): 62.58
Maltose M (g/L): 75.61
Volume V (L): 11.27

t = 72.00 h

Biomass X (g/L): 75.26
Glucose S (g/L): 54.49
Lactic Acid P (g/L): 70.16
Maltose M (g/L): 72.68
Volume V (L): 12.36

t = 73.00 h

Biomass X (g/L): 90.00
Glucose S (g/L): 50.69
Lactic Acid P (g/L): 69.28
Maltose M (g/L): 70.48
Volume V (L): 11.41

t = 74.00 h

Biomass X (g/L): 78.98
Glucose S (g/L): 58.90
Lactic Acid P (g/L): 73.88
Maltose M (g/L): 74.27
Volume V (L): 10.36

t = 75.00 h

Biomass X (g/L): 79.92
Glucose S (g/L): 61.33
Lactic Acid P (g/L): 66.09
Maltose M (g/L): 72.17

Volume V (L): 11.45

Batch 98

Initial Conditions:

Batch: 98.00

X0 (g/L): 9.90

S0 (g/L): 127.95

M0 (g/L): 7.20

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 0.06

F_vol (L/h): 0.03

S_feed (g/L): 284.02

M_feed (g/L): 15.98

V_final (L): 7.15

Batch 98 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 14.59
Glucose S (g/L): 129.45
Lactic Acid P (g/L): 10.12
Maltose M (g/L): 7.69
Volume V (L): 5.27

t = 1.00 h

Biomass X (g/L): 17.43
Glucose S (g/L): 126.41
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 7.26
Volume V (L): 5.07

t = 2.00 h

Biomass X (g/L): 15.95
Glucose S (g/L): 119.80
Lactic Acid P (g/L): 23.15
Maltose M (g/L): 6.10
Volume V (L): 5.13

t = 3.00 h

Biomass X (g/L): 19.27
Glucose S (g/L): 111.91
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 6.67
Volume V (L): 5.24

t = 4.00 h

Biomass X (g/L): 7.74
Glucose S (g/L): 91.92
Lactic Acid P (g/L): 15.18
Maltose M (g/L): 6.62
Volume V (L): 5.12

t = 5.00 h

Biomass X (g/L): 35.10
Glucose S (g/L): 102.49

Lactic Acid P (g/L): 15.42

Maltose M (g/L): 6.64

Volume V (L): 4.96

t = 6.00 h

Biomass X (g/L): 36.61

Glucose S (g/L): 108.26

Lactic Acid P (g/L): 35.79

Maltose M (g/L): 6.82

Volume V (L): 5.42

t = 7.00 h

Biomass X (g/L): 48.39

Glucose S (g/L): 88.68

Lactic Acid P (g/L): 11.60

Maltose M (g/L): 6.79

Volume V (L): 5.07

t = 8.00 h

Biomass X (g/L): 56.28

Glucose S (g/L): 85.48

Lactic Acid P (g/L): 34.83

Maltose M (g/L): 5.62

Volume V (L): 5.06

t = 9.00 h

Biomass X (g/L): 69.00

Glucose S (g/L): 79.46

Lactic Acid P (g/L): 31.87

Maltose M (g/L): 6.38

Volume V (L): 5.22

t = 10.00 h

Biomass X (g/L): 50.87

Glucose S (g/L): 87.05

Lactic Acid P (g/L): 53.26

Maltose M (g/L): 6.22

Volume V (L): 5.22

t = 11.00 h

Biomass X (g/L): 51.46

Glucose S (g/L): 75.85
Lactic Acid P (g/L): 32.67
Maltose M (g/L): 6.81
Volume V (L): 5.32

t = 12.00 h

Biomass X (g/L): 62.53
Glucose S (g/L): 87.52
Lactic Acid P (g/L): 48.13
Maltose M (g/L): 6.46
Volume V (L): 5.33

t = 13.00 h

Biomass X (g/L): 64.12
Glucose S (g/L): 81.38
Lactic Acid P (g/L): 38.13
Maltose M (g/L): 6.12
Volume V (L): 5.61

t = 14.00 h

Biomass X (g/L): 66.08
Glucose S (g/L): 75.21
Lactic Acid P (g/L): 50.89
Maltose M (g/L): 6.20
Volume V (L): 5.48

t = 15.00 h

Biomass X (g/L): 92.07
Glucose S (g/L): 73.80
Lactic Acid P (g/L): 47.44
Maltose M (g/L): 5.46
Volume V (L): 5.28

t = 16.00 h

Biomass X (g/L): 80.96
Glucose S (g/L): 64.47
Lactic Acid P (g/L): 49.27
Maltose M (g/L): 6.22
Volume V (L): 5.34

t = 17.00 h

Biomass X (g/L): 101.80
Glucose S (g/L): 62.82
Lactic Acid P (g/L): 34.92
Maltose M (g/L): 5.79
Volume V (L): 5.42

t = 18.00 h

Biomass X (g/L): 83.22
Glucose S (g/L): 71.67
Lactic Acid P (g/L): 60.82
Maltose M (g/L): 5.58
Volume V (L): 5.45

t = 19.00 h

Biomass X (g/L): 72.75
Glucose S (g/L): 71.82
Lactic Acid P (g/L): 47.25
Maltose M (g/L): 6.26
Volume V (L): 5.46

t = 20.00 h

Biomass X (g/L): 77.92
Glucose S (g/L): 69.23
Lactic Acid P (g/L): 48.31
Maltose M (g/L): 5.84
Volume V (L): 5.38

t = 21.00 h

Biomass X (g/L): 98.08
Glucose S (g/L): 60.05
Lactic Acid P (g/L): 45.19
Maltose M (g/L): 5.45
Volume V (L): 5.65

t = 22.00 h

Biomass X (g/L): 76.93
Glucose S (g/L): 59.75
Lactic Acid P (g/L): 51.43
Maltose M (g/L): 5.42
Volume V (L): 5.55

t = 23.00 h

Biomass X (g/L): 95.38
Glucose S (g/L): 55.23
Lactic Acid P (g/L): 68.25
Maltose M (g/L): 5.70
Volume V (L): 5.50

t = 24.00 h

Biomass X (g/L): 86.83
Glucose S (g/L): 66.46
Lactic Acid P (g/L): 68.80
Maltose M (g/L): 5.62
Volume V (L): 5.77

t = 25.00 h

Biomass X (g/L): 81.70
Glucose S (g/L): 46.82
Lactic Acid P (g/L): 71.92
Maltose M (g/L): 5.47
Volume V (L): 5.86

t = 26.00 h

Biomass X (g/L): 106.29
Glucose S (g/L): 44.63
Lactic Acid P (g/L): 55.77
Maltose M (g/L): 4.99
Volume V (L): 5.70

t = 27.00 h

Biomass X (g/L): 86.52
Glucose S (g/L): 45.61
Lactic Acid P (g/L): 69.16
Maltose M (g/L): 4.99
Volume V (L): 6.08

t = 28.00 h

Biomass X (g/L): 84.60
Glucose S (g/L): 44.29
Lactic Acid P (g/L): 61.59
Maltose M (g/L): 5.05
Volume V (L): 6.19

t = 29.00 h

Biomass X (g/L): 92.80
Glucose S (g/L): 68.46
Lactic Acid P (g/L): 78.20
Maltose M (g/L): 5.19
Volume V (L): 5.92

t = 30.00 h

Biomass X (g/L): 88.78
Glucose S (g/L): 64.23
Lactic Acid P (g/L): 73.58
Maltose M (g/L): 5.19
Volume V (L): 5.78

t = 31.00 h

Biomass X (g/L): 117.24
Glucose S (g/L): 50.14
Lactic Acid P (g/L): 63.34
Maltose M (g/L): 5.10
Volume V (L): 5.94

t = 32.00 h

Biomass X (g/L): 105.64
Glucose S (g/L): 57.61
Lactic Acid P (g/L): 81.11
Maltose M (g/L): 5.07
Volume V (L): 5.71

t = 33.00 h

Biomass X (g/L): 88.87
Glucose S (g/L): 68.48
Lactic Acid P (g/L): 76.34
Maltose M (g/L): 5.72
Volume V (L): 5.80

t = 34.00 h

Biomass X (g/L): 90.29
Glucose S (g/L): 61.63
Lactic Acid P (g/L): 57.81
Maltose M (g/L): 5.32

Volume V (L): 5.69

t = 35.00 h

Biomass X (g/L): 74.54

Glucose S (g/L): 72.86

Lactic Acid P (g/L): 55.50

Maltose M (g/L): 5.39

Volume V (L): 5.90

t = 36.00 h

Biomass X (g/L): 96.95

Glucose S (g/L): 57.24

Lactic Acid P (g/L): 59.29

Maltose M (g/L): 5.10

Volume V (L): 5.82

t = 37.00 h

Biomass X (g/L): 87.95

Glucose S (g/L): 53.45

Lactic Acid P (g/L): 90.04

Maltose M (g/L): 4.92

Volume V (L): 6.17

t = 38.00 h

Biomass X (g/L): 96.66

Glucose S (g/L): 50.48

Lactic Acid P (g/L): 63.64

Maltose M (g/L): 4.99

Volume V (L): 5.96

t = 39.00 h

Biomass X (g/L): 75.53

Glucose S (g/L): 58.24

Lactic Acid P (g/L): 93.28

Maltose M (g/L): 4.89

Volume V (L): 6.17

t = 40.00 h

Biomass X (g/L): 95.58

Glucose S (g/L): 64.68

Lactic Acid P (g/L): 64.03

Maltose M (g/L): 5.05

Volume V (L): 6.28

t = 41.00 h

Biomass X (g/L): 113.40

Glucose S (g/L): 58.90

Lactic Acid P (g/L): 63.38

Maltose M (g/L): 4.39

Volume V (L): 6.38

t = 42.00 h

Biomass X (g/L): 96.43

Glucose S (g/L): 55.79

Lactic Acid P (g/L): 84.52

Maltose M (g/L): 4.94

Volume V (L): 6.14

t = 43.00 h

Biomass X (g/L): 87.86

Glucose S (g/L): 67.18

Lactic Acid P (g/L): 85.38

Maltose M (g/L): 4.71

Volume V (L): 6.29

t = 44.00 h

Biomass X (g/L): 95.61

Glucose S (g/L): 62.21

Lactic Acid P (g/L): 82.03

Maltose M (g/L): 3.98

Volume V (L): 6.32

t = 45.00 h

Biomass X (g/L): 95.10

Glucose S (g/L): 57.85

Lactic Acid P (g/L): 83.61

Maltose M (g/L): 4.60

Volume V (L): 6.51

t = 46.00 h

Biomass X (g/L): 118.31

Glucose S (g/L): 53.42

Lactic Acid P (g/L): 69.06

Maltose M (g/L): 4.08

Volume V (L): 6.46

t = 47.00 h

Biomass X (g/L): 92.17

Glucose S (g/L): 57.15

Lactic Acid P (g/L): 85.90

Maltose M (g/L): 4.31

Volume V (L): 6.57

t = 48.00 h

Biomass X (g/L): 83.10

Glucose S (g/L): 62.40

Lactic Acid P (g/L): 80.25

Maltose M (g/L): 4.41

Volume V (L): 6.09

t = 49.00 h

Biomass X (g/L): 95.62

Glucose S (g/L): 53.80

Lactic Acid P (g/L): 74.46

Maltose M (g/L): 3.90

Volume V (L): 6.29

t = 50.00 h

Biomass X (g/L): 97.30

Glucose S (g/L): 50.81

Lactic Acid P (g/L): 79.41

Maltose M (g/L): 4.38

Volume V (L): 6.61

t = 51.00 h

Biomass X (g/L): 100.95

Glucose S (g/L): 56.27

Lactic Acid P (g/L): 89.46

Maltose M (g/L): 4.39

Volume V (L): 6.56

t = 52.00 h

Biomass X (g/L): 102.81

Glucose S (g/L): 59.47
Lactic Acid P (g/L): 83.01
Maltose M (g/L): 4.59
Volume V (L): 6.47

t = 53.00 h

Biomass X (g/L): 115.03
Glucose S (g/L): 69.55
Lactic Acid P (g/L): 57.57
Maltose M (g/L): 4.14
Volume V (L): 6.44

t = 54.00 h

Biomass X (g/L): 88.55
Glucose S (g/L): 52.11
Lactic Acid P (g/L): 88.61
Maltose M (g/L): 4.36
Volume V (L): 6.69

t = 55.00 h

Biomass X (g/L): 87.45
Glucose S (g/L): 62.01
Lactic Acid P (g/L): 88.11
Maltose M (g/L): 4.20
Volume V (L): 6.24

t = 56.00 h

Biomass X (g/L): 75.23
Glucose S (g/L): 58.70
Lactic Acid P (g/L): 76.32
Maltose M (g/L): 4.06
Volume V (L): 6.74

t = 57.00 h

Biomass X (g/L): 101.00
Glucose S (g/L): 59.27
Lactic Acid P (g/L): 81.58
Maltose M (g/L): 3.59
Volume V (L): 6.65

t = 58.00 h

Biomass X (g/L): 84.84
Glucose S (g/L): 71.85
Lactic Acid P (g/L): 81.79
Maltose M (g/L): 4.11
Volume V (L): 6.58

t = 59.00 h

Biomass X (g/L): 103.00
Glucose S (g/L): 61.88
Lactic Acid P (g/L): 60.92
Maltose M (g/L): 4.38
Volume V (L): 6.78

t = 60.00 h

Biomass X (g/L): 88.53
Glucose S (g/L): 55.15
Lactic Acid P (g/L): 77.08
Maltose M (g/L): 3.76
Volume V (L): 6.66

t = 61.00 h

Biomass X (g/L): 111.20
Glucose S (g/L): 67.99
Lactic Acid P (g/L): 77.70
Maltose M (g/L): 3.98
Volume V (L): 6.76

t = 62.00 h

Biomass X (g/L): 78.40
Glucose S (g/L): 67.71
Lactic Acid P (g/L): 79.98
Maltose M (g/L): 4.01
Volume V (L): 6.77

t = 63.00 h

Biomass X (g/L): 103.53
Glucose S (g/L): 55.25
Lactic Acid P (g/L): 76.50
Maltose M (g/L): 4.39
Volume V (L): 6.94

t = 64.00 h

Biomass X (g/L): 102.08
Glucose S (g/L): 54.15
Lactic Acid P (g/L): 88.70
Maltose M (g/L): 3.99
Volume V (L): 6.86

t = 65.00 h

Biomass X (g/L): 97.97
Glucose S (g/L): 65.60
Lactic Acid P (g/L): 74.06
Maltose M (g/L): 4.05
Volume V (L): 6.54

t = 66.00 h

Biomass X (g/L): 80.79
Glucose S (g/L): 62.75
Lactic Acid P (g/L): 91.61
Maltose M (g/L): 3.92
Volume V (L): 6.65

t = 67.00 h

Biomass X (g/L): 97.57
Glucose S (g/L): 59.93
Lactic Acid P (g/L): 88.84
Maltose M (g/L): 3.72
Volume V (L): 6.74

t = 68.00 h

Biomass X (g/L): 93.17
Glucose S (g/L): 70.86
Lactic Acid P (g/L): 96.55
Maltose M (g/L): 3.48
Volume V (L): 7.03

t = 69.00 h

Biomass X (g/L): 72.74
Glucose S (g/L): 59.58
Lactic Acid P (g/L): 100.51
Maltose M (g/L): 3.75
Volume V (L): 7.08

t = 70.00 h

Biomass X (g/L): 107.47
Glucose S (g/L): 66.92
Lactic Acid P (g/L): 100.76
Maltose M (g/L): 3.96
Volume V (L): 7.30

t = 71.00 h

Biomass X (g/L): 87.59
Glucose S (g/L): 72.44
Lactic Acid P (g/L): 71.42
Maltose M (g/L): 3.84
Volume V (L): 6.86

t = 72.00 h

Biomass X (g/L): 106.19
Glucose S (g/L): 62.10
Lactic Acid P (g/L): 87.04
Maltose M (g/L): 4.29
Volume V (L): 7.29

t = 73.00 h

Biomass X (g/L): 89.97
Glucose S (g/L): 51.94
Lactic Acid P (g/L): 76.50
Maltose M (g/L): 4.08
Volume V (L): 6.76

t = 74.00 h

Biomass X (g/L): 100.56
Glucose S (g/L): 62.16
Lactic Acid P (g/L): 117.01
Maltose M (g/L): 4.16
Volume V (L): 7.13

t = 75.00 h

Biomass X (g/L): 81.31
Glucose S (g/L): 66.56
Lactic Acid P (g/L): 90.28
Maltose M (g/L): 2.99

Volume V (L): 7.19

Batch 99

Initial Conditions:

Batch: 99.00

X0 (g/L): 2.48

S0 (g/L): 36.30

M0 (g/L): 110.01

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 3.03

F_vol (L/h): 0.09

S_feed (g/L): 74.43

M_feed (g/L): 225.57

V_final (L): 11.99

Batch 99 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 4.76
Glucose S (g/L): 32.06
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 103.29
Volume V (L): 5.00

t = 1.00 h

Biomass X (g/L): 0.32
Glucose S (g/L): 41.46
Lactic Acid P (g/L): 5.72
Maltose M (g/L): 110.45
Volume V (L): 5.31

t = 2.00 h

Biomass X (g/L): 5.35
Glucose S (g/L): 39.24
Lactic Acid P (g/L): 1.31
Maltose M (g/L): 110.61
Volume V (L): 5.13

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 42.04
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 106.62
Volume V (L): 5.31

t = 4.00 h

Biomass X (g/L): 15.26
Glucose S (g/L): 42.42
Lactic Acid P (g/L): 15.57
Maltose M (g/L): 104.40
Volume V (L): 5.36

t = 5.00 h

Biomass X (g/L): 11.41
Glucose S (g/L): 40.35

Lactic Acid P (g/L): 17.27
Maltose M (g/L): 111.04
Volume V (L): 5.22

t = 6.00 h

Biomass X (g/L): 14.83
Glucose S (g/L): 41.64
Lactic Acid P (g/L): 9.17
Maltose M (g/L): 105.72
Volume V (L): 5.66

t = 7.00 h

Biomass X (g/L): 11.54
Glucose S (g/L): 43.88
Lactic Acid P (g/L): 6.59
Maltose M (g/L): 115.49
Volume V (L): 6.00

t = 8.00 h

Biomass X (g/L): 14.24
Glucose S (g/L): 43.44
Lactic Acid P (g/L): 1.78
Maltose M (g/L): 109.86
Volume V (L): 5.51

t = 9.00 h

Biomass X (g/L): 25.77
Glucose S (g/L): 47.95
Lactic Acid P (g/L): 22.92
Maltose M (g/L): 112.90
Volume V (L): 6.31

t = 10.00 h

Biomass X (g/L): 39.57
Glucose S (g/L): 37.85
Lactic Acid P (g/L): 23.51
Maltose M (g/L): 108.66
Volume V (L): 6.33

t = 11.00 h

Biomass X (g/L): 15.31

Glucose S (g/L): 43.10
Lactic Acid P (g/L): 12.12
Maltose M (g/L): 110.55
Volume V (L): 5.53

t = 12.00 h

Biomass X (g/L): 18.78
Glucose S (g/L): 33.11
Lactic Acid P (g/L): 16.17
Maltose M (g/L): 108.15
Volume V (L): 5.58

t = 13.00 h

Biomass X (g/L): 38.82
Glucose S (g/L): 35.65
Lactic Acid P (g/L): 29.07
Maltose M (g/L): 110.70
Volume V (L): 6.46

t = 14.00 h

Biomass X (g/L): 36.94
Glucose S (g/L): 32.88
Lactic Acid P (g/L): 17.67
Maltose M (g/L): 101.22
Volume V (L): 5.98

t = 15.00 h

Biomass X (g/L): 44.84
Glucose S (g/L): 30.26
Lactic Acid P (g/L): 25.20
Maltose M (g/L): 109.68
Volume V (L): 5.81

t = 16.00 h

Biomass X (g/L): 46.81
Glucose S (g/L): 29.16
Lactic Acid P (g/L): 24.17
Maltose M (g/L): 108.66
Volume V (L): 7.45

t = 17.00 h

Biomass X (g/L): 56.04
Glucose S (g/L): 25.86
Lactic Acid P (g/L): 26.89
Maltose M (g/L): 105.81
Volume V (L): 6.04

t = 18.00 h

Biomass X (g/L): 50.35
Glucose S (g/L): 29.67
Lactic Acid P (g/L): 31.67
Maltose M (g/L): 104.66
Volume V (L): 5.83

t = 19.00 h

Biomass X (g/L): 44.29
Glucose S (g/L): 23.10
Lactic Acid P (g/L): 47.97
Maltose M (g/L): 104.60
Volume V (L): 6.33

t = 20.00 h

Biomass X (g/L): 50.65
Glucose S (g/L): 26.09
Lactic Acid P (g/L): 26.72
Maltose M (g/L): 105.95
Volume V (L): 6.99

t = 21.00 h

Biomass X (g/L): 58.79
Glucose S (g/L): 26.92
Lactic Acid P (g/L): 47.62
Maltose M (g/L): 108.35
Volume V (L): 7.00

t = 22.00 h

Biomass X (g/L): 50.46
Glucose S (g/L): 28.51
Lactic Acid P (g/L): 47.77
Maltose M (g/L): 102.67
Volume V (L): 8.11

t = 23.00 h

Biomass X (g/L): 63.78
Glucose S (g/L): 26.88
Lactic Acid P (g/L): 55.53
Maltose M (g/L): 98.73
Volume V (L): 6.71

t = 24.00 h

Biomass X (g/L): 66.35
Glucose S (g/L): 25.42
Lactic Acid P (g/L): 50.77
Maltose M (g/L): 110.89
Volume V (L): 7.37

t = 25.00 h

Biomass X (g/L): 63.91
Glucose S (g/L): 21.44
Lactic Acid P (g/L): 46.56
Maltose M (g/L): 105.00
Volume V (L): 7.96

t = 26.00 h

Biomass X (g/L): 70.63
Glucose S (g/L): 21.00
Lactic Acid P (g/L): 44.01
Maltose M (g/L): 101.36
Volume V (L): 7.06

t = 27.00 h

Biomass X (g/L): 54.66
Glucose S (g/L): 19.99
Lactic Acid P (g/L): 60.93
Maltose M (g/L): 105.66
Volume V (L): 7.91

t = 28.00 h

Biomass X (g/L): 62.03
Glucose S (g/L): 19.88
Lactic Acid P (g/L): 57.72
Maltose M (g/L): 104.44
Volume V (L): 6.59

t = 29.00 h

Biomass X (g/L): 61.28
Glucose S (g/L): 24.14
Lactic Acid P (g/L): 50.10
Maltose M (g/L): 102.65
Volume V (L): 8.69

t = 30.00 h

Biomass X (g/L): 81.31
Glucose S (g/L): 22.12
Lactic Acid P (g/L): 63.29
Maltose M (g/L): 99.28
Volume V (L): 8.35

t = 31.00 h

Biomass X (g/L): 95.62
Glucose S (g/L): 28.74
Lactic Acid P (g/L): 52.37
Maltose M (g/L): 102.25
Volume V (L): 7.51

t = 32.00 h

Biomass X (g/L): 66.33
Glucose S (g/L): 23.43
Lactic Acid P (g/L): 57.10
Maltose M (g/L): 103.06
Volume V (L): 8.45

t = 33.00 h

Biomass X (g/L): 73.13
Glucose S (g/L): 26.08
Lactic Acid P (g/L): 52.64
Maltose M (g/L): 94.13
Volume V (L): 8.81

t = 34.00 h

Biomass X (g/L): 69.46
Glucose S (g/L): 25.44
Lactic Acid P (g/L): 63.33
Maltose M (g/L): 99.51

Volume V (L): 7.45

t = 35.00 h

Biomass X (g/L): 78.68

Glucose S (g/L): 22.31

Lactic Acid P (g/L): 50.71

Maltose M (g/L): 99.26

Volume V (L): 7.93

t = 36.00 h

Biomass X (g/L): 71.38

Glucose S (g/L): 30.95

Lactic Acid P (g/L): 57.46

Maltose M (g/L): 98.68

Volume V (L): 9.11

t = 37.00 h

Biomass X (g/L): 80.28

Glucose S (g/L): 30.28

Lactic Acid P (g/L): 54.99

Maltose M (g/L): 97.54

Volume V (L): 8.49

t = 38.00 h

Biomass X (g/L): 84.33

Glucose S (g/L): 31.28

Lactic Acid P (g/L): 63.61

Maltose M (g/L): 100.50

Volume V (L): 8.79

t = 39.00 h

Biomass X (g/L): 66.91

Glucose S (g/L): 31.24

Lactic Acid P (g/L): 65.28

Maltose M (g/L): 95.81

Volume V (L): 8.68

t = 40.00 h

Biomass X (g/L): 81.38

Glucose S (g/L): 27.53

Lactic Acid P (g/L): 52.81

Maltose M (g/L): 95.96

Volume V (L): 9.23

t = 41.00 h

Biomass X (g/L): 62.21

Glucose S (g/L): 28.28

Lactic Acid P (g/L): 66.52

Maltose M (g/L): 96.44

Volume V (L): 9.11

t = 42.00 h

Biomass X (g/L): 77.01

Glucose S (g/L): 32.83

Lactic Acid P (g/L): 50.59

Maltose M (g/L): 94.47

Volume V (L): 8.98

t = 43.00 h

Biomass X (g/L): 76.64

Glucose S (g/L): 37.07

Lactic Acid P (g/L): 83.08

Maltose M (g/L): 90.36

Volume V (L): 7.83

t = 44.00 h

Biomass X (g/L): 77.77

Glucose S (g/L): 34.05

Lactic Acid P (g/L): 59.95

Maltose M (g/L): 94.69

Volume V (L): 9.46

t = 45.00 h

Biomass X (g/L): 77.19

Glucose S (g/L): 36.70

Lactic Acid P (g/L): 74.01

Maltose M (g/L): 94.25

Volume V (L): 9.60

t = 46.00 h

Biomass X (g/L): 75.65

Glucose S (g/L): 32.58

Lactic Acid P (g/L): 67.02

Maltose M (g/L): 91.99

Volume V (L): 8.60

t = 47.00 h

Biomass X (g/L): 77.97

Glucose S (g/L): 38.57

Lactic Acid P (g/L): 67.78

Maltose M (g/L): 92.21

Volume V (L): 8.73

t = 48.00 h

Biomass X (g/L): 82.87

Glucose S (g/L): 34.42

Lactic Acid P (g/L): 55.88

Maltose M (g/L): 82.56

Volume V (L): 9.76

t = 49.00 h

Biomass X (g/L): 83.25

Glucose S (g/L): 37.31

Lactic Acid P (g/L): 73.79

Maltose M (g/L): 95.29

Volume V (L): 9.68

t = 50.00 h

Biomass X (g/L): 81.93

Glucose S (g/L): 36.10

Lactic Acid P (g/L): 67.21

Maltose M (g/L): 93.89

Volume V (L): 8.56

t = 51.00 h

Biomass X (g/L): 75.51

Glucose S (g/L): 40.92

Lactic Acid P (g/L): 71.23

Maltose M (g/L): 86.68

Volume V (L): 9.30

t = 52.00 h

Biomass X (g/L): 91.18

Glucose S (g/L): 41.73
Lactic Acid P (g/L): 66.67
Maltose M (g/L): 88.20
Volume V (L): 10.02

t = 53.00 h

Biomass X (g/L): 95.44
Glucose S (g/L): 34.80
Lactic Acid P (g/L): 68.91
Maltose M (g/L): 94.12
Volume V (L): 10.32

t = 54.00 h

Biomass X (g/L): 68.92
Glucose S (g/L): 44.63
Lactic Acid P (g/L): 60.34
Maltose M (g/L): 85.82
Volume V (L): 9.71

t = 55.00 h

Biomass X (g/L): 82.66
Glucose S (g/L): 48.79
Lactic Acid P (g/L): 67.05
Maltose M (g/L): 90.27
Volume V (L): 10.05

t = 56.00 h

Biomass X (g/L): 76.70
Glucose S (g/L): 40.72
Lactic Acid P (g/L): 65.00
Maltose M (g/L): 86.88
Volume V (L): 9.73

t = 57.00 h

Biomass X (g/L): 89.85
Glucose S (g/L): 42.23
Lactic Acid P (g/L): 62.78
Maltose M (g/L): 89.13
Volume V (L): 11.06

t = 58.00 h

Biomass X (g/L): 80.99
Glucose S (g/L): 47.40
Lactic Acid P (g/L): 70.38
Maltose M (g/L): 87.58
Volume V (L): 10.51

t = 59.00 h

Biomass X (g/L): 81.59
Glucose S (g/L): 44.37
Lactic Acid P (g/L): 72.64
Maltose M (g/L): 86.12
Volume V (L): 10.13

t = 60.00 h

Biomass X (g/L): 85.73
Glucose S (g/L): 44.90
Lactic Acid P (g/L): 71.91
Maltose M (g/L): 87.68
Volume V (L): 9.95

t = 61.00 h

Biomass X (g/L): 78.22
Glucose S (g/L): 51.80
Lactic Acid P (g/L): 69.07
Maltose M (g/L): 84.58
Volume V (L): 9.98

t = 62.00 h

Biomass X (g/L): 96.81
Glucose S (g/L): 42.04
Lactic Acid P (g/L): 73.78
Maltose M (g/L): 80.24
Volume V (L): 11.79

t = 63.00 h

Biomass X (g/L): 85.73
Glucose S (g/L): 44.62
Lactic Acid P (g/L): 72.54
Maltose M (g/L): 82.36
Volume V (L): 10.78

t = 64.00 h

Biomass X (g/L): 89.65
Glucose S (g/L): 49.26
Lactic Acid P (g/L): 71.50
Maltose M (g/L): 81.55
Volume V (L): 10.74

t = 65.00 h

Biomass X (g/L): 78.71
Glucose S (g/L): 43.42
Lactic Acid P (g/L): 72.31
Maltose M (g/L): 79.43
Volume V (L): 10.94

t = 66.00 h

Biomass X (g/L): 79.56
Glucose S (g/L): 48.19
Lactic Acid P (g/L): 81.39
Maltose M (g/L): 84.93
Volume V (L): 10.80

t = 67.00 h

Biomass X (g/L): 88.30
Glucose S (g/L): 48.89
Lactic Acid P (g/L): 60.76
Maltose M (g/L): 82.85
Volume V (L): 11.19

t = 68.00 h

Biomass X (g/L): 87.27
Glucose S (g/L): 53.96
Lactic Acid P (g/L): 68.69
Maltose M (g/L): 84.00
Volume V (L): 11.36

t = 69.00 h

Biomass X (g/L): 68.68
Glucose S (g/L): 53.82
Lactic Acid P (g/L): 74.61
Maltose M (g/L): 84.06
Volume V (L): 11.44

t = 70.00 h

Biomass X (g/L): 81.57
Glucose S (g/L): 46.86
Lactic Acid P (g/L): 58.04
Maltose M (g/L): 84.34
Volume V (L): 10.82

t = 71.00 h

Biomass X (g/L): 84.74
Glucose S (g/L): 50.78
Lactic Acid P (g/L): 84.10
Maltose M (g/L): 82.85
Volume V (L): 12.12

t = 72.00 h

Biomass X (g/L): 78.93
Glucose S (g/L): 51.50
Lactic Acid P (g/L): 74.43
Maltose M (g/L): 78.25
Volume V (L): 11.40

t = 73.00 h

Biomass X (g/L): 83.63
Glucose S (g/L): 53.73
Lactic Acid P (g/L): 80.73
Maltose M (g/L): 82.52
Volume V (L): 11.44

t = 74.00 h

Biomass X (g/L): 79.11
Glucose S (g/L): 51.59
Lactic Acid P (g/L): 82.32
Maltose M (g/L): 81.01
Volume V (L): 11.51

t = 75.00 h

Biomass X (g/L): 58.62
Glucose S (g/L): 57.29
Lactic Acid P (g/L): 76.32
Maltose M (g/L): 86.51

Volume V (L): 11.40

Batch 100

Initial Conditions:

Batch: 100.00

X0 (g/L): 6.33

S0 (g/L): 15.80

M0 (g/L): 178.91

P0 (g/L): 0.00

V0 (L): 5.00

M/S ratio: 11.33

F_vol (L/h): 0.05

S_feed (g/L): 24.34

M_feed (g/L): 275.66

V_final (L): 8.81

Batch 100 - Time Course Measurements

t = 0.00 h

Biomass X (g/L): 8.87
Glucose S (g/L): 13.85
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 186.73
Volume V (L): 4.49

t = 1.00 h

Biomass X (g/L): 16.49
Glucose S (g/L): 17.10
Lactic Acid P (g/L): 2.42
Maltose M (g/L): 170.89
Volume V (L): 4.82

t = 2.00 h

Biomass X (g/L): 3.17
Glucose S (g/L): 20.08
Lactic Acid P (g/L): 1.64
Maltose M (g/L): 179.40
Volume V (L): 5.09

t = 3.00 h

Biomass X (g/L): 0.00
Glucose S (g/L): 26.40
Lactic Acid P (g/L): 7.12
Maltose M (g/L): 180.47
Volume V (L): 5.54

t = 4.00 h

Biomass X (g/L): 5.63
Glucose S (g/L): 23.34
Lactic Acid P (g/L): 0.00
Maltose M (g/L): 160.89
Volume V (L): 5.15

t = 5.00 h

Biomass X (g/L): 14.40
Glucose S (g/L): 24.88

Lactic Acid P (g/L): 13.84

Maltose M (g/L): 157.64

Volume V (L): 5.22

t = 6.00 h

Biomass X (g/L): 10.54

Glucose S (g/L): 17.60

Lactic Acid P (g/L): 12.42

Maltose M (g/L): 155.12

Volume V (L): 5.06

t = 7.00 h

Biomass X (g/L): 20.36

Glucose S (g/L): 14.80

Lactic Acid P (g/L): 12.96

Maltose M (g/L): 151.69

Volume V (L): 5.69

t = 8.00 h

Biomass X (g/L): 24.71

Glucose S (g/L): 20.54

Lactic Acid P (g/L): 26.12

Maltose M (g/L): 150.89

Volume V (L): 5.68

t = 9.00 h

Biomass X (g/L): 33.71

Glucose S (g/L): 16.35

Lactic Acid P (g/L): 10.94

Maltose M (g/L): 167.37

Volume V (L): 5.70

t = 10.00 h

Biomass X (g/L): 23.85

Glucose S (g/L): 18.03

Lactic Acid P (g/L): 21.26

Maltose M (g/L): 162.01

Volume V (L): 5.54

t = 11.00 h

Biomass X (g/L): 46.84

Glucose S (g/L): 10.95
Lactic Acid P (g/L): 28.84
Maltose M (g/L): 139.24
Volume V (L): 5.07

t = 12.00 h

Biomass X (g/L): 60.56
Glucose S (g/L): 15.76
Lactic Acid P (g/L): 21.05
Maltose M (g/L): 148.32
Volume V (L): 5.94

t = 13.00 h

Biomass X (g/L): 61.57
Glucose S (g/L): 11.90
Lactic Acid P (g/L): 12.99
Maltose M (g/L): 140.49
Volume V (L): 5.68

t = 14.00 h

Biomass X (g/L): 45.43
Glucose S (g/L): 10.21
Lactic Acid P (g/L): 23.16
Maltose M (g/L): 157.36
Volume V (L): 5.62

t = 15.00 h

Biomass X (g/L): 60.08
Glucose S (g/L): 10.09
Lactic Acid P (g/L): 35.37
Maltose M (g/L): 143.48
Volume V (L): 5.90

t = 16.00 h

Biomass X (g/L): 61.50
Glucose S (g/L): 12.98
Lactic Acid P (g/L): 35.74
Maltose M (g/L): 131.64
Volume V (L): 5.88

t = 17.00 h

Biomass X (g/L): 58.20
Glucose S (g/L): 13.84
Lactic Acid P (g/L): 39.80
Maltose M (g/L): 145.81
Volume V (L): 5.54

t = 18.00 h

Biomass X (g/L): 62.46
Glucose S (g/L): 9.48
Lactic Acid P (g/L): 35.72
Maltose M (g/L): 148.20
Volume V (L): 5.82

t = 19.00 h

Biomass X (g/L): 69.27
Glucose S (g/L): 12.15
Lactic Acid P (g/L): 31.43
Maltose M (g/L): 132.17
Volume V (L): 5.39

t = 20.00 h

Biomass X (g/L): 73.02
Glucose S (g/L): 12.24
Lactic Acid P (g/L): 46.36
Maltose M (g/L): 136.96
Volume V (L): 6.04

t = 21.00 h

Biomass X (g/L): 58.63
Glucose S (g/L): 13.91
Lactic Acid P (g/L): 45.09
Maltose M (g/L): 129.53
Volume V (L): 6.47

t = 22.00 h

Biomass X (g/L): 75.05
Glucose S (g/L): 15.08
Lactic Acid P (g/L): 53.93
Maltose M (g/L): 147.87
Volume V (L): 5.57

t = 23.00 h

Biomass X (g/L): 85.26
Glucose S (g/L): 18.48
Lactic Acid P (g/L): 42.56
Maltose M (g/L): 130.90
Volume V (L): 6.21

t = 24.00 h

Biomass X (g/L): 79.19
Glucose S (g/L): 10.30
Lactic Acid P (g/L): 32.16
Maltose M (g/L): 126.77
Volume V (L): 6.42

t = 25.00 h

Biomass X (g/L): 86.69
Glucose S (g/L): 17.92
Lactic Acid P (g/L): 28.03
Maltose M (g/L): 131.55
Volume V (L): 6.52

t = 26.00 h

Biomass X (g/L): 48.73
Glucose S (g/L): 13.33
Lactic Acid P (g/L): 49.44
Maltose M (g/L): 128.72
Volume V (L): 6.09

t = 27.00 h

Biomass X (g/L): 80.05
Glucose S (g/L): 9.27
Lactic Acid P (g/L): 55.43
Maltose M (g/L): 135.65
Volume V (L): 6.04

t = 28.00 h

Biomass X (g/L): 77.93
Glucose S (g/L): 11.45
Lactic Acid P (g/L): 53.18
Maltose M (g/L): 139.20
Volume V (L): 6.24

t = 29.00 h

Biomass X (g/L): 86.14
Glucose S (g/L): 16.19
Lactic Acid P (g/L): 51.13
Maltose M (g/L): 137.97
Volume V (L): 6.84

t = 30.00 h

Biomass X (g/L): 66.57
Glucose S (g/L): 17.74
Lactic Acid P (g/L): 42.67
Maltose M (g/L): 121.84
Volume V (L): 6.00

t = 31.00 h

Biomass X (g/L): 73.28
Glucose S (g/L): 11.82
Lactic Acid P (g/L): 40.85
Maltose M (g/L): 126.94
Volume V (L): 6.21

t = 32.00 h

Biomass X (g/L): 87.41
Glucose S (g/L): 13.39
Lactic Acid P (g/L): 62.47
Maltose M (g/L): 136.66
Volume V (L): 6.49

t = 33.00 h

Biomass X (g/L): 69.49
Glucose S (g/L): 14.91
Lactic Acid P (g/L): 57.84
Maltose M (g/L): 135.81
Volume V (L): 6.38

t = 34.00 h

Biomass X (g/L): 82.51
Glucose S (g/L): 14.34
Lactic Acid P (g/L): 57.40
Maltose M (g/L): 112.41

Volume V (L): 6.72

t = 35.00 h

Biomass X (g/L): 80.64

Glucose S (g/L): 11.22

Lactic Acid P (g/L): 47.34

Maltose M (g/L): 124.83

Volume V (L): 7.21

t = 36.00 h

Biomass X (g/L): 79.49

Glucose S (g/L): 10.65

Lactic Acid P (g/L): 53.36

Maltose M (g/L): 122.13

Volume V (L): 6.46

t = 37.00 h

Biomass X (g/L): 87.30

Glucose S (g/L): 18.26

Lactic Acid P (g/L): 63.35

Maltose M (g/L): 125.67

Volume V (L): 6.84

t = 38.00 h

Biomass X (g/L): 71.71

Glucose S (g/L): 21.37

Lactic Acid P (g/L): 66.22

Maltose M (g/L): 127.51

Volume V (L): 6.93

t = 39.00 h

Biomass X (g/L): 75.32

Glucose S (g/L): 23.62

Lactic Acid P (g/L): 62.74

Maltose M (g/L): 105.31

Volume V (L): 6.88

t = 40.00 h

Biomass X (g/L): 81.79

Glucose S (g/L): 16.81

Lactic Acid P (g/L): 77.54

Maltose M (g/L): 107.50
Volume V (L): 6.55

t = 41.00 h

Biomass X (g/L): 72.05
Glucose S (g/L): 17.63
Lactic Acid P (g/L): 60.16
Maltose M (g/L): 121.24
Volume V (L): 6.89

t = 42.00 h

Biomass X (g/L): 95.47
Glucose S (g/L): 21.47
Lactic Acid P (g/L): 77.84
Maltose M (g/L): 113.21
Volume V (L): 6.94

t = 43.00 h

Biomass X (g/L): 76.46
Glucose S (g/L): 24.23
Lactic Acid P (g/L): 68.55
Maltose M (g/L): 90.24
Volume V (L): 7.38

t = 44.00 h

Biomass X (g/L): 91.77
Glucose S (g/L): 28.78
Lactic Acid P (g/L): 68.07
Maltose M (g/L): 94.45
Volume V (L): 7.37

t = 45.00 h

Biomass X (g/L): 76.20
Glucose S (g/L): 28.47
Lactic Acid P (g/L): 76.14
Maltose M (g/L): 115.03
Volume V (L): 6.81

t = 46.00 h

Biomass X (g/L): 91.41
Glucose S (g/L): 25.19

Lactic Acid P (g/L): 62.31
Maltose M (g/L): 115.72
Volume V (L): 7.22

t = 47.00 h

Biomass X (g/L): 69.60
Glucose S (g/L): 28.70
Lactic Acid P (g/L): 65.48
Maltose M (g/L): 94.55
Volume V (L): 7.24

t = 48.00 h

Biomass X (g/L): 98.42
Glucose S (g/L): 24.98
Lactic Acid P (g/L): 57.71
Maltose M (g/L): 116.28
Volume V (L): 8.04

t = 49.00 h

Biomass X (g/L): 95.95
Glucose S (g/L): 29.21
Lactic Acid P (g/L): 77.97
Maltose M (g/L): 103.40
Volume V (L): 7.58

t = 50.00 h

Biomass X (g/L): 80.01
Glucose S (g/L): 29.49
Lactic Acid P (g/L): 72.87
Maltose M (g/L): 94.89
Volume V (L): 7.21

t = 51.00 h

Biomass X (g/L): 77.50
Glucose S (g/L): 35.78
Lactic Acid P (g/L): 80.27
Maltose M (g/L): 98.30
Volume V (L): 7.36

t = 52.00 h

Biomass X (g/L): 89.77

Glucose S (g/L): 33.15
Lactic Acid P (g/L): 72.44
Maltose M (g/L): 91.32
Volume V (L): 7.53

t = 53.00 h

Biomass X (g/L): 70.96
Glucose S (g/L): 29.20
Lactic Acid P (g/L): 81.06
Maltose M (g/L): 96.10
Volume V (L): 7.81

t = 54.00 h

Biomass X (g/L): 85.83
Glucose S (g/L): 30.02
Lactic Acid P (g/L): 64.41
Maltose M (g/L): 83.69
Volume V (L): 7.89

t = 55.00 h

Biomass X (g/L): 77.84
Glucose S (g/L): 32.40
Lactic Acid P (g/L): 80.69
Maltose M (g/L): 99.48
Volume V (L): 7.86

t = 56.00 h

Biomass X (g/L): 83.10
Glucose S (g/L): 28.57
Lactic Acid P (g/L): 68.29
Maltose M (g/L): 104.80
Volume V (L): 7.78

t = 57.00 h

Biomass X (g/L): 79.39
Glucose S (g/L): 34.35
Lactic Acid P (g/L): 72.29
Maltose M (g/L): 86.13
Volume V (L): 7.86

t = 58.00 h

Biomass X (g/L): 77.90
Glucose S (g/L): 35.44
Lactic Acid P (g/L): 79.27
Maltose M (g/L): 97.98
Volume V (L): 7.91

t = 59.00 h

Biomass X (g/L): 101.65
Glucose S (g/L): 38.50
Lactic Acid P (g/L): 94.26
Maltose M (g/L): 89.62
Volume V (L): 7.46

t = 60.00 h

Biomass X (g/L): 92.82
Glucose S (g/L): 39.81
Lactic Acid P (g/L): 80.69
Maltose M (g/L): 102.50
Volume V (L): 8.03

t = 61.00 h

Biomass X (g/L): 71.29
Glucose S (g/L): 38.64
Lactic Acid P (g/L): 81.84
Maltose M (g/L): 74.18
Volume V (L): 7.93

t = 62.00 h

Biomass X (g/L): 76.96
Glucose S (g/L): 32.91
Lactic Acid P (g/L): 68.69
Maltose M (g/L): 101.45
Volume V (L): 7.65

t = 63.00 h

Biomass X (g/L): 93.36
Glucose S (g/L): 35.20
Lactic Acid P (g/L): 75.41
Maltose M (g/L): 77.34
Volume V (L): 8.94

t = 64.00 h

Biomass X (g/L): 94.33
Glucose S (g/L): 38.25
Lactic Acid P (g/L): 73.34
Maltose M (g/L): 91.51
Volume V (L): 8.35

t = 65.00 h

Biomass X (g/L): 71.87
Glucose S (g/L): 43.71
Lactic Acid P (g/L): 87.40
Maltose M (g/L): 86.21
Volume V (L): 8.31

t = 66.00 h

Biomass X (g/L): 91.57
Glucose S (g/L): 44.31
Lactic Acid P (g/L): 73.36
Maltose M (g/L): 88.75
Volume V (L): 8.24

t = 67.00 h

Biomass X (g/L): 87.34
Glucose S (g/L): 41.79
Lactic Acid P (g/L): 87.15
Maltose M (g/L): 77.28
Volume V (L): 8.63

t = 68.00 h

Biomass X (g/L): 66.26
Glucose S (g/L): 52.29
Lactic Acid P (g/L): 63.20
Maltose M (g/L): 106.12
Volume V (L): 8.10

t = 69.00 h

Biomass X (g/L): 80.76
Glucose S (g/L): 50.02
Lactic Acid P (g/L): 88.24
Maltose M (g/L): 78.22
Volume V (L): 8.41

t = 70.00 h

Biomass X (g/L): 83.17
Glucose S (g/L): 49.52
Lactic Acid P (g/L): 67.68
Maltose M (g/L): 78.31
Volume V (L): 8.27

t = 71.00 h

Biomass X (g/L): 84.79
Glucose S (g/L): 48.21
Lactic Acid P (g/L): 83.65
Maltose M (g/L): 105.45
Volume V (L): 8.83

t = 72.00 h

Biomass X (g/L): 71.65
Glucose S (g/L): 48.00
Lactic Acid P (g/L): 68.06
Maltose M (g/L): 91.22
Volume V (L): 8.41

t = 73.00 h

Biomass X (g/L): 76.80
Glucose S (g/L): 48.42
Lactic Acid P (g/L): 81.88
Maltose M (g/L): 76.03
Volume V (L): 9.05

t = 74.00 h

Biomass X (g/L): 77.99
Glucose S (g/L): 40.90
Lactic Acid P (g/L): 67.54
Maltose M (g/L): 85.84
Volume V (L): 8.73

t = 75.00 h

Biomass X (g/L): 78.66
Glucose S (g/L): 44.40
Lactic Acid P (g/L): 86.10
Maltose M (g/L): 84.25

Volume V (L): 9.05